Heinrich Weber

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LEHRBUCH

DER

ALGEBRA

[TEXTBOOK OF ALGEBRA]

BY

HEINRICH WEBER
PROFESSOR OF MATHEMATICS AT THE UNIVERSITY OF STRASSBURG

SECOND EDITION

THIRD VOLUME

WITH TWO FIGURES IN THE TEXT

BRAUNSCHWEIG
PRINTED AND PUBLISHED BY FRIEDRICH VIEWEG AND SON
1908

ELLIPTIC

FUNCTIONS

AND

ALGEBRAIC NUMBERS

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TO RICHARD DEDEKIND,

DAVID HILBERT, HERMANN MINKOWSKI

IN CORDIAL FRIENDSHIP

DEDICATED.

PREFACE.

It has been granted to me, after manifold hindrances, to carry out after all the plan for a continuation of my Textbook of Algebra that I announced twelve years ago in the preface to the first edition of the second volume. Thanks to the obliging attitude of the publishing firm, this third volume of the Algebra appears at the same time as the second edition of the "Elliptic Functions and Algebraic Numbers", first printed in the year 1891.

It is chiefly concerned with the further development of the manifold applications of algebra, and in particular of the theory of quadratic fields, to the problems that have arisen from the elliptic functions. These problems furnish us with the first example, going beyond cyclotomy, of algebraic numbers whose laws are known to some extent. A more thorough treatment of the quadratic fields with negative discriminant serves as the foundation for this. Admittedly, here too not everything has been achieved that I had set myself as the ultimate goal. Thus the elaboration of the theory of relatively cyclic fields still had to be postponed — only for the time being, I hope.

On the other hand, in accordance with a wish that has repeatedly been put to me, I have added an outline of the theory of algebraic functions on an arithmetical foundation, which

essentially follows the paper by Dedekind and myself in the 92nd volume of Crelle's Journal. As it seems to me, however, it achieves a simplification through the application of the theory of functionals, on which I founded the theory of algebraic numbers in the second volume.

Strassburg, May 1908.

H. Weber.

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BOOK ONE.

ANALYTICAL PART.

First Chapter.

The Elliptic Integrals.

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§ 1. Definition of the Elliptic Integrals.

When the systematic exposition of the integral calculus has advanced to the point at which algebraic integrals containing the square root of a function of the first or second degree are reduced to integrals of rational functions, the learner meets at this place a barrier which he is not in a position to pass with the means that stand at his command up to that point. The striving after an extension of those means, so as to make the next class of integrals accessible to research as well, was historically the occasion of a more thorough study of the elliptic integrals and of the introduction of the elliptic functions; and it is also the most natural and most intelligible starting point for one who is to be introduced to the theory of these functions for the first time. It shall therefore be our next task to make ourselves acquainted with the elliptic integrals and with their most important properties.

The definition of an elliptic integral from which we wish to start is the following: let $f(x)$ denote an integral rational function of the third or fourth degree with four distinct roots

$$f(x) = a_0x^4 + 4a_1x^3 + 6a_2x^2 + 4a_3x + a_4, \tag{1}$$

in which a_0 and a_1 do not both vanish at once; let $\Phi(x, y)$ further be an arbitrary integral or fractional rational function of the two arguments x, y. Then

$$\int \Phi(x, \sqrt{f(x)})\, dx$$

is the general elliptic integral, and

$$\Phi(x, \sqrt{f(x)})\, dx$$

the general elliptic differential. But this general integral and differential can now be reduced to essentially simpler ones. First, Φ can be put in the form:

$$\frac{A + B\sqrt{f(x)}}{C + D\sqrt{f(x)}},$$

where A, B, C, D are integral rational functions of x, and, on multiplying numerator and denominator of this fraction by $C - D\sqrt{f(x)}$, in the form

$$\Psi(x) + \frac{\Phi(x)}{\sqrt{f(x)}},$$

where $\Psi(x)$ and $\Phi(x)$ are integral or fractional rational functions of x, namely

$$\Psi(x) = \frac{AC - BDf(x)}{C^2 - D^2f(x)},$$

$$\Phi(x) = \frac{(BC - AD)f(x)}{C^2 - D^2 f(x)}.$$

We now leave out of account the rational integral

$$\int \Psi(x)\, dx,$$

which leads to logarithms and algebraic functions, and occupy ourselves only with the elliptic integral

$$\int \frac{\Phi(x)\, dx}{\sqrt{f(x)}} \tag{2}$$

or with the corresponding elliptic differential

$$\Phi(x)\frac{dx}{\sqrt{f(x)}}. \tag{3}$$

It is sometimes useful to consider this differential in the homogeneous form; to this end we put for x the ratio of two variables $x : y$, and accordingly for dx

$$\frac{y\, dx - x\, dy}{y^2}.$$

If then

$$f(x, y) = a_0 x^4 + 4a_1 x^3 y + 6a_2 x^2 y^2 + 4a_3 x y^3 + a_4 y^4,$$

and $\Phi(x, y)$ is a homogeneous function of order 0, our elliptic differential assumes the shape

$$\Phi(x, y)\frac{y\, dx - x\, dy}{\sqrt{f(x, y)}}. \tag{4}$$

The task that will now occupy us first is a twofold one: by the introduction of new variables the differential

$$\frac{dx}{\sqrt{f(x)}} \quad \text{or} \quad \frac{y\, dx - x\, dy}{\sqrt{f(x, y)}} \tag{5}$$

is to be transformed into another of the same form, but one in which the function under the square-root sign is made as simple as possible and, in particular, made to depend on as small a number of parameters as possible (§ 3, 4, 5). Secondly, the general differential (3) or (4) is to be decomposed into other similar ones in which the function $\Phi(x)$ or $\Phi(x, y)$ is as simple as possible (§ 12).

The function $f(x)$ can be decomposed into linear factors (Bd. I, § 33). We accordingly set also

$$f(x) = a_0(x - x_1)(x - x_2)(x - x_3)(x - x_4).$$

Let us think of the quantities $x, x_1, x_2, \ldots$ as abscissas of points laid off on a straight line from an arbitrary origin, the positive ones towards the one side, the negative ones towards the other (for example, positive to the right, negative to the left). Then the differences $x - x_1, x - x_2, x - x_3, x - x_4$ represent the distances of the point x from x_1, x_2, x_3, x_4, and indeed reckoned positive when x lies towards the positive side of $x_1, x_2, \ldots$. In the same sense $x_2 - x_1$ is the distance of the point x_2 from x_1, and so on.

We also denote the differences more briefly by $(xx_1), (xx_2), \ldots (x_2x_1), \ldots$ We use this mode of representation only for the sake of an easier survey, and we do not exclude imaginary points of the line.

§ 2. Cross Ratios.

The simplest means for the simplification of the elliptic differential rests upon the linear transformation of the function $f(x)$ (Bd. I, § 67). Let $\alpha, \beta, \gamma, \delta$ denote four arbitrary constants whose determinant

$$\alpha\delta - \beta\gamma = r \tag{1}$$

is different from zero. We set

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}, \tag{2}$$

and call this a linear substitution. If we interpret x and x', as in § 1, as points on two straight lines L and L', then (2) fixes a mutually one-to-one relation between the points of L and L', that is, a mapping. To the values $x = \infty$ and $x' = \infty$ there correspond the infinitely distant parts of the straight lines, which we likewise regard as definite points. The point $x' = \infty$ then corresponds to the point $x = \alpha : \gamma$, and the point $x = \infty$ to the point $x' = -\delta : \gamma$; and only when $\gamma = 0$, that is, when the substitution (2) is integral, do the infinitely distant points on L and L' correspond to one another.

The substitution (2) is not altered if the four transformation numbers are multiplied by the same factor. The determinant r is multiplied by the square of this factor, and one can therefore choose this factor so that the determinant takes any prescribed value, for example the value 1.

If we apply the substitution (2) to any two points x_1, x_2 and the corresponding x_1', x_2', we obtain

$$(x_1 x_2) = \frac{r\,(x_1' x_2')}{(\gamma x_1' + \delta)(\gamma x_2' + \delta)},$$

and if we take in addition a second pair of points x_3, x_4 and the corresponding x_3', x_4':

$$(x_1 x_2)(x_3 x_4) = \frac{r^2 (x_1' x_2')(x_3' x_4')}{(\gamma x_1' + \delta)(\gamma x_2' + \delta)(\gamma x_3' + \delta)(\gamma x_4' + \delta)}.$$

If we here interchange x_2 with x_3 and form the quotient, there follows

$$\frac{(x_1 x_2)(x_3 x_4)}{(x_1 x_3)(x_2 x_4)} = \frac{(x_1' x_2')(x_3' x_4')}{(x_1' x_3')(x_2' x_4')}. \tag{3}$$

The expression on the left side is called the cross ratio of the pair of points $x_1 x_4$ to the pair of points $x_2 x_3$, and this formula therefore contains the theorem:

The cross ratio of two pairs of points is invariant under simultaneous linear transformation of the four points.

Four points can be decomposed into two pairs in six ways. But if we set

$$\begin{aligned} a &= (x_2 x_3)(x_1 x_4) \\ b &= (x_3 x_1)(x_2 x_4), \\ c &= (x_1 x_2)(x_3 x_4) \end{aligned}$$

then the identity holds:

$$a + b + c = 0 \tag{4}$$

and we obtain the six cross ratios

$$\begin{aligned} &-\frac{c}{b}, \quad -\frac{a}{c}, \quad -\frac{b}{a}, \\ &-\frac{b}{c}, \quad -\frac{c}{a}, \quad -\frac{a}{b}; \end{aligned} \tag{5}$$

if we set the first of them $-c : b = \varkappa^2$, then by (4) one obtains the six:

$$\varkappa^2, \quad 1 - \frac{1}{\varkappa^2}, \quad \frac{1}{1-\varkappa^2}, \quad \frac{1}{\varkappa^2}, \quad \frac{\varkappa^2}{\varkappa^2 - 1}, \quad 1 - \varkappa^2. \tag{6}$$

Thus the six cross ratios formed from the four points are expressed linearly by one among them.

If the x_1, x_2, x_3, x_4 are permuted among themselves, then the a, b, c are permuted among themselves and change their signs, yet in such a way that either all three signs are changed at once or all remain unchanged. For example the transpositions

$$(14) \text{ and } (23) \begin{pmatrix} a & b & c \\ -a & -c & -b \end{pmatrix}$$
$$(24) \text{ and } (31) \begin{pmatrix} a & b & c \\ -c & -b & -a \end{pmatrix}$$
$$(34) \text{ and } (12) \begin{pmatrix} a & b & c \\ -b & -a & -c \end{pmatrix}$$

If x_1, x_2, x_3, x_4, are real, then $\varkappa^2$ is a positive proper fraction then and only then, when $\varkappa^2$ and $1 - \varkappa^2$ are positive, that is, when either a and c are positive and b negative, or a and c negative and b positive. This occurs when x_1, x_2, x_3, x_4 follow one another in this order, ascending in magnitude, and in all those arrangements which arise from it by such permutations as leave b unchanged. These are the following eight:

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3
1	4	3	2
4	3	2	1
3	2	1	4
2	1	4	3

If we therefore imagine x_1, x_2, x_3, x_4 placed in this order on the circumference of a circle, we always obtain a positive properly fractional $\varkappa^2$ when the x_i follow one another in order of quantity in such a way that one begins with an arbitrary one as the smallest and then counts onward around the circle either to the right or to the left. We express this as follows:

In order that $\varkappa^2$ be a positive proper fraction, the quantities x_1, x_2, x_3, x_4 must follow one another cyclically in order of quantity.

§ 3. Linear Transformation of the Elliptic Differential.

If the two points x_3 and x_4 approach one and the same point x, then x_3' and x_4' approach the corresponding point x'. We then set $(x_3 x_4) : (x_3' x_4') = dx : dx'$ and obtain from (3), § 2, the relation between the differentials dx, dx':

$$\frac{(x_1 x_2)\, dx}{(x_1 x)(x_2 x)} = \frac{(x_1' x_2')\, dx'}{(x_1' x')(x_2' x')}. \tag{1}$$

In the same way, if in place of x_1, x_2 one puts two other points x_3, x_4, there results:

$$\frac{(x_3 x_4)\, dx}{(x_3 x)(x_4 x)} = \frac{(x_3' x_4')\, dx'}{(x_3' x')(x_4' x')}, \tag{2}$$

and if one multiplies and extracts the root:

$$\frac{\sqrt{(x_1 x_2)(x_3 x_4)}\, dx}{\sqrt{(x_1 x)(x_2 x)(x_3 x)(x_4 x)}} = \frac{\sqrt{(x_1' x_2')(x_3' x_4')}\, dx'}{\sqrt{(x_1' x')(x_2' x')(x_3' x')(x_4' x')}}, \tag{3}$$

hence a transformation of the elliptic differential by a linear substitution:

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}.$$

This substitution is completely determined when, for three arbitrary points x_1, x_2, x_3, the corresponding values x_1', x_2', x_3' are arbitrarily given, and one obtains it from the equality of the cross ratio:

$$\frac{(x_1x_2)(x_3x)}{(x_1x_3)(x_2x)} = \frac{(x_1'x_2')(x_3'x')}{(x_1'x_3')(x_2'x')}, \tag{4}$$

by solving for x; and the fourth value x_4', the one belonging to x_4, is obtained from

$$\frac{(x_1x_2)(x_3x_4)}{(x_1x_3)(x_2x_4)} = \frac{(x_1'x_2')(x_3'x_4')}{(x_1'x_3')(x_2'x_4')}. \tag{5}$$

If the function $f(x)$ is given in the elliptic differential

$$du = \frac{dx}{\sqrt{f(x)}}$$

then the x_1, x_2, x_3, x_4 are thereby given as well, and by the linear transformation (3) one can bring the differential into another form in which three of the quantities x_1', x_2', x_3', x_4' have arbitrarily given values. One obtains the normal form under the following assumptions:

$$x' = z, \quad x_1' = 0, \quad x_2' = 1, \quad x_3' = \frac{1}{\varkappa^2}, \quad x_4' = \infty$$

The quantity $\varkappa$ is called by Legendre the modulus of the elliptic differential, and $\sqrt{1-\varkappa^2} = \varkappa'$ the complement of the modulus. If, then, one assigns the points to one another in the following way:

$$\begin{array}{cccccc} z, & 0, & 1, & \frac{1}{\varkappa^2}, & \infty, \\ x, & x_1, & x_2, & x_3, & x_4, \end{array}$$

then from the equality of the cross ratios the relations follow easily:

$$\begin{aligned} z &= \frac{(xx_1)(x_2x_4)}{(xx_4)(x_2x_1)}, \\ 1-z &= \frac{(xx_2)(x_1x_4)}{(xx_4)(x_1x_2)}, \\ 1-\varkappa^2 z &= \frac{(xx_3)(x_1x_4)}{(xx_4)(x_1x_3)}, \end{aligned} \tag{6}$$

$$dz = \frac{(x_4x_1)(x_4x_2)}{(x_2x_1)} \frac{dx}{(xx_4)^2}, \tag{7}$$

$$\varkappa^2 = \frac{(x_3x_4)(x_1x_2)}{(x_1x_3)(x_2x_4)}, \quad \varkappa'^2 = \frac{(x_2x_3)(x_1x_4)}{(x_2x_4)(x_1x_3)} \tag{8}$$

and from (3):

(9) $$\frac{\sqrt{(x_3x_1)(x_4x_2)}\,dx}{\sqrt{-(x_1x)(x_2x)(x_3x)(x_4x)}} = \frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}.$$

According to § 2, $\varkappa^2$ becomes a positive proper fraction if the x_1, x_2, x_3, x_4, are real and follow one another cyclically in order of magnitude. This therefore gives eight different such transformations,

and among them one can select two in each case, for which the interval $z = 0$ to $z = 1$ corresponds to a given one of the intervals (x_1x_2), (x_2x_3), (x_3x_4), (x_4x_1); to increasing z there correspond, in one of these transformations, the increasing x, and in the other the decreasing x (with the passage from $+\infty$ to $-\infty$). In the first case the square roots in (9) have the same sign on both sides, in the second the opposite sign.

If one does not insist that $\varkappa^2$ be a positive proper fraction, then the points x_1, x_2, x_3, x_4 may be permuted in all ways, and one obtains 24 different linear transformations into the normal form, of which four at a time yield the same $\varkappa^2$; one obtains in all six different moduli, which are derived from one another according to § 2, (6).

One surveys the totality of these transformations most easily by assuming that the integral to be transformed already has the normal form:

$$\frac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}};$$

one has then to replace, in the formulae (6) to (9), the x_1, x_2, x_3, x_4 in all possible ways by 0, 1, $1 : \lambda^2$, ∞. We take as an example the following assignments:

	$x_1,$	$x_2,$	$x_3,$	$x_4,$
1)	$0,$	$\infty,$	$\frac{1}{\lambda^2},$	$1,$
2)	$0,$	$\frac{1}{\lambda^2},$	$1,$	$\infty,$
3)	$1,$	$0,$	$\infty,$	$\frac{1}{\lambda^2}.$

One then obtains from (6) and (8)

$$\begin{aligned}
&1) \quad z = \frac{-x}{1-x}, && \varkappa^2 = 1-\lambda^2 = \lambda'^2,\\
&2) \quad z = \lambda^2 x, && \varkappa^2 = \frac{1}{\lambda^2},\\
&3) \quad z = \frac{1-x}{1-\lambda^2 x}, && \varkappa^2 = \lambda^2
\end{aligned}$$

and for the differential

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

results in the three cases:

$$\begin{aligned}
&1) \quad \frac{dx}{\sqrt{-x(1-x)(1-\lambda^2 x)}},\\
&2) \quad \frac{\lambda\,dx}{\sqrt{x(1-x)(1-\lambda^2 x)}},\\
&3) \quad \frac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}}.
\end{aligned}$$

§ 4. The Legendre Normal Form.

If the function $f(x)$, which stands under the root sign in the elliptic differential, has real coefficients, then three cases are possible:

1. $f(x)$ has four real roots x_1, x_2, x_3, x_4;
2. $f(x)$ has two real roots x_1, x_2 and a pair of conjugate imaginary roots x_3, x_4;
3. $f(x)$ has two pairs of conjugate imaginary roots x_1, x_2 and x_3, x_4.

In all three cases the elliptic differential can be brought by a real linear transformation into the form:

$$\frac{dz}{\sqrt{(z^2-\alpha)(z^2-\beta)}}, \tag{1}$$

in which α and β are real (positive or negative) constants.

We determine the linear dependence between the variables x and z in such a way that the following values correspond to one another:

$$\begin{array}{cccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \sqrt{\alpha}, & -\sqrt{\alpha}, & \sqrt{\beta}, & -\sqrt{\beta}, \end{array} \tag{2}$$

and obtain a substitution of the form:

$$h\,\frac{x-x_1}{x-x_2} = \frac{z-\sqrt{\alpha}}{z+\sqrt{\alpha}}. \tag{3}$$

In order to determine h we set $x = x_3$, $x = x_4$ and correspondingly $z = \sqrt{\beta}$, $z = -\sqrt{\beta}$, and obtain

$$h\,\frac{(x_3x_1)}{(x_3x_2)} = \frac{\sqrt{\beta}-\sqrt{\alpha}}{\sqrt{\beta}+\sqrt{\alpha}}, \quad h\,\frac{(x_4x_1)}{(x_4x_2)} = \frac{\sqrt{\beta}+\sqrt{\alpha}}{\sqrt{\beta}-\sqrt{\alpha}}.$$

From this, by multiplication and division,

$$h = \sqrt{\frac{(x_3x_2)(x_4x_2)}{(x_3x_1)(x_4x_1)}}, \quad \frac{\sqrt{\alpha}-\sqrt{\beta}}{\sqrt{\alpha}+\sqrt{\beta}} = \sqrt{\frac{(x_3x_1)(x_4x_2)}{(x_3x_2)(x_4x_1)}}. \tag{4}$$

If we set, as in § 2

$$\begin{array}{c} a = (x_2x_3)(x_1x_4), \\ b = (x_3x_1)(x_2x_4), \\ c = (x_1x_2)(x_3x_4), \\ a+b+c = 0, \end{array} \tag{5}$$

then, since only the ratio of α to β matters, we can satisfy the last of the equations (4) by setting:

$$\sqrt{\alpha} = \sqrt{\pm a} + \sqrt{\mp b}, \quad \sqrt{\beta} = \sqrt{\pm a} - \sqrt{\mp b}, \tag{6}$$

and from (3) there results for z the expression

(7) $$z = \sqrt{\alpha}\,\frac{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} + (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} - (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}.$$

If beside (3) one sets up also the equation which follows from it:

$$h'\,\frac{x - x_3}{x - x_4} = \frac{z - \sqrt{\beta}}{z + \sqrt{\beta}}$$

then by logarithmic differentiation there results:

$$\frac{(x_1x_2)\,dx}{(xx_1)(xx_2)} = \frac{2\sqrt{\alpha}\,dz}{z^2 - \alpha}, \qquad \frac{(x_3x_4)\,dx}{(xx_3)(xx_4)} = \frac{2\sqrt{\beta}\,dz}{z^2 - \beta},$$

and from this by multiplication, with regard to (5) and (6):

(8) $$\frac{dx}{\sqrt{\pm(xx_1)(xx_2)(xx_3)(xx_4)}} = \frac{2\,dz}{\sqrt{(z^2 - \alpha)(z^2 - \beta)}}.$$

If now the four roots of $f(x)$ are real, then, as we have seen in § 2, there are eight ways of assigning these roots to the symbols x_1, x_2, x_3, x_4 in such a manner that a and b have opposite signs; and if therefore $\pm a$, $\mp b$ are positive, then $\sqrt{\alpha}$, $\sqrt{\beta}$ become real, hence α, β positive, and by (7) z then becomes real as well.

If, secondly, x_1, x_2 are real and x_3, x_4 conjugate imaginary, then a and $-b$ are conjugate imaginary; hence, if the signs of the square roots $\sqrt{\pm a}$, $\sqrt{\mp b}$ are suitably determined, $\sqrt{\alpha}$ becomes real and $\sqrt{\beta}$ purely imaginary, hence α positive, β negative and z real [because $(x_3x_1)(x_4x_1)$ and $(x_3x_2)(x_4x_2)$, as products of conjugate imaginary quantities, are positive].

If, finally, x_1, x_2 and x_3, x_4 are two conjugate imaginary pairs, then a and $-b$ are both positive. If we therefore take the lower signs in (6), then $\sqrt{\alpha}$, $\sqrt{\beta}$ become purely imaginary, hence α and β negative, and from (7) a real expression results for z.

If one then sets

(9) $$z^2 = y,$$

there results

(10) $$\frac{2\,dz}{\sqrt{(z^2 - \alpha)(z^2 - \beta)}} = \frac{dy}{\sqrt{y(y - \alpha)(y - \beta)}}$$

and one has thus obtained, by means of the quadratic substitution (9), an elliptic differential under whose root sign there stands a function of the third degree with real roots. By § 3 this differential can be brought by a linear substitution into the normal form

$$\frac{d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}}$$

and therein ζ and $\varkappa$ may be assumed to be positive proper fractions.

The Legendre normal form results from this if one sets:

$$\zeta = \sin^2\varphi, \qquad d\zeta = 2\sin\varphi\cos\varphi\,d\varphi$$

(11) $$\frac{d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}} = \frac{2\,d\varphi}{\sqrt{1 - \varkappa^2\sin^2\varphi}}.$$

§ 5. The Weierstrass Normal Form.

Weierstrass made another normal form of the elliptic differential the basis of his investigations, namely the form

$$du = \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}, \tag{1}$$

in which g_2, g_3 are constants that are called the first and second invariant of the differential. The general elliptic differential

$$\frac{dx}{\sqrt{f(x)}}, \tag{2}$$

in which

$$f(x) = a_0 x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4 \tag{3}$$

holds, can be brought into the Weierstrass normal form by a linear transformation, provided the roots x_1, x_2, x_3, x_4 of $f(x)$ are known. This is done most simply in the following way.

Let the roots of the cubic function

$$\varphi(z) = 4z^3 - g_2 z - y_3$$

be denoted by e_1, e_2, e_3. Then we have

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3)$$

and $e_1 + e_2 + e_3 = 0$.

We let the values of x and z correspond to one another in the following way:

$$\begin{array}{cccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \infty, & e_1, & e_2, & e_3, \end{array}$$

and, if we denote by m a constant factor which may be assumed arbitrarily, we obtain:

$$\begin{aligned} z - e_1 &= \frac{m(xx_2)}{(x_1x_2)(xx_1)} = m\left(\frac{1}{(x_1x_2)} - \frac{1}{(x_1x)}\right) \\ z - e_2 &= \frac{m(xx_3)}{(x_1x_3)(xx_1)} = m\left(\frac{1}{(x_1x_3)} - \frac{1}{(x_1x)}\right) \\ z - e_3 &= \frac{m(xx_4)}{(x_1x_4)(xx_1)} = m\left(\frac{1}{(x_1x_4)} - \frac{1}{(x_1x)}\right) \end{aligned} \tag{4}$$

The factor m must be the same in all three equations, in order that the differences $(z - e_3) - (z - e_2) = e_2 - e_3$ etc. become independent of x.

If we form these differences and put, as in Algebra Bd. 1, § 70

$$\begin{aligned} (x_1x_2)(x_3x_4) &= U, \\ (x_1x_3)(x_4x_2) &= V, \\ (x_1x_4)(x_2x_3) &= W, \end{aligned}$$

then it follows that

$$e_2 - e_3 = \frac{-mU}{(x_1x_2)(x_1x_3)(x_1x_4)}.$$

If we introduce a new arbitrary factor μ by setting

$$m = 3\mu a_0 (x_1x_2)(x_1x_3)(x_1x_4) \tag{5}$$

then it follows that

$$e_2 - e_3 = -3\mu a_0 U, \quad e_3 - e_1 = -3\mu a_0 V, \quad e_1 - e_2 = -3\mu a_0 W$$

and from this, because $e_1 + e_2 + e_3 = 0$,

$$\begin{aligned} e_1 &= \mu a_0(V - W), \\ e_2 &= \mu a_0(W - U), \\ e_3 &= \mu a_0(U - V). \end{aligned}$$

The quantities y_1, y_2, y_3 introduced in Algebra I, § 70 are therefore equal to $e_1{:}\mu$, $e_2{:}\mu$, $e_3{:}\mu$, and by formula (11) there one obtains for the e_1, e_2, e_3 the cubic equation:

$$z^3 - 3A\mu^2 z + \mu^3 B = 0.$$

Thus, if, with $\varphi(z) = z^3 - g_2 z - g_3$,

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3) = 4z^3 - g_2 z - g_3$$

is to hold,

$$g_2 = 12\mu^2 A, \qquad g_3 = -4\mu^3 B. \tag{6}$$

In these,

$$\begin{aligned} A &= a_2^2 - 3a_1 a_3 + 12 a_0 a_4 \\ B &= 27 a_1^2 a_4 + 27 a_0 a_3^2 + 2a_2^3 - 72 a_0 a_2 a_4 - 9 a_1 a_2 a_3 \end{aligned} \tag{7}$$

are the first and second invariants of the biquadratic form $f(x)$.

Logarithmic differentiation of the first two equations (4) gives

$$\frac{dz^2}{(z - e_1)(z - e_2)} = \frac{dx^2 (x_1 x_2)(x_1 x_3)}{(x x_1)^2 (x x_2)(x x_3)},$$

and with the help of the last equation (4), by (5):

$$\frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} = \frac{dx}{\sqrt{12\mu f(x)}}. \tag{8}$$

If we wish to apply these results to the case in which $f(x)$ has the normal form

$$f(x) = x(1 - x)(1 - \varkappa^2 x) = x - x^2(1 + \varkappa^2) + \varkappa^2 x^3$$

and the point $z = \infty$ corresponds to the point $x = 0$, then we let a_0 pass over into zero and x_4 into infinity, but the product $a_0 x_4$ into a finite value, which we may take to be $= 1$.

We then have

$$\begin{aligned} a_0 x_4 = 1, \qquad a_0 = 0, \qquad a_1 = \varkappa^2, \\ a_2 = -(1 + \varkappa^2), \qquad a_3 = 1, \qquad a_4 = 0 \end{aligned}$$

and consequently

$$\begin{aligned} A &= (1 + \varkappa^2)^2 - 3\varkappa^2 = 1 - \varkappa^2 - \varkappa^4 = 1 - \varkappa^2 \varkappa'^2 \\ B &= -(1 + \varkappa^2)[2(1 + \varkappa^2)^2 - 9\varkappa^2] \\ &= -(1 + \varkappa^2)(2 - \varkappa^2)(1 - 2\varkappa^2) \\ &= (1 + \varkappa^2)(1 + \varkappa'^2)(\varkappa^2 - \varkappa'^2) \\ &= (2 + \varkappa^2 \varkappa'^2)(\varkappa^2 - \varkappa'^2), \end{aligned}$$

when $\varkappa'^2 = 1 - \varkappa^2$ is put.

Thus, if we further append the discriminant

$$\Delta = 16 \,.\, 27\, \mu^6 (4A^3 - B^2)$$

we obtain

$$\begin{aligned} g_2 &= 12\mu^2(1 - \varkappa^2\varkappa'^2), \\ g_3 &= -4\mu^3(2 + \varkappa^2\varkappa'^2)(\varkappa^2 - \varkappa'^2), \\ \Delta &= g_2^3 - 27g_3^2 = 27^2 \,.\, 16\, \mu^6 \varkappa^4 \varkappa'^4, \end{aligned} \tag{9}$$

and from (6) one obtains the transformation

$$\frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} = \frac{dx}{\sqrt{12\mu\, x(1-x)(1-\varkappa^2 x)}}; \tag{10}$$

to find the substitution, we set $x_1 = 0$, $x_2 = 1$, $x_3 = 1{:}\varkappa^2$, $x_4 = \infty$, $a_0 z_4 = 1$ and obtain

$$a_0 U = 1, \qquad a_0 V = \frac{-1}{\varkappa^2}, \qquad a_0 W = \frac{\varkappa'^2}{\varkappa^2},$$

consequently

$$e_1 = \mu\, \frac{\varkappa^2 - 2}{\varkappa^2}, \qquad e_2 = \mu\, \frac{1 - 2\varkappa^2}{\varkappa^2}, \qquad e_3 = \mu\, \frac{1 + \varkappa^2}{\varkappa^2}$$

$$z = \frac{3\mu}{\varkappa^2}\left(\frac{\varkappa^2 + 1}{3} - \frac{1}{x}\right). \tag{11}$$

In this transformation of the elliptic differential into the Weierstrass normal form, the decomposition of the function $f(x)$ into its linear factors, and hence the solution of the biquadratic equation $f(x) = 0$, is presupposed. Another transformation, not linear, to be sure, which attains the same end without this assumption, has been given by Hermite (Crelles Journal, Bd. 52). In order to present it, we use the homogeneous form of the elliptic differential

$$\frac{y\, dx - x\, dy}{\sqrt{f(x,y)}}, \tag{12}$$

in which

$$f(xy) = a_0 x^4 + a_1 x^3 y + a_2 x^2 y^2 + a_3 x y^3 + a_4 y^4. \tag{13}$$

Besides the two invariants A, B, this biquadratic form has two further covariants:

$$\begin{aligned} H &= \frac{1}{3}\left[\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2\right] \\ T &= \frac{1}{12}\left(\frac{\partial f}{\partial x}\frac{\partial H}{\partial y} - \frac{\partial f}{\partial y}\frac{\partial H}{\partial x}\right), \end{aligned} \tag{14}$$

where H and T are forms of the fourth and sixth degree, whose coefficients are composed rationally, and with integral numerical coefficients, out of the coefficients of f. Between these forms there subsists the further relation

$$H^3 - 48\, AH f^2 - 64\, B f^3 = -27\, T^2 \tag{15}$$

(Bd. I, § 70, 72).

But by Euler's theorem on homogeneous functions

$$4f = \frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y, \qquad df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy,$$
$$4H = \frac{\partial H}{\partial x}x + \frac{\partial H}{\partial y}y, \quad dH = \frac{\partial H}{\partial x}dx + \frac{\partial H}{\partial y}dy,$$

whence

$$f\,dH - H\,df = -3\,T(y\,dx - x\,dy),$$

and so, if in accordance with (15)

$$3\sqrt{3}\,T = \sqrt{-H^3 + 48\,Af^2H + 64\,Bf^3} \tag{16}$$

is set:

$$\frac{y\,dx - x\,dy}{\sqrt{f}} = \frac{-\sqrt{3}\,(f\,dH - H\,df)}{\sqrt{-(H^3 - 48\,Af^2H - 64\,Bf^3)f}}.$$

If we now make the substitution

$$z = -\frac{\mu H}{4f} \tag{17}$$

with an indeterminate factor μ, there results

$$\frac{y\,dx - x\,dy}{\sqrt{3\mu f}} = \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}, \tag{18}$$

if, as before,

$$g_2 = 12\,\mu^2 A, \qquad g_3 = -4\,\mu^3 B. \tag{19}$$

We can apply this transformation to a differential which already has the normal form. If we set

$$f(x, y) = 4x^3y - g_2xy^3 - g_3y^4,$$

then we have to replace

$$a_0 \quad a_1 \quad a_2 \quad a_3 \quad a_4$$

by

$$0 \quad 4 \quad 0 \quad -g_2 \quad -g_3$$

and from (7) there results

$$A = 12\,g_2, \qquad B = -27 \cdot 16\,g_3.$$

If we therefore set $\mu = \frac{1}{12}$, the equations (19) pass over into the identities $g_2 = g_2$, $g_3 = g_3$.

If we then set $y = 1$, the transformation (18) gives

$$\frac{2\,dx}{\sqrt{4\,x^3 - g_2\,x - g_3}} = \frac{dz}{\sqrt{4\,z^3 - g_2\,z - g_3}}\,. \tag{20}$$

This transformation is mediated, according to (16), by

$$z = \frac{-H}{48\,f}$$

But from (14) it follows that:

$$H = -48\left[(x^2 + \tfrac{1}{4}g_2)^2 + 2\,g_3\,x\right],$$

and consequently we obtain

$$z = \frac{(x^2 + \tfrac{1}{4}g_2)^2 + 2\,g_3\,x}{4\,x^3 - g_2\,x - g_3}. \tag{21}$$

There follows further from (14)

$$T = 64\,x^6 - 80\,g_2\,x^4 - 320\,g_3\,x^3 - 20\,g_2^2\,x^2 - 16\,g_2\,g_3\,x + g_2^3 - 32\,g_3^2$$

and then, according to (16),

$$4\sqrt{4\,x^3 - g_2\,x - g_3}^{\,3}\ \sqrt{4\,z^3 - g_2\,z - g_3} = T. \tag{22}$$

By (20), (21), (22) the multiplication of the elliptic differential by 2 is accomplished. Not only is z thereby represented as a single-valued (rational) function of x, but one square root is also represented uniquely by the other; i. e., a point z is uniquely assigned to each point x.

§ 6. Elliptic Curves.

Every algebraic dependence between two variables x, y is expressed by an equation of the form

$$F(x, y) = 0, \tag{1}$$

in which $F(x, y)$ denotes an integral function of the two variables x, y. If one regards x and y as Cartesian coordinates of a point in the plane, then (1) is the equation of a curve of degree n, provided F, with respect to x and y taken together, is of the nth dimension. This curve is then the geometric image of the algebraic dependence, where, however, imaginary points must also be taken into account.

If then $\Phi(x, y)$ is an integral or fractional rational function of x and y, and y depends on x through equation (1), then

$$\Phi(x, y)\,dx, \qquad \int \Phi(x, y)\,dx \tag{2}$$

are the algebraic differentials and integrals belonging to the curve F.

For example, if $f(x)$ is a function of the third or fourth degree in x, then

$$y^2 - f(x) = 0$$

is the equation of a curve of the third or fourth degree, to which belong the elliptic differentials and integrals affected with the irrationality $\sqrt{f(x)}$.

We shall here call all curves whose differentials and integrals are reducible to elliptic ones elliptic curves. As the example shows, curves of the third and fourth degree occur among them.

Conics do not belong to the elliptic curves, because, when an equation of the second degree subsists between x and y, x and y can be represented as rational functions of a parameter t, whereby the algebraic differential can be reduced to a rational one in t. Such curves are called rational curves.

We shall see that all curves of the third degree without a double point or point of regression belong to the elliptic ones. Curves of higher degree can belong to them only when they have a certain number of singular points. This is a fundamental chapter in the general theory of algebraic functions, which does not belong to the plan of this work[1].

For the investigation of algebraic differentials from this point of view it is expedient to introduce homogeneous variables. We put $x = x_1{:}x_3$, $y = x_2{:}x_3$, $x_3^n F(x,y) = f(x_1,x_2,x_3)$; then $f(x_1,x_2,x_3)$ is a homogeneous function of order n in the three variables x_1,x_2,x_3, and

$$f(x_1,x_2,x_3) = 0 \tag{3}$$

is the equation of a curve of order n in homogeneous coordinates. There results

$$dx = \frac{x_3\,dx_1 - x_1\,dx_3}{x_3^2}$$

and

$$d\Omega = \Phi(x,y)\,dx = \frac{1}{x_3^2}\,\Phi\Big(\frac{x_1}{x_3},\frac{x_2}{x_3}\Big)\,(x_3\,dx_1 - x_1\,dx_3). \tag{4}$$

Now by Euler's theorem on homogeneous functions, if we denote by f_1,f_2,f_3 the partial derivatives of f with respect to x_1,x_2,x_3, we have

$$\begin{aligned} f_1x_1 + f_2x_2 + f_3x_3 &= 0,\\ f_1\,dx_1 + f_2\,dx_2 + f_3\,dx_3 &= 0, \end{aligned}$$

hence, if ϱ denotes a factor of proportionality,

$$\begin{aligned} f_1 &= \varrho(x_2\,dx_3 - x_3\,dx_2),\\ f_2 &= \varrho(x_3\,dx_1 - x_1\,dx_3),\\ f_3 &= \varrho(x_1\,dx_2 - x_2\,dx_1), \end{aligned}$$

and if one denotes by c_1,c_2,c_3 entirely arbitrary quantities, e. g. constants:

$$c_1f_1 + c_2f_2 + c_3f_3 = \varrho\,\Sigma \pm c_1x_2\,dx_3,$$

where, in the usual way, $\Sigma \pm c_1x_2\,dx_3$ denotes the determinant

$$\begin{vmatrix} c_1 & c_2 & c_3\\ x_1 & x_2 & x_3\\ dx_1 & dx_2 & dx_3 \end{vmatrix}$$

It follows therefore:

$$x_3\,dx_1 - x_1\,dx_3 = \frac{f_2\,\Sigma \pm c_1x_2\,dx_3}{c_1f_1 + c_2f_2 + c_3f_3},$$

and if one puts

$$\frac{f_2}{x_3^2}\,\Phi\Big(\frac{x_1}{x_4},\frac{x_2}{x_3}\Big) = \Psi$$

[1]The most important papers concerning this subject are:

Riemann, Theorie der Abelschen Funktionen [Crelles Journal, Bd. 54 (1857)]. Mathematische Werke, 2. Aufl., S. 88, 487. Nachträge, edited by Noether and Wirtinger (1902).

Aronhold, Monatsberichte der Berliner Akademie of 25 April 1861.

Clebsch, Über die Anwendung der Abelschen Funktionen in der Geometrie (Crelles Journal, Bd. 63).

Clebsch, Über diejenigen ebenen Kurven, deren Koordinaten rationale Funktionen eines Parameters sind (ibid., Bd. 64).

Clebsch, Über diejenigen Kurven, deren Koordinaten sich als elliptische Funktionen eines Parameters darstellen lassen (ibid., Bd. 64).

Brill, Über diejenigen Kurven, deren Koordinaten sich als hyperelliptische Funktionen eines Parameters darstellen lassen (ibid., Bd. 65).

Clebsch and Gordan, Theorie der Abelschen Funktionen (Teubner, Leipzig 1866).

is set, there results from (4) the expression for the most general algebraic differential belonging to the curve f:

$$d\Omega = \frac{\Psi\Sigma \pm c_1 x_2 dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}, \tag{5}$$

in which Ψ is an integral or fractional homogeneous function of order $(n-3)$. The arbitrary quantities c_1, c_2, c_3 occur only apparently in this expression. In reality it is entirely independent of them.

We consider again the case $n = 3$. Then the simplest case is that in which Ψ is a constant, and (5) then passes over into the elliptic differential of the first kind:

$$du = \frac{\Sigma \pm c_1 x_2 dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}. \tag{6}$$

This we now propose to transform, by means of the covariants of the ternary form of the third degree, into the Weierstrass form.

In Bd. II, § 107 we became acquainted with the fundamental covariants of the cubic form. These were, besides f itself:

$$\Delta = \frac{1}{6^3}\begin{vmatrix} f_{11}, & f_{12}, & f_{13} \\ f_{21}, & f_{22}, & f_{23} \\ f_{31}, & f_{32}, & f_{33} \end{vmatrix}, \quad J = \frac{1}{36}\begin{vmatrix} f_{11}, & f_{12}, & f_{13}, & \Delta_1 \\ f_{21}, & f_{22}, & f_{23}, & \Delta_2 \\ f_{31}, & f_{32}, & f_{33}, & \Delta_3 \\ \Delta_1, & \Delta_2, & \Delta_3, & 0 \end{vmatrix},$$

$$K = \frac{1}{9}\begin{vmatrix} f_1, & \Delta_1, & J_1 \\ f_2, & \Delta_2, & J_2 \\ f_3, & \Delta_3, & J_3 \end{vmatrix},$$

where the indices 1, 2, 3 denote differentiation with respect to the variables x_1, x_2, x_3.

It follows from the multiplication theorem of determinants, since f and $df = 0$, that:

$$\begin{vmatrix} c_1, & c_2, & c_3 \\ x_1, & x_2, & x_3 \\ dx_1, & dx_2, & dx_3 \end{vmatrix} \begin{vmatrix} f_1, & f_2, & f_3 \\ \Delta_1, & \Delta_2, & \Delta_3 \\ J_1, & J_2, & J_3 \end{vmatrix} = \begin{vmatrix} \Sigma x_i \Delta_i, & d\Delta \\ \Sigma x_i J_i, & dJ \end{vmatrix} \Sigma c_i f_i$$

$$= 3\Sigma c_i f_i (\Delta dJ - 2J d\Delta),$$

and from this

$$du = 3\frac{\Delta dJ - 2J d\Delta}{K}. \tag{7}$$

With the help of the covariants we have reduced the form $f(x)$ to the canonical form

$$\varphi(y) = y_1^3 + y_2^3 + y_3^3 + 6m y_1 y_2 y_3$$

We have, if r denotes the substitution determinant, set:

$$P = \frac{(2+m^3)m^3 f^2 + (2-5m^3)mr^2 f\Delta + 3m^2 r^4 \Delta^4 - r^6 J}{(1+8m^3)^2},$$

$$Q = \frac{(1+2m^3)f - 6mr^2\Delta}{1+8m^3},$$

$$R = \frac{m^2 f + r^2 \Delta}{1+8m^3}$$

and obtained y_1^3, y_2^3, y_3^3 as roots of the cubic equation

$$u^3 - Qu^2 + Pu - R^3 = 0,$$

and the discriminant of this cubic equation is

$$D_1 = \frac{r^{18}K^2}{(1+8m^3)^6}.$$

If one expresses this discriminant in terms of P, Q, R, one obtains K^2 represented rationally in terms of f, Δ, J. At the passage of the Algebra mentioned we did not give this expression explicitly. Now, however, we must form it, even if only under the assumption $f = 0$, i. e. only for the points of the curve. But we have [according to Bd. I, § 50, (10)]

$$D_1 = P^2Q^2 + 18PQR^3 - 4P^3 - 4Q^3R^3 - 27R^6.$$

In this one has to set:

$$P = \frac{3m^2r^4\Delta^2 - r^6J}{(1+8m^3)^2}, \quad Q = \frac{-6mr^2\Delta}{1+8m^3}, \quad R = \frac{r^2\Delta}{1+8m^3},$$

and one obtains, by a calculation that is not difficult, if one denotes by S and T the two invariants of the curve of the third order (Bd. II, § 108),

$$(1+8m^3)^6 D_1 = r^{18}(4J^3 + 108SJ\Delta^4 - 27T\Delta^6),$$

and consequently

$$K^2 = 4J^3 + 108SJ\Delta^4 - 27T\Delta^6. \tag{8}$$

This equation, however, is not satisfied identically, but only under the assumption $f = 0$. The complete expression of K^2 in terms of J, Δ, f becomes very much more complicated.

If one sets

$$\frac{J}{\Delta^2} = 3z, \quad dz = \frac{1}{3}\frac{\Delta dJ - 2Jd\Delta}{\Delta^3}, \tag{9}$$

then from (8) there results

$$K^2 = 27\Delta^6(4z^3 + 12Sz - T)$$

and from (7)

$$du = \frac{1}{\sqrt{3}}\frac{dz}{\sqrt{4z^3 + 12Sz - T}}, \tag{10}$$

and this is the Weierstrass normal form for $g_2 = -12S$, $g_3 = T$.

§ 7. Elliptic Space Curves of the Fourth Order.

Algebraic functions of one variable can also be represented by equations between several variables, provided one increases the number of equations correspondingly. If one takes, for example, three variables x, y, z and lets two equations

$$\varphi(x, y, z) = 0, \qquad \psi(x, y, z) = 0 \tag{1}$$

hold between them, then out of these two equations one can, for example, eliminate x and obtains an equation between y and z by which y is defined as an algebraic function of z. If certain exceptional

cases are excluded, x and every rational function of x, y, z can then be represented rationally by y and z. The integrals of the form

$$\int F(x, y, z)\, dz, \tag{2}$$

in which F denotes a rational function, then belong to the algebraic figure represented by (1). If one takes x, y, z as Cartesian coordinates in space, then (1) represents a space curve as the intersection of two surfaces $\varphi = 0$, $\psi = 0$, and the integrals (2) belong to this space curve. The curve is again called elliptic if these integrals are elliptic.

We wish to apply these considerations to the space curves of the fourth order of the first species, that is, to the curves of the fourth order which can be represented as the complete intersection of two surfaces of the second degree.

We again introduce homogeneous coordinates x_1, x_2, x_3, x_4 and take the equations of two given surfaces of the second degree in the form:

$$\begin{aligned} \varphi(x_1, x_2, x_3, x_4) &= \Sigma a_{ik} x_i x_k, \\ \psi(x_1, x_2, x_3, x_4) &= \Sigma b_{ik} x_i x_k. \end{aligned} \tag{3}$$

The two surfaces determine a pencil of surfaces of the second order $\xi\varphi = \eta\psi$. All surfaces of the pencil intersect in a space curve of the fourth order, which we call the fundamental curve. One obtains the same curve and the same pencil if the functions φ and ψ are replaced by φ', ψ', which are derived from φ, ψ by means of the linear substitution $\begin{pmatrix} m & n \\ p & q \end{pmatrix}$, that is, by

$$\begin{aligned} \varphi' &= m\varphi + n\psi, \\ \psi' &= p\varphi + q\psi \end{aligned} \tag{4}$$

and whose determinant $r = mq - np$ is different from zero. From this there results the identity

$$\xi\varphi + \eta\psi = \xi'\varphi' + \eta'\psi', \tag{5}$$

in which

$$\begin{aligned} \xi &= m\xi' + p\eta', \\ \eta &= n\xi' + q\eta' \end{aligned} \tag{6}$$

represents a linear transformation of the variables ξ, η. Besides this binary substitution there also comes into consideration a linear quaternary substitution of the variables x_1, x_2, x_3, x_4 (coordinate transformation), whose determinant we may assume $= 1$.

Every quadratic form possesses, with respect to these latter transformations, an invariant, namely the Hesse determinant (Bd. I, § 62, 63, 66), and we thus obtain as the invariant of the form (5):

$$f(\xi, \eta) = \Sigma \pm (\xi a_{11} + \eta b_{11})(\xi a_{22} + \eta b_{22})(\xi a_{33} + \eta b_{33})(\xi a_{44} + \eta b_{44}). \tag{7}$$

This is a binary biquadratic form in the variables ξ, η, which, developed, we represent thus:

$$f(\xi, \eta) = a_0\xi^4 + a_1\xi^3\eta + a_2\xi^2\eta^2 + a_3\xi\eta^3 + a_4\eta^4. \tag{8}$$

The coefficients of this biquadratic form a_0, a_1, a_2, a_3, a_4 do not change under a coordinate transformation. They are therefore called simultaneous invariants of the pair of forms φ, ψ (a_0 and a_4 are the determinants of φ and of ψ).

The a_i change when the ξ, η are subjected to a substitution (6). But if one forms the invariants of the form (8) (following Algebra, Bd. I, § 70), one obtains functions of the coefficients which

are invariant with respect to these substitutions as well, and which therefore belong not to the individual forms φ, ψ, but to the whole pencil and hence also to their intersection, the fundamental curve. We call them invariants of the fundamental curve. With respect to the system of forms φ, ψ they are also called combinants. The fundamental curve thus has two invariants:

$$\begin{aligned} A &= a_2^2 - 3\,a_1a_3 + 12\,a_0a_4, \\ B &= 27\,a_1^2a_4 + 27\,a_0a_3^2 + 2\,a_2^3 - 72\,a_0a_2a_4 - 9\,a_1a_2a_3, \end{aligned} \tag{9}$$

from which one derives the discriminant D by the following formula:

$$27\,D = 4\,A^3 - B^2 \tag{10}$$

With respect to the coefficients of φ and ψ the invariants A, B, D are of the degrees 8, 12, 24.

The system of forms φ, ψ has two simultaneous quadratic covariants, which are derived just as the covariant C is (in Algebra I, § 65). The first of them, if we put $\psi_1 = \frac{1}{2}\frac{\partial\psi}{\partial x_1}, \ldots,$ is:

$$\Phi = \begin{vmatrix} a_{11}, & a_{12}, & a_{13}, & a_{14}, & \psi_1 \\ a_{21}, & a_{22}, & a_{23}, & a_{24}, & \psi_2 \\ a_{31}, & a_{32}, & a_{33}, & a_{34}, & \psi_3 \\ a_{41}, & a_{42}, & a_{43}, & a_{44}, & \psi_4 \\ \psi_1, & \psi_2, & \psi_3, & \psi_4, & 0 \end{vmatrix} \tag{11}$$

and the second, Ψ, is obtained from it by interchanging a_{ik} with b_{ik} and ψ with φ.

The equation $\Phi = 0$ expresses a surface of the second degree (one that does not belong to the pencil), namely the locus of the points x whose polars with respect to ψ touch the surface φ.

We now assume that the discriminant D is different from zero. Then the two functions φ, ψ can be transformed simultaneously into the sum of four squares:

$$\begin{aligned} \varphi &= \alpha_1y_1^2 + \alpha_2y_2^2 + \alpha_3y_3^2 + \alpha_4y_4^2, \\ \psi &= \beta_1y_1^2 + \beta_2y_2^2 + \beta_3y_3^2 + \beta_4y_4^2, \end{aligned} \tag{12}$$

in which the y_i are linear functions of the x_i. This is the canonical form of the pair of functions. It then becomes:

$$f(\xi,\eta) = (\xi\alpha_1 + \eta\beta_1)(\xi\alpha_2 + \eta\beta_2)(\xi\alpha_3 + \eta\beta_3)(\xi\alpha_4 + \eta\beta_4), \tag{13}$$

and the functions Φ, Ψ take the canonical form:

$$\begin{aligned} \Phi &= \Sigma\,\beta_1^2\alpha_2\alpha_3\alpha_4\,y_1^2, \\ \Psi &= \Sigma\,\alpha_1^2\beta_2\beta_3\beta_4\,y_1^2. \end{aligned} \tag{14}$$

The determinant of the system (12), (14), regarded as linear equations for y_1^2, y_2^2, y_3^2, y_4^2, is

$$\begin{vmatrix} \alpha_1^2\beta_2\beta_3\beta_4, & \alpha_2^2\beta_1\beta_3\beta_4, & \alpha_3^2\beta_1\beta_2\beta_4, & \alpha_4^2\beta_1\beta_2\beta_3 \\ \beta_1^2\alpha_2\alpha_3\alpha_4, & \beta_2^2\alpha_1\alpha_3\alpha_4, & \beta_3^2\alpha_1\alpha_2\alpha_4, & \beta_4^2\alpha_1\alpha_2\alpha_3 \\ \alpha_1, & \alpha_2, & \alpha_3, & \alpha_4 \\ \beta_1, & \beta_2, & \beta_3, & \beta_4 \end{vmatrix}. \tag{15}$$

It vanishes when $\alpha_1 : \alpha_2 = \beta_1 : \beta_2$, because the first two columns then become proportional to one another, and likewise when $\alpha_i : \alpha_k = \beta_i : \beta_k$. Therefore (15), regarded as an integral function of the α, β, is divisible by

$$\alpha_i\beta_k - \alpha_k\beta_i = (\alpha_i\beta_k), \tag{16}$$

and consequently also by the product of all these factors

$$(\alpha_1\beta_2)(\alpha_1\beta_3)(\alpha_1\beta_4)(\alpha_2\beta_3)(\alpha_2\beta_4)(\alpha_3\beta_4) = \Delta. \tag{17}$$

From the comparison of the degrees it follows that the two functions (15), (17) can differ only by a numerical factor, and this factor is found, from the assumption $\beta_1 = 0$, $\alpha_2 = 0$, $\alpha_4 = 0$, to be equal to 1. The determinant (15) is therefore equal to the product Δ, and the square of Δ is the discriminant D of the function $f(\xi, \eta)$, hence equal to the function D, which we have assumed to be different from zero. The y_1^2, y_2^2, y_3^2, y_4^2 can therefore be represented linearly by φ, ψ, Φ, Ψ.

The covariants Φ, Ψ change when φ and ψ are replaced according to (4) by φ', ψ'. They therefore belong not to the fundamental curve, but to the pair of forms φ, ψ. But under this substitution Φ and Ψ, as linear functions of y_1^2, y_2^2, y_3^2, y_4^2, pass over into linear functions of φ, ψ, Φ, Ψ. These expressions are rather complicated. But we need them here only for the points of the fundamental curve, that is, for $\varphi = 0$, $\psi = 0$, and under this hypothesis they become very simple. For in that case we have

$$\begin{aligned} \Phi' &= \Sigma\,(p\alpha_1 + q\beta_1)^2(m\alpha_2 + n\beta_2)(m\alpha_3 + n\beta_3)(m\alpha_4 + n\beta_4)\,y_1^2, \\ \Psi' &= \Sigma\,(m\alpha_1 + n\beta_1)^2(p\alpha_2 + q\beta_2)(p\alpha_3 + q\beta_3)(p\alpha_4 + q\beta_4)\,y_1^2, \end{aligned} \tag{18}$$

and from this, by (12) and (14), there is to be deduced:

$$\begin{aligned} \Phi' &= M\Phi + N\Psi, \\ \Psi' &= P\Phi + Q\Psi, \end{aligned}$$

in which the coefficients M, N, P, Q are still to be determined. We can carry this out by calculation. Without calculation the result is obtained in the following way:

The M, N are integral functions of the second degree in p, q and of the third degree in m, n. If one takes $p : q = m : n$, and so $r = mq - np = 0$, then by (18) Φ' and $\dfrac{\partial\Phi'}{\partial p}$ pass over into linear combinations of φ and ψ, and therefore vanish. From this it follows that M and N are divisible by r^2, and that the quotients are linear functions of m and n. But since for the two cases $\begin{pmatrix} m & n \\ p & q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ Φ' must pass over into Φ and into Ψ, and since the same consideration can be applied to Ψ', it follows:

$$\begin{aligned} \Phi' &= r^2(m\Phi + n\Psi), \\ \Psi' &= r^2(p\Phi + q\Psi). \end{aligned} \tag{19}$$

Thus (apart from the factor r^2) the functions Φ, Ψ are transformed cogrediently with the φ, ψ. (On the rigorous justification of these inferences see Bd. I, § 20.)

From these results we can, in a simple way and always under the assumption $\varphi = 0$, $\psi = 0$, express the y_i^2 by means of Φ and Ψ, that is, solve the linear equations (12), (14). For if we apply the formulae (19) to the substitution

$$\begin{pmatrix} m, & n \\ p, & q \end{pmatrix} = \begin{pmatrix} \beta_1, & -\alpha_1 \\ 0, & 1 \end{pmatrix}$$

then $r = \beta_1$, the function ψ' remains unchanged $= \psi$, and φ' arises from φ when

$$\alpha_1, \quad \alpha_2, \quad \alpha_3, \quad \alpha_4$$

are replaced by

$$0, \quad (\alpha_2\beta_1), \quad (\alpha_3\beta_1), \quad (\alpha_4\beta_1)$$

Thus by (14)

$$\Phi' = \beta_1^2(\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1)\, y_1^2$$

and consequently by (19):

$$\begin{aligned}
(\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1)\, y_1^2 &= \alpha_1\Phi - \beta_1\Psi,\\
(\alpha_1\beta_2)(\alpha_3\beta_2)(\alpha_4\beta_2)\, y_2^2 &= \alpha_2\Phi - \beta_2\Psi,\\
(\alpha_1\beta_3)(\alpha_2\beta_3)(\alpha_4\beta_3)\, y_3^2 &= \alpha_3\Phi - \beta_3\Psi,\\
(\alpha_1\beta_4)(\alpha_2\beta_4)(\alpha_3\beta_4)\, y_4^2 &= \alpha_4\Phi - \beta_4\Psi.
\end{aligned} \tag{20}$$

The product of the constant factor on the left-hand side is the discriminant D, and so by (13) one obtains

$$\begin{aligned}
D\, y_1^2y_2^2y_3^2y_4^2 &= f(\Phi, -\Psi)\\
&= a_0\Phi^4 - a_1\Phi^3\Psi + a_2\Phi^2\Psi^2 - a_3\Phi\Psi^3 + a_4\Psi^4,
\end{aligned} \tag{21}$$

and the y_i^2 are therefore the linear factors of this biquadratic form.

There is yet a further simultaneous covariant of the form φ, ψ, namely the Jacobian functional determinant of the four functions φ, ψ, Φ, Ψ. We denote it thus:

$$K = \begin{vmatrix} \varphi_1, & \varphi_2, & \varphi_3, & \varphi_4\\ \psi_1, & \psi_2, & \psi_3, & \psi_4\\ \Phi_1, & \Phi_2, & \Phi_3, & \Phi_4\\ \Psi_1, & \Psi_2, & \Psi_3, & \Psi_4 \end{vmatrix}. \tag{22}$$

If one forms it for the canonical form, there results, by (12), (14), (15),

$$K = -\Delta\, y_1y_2y_3y_4,$$

and since $\Delta^2 = D$:

$$K^2 = D\, y_1^2y_2^2y_3^2y_4^2,$$

and so by (21) we have

$$K_2 = f(\Phi, -\Psi). \tag{23}$$

The biquadratic form $f(\Phi, -\Psi)$ has two covariants, which we denote, as in Bd. I, § 70, thus:

$$\begin{aligned}
H &= \frac{1}{3}\begin{vmatrix} \dfrac{\partial^2 f}{\partial\Phi^2}, & -\dfrac{\partial^2 f}{\partial\Phi\,\partial\Psi}\\[2ex] \dfrac{\partial^2 f}{\partial\Psi\,\partial\Phi}, & -\dfrac{\partial^2 f}{\partial\Psi^2} \end{vmatrix},\\
T &= \frac{1}{12}\begin{vmatrix} f'(\Phi), & f'(-\Psi)\\ H'(\Phi), & H'(-\Psi) \end{vmatrix}.
\end{aligned} \tag{24}$$

They are of the fourth and sixth degree in the Φ, Ψ, and hence of the eighth and twelfth degree in the x_i. They are covariants of the fundamental curve (combinants of φ and ψ), and between them there holds the relation:

$$H^3 - 48\, A\, H\, f^2 - 64\, B\, f^3 = -27\, T^2. \tag{25}$$

The most general integral (2) belonging to the fundamental curve passes, through the introduction of homogeneous variables $z = x_3 : x_4$, into the following form:

$$\int F'(x,y,z)\,\frac{x_3\,dx_4 - x_4\,dx_3}{x_4^2},$$

or, if one sets

$$F(x,y,z) = x_4^2\,\frac{F(x_1,x_2,x_3,x_4)}{\varphi_1\psi_2 - \varphi_2\psi_1}$$

it passes into

$$\int F(x_1,x_2,x_3,x_4)\,\frac{x_3\,dx_4 - x_4\,dx_3}{\varphi_1\psi_2 - \varphi_2\psi_1},$$

in which F is a homogeneous function of degree 0. The simplest case is $F = 1$, and so we consider the integral of the first kind:

$$u = \int \frac{x_3\,dx_4 - x_4\,dx_3}{\varphi_1\psi_2 - \varphi_2\psi_1}\,. \tag{26}$$

Now, by the multiplication theorem of determinants, we form the product $K(x_3\,dx_4 - x_2\,dx_4)$, and hence, since $\varphi = 0$, $\psi = 0$:

$$\begin{vmatrix} \varphi_1, & \varphi_2, & \varphi_3, & \varphi_4 \\ \psi_1, & \psi_2, & \psi_3, & \psi_4 \\ \Phi_1, & \Phi_2, & \Phi_3, & \Phi_4 \\ \Psi_1, & \Psi_2, & \Psi_3, & \Psi_4 \end{vmatrix} \begin{vmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ x_1, & x_2, & x_3, & x_4 \\ dx_1, & dx_2, & dx_3, & dx_4 \end{vmatrix} = \begin{vmatrix} \varphi_1, & \varphi_2, & 0, & 0 \\ \psi_1, & \psi_2, & 0, & 0 \\ \Phi_1, & \Phi_2, & \Phi, & d\Phi \\ \Psi_1, & \Psi_2, & \Psi, & d\Psi \end{vmatrix}$$

$$= (\varphi_1\psi_2 - \varphi_2\psi_1)(\Phi\,d\Psi - \Psi\,d\Phi),$$

and consequently there results from (26) and (23)

$$u = \int \frac{\Phi\,d\Psi - \Psi\,d\Phi}{K} = \int \frac{\Phi\,d\Psi - \Psi\,d\Phi}{\sqrt{f(\Phi,-\Psi)}}\,. \tag{27}$$

If one applies to this the transformation of § 5, replacing x, y by $\Phi, -\Psi$, there follows

$$u = \sqrt{3}\int \frac{f\,dH - H\,df}{\sqrt{-(H^3 - 48\,Af^2H - 64\,Bf^3)f}},$$

and if in § 5, (17) one takes $\mu = 3$, so that

$$z = -\frac{3H}{4f}$$

one obtains:

$$u = 3\int \frac{dz}{\sqrt{4z^3 - g_2z - g_3}},$$
$$g_2 = 12\cdot 9\,A, \qquad g_3 = -4\cdot 27\,B.$$

§ 8. **The Jacobi Transformation Principle.**

The Jacobi theory of transformation sets itself in general the task of reducing an elliptic differential, by an algebraic substitution, to another elliptic differential. In setting out here

the foundations of this theory, we make use of the representation of the elliptic differential in homogeneous variables [§ 1, (4)]:

$$\frac{x\,dy - y\,dx}{\sqrt{f(x,y)}}\,, \tag{1}$$

and substitute in it, for x, y, two coprime integral homogeneous functions of the same, but arbitrary, degree n in two new variables ξ, η:

$$x = U(\xi,\eta), \qquad y = V(\xi,\eta). \tag{2}$$

From the equations

$$nU = \xi\frac{\partial U}{\partial \xi} + \eta\frac{\partial U}{\partial \eta}, \quad dU = d\xi\frac{\partial U}{\partial \xi} + d\eta\frac{\partial U}{\partial \eta},$$
$$nV = \xi\frac{\partial V}{\partial \xi} + \eta\frac{\partial V}{\partial \eta}, \quad dV = d\xi\frac{\partial V}{\partial \xi} + d\xi\frac{\partial V}{\partial \eta}$$

there then follows

$$U\,dV - V\,dU = H(\xi\,d\eta - \eta\,d\xi), \tag{3}$$

if by H the functional determinant

$$H = \frac{1}{n}\left(\frac{\partial U}{\partial \xi}\frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi}\frac{\partial U}{\partial \eta}\right) \tag{4}$$

is meant.

Accordingly the elliptic differential (1) passes over into the following:

$$H\,\frac{\xi\,d\eta - \eta\,d\xi}{\sqrt{f(U,V)}}\,. \tag{5}$$

Now in order that this differential should again receive the form of an elliptic one, it is necessary that from the function $f(U,V)$, whose degree is the $4n$th, a square factor T^2 of degree $4n-4$ can be separated off; that is, if $\varphi(\xi,\eta)$ denotes a function of the fourth degree, that

$$f(U,V) = T^2\,\varphi(\xi,\eta) \tag{6}$$

should hold, whereby, if

$$\frac{T}{H} = M \tag{7}$$

is set, the differential (5) passes into

$$\frac{1}{M}\,\frac{\xi\,d\eta - \eta\,d\xi}{\sqrt{\varphi(\xi,\eta)}} \tag{8}$$

It can now be shown that, as soon as condition (6) is satisfied, H is divisible by T, and hence, since the degree of the two functions is the same, M becomes a constant. This constant is called the multiplier of the transformation.

For if we note that, since $f(x,y)$ contains no square factor, the two functions

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}$$

have no common divisor, and that in consequence, because U and V are coprime, the functions

$$\frac{\partial f}{\partial U}, \quad \frac{\partial f}{\partial V}$$

likewise have no common divisor (in relation to ξ, η), then two functions α, β of ξ, η can be so determined that

$$\alpha \frac{\partial f}{\partial U} + \beta \frac{\partial f}{\partial V}$$

is coprime to an arbitrarily given function T. One may, e. g., take α coprime to T and β divisible by those divisors of T which do not divide $\partial f/\partial U$, but indivisible by the common divisors of T and $\partial f/\partial U$. If we now differentiate equation (6), which we assume to be satisfied, with respect to ξ, η, there follows:

$$\frac{\partial f}{\partial U}\frac{\partial U}{\partial \xi} + \frac{\partial f}{\partial V}\frac{\partial V}{\partial \xi} = TX,$$
$$\frac{\partial f}{\partial U}\frac{\partial U}{\partial \eta} + \frac{\partial f}{\partial V}\frac{\partial V}{\partial \eta} = TY,$$

and from this, by solution with respect to $\partial f/\partial U$, $\partial f/\partial V$:

$$H\left(\alpha \frac{\partial f}{\partial U} + \beta \frac{\partial f}{\partial V}\right) = TZ,$$

in which X, Y, Z are integral homogeneous functions of ξ, η. But from the last equation one concludes that H must be divisible by T.

Our whole problem is accordingly contained in equation (6), which says nothing other than that the function $f(U, V)$, of degree $4n$, is to contain $2n - 2$ quadratic factors. To satisfy the $(2n - 2)$ conditions that follow from this one has at one's disposal the $2n + 2$ coefficients contained in U, V, so that four of these remain undetermined. This was to be foreseen, since arbitrary homogeneous linear functions of ξ, η may be introduced in place of ξ, η. These four supernumerary constants can be used to bring the differential (8) into a normal form. But since the equations of condition for the coefficients of U, V are not linear, there are several transformations for a given degree of transformation.

We shall meet this transformation problem again later from a quite different side, and shall have to go into it far more deeply. The details of the algebraic problem will therefore not be entered into more closely here; instead we wish to illustrate what has been said by two examples.

§ 9. The Transformation of the Second Degree.

In order to obtain the elliptic differentials in the normal form we set:

$$f(x, y) = xy(x - y)(x - \lambda^2 y),$$
$$\varphi(\xi, \eta) = \xi\eta(\xi - \eta)(\xi - \varkappa^2\eta),$$

and let U, V be of the second degree.

Equation (6), § 8, now becomes, since T is of the second degree

$$UV(U - V)(U - \lambda^2 V) = (a\xi + b\eta)^2(a'\xi + b'\eta)^2\, \xi\eta(\xi - \eta)(\xi - \varkappa^2\eta), \tag{1}$$

and two of the four factors of the second degree on the left side of this equation must be squares. We wish to restrict the great number of possibilities lying in this still further by assuming that η and y are to vanish simultaneously, so that V is divisible by η. This gives (apart from a constant factor) the following three possibilities for V:

$$1.\ V = \xi\eta, \qquad 2.\ V = \eta(\xi - \eta), \qquad 3.\ V = \eta(\xi - \varkappa^2\eta).$$

Each of these three cases now again comprises three subcases, since of the three remaining factors U, $U - V$, $U - \lambda^2 V$ any two may be assumed to be squares. Of these latter

three cases two pass into one another by interchange of V, λ^2 with $V : \lambda^2$, $1 : \lambda^2$.

We wish to carry out completely two of these transformations that are especially important for what follows.

1. The Gauss Transformation.

$$V = \xi\eta, \qquad U = (a\xi + b\eta)^2.$$

If now $U - \lambda^2 V$ is also to be a square, then we must put

$$\lambda^2 = 4ab$$

and it becomes

$$U - \lambda^2 V = (a\xi - b\eta)^2.$$

Since $U - V$ must then be divisible by $\xi - \eta$ and by $\xi - \varkappa^2\eta$, there follow from $U = V$, if one puts $\xi = \eta$, $\xi = \varkappa^2\eta$, the conditions

$$a + b = \pm 1, \qquad a\varkappa^2 + b = \pm\varkappa,$$

from which, if the upper signs are taken,

$$a = \frac{1}{1+\varkappa}, \qquad b = \frac{\varkappa}{1+\varkappa}, \qquad \lambda = \frac{2\sqrt{\varkappa}}{1+\varkappa},$$

$$V = \xi\eta, \qquad U = \left(\frac{\xi + \varkappa\eta}{1+\varkappa}\right)^2, \qquad U - \lambda^2 V = \left(\frac{\xi - \varkappa\eta}{1+\varkappa}\right)^2,$$

$$U - V = \frac{(\xi - \eta)(\xi - \varkappa^2\eta)}{(1+\varkappa)^2},$$

$$T = \frac{\xi^2 - \varkappa^2\eta^2}{(1+\varkappa)^3}, \qquad H = \frac{\xi^2 - \varkappa^2\eta^2}{(1+\varkappa)^2},$$

$$M = \frac{1}{1+\varkappa}.$$

If we therefore again put

$$\frac{y}{x} = z, \qquad \frac{\eta}{\xi} = \zeta,$$

then we have the result that by the substitution

$$z = \frac{(1+\varkappa)^2\zeta}{(1+\varkappa\zeta)^2}, \qquad \lambda^2 = \frac{4\varkappa}{(1+\varkappa)^2} \tag{2}$$

the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-\lambda^2 z)}} = \frac{(1+\varkappa)\,d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}} \tag{3}$$

is effected.

2. The Landen Transformation.

$$V = a\eta(\xi - \eta), \qquad U = \xi(\xi - \varkappa^2\eta).$$

The conditions that $U - V$, $U - \lambda^2 V$ be squares read

$$4a = (\varkappa^2 + a)^2, \qquad 4a\lambda^2 = (\varkappa^2 + a\lambda^2)^2,$$

from which

$$\lambda(\varkappa^2 + a) = \varkappa^2 + a\lambda^2,$$

or, divided by $\lambda - 1$,

$$a\lambda = \varkappa^2;$$

this, substituted into one of the above equations, gives

$$4\lambda = \varkappa^2(1 + \lambda)^2,$$

and by the solution of this quadratic equation, when $\varkappa' = \sqrt{1 - \varkappa^2}$ is set:

$$\lambda = \frac{1 - \varkappa'}{1 + \varkappa'}, \qquad a = (1 + \varkappa')^2,$$

$$\begin{aligned} U - V &= [\xi - (1 + \varkappa')\eta]^2, \\ U - \lambda^2 V &= [\xi - (1 - \varkappa')\eta]^2, \\ H &= (1 + \varkappa')^2[\xi - (1 + \varkappa')\eta][\xi - (1 - \varkappa')\eta], \\ T &= (1 + \varkappa')[(\xi - \eta)^2 - \varkappa'^2\eta^2], \\ M &= \frac{1}{1 + \varkappa'}, \end{aligned}$$

so that by the substitution

$$z = \frac{(1 + \varkappa')^2\zeta(1 - \zeta)}{1 - \varkappa^2\zeta}, \qquad \lambda = \frac{1 - \varkappa'}{1 + \varkappa'} \tag{4}$$

the transformation:

$$\frac{dz}{\sqrt{z(1 - z)(1 - \lambda^2 z)}} = \frac{(1 + \varkappa')\, d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}}. \tag{5}$$

If one applies to the left-hand side of this equation once more the Gauss transformation, replacing in the formulae (2), (3) $\zeta, \varkappa$ by z, λ and z by a variable η, then, as follows from (2) and (4), λ in (3) is to be replaced by $\varkappa$, and by combining (5) and (3) one finds:

$$\frac{d\eta}{\sqrt{\eta(1 - \eta)(1 - \varkappa^2\eta)}} = \frac{2\, d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}}, \tag{6}$$

and the combination of (2) and (4) yields, between the variables η and ζ, the connection:

$$\eta = \frac{4\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}{(1 - \varkappa^2\zeta^2)^2}. \tag{7}$$

The combination of the two transformations of the second degree therefore gives multiplication by 2.

§ 10. The Transformation of the Third Degree.

As a second example we consider the transformation of the third degree. The equation

$$UV(U-V)(U-\lambda^2 V) = T^2\xi\eta(\xi-\eta)(\xi-\varkappa^2\eta) \tag{1}$$

requires that each of the four factors of the third degree U, V, $U-V$, $U-\lambda^2 V$ be divisible by one of the linear factors ξ, η, $\xi-\eta$, $\xi-\varkappa^2\eta$, and that the quotient be a square. We therefore set

$$U = \xi(a\xi+b\eta)^2, \qquad V = \eta(a'\xi+b'\eta)^2, \tag{2}$$

and require further that

$$\frac{U-V}{\xi-\eta}, \qquad \frac{U-\lambda^2 V}{\xi-\varkappa^2\eta} \tag{3}$$

become the squares of linear functions.

Of the last two requirements the one follows from the other if we so arrange U, V that, apart from constant factors, U and V pass into one another under the interchange of ξ, η with $\varkappa^2\eta$, ξ. But under this interchange

$$\xi(a\xi+b\eta)^2, \qquad \eta(a'\xi+b'\eta)^2$$

pass over into

$$\varkappa^2\eta(b\xi+a\varkappa^2\eta)^2, \qquad \xi(b'\xi+a'\varkappa^2\eta)^2,$$

and our requirement is satisfied if

$$\varkappa^2 = \frac{bb'}{aa'} \tag{4}$$

holds. If then

$$\lambda^2 = \varkappa^2\left(\frac{ab}{a'b'}\right)^2 \tag{5}$$

holds, then by this interchange

$$\frac{U-V}{\xi-\eta} \text{ passes over into } \frac{b'^2}{a^2}\,\frac{U-\lambda^2 V}{\xi-\varkappa^2\eta} \tag{6}$$

Now that this has been laid down, only the condition remains to be satisfied that the first of the two quantities (3) becomes the square of a linear function, which we denote by $(\alpha^2\xi-\beta^2\eta)$, that is:

$$U-V = (\xi-\eta)(\alpha^2\xi-\beta^2\eta)^2 \tag{7}$$

or

$$U - \xi(\alpha^2\xi - \beta^2\eta)^2 = V - \eta(\alpha^2\xi - \beta^2\eta)^2,$$

and since U is divisible by ξ and V by η, this function must be divisible by $\xi\eta$; if we set it $= \xi\eta(m\xi + n\eta)$, then from (2) there results

$$(a\xi + b\eta)^2 = (\alpha^2\xi - \beta^2\eta)^2 + \eta(m\xi + n\eta),$$
$$(a'\xi + b'\eta)^2 = (\alpha^2\xi - \beta^2\eta)^2 + \xi(m\xi + n\eta).$$

The comparison of the coefficients gives

$$a = \alpha^2, \qquad b = -\beta^2 + \frac{m}{2\alpha^2}, \qquad b^2 = \beta^4 + n,$$

$$b' = \beta^2, \qquad a' = -\alpha^2 + \frac{n}{2\beta^2}, \qquad a'^2 = \alpha^4 + m,$$

and from the last two equations of each row:

$$m^2 = 4\alpha^2(n\alpha^2 + m\beta^2),$$
$$n^2 = 4\beta^2(n\alpha^2 + m\beta^2).$$

If, therefore, one sets

$$m = h\alpha, \qquad n = h\beta$$

then there results

$$h = 4\alpha\beta(\alpha + \beta).$$

Accordingly a, b, a', b' can be expressed by the two quantities α, β in the following way:

(8)
$$\begin{aligned} a &= \alpha^2, & b &= \beta^2 + 2\alpha\beta, \\ a' &= \alpha^2 + 2\alpha\beta, & b' &= \beta^2, \end{aligned}$$

and after that, from (4), (5),

(9)
$$\varkappa^2 = \frac{\beta^3}{\alpha^3}\,\frac{\beta + 2\alpha}{\alpha + 2\beta}, \qquad \frac{\lambda}{\varkappa} = \frac{\alpha}{\beta}\,\frac{\beta + 2\alpha}{\alpha + 2\beta},$$

or, by multiplication and division of these two equations:

(10)
$$\sqrt{\lambda\varkappa} = \frac{\beta}{\alpha}\,\frac{\beta + 2\alpha}{\alpha + 2\beta}, \qquad \frac{\varkappa^3}{\lambda} = \frac{\beta^4}{\alpha^4}.$$

By elimination of $\beta : \alpha$ one obtains an equation between $\varkappa, \lambda$, which is called the modular equation, and whose degree gives the number of the different transformations of the third degree. It takes the simplest shape when one sets:

(11)
$$\sqrt[4]{\varkappa} = u, \qquad \sqrt[4]{\lambda} = v.$$

Then the equations (10) become

$$u^2v^2 = \frac{\beta}{\alpha}\frac{\beta+2\alpha}{\alpha+2\beta}, \qquad \frac{u^3}{v} = \frac{\beta}{\alpha},$$

hence, by substituting the value of $\beta : \alpha$ into the first equation and removing the factor u^2:

$$v^4 - u^4 + 2u^3v^3 - 2uv = 0, \tag{12}$$

an equation of the fourth degree.

One obtains further [without lengthy calculation, by comparison of the coefficients of the highest power of ξ in (1)]

$$T = \frac{a}{b'}(a\xi + b\eta)(a'\xi + b'\xi)(\alpha^2\xi - \beta^2\eta)(\beta^2\xi - \varkappa^2\alpha^2\eta),$$
$$H = \frac{1}{3}\left(\frac{\partial U}{\partial \xi}\frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi}\frac{\partial U}{\partial \eta}\right) = \frac{a'}{a}T,$$

from which one easily finds by (8):

$$M = \frac{a}{a'} = \frac{\alpha}{\alpha+2\beta} = \frac{v}{v+2u^3}. \tag{13}$$

If one substitutes the expression resulting from this,

$$v = \frac{2u^3M}{1-M}$$

into equation (12), then for M there results an equation of the fourth degree, the multiplier equation, which can replace the modular equation. But one obtains this equation more simply in the following way. By (13),

$$\frac{1}{M} - 1 = 2\frac{\beta}{\alpha}, \quad \frac{1}{M} + 3 = \frac{2(\beta+2\alpha)}{\alpha},$$

hence by (9):

$$\left(\frac{1}{M} - 1\right)^3\left(\frac{1}{M} + 3\right) = 16\varkappa^2\frac{\alpha+2\beta}{\alpha},$$

and consequently by (13)

$$\frac{16\varkappa^2}{M} = \left(\frac{1}{M} - 1\right)^3\left(\frac{1}{M} + 3\right),$$

or, arranged in order,

$$\frac{1}{M^4} - \frac{6}{M^2} + \frac{8(1-2\varkappa^2)}{M} - 3 = 0. \tag{14}$$

If we therefore express our formulae in terms of u, v, the result of this consideration is the following:

By the substitution

$$z = -\frac{\xi[(v^2+2vu^3) + u^6\xi]^2}{[v^2 + (u^6+2vu^3)\xi]^2}$$

the transformation is effected

$$\frac{dz}{\sqrt{z(1-z)(1-v^8z)}} = \frac{v+2u^3}{v}\frac{d\xi}{\sqrt{\xi(1-\xi)(1-u^8\xi)}},$$

provided that the modular equation (12) holds between u, v.

For a given u the modular equation yields four values of v, and hence four different transformations of the third degree.

§11. **The three kinds of elliptic integrals.**

Let now

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 \tag{1}$$

be an integral function of the third or fourth degree, and let

$$\Omega(x) = \int \frac{\Phi(x)\,dx}{\sqrt{f(x)}} = \int R(x)\,dx, \tag{2}$$

where for brevity

$$R(x) = \frac{\Phi(x)}{\sqrt{f(x)}} \tag{3}$$

has been put, be an elliptic integral. In it x denotes a quantity varying without restriction (and complex as well), and $\sqrt{f(x)}$ has, for every value of x for which $f(x)$ does not vanish, two opposite values. By a point we understand not a value of x alone, but a pair of values of x and $\sqrt{f(x)}$ belonging together, that is, a value of x with a definite sign of $\sqrt{f(x)}$. If therefore x_0 is any value of x, then there are two points x_0 when $f(x_0)$ is different from zero, but only one when $f(x_0) = 0$. For $x = \infty$ we determine the points by the sign of $\sqrt{f(x)} : x^2$, and so we have two points $x = \infty$ when $f(x)$ is of the fourth degree, and only one when $f(x)$ is of the third.

By known theorems of the theory of functions there exists, if $f(x_0)$ is different from zero, a development by Taylor's theorem, convergent in a certain domain:

$$R(x) = A_m(x - x_0)^m + A_{m+1}(x - x_0)^{m+1} + A_{m+2}(x - x_0)^{m+2} + \cdots . \tag{4}$$

in which m is a positive or negative integer or else zero, while $A_m, A_{m+1}, A_{m+2}, \ldots$ denote constants, of which A_m does not vanish. This is a development in ascending powers. In addition, if $f(x)$ is of the fourth (not of the third) degree, there comes the development in descending powers

$$R(x) = C_mx^{-m-2} + C_{m+1}x^{-m-3} + C_{m+2}x^{-m-4} + \cdots , \tag{5}$$

of which we say that it holds in the neighbourhood of an infinitely distant point.

But if $f(x_0)$ vanishes, the development of $R(x)$ takes the form:

$$R(x) = A_m(x - x_0)^{m-1/2} + A_{m+1}(x - x_0)^{m-1/2+1} + A_{m+2}(x - x_0)^{m-1/2+2} + \cdots , \tag{6}$$

and if $f(x)$ is of the third degree, then in the neighbourhood of the infinitely distant point the development holds

$$R(x) = C_mx^{-m-3/2} + C_{m+1}x^{-m-5/2} + C_{m+2}x^{-m-7/2} + \cdots \tag{7}$$

From the developments (4) to (7) we can derive the developments of Ω, an additive constant remaining undetermined.

The development of Ω is then likewise a power series, to which, if m is negative, there is added in the cases (4) or (5) a further term $A_{-1}\log(x - x_0)$, $C_{-1}\log x$.

We now call x_0 a point of the first kind of Ω if m is not negative in these developments.

Then Ω is finite at the point x_0 (also in the case $x_0 = \infty$).

The point x_0 is called a point of the second kind of Ω, if in the developments (4), (5) m is negative and $A_{-1} = 0$, $C_{-1} = 0$, and in the case of the development (6), (7) with negative m.

Finally, x_0 is called a point of the third kind, if in the developments (4) or (5) A_{-1} or C_{-1} is different from zero, that is, if a logarithmic term occurs in the development of Ω.

In the integral (2) the zero points of $f(x)$ cannot be of the third kind; they can, however, become of the third kind if we consider the elliptic integral in the general form $\int \Phi(x, \sqrt{f(x)})\, dx$.

The integral Ω is called an integral of the first kind if it has only points of the first kind. Such an integral then has a finite value for all finite and infinite values of x.

Ω is called an integral of the second kind if it has only points of the first and second kind, and

Ω is called an integral of the third kind if, besides points of the first and second kind, it also has points of the third kind.

§ 12. Representation of the elliptic integrals by the simplest fundamental integrals.

In order to represent the integral $\Omega(x)$ by integrals of the simplest possible type, we decompose the rational function $\Phi(x)$ into an integral function and a sum of partial fractions; that is, we represent $\Phi(x)$ as a sum of expressions of the form

$$x^n, \quad \frac{1}{(x-\alpha)^n}$$

with constant coefficients, in which n denotes a positive integer (also zero) and α a constant value for which $\Phi(x)$ becomes infinite.

If we then set, for positive and for negative or vanishing n,

$$S_n(\alpha) = \int \frac{(x-\alpha)^n\, dx}{\sqrt{f(x)}}, \tag{1}$$

then $\Omega(x)$ becomes a sum of the form

$$\Omega(x) = \sum M S_n(\alpha),$$

where the sum may extend over several different values of n and of α, and the M are constants.

The functions S_n can be reduced to a small number by the mediation of algebraic functions. To this end we form

$$d(x-\alpha)^n\sqrt{f(x)} = \frac{n(x-\alpha)^{n-1}f(x) + (x-\alpha)^n \tfrac{1}{2}f'(x)}{\sqrt{f(x)}},$$

and if we set in it

$$f(x) = f(\alpha) + (x-\alpha)f'(\alpha) + \frac{(x-\alpha)^2}{2}f''(\alpha) + \frac{(x-\alpha)^3}{6}f'''(\alpha) + \frac{(x-\alpha)^4}{24}f''''(\alpha),$$

$$f'(x) = f'(\alpha) + (x-\alpha)f''(\alpha) + \frac{(x-\alpha)^2}{2}f'''(\alpha) + \frac{(x-\alpha)^3}{6}f''''(\alpha),$$

there follows

$$\begin{aligned}(x-\alpha)^n\sqrt{f(x)} = {} & nf(\alpha)S_{n-1} + \left(n+\frac{1}{2}\right)f'(\alpha)S_n \\ & + \frac{n+1}{2}f''(\alpha)S_{n+1} + \left(\frac{n}{6}+\frac{1}{4}\right)f'''(\alpha)S_{n+2} \\ & + \left(\frac{n}{24}+\frac{1}{12}\right)f''''(\alpha)S_{n+3}.\end{aligned} \tag{2}$$

We distinguish four cases:

1. $f''''(\alpha)$ and $f(\alpha)$ not $= 0$ [i. e. $f(x)$ of the fourth degree and α not a root of $f(x)$].

In this case formula (2) for $n = 0$ allows S_3 to be expressed by S_0, S_1, S_2, and hence S_n for every positive n by S_0, S_1, S_2. If, however, one puts $n = -1, -2, \ldots$, one obtains $S_{-2}, S_{-3}, \ldots$ expressed by S_{-1}, S_0, S_1, S_2. In this case, therefore, the four fundamental integrals remain

$$S_{-1}, \quad S_0, \quad S_1, \quad S_2.$$

2. $f''''(\alpha)$ not $= 0$, $f(\alpha) = 0$. In this case S_{-1} can still be expressed by S_0, S_1, S_2, and one obtains as fundamental integrals

$$S_0, \quad S_1, \quad S_2.$$

3. $f''''(\alpha) = 0$, $f(\alpha)$ not $= 0$. Here the fundamental integrals are

$$S_{-1}, \quad S_0, \quad S_1.$$

4. $f''''(\alpha) = 0$, $f(\alpha) = 0$. Here the fundamental integrals are

$$S_0, \quad S_1.$$

The result of this is therefore:

I. If $f(x)$ is of the fourth degree ($a_0 = 1$), then all integrals $\Omega(x)$ can be expressed by integrals of the form

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \quad S_1 = \int \frac{x\,dx}{\sqrt{f(x)}}, \quad S_2 = \int \frac{x^2\,dx}{\sqrt{f(x)}},$$
$$S_{-1} = \int \frac{dx}{(x-\alpha)\sqrt{f(x)}},$$

in which S_{-1} still depends on the parameter α.

II. If, however, $f(x)$ is of the third degree, then the three integrals suffice

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \quad S_1 = \int \frac{x\,dx}{\sqrt{f(x)}}, \quad S_{-1} = \int \frac{dx}{(x-\alpha)\sqrt{f(x)}}.$$

In both cases S_0 is of the first kind. If $f(x)$ is of the third degree, then S_1 is of the second and S_{-1} of the third kind. If on the other hand $f(x)$ is of the fourth degree, then S_1, S_2 and S_{-1} are of the third kind. But one can compose from S_1 and S_2 an integral of the second kind. For we have

$$\frac{1}{\sqrt{f(x)}} = (x^4 + a_1x^3 + a_2x^2 + a_3x^2 + a_4)^{-1/2}$$
$$= x^{-2} - \frac{a_1}{2}x^{-3} + \cdots,$$

and in the development of $\left(x^2 + \frac{a_1}{2}x\right) : \sqrt{f(x)}$ in descending powers no term with x^{-1} occurs. Consequently $S_2 + \frac{a_2}{2}S_1$ is an integral of the second kind.

By the mediation of a logarithmic function, however, the four integrals of Case I can be further reduced to three. To this end let $f(x)$ be brought into the form

$$x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = P^2 + ax + b,$$

in which $P = x^2 + px + q$ is a quadratic function and a and b are constants. There results:

$$p = \frac{a_1}{2}, \quad q = \frac{a_2}{2} - \frac{a_1^2}{8},$$

$$a = a_3 - \frac{a_1a_2}{2} + \frac{a_1^3}{8}, \quad b = a_4 - \left(\frac{a_2}{2} - \frac{a_1^2}{8}\right)^2.$$

Now:

$$d\log\frac{\sqrt{f}-P}{\sqrt{f}+P}=\frac{f'(x)P-2f(x)P'(x)}{f-P^2}\,\frac{dx}{\sqrt{f(x)}}, \tag{3}$$

$$f-P^2=ax+b,$$

$$f'(x)=2PP'+a$$

and consequently

$$f'(x)P-2f(x)P'(x)=a(P-2xP')-2bP'=Q$$

is a quadratic function of x, and accordingly there results from (3)

$$\log\frac{\sqrt{f}-P}{\sqrt{f}+P}=\int\frac{Q}{ax+b}\,\frac{dx}{\sqrt{f(x)}}. \tag{4}$$

The right-hand side, however, is a linear function of S_0, S_1 and S_{-1} (for $\alpha=-b/a$). In the particular case where $a=0$, Q becomes of the first degree, and a linear combination of S_0 and S_{-1} can be expressed by a logarithmic function.

If we assume a_1 and $a_3=0$, to which we can reduce the general case by § 4 by means of a linear transformation, we obtain as the fundamental integrals

$$S_0=\int\frac{dx}{\sqrt{f(x)}},\quad S_2=\int\frac{x^2\,dx}{\sqrt{f(x)}},\quad S_{-1}=\int\frac{dx}{(x-\alpha)\sqrt{f(x)}},$$

while $S_1=\int\frac{x\,dx}{\sqrt{f(x)}}$ is reduced by the substitution $x^2=y$ to the following non-elliptic integral:

$$\frac{1}{2}\int\frac{dy}{\sqrt{y^2+a_2y+a_4}}$$

Here S_0 is of the first, S_2 of the second, S_{-1} of the third kind. For the Legendre normal form we put $f(x)=(1-x^2)(1-\varkappa^2x^2)$ and take as the normal integral of the second kind not S_2 itself, but $S_0-\varkappa^2S_2$. This gives the normal integrals of the three kinds:

$$\int\frac{dx}{\sqrt{(1-x^2)(1-\varkappa^2x^2)}},\quad \int\sqrt{\frac{1-\varkappa^2x^2}{1-x^2}}\,dx,\quad \int\frac{dx}{(x-\alpha)\sqrt{(1-x^2)(1-\varkappa^2x^2)}}$$

and, if one puts $x=\sin\varphi$:

$$\int\frac{d\varphi}{\sqrt{1-\varkappa^2\sin^2\varphi}},\quad \int\sqrt{1-\varkappa^2\sin^2\varphi}\,d\varphi,\quad \int\frac{d\varphi}{(\sin\varphi-\alpha)\sqrt{1-\varkappa^2\sin^2\varphi}}.$$

The same is obtained if one takes the elliptic differential in the normal form

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2z)}}$$

and proceeds according to Case II.

§ 13. The Addition Theorem.

The addition theorem of the elliptic integrals, discovered by Euler, consists in the theorem, of fundamental importance for the whole of the theory that follows, that if, of the three pairs of values

$$x_1,\sqrt{f(x_1)};\quad x_2,\sqrt{f(x_2)};\quad x_3,\sqrt{f(x_3)}$$

two are assumed to be given, the third can be determined by algebraic means in such a way that the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0 \tag{1}$$

is satisfied. In this, $f(x)$ is a given function of the third or fourth degree. This is a special case of the great

Abel's theorem, and it is to be conceived and derived here in this way as well[2]. We preface the proof with the following elementary algebraic theorem:

If $F(x)$ is an integral rational function of degree n without multiple factors, $F'(x)$ its derivative, if further $x_1, x_2, \ldots, x_n$ are the roots of the equation $F(x) = 0$ and $\varphi(x)$ an integral function of x whose degree is at most $= n-2$, then

$$\frac{\varphi(x_1)}{F'(x_1)} + \frac{\varphi(x_2)}{F'(x_2)} + \cdots \frac{\varphi(x_n)}{F'(x_n)} = 0. \tag{2}$$

The proof of this theorem follows immediately from the decomposition of the rational fraction

$$\frac{x\,\varphi(x)}{F(x)}$$

into partial fractions

$$\frac{x_1\varphi(x_1)}{F'(x_1)(x-x_1)} + \frac{x_2\varphi(x_2)}{F'(x_2)(x-x_2)} + \cdots + \frac{x_n\varphi(x_n)}{F'(x_n)(x-x_n)},$$

if we there put $x = 0$ [Bd. I, § 15 (9)].

Now let P, Q denote integral functions of x of the degrees m and $m-2$, and we ask for the values of x, $\sqrt{f(x)}$, for which the function

$$P + Q\sqrt{f(x)} \tag{3}$$

vanishes. There are $2m$ such pairs of values (points), and one finds the values of x as the roots of the equation of degree $2m$:

$$F(x) = P^2 - Q^2 f(x) = 0, \tag{4}$$

and the corresponding values of $\sqrt{f(x)}$ from

$$P + Q\sqrt{f(x)} = 0. \tag{5}$$

We now assume that the coefficients in the functions P, Q are variable, either independent variables or dependent in some way on other variables.

The values x and $\sqrt{f(x)}$ determined by (4), (5) then also become functions of these variables, and (5) can be differentiated.

Let us denote by δ the differentiation with respect to the variables occurring in the coefficients of P, Q, where x is regarded as constant, and by dx the corresponding differential of x, when x is determined by (4) or (5) as a function of these coefficients; the differentiation of (4) then gives

$$2P\delta P - 2Q\delta Q f(x) + F'(x)\,dx = 0,$$

[2]This far-reaching theorem is found derived with magnificent simplicity in an article of barely two pages in the fourth volume of Crelles Journal "Démonstration d'une propriété générale d'une certaine classe de fonctions transcendentes"; Oeuvres complètes de N. H. Abel. Nouvelle édition, T. I, p. 515. A detailed presentation and applications are given in the great posthumous memoir: "Mémoire sur une propriété générale d'une classe très étendue de fonctions transcendentes".

from which, with the use of (5):

$$\frac{Q\,\delta P - P\,\delta Q}{F'(x)} = \frac{dx}{2\sqrt{f(x)}}, \tag{6}$$

and since the numerator on the left side is of degree $2m-2$, and $F(x)$ of degree $2m$, formula (2) can be applied, and it follows that

$$\sum \frac{dx}{\sqrt{f(x)}} = 0, \tag{7}$$

when the sum is extended over all the roots of equation (5).

We apply this theorem to the simplest case, namely $m = 2$, and then obtain the following theorem:

If

$$x_1, \sqrt{f(x_1)}; \quad x_2, \sqrt{f(x_2)}; \quad x_3, \sqrt{f(x_3)}; \quad x_4, \sqrt{f(x_4)} \tag{8}$$

are four pairs of values for which some function of the form

$$a + bx + cx^2 + \sqrt{f(x)} \tag{9}$$

vanishes, then these pairs of values are not entirely independent of one another, but there subsists between them the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} + \frac{dx_4}{\sqrt{f(x_4)}} = 0. \tag{10}$$

The dependence of these four pairs of values, and hence an integration of the differential equation (10), can however also be expressed in algebraic form, by setting the function (9) equal to zero for $x = x_1, x_2, x_3, x_4$ and then eliminating a, b, c. In this way one obtains the determinant equation:

$$\begin{vmatrix} 1, & x_1, & x_1^2, & \sqrt{f(x_1)} \\ 1, & x_2, & x_2^2, & \sqrt{f(x_2)} \\ 1, & x_3, & x_3^2, & \sqrt{f(x_3)} \\ 1, & x_4, & x_4^2, & \sqrt{f(x_4)} \end{vmatrix} = 0. \tag{11}$$

From this equation one can express x_1 and $\sqrt{f(x_1)}$ rationally in terms of $x_2, \sqrt{f(x_2)}$; $x_3, \sqrt{f(x_3)}$; $x_4, \sqrt{f(x_4)}$; for x_1 is the root of a biquadratic equation whose three other roots are x_2, x_3, x_4, and whose coefficients depend rationally on $x_2, \sqrt{f(x_2)}$; $x_3, \sqrt{f(x_3)}$; $x_4, \sqrt{f(x_4)}$, and $\sqrt{f(x_1)}$ can be represented rationally in terms of x_1 by means of (11). If one sets x_4 equal to an arbitrary constant, the differential equation (10) passes into (1), and (11) yields its integral.

We apply this theorem first to the normal form and set

$$f(x) = x(1-x)(1-\varkappa^2 x).$$

If we assume $x_4 = 0$ in (11), then this equation reduces, on division by $\sqrt{x_1 x_2 x_3}$, to:

$$\begin{vmatrix} \sqrt{x_1}, & x_1\sqrt{x_1}, & \sqrt{(1-x_1)(1-\varkappa^2 x_1)} \\ \sqrt{x_2}, & x_2\sqrt{x_2}, & \sqrt{(1-x_2)(1-\varkappa^2 x_2)} \\ \sqrt{x_3}, & x_3\sqrt{x_3}, & \sqrt{(1-x_3)(1-\varkappa^2 x_3)} \end{vmatrix} = 0, \tag{12}$$

or, if we develop the determinant according to the elements of the first row:

$$\sqrt{x_1}\,(a + bx_1) + c\sqrt{(1-x_1)(1-\varkappa^2 x_1)} = 0, \tag{13}$$

in which, for abbreviation, we have set:

$$
\begin{aligned}
a &= x_2\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2x_3)} - x_3\sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2x_2)},\\
b &= -\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2x_3)} + \sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2x_2)},\\
c &= -(x_2-x_3)\sqrt{x_2}\sqrt{x_3}.
\end{aligned}
\tag{14}
$$

If we make (13) rational, we obtain the cubic equation:

$$x(a+bx)^2 - c^2(1-x)(1-\varkappa^2x) = 0,$$

whose three roots are x_1, x_2, x_3. Accordingly we have identically

$$b^2(x-x_1)(x-x_2)(x-x_3) = x(a+bx)^2 - c^2(1-x)(1-\varkappa^2x), \tag{15}$$

and setting $x = 0, 1, 1:\varkappa^2$ in this identical equation, it follows that:

$$
\begin{aligned}
\sqrt{x_1x_2x_3} &= -\frac{c}{b}\\
\sqrt{(1-x_1)(1-x_2)(1-x_3)} &= \frac{b+a}{b}\\
\sqrt{(1-\varkappa^2x_1)(1-\varkappa^2x_2)(1-\varkappa^2x_3)} &= \frac{b+\varkappa^2a}{b}.
\end{aligned}
\tag{16}
$$

For the determination of the signs in these equations we obtain further from (13), if we replace x_1 by x_2 and by x_3 and multiply the results:

$$x_1x_2x_3(a+bx_1)(a+bx_2)(a+bx_3) = -c^3\sqrt{f(x_1)}\sqrt{f(x_2)}\sqrt{f(x_3)}$$

and (15) gives for $x = -a:b$

$$(a+bx_1)(a+bx_2)(a+bx_3) = \frac{c^2}{b}(b+a)(b+\varkappa^2a).$$

If one uses in addition the square of the first equation (16), there follows from this

$$\sqrt{f(x_1)}\sqrt{f(x_2)}\sqrt{f(x_3)} = -\frac{c}{b^3}(b+a)(b+\varkappa^2a).$$

But the same result is given by the multiplication of the three equations (16), and the signs in these formulae are therefore so chosen that the product of the left-hand sides gives the value of $\sqrt{f(x_1)}$ required by the differential equation (1). A further determination of the signs is, from the nature of the case, not possible in (16), and these equations serve for the unique definition of $\sqrt{x_1}, \sqrt{1-x_1}, \sqrt{1-\varkappa^2x_1}$, when the $\sqrt{x_2}, \sqrt{1-x_2}, \sqrt{1-\varkappa^2x_2}, \sqrt{x_3}, \sqrt{1-x_3}, \sqrt{1-\varkappa^2x_3}$ are presupposed as given.

In order to bring the equations (16) into the customary form, we first make the numerator of b in (14) rational, and obtain

$$b = -\frac{(x_2-x_3)(1-\varkappa^2x_2x_3)}{\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2x_3)} + \sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2x_2)}},$$

further, by (14):

$$
\begin{aligned}
b+a &= -\sqrt{(1-x_2)(1-x_3)}\left\{\sqrt{x_2(1-x_2)}\sqrt{1-\varkappa^2x_3} - \sqrt{x_3(1-x_3)}\sqrt{1-\varkappa^2x_2}\right\}\\
b+\varkappa^2a &= -\sqrt{(1-\varkappa^2x_2)(1-\varkappa^2x_3)}\left\{\sqrt{x_2(1-\varkappa^2x_2)}\sqrt{1-x_3} - \sqrt{x_3(1-\varkappa^2x_3)}\sqrt{1-x_2}\right\},
\end{aligned}
$$

by which the equations (16) are easily brought into the form:

$$
\begin{aligned}
\sqrt{x_1} &= -\frac{\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2 x_3)}+\sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2 x_2)}}{1-\varkappa^2 x_2 x_3},\\
\sqrt{1-x_1} &= \frac{\sqrt{1-x_2}\sqrt{1-x_3}-\sqrt{x_2 x_3}\sqrt{(1-\varkappa^2 x_2)(1-\varkappa^2 x_3)}}{1-\varkappa^2 x_2 x_3},\\
\sqrt{1-\varkappa^2 x_1} &= \frac{\sqrt{1-\varkappa^2 x_2}\sqrt{1-\varkappa^2 x_3}-\varkappa^2\sqrt{x_2 x_3}\sqrt{(1-x_2)(1-x_3)}}{1-\varkappa^2 x_2 x_3},
\end{aligned}
\tag{17}
$$

and our consideration teaches that, if x_1 and the roots $\sqrt{x_1}, \sqrt{1-x_2}, \sqrt{1-\varkappa^2 x_1}$ are determined by these equations as functions of x_2, x_3, the differential equation (1) is identically satisfied.

In order to obtain the addition theorem in the simplest possible shape for the Weierstrass normal form as well, we set

$$f(x) = 4x^3 - g_2 x - g_3,$$

and let x_4 become infinitely great in equation (11) (after division by x_4^2). There then results the integral of the differential equation (1)

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0$$

in the form

$$
\begin{vmatrix}
1, & x_1, & \sqrt{f(x_1)}\\
1, & x_2, & \sqrt{f(x_2)}\\
1, & x_3, & \sqrt{f(x_3)}
\end{vmatrix} = 0,
\tag{18}
$$

or

$$a + bx_1 + c\sqrt{f(x_1)} = 0, \tag{19}$$

if

$$
\begin{aligned}
a &= x_2\sqrt{f(x_3)} - x_3\sqrt{f(x_2)}\\
b &= \sqrt{f(x_2)} - \sqrt{f(x_3)}\\
c &= x_3 - x_2.
\end{aligned}
\tag{20}
$$

Then x_1, x_2, x_3 become the roots of the cubic equation

$$(a+bx)^2 - c^2 f(x) = 0,$$

and if we here set the coefficient of x^2, divided by that of x^3, equal to the negative sum of the roots, we obtain

$$x_1 + x_2 + x_3 = \frac{1}{4}\left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3}\right)^2, \tag{21}$$

and from (19)

$$\sqrt{f(x_1)} = \frac{1}{4}\left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3}\right)^3 - (x_2+x_3)\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3} - \frac{x_3\sqrt{f(x_2)} - x_2\sqrt{f(x_3)}}{x_2 - x_3}. \tag{22}$$

§ 14. Origin of the elliptic functions.

If in the elliptic differential of the first kind in the Legendre normal form

$$\frac{1}{2}\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

the substitution

$$z = \sin^2\varphi \tag{1}$$

is made, and the integration is then carried out from $\varphi = 0$ onwards, there arises the elliptic integral of the first kind

$$\int_0^{\varphi} \frac{d\varphi}{\sqrt{1-\varkappa^2\sin^2\varphi}} = u. \tag{2}$$

Jacobi calls the upper limit φ of this integral its amplitude and writes

$$\varphi = \operatorname{am} u. \tag{3}$$

The trigonometric functions of this arc

$$\sin\varphi,\ \cos\varphi,\ \sqrt{1-\varkappa^2\sin^2\varphi} = \Delta\varphi \tag{4}$$

are now those which, regarded as functions of u, are called elliptic functions and are denoted by Jacobi as

$$\operatorname{sinam} u, \quad \operatorname{cosam} u, \quad \Delta\operatorname{am} u,$$

or, following Gudermann, more briefly as follows:

$$\operatorname{sn} u, \quad \operatorname{cn} u, \quad \operatorname{dn} u \tag{5}$$

These functions depend on the two arguments u and $\varkappa^2$, and when a more exact notation is necessary, one also writes

$$\operatorname{sn}(u,\varkappa^2), \quad \operatorname{cn}(u,\varkappa^2), \quad \operatorname{dn}(u,\varkappa^2)$$

instead.

If in formula (1) φ is changed into $-\varphi$, then u passes into $-u$; thus $\operatorname{am}(-u) = -\operatorname{am} u$, and consequently

$$\operatorname{sn}(-u) = -\operatorname{sn} u, \quad \operatorname{cn}(-u) = \operatorname{cn} u, \quad \operatorname{dn}(-u) = \operatorname{dn} u,$$

that is, sn u is an odd function, cn u, dn u sin even functions. If one therefore puts

$$\frac{d\varphi_2}{\Delta\varphi_2} = du, \quad \frac{d\varphi_3}{\Delta\varphi_3} = dv, \quad \frac{d\varphi_1}{\Delta\varphi_1} = -d(u+v),$$

then the formulae (17), § 13 can be applied, and one obtains the addition theorems:

$$\begin{aligned}
\operatorname{sn}(u+v) &= \frac{\operatorname{sn} u\operatorname{cn} v\operatorname{dn} v + \operatorname{sn} v\operatorname{cn} u\operatorname{dn} u}{1-\varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v},\\
\operatorname{cn}(u+v) &= \frac{\operatorname{cn} u\operatorname{cn} v - \operatorname{sn} u\operatorname{sn} v\operatorname{dn} u\operatorname{dn} v}{1-\varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v},\\
\operatorname{dn}(u+v) &= \frac{\operatorname{dn} u\operatorname{dn} v - \varkappa^2\operatorname{sn} u\operatorname{sn} v\operatorname{cn} u\operatorname{cn} v}{1-\varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v}.
\end{aligned} \tag{6}$$

From these the three elliptic functions can be calculated uniquely for arbitrary values of the argument, provided they are determined for some region about the origin, however small. Weierstrass takes this theorem as the starting point of the theory of elliptic functions.

If one puts

$$\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\Delta\varphi} = K,$$

then $\operatorname{sn} K = 1$, $\operatorname{cn} K = 0$, $\operatorname{dn} K = \sqrt{1-\varkappa^2} = \varkappa'$, and the formulae (6) accordingly yield, on putting $v = K$:

$$\begin{aligned} \operatorname{sn}(u+K) &= \frac{\operatorname{cn} u}{\operatorname{dn} u}, \\ \operatorname{cn}(u+K) &= \frac{-\varkappa' \operatorname{sn} u}{\operatorname{dn} u}, \\ \operatorname{dn}(u+K) &= \frac{\varkappa'}{\operatorname{dn} u}, \end{aligned} \tag{7}$$

and if this formula is applied once more:

$$\begin{aligned} \operatorname{sn}(u+2K) &= -\operatorname{sn} u, \\ \operatorname{cn}(u+2K) &= -\operatorname{cn} u, \\ \operatorname{dn}(u+2K) &= \operatorname{dn} u, \end{aligned} \tag{8}$$

and applied once again:

$$\begin{aligned} \operatorname{sn}(u+4K) &= \operatorname{sn} u, \\ \operatorname{cn}(n+4K) &= \operatorname{cn} u, \\ \operatorname{dn}(u+4K) &= \operatorname{dn} u. \end{aligned} \tag{9}$$

The elliptic functions therefore have the property of periodicity in common with the trigonometric functions. They have the period $4K$ (dn u also the period $2K$).

In § 3 we derived, as an example, the transformation, formula (1):

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}} = \frac{dx}{i\sqrt{x(1-x)(1-\varkappa'^2 x)}}, \tag{10}$$

when

$$z = \frac{-x}{1-x} \tag{11}$$

held. If we therefore set

$$\frac{dx}{\sqrt{x(1-x)(1-\varkappa'^2 x)}} = i\,du,$$

then, if x and u vanish together, $x = \operatorname{sn}^2(iu, \varkappa')$, $1 - x = \operatorname{cn}^2(iu, \varkappa')$, and from (10) it follows that

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}} = du,$$

and hence

$$z = \operatorname{sn}^2(u, \varkappa).$$

Accordingly (11) gives

$$\operatorname{sn}(u, \varkappa) = \frac{-i \operatorname{sn}(iu, \varkappa')}{\operatorname{cn}(iu, \varkappa')} \tag{12}$$

(in which the sign is to be determined from the special value $u = 0$).

If we now set

$$\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1-\varkappa'^2 \sin^2\varphi}} = K',$$

then from (8) there results

$$\operatorname{sn}(iu + 2K', \varkappa') = -\operatorname{sn}(iu, \varkappa'),$$
$$\operatorname{cn}(iu + 2K', \varkappa') = -\operatorname{cn}(iu, \varkappa'),$$

and consequently from (12)

$$\operatorname{sn}(u - 2iK', \varkappa) = \operatorname{sn}(u, \varkappa)$$

or else, if one transforms u into $u + 2iK'$ and again drops the notation $\varkappa$:

$$\operatorname{sn}(u + 2iK') = \operatorname{sn} u. \tag{13}$$

The function sn u therefore has a double periodicity. One of these periods, $4K$, is real; the other, $2iK$, purely imaginary (at least when the modulus $\varkappa$ is a positive proper fraction). This double periodicity is a property which occurs nowhere in the region of the elementary functions, and which thus showed most clearly that one has here to do with a new kind of function. At the same time it is the consequences derived from the double periodicity which, as experience has shown, lead by far the most quickly and with satisfying rigour into the interior of the theory, so that the most suitable starting point for a methodical development is to be sought here. Our next considerations will therefore be devoted to the doubly periodic functions in general.

Second Section.

Theta Functions.

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§ 15. Assumptions from the Theory of Functions.

The concept of doubly periodic functions can be rightly grasped only if one regards them as functions of a complex argument $u = v + iw$, in which i has, as usual, the meaning of $\sqrt{-1}$. The variable u counts for us as an independent and unrestrictedly variable quantity, which, following the precedent of Gauss, is geometrically represented by the points of a plane in such a way that the point whose coordinates in a rectangular system are v, w is regarded as the carrier of the value $u = v + iw$ and is designated briefly as the point u.

We here restrict our consideration to single-valued analytic functions $\varphi(u)$ which have no essentially singular points in the finite region. What happens at infinity we leave undecided. We therefore assume the following of the functions under consideration:

1. In every finite portion of surface there lies a finite number of points at which the function $\varphi(u)$ becomes infinite; these points are called points of discontinuity.

The number of points of discontinuity altogether, i. e. in the whole infinite plane, may of course also be infinitely great.

2. If u_0 is a point not belonging to the points of discontinuity, then the function is developable in a series proceeding by integral ascending positive powers of $u - u_0$, which is convergent in a circle having the point u_0 as its centre and reaching as far as the nearest point of discontinuity. One says that the function has, in the neighbourhood of the point u_0, the character of an integral function.

3. If u_0 is a point of discontinuity, then there is a positive integer m of such a kind that $(u - u_0)^m \varphi(u)$ remains finite, continuous and different from zero at the point u_0, and is therefore

developable in integral positive powers of $u - u_0$. In this case the point of discontinuity is said to be of the mth order. From this there results a development of $\varphi(u)$ in ascending powers of $u - u_0$ which begins with $(u - u_0)^{-m}$ and converges within a circle described about u_0 that reaches as far as the nearest point of discontinuity. The coefficient of $(u - u_0)^{-1}$ in this development is called the residue of this function for the point u_0.

4. Cauchy's theorem. The integral

$$\frac{1}{2\pi i} \int \varphi(u)\, du,$$

extended in the positive sense over the boundary of a finite piece of surface which contains no points of discontinuity on its boundary line, is equal to the sum of the residues for the points of discontinuity lying in the interior of the piece of surface. The positive direction of integration is here to be taken as that one which lies with respect to the interior of the piece of surface as the v-axis lies to the w-axis, so that, with the usual mode of determination of these axes, the interior of the surface remains on the left when the boundary is traversed in the positive sense.

5. The function

$$\psi(u) = \frac{d \log \varphi(u)}{du} = \frac{1}{\varphi(u)} \frac{d\varphi(u)}{du}$$

has only points of discontinuity of the first order, and if at a point u_0 the product $(u - u_0)^{-m}\varphi(u)$ is finite and different from zero, then m is the residue of $\psi(u)$ for this point. If m is positive, then u_0 is called a zero point of the mth order of $\varphi(u)$.

Accordingly the following is an immediate consequence of Cauchy's theorem:

6. The integral

$$\frac{1}{2\pi i} \int d \log \varphi(u),$$

extended in the positive direction over the boundary of a piece of surface, is equal to the number of zero points, diminished by the number of points of discontinuity which lie in the interior of this

piece of surface, where zero points and points of discontinuity of order m are to be counted as m such points of the first order. From this it follows that in the interior of a finite piece of surface there also lies only a finite number of zero points.

In the same sense there results

7. The integral

$$\frac{1}{2\pi i} \int u\, d \log \varphi(u)$$

is equal to the sum of the values of u for which $\varphi(u)$ vanishes, diminished by the sum of the values of u for which $\varphi(u)$ becomes infinite.

8. A function which possesses no points of discontinuity at all in the finite region is called an integral function. An integral function which also remains finite at infinity is a constant. An integral function which also has no essentially singular point at infinity, for which there is therefore an exponent m such that $u^{-m}\varphi(u)$ does not become infinite for $u = \infty$, is a rational function.

9. An integral function whose reciprocal is likewise integral is called a unit function. Such a function becomes neither infinite nor zero in the finite region. Its logarithm is therefore likewise an integral function, and from this it follows that every unit function can be represented in the form $e^{g(u)}$, in which $g(u)$ is an integral function. Conversely, every expression of this form is a unit function.

10. According to the fundamental investigations of Weierstrass on single-valued analytic functions (Abhandlungen der Berliner Akademie von 1876) there is always an integral function $G(u)$ which vanishes at the points of discontinuity of a function $\varphi(u)$ and only at these, and this function $G(u)$ is determined by the points of discontinuity of $\varphi(u)$ up to a unit function as a factor.

Then $\varphi(u)G(u) = G_1(u)$ is likewise an integral function, and one can therefore represent every function $\varphi(u)$ as the quotient of two integral functions

$$\varphi(u) = \frac{G_1(u)}{G(u)}$$

in which $G(u)$ and $G_1(u)$ have no zero points in common.

11. Numerator and denominator of this representation are uniquely determined by $\varphi(u)$ itself, apart from unit factors.

The functions $G(u)$, $G_1(u)$ can be decomposed into prime factors, i.e. into factors which become zero at only one point. If one pursues this path in the case of the doubly periodic functions, one arrives at the Weierstrass σ-functions. We shall, however, take another path, on which we shall not meet the σ-functions until a later place.

§ 16. Periodicity.

A function $\varphi(u)$ of u which has the property that, for a constant ω and for every value of u,

$$\varphi(u + \omega) = \varphi(u) \tag{1}$$

holds, is called periodic, and ω is called the period.

The subject of our investigation is functions $\varphi(u)$ with two periods ω_1, ω_2, of which we shall presuppose once for all that their ratio $\omega_2 : \omega_1$ is not real and that the imaginary part of this ratio is positive. In the latter assumption there lies only a stipulation concerning the notation ω_1, ω_2. For if $\omega_2 : \omega_1$ has a negative imaginary part, then that of $\omega_1 : \omega_2$ is positive.

Every combination $m_1\omega_1 + m_2\omega_2$ is likewise a period, if m_1, m_2 are integers.

We take an arbitrary point u_0 in the plane u and denote the four points

$$u_0, \quad u_0 + \omega_1, \quad u_0 + \omega_1 + \omega_2, \quad u_0 + \omega_2;$$

if we join these four points in order by straight lines, returning from the last to the first, we obtain a parallelogram, which is called the period parallelogram. The rectilinear sides of the period parallelogram may also be replaced by curvilinear paths, provided only that the opposite ones come into coincidence by parallel displacement and that the individual paths form no loops.

If $\omega_1 = \alpha_1 + \beta_1 i$, $\omega_2 = \alpha_2 + \beta_3 i$ and $\alpha_1, \alpha_2, \beta_1, \beta_3$ are real, then by hypothesis $(\alpha_1\beta_2 - \alpha_2\beta_1)$ is positive, and the geometrical meaning of this quantity is (by known theorems

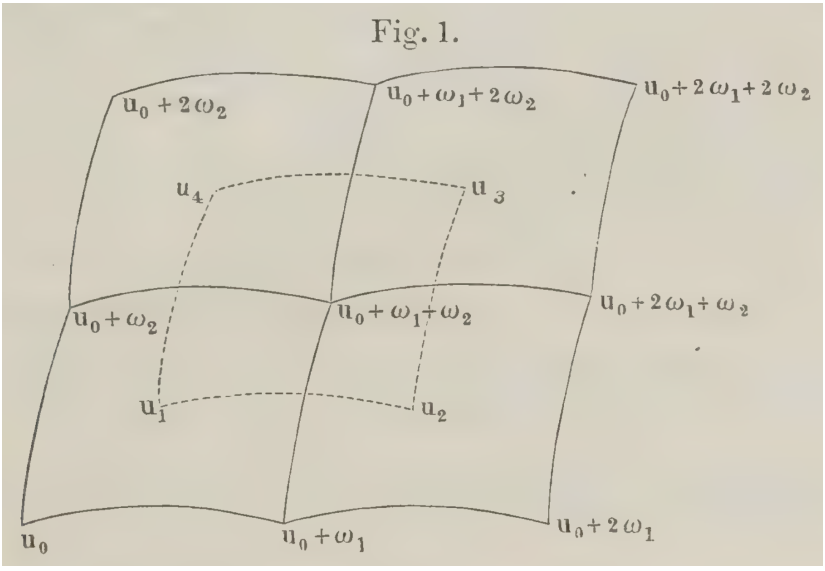
of analytic geometry) the area of the period parallelogram.

By ranging congruent period parallelograms one against another, the whole u-plane can be covered simply and without gaps. Corresponding points of two different ones among these parallelograms are the representatives of u-values that differ by integral multiples of the periods, and that are called congruent with respect to the modulus (ω_1, ω_2) or, when no doubt is possible, congruent for short. The symbol of congruence is

$$u \equiv u' \pmod{\omega_1,\ \omega_2},$$

and it says that u' has the form $u + m_1\omega_1 + m_2\omega_2$, if m_1, m_2 are integers.

Fig. 1.



Thus Fig. 1 represents four period parallelograms laid one against another; the points u_1, u_2, u_3, u_4 are congruent:

$$u_2 = u_1 + \omega_1\,, \quad u_3 = u_1 + \omega_1 + \omega_2\,, \quad u_4 = u_1 + \omega_2\,,$$

and u_1, u_2, u_3, u_4 likewise form the corners of a period parallelogram.

The doubly periodic function $\varphi(u)$ has the same value at all congruent points, and the whole range of values of this function is therefore exhausted within a single period parallelogram, provided that of two opposite sides only the one is reckoned as belonging to the parallelogram.

From our general assumptions we draw the conclusion:

1. There exists (apart from the constant) no doubly periodic function that is free of points of discontinuity in the period parallelogram; for such a function would be finite in the whole u-plane and would therefore, by § 15, 8, have to be a constant.

If $\varphi(u)$ is a doubly periodic function with the periods ω_1, ω_2, then the integral taken over the boundary of the period parallelogram is [because of (1)]

$$\int d\log\varphi(u) = \int_{u_0}^{u_0+\omega_1} [d\log\varphi(u) - d\log\varphi(u+\omega_2)]$$
$$- \int_{u_0}^{u_0+\omega_2} [d\log\varphi(u) - d\log\varphi(u+\omega_1)] = 0\,.$$

From this there follows, by § 15, 6., the theorem:

2. A doubly periodic function becomes zero in a period parallelogram at just as many points as it becomes infinite. If m is this number, then m is called the order of the doubly periodic function. By 1. there are no doubly periodic functions of order zero.

If in the proof one replaces the function $\varphi(u)$ by $\varphi(u) - c$, it follows that a doubly periodic function of order m takes any arbitrary value c at equally many points.

If we take the integral of the formula in 7., § 15 over the boundary of the period parallelogram, there results:

$$\int u\, d\log\varphi(u) = \int_{u_0}^{u_0+\omega_1} [u\, d\log\varphi(u) - (u+\omega_2)\, d\log\varphi(u+\omega_2)]$$

$$-\int_{u_0}^{u_0+\omega_2} [u\,d\log\varphi(u) - (u+\omega_1)\,d\log\varphi(u+\omega_1)]$$

$$= \omega_1 \int_{u_0}^{u_0+\omega_2} d\log\varphi(u) - \omega_2 \int_{u_0}^{u_0+\omega_1} d\log\varphi(u)\,.$$

Now, because of the many-valuedness of the logarithm,

$$\int_{u_0}^{u_0+\omega_2} d\log\varphi(u) = \log\varphi(u+\omega_2) - \log\varphi(u) = 2\pi i\, m_1$$

$$\int_{u_0}^{u_0+\omega_1} d\log\varphi(u) = \log\varphi(u+\omega_1) - \log\varphi(u) = -2\pi i\, m_2\,,$$

where m_1 and m_2 are indeterminate integers.

If therefore the function $\varphi(u)$ becomes infinite at the points α_1, α_2, …, α_m and zero at the points β_1, β_2, …, β_m, then by § 15, 7.:

$$\Sigma\alpha_i \equiv \Sigma\beta_i \pmod{\omega_1,\ \omega_2}\,.$$

If one replaces $\varphi(u)$ by $\varphi(u) - c$, where c denotes an arbitrary value, the important theorem follows:

3. The sum of the argument values at which a doubly periodic function takes one and the same value c in the interior of a period parallelogram is independent of c, or under continuous variation of c changes at most by jumps of a period.

4. From this it follows that there is also no doubly periodic function of the first order.

For such a function would have to take every value c at a single point u. If, then, it is equal to c' at a point u' and equal to c'' at a point u'' different from it, it would follow from 3. that $u' \equiv u''$, contrary to the assumption.

By a doubly periodic function of the second kind one understands, following Hermite, a function $\psi(u)$ which behaves with respect to the periods ω_1, ω_2 according to the equations:

$$\psi(u+\omega_1) = a_1\,\psi(u)\,, \quad \psi(u+\omega_2) = a_2\,\psi(u)\,,$$

where a_1, a_2 are constants.

5. For a doubly periodic function of the second kind also the theorem holds that it becomes zero and infinite at equally many points of the period parallelogram. If the number of these points is m, then here too the function is said to be of the mth order.

The proof is the same as that of theorem 2. But the extension no longer holds here, that the function $\psi(u)$ takes every value equally often, because $\psi(u) - c$ is now no longer periodic.

6. A doubly periodic function $\psi(u)$ of the second kind and of order zero is necessarily a constant or an exponential function. For if $\psi(u)$ does not become zero, then

$$\varphi(u) = \frac{\psi'(u)}{\psi(u)}$$

is a doubly periodic function of the first kind which does not become infinite, and hence, by 1., a constant a. From this it follows:

$$\psi(u) = A\, e^{a\, u}\,,$$

where A is a new constant.

§ 17. The Functions T.

If doubly periodic functions $\varphi(u)$ of the mth order exist, they must be fractional functions. If we therefore set them, according to § 15, 10., in the form

$$\varphi(u) = \frac{G_1(u)}{G(u)}, \tag{1}$$

where $G_1(u)$ and $G(u)$ are integral functions without a common zero point, then

$$\frac{G_1(u+\omega_1)}{G(u+\omega_1)} = \frac{G_1(u+\omega_2)}{G(u+\omega_2)} = \frac{G_1(u)}{G(u)},$$

and in this the three functions $G(u)$, $G(u+\omega_1)$, $G(u+\omega_2)$ have the same m zero points, namely the points of discontinuity of $\varphi(u)$. Their quotients are therefore unit functions, and hence by § 15, 9., if $g_1(u)$, $g_2(u)$ are integral functions:

$$\begin{aligned} G(u+\omega_1) &= e^{g_1(u)}\, G(u), \\ G(u+\omega_2) &= e^{g_2(u)}\, G(u), \end{aligned} \tag{2}$$

and the function $G_1(u)$ satisfies the same equations.

If

$$\varphi(u) = \frac{T_1(u)}{T(u)} \tag{3}$$

is a second representation of $\varphi(u)$ of the form (1), then

$$T(u) = e^{g(u)}\, G(u), \tag{4}$$

where $g(u)$ is again an integral function, and if therefore, by (2),

$$\begin{aligned} T(u+\omega_1) &= e^{l_1(u)}\, T(u), \\ T(u+\omega_2) &= e^{l_2(u)}\, T(u) \end{aligned} \tag{5}$$

holds, then

$$\begin{aligned} g_1(u) &= l_1(u) + g(u) - g(u+\omega_1), \\ g_2(u) &= l_2(u) + g(u) - g(u+\omega_2). \end{aligned}$$

Everything therefore depends on our being able, under some suitable assumption about the integral functions $l_1(u)$, $l_2(u)$, to form integral functions $T(u)$ which satisfy the conditions (5) and vanish at m arbitrarily given points of the period parallelogram. By the quotients of two such functions with the same $l_1(u)$, $l_2(u)$ we can then represent all doubly periodic functions $\varphi(u)$, and

the more general functions $G(u)$ which accomplish the same are obtained, according to (4), by the addition of an arbitrary unit factor.

If we wished to assume $l_1(u)$, $l_2(u)$ to be constants, then $T(u)$ would be an integral doubly periodic function of the second kind, and consequently, by § 16, 6., an exponential function, from which no doubly periodic functions can be formed. The simplest assumption that then remains to us is that $l_1(u)$, $l_2(u)$ are linear functions of u, and this includes as a special case that one of the two functions l_1, l_2 is constant.

We therefore set up the following definition:

1. A T-function, $T(u)$, is an integral function of u which satisfies the following two equations:

$$\begin{aligned} T(u+\omega_1) &= e^{-\pi i[a_1(2u+\omega_1)+b_1]}\, T(u), \\ T(u+\omega_2) &= e^{-\pi i[a_2(2u+\omega_2)+b_2]}\, T(u). \end{aligned} \tag{6}$$

Here a_1, b_1, a_2, b_2 are constants, that is, quantities independent of u. The periods ω_1, ω_2 are here taken throughout as constants given once and for all, whose ratio $\omega_2 : \omega_1$ has a positive imaginary part.

The factors

$$\begin{aligned} e_1 = e_1(u) &= e^{-\pi i[a_1(2u+\omega_1)+b_1]}, \\ e_2 = e_2(u) &= e^{-\pi i[a_2(2a+\omega_2)+b_2]} \end{aligned}$$

are called the two periodicity factors of the function T.

The periodicity factors do not change when b_1, b_2 are altered by even integers. That functions of this kind really exist will appear only in the further course of the investigation.

2. If the function $T(u)$ vanishes at any point, then it vanishes also at all congruent points. If it vanishes at m points of the period parallelogram, it is said to be of the mth order.

From this definition it follows at once:

3. By multiplication of two T-functions T, T' of order m, m' one obtains a T-function of order $m + m'$. The periodicity factors of the product T, T' are the products of the periodicity factors of T and T'.

We ask first about the possibility of T-functions of order zero. If $L(u)$ is such a function and $L'(u)$ its derivative, then twofold differentiation of (6) gives

$$\frac{d}{du}\frac{L'(u+\omega_1)}{L(u+\omega_1)} = \frac{d}{du}\frac{L'(u)}{L(u)},$$

and correspondingly from the second equation (6). Accordingly $d^2 \log L(u)/du^2$, being an integral doubly periodic function, is a constant, and consequently $\log L(u)$ is an integral function of the second degree in u. We therefore obtain:

4. A T-function of order zero is of the form

$$L(u) = C\, e^{-\pi i(\lambda u^2 + \mu u)}, \tag{7}$$

where λ, μ, C are constants.

That every function of this form is a T-function of order zero is seen at once.

The product

$$T'(u) = L(u)\, T(u) \tag{8}$$

is, just like $T(u)$, a T-function of order m, and vanishes at the same points.

The periodicity factors of T' are obtained from those of T by replacing the exponents a_1, a_2, b_1, b_2 by

$$\begin{aligned} a_1' &= a_1 + \lambda\omega_1, \quad b_1' = b_1 + \mu\omega_1 \\ a_2' &= a_2 + \lambda\omega_2, \quad b_2' = b_2 + \mu\omega_2 \end{aligned} \tag{9}$$

in which, however, the b, b' are determined only up to additive even integers.

5. By (9), μ can always be determined, and in only one way, so that b_1', b_2' become real.

For if the imaginary parts of b_1', b_2' are set equal to zero, one obtains two linear equations for the imaginary and the real part of μ, whose determinant is, by § 16, the area of the period parallelogram, and hence different from zero.

The periodicity factors and the zero points of a T-function of order m stand in a certain dependence upon one another, which we obtain without difficulty from the theorems of § 15. First of all, the order m results if one takes the integral

$$\frac{1}{2\pi i}\int d\log T(u) = m$$

extends over the boundary of the period parallelogram. If, for the sake of simplicity, we place the vertex u_0 (Fig. 1) at the origin of coordinates, this integral can be decomposed as follows:

$$\begin{aligned} 2\pi i m = &\int_0^{\omega_1} d\big[\log T(u) - \log T(u+\omega_2)\big] \\ &- \int_0^{\omega_2} d\big[\log T(u) - \log T(u+\omega_1)\big] \end{aligned}$$

and by (6) we have

$$\begin{aligned} d\big[\log T(u) - \log T(u+\omega_2)\big] &= 2\pi i\, a_2, \\ d\big[\log T(u) - \log T(u+\omega_1)\big] &= 2\pi i\, a_1. \end{aligned}$$

From this there results

$$a_2\omega_1 - a_1\omega_2 = m. \tag{10}$$

If we denote further by α_1, α_2, …, α_m the zero points of a T-function in the interior of a period parallelogram, we obtain from § 15, 7.

$$2\pi i \Sigma\alpha = \int u\, d\log Tu,$$

if the integral is again extended over the boundary of the parallelogram. But we have

$$\begin{aligned} \int u\, d\log Tu = &\int_0^{\omega_1} \big[u\, d\log Tu - (u+\omega_2)\, d\log T(u+\omega_2)\big] \\ &- \int_0^{\omega_2} \big[u\, d\log Tu - (u+\omega_1)\, d\log T(u+\omega_1)\big], \end{aligned}$$

and the first of these two integrals is, by (6),

$$\begin{aligned} &- \omega_2 \int_0^{\omega_1} d\log T(u) + 2\pi i\, a_2 \int_0^{\omega_1} (u+\omega_2)\, du \\ &= -\omega_2\big[\log T(\omega_1) - \log T(0)\big] + \pi i\, a_2\omega_1(\omega_1 + 2\omega_2) \\ &= \pi i\, \omega_2(a_1\omega_1 + b_1) + \pi i\, a_2(\omega_1^2 + 2\omega_1\omega_2) + 2\pi i\, N_2\omega_2, \end{aligned}$$

in which N_2 is (because of the many-valuedness of the logarithm) an integer not more closely determined. In the same way there results for the second of the integrals:

$$-\pi i\,\omega_1(a_2\omega_2+b_2)-\pi i\,a_1(\omega_2^2+2\omega_1\omega_2)+2\pi i\,N_1\omega_1,$$

and consequently (written as a congruence)

$$\Sigma\alpha\equiv\frac{1}{2}(b_1\omega_2-b_2\omega_1)+\frac{m}{2}(\omega_1+\omega_2). \tag{11}$$

This sum, or rather the totality of the numbers congruent to it with respect to the modulus (ω_1,ω_2), is called the character of the T-function.

The character of the function T does not change if T is replaced by one of the functions LT (by 8.). One can therefore, following 5., also represent the character in the form:

$$\Sigma\alpha\equiv\frac{1}{2}(g_1\omega_2-g_2\omega_1)+\frac{m}{2}(\omega_1+\omega_2), \tag{12}$$

in which g_1, g_2 are real. These real numbers are determined by the character up to multiples of 2, and can take any value between 0 and 2. The symbol (g_1,g_2) is called the characteristic of the T-function. From the characteristic the character of the T-function is determined by (12).

If γ is the character of a function $T(u)$ of order m, and v is an arbitrary constant, then $T(u+v)$ has the character

$$\gamma'\equiv\gamma-vm.$$

Accordingly, when $m>0$, the different characters can be reduced to one another.

With the help of equation (10) the two equations (6) can be combined into a single general one:

If in the first equation (6) one repeatedly changes u into $u+\omega_1$, one obtains the system:

$$\begin{aligned}
T(u+\omega_1)&=e^{-\pi i[a_1(2u+\omega_1)+b_1]}T(u),\\
T(u+2\omega_1)&=e^{-\pi i[a_1(2u+3\omega_1)+b_1]}T(u+\omega_1),\\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\
T(u+n_1\omega_1)&=e^{-\pi i[a_1(2u+(2n_1-1)\omega_1)+b_1]}T[u+(n_1-1)\omega_1],
\end{aligned}$$

and by multiplication of all these formulae:

$$T(u+n_1\omega_1)=e^{-\pi i[a_1(2n_1u+n_1^2\omega_1)+n_1b_1]}T(u). \tag{13}$$

This formula has been derived in the first instance for a positive integer n_1. But if one replaces u in it by $u-n_1\omega_1$, its correctness follows for negative n_1 as well.

In the same way one obtains:

$$T(u+n_2\omega_2)=e^{-\pi i[a_2(2n_2u+n_2^2\omega_2)+n_2b_2]}T(u), \tag{14}$$

and if in this formula one changes u into $u+n_1\omega_1$ and applies (13) again:

$$\begin{aligned}
&T(u+n_1\omega_1+n_2\omega_2)\\
&=e^{-\pi i[2(a_1n_1+a_2n_2)u+2a_2n_1n_2\omega_1+a_1n_1^2\omega_1+a_2n_2^2\omega_2+n_1b_1+n_2b_2]}T(u).
\end{aligned} \tag{15}$$

But by (10) we have

$$2a_2n_1n_2\omega_1=a_1n_1n_2\omega_1+a_2n_1n_2\omega_2+mn_1n_2,$$

and accordingly this last formula becomes:

$$(16)\qquad \begin{aligned} &T(u+n_1\omega_1+n_2\omega_2)\\ &=e^{-\pi i[(n_1a_1+n_2a_2)(2u+n_1\omega_1+n_2\omega_2)+\pi i(b_1n_1+b_2n_2+mn_1n_2)]}T(u). \end{aligned}$$

And here n_1 and n_2 are arbitrary positive or negative integers.

If two T-functions of orders m and m', with the characters γ, γ', are multiplied together, there arises a new T-function whose order is $m+m'$ and whose character is $\gamma+\gamma'$. If (g_1, g_2) and (g_1', g_2') are the characteristics of the two factors, then (g_1+g_1', g_2+g_2') is the characteristic of the product.

§ 18. Relations between related T-Functions.

Two T-functions with the same periods, the same order and the same character we shall call related. If $T'(u)$ is a T-function related to $T(u)$, and if a_1', a_2', b_1', b_2' have the same meaning for T' as a_1, a_2, b_1, b_2 have for T, then in consequence of the assumed relationship [§ 17, (10), (11)]:

$$(1)\qquad a_1'\omega_2-a_2'\omega_1=a_1\omega_2-a_2\omega_1,$$

$$(2)\qquad b_1'\omega_2-b_2'\omega_1=b_1\omega_2-b_2\omega_1+2n_1\omega_2-2n_2\omega_1,$$

where n_1, n_2 are integers.

Accordingly λ and μ can be determined by § 17, (9) in such a way that the periodicity factors of the function

$$e^{-\pi i(\lambda u^2+\mu u)}T(u)$$

become the same as those of the function $T'(u)$, and hence we have the theorem:

1. Related T-functions can, by the addition of a T-function of order zero:

$$L=Ce^{-\pi i(\lambda u^2+\mu u)}$$

as a factor, be given the same periodicity factors.

Further, from the equality of the characters there follows immediately:

2. If related T-functions of order m have $m-1$ common zero points in the period parallelogram, then they have the mth in common as well; and from this:

3. Between at most $m+1$ related T-functions of order m, $T, T_1, \ldots, T_m$, there exists an identical equation of the form

$$LT+L_1T_1+\cdots+L_mT_m=0.$$

For if this relation does not already hold for $L=0$, then among the constants available in $L_1, L_2, \ldots, L_m$ one can first determine $\lambda_1, \lambda_2 \ldots, \lambda_m, \mu_1, \mu_2 \ldots, \mu_m$ in such a way that all the products $L_1T_1, L_2T_2, \ldots, L_mT_m$ receive the same periodicity factors, and then the $C_1, C_2, \ldots, C_m$ in such a way that $m-1$ of the zero points of the function $L_1T_1+L_2T_2+\cdots+L_mT_m$, and consequently all of them, coincide with the zero points of the function T. From this, however, it follows that the ratio of $L_1T_1+L_2T_2+\cdots+L_mT_m$ to T is equal to a T^0-function $-L$, which is what was to be proved.

4. The quotients of related T-functions can be transformed into doubly periodic functions by the addition of a factor L.

5. If we assume the existence of a T-function of the first order $t(u)$, then every T-function of order m can be derived from it.

For let γ be the character of $t(u)$, and let $\alpha_1, \alpha_2, \ldots, \alpha_m$ be arbitrarily given values. Then the function

$$t(u-\alpha_i+\gamma)=t_i(u), \qquad i=1,2\ldots m$$

vanishes at the point α_i and at all points congruent to α_i, but at no other.

The product

$$T(u) = t_1(u)t_2(u)\dots t_m(u)$$

is therefore a T-function of order m with the arbitrarily given zero points $\alpha_1, \alpha_2, \dots, \alpha_m$.

6. From this it can easily be proved that every doubly periodic function which has only a finite number of points of discontinuity in a period parallelogram can be represented as the quotient of two T-functions.

For let $\varphi(u)$ be a doubly periodic function with the periods ω_1, ω_2 and the points of discontinuity $\alpha_1, \alpha_2, \dots, \alpha_m$, which may also in part coincide in points of discontinuity of higher order.

If, according to 5., one determines a function $T(u)$ with the zero points $\alpha_1, \alpha_2, \dots, \alpha_m$, then:

$$T(u)\varphi(u) = T_1(u)$$

is a T-function of order m, and

$$\varphi(u) = \frac{T_1(u)}{T(u)} \qquad \text{q. e. d.}$$

The functions $T(u)$, $T_1(u)$ are related, since they have the same periodicity factors and therefore also the same characteristic.

The zero points of $T_1(u)$ are at the same time the zero points of $\varphi(u)$, and the sum of these zero values is therefore congruent with the sum of the values of discontinuity.

§ 19. T-Functions of the First Order.

We now proceed to determine more closely the T-functions of the first order, which we denote by $t(u)$, and from which, as we have seen, the remaining T-functions can be derived.

From 3. of the preceding section it follows that two t-functions of the same characteristic differ from one another only by a factor of the form

$$Ce^{-\pi i(\lambda u^2 + \mu u)}$$

and we now wish, by an extension of the definition, to determine this factor still more closely.

This is to be done — which by § 17, (9) is possible without restriction of generality — in such a way that in the periodicity factors

$$a_1 = 0, \quad b_1 = g_1, \quad b_2 = g_2$$

holds, where (g_1, g_2) is the characteristic; on account of the relation

$$a_2\omega_1 - a_1\omega_2 = 1$$

we then have

$$a_2 = \frac{1}{\omega_1},$$

and the conditions for this t-function then read (§ 17, 1.):

$$\begin{aligned} t(u+\omega_1) &= e^{-\pi i g_1} t(u), \\ t(u+\omega_2) &= e^{-\pi i \frac{2u+\omega_2}{\omega_1}} e^{-\pi i g_2} t(u), \end{aligned} \tag{1}$$

and by this the function $t(u)$ is determined up to a factor independent of u. In order to determine this factor still more closely, we

determine, we take into view the dependence of the function t upon ω_1, ω_2 as well, and, regarding g_1, g_2 as given numbers independent of ω_1, ω_2, we denote it by $t(u, \omega_1, \omega_2)$. It then appears that, if h is an arbitrary factor,

$$t(h\,u,\ h\,\omega_1,\ h\,\omega_2)$$

likewise satisfies the conditions (1), and that accordingly

$$\frac{t(hu,\ h\omega_1,\ h\omega_2)}{t(u,\ \omega_1,\ \omega_2)} = f(\omega_1,\ \omega_2) \tag{2}$$

is independent of u.

But the function $f(\omega_1, \omega_2)$ can always be put into the form:

$$f(\omega_1,\ \omega_2) = \frac{\varphi(h\omega_1,\ h\omega_2)}{\varphi(\omega_1,\ \omega_2)};$$

one has only to put, if $t(u)$ does not vanish for $u = 0$:

$$\varphi(\omega_1,\ \omega_2) = t(0,\ \omega_1,\ \omega_2),$$

and, if $t(0) = 0$, say

$$\varphi(\omega_1,\ \omega_2) = \omega_1\, t'(0,\ \omega_1,\ \omega_2)$$

where t' denotes the derivative of t with respect to u. If, then, one now denotes

$$\frac{t(u,\ \omega_1,\ \omega_2)}{\varphi(\omega_1,\ \omega_2)} \tag{3}$$

again by $t(u, \omega_1, \omega_2)$, one may add to the conditions (1) the further one that for an arbitrary h

$$t(u,\ \omega_1,\ \omega_2) = t(hu,\ h\omega_1\ h\omega_2). \tag{4}$$

The function t thus depends now upon only two variables, namely upon the ratios $u : \omega_1 : \omega_2$. In (4) we put

$$h = \frac{1}{\omega_1}, \quad \frac{u}{\omega_1} = v, \quad \frac{\omega_2}{\omega_1} = \omega,$$

and obtain

$$t(u,\ \omega_1,\ \omega_2) = t\left(\frac{u}{\omega_1},\ 1,\ \frac{\omega_2}{\omega_1}\right) = \vartheta(v, \omega), \tag{5}$$

wherein ϑ is a new function symbol.

In this way, then, a new function $\vartheta(v)$ is defined, of only two variables v, ω, the first of which is unrestrictedly variable, while ω, by the assumption made in § 16, has a positive imaginary part; v is called the argument, ω the modulus of the ϑ-function.

§ 20. The ϑ-Function.

The function $\vartheta(v)$ is an integral function of v which satisfies the conditions:

$$\begin{aligned} \vartheta(v+1) &= e^{-\pi i g_1}\vartheta(v), \\ \vartheta(v+\omega) &= e^{-\pi i(2v+\omega)}e^{-\pi i g_2}\vartheta(v), \end{aligned} \tag{1}$$

and it is thereby determined up to a factor independent of v. It is therefore contained among the t-functions as a special case, while on the other hand the general t-function can be expressed by the ϑ-function in the following manner:

$$t(u) = C\, e^{-\pi i(\lambda u^2 + \mu u)}\, \vartheta\left(\frac{u}{\omega_1},\ \frac{\omega_2}{\omega_1}\right),$$

in which C, λ, μ are arbitrary quantities, either constant or dependent on ω_1, ω_2.

The function ϑ depends further on the numbers g_1, g_2 occurring in the characteristic, and if a notation for this dependence is required, then for $\vartheta(v, \omega)$ we shall set:

$$\vartheta_{g_1, g_2}(v, \omega)$$

where g_1, g_2 may be arbitrary numbers. According to what has gone before, it suffices to assume them real and between 0 and 2.

Corresponding to the T-functions of higher order, we shall also introduce Θ-functions of the mth order, and understand by these an integral function of v which satisfies the conditions:

$$\begin{aligned} \Theta(v+1) &= e^{-\pi i g_1}\Theta(v) \\ \Theta(v+\omega) &= e^{-m\pi i(2v+\omega)}e^{-\pi i g_2}\Theta(v). \end{aligned} \tag{2}$$

Here too the characteristic g_1, g_2 can be taken up into the notation:

$$\Theta_{g_1, g_2}(v, \omega).$$

These functions are contained as special cases among the T-functions, and one can form general T-functions in the following manner:

$$C\, e^{-\pi i(\lambda u^2 + \mu u)}\, \Theta\left(\frac{u}{\omega_1},\ \frac{\omega_2}{\omega_1}\right). \tag{3}$$

There follows therefore from § 18, 3. the fundamental theorem for the Θ-functions, which we state as follows:

If we call Θ-functions with the same periods, of the same order and the same characteristic, related, and if we further designate a system of functions as linearly dependent or independent according as a homogeneous linear relation with constant (not all vanishing) coefficients does or does not subsist between these functions, then the theorem holds:

$m + 1$ related Θ-functions of order m are always linearly dependent;

or:

From m linearly independent related Θ-functions of order m every other related Θ-function can be composed linearly (with constant coefficients).

This theorem is of the most fundamental importance for the considerations that follow; from it there results, by § 18, 3., that every T-function of order m can be composed, with the use of formula (3), from m linearly independent Θ-functions.

By § 17, (11), the sum of the m argument values for which the function $\Theta_{g_1, g_2}(v)$ vanishes in the period parallelogram is congruent, with respect to the modulus $(1, \omega)$, to

$$m\,\frac{1+\omega}{2} + \frac{g_1\omega - g_2}{2}.$$

The definition of the function ϑ given up to now leaves undetermined a factor which may be a function of ω. But by an addition to the definition we can determine this factor still more closely.

If the defining equations (1) are differentiated twice with respect to v and once with respect to ω, regarding g_1, g_2 as constant, then a simple calculation shows that the function

$$\frac{\partial^2\vartheta(v,\omega)}{\partial v^2} - 4\pi i\,\frac{\partial\vartheta(v,\omega)}{\partial\omega}$$

itself satisfies the conditions (1) and therefore has the form

$$\varphi(\omega)\,\vartheta(v,\omega)$$

in which $\varphi(\omega)$ is constant with respect to v, and hence a function of ω alone. If one now

$$e^{-\frac{1}{4\pi i}\int \varphi(\omega)\, d\omega}\ \vartheta(v,\omega)$$

equal to a new ϑ, there results for the latter the partial differential equation

$$\frac{\partial^2\vartheta(v,\omega)}{\partial v^2} - 4\pi i\,\frac{\partial\vartheta(v,\omega)}{\partial\omega} = 0, \tag{4}$$

and by (1) and (4) the function ϑ is now defined up to a factor that is independent of v and ω and therefore still depends only on g_1, g_2.

§ 21. The Theta Functions of Different Characteristics. Principal Characteristics.

Before we go on to determine the factor in the ϑ-functions that still remains and is independent of v and ω, we now wish to reduce the different characteristics to one another, which by § 17 is always possible. If we denote by $\vartheta(v)$ the ϑ-function belonging to the characteristic $(0,0)$, then

$$\Phi(v,\omega) = e^{-\pi i g_1 v}\,\vartheta\left(v - \frac{g_1\omega - g_2}{2}\right)$$

is a function satisfying the conditions (1) §20. But now, in view of the differential equation (4):

$$\frac{\partial^2\Phi}{\partial v^2} - 4\pi i\,\frac{\partial\Phi}{\partial\omega} = -\pi^2 g_1^2\,\Phi,$$

and so, if we put

$$\vartheta_{g_1,\,g_2}(v) = e^{\frac{\pi i\omega g_1^2}{4} - \pi i g_1 v + \frac{g_1 g_2 \pi i}{2}}\ \vartheta\left(v - \frac{g_1\omega - g_2}{2}\right) \tag{1}$$

then the differential equation (4) §20 is satisfied by all these functions. In this way, then, the functions $\vartheta_{g_1,\,g_2}$ are determined up to a numerical factor common to them all.

From formula (1) a more general one can be derived by replacing v by $v - \dfrac{g_1'\omega - g_2'}{2}$. One obtains in this way

$$\vartheta_{g_1,\,g_2}\left(v - \frac{g_1'\omega - g_2'}{2}\right) = e^{-\frac{\pi i\omega}{4}g_1'^2 + \pi i g_1' v - \pi i g_1 g_2' - \frac{\pi i}{2}g_1'(g_2+g_2')}\ \vartheta_{g_1+g_1',\,g_2+g_2'}(v). \tag{2}$$

From (2) there follows further, with the use of (1) § 20, if one puts $g_1' = 2$, $g_2' = 0$, or $g_1' = 0$, $g_2' = 2$:

$$\begin{aligned}\vartheta_{g_1+2,\,g_2}(v) &= e^{2\pi i g_2}\,\vartheta_{g_1,g_2}(v),\\ \vartheta_{g_1,\,g_2+2}(v) &= e^{\pi i g_1}\,\vartheta_{g_1,g_2}(v),\end{aligned} \tag{3}$$

and thus by (2) the periodicity of the functions $\vartheta_{g_1,g_2}(v)$ is at the same time completely expressed. For example, if μ and ν are integers, there follows:

$$\vartheta_{00}\left(v - \frac{(2\nu-1)\,\omega + 2\mu - 1}{2}\right) = (-1)^{\nu+1}\,i\,e^{\frac{-\pi i\omega}{4}(2\nu-1)^2}\,e^{\pi i(2\nu-1)v}\,\vartheta_{11}(v). \tag{4}$$

(5) $$\vartheta_{11}(v-\nu\omega-\mu) = (-1)^{\mu+\nu}\, e^{-\pi i\omega\nu^2}\, e^{2\pi i\nu v}\, \vartheta_{11}(v).$$

Formula (2) is valid for entirely arbitrary, even complex values of g_1, g_2.

If in the defining formulae of the ϑ-functions [§ 20, (1)] one replaces v by $-v-1$, and by $-v-\omega$, and further g_1, g_2 by $-g_1, -g_2$, it appears that $\vartheta_{-g_1,-g_2}(-v)$ satisfies the same conditions as $\vartheta_{g_1,g_2}(v)$, and that accordingly

$$\vartheta_{g_1,g_2}(v), \quad \vartheta_{-g_1,-g_2}(-v)$$

are identical up to a constant factor; in particular, therefore, as follows from $v=0$:

$$\vartheta_{0,0}(v) = \vartheta_{0,0}(-v),$$

and thereupon formula (1) yields, if one replaces v, g_1, g_2 by $-v, -g_1, -g_2$:

(6) $$\vartheta_{g_1,g_2}(v) = \vartheta_{-g_1,-g_2}(-v).$$

If the elements g_1, g_2 are integers, then (g_1, g_2) is called a principal characteristic. There are four essentially different ones, namely:

$$(0,0), \quad (0,1), \quad (1,0), \quad (1,1),$$

and accordingly also four essentially different principal ϑ-functions:

(7) $$\vartheta_{00}(v), \quad \vartheta_{01}(v), \quad \vartheta_{10}(v), \quad \vartheta_{11}(v),$$

of which the first is also denoted by $\vartheta(v)$.

In what follows, by characteristics and ϑ-functions only principal characteristics and principal functions will be understood.

Formula (2), applied to this system of functions, yields the following, frequently used table:

(8)
$$\begin{aligned}
\vartheta_{00}(v) &= \vartheta_{01}\Big(v+\tfrac{1}{2}\Big) = \varepsilon\,\vartheta_{10}(v+\tfrac{\omega}{2}) = \varepsilon\,\vartheta_{11}\Big(v+\tfrac{1+\omega}{2}\Big),\\
\vartheta_{01}(v) &= \vartheta_{00}\Big(v+\tfrac{1}{2}\Big) = -i\varepsilon\,\vartheta_{11}(v+\tfrac{\omega}{2}) = i\varepsilon\,\vartheta_{10}\Big(v+\tfrac{1+\omega}{2}\Big),\\
\vartheta_{10}(v) &= \vartheta_{11}\Big(v+\tfrac{1}{2}\Big) = \varepsilon\,\vartheta_{00}(v+\tfrac{\omega}{2}) = \varepsilon\,\vartheta_{01}\Big(v+\tfrac{1+\omega}{2}\Big),\\
\vartheta_{11}(v) &= -\vartheta_{10}\Big(v+\tfrac{1}{2}\Big) = -i\varepsilon\,\vartheta_{01}(v+\tfrac{\omega}{2}) = -i\varepsilon\,\vartheta_{11}\Big(v+\tfrac{1+\omega}{2}\Big),
\end{aligned}$$

in which, for brevity,

$$\varepsilon = e^{\frac{\pi i\omega}{4}+\pi iv}$$

has been set.

According to (3), (4) we have

(9)
$$\begin{aligned}
&\vartheta_{00}(-v) = \vartheta_{00}(v), \quad \vartheta_{01}(-v) = \vartheta_{01}(v),\\
&\vartheta_{10}(-v) = \vartheta_{10}(v), \quad \vartheta_{11}(-v) = -\vartheta_{11}(v),
\end{aligned}$$

that is, $\vartheta_{00}(v)$, $\vartheta_{01}(v)$, $\vartheta_{10}(v)$ are even functions, $\vartheta_{11}(v)$ is an odd function.

According to § 17 (12) the four functions (7) vanish, respectively, for the following values of the argument

$$\frac{1+\omega}{2}, \quad \frac{\omega}{2}, \quad \frac{1}{2}, \quad 0,$$

hence for certain half periods. Of importance are the values which all the functions (7) assume for these values of the argument, and these can be reduced, by means of the table (8), to the three constants:

$$\vartheta_{00}(0) = \vartheta_{00}, \quad \vartheta_{01}(0) = \vartheta_{01}, \quad \vartheta_{10}(0) = \vartheta_{10}$$

One thus obtains from (8) the following system of formulae:

$$\begin{aligned}
\vartheta_{00} &= \vartheta_{01}\Big(\tfrac{1}{2}\Big) = \varepsilon_0\,\vartheta_{10}\big(\tfrac{\omega}{2}\big) = \varepsilon_0\,\vartheta_{11}\Big(\tfrac{1+\omega}{2}\Big), \\
\vartheta_{01} &= \vartheta_{00}\Big(\tfrac{1}{2}\Big) = -i\varepsilon_0\,\vartheta_{11}\big(\tfrac{\omega}{2}\big) = i\varepsilon_0\,\vartheta_{10}\Big(\tfrac{1+\omega}{2}\Big), \\
\vartheta_{10} &= \vartheta_{11}\Big(\tfrac{1}{2}\Big) = \varepsilon_0\,\vartheta_{00}\big(\tfrac{\omega}{2}\big) = \varepsilon_0\,\vartheta_{01}\Big(\tfrac{1+\omega}{2}\Big),
\end{aligned} \tag{10}$$

in which

$$\varepsilon_0 = e^{\frac{\pi i\omega}{4}}$$

and

$$\vartheta_{11}(0) = 0, \quad \vartheta_{10}\Big(\tfrac{1}{2}\Big) = 0, \quad \vartheta_{01}\big(\tfrac{\omega}{2}\big) = 0, \quad \vartheta_{00}\Big(\tfrac{1+\omega}{2}\Big) = 0. \tag{11}$$

The squares of the functions (7):

$$\vartheta_{00}^2(v), \quad \vartheta_{01}^2(v), \quad \vartheta_{10}^2(v), \quad \vartheta_{11}^2(v) \tag{12}$$

are Θ_{00}-functions of the second order, and consequently there exist two linear relations between them. If we take these relations in the form:

$$\begin{aligned}
\vartheta_{10}^2(v) &= A\,\vartheta_{01}^2(v) + B\,\vartheta_{11}^2(v), \\
\vartheta_{00}^2(v) &= A'\vartheta_{01}^2(v) + B'\vartheta_{11}^2(v),
\end{aligned}$$

then one can easily determine the coefficients on the basis of formulae (10), (11), by setting $v = 0$ and $v = \dfrac{\omega}{2}$. In this way one obtains

$$\begin{aligned}
\vartheta_{01}^2\,\vartheta_{10}^2(v) &= \vartheta_{10}^2\,\vartheta_{01}^2(v) - \vartheta_{00}^2\,\vartheta_{11}^2(v), \\
\vartheta_{01}^2\,\vartheta_{00}^2(v) &= \vartheta_{00}^2\,\vartheta_{01}^2(v) - \vartheta_{10}^2\,\vartheta_{11}^2(v),
\end{aligned} \tag{13}$$

and from this further, on setting $v = \frac{1}{2}$:

$$\vartheta_{00}^4 = \vartheta_{01}^4 + \vartheta_{10}^4. \tag{14}$$

By means of any two of the squares (12) all functions Θ_{00} of the second order can be represented linearly; but the remaining Θ-functions of the second order can also be composed from the functions

ϑ_{g_1,g_2}, for one obtains for each characteristic two linearly independent products, of which the one is an even and the other an odd function, namely:

$$
\begin{array}{lll}
\vartheta_{00}(v)\,\vartheta_{01}(v), & \vartheta_{10}(v)\,\vartheta_{11}(v) & \text{Characteristic } (0,1) \\
\vartheta_{00}(v)\,\vartheta_{10}(v), & \vartheta_{01}(v)\,\vartheta_{11}(v) & \text{„} \quad (1,0) \\
\vartheta_{10}(v)\,\vartheta_{01}(v), & \vartheta_{00}(v)\,\vartheta_{11}(v) & \text{„} \quad (1,1)
\end{array} \tag{15}
$$

According to the same principle all Θ-functions of arbitrary order and of arbitrary principal characteristic can now be formed from the functions ϑ. To establish this, we denote by Θ_0, Θ_1 any two of the ϑ-squares (12), and by $F^{(n)}(\Theta_0, \Theta_1)$ an integral rational and homogeneous function of order n of the two arguments Θ_0, Θ_1. In accordance with this one obtains for the Θ-functions of order m, $\Theta^{(m)}(v)$, the following expressions:

m even **even functions**

$$
\begin{aligned}
\Theta_{00}^{(m)}(v) &= F^{\left(\frac{m}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{01}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{01}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1) \\
\Theta_{10}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{10}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1) \\
\Theta_{11}^{(m)}(v) &= \vartheta_{10}(v)\,\vartheta_{01}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1)
\end{aligned}
$$

odd functions

$$
\begin{aligned}
\Theta_{00}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{10}(v)\,\vartheta_{01}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m-4}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{01}^{(m)}(v) &= \vartheta_{10}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1) \\
\Theta_{10}^{(m)}(v) &= \vartheta_{01}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1) \\
\Theta_{11}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m}{2}-1\right)}(\Theta_0, \Theta_1)
\end{aligned} \tag{16}
$$

m odd **even functions**

$$
\begin{aligned}
\Theta_{00}^{(m)}(v) &= \vartheta_{00}(v)\,F^{\left(\frac{m-1}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{01}^{(m)}(v) &= \vartheta_{01}(v)\,F^{\left(\frac{m-1}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{10}^{(m)}(v) &= \vartheta_{10}(v)\,F^{\left(\frac{m-1}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{11}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{01}(v)\,\vartheta_{10}(v)\,F^{\left(\frac{m-3}{2}\right)}(\Theta_0, \Theta_1)
\end{aligned}
$$

odd functions

$$
\begin{aligned}
\Theta_{00}^{(m)}(v) &= \vartheta_{01}(v)\,\vartheta_{10}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m-3}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{01}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{10}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m-3}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{10}^{(m)}(v) &= \vartheta_{00}(v)\,\vartheta_{01}(v)\,\vartheta_{11}(v)\,F^{\left(\frac{m-3}{2}\right)}(\Theta_0, \Theta_1) \\
\Theta_{11}^{(m)}(v) &= \vartheta_{11}(v)\,F^{\left(\frac{m-1}{2}\right)}(\Theta_0, \Theta_1).
\end{aligned} \tag{17}
$$

That all Θ-functions are representable in this form follows, on the basis of § 20, from the following three considerations.

1. Between even and odd functions no linear dependence can subsist unless a linear dependence already subsists among the even functions by themselves or among the odd ones by themselves.

2. Since two of the ϑ-squares (12) do not stand in a constant ratio, there is also no equation of the form $F^{(n)}(\Theta_0, \Theta_1) = 0$.

3. The total number of the constants which, by (16), (17), occur in the even and odd functions belonging to one and the same characteristic is exactly equal to the order m.

§ 22. The Addition Theorem.

If u, v are two variables, then the products

$$\vartheta_{g_1,g_2}(u+v)\,\vartheta_{g_1',g_2'}(u-v)$$

belong to the Θ-functions of the second order, with the characteristic

$$(g_1+g_1',\ g_2+g_2'),$$

and indeed for each of the two variables u, v; as functions of v they can therefore be represented linearly by the functions (12) and (15) in § 21, so that the variable u occurs in the coefficients. When the characteristic is $(0,0)$, this representation is possible in several ways, since one may choose any two of the functions (12); in the other cases it is possible in only one way. One obtains in this manner 16 formulae, of which, however, 6 are to be derived from the others by mere interchange of v with $-v$, so that only 10 essentially different ones remain. These formulae are derived very easily by first setting them up with indeterminate coefficients and then determining these by putting for v such special values as make one or another of the functions $\vartheta(v)$ vanish. Thus, for example:

$$\vartheta_{00}(u+v)\,\vartheta_{00}(u-v) = A\,\vartheta_{00}^2(v) + B\,\vartheta_{11}^2(v),$$

and if one puts $v=0$ and $v=\dfrac{1+\omega}{2}$, then, making use of the formulae of § 21, one obtains:

$$A\,\vartheta_{00}^2 = \vartheta_{00}^2(u), \quad B\,\vartheta_{00}^2 = \vartheta_{11}^2(u),$$

hence

$$\vartheta_{00}^2\,\vartheta_{00}(u+v)\,\vartheta_{00}(u-v) = \vartheta_{00}^2(u)\,\vartheta_{00}^2(v) + \vartheta_{11}^2(u)\,\vartheta_{11}^2(v), \tag{1}$$

and if in this one replaces u by

$$u+\frac{1}{2}, \quad u+\frac{\omega}{2}, \quad u+\frac{1+\omega}{2}$$

then one forms three further relations, which could also have been derived directly in the same way as (1):

$$\vartheta_{01}^2\,\vartheta_{01}(u+v)\,\vartheta_{01}(u-v) = \vartheta_{01}^2(u)\,\vartheta_{01}^2(v) - \vartheta_{11}^2(u)\,\vartheta_{11}^2(v). \tag{2}$$

$$\vartheta_{10}^2\,\vartheta_{10}(u+v)\,\vartheta_{10}(u-v) = \vartheta_{10}^2(u)\,\vartheta_{10}^2(v) - \vartheta_{11}^2(u)\,\vartheta_{11}^2(v). \tag{3}$$

$$\vartheta_{01}^2\,\vartheta_{11}(u+v)\,\vartheta_{11}(u-v) = \vartheta_{11}^2(u)\,\vartheta_{01}^2(v) - \vartheta_{01}^2(u)\,\vartheta_{11}^2(v). \tag{4}$$

These four equations can, as has already been mentioned, be transformed in manifold ways by applying the formulae (13) of the preceding section. The six formulae that follow have only one form. Beginning again with an arbitrary example, we have:

$$\vartheta_{00}(u+v)\,\vartheta_{11}(u-v) = A\,\vartheta_{01}(v)\,\vartheta_{10}(v) + B\,\vartheta_{00}(v)\,\vartheta_{11}(v),$$

and if one sets $v = 0$:

$$A\,\vartheta_{01}\vartheta_{10} = \vartheta_{00}(u)\,\vartheta_{11}(u),$$

from which one obtains B by interchanging u with v. In this way the three following formulae are derived:

$$\begin{aligned}(5)\qquad &\vartheta_{01}\vartheta_{10}\,\vartheta_{00}(u+v)\,\vartheta_{11}(u-v)\\ &= \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v) - \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v),\end{aligned}$$

$$\begin{aligned}(6)\qquad &\vartheta_{00}\vartheta_{01}\,\vartheta_{10}(u+v)\,\vartheta_{11}(u-v)\\ &= \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(v)\vartheta_{11}(v),\end{aligned}$$

$$\begin{aligned}(7)\qquad &\vartheta_{00}\vartheta_{10}\,\vartheta_{01}(u+v)\,\vartheta_{11}(u-v)\\ &= \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{10}(v) - \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{11}(v),\end{aligned}$$

which can be verified immediately by setting first u, then $v = 0$. From these the three others follow by increasing u by

$$\frac{1}{2}, \quad \frac{\omega}{2}, \quad \frac{1}{2}:$$

$$\begin{aligned}(8)\qquad &\vartheta_{01}\vartheta_{10}\,\vartheta_{01}(u+v)\,\vartheta_{10}(u-v)\\ &= \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{10}(v) + \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{11}(v),\end{aligned}$$

$$\begin{aligned}(9)\qquad &\vartheta_{00}\vartheta_{01}\,\vartheta_{00}(u+v)\,\vartheta_{01}(u-v)\\ &= \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{10}(v)\vartheta_{11}(v),\end{aligned}$$

$$\begin{aligned}(10)\qquad &\vartheta_{00}\vartheta_{10}\,\vartheta_{00}(u+v)\,\vartheta_{10}(u-v)\\ &= \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{10}(v) + \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{11}(v).\end{aligned}$$

These formulae, whose totality is designated by the name of the addition theorem, admit of generalisation in manifold ways, of which the following may serve as an example.

If u, v are two variables, then the four products

$$\begin{aligned}&\vartheta_{00}(u)\,\vartheta_{00}(u+v), \quad \vartheta_{01}(u)\,\vartheta_{01}(u+v),\\ &\vartheta_{10}(u)\,\vartheta_{10}(u+v), \quad \vartheta_{11}(u)\,\vartheta_{11}(u+v),\end{aligned}$$

considered as functions of u, related T-functions of the second order with the same periodicity factors, and hence any three of them are linearly dependent. We therefore have

$$A\,\vartheta_{00}(u)\,\vartheta_{00}(u+v) + B\,\vartheta_{10}(u)\,\vartheta_{10}(u+v) + C\,\vartheta_{11}(u)\,\vartheta_{11}(u+v) = 0,$$

from which, for $u = 0$, $u = \dfrac{1}{2}$:

$$\begin{aligned}&A\,\vartheta_{00}\vartheta_{00}(v) + B\,\vartheta_{10}\vartheta_{10}(v) = 0,\\ &A\,\vartheta_{01}\vartheta_{01}(v) + C\,\vartheta_{10}\vartheta_{10}(v) = 0\end{aligned}$$

and from these:

$$\begin{aligned}(11)\qquad &\vartheta_{10}\vartheta_{10}(v)\,\vartheta_{00}(u)\,\vartheta_{00}(u+v) - \vartheta_{00}\vartheta_{00}(v)\,\vartheta_{10}(u)\,\vartheta_{10}(u+v)\\ &\qquad - \vartheta_{01}\vartheta_{01}(v)\,\vartheta_{11}(u)\,\vartheta_{11}(u+v) = 0.\end{aligned}$$

In the same way, if w is a third variable, the products contained in the form

$$\vartheta_{g_1,g_2}(u+v)\,\vartheta_{g_1,g_2}(u+w), \quad \vartheta_{g_1,g_2}(u)\,\vartheta_{g_1,g_2}(u+v+w)$$

are, considered as functions of u, related T-functions of the second order with the same periodicity factors, so that between any three of them there exists a linear dependence. The coefficients are determined, as above, by special values of u. Thus the system of formulae follows:

$$\begin{aligned}(12)\qquad &\vartheta_{00}\vartheta_{00}(v+w)\,\vartheta_{00}(u+v)\,\vartheta_{00}(u+w)\\ &=\vartheta_{00}(u)\vartheta_{00}(v)\vartheta_{00}(w)\vartheta_{00}(u+v+w)\\ &-\vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w).\end{aligned}$$

$$\begin{aligned}(13)\qquad &\vartheta_{01}\vartheta_{01}(v+w)\,\vartheta_{01}(u+v)\,\vartheta_{01}(u+w)\\ &=\vartheta_{01}(u)\vartheta_{01}(v)\vartheta_{01}(w)\vartheta_{01}(u+v+w)\\ &+\vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w).\end{aligned}$$

$$\begin{aligned}(14)\qquad &\vartheta_{10}\vartheta_{10}(v+w)\,\vartheta_{10}(u+v)\,\vartheta_{10}(u+w)\\ &=\vartheta_{10}(u)\vartheta_{10}(v)\vartheta_{10}(w)\vartheta_{10}(u+v+w)\\ &+\vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w).\end{aligned}$$

All these formulae can be regarded as special cases of a general formula which Jacobi derived from the series developments and employed as the foundation of the theory of elliptic functions[3]. This formula can also be obtained in the following way from the concept of the theta functions.

The four functions

$$(15)\qquad \vartheta_{00}(2v), \quad \vartheta_{01}(2v), \quad \vartheta_{10}(2v), \quad \vartheta_{11}(2v),$$

are Θ_{00}-functions of the fourth order in v, and since they are linearly independent, all Θ_{00}-functions of the fourth order can be expressed linearly by means of them. But such a function is also the product

$$(16)\qquad \vartheta(v+a_1)\,\vartheta(v+a_2)\,\vartheta(v+a_3)\,\vartheta(v+a_4),$$

provided that the quantities a satisfy the condition:

$$(17)\qquad a_1+a_2+a_3+a_4=0$$

We therefore obtain, if we denote by A_1, A_2, A_3, A_4 constants (with respect to v):

$$\begin{aligned}&\vartheta(v+a_1)\,\vartheta(v+a_2)\,\vartheta(v+a_3)\,\vartheta(v+a_4)\\ &=A_1\,\vartheta_{00}(2v)+A_2\,\vartheta_{01}(2v)+A_3\,\vartheta_{10}(2v)+A_4\,\vartheta_{11}(2v),\end{aligned}$$

and from this one obtains, by increasing v by $\frac{1}{2}(g_1\omega+g_2)$, four formulae:

$$\begin{aligned}&\vartheta_{g_1,g_2}(v+a_1)\,\vartheta_{g_1,g_2}(v+a_2)\,\vartheta_{g_1,g_2}(v+a_3)\,\vartheta_{g_1,g_2}(v+a_4)\\ &=A_1\,\vartheta_{00}(2v)+(-1)^{g_2}A_2\,\vartheta_{01}(2v)+(-1)^{g_1}A_3\,\vartheta_{10}(2v)\\ &\quad+(-1)^{g_1+g_2}A_4\,\vartheta_{11}(2v).\end{aligned}$$

[3]Theorie der elliptischen Funktionen, aus den Eigenschaften der Thetareihen abgeleitet. Jacobis gesammelte Werke, Bd. I, S. 497.

If these four formulae are added, it follows that:

$$4A_1\,\vartheta_{00}(2v)$$

$$= \sum^{g_1,g_2} \vartheta_{g_1,g_2}(v+a_1)\,\vartheta_{g_1,g_2}(v+a_2)\,\vartheta_{g_1,g_2}(v+a_3)\,\vartheta_{g_1,g_2}(v+a_4), \tag{18}$$

where in the sum all combinations of 0 and 1 are to be put for g_1 and g_2.

We now introduce a somewhat different notation, putting:

$$v+a_1=v_1', \quad v+a_2=v_2', \quad v+a_3=v_3', \quad v+a_4=v_4',$$

hence, on account of (17),

$$4v=v_1'+v_2'+v_3'+v_4',$$

and we now define the four quantities v_1, v_2, v_3, v_4 by the equations

$$\begin{aligned} v_1&=\tfrac{1}{2}(v_1'+v_2'+v_3'+v_4'),\\ v_2&=\tfrac{1}{2}(v_1'+v_2'-v_3'-v_4'),\\ v_3&=\tfrac{1}{2}(v_1'-v_2'+v_3'-v_4'),\\ v_4&=\tfrac{1}{2}(v_1'-v_2'-v_3'+v_4'). \end{aligned} \tag{19}$$

From this, however, one obtains:

$$\begin{aligned} v_1'&=\tfrac{1}{2}(v_1+v_2+v_3+v_4), & a_1&=\tfrac{1}{2}(\ \ v_2+v_3+v_4),\\ v_2'&=\tfrac{1}{2}(v_1+v_2-v_3-v_4), & a_2&=\tfrac{1}{2}(\ \ v_2-v_3-v_4),\\ v_3'&=\tfrac{1}{2}(v_1-v_2+v_3-v_4), & a_3&=\tfrac{1}{2}(-v_2+v_3-v_4),\\ v_4'&=\tfrac{1}{2}(v_1-v_2-v_3+v_4), & a_4&=\tfrac{1}{2}(-v_2-v_3+v_4), \end{aligned} \tag{20}$$

and formula (18) then gives:

$$4\,A_1\,\vartheta_{00}(v_1)=\sum \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4'),$$

in which now either the v_i or the v_i' may be regarded as independent variables.

If one regards the v_i as independent variables, then in this formula A_1 is independent of v_1, but still depends on v_2, v_3, v_4. If we therefore set $4\,A_1=c\,\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4)$, there results

$$c\,\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4)=\sum \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4'),$$

and in this c is in any case independent of v_1. But since the right-hand side of this formula remains unchanged under arbitrary permutations of v_1, v_2, v_3, v_4, c is independent of all the v_i, and one obtains its value by setting all $v_i=0$:

$$c\,\vartheta_{00}^4=\vartheta_{00}^4+\vartheta_{01}^4+\vartheta_{10}^4,$$

from which, by § 21, (14), it follows that $c=2$.

We thus have the formula:

$$2\,\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4)=\sum^{g_1,g_2} \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4').$$

If in it each of the variables v_1, v_2, v_3, v_4 is increased by a half period $-\frac{1}{2}(g_1'\omega_1-g_2')$, then v_1' is increased by a whole period $-(g_1'\omega_1-g_2')$, while v_2', v_3', v_4' remain unchanged, and accordingly one obtains, with the help of § 21, (2) and (3), a system of four formulae:

$$(21)\qquad \begin{aligned} &2\,\vartheta_{g_1',g_2'}(v_1)\vartheta_{g_1',g_2'}(v_2)\vartheta_{g_1',g_2'}(v_3)\vartheta_{g_1',g_2'}(v_4)\\ &=\sum^{g_1,g_2}(-1)^{g_1g_2'+g_2g_1'+g_1'g_2'}\,\vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_\nu}(v_4'). \end{aligned}$$

These are the Jacobi formulae. If one sets, for example,

$$\begin{aligned} &v_1=0, && v_2=v+w, && v_3=u+v, && v_4=u+w,\\ &v_1'=u+v+w, && v_2'=-u, && v_3'=-w, && v_4'=-v, \end{aligned}$$

then from (21), for $g_1', g_2' = 1, 1$, there results:

$$0=\sum^{g_1,g_2}(-1)^{g_1+g_2}\,\vartheta_{g_1,g_2}(u)\vartheta_{g_1,g_2}(v)\vartheta_{g_1,g_2}(w)\vartheta_{g_1,g_2}(u+v+w),$$

and from this one then obtains the formulae (12), (13), (14), if in (21) one puts the three other characteristics for g_1', g_2'.

§ 23. The Derivatives of the ϑ-Functions.

In what follows we denote by $\vartheta'_{g_1,g_2}(v)$, $\vartheta''_{g_1,g_1}(v)$ the derivatives of the function $\vartheta_{g_1,g_2}(v)$ taken with respect to v, and by ϑ'_{g_1,g_2}, ϑ''_{g_1,g_2} the values of this function for $v=0$. Then ϑ'_{01}, ϑ'_{10}, ϑ'_{00} vanish, because $\vartheta_{01}(v)$, $\vartheta_{10}(v)$, $\vartheta_{00}(v)$ are even functions of v, and likewise ϑ''_{11} vanishes.

The six functions

$$\vartheta_{g_1,g_2}(v)\vartheta'_{g_1',g_2'}(v)-\vartheta_{g_1',g_2'}(v)\vartheta'_{g_1,g_2}(v)$$

are, as follows from the fundamental equations § 20, (1), ϑ-functions of the second order with the characteristic

$$(g_1+g_1',\,g_2+g_2'),$$

and at the same time either even or odd functions, so that by § 21 they can be represented by ϑ-functions. A constant factor is determined by a special value of v ($v=0$). Thus we have

$$(1)\qquad \vartheta'_{11}(v)\vartheta_{01}(v)-\vartheta'_{01}(v)\vartheta_{11}(v)=A\,\vartheta_{10}(v)\vartheta_{00}(v).$$

$$(2)\qquad A=\frac{\vartheta'_{11}\,\vartheta_{01}}{\vartheta_{10}\,\vartheta_{00}}.$$

But this expression for A can still be simplified.

If we differentiate (1) twice with respect to v and then put $v=0$, it follows that

$$\vartheta'''_{11}\vartheta_{01}-\vartheta''_{01}\vartheta'_{11}=A(\vartheta''_{10}\vartheta_{00}+\vartheta_{10}\vartheta''_{00}),$$

and if one introduces for A the value (2):

$$(3)\qquad \frac{\vartheta'''_{11}}{\vartheta'_{11}}=\frac{\vartheta''_{01}}{\vartheta_{01}}+\frac{\vartheta''_{10}}{\vartheta_{10}}+\frac{\vartheta''_{00}}{\vartheta_{00}}.$$

But now the four functions ϑ'_{11}, ϑ_{01}, ϑ_{10}, ϑ_{00} satisfy [by § 20, (4)] the differential equation

$$(4)\qquad \vartheta''=4\pi i\,\frac{\partial\vartheta}{\partial\omega},$$

and accordingly (3) can be written thus:

$$\frac{d\log\vartheta'_{11}}{d\omega}=\frac{d\log\vartheta_{00}\vartheta_{10}\vartheta_{01}}{d\omega},$$

or by integration:

$$\vartheta'_{11} = c\,\vartheta_{00}\vartheta_{10}\vartheta_{01},$$

in which c is independent of ω. By § 20, 21 the ϑ-functions were determined up to a factor independent of v, ω and common to them all. This factor shall now be disposed of in such a way that the constant c receives the value π, so that the formula holds:

$$\vartheta'_{11} = \pi\,\vartheta_{00}\vartheta_{10}\vartheta_{01}, \tag{5}$$

and thereby the ϑ-functions are now completely defined up to the common sign $\pm$. By applying (5), formula (1) can be given the shape:

$$\vartheta'_{11}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{11}(v) = \pi\,\vartheta_{01}^2\,\vartheta_{10}(v)\vartheta_{00}(v), \tag{6}$$

and in the same way one obtains, using § 21, (8):

$$\vartheta'_{11}(v)\vartheta_{00}(v) - \vartheta'_{00}(v)\vartheta_{11}(v) = \pi\,\vartheta_{00}^2\,\vartheta_{10}(v)\vartheta_{01}(v), \tag{7}$$

$$\vartheta'_{11}(v)\vartheta_{10}(v) - \vartheta'_{10}(v)\vartheta_{11}(v) = \pi\,\vartheta_{10}^2\,\vartheta_{01}(v)\vartheta_{00}(v), \tag{8}$$

$$\vartheta'_{10}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{10}(v) = -\pi\,\vartheta_{00}^2\,\vartheta_{00}(v)\vartheta_{11}(v), \tag{9}$$

$$\vartheta'_{00}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{00}(v) = -\pi\,\vartheta_{10}^2\,\vartheta_{10}(v)\vartheta_{11}(v), \tag{10}$$

$$\vartheta'_{00}(v)\vartheta_{10}(v) - \vartheta'_{10}(v)\vartheta_{00}(v) = \pi\,\vartheta_{01}^2\,\vartheta_{01}(v)\vartheta_{11}(v). \tag{11}$$

If we differentiate equations (9), (10), (11) with respect to v and set $v = 0$, we obtain by means of (4) and (5) the relations

$$4\,\frac{d}{d\omega}\log\frac{\vartheta_{10}}{\vartheta_{01}} = i\pi\,\vartheta_{00}^4, \tag{12}$$

$$4\,\frac{d}{d\omega}\log\frac{\vartheta_{00}}{\vartheta_{01}} = i\pi\,\vartheta_{10}^4, \tag{13}$$

$$4\,\frac{d}{d\omega}\log\frac{\vartheta_{10}}{\vartheta_{00}} = i\pi\,\vartheta_{01}^4. \tag{14}$$

If one differentiates the fundamental formulae for the ϑ-functions twice logarithmically, one recognises that the functions

$$\vartheta^2(v)\,\frac{d^2\log\vartheta(v)}{dv^2} = \vartheta''(v)\vartheta(v) - \vartheta'(v)\vartheta'(v)$$

are Θ-functions of the second order with the characteristic $(0,0)$, and one can therefore express them linearly by means of the functions $\vartheta^2(v)$. In this way there results

$$\vartheta_{00}^2\,\vartheta_{00}^2(v)\,\frac{d^2\log\vartheta_{00}(v)}{dv^2} = \vartheta_{00}\vartheta''_{00}\vartheta_{00}^2(v) + \vartheta'^2_{11}\vartheta_{11}^2(v), \tag{15}$$

$$\vartheta_{01}^2\,\vartheta_{01}^2(v)\,\frac{d^2\log\vartheta_{01}(v)}{dv^2} = \vartheta_{01}\vartheta''_{01}\vartheta_{01}^2(v) - \vartheta'^2_{11}\vartheta_{11}^2(v), \tag{16}$$

(17) $$\vartheta_{10}^2\,\vartheta_{10}^2(v)\,\frac{d^2\log\vartheta_{10}(v)}{dv^2}=\vartheta_{10}\vartheta''_{10}\vartheta_{10}^2(v)-\vartheta'^2_{11}\vartheta_{11}^2(v),$$

(18) $$\vartheta_{00}^2\,\vartheta_{11}^2(v)\,\frac{d^2\log\vartheta_{11}(v)}{dv^2}=\vartheta_{00}\vartheta''_{00}\vartheta_{11}^2(v)-\vartheta'^2_{11}\vartheta_{00}^2(v).$$

From these, further relations can still be derived by continued differentiation. We adduce one more of these formulae, which is obtained if one differentiates (15) twice more with respect to v and then sets $v=0$. If the differentiations with respect to v are expressed by those with respect to ω by means of the partial differential equation (4), it follows that:

$$\frac{d^2\log\vartheta_{00}}{d\omega^2}-2\left(\frac{d\log\vartheta_{00}}{d\omega}\right)^2=-\frac{\pi^2}{8}\,\vartheta_{10}^4\vartheta_{01}^4,$$

or

(19) $$\frac{d}{d\omega}\left(\frac{1}{\vartheta_{00}^4}\,\frac{d\vartheta_{00}^2}{d\omega}\right)=-\frac{\pi^2}{4}\,\frac{\vartheta_{10}^4\vartheta_{01}^4}{\vartheta_{00}^2}.$$

§ 24. Representation of the ϑ-Functions by Infinite Products.

The theory of the ϑ-functions has now been carried far enough for their representation to follow very easily. With this the existence of these functions is then established, and the considerations made so far first receive their secure foundation. Two ways to this goal stand open to us. The first proceeds from the zero points of the ϑ-functions already known to us and composes infinite products out of them.

The function $\vartheta_{00}(v)$ vanishes, according to § 21, when

$$2v=(2\nu-1)\omega+(2\mu-1)$$

holds, where ν,μ are integers, or when

$$e^{\pm 2\pi i v}=-e^{\pi i\omega(2\nu-1)},$$

where we can now restrict ν to positive numbers. If we therefore set, for brevity,

$$e^{\pi i\omega}=q,$$

then q is a quantity whose absolute value is a proper fraction. The convergent infinite product

$$P(v)=\prod_{1,\infty}^{\nu}(1+q^{2\nu-1}e^{2\pi iv})(1+q^{2\nu-1}e^{-2\pi iv})$$

therefore vanishes at all and only those points at which $\vartheta_{00}(v)$ vanishes, and moreover

$$P(v+1)=P(v)$$
$$P(v+\omega)=\frac{1+q^{-1}e^{-2\pi iv}}{1+qe^{2\pi iv}}P(v)=q^{-1}e^{-2\pi iv}P(v).$$

But by § 20, (1) this is the period property of the function $\vartheta_{00}(v)$, and we therefore have:

(1) $$\vartheta_{00}(v)=Q\prod_{1,\infty}^{\nu}(1+q^{2\nu-1}e^{2\pi iv})(1+q^{2\nu-1}e^{-2\pi iv}),$$

where Q is a factor independent of v. By the formulae (8) of § 21 one obtains from this, on putting in place of v

$$v + \frac{1}{2}, \qquad v + \frac{\omega}{2}, \qquad v + \frac{1+\omega}{2}$$

respectively:

$$\vartheta_{01}(v) = Q \prod_{1,\infty}^{\nu} (1 - q^{2\nu-1} e^{2\pi i v})(1 - q^{2\nu-1} e^{-2\pi i v}), \tag{2}$$

$$\vartheta_{10}(v) = Q q^{\frac{1}{4}} (e^{\pi i v} + e^{-\pi i v}) \prod_{1,\infty}^{\nu} (1 + q^{2\nu} e^{2\pi i v})(1 + q^{2\nu} e^{-2\pi i v}), \tag{3}$$

$$\vartheta_{11}(v) = -iQ q^{\frac{1}{4}} (e^{\pi i v} - e^{-\pi i v}) \prod_{1,\infty}^{\nu} (1 - q^{2\nu} e^{2\pi i v})(1 - q^{2\nu} e^{-2\pi i v}). \tag{4}$$

If one sets $v = 0$ in these formulae, in the last one after a single differentiation, there follows

$$\begin{aligned} \vartheta_{00} &= Q \prod_{1,\infty}^{\nu} (1 + q^{2\nu-1})^2, \\ \vartheta_{01} &= Q \prod_{1,\infty}^{\nu} (1 - q^{2\nu-1})^2, \\ \vartheta_{10} &= 2Q q^{\frac{1}{4}} \prod_{1,\infty}^{\nu} (1 + q^{2\nu})^2, \\ \vartheta'_{11} &= 2\pi Q q^{\frac{1}{4}} \prod_{1,\infty}^{\nu} (1 - q^{2\nu})^2. \end{aligned} \tag{5}$$

Accordingly, it is possible by means of (5), § 23:

$$\vartheta'_{11} = \pi \vartheta_{00} \vartheta_{01} \vartheta_{10}$$

determine the factor Q. One obtains first

$$1 = Q^2 \prod^{\nu} \frac{(1 - q^{2\nu-1})^2 (1 + q^{2\nu-1})^2 (1 + q^{2\nu})^2}{(1 - q^{2\nu})^2},$$

or, by disposing of the still undetermined sign of Q and thereby of the signs of the ϑ-functions:

$$Q = \prod_{1,\infty}^{\nu} \frac{(1 - q^{2\nu})}{(1 + q^{2\nu})(1 + q^{2\nu-1})(1 - q^{2\nu-1})}.$$

In the denominator one may put $\prod(1 + q^{\nu})$ in place of $\prod(1 + q^{2\nu})(1 + q^{2\nu-1})$, and the numerator $\prod(1 - q^{2\nu})$ can be decomposed into

$$\prod (1 - q^{\nu})(1 + q^{\nu}) = \prod (1 - q^{2\nu})(1 - q^{2\nu-1})(1 + q^{\nu}).$$

In this way there results finally

$$Q = \prod^{\nu} (1 - q^{2\nu}). \tag{6}$$

The expressions (1), (2), (3), (4) can be represented in real form if one carries out the multiplication of the conjugate imaginary factors. One thus obtains:

$$
\begin{aligned}
\vartheta_{00}(v) &= \prod_{1,\infty}^{\nu}(1-q^{2\nu})(1+2q^{2\nu-1}\cos 2\pi v+q^{4\nu-2}),\\
\vartheta_{01}(v) &= \prod_{1,\infty}^{\nu}(1-q^{2\nu})(1-2q^{2\nu-1}\cos 2\pi v+q^{4\nu-1}),\\
\vartheta_{10}(v) &= 2q^{\frac{1}{4}}\cos\pi v\prod_{1,\infty}^{\nu}(1-q^{2\nu})(1+2q^{2\nu}\cos 2\pi v+q^{4\nu}),\\
\vartheta_{11}(v) &= 2q^{\frac{1}{4}}\sin\pi v\prod_{1,\infty}^{\nu}(1-q^{2\nu})(1-2q^{2\nu}\cos 2\pi v+q^{4\nu}).
\end{aligned}
\tag{7}
$$

If we introduce the expression (6) into the last equation (5), there results

$$\vartheta_{11}' = 2\pi q^{\frac{1}{4}}\prod(1-q^{2\nu})^3,$$

and if we therefore put

$$\eta(\omega) = q^{\frac{1}{12}}\prod(1-q^{2\nu}) \tag{8}$$

then we have

$$\vartheta_{11}' = 2\pi\eta(\omega)^3. \tag{9}$$

By eliminating Q with the help of the relation

$$Q = q^{-\frac{1}{12}}\eta(\omega)$$

from (5), we set

$$
\begin{aligned}
\vartheta_{00} &= \eta(\omega)f(\omega)^2,\\
\vartheta_{01} &= \eta(\omega)f_1(\omega)^2,\\
\vartheta_{10} &= \eta(\omega)f_2(\omega)^2,
\end{aligned}
\tag{10}
$$

in which the functions $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ are defined as follows:

$$
\begin{aligned}
f(\omega) &= q^{-\frac{1}{24}}\prod_{1,\infty}^{\nu}(1+q^{2\nu-1}),\\
f_1(\omega) &= q^{-\frac{1}{24}}\prod_{1,\infty}^{\nu}(1-q^{2\nu-1}),\\
f_2(\omega) &= \sqrt{2}\,q^{\frac{1}{12}}\prod_{1,\infty}^{\nu}(1+q^{2\nu}).
\end{aligned}
\tag{11}
$$

The functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ will play an important part in our later investigations. From § 21, (14) there follows, by (10), the relation

$$f(\omega)^8 = f_1(\omega)^8 + f_2(\omega)^8, \tag{12}$$

and from § 23 (5), by (9),

$$f(\omega)f_1(\omega)f_2(\omega) = \sqrt{2}. \tag{13}$$

The last formula also follows from (11) by the use of the identical relation:

$$\prod(1+q^{\nu})(1-q^{2\nu-1}) = \prod\frac{(1+q^{\nu})(1-q^{\nu})}{1-q^{2\nu}} = 1.$$

§ 25. Representation of the ϑ-Functions by Infinite Series.

The second way of arriving at the representation of the ϑ-functions consists in seeking to form a convergent infinite series that satisfies the fundamental equations.

Let us first observe that, on account of the condition

$$\vartheta_{00}(v+1) = \vartheta_{00}(v)$$

the function $\vartheta_{00}(v)$ can be regarded as a one-valued function of $e^{2\pi i v}$, and let us accordingly put

$$\vartheta_{00}(v) = \sum_{-\infty,\infty}^{\nu} A_\nu e^{2\pi i v \nu},$$

the differential equation § 20, (4):

$$\frac{\partial^2 \vartheta}{\partial v^2} - 4\pi i \frac{\partial \vartheta}{\partial \omega} = 0$$

then gives for A_ν the condition

$$\frac{dA_\nu}{d\omega} = \pi i \nu^2 A_\nu, \qquad A_\nu = c_\nu e^{\pi i \omega \nu^2} = c_\nu q^{\nu^2},$$

in which c_ν is constant with respect to ω. Thus we obtain

$$\vartheta_{00}(v) = \sum c_\nu q^{\nu^2} c^{2\pi i v \nu},$$

and from this:

$$\begin{aligned}\vartheta_{00}(v+\omega) &= \sum c_\nu q^{\nu^2+2\nu} e^{2\pi i v \nu} \\ &= q^{-1} e^{-2\pi i v} \sum c_\nu q^{(\nu+1)^2} e^{2\pi i v(\nu+1)}.\end{aligned}$$

Since ν runs from $-\infty$ to $+\infty$, it is permissible to put $\nu - 1$ in place of ν in this formula, and if one does so, there results

$$\vartheta_{00}(v+\omega) = q^{-1} e^{-2\pi i v} \sum c_{\nu-1} q^{\nu^2} e^{2\pi i v \nu}.$$

By the second of the fundamental equations [§ 20, (1)] the two series

$$\sum c_\nu q^{\nu^2} e^{2\pi i v \nu} \quad \text{and} \quad \sum c_{\nu-1} q^{\nu^2} e^{2\pi i v \nu}$$

must therefore agree with one another, that is, we must have

$$c_{\nu-1} = c_\nu$$

must hold. The c_ν are therefore all equal to one another; that they have the value 1 is shown by the comparison of the two developments

$$\sum_{-\infty,\infty}^{\nu} q^{\nu^2} e^{2\pi i v \nu}, \qquad \prod_{1,\infty}^{\nu} (1-q^{2\nu})(1+q^{2\nu-1}e^{2\pi i v})(1+q^{2\nu-1}e^{-2\pi i v})$$

for the value $q = 0$.

One may also give the development thus found for $\vartheta_{00}(v)$ the two formulae:

(1)
$$\begin{aligned}\vartheta_{00}(v) &= \sum q^{\nu^2} e^{2\nu\pi i v} \\ &= 1 + 2q\cos 2\pi v + 2q^4 \cos 4\pi v + 2q^9 \cos 6\pi v + \cdots,\end{aligned}$$

and from this one obtains, by § 21, (8), through the increase of v by $\frac{1}{2}$, $\frac{\omega}{2}$, $\frac{1+\omega}{2}$, the developments for the three remaining ϑ-functions:

$$\begin{aligned}\vartheta_{01}(v) &= \sum (-1)^{\nu} q^{\nu^2} e^{2\nu\pi i v} \\ &= 1 - 2q\cos 2\pi v + 2q^4 \cos 4\pi v - 2q^9 \cos 6\pi v + \cdots\end{aligned} \tag{2}$$

$$\begin{aligned}\vartheta_{10}(v) &= \sum q^{\frac{(2\nu+1)^2}{4}} e^{(2\nu+1)\pi i v} \\ &= 2q^{\frac{1}{4}} \cos \pi v + 2q^{\frac{9}{4}} \cos 3\pi v + 2q^{\frac{25}{4}} \cos 5\pi v + \cdots\end{aligned} \tag{3}$$

$$\begin{aligned}\vartheta_{11}(v) &= -i \sum (-1)^{\nu} q^{\frac{(2\nu+1)^2}{4}} e^{(2\nu+1)\pi i v} \\ &= 2q^{\frac{1}{4}} \sin \pi v - 2q^{\frac{9}{4}} \sin 3\pi v + 2q^{\frac{25}{4}} \sin 5\pi v + \cdots\end{aligned} \tag{4}$$

For later applications we draw from these developments the conclusions:

When the imaginary part of ω grows to infinity, so that the absolute value of q vanishes, then

$$\vartheta_{00}(v) = 1, \quad \vartheta_{01}(v) = 1, \quad q^{-\frac{1}{4}} \vartheta_{10}(v) = 2\cos \pi v,$$

$$q^{-\frac{1}{4}} \vartheta_{11}(v) = 2 \sin \pi v.$$

If we take ω purely imaginary, and hence q real, positive and properly fractional, then for a real v:

1. $\vartheta_{00}(v)$, $\vartheta_{01}(v)$, $\vartheta_{10}(v)$, $\vartheta_{11}(v)$ real, and when v lies between 0 and $\frac{1}{2}$, positive [by § 24 (7)], consequently also ϑ_{00}, ϑ_{01}, ϑ_{10}, ϑ'_{11} positive;

2. $\vartheta_{00}(iv)$, $\vartheta_{01}(iv)$, $\vartheta_{10}(iv)$, $-i\vartheta_{11}(iv)$ real, and as long as v lies between 0 and $\dfrac{-i\omega}{2}$, positive; further, with the aid of the formulae § 21, (8)

3. $\vartheta_{00}(\frac{1}{2}+iv)$, $\vartheta_{01}(\frac{1}{2}+iv)$, $i\vartheta_{10}(\frac{1}{2}+iv)$, $\vartheta_{11}(\frac{1}{2}+iv)$ real, and as long as v lies between 0 and $\dfrac{-i\omega}{2}$, positive;

4. $$q^{\frac{1}{4}} e^{\pi i v} \vartheta_{00}\left(\frac{\omega}{2}+v\right), \quad -iq^{\frac{1}{4}} e^{\pi i v} \vartheta_{01}\left(\frac{\omega}{2}+v\right),$$
$$q^{\frac{1}{4}} e^{\pi i v} \vartheta_{10}\left(\frac{\omega}{2}+v\right), \quad -iq^{\frac{1}{4}} e^{\pi i v} \vartheta_{11}\left(\frac{\omega}{2}+v\right)$$

real, and as long as v lies between 0 and $\frac{1}{2}$, positive.

§ 26. Development of ϑ-Quotients.

If v denotes a variable and a an arbitrary constant, then the quotients

$$\frac{\vartheta_{g_1,g_2}(v+a)}{\vartheta_{g'_1,g'_2}(v)}$$

are doubly periodic functions of the second kind and of the first order, with the periodicity factors $(-1)^{g_1+g'_1}$, $(-1)^{g_2+g'_2} e^{-2\pi i a}$; and if one adds an exponential factor $e^{\lambda v}$, then λ and a can be so determined that the periodicity factors become arbitrarily given quantities. By choosing the principal characteristics $(g_1, g_2), (g'_1, g'_2)$ in all possible ways, one obtains 16 such functions. We start out from one among them, for which, with the addition of a constant factor, we

$$F = \frac{\vartheta'_{11} \vartheta_{11}(v+a)}{2i\pi \vartheta_{11}(v) \vartheta_{11}(a)} \tag{1}$$

choose. This is a single-valued function $F(z)$ of the variable

$$z = e^{2\pi i v},$$

and if, as before, $q = e^{\pi i \omega}$, then it has the periodicity property:

$$F(q^2 z) = e^{-2\pi i a} F(z), \tag{2}$$

from which, for every integer ν, it follows that:

$$F(q^{2\nu} z) = e^{-2\pi i a \nu} F(z).$$

The function $F(z)$ becomes infinite for a finite z only when one of the factors $q^{2\nu} z - 1$ vanishes, and in particular

$$(z-1)F(z) = 1 \quad (\text{for } z = 1). \tag{3}$$

But by (2)

$$(q^{2\nu} z - 1) F(q^{2\nu} z) = e^{-2\pi i a \nu} F(z)(q^{2\nu} z - 1)$$

and consequently, by (3),

$$F(z)(q^{2\nu} z - 1) = e^{2\pi i a \nu} \quad (\text{for } z = q^{-2\nu}). \tag{4}$$

If we set, under the assumption of convergence that is presently to be examined more closely,

$$S(z) = \sum_{-\infty,\infty}^{\nu} \frac{e^{2\pi i a \nu}}{q^{2\nu} z - 1}, \tag{5}$$

then, replacing z by $q^2 z$ and ν by $\nu - 1$, we obtain:

$$S(q^2 z) = e^{-2\pi i a} S(z), \tag{6}$$

and the difference $F(z) - S(z)$ would therefore be, considered as a function of v, by (3) and (4) an integral doubly periodic function of the second kind. But by § 16, 6. such a function must be an exponential function, and hence

$$F(z) = S(z) + C e^{-2\pi i m v},$$

where m is an integer and C a constant. But from (2) and (6), unless $C = 0$, it follows that:

$$e^{-2\pi i m \omega} = e^{-2\pi i a};$$

hence $a = m\omega + n$, where m, n are integers. But if a is a period, then $F(z)$ reduces to a constant or an exponential function. Otherwise $C = 0$ must hold, and there follows the development:

$$\frac{\vartheta_{11}' \vartheta_{11}(v+a)}{2i\pi \vartheta_{11}(v)\vartheta_{11}(a)} = \sum_{-\infty,+\infty}^{\nu} \frac{e^{2\pi i \nu a}}{q^{2\nu} e^{2\pi i v} - 1}. \tag{7}$$

We have assumed the convergence of this series for all values of v; in order to judge of this, we observe that the general term of this series becomes

for	$\nu = +\infty$	equal to	$-e^{2\pi i \nu a}$
"	$\nu = -\infty$	"	$e^{-2\pi i v} e^{-2\pi i \nu(\omega - a)}$

Convergence will therefore take place if the imaginary part of a and of $\omega - a$ is positive. If we therefore set $\omega = \omega' + i\omega''$, $a = a' + ia''$, then the condition of convergence is

(8) $$0 < a'' < \omega''.$$

If in (7) one replaces a by $a + \frac{1}{2}$, there results the development for a second of the 16 functions:

(9) $$\frac{\vartheta_{11}'\vartheta_{10}(v+a)}{2\pi i\vartheta_{11}(v)\vartheta_{10}(a)} = \sum^{\nu} \frac{(-1)^{\nu}e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}-1},$$

and if in (7) and (9) one changes a into $a + \frac{1}{2}\omega$:

(10) $$\frac{\vartheta_{11}'\vartheta_{01}(v+a)}{2\pi i\vartheta_{11}(v)\vartheta_{01}(a)} = \sum^{\nu} \frac{e^{2\pi i a\nu}q^{\nu}e^{\pi iv}}{q^{2\nu}e^{2\pi iv}-1},$$

(11) $$\frac{\vartheta_{11}'\vartheta_{00}(v+a)}{2\pi i\vartheta_{11}(v)\vartheta_{00}(a)} = \sum^{\nu} \frac{(-1)^{\nu}e^{2\pi i a\nu}q^{\nu}e^{\pi iv}}{q^{2\nu}e^{2\pi iv}-1},$$

where it is however to be noted that in the last two formulae the condition of convergence is altered, namely:

$$-\frac{\omega''}{2} < a'' < \frac{\omega''}{2}.$$

From the four formulae the remaining twelve follow if we replace v by $v + \frac{1}{2}$, $v + \frac{1}{2}\omega$, $v + \frac{1}{2}(1+\omega)$, whereby the domains of convergence are not further altered. In this way the formulae of Table I at the end of the volume are derived.

The series so obtained do not all converge for real values of a; the series (7), for example, does not, since in it the part running towards the side of the positive ν

(12) $$\sum_{1,\infty}^{\nu} \frac{e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi iv}-1}$$

ceases to converge when the imaginary part of a becomes equal to zero, while the other part is still convergent even there

remains. But from (12) one obtains an expression which still converges for a real a as well, if one adds the sum

(13) $$\sum_{1,\infty}^{\nu} e^{2\pi i\nu a} = -\frac{e^{2\pi ia}}{e^{2\pi ia}-1}$$

to it. For thereby it passes over into

$$\sum_{1,\infty}^{\nu} \frac{e^{2\pi i\nu a}q^{2\nu}e^{2\pi iv}}{q^{2\nu}e^{2\pi iv}-1},$$

and this series remains convergent as long as the imaginary part a'' of a lies between $-\omega''$ and $+\omega''$.

Accordingly there results from (7), if we separate off the term corresponding to the value $\nu = 0$ and replace the negative ν by $-\nu$:

$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \frac{2i}{e^{2\pi iv}-1} + \frac{2ie^{2\pi ia}}{e^{2\pi ia}-1}$$

$$+2i\sum_{1,\infty}^{\nu} \frac{e^{-2\pi i\nu a}}{q^{-2\nu}e^{2\pi iv}-1} + 2i\sum_{1,\infty}^{\nu} \frac{q^{2\nu}e^{2\pi i\nu a}e^{2\pi iv}}{q^{2\nu}e^{2\pi iv}-1}.$$

But we have

$$\frac{2i}{e^{2\pi iv}-1} = \frac{e^{-\pi iv}}{\sin\pi v} = \operatorname{cotg}\pi v - i,$$

$$\frac{2ie^{2\pi ia}}{e^{2\pi ia}-1}=\frac{e^{\pi ia}}{\sin\pi a}=\operatorname{cotg}\pi a+i,$$

and accordingly this development can also be represented thus:

$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)}=\operatorname{cotg}\pi v+\operatorname{cotg}\pi a-2i\sum\left(\frac{q^m e^{m\pi ia}}{e^{-2\pi iv}-q^m}-\frac{q^m e^{-m\pi ia}}{e^{2\pi iv}-q^m}\right), \tag{14}$$

in which m runs through the sequence of the even numbers

$$m=2,4,6,8,\ldots$$

The domain of validity of this development is

$$-\omega''<a''<\omega''.$$

In Table II these 16 developments are compiled.

From these formulae three others can be derived, by increasing v or a or both by $\frac{1}{2}$. Three others, however, which correspond to the increase of v and a by $\frac{1}{2}\omega$, must be derived directly. They exhibit real form when q, a, v are real.

A third kind of development, in which the symmetry of the functions with respect to the two variables v, a finds expression, arises if the fractions that occurred in the previous developments are developed in powers of q. In this way one obtains

$$\frac{q^m}{e^{-2\pi iv}-q^m}=\Sigma\, q^{m\nu}e^{2\pi i\nu v},$$
$$\frac{q^m}{e^{+2\pi iv}-q^m}=\Sigma\, q^{m\nu}e^{-2\pi i\nu v},$$
$$\nu=1,\,2,\,3,\,\ldots$$

and these developments hold as long as the absolute value of $q\,e^{2\pi iv}$ is a proper fraction, hence, if we put $v=v'+iv''$, as long as

$$-\omega''<v<\omega''$$

holds. Accordingly there follows from (14)

$$\frac{\vartheta_{11}'\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{11}(v)\,\vartheta_{11}(a)}=\operatorname{cotg}\pi v+\operatorname{cotg}\pi a$$
$$+4\Sigma\, q^{\frac{mm'}{2}}\sin(ma+m'v)\pi,$$

in which m, m' run, independently of one another, through the even numbers 2, 4, 6, 8, ... This formula is valid for real a and v, and it holds beyond that as well, as long as the imaginary part, of v as well as of a, is in absolute value less than ω''.

In Table III the 16 formulae of this kind are collected[1]).

[1]) Such developments have been considered by Jacobi (sur la rotation d'un corps) (ges. Werke Bd. II), thereupon by Hermite (Annales de l'école normale 1885) and by Kronecker (Berliner Akademie 1885). Cf. also the Strasbourg dissertation of L. Vockerodt (1905).

Third Section.
Transformation of the Theta-Functions.

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§ 27. The Principle of Transformation.

We return to the defining equations of the T-functions of order m given in § 17, and there, for a reason that will be evident at once, we furnish the letters ω, a, b, m with accents, so that these equations read:

$$\begin{aligned} T(u+\omega_1') &= e^{-\pi i[a_1'(2u+\omega_1')+b_1']}T(u), \\ T(u+\omega_2') &= e^{-\pi i a_2'(2u+\omega_2')+b_2'}T(u). \end{aligned} \tag{1}$$

$$a_2'\omega_1' - a_1'\omega_2' = m'. \tag{2}$$

If a, c are any integers whatever (positive or negative), then, on replacing n_1, n_2 in § 17, (16) by $-c$, a, there results:

$$\begin{aligned} &T(u - c\omega_1' + a\omega_2') \\ &= e^{-\pi i(-ca_1'+aa_2')(2u-c\omega_1'+a\omega_2')-\pi i(-cb_1'+ab_2'-m'ac)}T(u), \end{aligned} \tag{3}$$

an equation which also follows from one of the equations (1) if in it a', b', ω' are replaced by

$$-ca_1' + aa_2', \quad -cb_1' + ab_2' - m'ac, \quad -c\omega_1' + a\omega_2'$$

respectively.

Herein is contained the principle of the transformation of the T-functions.

Let b, ∂ be two other integers for which the determinant

$$n = a\partial - bc \tag{4}$$

has a positive value. We put

$$\begin{aligned} \omega_1 &= +\partial\omega_1' - b\omega_2', \\ \omega_2 &= -c\omega_1' + a\omega_2', \end{aligned} \tag{5}$$

and consequently

$$\begin{aligned} n\omega_1' &= a\omega_1 + b\omega_2 \\ n\omega_2' &= c\omega_1 + \partial\omega_2. \end{aligned} \tag{6}$$

Then, as one recognises from

$$\frac{\omega_2'}{\omega_1'} = \frac{c\omega_1 + \partial\omega_2}{a\omega_1 + b\omega_2}$$

by separation of the real from the imaginary part, the imaginary part of $\omega_2 : \omega_1$ has the same sign as that of $\omega_2' : \omega_1'$ (the positive one).

If we set

$$\begin{aligned} a_1 &= \partial a_1' - ba_2' \\ a_2 &= -ca_1' + aa_2' \end{aligned} \tag{7}$$

$$\begin{aligned} b_1 &= \partial b_1' - bb_2' - m'b\partial \\ b_2 &= -cb_1' + ab_2' - m'ac, \end{aligned} \tag{8}$$

then one concludes from (3) that the function $T(u)$ satisfies not only the conditions (1), but also the equations that arise from (1) by the interchange of ω_1', ω_2', a_1', a_2', b_1', b_2' with ω_1, ω_2, a_1, a_2, b_1, b_2, that is, the equations (1), § 17. It is therefore at the same time a T-function of the periods ω_1', ω_2' and of the periods ω_1, ω_2, which we indicate by the following equation:

$$T'(u, \omega_1'\, \omega_2') = T(u, \omega_1, \omega_2). \tag{9}$$

But by (5) and (7) we have

$$a_2\omega_1 - a_1\omega_2 = (a_2'\omega_1' - a_1'\omega_2')(a\partial - bc),$$

hence, if m is the order of T,

$$m = m'n. \tag{10}$$

By (8), if (g_1', g_2') is the characteristic of T' (§ 17), the characteristic (g_1, g_2) of T is

$$(g_1, g_2) = (\partial g_1' - bg_2' - m'b\partial,\ -cg_1' + ag_2' - m'ac). \tag{11}$$

By the transformation of the T-functions one understands the representation of the functions T' with the periods ω_1', ω_2' by T-functions with the periods ω_1, ω_2.

The numbers a, b, c, ∂ are called the transformation numbers and $n = a\partial - bc$ the degree of transformation.

In order to survey the form of this representation more clearly, we shall seek the conditions under which $T'(u, \omega_1, \omega_2)$ becomes a Θ-function of the mth order $\Theta(u, \omega)$ (§ 20). We take b_1', b_2', and consequently also b_1, b_2, to be integers, so that b_1, b_2, b_1', b_2' can be replaced by g_1, g_2, g_1', g_2'. We then have

$$\omega_1 = 1, \quad \omega_2 = \omega, \quad a_1 = 0, \quad a_2 = m$$

to be set, and accordingly [by (6), (7), (10)] we obtain

$$\omega_1' = \frac{a + b\omega}{n}, \quad \omega_2' = \frac{c + \partial\omega}{n},$$
$$a_1' = m'b, \quad a_2' = m'\partial.$$

The function $T'(u, \omega_1', \omega_2')$ therefore satisfies the conditions (1):

$$T(u + \omega_1') = (-1)^{g_1'} e^{-\pi i m' b(2u + \omega_1')} T(u),$$
$$T(u + \omega_2') = (-1)^{g_2'} e^{-\pi i m' \partial(2u + \omega_2')} T(u),$$

and from this it follows that the product

$$e^{\frac{\pi i m' b u^2}{\omega_1'}} T'(u, \omega_1', \omega_2')$$

is a Θ-function of the order m', with the arguments

$$\frac{u}{\omega_1'}, \quad \frac{\omega_2'}{\omega_1'}$$

and the characteristic (g_1', g_2'). We can express this in the equation:

$$e^{-\frac{\pi i m' n b u^2}{a + b\omega}} \Theta_{g_1', g_2'}^{(m')} \left(\frac{nu}{a + b\omega}, \frac{c + \partial\omega}{a + b\omega} \right) = \Theta_{g_1, g_2}^{(m'n)}(u, \omega), \tag{12}$$

in which the characteristics are determined by (11). The means for the representation of these functions are contained in § 21.

We denote the transformation of T and T' by a single letter S, or by (T', T), thus:

$$S = (T', T).$$

If S' denotes a second transformation by which T' passes into T'', that is,

$$S' = (T'', T'),$$

then we can derive from these a new transformation S'' by which T passes into T''. This one is said to be composed from S and S', and is denoted thus:

$$S'' = S'S$$

or

$$(T'', T) = (T'', T')(T', T).$$

In this composition the commutative law does not in general hold; thus SS' is different from $S'S$. The associative law, however, does hold, and it is expressed in the formula:

$$(T''', T'')[(T'', T')(T', T)] = [(T''', T'')(T'', T')](T', T) = (T''', T).$$

§ 28. Composition of the Transformations.

A transformation [§ 27, (9)] is completely determined by the transformation numbers a, b, c, ∂, and these four integers may be given arbitrarily, provided only that their determinant $n = a\partial - bc$ is positive. If one gives these four numbers the opposite sign, then ω_1', ω_2' pass over, by § 27, (6), into $-\omega_1', -\omega_2'$, and if one replaces a, b, c, d by $ma, mb, mc, m\partial$, where m is an arbitrary natural number, then ω_1', ω_2' pass over into ω_1'/m, ω_2'/m and n into $m^2 n$. The ratio of the periods $\omega' = \omega_2'/\omega_1'$ remains unchanged in these two cases. For the present, however, we shall always regard two transformations as different when the transformation numbers are different. In accordance with this stipulation we can represent a transformation unambiguously by a matrix

$$S = \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} \tag{1}$$

The determinant

$$n = a\partial - bc \tag{2}$$

is the degree of the transformation.

With this notation we represent the relations (6), § 27 in this way as well:

$$n(\omega_1', \omega_2') = \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} (\omega_1, \omega_2). \tag{3}$$

If one puts

$$S' = \begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix}, \quad a'\partial' - b'c' = n',$$

then

$$n'(\omega_1'', \omega_2'') = \begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix} (\omega_1', \omega_2'), \tag{4}$$

and if in (4) one expresses ω_1', ω_2' in terms of ω_1, ω_2 according to (3), one obtains

$$nn'(\omega_1'', \omega_2'') = \begin{pmatrix} a'a + b'c, & a'b + b'\partial \\ c'a + \partial'c, & c'b + \partial'\partial \end{pmatrix} (\omega_1, \omega_2). \tag{5}$$

If we therefore put

$$S'' = S'S, \tag{6}$$

then

$$S'' = \begin{pmatrix} a'a + b'c, & a'b + b'\partial \\ c'a + \partial'c, & c'b + \partial'\partial \end{pmatrix} = \begin{pmatrix} a'', & b'' \\ c'', & \partial'' \end{pmatrix}, \quad a''\partial'' - b''c'' = n'' = nn'.$$

The transformations S therefore compose according to the same rule as the linear substitutions and matrices which we considered in the sixth section of the second volume. The degree of a composed transformation is equal to the product of the degrees of the components. These matrices are here bound to the assumption that their elements are integers and their determinant is positive.

These properties are preserved under the composition of the transformations. Nevertheless the totality $\mathfrak{S}$ of the transformations S forms no group, any more than the totality of the natural numbers forms a group under composition by multiplication; for when S', S'' are given it is not always possible to determine an S that satisfies condition (6), which yet (according to Bd. II, § 1, 4.) would be required for a group.

By the special transformation of degree m^2:

$$M = \begin{pmatrix} \pm m, & 0 \\ 0, & \pm m \end{pmatrix}$$

the periods ω_1, ω_2 pass into $\omega_1' = \omega_1/m$, $\omega_2' = \omega_2/m$, and the period ratio $\omega = \omega_2/\omega_1$ remains unchanged. These transformations are called multiplications (similarity transformations, Bd. II, § 41). Among them is contained the identical substitution

$$1 = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$$

which leaves everything unchanged and plays the role of the unit under composition.

The multiplications are permutable under composition with every transformation S:

$$SM = MS. \tag{7}$$

If in SM or MS one holds the transformation S fixed and lets M run through the totality $\mathfrak{M}$ of the multiplications, one obtains a system $\mathfrak{M}S$, which according to Bd. II, § 46 would have to be designated a collineation. If S_1 and S_2 belong to a collineation C, and S_1' and S_2' to a collineation C', then $S_1'S_1$ and $S_2'S_2$ also belong to the same collineation C''. In this way, by setting $C'' = C'C$, one can compose the collineations. Under this composition the collineation $\mathfrak{M}$ plays the role of the unit. If $S = \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$ is an arbitrary transformation, then

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} \begin{pmatrix} \partial, & -b \\ -c, & a \end{pmatrix} = \begin{pmatrix} n, & 0 \\ 0, & n \end{pmatrix}$$

is a multiplication. The two collineations

$$C = \mathfrak{M} \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}, \quad C^{-1} = \mathfrak{M} \begin{pmatrix} \partial, & -b \\ -c, & a \end{pmatrix}$$

therefore give, on composition, $CC^{-1} = C^{-1}C = \mathfrak{M}$, and so are reciprocal to one another.

Accordingly the totality of the collineations forms a group.

The transformations of degree 1 are called linear transformations. For these we denote the transformation numbers, for distinction, by the Greek letters $\alpha, \beta, \gamma, \delta$, so that

$$L = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1$$

means a linear transformation. The system $\mathfrak{L}$ of the linear transformations is a group contained in $\mathfrak{S}$; for if

$$L = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad L' = \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix},$$

then

$$L'' = L'L = \begin{pmatrix} \alpha'\alpha + \beta'\gamma, & \alpha'\beta + \beta'\delta \\ \gamma'\alpha + \delta'\gamma, & \gamma'\beta + \delta'\delta \end{pmatrix} = \begin{pmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{pmatrix}$$

is likewise linear, and, for given L', L'', one can determine L from the equations

$$\begin{aligned} \alpha'\alpha + \beta'\gamma = \alpha'', \quad \alpha'\beta + \beta'\delta = \beta'', \\ \gamma'\alpha + \delta'\gamma = \gamma'', \quad \gamma'\beta + \delta'\delta = \delta'' \end{aligned}$$

uniquely. The unit of the group $\mathfrak{L}$ is the identical substitution $\begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$, and every substitution L has its reciprocal L^{-1}, as the composition

$$LL^{-1} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} \delta, & -\beta \\ -\gamma, & \alpha \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$$

shows. The group $\mathfrak{L}$ is infinite and is not commutative.

From the equation

$$\alpha\delta - \beta\gamma = 1 \tag{8}$$

it follows that neither α, β nor α, γ, nor δ, β, nor δ, γ can have a common factor. But if α, β have been assumed arbitrarily without a common divisor, then γ, δ can still be determined from (8) in infinitely many ways. If γ, δ is one of these determinations, then they are all contained in the form

$$\gamma + \lambda\alpha, \quad \delta + \lambda\beta \tag{9}$$

in which λ is an arbitrary integer (Bd. I, § 126).

§ 29. Composition of the transformations from simpler ones.

In the system $\mathfrak{S}$ of all transformations S there is contained a system $\mathfrak{S}_0$ which consists of all those transformations S_0 whose second transformation number is $b = 0$, while a and ∂ are positive:

$$S_0 = \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}. \tag{1}$$

When two S_0 are composed there arises again an S_0; but $\mathfrak{S}_0$ is nevertheless just as little a group as $\mathfrak{S}$.

One can reduce any arbitrary transformation

$$S = \begin{pmatrix} p, & q \\ r, & s \end{pmatrix}$$

to an S_0 by a composition LS. For if

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} p, & q \\ r, & s \end{pmatrix} = \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \tag{2}$$

is to hold, then α, β must satisfy the condition:

$$\alpha q + \beta s = 0,$$

and accordingly, if ∂ is the greatest common divisor of q and s, one sets

$$\partial\alpha = s, \quad \partial\beta = -q,$$

and, after α and β have hereby been ascertained as relatively prime numbers, one determines γ and δ from formula (8), § 28. Then (2) is satisfied if

$$a\partial = n, \quad c = \gamma p + \delta r$$

is put; a and ∂ are hereby uniquely determined, but c can, for another choice of γ and δ, be replaced by $c + \lambda a$. One can therefore dispose of λ so that c lies in the sequence

of the numbers 0, 1, 2 $\ldots a - 1$, and thereby the substitution S_0 is then completely determined.

If the four transformation numbers have a common factor, then this factor can be separated off by means of the formula

$$\begin{pmatrix} m, & 0 \\ 0, & m \end{pmatrix} \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} = \begin{pmatrix} ma, & mb \\ mc, & m\partial \end{pmatrix}$$

through composition with the multiplication, and we accordingly now assume that a, b, c, ∂ have no common divisor. One can then always determine the two integers ξ, η in such a way that

$$\begin{aligned} a\eta - c\xi &= \alpha \\ b\eta - \partial\xi &= \beta \end{aligned} \tag{3}$$

are without a common divisor.

To see this, we first assume ξ, η to be relatively prime. Then every common divisor of α, β is necessarily a divisor of n, as the solutions of (3)

$$\begin{aligned} n\xi &= b\alpha - a\beta \\ n\eta &= \partial\alpha - c\beta \end{aligned} \tag{4}$$

show. If one therefore takes ξ not divisible by all the primes that divide a and b at once, but ξ divisible and η indivisible by all the other primes that divide n, and moreover ξ, η relatively prime, which is always possible, then α and β have no common divisor. Thereupon one determines γ, δ so that

$$\alpha\delta - \beta\gamma = 1.$$

By (3) it is then also the case that

$$(a\delta - b\gamma)\eta - (c\delta - \partial\gamma)\xi = 1,$$

and the following composition results, as is easily recognised by use of (4):

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} = \begin{pmatrix} a\delta - b\gamma, & \xi \\ c\delta - \partial\gamma, & \eta \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}. \tag{5}$$

If we therefore call

$$\begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \tag{6}$$

the principal transformation of degree n, then it is thereby proved that all transformations of degree n can be composed of a multiplication, a principal transformation and linear transformations.

From the composition

$$\begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 0, & m \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & mn \end{pmatrix}, \quad \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} n, & 0 \\ 0, & 1 \end{pmatrix} = \begin{pmatrix} n, & 0 \\ 0, & n \end{pmatrix} \tag{7}$$

we can conclude further that every transformation of degree n can be composed of such transformations as have prime degree. If one decomposes $n = pq$ into two factors p and q that are relatively prime to one another, then, determining the numbers β, δ from

$$p\delta - q\beta = 1$$

one obtains the composition

$$\begin{pmatrix} p, & \beta \\ q, & \delta \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} p\delta, & -q\beta \\ -1, & 1 \end{pmatrix} = \begin{pmatrix} p, & 0 \\ 0, & q \end{pmatrix},$$

from which it is to be seen that in place of the principal transformation one may also use any of these transformations $\begin{pmatrix} p, & 0 \\ 0, & q \end{pmatrix}$ for the derivation of all the others.

§ 30. The linear fundamental transformations.

The whole group $\mathfrak{L}$ of linear transformations can be derived by the repetition of two among them, which we call the linear fundamental transformations. If

$$L = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1$$

is an arbitrary linear transformation, then

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} \delta, & -\beta \\ -\gamma, & \alpha \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}, \tag{1}$$

and since the identical transformation is the unit in the group L,

$$L^{-1} = \begin{pmatrix} \delta, & -\beta \\ -\gamma, & \alpha \end{pmatrix}$$

is the transformation reciprocal to L.

By the power $L^{\pm m}$ we denote that which arises from the m-fold repetition of L or of L^{-1}, and we now wish to show that by the powers of the fundamental transformations

(2) $$A = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

every substitution L of the group $\mathfrak{L}$ can be composed. First of all we have

(3) $$A^{-1} = \begin{pmatrix} 1, & 0 \\ -1, & 1 \end{pmatrix}, \quad A^{\lambda} = \begin{pmatrix} 1, & 0 \\ \lambda, & 1 \end{pmatrix}$$

(for every positive or negative integer λ)

(4) $$B^{-1} = \begin{pmatrix} 0, & -1 \\ 1, & 0 \end{pmatrix}, \quad B^2 = \begin{pmatrix} -1, & 0 \\ 0, & -1 \end{pmatrix}, \quad B^3 = B^{-1}.$$

We further put

(5) $$C = \begin{pmatrix} 1, & -1 \\ 0, & 1 \end{pmatrix} = B^{-1}AB, \quad C^{\lambda} = \begin{pmatrix} 1, & -\lambda \\ 0, & 1 \end{pmatrix},$$

C is therefore derivable from A and B.

Now let $L = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ be an arbitrary linear transformation. From it we derive the sequence:

$$L' = LA^{\lambda}, \quad L'' = L'C^{\lambda'}, \quad L''' = L''A^{\lambda''}, \quad L'''' = L'''C^{\lambda'''},$$

whose first and second elements are formed thus:

$$\alpha' = \alpha + \lambda\beta, \quad \beta'' = \beta' - \lambda'\alpha', \quad \alpha''' = \alpha'' + \lambda''\beta'',$$
$$\beta'''' = \beta''' - \lambda'''\alpha''', \ldots,$$

and one may dispose of $\lambda, \lambda', \lambda'', \lambda''', \ldots$ in such a way that, in absolute value,

$$\alpha' \leqq \tfrac{1}{2}\beta, \quad \beta'' \leqq \tfrac{1}{2}\alpha', \quad \alpha''' \leqq \tfrac{1}{2}\beta'', \quad \beta'''' \geqq \tfrac{1}{2}\alpha''' \ldots,$$

as long as none of these numbers vanishes. The numbers

$$\beta, \alpha', \beta'', \alpha''', \beta'''', \ldots$$

therefore form a sequence of numbers decreasing in absolute value, and after a finite number of compositions of this kind a number of this sequence must vanish.

If $\beta^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} \pm 1, & 0 \\ \gamma^{(\nu)}, & \pm 1 \end{pmatrix} = \begin{pmatrix} \pm 1, & 0 \\ 0, & \pm 1 \end{pmatrix} A^{\pm\gamma^{(\nu)}}$$

and if $\alpha^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} 0, & \pm 1 \\ \mp 1, & \delta^{(\nu)} \end{pmatrix} = A^{\pm\delta^{\nu}} B^{\pm 1},$$

and since

$$L = L'A^{-\lambda} = L''C^{-\lambda'}A^{-\lambda}$$
$$= L'''A^{-\lambda''}C^{-\lambda'}A^{-\lambda} = L''''C^{-\lambda'''}A^{-\lambda''}C^{-\lambda'}A^{-\lambda}, \ldots$$

holds, the theorem is proved.

§ 31. The linear fundamental transformations of the ϑ-functions.

In the application to the transformation of the Θ-functions [§ 27, (12)] the transformed modulus

$$\omega' = \frac{c + \partial\omega}{a + b\omega}$$

first comes into consideration. This does not alter if the four transformation numbers receive a common factor m. The transformation is called a proper one if a, b, c, ∂ have no common factor.

The transformed argument

$$u' = \frac{nu}{a + b\omega}$$

passes over into mu' if a, b, c, ∂ are replaced by $ma, mb, mc, m\partial$, and hence n by $m^2 n$. The multiplication $\begin{pmatrix} m, & 0 \\ 0, & m \end{pmatrix}$ leaves the modulus ω unchanged and changes u into mu.

According to the results of the two preceding sections, the whole system of proper transformations can be derived by repeated application of the three transformations

$$\begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \quad \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \quad \begin{pmatrix} n, & 0 \\ 0, & 1 \end{pmatrix},$$

and so we consider first the linear fundamental transformations of the ϑ-functions.

I. $\begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}$ or $(\omega, \omega + 1)$.

By § 27, (11), (12) we have

$$\vartheta_{11}(u, \omega + 1) = A\vartheta_{11}(u, \omega), \tag{1}$$

in which A is independent of u. If one replaces u by

$$u + \tfrac{1}{2}, \quad u + \tfrac{\omega}{2}, \quad u + \tfrac{1+\omega}{2},$$

then the formulae § 21, (8) give

$$\vartheta_{10}(u, \omega + 1) = A\vartheta_{10}(u, \omega) \tag{2}$$

$$e^{\frac{\pi i}{4}}\vartheta_{00}(u, \omega + 1) = A\vartheta_{01}(u, \omega) \tag{3}$$

$$e^{\frac{\pi i}{4}}\vartheta_{01}(u, \omega + 1) = A\vartheta_{00}(u, \omega). \tag{4}$$

To determine the constant A we employ, as we shall often do in what follows, the device of differentiating (1), setting $u = 0$, and then using formula (5), § 23, on both sides:

$$\vartheta'_{11} = \pi\,\vartheta_{00}\,\vartheta_{10}\,\vartheta_{01} \tag{5}$$

It follows that

$$A^2 = e^{\frac{\pi i}{2}}, \quad A = e^{\frac{\pi i}{4}};$$

that the positive sign stands with A follows from one of the formulae (3), (4), according to the closing remark of § 25, if one lets ω become infinite.

Accordingly one obtains

$$\begin{aligned} \vartheta_{11}(u, \omega + 1) &= e^{\frac{\pi i}{4}}\vartheta_{11}(u) \\ \vartheta_{10}(u, \omega + 1) &= e^{\frac{\pi i}{4}}\vartheta_{10}(u) \\ \vartheta_{01}(u, \omega + 1) &= \vartheta_{00}(u) \\ \vartheta_{00}(u, \omega + 1) &= \vartheta_{01}(u). \end{aligned} \tag{6}$$

II. $\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$ or $\left(\omega, \frac{-1}{\omega}\right)$.

Again we have, by § 27, (12),

$$e^{-\frac{\pi i u^2}{\omega}} \vartheta_{11}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = A\vartheta_{11}(u, \omega), \tag{7}$$

and, on increasing u by $\frac{1}{2}$, $\frac{\omega}{2}$, $\frac{1+\omega}{2}$,

$$e^{-\frac{\pi i u^2}{\omega}} \vartheta_{01}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{10}(u, \omega), \tag{8}$$

$$e^{-\frac{\pi i u^2}{\omega}} \vartheta_{10}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{01}(u, \omega), \tag{9}$$

$$e^{-\frac{\pi i u^2}{\omega}} \vartheta_{00}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{00}(u, \omega), \tag{10}$$

from which one obtains, as above:

$$A = \pm i\sqrt{-i\omega}.$$

From $u = 0$ and a purely imaginary ω one concludes that, if $\sqrt{-i\omega}$ is taken so that the real part is positive, the lower sign must stand, and there results therefore:

$$\begin{aligned}
e^{-\frac{\pi i u^2}{\omega}} \vartheta_{11}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= -i\sqrt{-i\omega}\,\vartheta_{11}(u), \\
e^{-\frac{\pi i u^2}{\omega}} \vartheta_{01}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{10}(u), \\
e^{-\frac{\pi i u^2}{\omega}} \vartheta_{10}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{01}(u), \\
e^{-\frac{\pi i u^2}{\omega}} \vartheta_{00}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{00}(u).
\end{aligned} \tag{11}$$

§ 32. The principal transformations of the second order of the ϑ-functions.

The two principal transformations of order n

$$\begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix}, \quad \begin{pmatrix} n, & 0 \\ 0, & 1 \end{pmatrix}$$

transform, by § 27, (11), the characteristic (g_1, g_2) into

$$(ng_1, g_2), \quad (g_1, ng_2),$$

that is, for odd n the characteristic remains unchanged, while for even n it passes over into

$$(0, g_2) \text{ or } (g_1, 0). \tag{1}$$

Since the transformation of even degree accordingly behaves essentially otherwise than that of odd degree, we consider first the case $n = 2$. The first and second principal transformations of the second degree are called the Landen and the Gauss transformations.

By § 27, (12),

$$\begin{aligned}\vartheta_{g_1,g_2}\left(u,\frac{\omega}{2}\right)&=\Theta_{g_1,0}(u,\omega)\\ \vartheta_{g_1,g_2}(2u,2\omega)&=\Theta_{0,g_2}(u,\omega)\end{aligned}\tag{2}$$

are Θ-functions of the second order in u,ω, which admit of representation according to § 21.

We obtain first the two pairs of formulae in which A,B are independent of u:

$$\begin{aligned}A\vartheta_{11}\left(u,\frac{\omega}{2}\right)&=\vartheta_{01}(u,\omega)\vartheta_{11}(u,\omega),\\ A\vartheta_{10}\left(u,\frac{\omega}{2}\right)&=\vartheta_{00}(u,\omega)\vartheta_{10}(u,\omega),\end{aligned}\tag{3}$$

$$\begin{aligned}B\vartheta_{11}(2u,2\omega)&=\vartheta_{10}(u,\omega)\vartheta_{11}(u,\omega),\\ B\vartheta_{01}(2u,2\omega)&=\vartheta_{00}(u,\omega)\vartheta_{01}(u,\omega),\end{aligned}\tag{4}$$

of which in each case the second can be derived from the first by increasing the argument by a half period.

If one sets $u=0$ in these equations, it follows that

$$\begin{aligned}A\vartheta'_{11}\left(0,\frac{\omega}{2}\right)&=\vartheta_{01}\vartheta'_{11},\quad 2B\vartheta'_{11}(0,2\omega)=\vartheta_{10}\vartheta'_{11},\\ A\vartheta_{10}\left(0,\frac{\omega}{2}\right)&=\vartheta_{00}\vartheta_{10},\qquad B\vartheta_{01}(0,2\omega)=\vartheta_{00}\vartheta_{01},\end{aligned}\tag{5}$$

from which, by division, and with the use of the relation

$$\vartheta'_{11}=\pi\,\vartheta_{00}\,\vartheta_{11}\,\vartheta_{01},$$

$$\vartheta_{00}\left(0,\frac{\omega}{2}\right)\vartheta_{01}\left(0,\frac{\omega}{2}\right)=\vartheta_{01}^2,\tag{6}$$

$$2\vartheta_{00}(0,2\omega)\vartheta_{10}(0,2\omega)=\vartheta_{10}^2,\tag{7}$$

and, if in (6) one replaces ω by 2ω and in (7) ω by ω:2:

$$\vartheta_{01}(0,2\omega)^2=\vartheta_{00}\vartheta_{01},\tag{8}$$

$$\vartheta_{10}\left(0,\frac{\omega}{2}\right)^2=2\vartheta_{00}\vartheta_{10}.\tag{9}$$

By (8) and (9) there results from the second equations (5):

$$2A=\vartheta_{10}\left(0,\frac{\omega}{2}\right),\quad B=\vartheta_{01}(0,2\omega),$$

and one thus obtains for the Gauss transformation:

$$\begin{aligned}\vartheta_{10}\left(0,\frac{\omega}{2}\right)\vartheta_{11}\left(u,\frac{\omega}{2}\right)&=2\vartheta_{01}(u,\omega)\vartheta_{11}(u,\omega),\\ \vartheta_{10}\left(0,\frac{\omega}{2}\right)\vartheta_{10}\left(u,\frac{\omega}{2}\right)&=2\vartheta_{00}(u,\omega)\vartheta_{10}(u,\omega),\end{aligned}\tag{10}$$

and for the Landen transformation:

$$\begin{aligned}\vartheta_{01}(0,2\omega)\vartheta_{11}(2u,2\omega) &= \vartheta_{10}(u,\omega)\vartheta_{11}(u,\omega),\\ \vartheta_{01}(0,2\omega)\vartheta_{01}(2u,2\omega) &= \vartheta_{00}(u,\omega)\vartheta_{01}(u,\omega).\end{aligned} \tag{11}$$

For each of the two transformations there remain two further ϑ-functions to be expressed. These expressions can be derived from (10), (11) after § 21, (13), but one also arrives at them directly in the following way. The functions

$$\vartheta_{01}\left(u,\frac{\omega}{2}\right),\quad \vartheta_{10}(2u,2\omega) \tag{12}$$

vanish for

$$w=\frac{\omega}{4},\quad u=\frac{1}{4}.$$

On the other hand, from the formulae (8) of § 21 there follows, if we there set $v=-\dfrac{\omega}{4}$ and $=-\dfrac{1}{4}$,

$$\begin{aligned}\vartheta_{11}\left(\frac{\omega}{4}\right) &= i\vartheta_{01}\left(\frac{\omega}{4}\right)\\ \vartheta_{11}\left(\frac{1}{4}\right) &= \vartheta_{10}\left(\frac{1}{4}\right)\end{aligned}$$

and accordingly the two functions (12), which are linearly expressible by two ϑ-squares, agree, apart from constant factors, with

$$\vartheta_{01}^2(u)+\vartheta_{11}^2(u),\quad \vartheta_{10}^2(u)-\vartheta_{11}^2(u).$$

The constant factors are obtained immediately, for $u=0$, from the relations (6), (7):

$$\vartheta_{00}\left(0,\frac{\omega}{2}\right)\vartheta_{01}\left(u,\frac{\omega}{2}\right)=\vartheta_{01}^2(u)+\vartheta_{11}^2(u), \tag{13}$$

$$2\vartheta_{00}(0,2\omega)\vartheta_{10}(2u,2\omega)=\vartheta_{10}^2(u)-\vartheta_{11}^2(u). \tag{14}$$

From these one obtains the last two formulae by changing u into $u+\dfrac{1}{2}$ and $u+\dfrac{\omega}{2}$ [or else in the same way as (13), (14)]:

$$\vartheta_{00}\left(0,\frac{\omega}{2}\right)\vartheta_{00}\left(u,\frac{\omega}{2}\right)=\vartheta_{00}^2(u)+\vartheta_{10}^2(u), \tag{15}$$

$$2\vartheta_{00}(0,2\omega)\vartheta_{00}(2u,2\omega)=\vartheta_{00}^2(u)+\vartheta_{01}^2(u). \tag{16}$$

From these, manifold relations between the zero values of the ϑ-functions can be derived, of which we adduce only the three following ones. The first two of them flow from (8), (9), while the last follows from the first equation (11) if one replaces ω by ω:2, assumes $u=1/4$, and takes into account that $\vartheta_{11}(1/4)=\vartheta_{10}(1/4)$.

$$\begin{aligned}\sqrt{\vartheta_{00}\vartheta_{01}} &= \vartheta_{01}(0,2\omega),\\ \sqrt{\vartheta_{00}\vartheta_{10}} &= \frac{1}{\sqrt{2}}\vartheta_{10}\left(0,\frac{\omega}{2}\right),\\ \sqrt{\vartheta_{01}\vartheta_{10}} &= \vartheta_{10}\left(\frac{1}{4},\frac{\omega}{2}\right).\end{aligned} \tag{17}$$

These formulae are of interest because they represent the square roots as single-valued functions of ω.

We make one further application of the transformation of the second order, to the proof of a formula which is necessary for the transformation of odd order.

In the second equation (10) we replace ω by 2ω, so that:

$$2\vartheta_{00}(u,\,2\omega)\vartheta_{10}(u,\,2\omega) = \vartheta_{10}\vartheta_{10}(u).$$

By this we multiply the second equation (11), so that we obtain

$$2\vartheta_{01}(0,\,2\omega)\vartheta_{01}(2u,\,2\omega)\vartheta_{00}(u,\,2\omega)\vartheta_{10}(u,\,2\omega) = \vartheta_{10}\vartheta_{10}(u)\vartheta_{00}(u)\vartheta_{01}(u).$$

This we divide by the product of the two equations (7), (8):

$$2\vartheta_{00}(0,\,2\omega)\vartheta_{10}(0,\,2\omega)\vartheta_{01}(0,\,2\omega)^2 = \vartheta_{10}^2\vartheta_{00}\vartheta_{01}$$

and obtain

$$\frac{\vartheta_{00}(u,\,2\omega)\vartheta_{10}(u,\,2\omega)\vartheta_{01}(2u,\,2\omega)}{\vartheta_{00}(0,\,2\omega)\vartheta_{10}(0,\,2\omega)\vartheta_{01}(0,\,2\omega)} = \frac{\vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(u)}{\vartheta_{00}\vartheta_{10}\vartheta_{01}}. \tag{18}$$

If now n denotes any odd integer, then the function

$$\vartheta_{01}\left(\frac{\nu}{n}\right),$$

remains unchanged when ν increases by a multiple of n, and consequently the product

$$\prod^{\nu}\vartheta_{01}\left(\frac{\nu}{n}\right),$$

when it is taken over a complete system of residues with respect to the modulus n, is independent of the particular choice of this system of residues. Therefore, since 2ν runs through such a system of residues at the same time as ν,

$$\prod_{1,\,n-1}^{\nu}\vartheta_{01}\left(\frac{\nu}{n}\right) = \prod_{1,\,n-1}^{\nu}\vartheta_{01}\left(\frac{2\nu}{n}\right). \tag{19}$$

If therefore we set $u = \nu : n$ in (18), form the product, and make use of formula (19) in the last factor of the left side, it follows that the product

$$\frac{\prod_{1,\,n-1}^{\nu}\vartheta_{00}\left(\frac{\nu}{n}\right)\vartheta_{10}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}\right)}{\vartheta_{00}^{n-1}\vartheta_{10}^{n-1}\vartheta_{01}^{n-1}} \tag{20}$$

remains unchanged when ω is replaced by 2ω.

The values of ν occurring in (20) can be arranged in pairs as follows:

$$\nu,\ n-\nu, \qquad \nu = 1,\ 2,\ \cdots\ \frac{n-1}{2},$$

and since

$$\vartheta_{00}\left(\frac{\nu}{n}\right) = \vartheta_{00}\left(\frac{n-\nu}{n}\right), \quad \vartheta_{10}\left(\frac{\nu}{n}\right) = -\vartheta_{10}\left(\frac{n-\nu}{n}\right),$$
$$\vartheta_{01}\left(\frac{\nu}{n}\right) = \vartheta_{01}\left(\frac{n-\nu}{n}\right),$$

(20) agrees, apart from the sign, with the square of

$$\frac{\prod\limits_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{\nu}{n}\right)\vartheta_{10}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}\right)}{\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}}}. \tag{21}$$

The last quotient, which is a continuous function of ω different from zero as long as the imaginary part of ω is positive, therefore likewise remains unchanged when ω is replaced by 2ω, and hence also by 4ω, 8ω, ... The value of this expression can be determined by taking the imaginary part of ω to be infinite, and hence assuming $q = 0$. But for $q = 0$ we have (by § 25):

$$\vartheta_{00}(v) = 1, \quad \vartheta_{01}(v) = 1, \quad \frac{\vartheta_{10}(v)}{\vartheta_{10}} = \cos \pi v,$$

and hence the value of (21) is:

$$\prod_{1,\,\frac{n-1}{2}}^{\nu} \cos\frac{\nu\pi}{n}. \tag{22}$$

Since here $\nu\pi/n$ is less than $\frac{1}{2}\pi$, this product has a positive value. But we have

$$\cos\frac{\nu\pi}{n} = -\cos\frac{(n-\nu)\pi}{n}$$

and by Bd. I, § 144 we have

$$2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \cos\frac{\nu\pi}{n} = 1.$$

This proves the formula:

$$2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{\nu}{n}\right)\vartheta_{10}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}\right) = \vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}}. \tag{23}$$

With regard to the formula Bd. I, § 145, (3), one may also give this relation the form:

$$2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{2\nu}{n}\right)\vartheta_{10}\left(\frac{2\nu}{n}\right)\vartheta_{01}\left(\frac{2\nu}{n}\right) = (-1)^{\frac{n^2-1}{8}}\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}}. \tag{24}$$

§ 33. The Principal Transformations of Odd Order.

The formula last proved is of use to us in carrying out the transformation of odd order n. We consider first the first principal transformation. By § 27, (12),

$$\vartheta_{11}(nu,\, n\omega) \tag{1}$$

is a Θ_{11}-function of order n in u and ω, and this can easily be formed from its zero points, of which there are n in the period parallelogram. The zero points of (1) are the values

$$u = \frac{\nu + \mu n\omega}{n} = \frac{\nu}{n} + \mu\omega,$$

when ν and μ are integers, and one obtains all the incongruent ones among these values by holding μ fixed and letting ν run through a complete system of residues with respect to the modulus n. We choose the system of residues

$$0,\ \pm 1,\ \pm 2,\ \ldots,\ \pm\frac{(n-1)}{2},$$

and accordingly obtain, if C denotes a factor independent of u,

$$C\vartheta_{11}(nu,\, n\omega) = \vartheta_{11}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{11}\left(\frac{\nu}{n}+u\right)\vartheta_{11}\left(\frac{\nu}{n}-u\right), \tag{2}$$

a formula which, by § 22, (4), can also be expressed in the following way through the functions $\vartheta_{11}(u)$, $\vartheta_{01}(u)$:

$$C\vartheta_{01}^{n-1}\vartheta_{11}(nu,\, n\omega) = \vartheta_{11}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \left[\vartheta_{11}^2\left(\frac{\nu}{n}\right)\vartheta_{01}^2(u) - \vartheta_{01}^2\left(\frac{\nu}{n}\right)\vartheta_{11}^2(u)\right].$$

We replace u by

$$u+\frac{1}{2}, \quad u+\frac{\omega}{2}, \quad u+\frac{1+\omega}{2}$$

and obtain from (8) (§ 21):

$$C\vartheta_{10}(nu,\, n\omega) = \vartheta_{10}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{10}\left(\frac{\nu}{n}+u\right)\vartheta_{10}\left(\frac{\nu}{n}-u\right),$$

$$C\vartheta_{01}(nu,\, n\omega) = \vartheta_{01}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{01}\left(\frac{\nu}{n}+u\right)\vartheta_{01}\left(\frac{\nu}{n}-u\right), \tag{3}$$

$$C\vartheta_{00}(nu,\, n\omega) = \vartheta_{00}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{\nu}{n}+u\right)\vartheta_{00}\left(\frac{\nu}{n}-u\right).$$

But from this there follows, for $u=0$, by the formula $\vartheta_{11}' = \pi\vartheta_{00}\vartheta_{01}\vartheta_{10}$:

$$C \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{11}\left(\frac{\nu}{n}\right) = \sqrt{n} \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}\right), \tag{4}$$

or, using (23) of the preceding section:

$$C\, 2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{11}\left(\frac{\nu}{n}\right) = \sqrt{n}\, \vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}}. \tag{5}$$

The sign follows from the circumstance, which results from one of the formulae (3), that $C=1$ for $q=0$.

After this determination of C the formulae (2), (3) can be written as follows:

$$\sqrt{n}\,\vartheta_{11}(nu,\, n\omega)\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}} = 2^{\frac{n-1}{2}}\vartheta_{11}(u) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{11}\left(\frac{\nu}{n}\right)\vartheta_{11}\left(\frac{\nu}{n}+u\right)\vartheta_{11}\left(\frac{\nu}{n}-u\right). \tag{6}$$

$$(7)\qquad \sqrt{n}\,\vartheta_{10}(nu,\,n\omega)\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}} = 2^{\frac{n-1}{2}}\vartheta_{10}(u)\prod_{1,\,\frac{n-1}{2}}^{\nu}\vartheta_{11}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}+u\right)\vartheta_{01}\left(\frac{\nu}{n}-u\right).$$

$$(8)\qquad \sqrt{n}\,\vartheta_{01}(nu,\,n\omega)\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}} = 2^{\frac{n-1}{2}}\vartheta_{01}(u)\prod_{1,\,\frac{n-1}{2}}^{\nu}\vartheta_{11}\left(\frac{\nu}{n}\right)\vartheta_{01}\left(\frac{\nu}{n}+u\right)\vartheta_{01}\left(\frac{\nu}{n}-u\right).$$

$$(9)\qquad \sqrt{n}\,\vartheta_{00}(nu,\,n\omega)\vartheta_{00}^{\frac{n-1}{2}}\vartheta_{10}^{\frac{n-1}{2}}\vartheta_{01}^{\frac{n-1}{2}} = 2^{\frac{n-1}{2}}\vartheta_{00}(u)\prod_{1,\,\frac{n-1}{2}}^{\nu}\vartheta_{11}\left(\frac{\nu}{n}\right)\vartheta_{00}\left(\frac{\nu}{n}+u\right)\vartheta_{00}\left(\frac{\nu}{n}-u\right).$$

§ 34. The functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$.

The functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ have already been mentioned in § 24, where, in the representation of the ϑ-functions by infinite products, they presented themselves almost of their own accord as single-valued functions of ω, whereas they did not arise in the second representation by infinite series. Connected with this is a remarkable circumstance: many of our formulae, for example those belonging to the transformation of the second order, (17), § 32, or formula (23), § 32, can easily be verified by means of the infinite products, but with difficulty or not at all by means of the infinite series. For that reason it was of interest to us to derive these results from the theory of transformation without making use of either of these expressions, and in the same way the functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ and their fundamental properties shall now likewise be obtained from this source.

Formula (6) of the preceding section gives for $n = 3$:

$$\sqrt{3}\,\vartheta_{11}(3u,\,3\omega)\,\vartheta_{00}\,\vartheta_{10}\,\vartheta_{01} = 2\,\vartheta_{11}\left(\tfrac{1}{3}\right)\vartheta_{11}(u)\,\vartheta_{11}\left(\tfrac{1}{3}+u\right)\vartheta_{11}\left(\tfrac{1}{3}-u\right),$$

and if one differentiates and then sets $u = 0$, one obtains by (4):

$$3\sqrt{3}\,\vartheta'_{11}(0,\,3\omega) = 2\pi\,\vartheta_{11}\left(\tfrac{1}{3}\right)^3,$$

or, replacing ω by $\dfrac{\omega}{3}$,

$$3\sqrt{3}\,\vartheta'_{11} = 2\pi\left[\vartheta_{11}\left(\tfrac{1}{3},\,\tfrac{\omega}{3}\right)\right]^3.$$

If one therefore puts

$$(1)\qquad \eta(\omega) = \frac{1}{\sqrt{3}}\,\vartheta_{11}\left(\frac{1}{3},\,\frac{\omega}{3}\right),$$

then it follows, in the notation agreeing with § 24 (9):

$$(2)\qquad \vartheta'_{11} = \pi\,\vartheta_{00}\,\vartheta_{01}\,\vartheta_{10} = 2\pi\,\eta(\omega)^3$$

and from § 25, (4) one obtains for $\eta(\omega)$ the series development

$$(3)\qquad \eta(\omega) = q^{\frac{1}{12}}\sum_{-\infty,\,\infty}^{\nu}(-1)^\nu\,q^{3\nu^2+\nu}.$$

The function $\eta(\omega)$ is real for a purely imaginary ω (real q), and for an infinitely large ω (i. e. vanishing q) we have

$$q^{-\frac{1}{12}}\,\eta(\omega) = 1.$$

Accordingly, if one applies the formulae § 31, (6), (11) in (2), one finds for the linear fundamental transformations of $\eta(\omega)$

$$\eta(\omega+1) = e^{\frac{\pi i}{12}}\eta(\omega), \tag{4}$$

$$\eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\,\eta(\omega), \tag{5}$$

of which the first also follows immediately from the series representation (3), while the second cannot be proved so easily by direct transformation of the series.

We pass on to the consideration of the two principal transformations of the second order of the η-function.

From § 32, (11) one obtains, by differentiation and setting u equal to zero,

$$2\vartheta_{01}(0,2\omega)\eta(2\omega)^3 = \vartheta_{10}\eta(\omega)^3,$$

hence, if one raises this to the square and applies § 32, (8):

$$4\vartheta_{00}\vartheta_{10}\vartheta_{01}\eta(2\omega)^6 = \vartheta_{10}^3\eta(\omega)^6,$$

and, with the application of (2) and extraction of the cube root:

$$2\eta(2\omega)^2 = \vartheta_{10}\eta(\omega). \tag{6}$$

If in (6) one replaces ω by $\frac{\omega}{2}$, it follows that

$$2\eta(\omega)^2 = \vartheta_{10}\left(0,\frac{\omega}{2}\right)\eta\left(\frac{\omega}{2}\right),$$

and if one squares and applies the formula § 32 (9):

$$2\eta(\omega)^4 = \eta\left(\frac{\omega}{2}\right)^2\vartheta_{00}\vartheta_{10}.$$

If one multiplies both sides by ϑ_{01}, it follows by (2):

$$\eta\left(\frac{\omega}{2}\right)^2 = \vartheta_{01}\eta(\omega) \tag{7}$$

and by changing ω into $\omega+1$ [(4) and § 31, (6)]:

$$e^{-\frac{\pi i}{12}}\eta\left(\frac{\omega+1}{2}\right)^2 = \vartheta_{00}\eta(\omega). \tag{8}$$

Accordingly we introduce the three functions:

$$\begin{aligned} f(\omega) &= \frac{e^{-\frac{\pi i}{24}}\eta\left(\frac{\omega+1}{2}\right)}{\eta(\omega)}, \\ f_1(\omega) &= \frac{\eta\left(\frac{\omega}{2}\right)}{\eta(\omega)}, \\ f_2(\omega) &= \sqrt{2}\,\frac{\eta(2\omega)}{\eta(\omega)}. \end{aligned} \tag{9}$$

Then by (6), (7), (8)

$$\begin{aligned} \vartheta_{00} &= \eta(\omega) f(\omega)^2, \\ \vartheta_{01} &= \eta(\omega) f_1(\omega)^2, \\ \vartheta_{10} &= \eta(\omega) f_2(\omega)^2, \end{aligned} \tag{10}$$

and from (8) and (9) it follows that $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ are real and positive for a purely imaginary ω. (10) agrees with § 24, (10), and the functions f, f_1, f_2 introduced there are the same as these. But from (10) the relations follow easily [(2) and § 21 (14)]:

$$\begin{aligned} f(\omega)^8 &= f_1(\omega)^8 + f_2(\omega)^8, \\ \sqrt{2} &= f(\omega) f_1(\omega) f_2(\omega). \end{aligned} \tag{11}$$

With the help of these formulae one can express each of the three functions $f(\omega)^8$, $f_1(\omega)^8$, $f_2(\omega)^8$ by square roots in terms of any other. The formula flowing from (11) leads to this:

$$\begin{aligned} [f_1(\omega)^8 - f_2(\omega)^8]^2 &= f(\omega)^{16} - \frac{64}{f(\omega)^8}, \\ [f(\omega)^8 + f_2(\omega)^8]^2 &= f_1(\omega)^{16} + \frac{64}{f_1(\omega)^8}, \\ [f(\omega)^8 + f_1(\omega)^8]^2 &= f_2(\omega)^{16} + \frac{64}{f_2(\omega)^8}. \end{aligned} \tag{12}$$

For the linear fundamental transformations of the functions f one obtains first of all from (4) and (9):

$$\begin{aligned} f(\omega + 1) &= e^{-\frac{\pi i}{24}} f_1(\omega), \\ f_1(\omega + 1) &= e^{-\frac{\pi i}{24}} f(\omega), \\ f_2(\omega + 1) &= e^{\frac{\pi i}{12}} f_2(\omega). \end{aligned} \tag{13}$$

Furthermore, there follows from (5) and from the last two equations (9):

$$f_1\left(-\frac{1}{\omega}\right) = f_2(\omega), \qquad f_2\left(-\frac{1}{\omega}\right) = f_1(\omega), \tag{14}$$

and if one makes use of this in the second equation (11):

$$f\left(-\frac{1}{\omega}\right) = f(\omega). \tag{15}$$

For the transformation of the second order there follows immediately from the last two equations (9):

$$f_1(2\,\omega) f_2(\omega) = \sqrt{2}, \tag{16}$$

from which, by applying (12), there follows further:

$$f_2(\omega)^4 \left[f(2\,\omega)^8 + f_2(2\,\omega)^8 \right] = 2 \left[f(\omega)^8 + f_1(\omega)^3 \right]. \tag{17}$$

If in (16) one substitutes, in accordance with (13), (14)

$$f_1(2\,\omega) = e^{-\frac{\pi i}{24}} f(2\,\omega - 1),$$
$$f_2(\omega) = e^{\frac{\pi i}{24}} f\left(1 - \frac{1}{\omega}\right),$$

then there follows:

$$f\left(1 - \frac{1}{\omega}\right) f(2\,\omega - 1) = \sqrt{2},$$

and on setting $2\,\omega - 1$ equal to a new ω:

$$f(\omega) f\left(\frac{\omega - 1}{\omega + 1}\right) = \sqrt{2}. \tag{18}$$

If in (16) one replaces ω by $\frac{\omega}{2}$ and $\frac{\omega + 1}{2}$, there follows:

$$\begin{aligned} f_1(\omega) f_2\left(\frac{\omega}{2}\right) &= \sqrt{2}, \\ f(\omega) f_2\left(\frac{\omega + 1}{2}\right) &= e^{\frac{\pi i}{24}} \sqrt{2}. \end{aligned} \tag{19}$$

Finally we also set up, for an odd n, the formulae for the principal transformation of order n of the functions η, f. For $\eta(n\,\omega)$ one obtains this easily by setting $u = 0$ in the formulae (6), § 33 after the differentiation and extracting the third root:

$$\sqrt{n}\,\eta(n\,\omega)\,\eta(\omega)^{\frac{n-3}{2}} = \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{11}\left(\frac{\nu}{n}\right). \tag{20}$$

If one further sets $u = 0$ in (7), (8), (9), § 33, then uses the relations (10) and (20) and extracts the square root, one finds

$$\begin{aligned} f(n\,\omega)\,\eta(\omega)^{\frac{n-1}{2}} &= f(\omega) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{00}\left(\frac{\nu}{n}\right), \\ f_1(n\,\omega)\,\eta(\omega)^{\frac{n-1}{2}} &= f_1(\omega) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{01}\left(\frac{\nu}{n}\right), \\ f_2(n\,\omega)\,\eta(\omega)^{\frac{n-1}{2}} &= f_2(\omega) \prod_{1,\,\frac{n-1}{2}}^{\nu} \vartheta_{10}\left(\frac{\nu}{n}\right). \end{aligned} \tag{21}$$

The signs follow here from the assumption of an infinitely small q.

§ 35. The Weierstrass σ-Function.

By the general linear substitution

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1,$$

applied to the variables ω_1, ω_2, every t-function passes again into a t-function, whereby the characteristic has changed. If (g_1, g_2) is the characteristic of the original function and (g_1', g_2') that of the transformed function, then by § 27, (11)

$$(g_1, g_2) = (\delta g_1' - \beta g_2' - \beta\delta,\ -\gamma g_1' + \alpha g_2' - \alpha\gamma).$$

If one solves the two linear equations contained in this for g_1', g_2', and observes that $\alpha\beta(\gamma+\delta+1)$, $\gamma\delta(\alpha+\beta+1)$ are necessarily even numbers, and that a characteristic does not change when its elements change by multiples of 2, then one may also write in place of it:

(1) $$(g_1', g_2') = (\alpha g_1 + \beta g_2 + \alpha\beta,\ \gamma g_1 + \delta g_2 + \gamma\delta).$$

The ϑ-functions therefore pass into one another, apart from exponential factors. It is however of importance to form a function which is absolutely invariant with respect to the linear transformations, and such a function is the σ-function introduced into the theory by Weierstrass, to whose definition we now proceed.

If we apply formula (1) to the four principal characteristics: $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$, then these pass over, in order,

into $(\alpha\beta, \gamma\delta)$, $[(\alpha+1)\beta, (\gamma+1)\delta]$, $[\alpha(\beta+1), \gamma(\delta+1)]$, $[(\alpha+1)(\beta+1)-1,\ (\gamma+1)(\delta+1)-1) = (1,1)]$. Thus only the last characteristic $(1,1)$ remains unchanged under all linear transformations, since neither α and β nor γ and δ can be even numbers at the same time. Let $t(u, \omega_1, \omega_2)$ therefore be a t-function of the characteristic $(1,1)$, which by § 17, (12) must vanish for $u = 0$.

If

$$(\omega_1', \omega_2') = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (\omega_1, \omega_2),$$

then, by our principle of transformation,

$$t(u, \omega_1, \omega_2) \quad \text{and} \quad t(u, \omega_1', \omega_2')$$

are related T-functions of the first order, and consequently, if C, λ, μ are quantities independent of u,

(2) $$t(u, \omega_1', \omega_2') = Ce^{\lambda u^2 + \mu u} t(u, \omega_1.\omega_2).$$

From this there follows by logarithmic differentiation

(3) $$2\lambda u + \mu = \frac{d \log t(u, \omega_1', \omega_2')}{du} - \frac{d \log t(u, \omega_1, \omega_2)}{du}.$$

Now if we develop in powers of u by Taylor's theorem and denote by t', t'', t''' the first, second and third derivative of t with respect to u for $u = 0$, then

$$\frac{d \log t(u, \omega_1, \omega_2)}{du} = \frac{1}{u} + \frac{t''}{2t'} + u\left(\frac{t'''}{3t'} - \frac{t''^2}{4t'^2}\right) + \cdots,$$

from which it follows by comparison with (3) that λ and μ can be put into the form:

$$\lambda = \varphi(\omega_1', \omega_2') - \varphi(\omega_1, \omega_2),$$
$$\mu = \psi(\omega_1', \omega_2') - \psi(\omega_1, \omega_2),$$

in which

$$\varphi(\omega_1,\omega_2)=\frac{t'''}{6t'}-\frac{t''^2}{8t'^2},\qquad \psi(\omega_1,\omega_2)=\frac{t''}{2t'}. \tag{4}$$

If one determines further the factor C in (2) from the special value $u=0$, it follows that

$$C=\frac{t'(\omega_1',\omega_2')}{t'(\omega_1,\omega_2)}. \tag{5}$$

Formula (2) can accordingly be expressed in the following theorem:

The function

$$\sigma(u,\omega_1,\omega_2)=e^{-u^2\left(\frac{t'''}{6t'}-\frac{t''^2}{8t'^2}\right)-u\frac{t''}{2t'}}\frac{t(u,\omega_1,\omega_2)}{t'} \tag{6}$$

remains unchanged if one replaces ω_1, ω_2 by ω_1', ω_2', that is to say, if one applies any linear transformation whatever, or:

$$\sigma(u,\omega_1',\omega_2')=\sigma(u,\omega_1,\omega_2). \tag{7}$$

The function defined by (6) remains unchanged, as a simple calculation shows, if one replaces t by any related t-function.

The function $t(u)$ has by hypothesis the characteristic $(1,1)$, and its zero points are therefore congruent with 0. Accordingly $t(-u)$ has the same zero points, and hence also the same character as $t(u)$. The two functions therefore differ from one another only by an exponential factor, and it follows easily, if one determines a constant factor from $u=0$:

$$t(-u)=-e^{2\pi i\mu u}t(u),$$

in which μ is a constant.

If one differentiates this equation twice with respect to u and then sets $u=0$, it follows that

$$2\pi i\mu=-\frac{t''}{t'},$$

and from this it follows by (6) that the function $\sigma(u)$ satisfies the condition

$$\sigma(-u)=-\sigma(u) \tag{8}$$

and is therefore an odd function of u.

Since $\sigma(u)$ is a t-function of the first order, it satisfies the two conditions:

$$\sigma(u+\omega_1)=c_1e^{\eta_1(2u+\omega_1)}\sigma(u),$$
$$\sigma(u+\omega_2)=c_2e^{\eta_2(2u+\omega_2)}\sigma(u),$$

in which c_1, c_2, η_1, η_2 are constants. The function $\sigma(u)$ vanishes for $u=0$, but not for $u=\frac{1}{2}\omega_1$, $u=\frac{1}{2}\omega_2$; and so if one sets $u=-\frac{1}{2}\omega_1$, $-\frac{1}{2}\omega_2$ in the foregoing formulae, it follows by (8) that $c_1=-1$, $c_2=-1$, and hence the formulae:

$$\sigma(u+\omega_1)=-e^{\eta_1(2u+\omega_1)}\sigma(u),$$
$$\sigma(u+\omega_2)=-e^{\eta_2(2u+\omega_2)}\sigma(u). \tag{9}$$

The η_1, η_2 introduced in this way are functions of ω_1, ω_2 which, by the relation § 17, (10) (in which $\pi i a_1$, $\pi i a_2$, m are to be replaced by $-\eta_1$, $-\eta_2$, 1), satisfy the equation:

$$\eta_1\omega_2 - \eta_2\omega_1 = \pi i. \tag{10}$$

By repeated application of (9) it follows, if a, b are integers [cf. § 17, (16)]:

$$\sigma(u + a\omega_1 + b\omega_2) = (-1)^{a+b+ab}\, e^{(a\eta_1+b\eta_2)(2u+a\omega_1+b\omega_2)}\, \sigma(u). \tag{11}$$

If one develops the function $\sigma(u)$, as it is given by formula (6), in powers of u, it is found that not only the second but also the third power of u fails to occur in the development, and one has therefore:

$$\sigma(0) = 0, \quad \sigma'(0) = 1, \quad \sigma''(0) = 0, \quad \sigma'''(0) = 0. \tag{12}$$

If to ω_1, ω_2 a linear substitution

$$(\omega_1', \omega_2') = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (\omega_1, \omega_2) \tag{13}$$

is applied, then the quantities η_1, η_2 undergo the corresponding substitution

$$(\eta_1', \eta_2') = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (\eta_1, \eta_2), \tag{14}$$

as one infers immediately from the relations (7) and (11).

If the formulae (9) are differentiated logarithmically with respect to u, there follows

$$\begin{aligned} \frac{\sigma'(u+\omega_1)}{\sigma(u+\omega_1)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_1, \\ \frac{\sigma'(u+\omega_2)}{\sigma(u+\omega_2)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_2, \end{aligned} \tag{15}$$

and by setting, in the first, $u = -\dfrac{\omega_1}{2}$, in the second $u = -\dfrac{\omega_2}{2}$, and observing that $\sigma(u)$ is an odd and consequently $\sigma'(u)$ an even function of u:

$$\eta_1 = \frac{\sigma'\left(\dfrac{\omega_1}{2}\right)}{\sigma\left(\dfrac{\omega_1}{2}\right)}, \quad \eta_2 = \frac{\sigma'\left(\dfrac{\omega_2}{2}\right)}{\sigma\left(\dfrac{\omega_2}{2}\right)}. \tag{16}$$

§ 36. The Functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$.

It still remains for us to obtain the analogous results for the remaining characteristics. We start from the following remark: if x_1, x_2; y_1, y_2 are any two pairs of quantities which are transformed by one and the same linear substitution S into x_1', x_2'; y_1', y_2', then we have:

$$(x_1', x_2') = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (x_1, x_2); \quad (y_1', y_2') = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (y_1, y_2),$$

then also

$$x_1y_2 - x_2y_1 = x_1'y_2' - x_2'y_1',$$

as follows from the multiplication of the determinants. Accordingly, whatever x_1, x_2 may be, we have by (7), § 35:

$$\sigma(u + x_2\omega_1 - x_1\omega_2,\ \omega_1, \omega_2) = \sigma(u + x_2'\omega_1' - x_1'\omega_2',\ \omega_1', \omega_2'). \tag{1}$$

This function changes in accordance with formula (11) of the preceding section when x_1, x_2 are altered by integers. This change can, however, be avoided if instead one considers the function

$$e^{-2(\eta_1 x_2 - \eta_2 x_1)u}\,\frac{\sigma(u + x_2\omega_1 - x_1\omega_2)}{\sigma(x_2\omega_1 - x_1\omega_2)}$$

which we shall (for the moment) denote by $\sigma(u, x_1, x_2, \omega_1, \omega_2)$. This function satisfies the conditions:

$$\sigma(u, x_1', x_2', \omega_1', \omega_2') = \sigma(u, x_1, x_2, \omega_1, \omega_2), \tag{2}$$

$$\begin{aligned} \sigma(u, x_1 + 1, x_2, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \\ \sigma(u, x_1, x_2 + 1, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \end{aligned} \tag{3}$$

$$\begin{aligned} \sigma(u + \omega_1) &= -e^{-2\pi i x_1}\, e^{\eta_1(2u+\omega_1)}\, \sigma(u), \\ \sigma(u + \omega_2) &= -e^{-2\pi i x_2}\, e^{\eta_2(2u+\omega_2)}\, \sigma(u), \end{aligned} \tag{4}$$

and it therefore remains unchanged when x_1 and x_2 are altered by integers.

Restricting ourselves now once more to the principal characteristics, we set $(x_1, x_2) = \left(-\frac{1}{2}, \frac{1}{2}\right)$, $\left(0, \frac{1}{2}\right)$, $\left(-\frac{1}{2}, 0\right)$ and obtain in this way the following three functions:

$$\begin{aligned} \sigma_{00}(u) &= \sigma\left(u, -\frac{1}{2}, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-(\eta_1+\eta_2)u}\,\frac{\sigma\left(u + \dfrac{\omega_1+\omega_2}{2}\right)}{\sigma\left(\dfrac{\omega_1+\omega_2}{2}\right)}, \\ \sigma_{10}(u) &= \sigma\left(u, 0, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-\eta_1 u}\,\frac{\sigma\left(u + \dfrac{\omega_1}{2}\right)}{\sigma\left(\dfrac{\omega_1}{2}\right)}, \\ \sigma_{01}(u) &= \sigma\left(u, -\frac{1}{2}, 0, \omega_1, \omega_2\right) = e^{-\eta_2 u}\,\frac{\sigma\left(u + \dfrac{\omega_2}{2}\right)}{\sigma\left(\dfrac{\omega_2}{2}\right)}, \end{aligned} \tag{5}$$

and the function $\sigma(u)$ itself may accordingly also be denoted by $\sigma_{11}(u)$.

For these functions there follow from (4) the characteristic period equations

$$\begin{aligned} \sigma_{g_1,g_2}(u + \omega_1) &= (-1)^{g_1}\, e^{\eta_1(2u+\omega_1)}\, \sigma_{g_1,g_2}(u), \\ \sigma_{g_1,g_2}(u + \omega_2) &= (-1)^{g_2}\, e^{\eta_2(2u+\omega_2)}\, \sigma_{g_1,g_2}(u), \\ \sigma_{g_1,g_2}(u + a\omega_1 + b\omega_2) &= (-1)^{g_1 a + g_2 b + ab}\, e^{(a\eta_1 + b\eta_2)(2u + a\omega_1 + b\omega_2)}\, \sigma_{g_1,g_2}(u). \end{aligned} \tag{6}$$

By the application of a linear transformation the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ are permuted among one another, as formula (2) teaches [or § 35, (1)]. According to this permutation the linear transformations split into six classes, the first of which comprises all those transformations that leave the functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ unchanged. These are characterized by the fact that α, δ are odd and β, γ even numbers, which we write briefly thus:

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} (\text{mod } 2).$$

Accordingly the six classes of linear transformations are to be characterized as follows:

(7)
$$\begin{aligned}
&\text{I.} \quad \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} (\text{mod } 2) \quad (00, 01, 10), \\
&\text{II.} \quad „ \equiv \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \quad „ \quad (00, 10, 01), \\
&\text{III.} \quad „ \equiv \begin{pmatrix} 1, & 1 \\ -1, & 0 \end{pmatrix} \quad „ \quad (10, 00, 01), \\
&\text{IV.} \quad „ \equiv \begin{pmatrix} 1, & 1 \\ 0, & 1 \end{pmatrix} \quad „ \quad (10, 01, 00), \\
&\text{V.} \quad „ \equiv \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} \quad „ \quad (01, 00, 10), \\
&\text{VI.} \quad „ \equiv \begin{pmatrix} 0, & 1 \\ -1, & 1 \end{pmatrix} \quad „ \quad (01, 10, 00),
\end{aligned}$$

where in the last column the permutation of the characteristics 00, 01, 10 occurring in each case is set out.

The transformations of the first class form a group $\mathfrak{A}$ contained in the group of all linear substitutions, and hence a subgroup of $\mathfrak{L}$ (§ 28). If we set

$$\alpha_1 = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \quad \alpha_3 = \begin{pmatrix} 1, & 1 \\ -1, & 0 \end{pmatrix},$$
$$\alpha_4 = \begin{pmatrix} 1, & 1 \\ 0, & 1 \end{pmatrix}, \quad \alpha_5 = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \quad \alpha_6 = \begin{pmatrix} 0, & 1 \\ -1, & 1 \end{pmatrix},$$

then $\alpha_1\mathfrak{A}, \alpha_2\mathfrak{A}, \alpha_3\mathfrak{A}, \alpha_4\mathfrak{A}, \alpha_5\mathfrak{A}, \alpha_6\mathfrak{A}$ are the cosets of $\mathfrak{A}$, and we have

$$\mathfrak{L} = \alpha_1\mathfrak{A} + \alpha_2\mathfrak{A} + \alpha_3\mathfrak{A} + \alpha_4\mathfrak{A} + \alpha_5\mathfrak{A} + \alpha_6\mathfrak{A}$$

and $\mathfrak{A}$ is a subgroup of $\mathfrak{L}$ of finite index 6:

$$(\mathfrak{L}, \mathfrak{A}) = 6 \quad (\text{Bd. II, § 2}).$$

The totality $\alpha_1\mathfrak{A} + \alpha_2\mathfrak{A}$ is likewise a group $\mathfrak{A}'$, and we have

$$\mathfrak{L} = \alpha_1\mathfrak{A}' + \alpha_3\mathfrak{A}' + \alpha_4\mathfrak{A}'$$

and $(\mathfrak{L}, \mathfrak{A}') = 3$.

Just as all the linear transformations may be derived from the two fundamental transformations, so the transformations of the first class may be derived from repeated application of

$$\begin{pmatrix} 1, & 0 \\ 2, & 1 \end{pmatrix}, \quad \begin{pmatrix} 1, & 2 \\ 0, & 1 \end{pmatrix}, \quad \begin{pmatrix} -1, & 0 \\ 0, & -1 \end{pmatrix}$$

which can be proved by the method of § 30.

§ 37. Representation of the σ-functions by ϑ-functions.

In order to represent the function $\sigma(u)$ as a ϑ-function, one can simply apply formula (6) of § 35 to the function

$$t(u) = \vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)$$

and thus set

$$\sigma(u, \omega_1, \omega_2) = \omega_1\, e^{-\frac{u^2}{6\omega_1^2}\frac{\vartheta'''_{11}}{\vartheta'_{11}}}\, \frac{\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta'_{11}} \tag{1}$$

But in consequence of the differential equation § 20, (4), we have:

$$\vartheta'''_{11} = 4\pi i\, \frac{d\vartheta'_{11}}{d\omega}$$

and hence, by the definition of the function $\eta(\omega)$ in § 34, (2):

$$\frac{\vartheta'''_{11}}{\vartheta'_{11}} = 4\pi i\, \frac{d\log\vartheta'_{11}}{d\omega} = 12\pi i\, \frac{d\log\eta(\omega)}{d\omega}. \tag{2}$$

By logarithmic differentiation of (1) one obtains, by means of (2):

$$\frac{d\log\sigma(u)}{du} = -\frac{4\pi i u}{\omega_1^2}\, \frac{d\log\eta(\omega)}{d\omega} + \frac{d\log\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{du},$$

and if one here sets $u = \frac{\omega_1}{2}, \frac{\omega_2}{2}$ and applies the formulae (8) of § 21, then by § 35, (16):

$$\begin{aligned} \eta_1 &= -\frac{2\pi i}{\omega_1}\, \frac{d\log\eta(\omega)}{d\omega}, \\ \eta_2 &= -\frac{2\pi i\omega_2}{\omega_1^2}\, \frac{d\log\eta(\omega)}{d\omega} - \frac{\pi i}{\omega_1}. \end{aligned} \tag{3}$$

Accordingly one may set for σ:

$$\sigma(u) = \omega_1\, e^{\frac{\eta_1 u^2}{\omega_1}}\, \frac{\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta'_{11}}, \tag{4}$$

and for the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ one obtains by § 36, (5), when one determines a constant factor from

$$\sigma_{00}(0) = 1, \quad \sigma_{01}(0) = 1, \quad \sigma_{10}(0) = 1 \tag{5}$$

the following:

$$
\begin{aligned}
\sigma_{00}(u) &= e^{\frac{\eta_1 u^2}{\omega_1}}\,\frac{\vartheta_{00}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}},\\
\sigma_{01}(u) &= e^{\frac{\eta_1 u^2}{\omega_1}}\,\frac{\vartheta_{01}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)}{\vartheta_{01}},\\
\sigma_{10}(u) &= e^{\frac{\eta_1 u^2}{\omega_1}}\,\frac{\vartheta_{10}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)}{\vartheta_{10}}.
\end{aligned}
\tag{6}
$$

The functions $\sigma_{00}(u)$, $\sigma_{01}(u)$, $\sigma_{10}(u)$ are even functions of u, and by twofold differentiation of the logarithms of (6) there results further [by (3) and § 34, (2)]:

$$
\sigma''_{00}(0)+\sigma''_{01}(0)+\sigma''_{10}(0)=0. \tag{7}
$$

The expressions (4), (6) reveal at first glance an important property of the σ-functions, namely that $\sigma_{00},\sigma_{01},\sigma_{10}$ depend only on the ratios $u:\omega_1:\omega_2$, while for σ the same holds apart from the factor ω_1. Thus σ, σ_{00}, σ_{01}, σ_{10} are homogeneous functions of the three variables u, ω_1, ω_2, the former of the first order, the three others of the zeroth order, or, in symbols, if λ denotes an arbitrary factor:

$$
\begin{aligned}
\sigma(\lambda u_1,\lambda\omega_1,\lambda\omega_2) &= \lambda\sigma(u,\omega_1,\omega_2),\\
\sigma_{00}(\lambda u_1,\lambda\omega_1,\lambda\omega_2) &= \sigma_{00}(u,\omega_1,\omega_2),\\
\sigma_{01}(\lambda u_1,\lambda\omega_1,\lambda\omega_2) &= \sigma_{01}(u,\omega_1,\omega_2),\\
\sigma_{10}(\lambda u_1,\lambda\omega_1,\lambda\omega_2) &= \sigma_{10}(u,\omega_1,\omega_2).
\end{aligned}
\tag{8}
$$

§ 38. Linear Transformations of the Function $\eta(\omega)$.

The formulae (3), § 37 lead to the linear transformation of the function $\eta(\omega)$. For if

$$
(\omega'_1,\omega'_2)=\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}(\omega_1,\omega_2) \tag{1}
$$

is set, then (η_1,η_2) passes over into

$$
(\eta'_1,\eta'_2)=\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}(\eta_1,\eta_2) \tag{2}
$$

[§ 35, (14)], and $\omega=\omega_2:\omega_1$ into

$$
\omega'=\frac{\gamma+\delta\omega}{\alpha+\beta\omega}. \tag{3}
$$

Now from (2), having regard to (3) of the preceding section, there follows:

$$
\begin{aligned}
\eta'_1 &= -\frac{2\pi i}{\omega'_1}\,\frac{d\log\eta(\omega')}{d\omega'} = -\frac{2\pi i\omega'_1}{\omega_1^2}\,\frac{d\log\eta(\omega)}{d\omega}-\frac{\pi i\beta}{\omega_1},\\
\eta'_2 &= -\frac{2\pi i\omega'_2}{{\omega'_1}^2}\,\frac{d\log\eta(\omega')}{d\omega'}-\frac{\pi i}{\omega'_1} = -\frac{2\pi i\omega'_2}{\omega_1^2}\,\frac{d\log\eta(\omega)}{d\omega}-\frac{\pi i\delta}{\omega_1},
\end{aligned}
$$

and these two relations agree in giving

$$
\frac{d\log\eta(\omega')}{{\omega'_1}^2\,d\omega'}=\frac{d\log\eta(\omega)}{\omega_1^2\,d\omega}+\frac{\beta}{2\omega_1\omega'_1},
$$

or finally, since by (3)

$$d\omega' = \frac{d\omega}{(\alpha+\beta\omega)^2}, \quad \omega_1'^{\,2}\, d\omega' = \omega_1^2\, d\omega,$$

$$\frac{d\log\eta(\omega')}{d\omega} = \frac{d\log\eta(\omega)}{d\omega} + \frac{\beta}{2(\alpha+\beta\omega)}.$$

From this it follows by integration:

$$\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varepsilon\sqrt{\alpha+\beta\omega}\;\eta(\omega), \tag{4}$$

in which ε is a quantity independent of ω, and therefore depending only on the numbers α, β, γ, δ.

The exact determination of this constant ε, especially with regard to the sign as well, is a well-known and important problem which presents peculiar difficulties; its solution, however, is indispensable for us. Solutions have been given along various routes by Hermite[4] and Dedekind[5], and more recently also by Mertens and Scheibner[6]. We wish here to follow a route which has the advantage of great simplicity, but which for that very reason contains no derivation, but only a proof of the finished formula.

For two special cases we have already carried out this determination earlier [§ 34, (4), (5)], and it is upon this result that we shall rely here. They are the formulae:

$$\eta(\omega \pm 1) = e^{\pm\frac{\pi i}{12}}\,\eta(\omega) \tag{5}$$

and

$$\eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\;\eta(\omega), \tag{6}$$

where $\sqrt{-i\omega}$ is to be taken with positive real part.

We now set

$$\frac{\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right)}{\eta(\omega)} = E\left(\begin{smallmatrix}\alpha,\ \beta\\ \gamma,\ \delta\end{smallmatrix},\omega\right), \tag{7}$$

and have to determine the value of this symbol. By (5), (6), (7) we have:

$$E\left(\begin{smallmatrix}-\alpha,\ -\beta\\ -\gamma,\ -\delta\end{smallmatrix},\omega\right) = E\left(\begin{smallmatrix}\alpha,\ \beta\\ \gamma,\ \delta\end{smallmatrix},\omega\right), \tag{8}$$

$$E\left(\begin{smallmatrix}1,\ 0\\ 0,\ 1\end{smallmatrix},\omega\right) = 1, \tag{9}$$

$$E\left(\begin{smallmatrix}1,\ \ 0\\ \pm1,\ 1\end{smallmatrix},\omega\right) = e^{\pm\frac{\pi i}{12}}, \tag{10}$$

$$E\left(\begin{smallmatrix}0,\ 1\\ -1,\ 0\end{smallmatrix},\omega\right) = \sqrt{-i\omega}. \tag{11}$$

[4] Liouvilles Journal, Ser. II, T. III, 1858. Oeuvres de Charles Hermite, p. 487.

[5] Erläuterungen zu Nr. XXVIII von Riemanns Werken, second edition, and „Über die elliptischen Modulfunktionen“, Crelles Journal, Bd. 83, S. 265. Cf. also the author's memoir „Zur Theorie der elliptischen Funktionen“, Acta Mathematica, Bd. 6, S. 341 ff.

[6] Mertens, Zur linearen Transformation der ϑ-Reihen, Transactions of the American mathematical society. July 1901. — Scheibner, Zur linearen Transformation der Theta-Funktionen und elliptischen Modulfunktionen, Berichte der Sächs. Gesellschaft der Wissenschaften. Oktober 1906.

We now consider two substitutions and the one composed of both, thus:

$$\begin{pmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{pmatrix} = \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}.$$

If then

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

there results from the definition (7):

$$E\left(\begin{smallmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{smallmatrix}, \omega\right) = E\left(\begin{smallmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{smallmatrix}, \omega'\right) E\left(\begin{smallmatrix} \alpha, & \beta \\ \gamma, & \delta \end{smallmatrix}, \omega\right), \tag{12}$$

and from this two special cases, by putting

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \pm 1, & 1 \end{pmatrix}, \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

and writing α, β, γ, δ again in place of α', β', γ', δ', thus

$$\begin{pmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{pmatrix} = \begin{pmatrix} \alpha \pm \beta, & \beta \\ \gamma \pm \delta, & \delta \end{pmatrix}, \begin{pmatrix} -\beta, & \alpha \\ -\delta, & \gamma \end{pmatrix}:$$

$$E\left(\begin{smallmatrix} \alpha \pm \beta, & \beta \\ \gamma \pm \delta, & \delta \end{smallmatrix}, \omega\right) = e^{\pm \frac{\pi i}{12}} E\left(\begin{smallmatrix} \alpha, & \beta \\ \gamma, & \delta \end{smallmatrix}, \omega \pm 1\right), \tag{13}$$

$$E\left(\begin{smallmatrix} -\beta, & \alpha \\ -\delta, & \gamma \end{smallmatrix}, \omega\right) = \sqrt{-i\omega}\; E\left(\begin{smallmatrix} \alpha, & \beta \\ \gamma, & \delta \end{smallmatrix}, \frac{-1}{\omega}\right). \tag{14}$$

It has now been shown in § 30 that all linear transformations can be composed by repeated application of the two fundamental transformations

$$\begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

and of their inverse transformations, and from this it follows on the basis of (12) that the symbol E is completely defined by the formulae (8) to (14).

If, then, we know an expression satisfying these conditions, it must agree with E. In order to set up such an expression, we distinguish two cases. Since α, β are relatively prime numbers, one of them is certainly odd. We put:

1. α odd and positive:

$$E\left(\begin{smallmatrix} \alpha, & \beta \\ \gamma, & \delta \end{smallmatrix}, \omega\right) = \left(\frac{\beta}{\alpha}\right) i^{\frac{\alpha-1}{2}} e^{\frac{\pi i}{12}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]} \sqrt{(\alpha+\beta\omega)}; \tag{15}$$

2. β odd and positive:

$$E\left(\begin{smallmatrix} \alpha, & \beta \\ \gamma, & \delta \end{smallmatrix}, \omega\right) = \left(\frac{\alpha}{\beta}\right) i^{\frac{1-\beta}{2}} e^{\frac{\pi i}{12}[\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma]} \sqrt{-i(\alpha+\beta\omega)},$$

to which the following is still to be remarked: the roots $\sqrt{\alpha+\beta\omega}$, $\sqrt{-i(\alpha+\beta\omega)}$ are to be taken with positive real part. That one of them should be purely imaginary is excluded by the assumption that ω has a positive imaginary part and that α, or β respectively, is positive; for accordingly $\alpha+\beta\omega$ or $-i(\alpha+\beta\omega)$ cannot be real and negative; $\left(\frac{\beta}{\alpha}\right)$ and $\left(\frac{\alpha}{\beta}\right)$ is the Legendre-Jacobi symbol from the theory of quadratic residues, with the extension that $\left(\frac{\beta}{1}\right)$ and $\left(\frac{0}{1}\right) = 1$

shall hold. If in the first case α, or in the second β, is negative, then on the right all the signs of $\alpha, \beta, \gamma, \delta$ must be reversed. If both α and β are odd, then (15), 1. as well as (15), 2. may be applied, and both yield, as one easily proves on the basis of the law of reciprocity for quadratic residues, the same result. For, if α and β are odd, α is taken positive, and the upper or lower sign holds according as β is positive or negative, we have:

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\pm\alpha}{\pm\beta}\right) = \pm\, e^{-\frac{\pi i}{4}(\alpha\mp 1)(\beta-1)},$$

$$\sqrt{\mp i(\alpha+\beta\omega)} = e^{\mp\frac{\pi i}{4}}\sqrt{\alpha+\beta\omega}$$

and the identity of (15), 1., 2. then follows from the congruences

$$-3\alpha\beta + \beta(\alpha+\delta) - (\beta^2-1)\alpha\gamma \equiv \alpha(\gamma-\beta) - (\alpha^2-1)\beta\delta \pmod{24},$$
$$\alpha\gamma + \alpha\beta \equiv \beta\delta \pmod{8},$$

of which the second holds also for the modulus 24 whenever α and β are both not divisible by 3, so that $\alpha^2 \equiv \beta^2 \equiv 1 \pmod 3$.

That the formulae (8) to (11) are satisfied by (15) is immediately evident, and it still remains to be shown that (13) and (14) are fulfilled. We begin with (14), where it may be assumed that α is odd (and positive); for if one interchanges ω in (14) with $-1 : \omega$, then α and $-\beta$ are interchanged, and these cannot both be even.

We have

$$\sqrt{-i\omega}\,\sqrt{\alpha - \frac{\beta}{\omega}} = \sqrt{-i(-\beta+\alpha\omega)};$$

for if we set for the moment

$$-i\omega = r\,e^{i\varphi}, \qquad \alpha - \frac{\beta}{\omega} = \varrho\, e^{i\psi}$$
$$-i(-\beta+\alpha\omega) = r\varrho\, e^{i(\varphi+\psi)},$$

then, since the real parts of $-i\omega$, $-i(-\beta+\alpha\omega)$ are positive,

$$-\frac{\pi}{2} < \varphi < \frac{\pi}{2}, \quad -\pi < \psi < \pi, \quad -\frac{\pi}{2} < \varphi+\psi < \frac{\pi}{2},$$

$$\sqrt{-i\omega} = \sqrt{r}\, e^{\frac{i\varphi}{2}}, \qquad \sqrt{\alpha - \frac{\beta}{\omega}} = \sqrt{\varrho}\, e^{\frac{i\psi}{2}},$$

and the real part of

$$\sqrt{-i\omega}\,\sqrt{\alpha - \frac{\beta}{\omega}} = \sqrt{-i(-\beta+\alpha\omega)} = \sqrt{r\varrho}\, e^{\frac{i(\varphi+\psi)}{2}}$$

is positive. Accordingly there results from (15), 1:

$$\sqrt{-i\omega}\, E\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}, \frac{-1}{\omega}\Bigg) = \left(\frac{\beta}{\alpha}\right) i^{\frac{\alpha-1}{2}} e^{\frac{\pi i}{12}\left[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta\right]}\sqrt{-i(-\beta+\alpha\omega)}.$$

The same formula, however, is obtained, since $\left(\frac{-\beta}{\alpha}\right) = i^{(\alpha-1)}\left(\frac{\beta}{\alpha}\right)$, from (15), 2. for

$$E\begin{pmatrix}-\beta, & \alpha\\ -\delta, & \gamma\end{pmatrix}, \omega\Bigg),$$

and with that (14) is proved.

There still remains formula (13). It is enough to take account of the upper signs alone, that is, to prove the formula

$$E\begin{pmatrix}\alpha+\beta, & \beta\\ \gamma+\delta, & \delta\end{pmatrix},\omega\Big) = e^{\frac{\pi i}{12}}E\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix},\omega+1\Big)$$

since the other case reduces to this one by the interchange of α,γ,ω with $\alpha+\beta,\gamma+\delta,\omega+1$. If in the first place β is odd (and positive), then (13) follows from (15), 2. on the basis of the congruence

$$(\beta^2-1)(1-\beta\gamma-\alpha\delta-\beta\delta)\equiv -\beta(\beta^2-1)(2\gamma+\delta)\equiv 0 \pmod{24}.$$

If β is even, then α is odd. If we take α positive, then $\alpha+\beta$ can be positive or negative. If in the first case the upper signs hold, in the second the lower, then (15), 1 gives us

$$e^{\frac{i\pi}{12}}E\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix},\omega+1\Big) = \left(\frac{\beta}{\alpha}\right) i^{-\frac{1-\alpha}{2}} e^{\frac{\pi i}{12}\left[1+\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta\right]}\sqrt{\alpha+\beta+\beta\omega}, \tag{16}$$

$$E\begin{pmatrix}\alpha+\beta, & \beta\\ \gamma+\delta, & \delta\end{pmatrix},\omega\Big) = \left(\frac{\pm\beta}{\pm(\alpha+\beta)}\right) i^{-\frac{1\mp(\alpha+\beta)}{2}} e^{\frac{\pi i}{12}\left\{(\alpha+\beta)(\gamma+\delta-\beta)-[(\alpha+\beta)^2-1]\beta\delta\right\}}\sqrt{\pm(\alpha+\beta+\beta\omega)}. \tag{17}$$

Now, if the lower signs hold, β is negative, and if accordingly

$$-(\alpha+\beta+\beta\omega) = r\,e^{i\varphi}$$

$$i(\alpha+\beta+\beta\omega) = r\,e^{i\left(\varphi-\frac{\pi}{2}\right)}$$

is set, then $\frac{\varphi}{2}$ and $\frac{\varphi}{2}-\frac{\pi}{4}$ lie between $-\frac{\pi}{2}$ and $+\frac{\pi}{2}$, and from this it follows that we have to set:

$$\sqrt{-(\alpha+\beta+\beta\omega)} = i\sqrt{\alpha+\beta+\beta\omega},$$

further, by the law of reciprocity for quadratic residues:

$$\left(\frac{\beta}{\alpha+\beta}\right) = (-1)^{\frac{\beta(\alpha+1)}{4}}\left(\frac{\beta}{\alpha}\right),$$

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = (-1)^{\frac{\alpha-1}{2}}(-1)^{\frac{\beta(\alpha-1)}{4}}\left(\frac{\beta}{\alpha}\right)\ ^{1}),$$

and from this follows the agreement of the two expressions (16), (17), and consequently the correctness of formula (13) in accordance with the congruence

$$\beta(3\alpha-2\gamma+\beta-\delta+\beta^2\delta+2\alpha\beta\delta)\equiv 0 \pmod{24},$$

which, since β is assumed to be even, follows from $\alpha\delta-\beta\gamma=1$.

Thus the formulae (15) are proved to be correct.

---

[1]) By the law of reciprocity we have, if $\beta=\pm 2^\lambda\beta'$ is set and β' is assumed to be odd and positive, when $\alpha+\beta$ is positive

$$\left(\frac{\beta}{\alpha+\beta}\right) = \left(\frac{\pm 2^\lambda}{\alpha+\beta}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}},$$

$$\left(\frac{\beta}{\alpha}\right) = \left(\frac{\pm 2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

If $\lambda \geqq 2$, these two values are equal to one another; if $\lambda = 1$, they differ by the factor $(-1)^{\frac{\alpha+1}{2}}$, in agreement with the first of the formulae above. The correctness of the second formula follows, when $\alpha + \beta$ is negative, from

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = \left(\frac{2^\lambda}{-(\alpha+\beta)}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}},$$

$$\left(\frac{\beta}{\alpha}\right) = \left(\frac{2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{\alpha-1}{2}}(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

If we set

$$E\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}, \omega\Big) = \varepsilon\sqrt{\alpha+\beta\omega}, \tag{18}$$

then ε is a 24th root of unity, whose product with $\sqrt{\alpha+\beta\omega}$ is completely determined by (15), and there results the transformation of the η-function

$$\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varepsilon\sqrt{\alpha+\beta\omega}\;\eta(\omega). \tag{19}$$

For ε^{12} one finds

$$\varepsilon^{12} = (-1)^{\alpha\beta+\gamma\delta+\beta\gamma}. \tag{20}$$

§ 39. Linear Transformation of the ϑ-Functions.

The transformation formulae of the ϑ-functions are consequences of the fundamental properties of the σ-function, namely of remaining, under the substitution

$$(\omega_1', \omega_2') = \begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}(\omega_1, \omega_2)$$

unchanged. From § 37, (4) there follows accordingly, if one replaces $\omega_1, \omega_2, \eta_1$ by $\omega_1', \omega_2', \eta_1'$:

$$\frac{\omega_1\, e^{\frac{\eta_1 u^2}{\omega_1}}\,\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2}{\omega_1}\right)} = \frac{\omega_1'\, e^{\frac{\eta_1' u^2}{\omega_1'}}\,\vartheta_{11}\left(\frac{u}{\omega_1'}, \frac{\omega_2'}{\omega_1'}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2'}{\omega_1'}\right)},$$

or, since by § 35, (10), (13), (14)

$$\frac{\eta_1}{\omega_1} - \frac{\eta_1'}{\omega_1'} = \frac{\omega_1'\eta_1 - \eta_1'\omega_1}{\omega_1\omega_1'} = \beta\,\frac{\omega_2\eta_1 - \omega_1\eta_2}{\omega_1\omega_1'} = \frac{\pi i\beta}{\omega_1\omega_1'} \tag{1}$$

holds, if we set:

$$\begin{aligned} v &= \frac{u}{\omega_1}, & \omega &= \frac{\omega_2}{\omega_1},\\ v' &= \frac{u}{\omega_1'} = \frac{v}{\alpha+\beta\omega}, & \omega' &= \frac{\omega_2'}{\omega_1'} = \frac{\gamma+\delta\omega}{\alpha+\beta\omega} \end{aligned} \tag{2}$$

then:

$$\frac{\vartheta_{11}(v', \omega')}{\vartheta_{11}'(0, \omega')} = \frac{e^{\pi i\beta vv'}}{\alpha+\beta\omega}\;\frac{\vartheta_{11}(v, \omega)}{\vartheta_{11}'(0, \omega)}.$$

Furthermore we have

$$\vartheta'_{11}(0,\omega) = 2\pi\,\eta(\omega)^3, \qquad \vartheta'_{11}(0,\omega') = 2\pi\,\eta(\omega')^3,$$

from which, by § 38, (19):

$$\vartheta'_{11}(0,\omega') = \varepsilon^3\sqrt{\alpha+\beta\omega}^{\,3}\;\vartheta'_{11}(0,\omega),$$

and from this finally:

$$e^{-\pi i\beta vv'}\vartheta_{11}(v',\omega') = \varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{11}(v,\omega). \tag{3}$$

The transformation formulae of the three remaining functions $\vartheta_{00},\vartheta_{01},\vartheta_{10}$ are different in the six classes of § 36, and can be derived from the formulae (5), § 36 in the same way. But the six cases can also be combined into a single system of formulae, which one obtains from (3) by replacing v by

$$v+\frac{\alpha+\beta\omega}{2}, \quad v+\frac{\gamma+\delta\omega}{2}, \quad v+\frac{\alpha+\gamma+(\beta+\delta)\omega}{2},$$

and therefore v' by

$$v'+\frac{1}{2}, \quad v'+\frac{\omega'}{2}, \quad v'=\frac{1+\omega'}{2},$$

and then applying the formulae (2), (3), (8) of § 21 to the right and the left side of (3). Thus one obtains:

$$\begin{aligned} &e^{-\pi i\beta vv'}\vartheta_{10}(v',\omega') \\ &= i^{\beta}e^{-\frac{\pi i\alpha\beta}{4}}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\beta,\,1-\alpha}(v,\omega), \end{aligned} \tag{4}$$

$$\begin{aligned} &e^{-\pi i\beta vv'}\vartheta_{01}(v',\omega') \\ &= i^{\delta-1}e^{-\frac{\pi i\gamma\delta}{4}}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\delta,\,1-\gamma}(v,\omega), \end{aligned} \tag{5}$$

$$\begin{aligned} &e^{-\pi i\beta vv'}\vartheta_{00}(v',\omega') \\ &= i^{\delta+\beta-\alpha\delta}e^{-\frac{\pi i}{4}(\alpha\beta+\gamma\delta)}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\beta+\delta,\,1-\alpha-\gamma}(v,\omega). \end{aligned} \tag{6}$$

The fourth powers of these functions admit of a simpler expression by

$$e^{-4\pi i\beta vv'}\vartheta^4_{g_1,g_2}(v',\omega') = (-1)^{g_2\alpha\beta+g_1\gamma\delta+\beta\gamma}(\alpha+\beta\omega)^2\,\vartheta^4_{g'_1,g'_2}(v), \tag{7}$$

in which, in the six classes, the characteristics (g'_1,g'_2) belonging to the characteristics $(g_1,g_2) = (00),(01),(10)$ are to be taken from the last column of the table in § 36.

A simpler transformation formula is obtained from the σ-functions for the product of the three ϑ-functions (4), (5), (6). For if one forms the product of the three functions § 37, (6), we have:

$$\sigma_{00}(u)\,\sigma_{01}(u)\,\sigma_{10}(u) = e^{\frac{3\eta_1u^2}{\omega_1}}\;\frac{\vartheta_{00}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)\vartheta_{01}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)\vartheta_{10}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}\,\vartheta_{01}\,\vartheta_{10}}$$

a function which, by § 36, remains completely unchanged under linear transformation. If one makes further use of the relation (2):

$$\eta'_1\omega_1-\eta_1\omega'_1 = -\pi i\beta,$$

then it follows that

$$\frac{\vartheta_{00}(v,\omega)\vartheta_{01}(v,\omega)\vartheta_{10}(v,\omega)}{\vartheta_{00}\vartheta_{01}\vartheta_{10}} = e^{-3\pi i\beta vv'}\,\frac{\vartheta_{00}(v',\omega')\vartheta_{01}(v',\omega')\vartheta_{10}(v',\omega')}{\vartheta_{00}(0,\omega')\vartheta_{01}(0,\omega')\vartheta_{10}(0,\omega')}, \tag{8}$$

in which one may still, by (19) of the preceding section,

$$\vartheta_{00}(0,\omega')\vartheta_{01}(0,\omega')\vartheta_{10}(0,\omega') = \varepsilon^3\sqrt{\alpha+\beta\omega}^{\,3}\,\vartheta_{00}\vartheta_{01}\vartheta_{10}$$

set.

But we draw another conclusion from (8): namely, if one puts

$$v' = \frac{h}{n}, \quad v = \frac{h(\alpha+\beta\omega)}{n},$$

and takes the product for $h = 1, 2, \cdots, \dfrac{n-1}{2}$, then there results, by means of the known relation (Bd. I, § 11)

$$\sum_{1,\ \frac{n-1}{2}}^{h} h^2 = n\,\frac{n^2-1}{24},$$

if one applies formula (23), § 32 to the right-hand side of (8):

$$\begin{aligned}(9)\qquad & 2^{\frac{n-1}{2}} \prod_{1,\ \frac{n-1}{2}}^{h} \vartheta_{00}\Big(\frac{h(\alpha+\beta\omega)}{n}\Big)\,\vartheta_{01}\Big(\frac{h(\alpha+\beta\omega)}{n}\Big)\,\vartheta_{10}\Big(\frac{h(\alpha+\beta\omega)}{n}\Big)\\ & = e^{-\frac{\pi i}{8}\frac{n^2-1}{n}\beta(\alpha+\beta\omega)}\ \vartheta_{00}^{\frac{n-1}{2}}\ \vartheta_{01}^{\frac{n-1}{2}}\ \vartheta_{01}^{\frac{n-1}{2}},\end{aligned}$$

in which now α, β may be any pair of relatively prime numbers.

§ 40. Linear Transformation of the Functions $f(\omega), f_1(\omega), f_2(\omega)$.

By § 34, (9), the formulae for the linear transformation of the f-functions are a simple consequence of the transformation of the η-function.

If we set in the linear transformation

$$\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}$$

first that β is even and consequently that α, δ are odd, then

$$(1)\qquad \begin{pmatrix}1, & 0\\ 0, & 2\end{pmatrix}\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix} = \begin{pmatrix}\alpha, & \frac{1}{2}\beta\\ 2\gamma, & \delta\end{pmatrix}\begin{pmatrix}1, & 0\\ 0, & 2\end{pmatrix},$$

that is,

$$2\,\frac{\gamma+\delta\omega}{\alpha+\beta\omega} = \frac{2\gamma+\delta.2\omega}{\alpha+\frac{1}{2}\beta.2\omega}.$$

But by § 34, (9) we have:

$$f_2(\omega) = \sqrt{2}\,\frac{\eta(2\omega)}{\eta(\omega)},$$

and if we therefore carry out the substitution

$$\omega,\ \frac{\gamma+\delta\omega}{\alpha+\beta\omega}$$

then, using the notation of § 38, we obtain

$$(2)\qquad f_2\Big(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\Big) = \frac{E\Big(\begin{smallmatrix}\alpha, & \frac{1}{2}\beta\\ 2\gamma, & \delta\end{smallmatrix}, 2\omega\Big)}{E\Big(\begin{smallmatrix}\alpha, & \beta\\ \gamma, & \delta\end{smallmatrix}, \omega\Big)}\,f_2(\omega).$$

Here the E-functions are to be determined according to § 38, (15), 1., from which it follows:

$$\frac{E\left(\begin{smallmatrix}\alpha, & \frac{1}{2}\beta\\ 2\gamma, & \delta\end{smallmatrix}, 2\omega\right)}{E\left(\begin{smallmatrix}\alpha, & \beta\\ \gamma, & \delta\end{smallmatrix}, \omega\right)} = \left(\frac{2}{\alpha}\right) e^{\frac{\pi i}{12}\left(\alpha\gamma+\frac{\alpha\beta}{2}+(\alpha^2-1)\frac{\beta\delta}{2}\right)}.$$

But $\dfrac{1}{24} = \dfrac{3}{8} - \dfrac{1}{3}$, and

$$\alpha\gamma + \frac{\alpha\beta}{2} + (\alpha^2-1)\frac{\beta\delta}{2} \equiv \frac{\alpha(2\gamma+\beta)}{2} \pmod 8$$

$$\equiv \alpha(\gamma-\beta) - (\alpha^2-1)\beta\delta \pmod 3,$$

and if we therefore put, for brevity,

$$\varrho = e^{-\frac{2\pi i}{3}\left[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta\right]} \tag{3}$$

then from (2) we obtain:

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\alpha}\right)\varrho\, e^{\frac{3\pi i}{8}\alpha(2\gamma+\beta)}\, f_2(\omega), \qquad \beta \equiv 0 \pmod 2. \tag{4}$$

In the same way all the other formulae of this kind may be derived; but they are obtained more simply from (4) itself by using the fundamental transformations § 34, (13), (14), (15). Thus, if in (4) one replaces ω by $-1:\omega$ and then interchanges $\alpha, \beta, \gamma, \delta$ with $\beta, -\alpha, \delta, -\gamma$ (whereby ϱ remains unchanged), it follows that:

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\beta}\right)\varrho\, e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)}\, f_1(\omega), \qquad \alpha \equiv 0 \pmod 2, \tag{5}$$

and if in this we replace ω by $\omega+3$ and γ, α by $\gamma-3\delta, \alpha-3\beta$, it follows that

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\varrho\, e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)}\, f(\omega), \qquad \alpha-\beta \equiv 0 \pmod 2. \tag{6}$$

If in (4), (5), (6) we set

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = f_1\left(-\frac{\alpha+\beta\omega}{\gamma+\delta\omega}\right),$$

and then interchange $\alpha, \beta, \gamma, \delta$ with $-\gamma, -\delta, \alpha, \beta$, there results

$$f_1\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\gamma}\right)\varrho\, e^{-\frac{3\pi i}{8}\gamma(2\alpha-\delta)}\, f_2(\omega), \qquad \delta \equiv 0 \pmod 2, \tag{7}$$

$$= \left(\frac{2}{\delta}\right)\varrho\, e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)}\, f_1(\omega), \qquad \gamma \equiv 0 \pmod 2, \tag{8}$$

$$= -\varrho\, e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)}\, f(\omega), \qquad \gamma-\delta \equiv 0 \pmod 2. \tag{9}$$

From (9) and (6) one obtains, on replacing ω by $\dfrac{-\gamma+\alpha\omega}{\delta-\beta\omega}$ and then $\alpha, \beta, \gamma, \delta$ by $\delta, -\beta, -\gamma, \alpha$:

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\varrho\, e^{-\frac{3\pi i}{8}\alpha(2\beta+\gamma)}\, f_1(\omega), \qquad \alpha-\gamma \equiv 0 \pmod 2, \tag{10}$$

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\varrho\, e^{\frac{3\pi i}{8}\beta(2\alpha-\delta)}\, f_2(\omega), \qquad \beta-\delta \equiv 0 \pmod 2, \tag{11}$$

and if finally in (10) one replaces ω by $\omega + 9$ and γ, α by $\gamma - 9\delta, \alpha - 9\beta$:

$$f\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \varrho\left(\frac{2}{\alpha - \beta}\right) e^{-\frac{3\pi i}{8}(\alpha-\beta)(\alpha+\beta+\gamma-\delta)}\, f(\omega), \tag{12}$$

$$\alpha + \beta + \gamma - \delta \equiv 0 \pmod 2.$$

[7]

Fourth Section.

The Elliptic Functions.

---

§ 41. Connection of the ϑ-functions with the elliptic integrals.

According to § 21 there exist between the squares of the four ϑ-functions two linear equations independent of one another, and one can therefore express two of these squares by the other two, or indeed all four by two independent variables ξ, η. In doing the latter, we denote by ξ_1, η_1; ξ_2, η_2; ξ_3, η_3; ξ_4, η_4 constants, and, connecting with the mode of notation of Bd. I, § 67, we set:

$$\begin{aligned} \vartheta_{01}^2(u) &= \xi\eta_1 - \eta\xi_1 = (\xi\eta_1), \\ \vartheta_{11}^2(u) &= \xi\eta_2 - \eta\xi_2 = (\xi\eta_2), \\ \vartheta_{10}^2(u) &= \xi\eta_3 - \eta\xi_3 = (\xi\eta_3), \\ \vartheta_{00}^2(u) &= \xi\eta_4 - \eta\xi_4 = (\xi\eta_4). \end{aligned} \tag{1}$$

Between the constants ξ_i, η_i and the values ϑ_{01}, ϑ_{10}, ϑ_{00} there exist four relations, which can be derived from the equations (13) of § 21. These equations we can write, following the notation (1), in the form:

$$\begin{aligned} (\xi\eta_3)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{10}^2 - (\xi\eta_2)\vartheta_{00}^2, \\ (\xi\eta_4)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{00}^2 - (\xi\eta_2)\vartheta_{10}^2. \end{aligned} \tag{2}$$

By setting in (2) ξ, $\eta = \xi_1$, η_1 and $= \xi_2$, η_2, one can give these relations the form:

$$\begin{aligned} (\xi_1\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2, \\ (\xi_2\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2, \\ (\xi_1\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2, \\ (\xi_2\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2, \end{aligned} \tag{3}$$

[7] The functions $\varphi(\omega), \psi(\omega), \chi(\omega)$ introduced by Hermite (Sur la théorie des équations modulaires, Paris 1859) are connected with the functions $f(\omega), f_1(\omega), f_2(\omega)$ by the equations

$$f(\omega) = \frac{\sqrt[6]{2}}{\chi(\omega)}, \quad f_1(\omega) = \sqrt[6]{2}\,\frac{\psi(\omega)}{\chi(\omega)}, \quad f_2(\omega) = \sqrt[6]{2}\,\frac{\varphi(\omega)}{\chi(\omega)}$$

shown above. The transformation formulae of the f-functions can be recast in manifold ways by using the relation $\alpha\delta - \beta\gamma = 1$. Thus the formulae given by Hermite loc. cit. are not at once to be recognised as identical with ours. A simple calculation, however, shows their agreement. The formulae given above have the advantage that, without any loss of simplicity, they combine two of the six transformation classes at a time into a single expression.

to which one can further add, by setting in (2) ξ, $\eta = \xi_4$, η_4 and using (3) and § 21, (14):

$$(\xi_4\eta_3) = (\xi_2\eta_1). \tag{4}$$

From (3) and (4) there follows for the cross ratios:

$$\frac{(\xi_2\eta_3)(\xi_1\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{5}$$

$$\frac{(\xi_1\eta_2)(\xi_3\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{01}^4}{\vartheta_{00}^4}. \tag{6}$$

If we now differentiate two of the equations (1), say the first two, there follows:

$$2\vartheta_{01}(u)\vartheta'_{01}(u)\,du = \eta_1\,d\xi - \xi_1\,d\eta = (\eta_1\,d\xi),$$
$$2\vartheta_{11}(u)\vartheta'_{11}(u)\,du = \eta_2\,d\xi - \xi_2\,d\eta = (\eta_2\,d\xi),$$

and from this, with the use of (1),

$$\begin{aligned} 2\vartheta_{01}(u)\vartheta_{11}(u)\big[\vartheta'_{01}(u)\vartheta_{11}(u) - \vartheta'_{11}(u)\vartheta_{01}(u)\big]\,du \\ = (\eta_1\,d\xi)\vartheta_{11}^2(u) - (\eta_2\,d\xi)\vartheta_{01}^2(u) \\ = (\eta_1\,d\xi)(\xi\eta_2) - (\eta_2\,d\xi)(\xi\eta_1) = (\xi_2\eta_1)(\xi\,d\eta). \end{aligned}$$

The last expression can be derived without calculation by regarding the next to last one as a linear function of $d\xi$, $d\eta$ which vanishes for $d\xi : d\eta = \xi : \eta$ and is therefore divisible by $(\xi\,d\eta)$. The quotient $(\xi_2\eta_1)$ is obtained by setting $d\xi : d\eta = \xi_2 : \eta_2$. With the use of § 23, (6) one then obtains

$$2\pi\vartheta_{01}^2\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{10}(u)\vartheta_{01}(u)\,du = (\xi_1\eta_2)(\xi\,d\eta). \tag{7}$$

If one introduces here, according to (3) (first and last formula),

$$(\xi_1\eta_2)\vartheta_{00}^2 = \sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\;\vartheta_{01}^2 \tag{8}$$

and substitutes for the ϑ-functions the expressions (1), there finally follows

$$2\pi\vartheta_{00}^2\,du = \frac{\sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\,(\xi\,d\eta)}{\sqrt{(\xi\eta_1)(\xi\eta_2)(\xi\eta_3)(\xi\eta_4)}}, \tag{9}$$

by which du is represented as an elliptic differential of the first kind in homogeneous variables [§ 1, (5)].

By (1) the ratio $\xi : \eta$ is determined as a doubly periodic function of u. But likewise the ratios of the square roots

$$\sqrt{(\xi\eta_1)}, \qquad \sqrt{(\xi\eta_2)}, \qquad \sqrt{(\xi\eta_3)}, \qquad \sqrt{(\xi\eta_4)} \tag{10}$$

are also defined as single-valued doubly periodic functions, and the sign of the square roots in (9) is thereby, and by (8), likewise uniquely determined.

If ω is the modulus of the ϑ-functions, then, as follows by (1) from the vanishing of the four ϑ-functions, the following values belong together:

$$\begin{aligned} u &= \tfrac{1}{2}\omega, & \xi : \eta &= \xi_1 : \eta_1, \\ u &= 0, & \xi : \eta &= \xi_2 : \eta_2, \\ u &= \tfrac{1}{2}, & \xi : \eta &= \xi_3 : \eta_3, \\ u &= \frac{1+\omega}{2}, & \xi : \eta &= \xi_4 : \eta_4. \end{aligned} \tag{11}$$

§ 42. Jacobi's elliptic functions.

Since the variables ξ, η can be replaced by two new variables by means of a linear substitution in which four coefficients are at our disposal, one can assign arbitrary values to four of the quantities ξ_i, η_i, or to three of their ratios, without impairing the generality.

We shall set

$$\xi_1 = 0, \qquad \eta_2 = 0, \qquad \xi_3 = \eta_3, \tag{1}$$

and we further introduce $\varkappa^2$, $\varkappa'^2$, ζ by the equations:

$$\xi_4 = \varkappa^2 \eta_4, \qquad \varkappa'^2 = 1 - \varkappa^2, \qquad \frac{\eta}{\xi} = \zeta. \tag{2}$$

From § 41, (3), (5), (6) there then follows

$$\begin{aligned} &\frac{\xi_2}{\eta_1} = -\frac{\vartheta_{10}^2}{\vartheta_{00}^2}, \qquad \frac{\eta_3}{\eta_1} = \frac{\vartheta_{10}^2}{\vartheta_{01}^2}, \quad \frac{\eta_4}{\eta_1} = \frac{\vartheta_{00}^2}{\vartheta_{01}^2}, \\ &\sqrt{\varkappa} = \frac{\vartheta_{10}}{\vartheta_{00}}, \qquad \sqrt{\varkappa'} = \frac{\vartheta_{01}}{\vartheta_{00}}, \end{aligned} \tag{3}$$

and from (1), (9), § 41 one finds:

$$\begin{aligned} \frac{\vartheta_{00}}{\vartheta_{10}} \frac{\vartheta_{11}(u)}{\vartheta_{01}(u)} &= \sqrt{\zeta}, \\ \frac{\vartheta_{01}}{\vartheta_{10}} \frac{\vartheta_{10}(u)}{\vartheta_{01}(u)} &= \sqrt{1-\zeta}, \\ \frac{\vartheta_{01}}{\vartheta_{00}} \frac{\vartheta_{00}(u)}{\vartheta_{01}(u)} &= \sqrt{1-\varkappa^2\zeta}. \end{aligned} \tag{4}$$

$$2\pi\vartheta_{00}^2 u = \int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{5}$$

in which the path of integration and the meaning of the root symbols are determined by the formulae (4) themselves, when the transition

from 0 to u in the u-plane is given. But the path of integration in (5) may then also be altered, provided only that in doing so none of the singular points ∞, 1, $\frac{1}{\varkappa^2}$ is crossed.

Equation (5) can be written in the following three forms:

$$\begin{aligned} d\sqrt{\zeta} &= \pi\vartheta_{00}^2 \sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\, du, \\ d\sqrt{1-\zeta} &= -\pi\vartheta_{00}^2 \sqrt{\zeta(1-\varkappa^2\zeta)}\, du, \\ d\sqrt{1-\varkappa^2\zeta} &= -\pi\vartheta_{00}^2 \varkappa^2 \sqrt{\zeta(1-\zeta)}\, du. \end{aligned} \tag{6}$$

If we introduce a new variable v by the equation

$$\pi\vartheta_{00}^2 u = v, \tag{7}$$

then equations (5), (6) become:

$$v = \frac{1}{2} \int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{8}$$

$$
\begin{aligned}
d\sqrt{\zeta} &= \sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\,dv,\\
d\sqrt{1-\zeta} &= -\sqrt{\zeta(1-\varkappa^2\zeta)}\,dv,\\
d\sqrt{1-\varkappa^2\zeta} &= -\varkappa^2\sqrt{\zeta(1-\zeta)}\,dv,
\end{aligned}
\tag{9}
$$

and now we regard the three quantities $\sqrt{\zeta} = x$, $\sqrt{1-\zeta} = y$, $\sqrt{1-\varkappa^2\zeta} = z$, defined by (4) and (7), as functions of the variable v. By (4) they are one-valued, doubly periodic and, apart from isolated points at which they become infinite, continuous functions of v. After Jacobi they are called sinus amplitudinis, cosinus amplitudinis, Δ amplitudinis of v, and are denoted by $\operatorname{sinam} v$, $\operatorname{cosam} v$, $\Delta\operatorname{am} v$. Here we wish to make use of the shorter Gudermann notation $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$, already mentioned in § 14, according to which

$$
\begin{aligned}
\frac{\vartheta_{11}(u)}{\vartheta_{01}(u)} &= \sqrt{\varkappa}\operatorname{sn} v,\\
\frac{\vartheta_{10}(u)}{\vartheta_{01}(u)} &= \sqrt{\frac{\varkappa}{\varkappa'}}\operatorname{cn} v,\\
\frac{\vartheta_{00}(u)}{\vartheta_{01}(u)} &= \frac{1}{\sqrt{\varkappa'}}\operatorname{dn} v.
\end{aligned}
\tag{10}
$$

These functions satisfy, by (9), the differential equations

$$
\begin{aligned}
\frac{dx}{dv} &= yz,\\
\frac{dy}{dv} &= -zx,\\
\frac{dz}{dv} &= -\varkappa^2 xy,
\end{aligned}
\tag{11}
$$

and the subsidiary conditions:

$$
\operatorname{sn} 0 = 0, \qquad \operatorname{cn} 0 = 1, \qquad \operatorname{dn} 0 = 1. \tag{12}
$$

Between them there subsist the relations

$$
y^2 = 1 - x^2, \qquad z^2 = 1 - \varkappa^2 x^2. \tag{13}
$$

From the fundamental properties of the ϑ-functions there follow the first properties of the elliptic functions:

$$
\operatorname{sn} v = -\operatorname{sn}(-v), \qquad \operatorname{cn} v = \operatorname{cn}(-v), \qquad \operatorname{dn} v = \operatorname{dn}(-v),
$$

that is, $\operatorname{sn} v$ is an odd function, while $\operatorname{cn} v$, $\operatorname{dn} v$ are even functions.

If we put further

$$
\pi\vartheta_{00}^2 = 2K, \qquad \pi\vartheta_{00}^2\omega = 2iK', \tag{14}
$$

and consequently

$$
\pi\vartheta_{01}^2 = 2\varkappa' K, \qquad \pi\vartheta_{10}^2 = 2\varkappa K, \qquad \omega = \frac{iK'}{K}, \tag{15}
$$

then v takes the values K, iK', $K + iK'$ when $u = \frac{1}{2}, \frac{\omega}{2}, \frac{1+\omega}{2}$, and it follows from (4), having regard to (3) and § 21, (10):

$$
\begin{aligned}
&\operatorname{sn} K = 1, \quad \operatorname{sn}(K + iK') = \frac{1}{\varkappa},\\
&\operatorname{cn} K = 0, \quad \operatorname{cn}(K + iK') = -\frac{i\varkappa'}{\varkappa},\\
&\operatorname{dn} K = \varkappa', \quad \operatorname{dn}(K + iK') = 0,\\
&\operatorname{sn} iK', \quad \operatorname{cn} iK', \quad \operatorname{dn} iK' = \infty.
\end{aligned}
\tag{16}
$$

Now K, K' can be expressed by means of (8) in terms of definite integrals, and one obtains, if one displaces v from one vertex of the parallelogram $0, K, K+iK', iK'$ to the next along the periphery, and hence u along

$$0, \qquad \frac{1}{2}, \qquad \frac{1+\omega}{2}, \qquad \frac{\omega}{2}.$$

$$K = \frac{1}{2}\int_0^1 \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}, \tag{17}$$

$$iK' = \frac{1}{2}\int_1^{\frac{1}{\varkappa^2}} \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}, \tag{18}$$

$$-K = \frac{1}{2}\int_{\frac{1}{\varkappa^2}}^{\infty} \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}, \tag{19}$$

$$-iK' = \frac{1}{2}\int_{\infty}^{0} \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}. \tag{20}$$

The values of $\sqrt{\xi}$, $\sqrt{1-\xi}$, $\sqrt{1-\varkappa^2\xi}$ are determined in these integrations by the formulae (4). If, however, ω is purely imaginary, and consequently $\varkappa^2$ a positive proper fraction and K, K' are real and positive, then the paths for v run parallel to the real and the imaginary axis, and having regard to the remarks at the close of § 25 the formulae (4) show the following:

In (17) are	$\sqrt{\xi}$,	$\sqrt{1-\xi}$,	$\sqrt{1-\varkappa^2\xi}$	real and positive,
„ (18) „	$\sqrt{\xi}$,	$i\sqrt{1-\xi}$,	$\sqrt{1-\varkappa^2\xi}$	„ „ „
„ (19) „	$\sqrt{\xi}$,	$i\sqrt{1-\xi}$,	$i\sqrt{1-\varkappa^2\xi}$	„ „ „
„ (20) „	$-i\sqrt{\xi}$,	$\sqrt{1-\xi}$,	$\sqrt{1-\varkappa^2\xi}$	„ „ „

and $\sqrt{\xi}$ has on none of the paths of integration a maximum or minimum. Hence K, K' can be defined by the four following real integrals with positive square root:

$$K = \frac{1}{2}\int_0^1 \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}} = \frac{1}{2}\int_{\frac{1}{\varkappa^2}}^{\infty} \frac{d\xi}{\sqrt{\xi(\xi-1)(\varkappa^2\xi-1)}},$$

$$K' = \frac{1}{2}\int_{-\infty}^0 \frac{d\xi}{\sqrt{-\xi(1-\xi)(1-\varkappa^2\xi)}} = \frac{1}{2}\int_1^{\frac{1}{\varkappa^2}} \frac{d\xi}{\sqrt{\xi(\xi-1)(1-\varkappa^2\xi)}}.$$

§ 43. The Jacobian Functions $\Theta(v)$, $H(v)$.

If one regards the ϑ-functions of the variable u as functions of v, one obtains the Jacobian Θ-functions. We put

$$\begin{aligned} \vartheta_{01}\left(\frac{v}{2K}, \omega\right) &= \Theta(v), \\ \vartheta_{11}\left(\frac{v}{2K}, \omega\right) &= H(v). \end{aligned} \tag{1}$$

Then $\Theta(v)$, $H(v)$ are to be designated as functions $t(v, 2K, 2K')$, and according to § 25 we have

$$\begin{aligned} \Theta(v) &= 1 - 2q\cos\frac{\pi v}{K} + 2q^4\cos\frac{2\pi v}{K} - 2q^9\cos\frac{3\pi v}{K} + \cdots \\ H(v) &= 2q^{\frac{1}{4}}\sin\frac{\pi v}{2K} - 2q^{\frac{9}{4}}\sin\frac{3\pi v}{2K} + 2q^{\frac{25}{4}}\sin\frac{5\pi v}{2K} - \cdots \\ q &= e^{-\frac{\pi K'}{K}}. \end{aligned} \tag{2}$$

According to § 21 (8) we have

$$
\begin{aligned}
H(v) &= -ie^{-\frac{\pi K'}{4K}+\frac{\pi i v}{2K}}\Theta(v+iK'),\\
\Theta(v) &= -ie^{-\frac{\pi K'}{4K}+\frac{\pi i v}{2K}}H(v+iK').
\end{aligned}
\tag{3}
$$

Further:

$$
\begin{aligned}
\vartheta_{00}\left(\frac{v}{2K},\omega\right) &= \Theta(v+K),\\
\vartheta_{10}\left(\frac{v}{2K},\omega\right) &= H(v+K),
\end{aligned}
\tag{4}
$$

whereupon, by means of formulae (3), (10), (15) of the preceding section and (5), § 23:

$$
\sqrt{\varkappa}=\frac{H(K)}{\Theta(K)},\qquad \sqrt{\varkappa'}=\frac{\Theta(0)}{\Theta(K)},
$$

$$
\begin{aligned}
&\Theta(K)=\sqrt{\frac{2K}{\pi}},\quad \Theta(0)=\sqrt{\frac{2\varkappa' K}{\pi}},\quad H(K)=\sqrt{\frac{2\varkappa K}{\pi}},\\
&H'(0)=\frac{\Theta(0)H(K)}{\Theta(K)}=\sqrt{\frac{2\varkappa\varkappa' K}{\pi}};
\end{aligned}
\tag{5}
$$

finally the representation of the elliptic functions:

$$
\begin{aligned}
\frac{H(v)}{\Theta(v)} &= \sqrt{\varkappa}\,\mathrm{sn}\,v,\\
\frac{H(v+K)}{\Theta(v)} &= \sqrt{\frac{\varkappa}{\varkappa'}}\,\mathrm{cn}\,v,\\
\frac{\Theta(v+K)}{\Theta(v)} &= \frac{1}{\sqrt{\varkappa'}}\,\mathrm{dn}\,v.
\end{aligned}
\tag{6}
$$

The functions Θ, H have, as follows easily from (3), the periodicity expressed by the following equations:

$$
\begin{aligned}
\Theta(v+2nK) &= \Theta(v),\\
\Theta(v+2niK') &= (-1)^n e^{\frac{n^2\pi K'}{K}-\frac{\pi i n v}{K}}\Theta(v),
\end{aligned}
\tag{7}
$$

$$
\begin{aligned}
H(v+2nK) &= (-1)^n H(v),\\
H(v+2niK') &= (-1)^n e^{\frac{n^2\pi K'}{K}-\frac{\pi i n v}{K}}H(v).
\end{aligned}
\tag{8}
$$

It is sometimes useful to combine the equations (3), (7), (8) in the following way:

$$
\Phi(v+niK')=i^n e^{\frac{n^2\pi i}{4}\frac{K'}{K}-\frac{\pi i n v}{2K}}\Psi(v),
\tag{9}
$$

in which, if n is even, Φ and Ψ are both to be set equal to Θ or both equal to H; if n is odd, $\Phi=\Theta$, $\Psi=H$ or $\Phi=H$, $\Psi=\Theta$.

From § 22 (2), (4) there result the addition formulae:

$$
\begin{aligned}
\Theta^2(0)\Theta(u+v)\Theta(u-v) &= \Theta^2(u)\Theta^2(v)-H^2(u)H^2(v),\\
\Theta^2(0)H(u+v)H(u-v) &= H^2(u)\Theta^2(v)-\Theta^2(u)H^2(v),
\end{aligned}
\tag{10}
$$

for which one may also write:

$$(11)\qquad \begin{aligned} \Theta^2(0)\Theta(u+v)\Theta(u-v) &= \Theta^2(u)\Theta^2(v)(1-\varkappa^2\,\mathrm{sn}^2\,u\,\mathrm{sn}^2\,v),\\ \Theta^2(0)H(u+v)H(u-v) &= \Theta^2(u)\Theta^2(v)\varkappa(\mathrm{sn}^2\,u-\mathrm{sn}^2\,v). \end{aligned}$$

We now proceed to carry over the theorems on the ϑ-functions to the elliptic functions, and we begin with the addition theorem.

§ 44. Addition theorem of the elliptic functions.

From the four ϑ-functions twelve quotients in all can be formed, which by § 42 are expressible by elliptic functions. If one forms these quotients for the argument $u+v$, then by § 22 one can represent them by ϑ-functions of u and of v separately, and hence also by the corresponding elliptic functions, and indeed always in such a way that the denominator

contains only squares of ϑ, and is therefore expressed rationally by sn^2u, sn^2v. To carry an example through, let us take formula (5), § 22, and divide it once by (1) and then by (4) (the first time converting v into $-v$ in (5)); it then follows that:

$$\begin{aligned} \frac{\vartheta_{01}\vartheta_{10}}{\vartheta_{00}^2}\,\frac{\vartheta_{11}(u+v)}{\vartheta_{00}(u+v)} &= \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)+\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{00}^2(u)\vartheta_{00}^2(v)+\vartheta_{11}^2(u)\vartheta_{11}^2(v)},\\ \frac{\vartheta_{10}}{\vartheta_{01}}\,\frac{\vartheta_{00}(u+v)}{\vartheta_{11}(u+v)} &= \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)-\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{11}^2(u)\vartheta_{01}^2(v)-\vartheta_{01}^2(u)\vartheta_{11}^2(v)}, \end{aligned}$$

and if u, v are replaced by $u:2K$, $v:2K$, then according to § 42 (10) everything can be expressed by elliptic functions. One obtains in this way

$$(1)\qquad \frac{\mathrm{sn}(u+v)}{\mathrm{dn}(u+v)} = \frac{\mathrm{dn}\,u\,\mathrm{sn}\,u\,\mathrm{cn}\,v+\mathrm{dn}\,v\,\mathrm{sn}\,v\,\mathrm{cn}\,u}{\mathrm{dn}^2u\,\mathrm{dn}^2v+\varkappa^2\varkappa'^2\mathrm{sn}^2u\,\mathrm{sn}^2v},$$

$$(2)\qquad \frac{\mathrm{dn}(u+v)}{\mathrm{sn}(u+v)} = \frac{\mathrm{dn}\,u\,\mathrm{sn}\,u\,\mathrm{cn}\,v-\mathrm{dn}\,v\,\mathrm{sn}\,v\,\mathrm{cn}\,u}{\mathrm{sn}^2u-\mathrm{sn}^2v}.$$

In the same way one obtains from formulae (6), (3), (4); (7), (2), (4); (8), (2), (3); (9), (1), (2); (10), (1), (3) of § 22:

$$(3)\qquad \frac{\mathrm{sn}(u+v)}{\mathrm{cn}(u+v)} = \frac{\mathrm{cn}\,u\,\mathrm{sn}\,u\,\mathrm{dn}\,v+\mathrm{cn}\,v\,\mathrm{sn}\,v\,\mathrm{dn}\,u}{\mathrm{cn}^2u\,\mathrm{cn}^2v-\varkappa'^2\mathrm{sn}^2u\,\mathrm{sn}^2v},$$

$$(4)\qquad \frac{\mathrm{cn}(u+v)}{\mathrm{sn}(u+v)} = \frac{\mathrm{cn}\,u\,\mathrm{sn}\,u\,\mathrm{dn}\,v-\mathrm{cn}\,v\,\mathrm{sn}\,v\,\mathrm{dn}\,u}{\mathrm{sn}^2u-\mathrm{sn}^2v},$$

$$(5)\qquad \mathrm{sn}(u+v) = \frac{\mathrm{sn}\,u\,\mathrm{cn}\,v\,\mathrm{dn}\,v+\mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{dn}\,u}{1-\varkappa^2\mathrm{sn}^2u\,\mathrm{sn}^2v},$$

$$(6)\qquad \frac{1}{\mathrm{sn}(u+v)} = \frac{\mathrm{sn}\,u\,\mathrm{cn}\,v\,\mathrm{dn}\,v-\mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{dn}\,u}{\mathrm{sn}^2u-\mathrm{sn}^2v},$$

$$(7)\qquad \mathrm{cn}(u+v) = \frac{\mathrm{cn}\,u\,\mathrm{cn}\,v-\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{dn}\,u\,\mathrm{dn}\,v}{1-\varkappa^2\mathrm{sn}^2u\,\mathrm{sn}^2v},$$

$$(8)\qquad \frac{1}{\mathrm{cn}(u+v)} = \frac{\mathrm{cn}\,u\,\mathrm{cn}\,v+\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{dn}\,u\,\mathrm{dn}\,v}{\mathrm{cn}^2u\,\mathrm{cn}^2v-\varkappa'^2\mathrm{sn}^2u\,\mathrm{sn}^2v},$$

$$\mathrm{dn}(u+v) = \frac{\mathrm{dn}\,u\,\mathrm{dn}\,v - \varkappa^2 \mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{cn}\,v}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \tag{9}$$

$$\frac{1}{\mathrm{dn}(u+v)} = \frac{\mathrm{dn}\,u\,\mathrm{dn}\,v + \varkappa^2 \mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{cn}\,v}{\mathrm{dn}^2 u\,\mathrm{dn}^2 v + \varkappa^2 \varkappa'^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \tag{10}$$

$$\frac{\mathrm{dn}(u+v)}{\mathrm{cn}(u+v)} = \frac{\mathrm{dn}\,u\,\mathrm{dn}\,v\,\mathrm{cn}\,u\,\mathrm{cn}\,v + \varkappa'^2 \mathrm{sn}\,u\,\mathrm{sn}\,v}{\mathrm{cn}^2 u\,\mathrm{cn}^2 v - \varkappa'^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \tag{11}$$

$$\frac{\mathrm{cn}(u+v)}{\mathrm{dn}(u+v)} = \frac{\mathrm{dn}\,u\,\mathrm{dn}\,v\,\mathrm{cn}\,u\,\mathrm{cn}\,v - \varkappa'^2 \mathrm{sn}\,u\,\mathrm{sn}\,v}{\mathrm{dn}^2 u\,\mathrm{dn}^2 v + \varkappa^2 \varkappa'^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}. \tag{12}$$

Many other formulae besides can be derived in the same way, among which we adduce the following three, which result from the division of § 22 (4), (3), (1) by (2).

$$\mathrm{sn}(u+v)\,\mathrm{sn}(u-v) = \frac{\mathrm{sn}^2 u - \mathrm{sn}^2 v}{1 - \varkappa^2 \mathrm{ns}^2 u\,\mathrm{sn}^2 v}, \tag{13}$$

$$\mathrm{cn}(u+v)\,\mathrm{cn}(u-v) = \frac{\mathrm{cn}^2 u\,\mathrm{cn}^2 v - \varkappa'^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \tag{14}$$

$$\mathrm{dn}(u+v)\,\mathrm{dn}(u-v) = \frac{\mathrm{dn}^2 u\,\mathrm{dn}^2 v + \varkappa^2 \varkappa'^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}. \tag{15}$$

The most important among these formulae are (5), (7), (9), from which all the rest can be derived, even if by lengthy calculations. For the sake of a clear view we set them down here once more:

$$\begin{aligned}
\mathrm{sn}(u+v) &= \frac{\mathrm{sn}\,u\,\mathrm{cn}\,v\,\mathrm{dn}\,v + \mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{dn}\,u}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \\
\mathrm{cn}(u+v) &= \frac{\mathrm{cn}\,u\,\mathrm{cn}\,v - \mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{dn}\,u\,\mathrm{dn}\,v}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}, \\
\mathrm{dn}(u+v) &= \frac{\mathrm{dn}\,u\,\mathrm{dn}\,v - \varkappa^2 \mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{cn}\,u\,\mathrm{cn}\,v}{1 - \varkappa^2 \mathrm{sn}^2 u\,\mathrm{sn}^2 v}.
\end{aligned} \tag{16}$$

By combining these equations two at a time, one can give them, among others, the forms:

$$\begin{aligned}
\mathrm{cn}(u+v)\,\mathrm{dn}\,u\,\mathrm{dn}\,v - \mathrm{dn}(u+v)\,\mathrm{cn}\,u\,\mathrm{cn}\,v &= -\varkappa'^2 \mathrm{sn}\,u\,\mathrm{sn}\,v \\
\mathrm{dn}(u+v)\,\mathrm{sn}\,u\,\mathrm{cn}\,v - \mathrm{sn}(u+v)\,\mathrm{dn}\,u &= -\mathrm{cn}\,u\,\mathrm{sn}\,v, \\
\mathrm{sn}(u+v)\,\mathrm{cn}\,u - \mathrm{cn}(u+v)\,\mathrm{sn}\,u\,\mathrm{dn}\,v &= \mathrm{dn}\,u\,\mathrm{sn}\,v, \\
\mathrm{cn}(u+v)\,\mathrm{cn}\,u + \mathrm{sn}(u+v)\,\mathrm{sn}\,u\,\mathrm{dn}\,v &= \mathrm{cn}\,v.
\end{aligned} \tag{17}$$

We apply the formulae (16) first of all to establish the periodicity of the elliptic functions, which can of course also be derived from § 21. If in (16) one puts $v = \pm K$, $v = K + iK'$, there follows from § 42 (16):

$$\begin{aligned}
\mathrm{sn}(u \pm K) &= \pm \frac{\mathrm{cn}\,u}{\mathrm{dn}\,u}, \\
\mathrm{cn}(u \pm K) &= \mp \frac{\varkappa'\,\mathrm{sn}\,u}{\mathrm{dn}\,u}, \\
\mathrm{dn}(u \pm K) &= \frac{\varkappa'}{\mathrm{dn}\,u},
\end{aligned} \tag{18}$$

$$
\begin{aligned}
\operatorname{sn}(u+K+iK') &= \frac{1}{\varkappa}\frac{\operatorname{dn} u}{\operatorname{cn} u},\\
\operatorname{cn}(u+K+iK') &= \frac{i\varkappa'}{\varkappa}\frac{1}{\operatorname{cn} u},\\
\operatorname{dn}(u+K+iK') &= \frac{i\varkappa'\operatorname{sn} u}{\operatorname{cn} u},
\end{aligned}
\tag{19}
$$

and if one changes u into $u-K$ in (19):

$$
\begin{aligned}
\operatorname{sn}(u+iK') &= \frac{1}{\varkappa}\frac{1}{\operatorname{sn} u},\\
\operatorname{cn}(u+iK') &= -\frac{i}{\varkappa}\frac{\operatorname{dn} u}{\operatorname{sn} u},\\
\operatorname{dn}(u+iK') &= -i\frac{\operatorname{cn} u}{\operatorname{sn} u}.
\end{aligned}
\tag{20}
$$

The formulae (20) show that the three functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$ become infinite for $u = iK'$. If one forms the quotients of each two of them, there follows:

$$
\begin{aligned}
\lim\frac{\operatorname{sn} u}{\operatorname{cn} u} &= i\\
\lim\frac{\operatorname{sn} u}{\operatorname{dn} u} &= i
\end{aligned}
\qquad \text{for } u = iK'.
\tag{21}
$$

By means of the relations (18), (19), (20) one can derive three others from every addition formula, by replacing u by $u+K$, $u+K+iK'$, $u+iK'$, that is, by making the permutations that are collected in the following table:

$\operatorname{sn} u,$	$\operatorname{cn} u,$	$\operatorname{dn} u,$
$\dfrac{\operatorname{cn} u}{\operatorname{dn} u},$	$\dfrac{-\varkappa'\operatorname{sn} u}{\operatorname{dn} u},$	$\dfrac{\varkappa'}{\operatorname{dn} u},$
$\dfrac{1}{\varkappa}\dfrac{\operatorname{dn} u}{\operatorname{cn} u},$	$\dfrac{i\varkappa'}{\varkappa\operatorname{cn} u},$	$\dfrac{i\varkappa'\operatorname{sn} u}{\operatorname{cn} u},$
$\dfrac{1}{\varkappa\operatorname{sn} u},$	$\dfrac{-i\operatorname{dn} u}{\varkappa\operatorname{sn} u},$	$\dfrac{-i\operatorname{cn} u}{\operatorname{sn} u},$

and applies the same permutations to $\operatorname{sn}(u\pm v)$, $\operatorname{cn}(u\pm v)$, $\operatorname{dn}(u\pm v)$ as well. In this way, e. g., the twelve formulae (1) to (12) follow from the three formulae (16).

If these permutations are applied to the relations (18), (19), (20) themselves, there results:

$$
\begin{aligned}
\operatorname{sn}(u+2K) &= -\operatorname{sn} u, \quad & \operatorname{sn}(u+2iK') &= \operatorname{sn} u,\\
\operatorname{cn}(u+2K) &= -\operatorname{cn} u, \quad & \operatorname{cn}(u+2iK') &= -\operatorname{cn} u,\\
\operatorname{dn}(u+2K) &= \operatorname{dn} u, \quad & \operatorname{dn}(u+2iK') &= -\operatorname{dn} u.
\end{aligned}
\tag{22}
$$

The common periods of these three functions are therefore $4K$, $4iK'$; besides these, however, $\operatorname{sn} u$ has the period $2iK'$, $\operatorname{dn} u$ the period $2K$, $\operatorname{cn} u$ the period $2K+2iK'$.

We give the theorem on Θ-functions (§ 21) only another expression, by stating the following theorem:

Every doubly periodic function of v with the periods $2K$, $2iK'$ (which everywhere has the character of a rational function) is, if it is an even function, a rational function of $\operatorname{sn}^2 v$, and if it is an odd function, the product of $\operatorname{sn} v \operatorname{cn} v \operatorname{dn} v$ with a rational function of $\operatorname{sn}^2 v$.

In this way every theorem that refers to homogeneous functions of order zero of ϑ-functions can be converted into a theorem on elliptic functions, and from this one obtains in turn the corresponding theorems on elliptic integrals. One sees at once that the addition formulae (16) are

no others than the formulae (17) of § 13 in the first section. In the same way the transformation theory of the ϑ-functions furnishes us a solution of the Jacobi transformation problem of the elliptic integrals, as we have set it out in § 8.

We will verify this for the two principal transformations of the second order, which in § 32, as also in § 9, we have designated as the Gauss and the Landen transformation. According to § 32, for the Gauss transformation

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right)\vartheta_{11}\left(u, \frac{\omega}{2}\right)}{\vartheta_{00}\left(0, \frac{\omega}{2}\right)\vartheta_{01}\left(u, \frac{\omega}{2}\right)} = \frac{2\,\vartheta_{01}(u)\,\vartheta_{11}(u)}{\vartheta_{01}^2(u) + \vartheta_{11}^2(u)}, \tag{23}$$

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right)^2}{\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2} = \frac{2\,\vartheta_{10}\,\vartheta_{00}}{\vartheta_{00}^2 + \vartheta_{10}^2} = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \tag{24}$$

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2 = \vartheta_{00}^2 + \vartheta_{10}^2 = (1+\varkappa)\,\vartheta_{00}^2. \tag{25}$$

According to § 42 we thus obtain a relation between elliptic functions with two different moduli, and if λ, L, L' arise from $\varkappa$, K, K' by the interchange $\left(\omega, \frac{\omega}{2}\right)$, then by (23), (24), (25)

$$\lambda = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \quad L = (1+\varkappa)\,K, \quad L' = (1+\varkappa)\,K' \tag{26}$$

$$\operatorname{sn}\big[(1+\varkappa)\,v,\ \lambda\big] = \frac{(1+\varkappa)\operatorname{sn} v}{1+\varkappa\operatorname{sn}^2 v}, \tag{27}$$

if for the sake of clarity the modulus is also given as second argument under the function sign sn.

For the Landen transformation we have

$$\frac{\vartheta_{11}(2u,\,2\omega)}{\vartheta_{01}(2u,\,2\omega)} = \frac{\vartheta_{10}(u)\,\vartheta_{11}(u)}{\vartheta_{00}(u)\,\vartheta_{01}(u)},$$

$$\frac{\vartheta_{01}(0,\,2\omega)^2}{\vartheta_{00}(0,\,2\omega)^2} = \frac{2\,\vartheta_{00}\vartheta_{01}}{\vartheta_{00}^2 + \vartheta_{01}^2} = \frac{2\sqrt{\varkappa'}}{1+\varkappa'},$$

$$2\,\vartheta_{00}^2(0,\,2\omega) = (1+\varkappa')\,\vartheta_{00}^2,$$

and from this, if λ, λ', L, L' arise by the interchange $(\omega, 2\omega)$ from $\varkappa$, $\varkappa'$, K, K':

$$\lambda' = \frac{2\sqrt{\varkappa'}}{1+\varkappa'}, \qquad \sqrt{\lambda} = \frac{\varkappa}{1+\varkappa'}, \qquad \lambda = \frac{1-\varkappa'}{1+\varkappa'}, \tag{28}$$

$$2L = (1+\varkappa')\,K, \qquad\qquad L' = (1+\varkappa')\,K',$$

$$\operatorname{sn}\big[(1+\varkappa')\,v,\ \lambda\big] = \frac{(1+\varkappa')\operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \tag{29}$$

and in (26) to (29) one recognises, by § 42, the formulae (2) and (4) of § 9.

§ 45. The linear transformation of the elliptic functions.

The effect of the linear transformation upon the elliptic functions is best surveyed from the representation by the σ-functions. From (4) or (10), § 42, and the formulae (4), (6) of § 37 one obtains:

$$\begin{aligned} \frac{\omega_1}{2K}\,\mathrm{sn}\,\frac{2Ku}{\omega_1} &= \frac{\sigma(u)}{\sigma_{01}(u)}, \\ \mathrm{cn}\,\frac{2Ku}{\omega_1} &= \frac{\sigma_{10}(u)}{\sigma_{01}(u)}, \\ \mathrm{dn}\,\frac{2Ku}{\omega_1} &= \frac{\sigma_{00}(u)}{\sigma_{01}(u)}, \end{aligned} \tag{1}$$

or also, by making use of the homogeneity of the σ-function [§ 37 (8)]:

$$\begin{aligned} \mathrm{sn}\,v &= \frac{\sigma(v,\,2K,\,2iK')}{\sigma_{01}(v,\,2K,\,2iK')}, \\ \mathrm{cn}\,v &= \frac{\sigma_{10}(v,\,2K,\,2iK')}{\sigma_{01}(v,\,2K,\,2iK')}, \\ \mathrm{dn}\,v &= \frac{\sigma_{00}(v,\,2K,\,2iK')}{\sigma_{01}(v,\,2K,\,2iK')}. \end{aligned} \tag{2}$$

The σ-functions occurring on the right-hand side of (2) depend only on the two variables v, ω, and it is first of all the change of these functions under application of the fundamental transformations

$$(\omega,\ \omega+1), \qquad \left(\omega,\ -\frac{1}{\omega}\right)$$

that has to be determined. Because of

$$\pi\,\vartheta_{00}^2 = 2K, \qquad 2iK' = 2\omega K$$

one obtains from § 31 (6) and (11) the corresponding changes:

$$\begin{array}{llllll} \omega, & 2K, & 2iK', & \varkappa, & \varkappa', \\ \omega+1, & 2\varkappa' K, & 2\varkappa'(K+iK'), & \dfrac{i\varkappa}{\varkappa'}, & \dfrac{1}{\varkappa'}, \\ -\dfrac{1}{\omega}, & 2K', & 2iK, & \varkappa', & \varkappa. \end{array}$$

If one sets for the moment:

$$f(v,\omega) = \sigma(v,\,2K,\,2iK'),$$

there results:

$$\begin{aligned} f(v,\omega+1) &= \sigma\big[v,\,2\varkappa' K,\,2\varkappa'(K+iK')\big] \\ &= \sigma(v,\,2\varkappa' K,\,2\varkappa' iK') \text{ [by § 35 (7)]} \\ &= \varkappa'\,\sigma\left(\frac{v}{\varkappa'},\,2K,\,2iK'\right) \text{ [by § 37 (8)]}, \\ f\left(v,-\frac{1}{\omega}\right) &= \sigma(v,\,2K',\,2iK) \\ &= \sigma(v,\,-2iK,\,2K') \text{ [by § 35 (7)]} \\ &= -i\,\sigma(iv,\,2K,\,2iK') \text{ [by § 37 (8)]}, \end{aligned}$$

and if one does the corresponding thing for the three remaining σ-functions and takes § 36 (7) into account, one obtains the following interchanges that belong together:

$$
\begin{array}{lllll}
\omega, & \sigma(v), & \sigma_{00}(v), & \sigma_{01}(v), & \sigma_{10}(v), \\
\omega+1, & \varkappa'\sigma\left(\dfrac{v}{\varkappa'}\right), & \sigma_{01}\left(\dfrac{v}{\varkappa'}\right), & \sigma_{00}\left(\dfrac{v}{\varkappa'}\right), & \sigma_{10}\left(\dfrac{v}{\varkappa'}\right), \\
-\dfrac{1}{\omega}, & -i\,\sigma(iv), & \sigma_{00}(iv), & \sigma_{10}(iv), & \sigma_{01}(iv),
\end{array} \tag{3}
$$

in which the periods are $2K$, $2iK'$. From this there follow, according to (2), the first two linear transformations of the elliptic functions:

$$
\begin{aligned}
\operatorname{sn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \varkappa' \frac{\operatorname{sn} v}{\operatorname{dn} v}, \\
\operatorname{cn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \frac{\operatorname{cn} v}{\operatorname{dn} v}, \\
\operatorname{dn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \frac{1}{\operatorname{dn} v}.
\end{aligned} \tag{4}
$$

$$
\begin{aligned}
\operatorname{sn}(iv, \varkappa') &= i\,\frac{\operatorname{sn} v}{\operatorname{cn} v}, \\
\operatorname{cn}(iv, \varkappa') &= \frac{1}{\operatorname{cn} v}, \\
\operatorname{dn}(iv, \varkappa') &= \frac{\operatorname{dn} v}{\operatorname{cn} v}.
\end{aligned} \tag{5}
$$

The first formula (5) shows, by § 44 (21), that $\operatorname{sn}(v, \varkappa') = 1$ when $v = K'$. From this there results, by § 42 (17), the second frequently used representation for K':

$$
K' = \frac{1}{2}\int_0^1 \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa'^2\xi)}}. \tag{6}
$$

From (4), (5) one obtains the remaining cases of the linear transformation by repeated application. If in (4) one puts iv, $\varkappa'$ in place of v, $\varkappa$ and applies (5), then there follows:

$$
\begin{aligned}
\operatorname{sn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= i\varkappa \frac{\operatorname{sn} v}{\operatorname{dn} v}, \\
\operatorname{cn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{1}{\operatorname{dn} v}, \\
\operatorname{dn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{\operatorname{cn} v}{\operatorname{dn} v}.
\end{aligned} \tag{7}
$$

If, conversely, one replaces v, $\varkappa$, $\varkappa'$ in (5) by $\varkappa' v$, $\dfrac{i\varkappa}{\varkappa'}$, $\dfrac{1}{\varkappa'}$ and applies (4), it follows that:

$$
\begin{aligned}
\operatorname{sn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= i\varkappa' \frac{\operatorname{sn} v}{\operatorname{cn} v}, \\
\operatorname{cn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{\operatorname{dn} v}{\operatorname{cn} v}, \\
\operatorname{dn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{1}{\operatorname{cn} v}.
\end{aligned} \tag{8}
$$

If one here again replaces v by iv, and $\varkappa$, $\varkappa'$ by $\varkappa'$, $\varkappa$, and applies (5), one finds:

$$\begin{aligned} \operatorname{sn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \varkappa \operatorname{sn} v, \\ \operatorname{cn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{dn} v, \\ \operatorname{dn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{cn} v, \end{aligned} \tag{9}$$

with which the six classes of linear transformations are exhausted, since a still more frequent repetition gives occasion to no new formulae.

§ 46. The Weierstrass $\wp$-Function.

The linear transformations of the elliptic functions suggest that we look for an elliptic function which, like the σ-function itself, remains unchanged under the linear transformations. In order to form such a function, we first put the equations (1) of the preceding section into the following form:

$$\begin{aligned} \frac{\sigma_{10}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{\operatorname{cn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{00}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{\operatorname{dn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{01}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{1}{\operatorname{sn} v}, \end{aligned} \tag{1}$$

in which, for brevity,

$$v = \frac{2Ku}{\omega_1} \tag{2}$$

has been set. From this one concludes, with the help of the relations

$$\operatorname{cn}^2 v = 1 - \operatorname{sn}^2 v, \qquad \operatorname{dn}^2 v = 1 - \varkappa^2 \operatorname{sn}^2 v,$$

$$\frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{10}^2(u)}{\sigma^2(u)} = \frac{4K^2}{\omega_1^2},$$

$$\frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{00}^2(u)}{\sigma^2(u)} = \frac{4K^2}{\omega_1^2}\varkappa^2.$$

Three quantities e_1, e_2, e_3 independent of u (and therefore depending only on ω_1, ω_2) can accordingly be determined in such a way that

$$\frac{\sigma_{10}^2(u)}{\sigma^2(u)} + e_1 = \frac{\sigma_{00}^2(u)}{\sigma^2(u)} + e_2 = \frac{\sigma_{01}^2(u)}{\sigma^2(u)} + e_3 = \wp(u), \tag{3}$$

where $\wp(u)$ is a doubly periodic function newly defined by (3). The quantities e_1, e_2, e_3 may be arbitrary, provided only that they satisfy the conditions

$$e_1 - e_3 = \frac{4K^2}{\omega_1^2}, \quad e_2 - e_3 = \frac{4K^2}{\omega_1^2}\varkappa^2 \tag{4}$$

We are therefore free to add, for their complete determination, the further condition

$$e_1 + e_2 + e_3 = 0 \tag{5}$$

The following expressions then result for e_1, e_2, e_3:

$$
\begin{aligned}
e_1 &= \frac{4K^2}{\omega_1^2}\frac{1+\varkappa'^2}{3},\\
e_2 &= -\frac{4K^2}{\omega_1^2}\frac{\varkappa'^2-\varkappa^2}{3},\\
e_3 &= -\frac{4K^2}{\omega_1^2}\frac{1+\varkappa^2}{3}.
\end{aligned}
\tag{6}
$$

The function $\wp(u)$, which possesses the periods ω_1, ω_2, is hereby expressed by means of (1):

$$
\wp(u) = \frac{4K^2}{\omega_1^2}\left(\frac{1}{\mathrm{sn}^2 v} - \frac{1+\varkappa^2}{3}\right).
\tag{7}
$$

The function $\wp(u)$ has the desired property of invariability under linear transformations, as one recognises from the expression

$$
\wp(u) = \frac{1}{3}\frac{\sigma_{00}^2(u)+\sigma_{01}^2(u)+\sigma_{10}^2(u)}{\sigma^2(u)}
\tag{8}
$$

In consequence of (3) the e_1, e_2, e_3 are permuted under a linear transformation in the same way as the σ_{10}, σ_{00}, σ_{01}; a symmetric function of them is therefore, under a linear trans-

formation unchanged and is called an invariant. There are two fundamental ones, which we denote by g_2, g_3:

$$
\begin{aligned}
g_2 &= -4(e_2e_3+e_3e_1+e_1e_2) = 2(e_1^2+e_2^2+e_3^2)\\
&= \frac{64}{3}\frac{K^4}{\omega_1^4}(1-\varkappa^2\varkappa'^2),
\end{aligned}
\tag{9}
$$

$$
g_3 = 4e_1e_2e_3 = \frac{2^8}{27}\frac{K^6}{\omega_1^6}(2+\varkappa^2\varkappa'^2)(\varkappa'^2-\varkappa^2),
\tag{10}
$$

and we add besides the function known under the name of the discriminant:

$$
\begin{aligned}
G &= (e_2-e_3)^2(e_3-e_1)^2(e_1-e_2)^2 = \frac{1}{16}(g_2^3-27g_3^2)\\
&= \frac{(2K)^{12}}{\omega_1^{12}}\varkappa^4\varkappa'^4,
\end{aligned}
\tag{11}
$$

for which, by § 43 (3), (15) and § 34 (2), we may also put:

$$
G = \frac{2^8\pi^{12}\eta(\omega)^{24}}{\omega_1^{12}}.
\tag{12}
$$

$\wp(u)$, g_2, g_3, G are homogeneous functions, as the following relations show, in which λ is an arbitrary factor.

$$
\begin{aligned}
\wp(\lambda u,\lambda\omega_1,\lambda\omega_2) &= \lambda^{-2}\wp(u,\omega_1,\omega_2),\\
\wp'(\lambda u,\lambda\omega_1,\lambda\omega_2) &= \lambda^{-3}\wp'(u,\omega_1,\omega_2),\\
g_2(\lambda\omega_1,\lambda\omega_2) &= \lambda^{-4}g_2(\omega_1,\omega_2),\\
g_3(\lambda\omega_1,\lambda\omega_2) &= \lambda^{-6}g_3(\omega_1,\omega_2),\\
G(\lambda\omega_1,\lambda\omega_2) &= \lambda^{-12}G(\omega_1,\omega_2).
\end{aligned}
\tag{13}
$$

If we put, as in § 45 (2)

$$
\omega_1 = 2K, \qquad \omega_2 = 2iK',
$$

we then have

$$
\begin{aligned}
&e_1 = \frac{1+\varkappa'^2}{3}, \quad e_2 = -\frac{\varkappa'^2-\varkappa^2}{3}, \quad e_3 = -\frac{1+\varkappa^2}{3},\\
&g_2 = \frac{4}{3}(1-\varkappa^2\varkappa'^2), \quad g_3 = \frac{4}{27}(\varkappa'^2-\varkappa^2)(2+\varkappa^2\varkappa'^2),\\
&G = \varkappa^4\varkappa'^4.
\end{aligned} \tag{14}
$$

If we eliminate ω_1 from the general expressions (9), (10), (11) for g_2, g_3, G, we obtain homogeneous functions of order zero, that is, functions of ω alone, which remain unchanged under linear transformations. Such functions are $g_2^3 : G$, $g_3^2 : G$. Among these functions, which can all be reduced to one another, we single out one, which we denote by

$j(\omega)$ and call it simply the invariant, and define it thus:

$$
j(\omega) = 2^8 \frac{(1-\varkappa^2\varkappa'^2)^3}{\varkappa^4\varkappa'^4}, \tag{15}
$$

from which we obtain:

$$
\begin{aligned}
&\frac{4.27g_2^3}{G} = j(\omega),\\
&\frac{4.27.27g_3^2}{G} = j(\omega) - 27.64 = \frac{64.(2+\varkappa^2\varkappa'^2)^2(\varkappa'^2-\varkappa^2)^2}{\varkappa^4\varkappa'^4}.
\end{aligned} \tag{16}
$$

Regarded as a function of $\varkappa^2$, the function $j(\omega)$ has the property of remaining unchanged if in place of $\varkappa^2$ one of the six values

$$
\varkappa^2, \quad \varkappa'^2, \quad \frac{1}{\varkappa^2}, \quad \frac{1}{\varkappa'^2}, \quad -\frac{\varkappa^2}{\varkappa'^2}, \quad -\frac{\varkappa'^2}{\varkappa^2}
$$

is substituted.

If we form the differential quotient of the function $\wp(u)$ according to (7), we obtain

$$
\wp'(u) = -\frac{16K^3}{\omega_1^3}\frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn}^3 v} = -2\frac{\sigma_{00}(u)\sigma_{10}(u)\sigma_{01}(u)}{\sigma(u)^3} \tag{17}
$$

and from this, by (3):

$$
\begin{aligned}
\wp'(u)^2 &= 4[\wp(u)-e_1][\wp(u)-e_2][\wp(u)-e_3]\\
&= 4\wp(u)^3 - g_2\wp(u) - g_3,
\end{aligned} \tag{18}
$$

or finally

$$
du = \frac{d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}}, \tag{19}
$$

from which one sees that the function $\wp(u)$ stands in the same relation to the Weierstrass normal form of the elliptic differential as the function $\operatorname{sn} v$ does to the Legendre one.

§ 47. The elliptic transcendents of the second kind.

Jacobi introduced as the transcendent of the second kind the function

$$
Z(v) = \frac{\Theta'(v)}{\Theta(v)} = \frac{d\log\Theta(v)}{dv} \tag{1}
$$

The relation of this function to the elliptic integrals of the second kind follows from formula (16) of § 23; and it may be remarked at once that quite similar considerations could be attached to the formulae (15), (17), (18) there, which, however, do not lead to essentially new results.

If we there put $v : 2K$ in place of v, then from § 42 and 43 we obtain:

$$\begin{aligned}\frac{d^2 \log \Theta(v)}{dv^2} &= \frac{1}{4K^2}\frac{\vartheta_{01}''}{\vartheta_{01}} - \varkappa^2 \mathrm{sn}^2 v \\ &= \frac{1}{4K^2}\frac{\vartheta_{01}''}{\vartheta_{01}} - 1 + \mathrm{dn}^2 v.\end{aligned} \tag{2}$$

We set

$$1 - \frac{\vartheta_{01}''}{4K^2 \vartheta_{01}} = \frac{E}{K}, \tag{3}$$

and obtain by integration of (2)

$$\frac{d \log \Theta(v)}{dv} = Z(v) = -\frac{E}{K} v + \int_0^v \mathrm{dn}^2 v \, dv,$$

or, if we set

$$E(v) = \int_0^v \mathrm{dn}^2 v \, dv \tag{4}$$

we have:

$$Z(v) = E(v) - \frac{E}{K} v, \qquad \frac{dZ(v)}{dv} = \mathrm{dn}^2 v - \frac{E}{K}. \tag{5}$$

The functions $\Theta(v)$ and $\Theta(v + K)$ are even functions, so both derivatives vanish: $\Theta'(0)$ and $\Theta'(K) = 0$. If, then, we set $v = K$ in (5), it follows that:

$$E = \int_0^K \mathrm{dn}^2 v \, dv. \tag{6}$$

If one further introduces for dv the differential

$$\frac{1}{2} \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2 \xi)}}$$

[§ 42 (8)], one obtains for $E(v)$ an elliptic integral of the second kind (§ 11):

$$E(v) = \frac{1}{2} \int_0^\xi \frac{d\xi (1 - \varkappa^2 \xi)}{\sqrt{\xi(1-\xi)(1-\varkappa^2 \xi)}}, \tag{7}$$

$$E = \frac{1}{2} \int_0^1 \frac{d\xi (1 - \varkappa^2 \xi)}{\sqrt{\xi(1-\xi)(1-\varkappa^2 \xi)}}, \tag{8}$$

where the latter integral is to be taken in the same sense as § 42 (17).

$E(v)$ and $Z(v)$ are odd functions of the argument.

From the addition theorem of the Θ-function [§ 43 (11)] there follows, by logarithmic differentiation:

$$Z(u+v) + Z(u-v) = 2\, Z(u) - \frac{2\, \varkappa^2 \, \mathrm{sn}^2 v \, \mathrm{sn}\, u \, \mathrm{cn}\, u \, \mathrm{dn}\, u}{1 - \varkappa^2 \, \mathrm{sn}^2 u \, \mathrm{sn}^2 v}, \tag{9}$$

$$Z(u+v) - Z(u-v) = 2\, Z(v) - \frac{2\, \varkappa^2 \, \mathrm{sn}^2 u \, \mathrm{sn}\, v \, \mathrm{cn}\, v \, \mathrm{dn}\, v}{1 - \varkappa^2 \, \mathrm{sn}^2 u \, \mathrm{sn}^2 v}. \tag{10}$$

In these, Z may also be replaced by E, and by addition there results [§ 44 (16)]:

$$E(u) + E(v) - E(u+v) = \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u+v). \tag{11}$$

From this one obtains the period properties of the function $E(u)$ by setting $v = \pm K$, $K + iK'$. But one arrives at them in the following way as well.

From the first equation § 43 (3) it follows:

$$\frac{d \log H(v)}{dv} = Z(v + iK') + \frac{i\pi}{2K},$$

and accordingly, from § 43 (6) and § 42 (11):

$$\begin{aligned} Z(v + iK') &= Z(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} - \frac{i\pi}{2K}, \\ Z(v + K) &= Z(v) - \frac{\varkappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \end{aligned} \tag{12}$$

and if in the first of these formulae one sets $v = K$:

$$Z(K + iK') = -\frac{i\pi}{2K}. \tag{13}$$

By twofold application of the formulae (12) [with regard to § 44 (18), (20)]:

$$\begin{aligned} Z(v + 2\,iK') &= Z(v) - \frac{i\pi}{K}, \\ Z(v + 2K) &= Z(v). \end{aligned} \tag{14}$$

If one carries these equations over to the function $E(v)$, there follows:

$$\begin{aligned} E(v + iK') &= E(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} + \frac{i\,E\,K'}{K} - \frac{i\pi}{2K}, \\ E(v + K) &= E(v) + E - \frac{\varkappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}. \end{aligned} \tag{15}$$

$$\begin{aligned} E(v + 2\,iK') &= E(v) + \frac{2\,i\,E\,K'}{K} - \frac{i\pi}{K}, \\ E(v + 2K) &= E(v) + 2\,E. \end{aligned} \tag{16}$$

If in (15) we set $v = K$ and define a quantity E' by the equation:

$$E(K + iK') = E + i(K' - E'), \tag{17}$$

then one obtains the formula known under the name of Legendre's relation

$$EK' + KE' - KK' = \frac{\pi}{2}. \tag{18}$$

The significance of the quantity E' introduced here is recognised if one derives Legendre's relation by a second route, making use of one of the linear transformations.

It follows, if one sets in § 31 (11):

$$u = \frac{v}{2K}, \qquad \omega = \frac{iK'}{K}, \qquad \frac{u}{\omega} = \frac{v}{2iK'},$$

by § 43 (1), (4):

$$\sqrt{K}\, e^{-\frac{\pi v^2}{4KK'}}\, \Theta(iv, \varkappa') = \sqrt{K'}\, H(v+K), \tag{19}$$

and from this, by logarithmic differentiation,

$$iZ(iv, \varkappa') = \frac{\pi v}{2KK'} + \frac{H'(v+K)}{H(v+K)} \tag{20}$$

or, by § 43 (6),

$$iZ(iv, \varkappa') = \frac{\pi v}{2KK'} + Z(v) + \frac{d \log \operatorname{cn} v}{dv}. \tag{21}$$

But by (5) we have

$$i\, \frac{dZ(iv, \varkappa')}{dv} = -\operatorname{dn}^2(iv, \varkappa') + \frac{E'}{K'},$$

if E' now has the meaning:

$$E' = \int_v^{K'} \operatorname{dn}^2(v, \varkappa')\, dv, \tag{22}$$

so that it arises from E by the interchange of $\varkappa$ with $\varkappa'$. But if one sets $v = 0$ in (21), after having first differentiated, the relation (18) again results, from which it follows that E' is in both cases the same quantity.

§ 48. The elliptic transcendents of the third kind.

If we take the formula (10) of the preceding section:

$$\frac{\Theta'(u+v)}{\Theta(u+v)} - \frac{\Theta'(u-v)}{\Theta(u-v)} = 2Z(v) - \frac{2\varkappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}$$

and integrate it with respect to u, then it follows:

$$\frac{1}{2} \log \frac{\Theta(u-v)}{\Theta(u+v)} + u\, Z(v) = \int_0^u \frac{\varkappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}\, du, \tag{1}$$

and we can further add the formula, which is to be derived in the same way [from § 43 (11)]:

$$\frac{1}{2} \log \frac{H(v-u)}{H(v+u)} + u\, Z(v) = \int_0^u \frac{\operatorname{sn} v \operatorname{cn} v \operatorname{dn} v\, du}{\operatorname{sn}^2 u - \operatorname{sn}^2 v} \tag{2}$$

which, moreover, can also be derived from (1). We now set, with Jacobi,

$$\Pi(u, v) = \int_0^u \frac{\varkappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}\, du, \tag{3}$$

and call $\Pi(u, v)$ the transcendent of the third kind with the argument u and the parameter v. If one replaces du by the expression:

$$\frac{d\xi}{2\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}$$

and $\mathrm{sn}^2 u$ by ξ, then both (1) and (2) yield an elliptic integral of the third kind, such as we became acquainted with in § 11.

From (1) there follows first of all

$$\Pi(u, v) - u\,Z(v) = \Pi(v, u) - v\,Z(u), \tag{4}$$

or Jacobi's theorem on the interchange of argument and parameter.

The function $\Pi(u, v)$ vanishes when $v = K$ or $v = K + iK'$, because for the former value $\mathrm{cn}\, v$ is equal to zero, and for the second $\mathrm{dn}\, v$. If we therefore substitute these values for u in (4), there follows

$$\begin{aligned} \Pi(K, v) &= K\,Z(v) \\ \Pi(K + iK', v) &= (K + iK')\,Z(v) + \frac{i\pi v}{2K}\ [\S\ 47,\ (13)], \end{aligned} \tag{5}$$

by which the complete integrals of the third kind are reduced to the second kind.

We wish also to derive the addition theorem of the Π-function, which Jacobi gives in the Fund. nova art. 53–55[8] in various

forms that can be reduced to one another only with difficulty. First we obtain from the definition (1), if we now denote the parameter by a:

$$\begin{aligned} &\Pi(u + v, a) - \Pi(u, a) - \Pi(v, a) \\ &\quad = \frac{1}{2}\log\frac{\Theta(u + v - a)}{\Theta(u + v + a)}\,\frac{\Theta(u + a)\,\Theta(v + a)}{\Theta(u - a)\,\Theta(v - a)}, \end{aligned} \tag{6}$$

and it remains only to represent the expression standing under the logarithm by means of elliptic functions.

This is first done easily, as at the passage of the Fundamenta already mentioned, by means of the formula [§ 43, (11)]:

$$\Theta^2(0)\,\Theta(u + v)\,\Theta(u - v) = \Theta^2(u)\,\Theta^2(v)\,[1 - \varkappa^2\,\mathrm{sn}^2 u\,\mathrm{sn}^2 v],$$

if one puts therein for u, v the values $u \pm a$, $v \pm a$, then $\pm a$, $u + v \pm a$.

One thus obtains:

$$\begin{aligned} \Theta^2(0)\,\Theta(u + v \pm 2a)\,\Theta(u - v) &= \Theta^2(u \pm a)\,\Theta^2(v \pm a)\,[1 - \varkappa^2\,\mathrm{sn}^2(u \pm a)\,\mathrm{sn}^2(v \pm a)], \\ \Theta^2(0)\,\Theta(u + v \pm 2a)\,\Theta(u + v) &= \Theta^2(a)\,\Theta^2(u + v \pm a)\,[1 - \varkappa^2\,\mathrm{sn}^2 a\,\mathrm{sn}^2(u + v \pm a)], \end{aligned}$$

from which it follows:

$$\frac{\Theta(u + v - a)\,\Theta(u + a)\,\Theta(v + a)}{\Theta(u + v + a)\,\Theta(u - a)\,\Theta(v - a)} = \sqrt{\frac{[1 - \varkappa^2\,\mathrm{sn}^2(u - a)\,\mathrm{sn}^2(v - a)]\,[1 - \varkappa^2\,\mathrm{sn}^2 a\,\mathrm{sn}^2(u + v + a)]}{[1 - \varkappa^2\,\mathrm{sn}^2(u + a)\,\mathrm{sn}^2(v + a)]\,[1 - \varkappa^2\,\mathrm{sn}^2 a\,\mathrm{sn}^2(u + v - a)]}}. \tag{7}$$

[8] C. G. J. Jacobis gesammelte Werke, Berlin 1881, Bd. I, S. 204.

A second expression is obtained from the addition formula (13), § 22, which can be put in the form:

$$\Theta(0)\,\Theta(u\pm a)\,\Theta(v\pm a)\,\Theta(u+v)=\Theta(u)\,\Theta(v)\,\Theta(a)\,\Theta(u+v\pm a)\,[1\pm\varkappa^2\,\mathrm{sn}\,a\,\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{sn}(u+v\pm a)],$$

from which by division:

$$\frac{\Theta(u+v-a)\,\Theta(u+a)\,\Theta(v+a)}{\Theta(u+v+a)\,\Theta(u-a)\,\Theta(v-a)}=\frac{1+\varkappa^2\,\mathrm{sn}\,a\,\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{sn}(u+v+a)}{1-\varkappa^2\,\mathrm{sn}\,a\,\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{sn}(u+v-a)}. \tag{8}$$

Jacobi gives yet a third expression for the Θ-quotients in (6):

$$\frac{\left\{1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u-v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}+a\right)\right\}\left\{1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u+v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}-a\right)\right\}}{\left\{1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u-v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}-a\right)\right\}\left\{1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u+v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}+a\right)\right\}}. \tag{9}$$

The direct transformation of the three expressions (7), (8), (9) into one another is most easily accomplished if one starts from the formula last given by Jacobi:

$$1-\varkappa^2\,\mathrm{sn}(a+u)\,\mathrm{sn}(a-u)\,\mathrm{sn}(a+v)\,\mathrm{sn}(a-v)=\frac{(1-\varkappa^2\,\mathrm{sn}^4\,a)(1-\varkappa^2\,\mathrm{sn}^2\,u\,\mathrm{sn}^2\,v)}{(1-\varkappa^2\,\mathrm{sn}^2\,a\,\mathrm{sn}^2\,u)(1-\varkappa^2\,\mathrm{sn}^2\,a\,\mathrm{sn}^2\,v)}. \tag{10}$$

The verification of this formula is easy for this reason, that with the help of § 44, (13) one obtains on the right and on the left rational functions of $\mathrm{sn}^2\,u$, $\mathrm{sn}^2\,v$, whose identity is immediately evident.

If in this formula one replaces

$$u,\quad v,\quad a$$

by

$$\frac{u-v}{2},\quad \frac{u+v}{2}\pm a,\quad \frac{u+v}{2},$$

there results

$$1\pm\varkappa^2\,\mathrm{sn}\,u\,\mathrm{sn}\,v\,\mathrm{sn}\,a\,\mathrm{sn}(u+v\pm a)$$
$$=\frac{\left[1-\varkappa^2\,\mathrm{sn}^4\left(\frac{u+v}{2}\right)\right]\left[1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u-v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}\pm a\right)\right]}{\left[1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u+v}{2}\right)\mathrm{sn}^2\,\frac{u-v}{2}\right]\left[1-\varkappa^2\,\mathrm{sn}^2\left(\frac{u+v}{2}\right)\mathrm{sn}^2\left(\frac{u+v}{2}\pm a\right)\right]},$$

and if one divides the two formulae contained herein by one another, there results the agreement of (8) and (9).

If one puts $v=u$ in (10), there results:

$$1-\varkappa^2\,\mathrm{sn}^2(a+u)\,\mathrm{sn}^2(a-u)=\frac{(1-\varkappa^2\,\mathrm{sn}^4\,a)(1-\varkappa^2\,\mathrm{sn}^4\,u)}{(1-\varkappa^2\,\mathrm{sn}^2\,a\,\mathrm{sn}^2\,u)^2}. \tag{11}$$

If one replaces herein

$$a\quad u$$

first by

$$\frac{u+v}{2}\pm a,\quad -\frac{u+v}{2},$$

and then by

$$\frac{u+v}{2} \pm a, \quad \frac{u-v}{2},$$

there result four formulae:

$$1 - \varkappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u+v\pm a) = \frac{\left[1 - \varkappa^2 \operatorname{sn}^4\left(\frac{u+v}{2} \pm a\right)\right]\left(1 - \varkappa^2 \operatorname{sn}^4 \frac{u+v}{2}\right)}{\left[1 - \varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right) \operatorname{sn}^2 \frac{u+v}{2}\right]^2},$$

$$1 - \varkappa^2 \operatorname{sn}^2(u \pm a) \operatorname{sn}^2(v \pm a) = \frac{\left[1 - \varkappa^2 \operatorname{sn}^4\left(\frac{u+v}{2} \pm a\right)\right]\left(1 - \varkappa^2 \operatorname{sn}^4 \frac{u-v}{2}\right)}{\left[1 - \varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right) \operatorname{sn}^2\left(\frac{u-v}{2}\right)\right]^2},$$

from which the agreement of (9) with (7) follows.

§ 49. The transcendents of the second and third kind of Weierstrass.

It remains now to set up the transcendents of the second and third kind in the Weierstrass form. We reach the former much as in § 47, by twofold differentiation of $\log \sigma(u)$. In this way there follows from

$$\sigma(u) = \omega_1 e^{\frac{\eta_1 u^2}{\omega_1}} \frac{\vartheta_{11}\left(\frac{u}{\omega_1}\right)}{\vartheta'_{11}} \quad [\S\,37,\ (4)], \tag{1}$$

$$\frac{d^2 \log \sigma(u)}{du^2} = \frac{2\eta_1}{\omega_1} + \frac{d^2 \log \vartheta_{11}\left(\frac{u}{\omega_1}\right)}{du^2},$$

$$= \frac{2\eta_1}{\omega_1} + \frac{1}{\omega_1^2}\frac{\vartheta''_{00}}{\vartheta_{00}} - \frac{1}{\omega_1^2}\frac{\vartheta'^2_{11}}{\vartheta^2_{00}}\frac{\vartheta^2_{00}\left(\frac{u}{\omega_1}\right)}{\vartheta^2_{11}\left(\frac{u}{\omega_1}\right)} \quad [\S\,23,\ (18)]$$

and hence, if one expresses the ϑ-quotient by elliptic functions according to § 42, and these by the $\wp$-function according to § 46:

$$\frac{d^2 \log \sigma(u)}{du^2} = A - \wp(u),$$

in which A is a constant. This constant is found, however, if one writes the equation, following § 46, (8), in this way:

$$\sigma''(u)\sigma(u) - \sigma'(u)\sigma'(u) = A\sigma^2(u) - \tfrac{1}{3}\left[\sigma^2_{00}(u) + \sigma^2_{01}(u) + \sigma^2_{10}(u)\right].$$

If one differentiates this twice and then sets $u = 0$, it follows that $A = 0$ [§ 35, (12), § 37, (5), (7)], and we obtain:

$$\frac{d^2 \log \sigma(u)}{du^2} = -\wp(u), \tag{2}$$

a formula which in Weierstrass serves as the definition of the $\wp$-function.

If therefore we set

$$\frac{\sigma'(u)}{\sigma(u)} = \zeta(u) \tag{3}$$

then $\zeta(u)$ is a single-valued function of u, and indeed an elliptic integral of the second kind:

$$\zeta(u) = -\int \wp(u)\,du = -\int \frac{\wp\,d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}} \text{ [§ 46, (19)]}, \tag{4}$$

in which the additive constant is so determined that $\zeta(u)$ must be an odd function. The periodicity of the ζ-function follows from the period equations of the σ-function [§ 35, (9)]:

$$\begin{aligned} \zeta(u+\omega_1) - \zeta(u) &= 2\eta_1, \\ \zeta(u+\omega_2) - \zeta(u) &= 2\eta_2. \end{aligned} \tag{5}$$

Thus $2\eta_1$ and $2\eta_2$ are represented as complete integrals of the second kind (analogous to the E, E'):

$$2\eta_1 = -\int_u^{u+\omega_1} \wp(u)\,du, \quad 2\eta_2 = -\int_u^{u+\omega_2} \wp(u)\,du \tag{6}$$

with the equation corresponding to the Legendre relation

$$\eta_1\omega_2 - \eta_2\omega_1 = \pi i \text{ [§ 35, (10)]}. \tag{7}$$

In order to form the addition theorems for the Weierstrass transcendents, we start from the σ-function. For this function we obtain from (1), using the addition formula for ϑ_{11}, § 22, (4):

$$\frac{\sigma(u+v)\sigma(u-v)}{\sigma(u)^2\sigma(v)^2} = \wp(v) - \wp(u), \tag{8}$$

and from this, by logarithmic differentiation:

$$\begin{aligned} \zeta(u+v) + \zeta(u-v) - 2\zeta(u) &= \frac{\wp'(u)}{\wp(u) - \wp(v)}, \\ \zeta(u+v) - \zeta(u-v) - 2\zeta(v) &= \frac{-\wp'(v)}{\wp(u) - \wp(v)}, \end{aligned} \tag{9}$$

$$\zeta(u+v) = \zeta(u) + \zeta(v) + \frac{1}{2}\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}. \tag{10}$$

From this one derives, by a further differentiation, the addition theorem for the $\wp$-function. From (9) one obtains, by differentiation with respect to u and v:

$$\begin{aligned} \wp(u+v) + \wp(u-v) - 2\wp(u) &= -\frac{\wp''(u)}{\wp(u) - \wp(v)} + \frac{\wp'(u)^2}{[\wp(u) - \wp(v)]^2}, \\ 2\wp(u+v) - 2\wp(u-v) &= -\frac{2\wp'(u)\wp'(v)}{[\wp(u) - \wp(v)]^2}, \\ \wp(u+v) + \wp(u-v) - 2\wp(v) &= \frac{\wp''(v)}{\wp(u) - \wp(v)} + \frac{\wp'(v)^2}{[\wp(u) - \wp(v)]^2}, \end{aligned}$$

and on adding:

$$4\wp(u+v) - 2\wp(u) - 2\wp(v) = \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}\right)^2 - \frac{\wp''(u) - \wp''(v)}{\wp(u) - \wp(v)}.$$

But we have

$$\begin{aligned} \wp'(u)^2 &= 4\wp(u)^3 - g_2\wp(u) - g_3, \\ \wp''(u) &= 6\wp(u)^2 - \tfrac{1}{2}g_2, \\ \wp''(u) - \wp''(v) &= 6\left[\wp^2(u) - \wp^2(v)\right], \end{aligned}$$

from which one obtains:

$$\wp(u+v)+\wp(u)+\wp(v)=\frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2, \tag{11}$$

in agreement with formula (21), § 13.

Finally, by integrating the second equation (9) with respect to u, one obtains further:

$$\frac{1}{2}\log\frac{\sigma(v-u)}{\sigma(v+u)}+u\,\zeta(v)=\frac{1}{2}\int_0^u\frac{\wp'(v)\,du}{\wp(u)-\wp(v)}, \tag{12}$$

by which an integral of the third kind with argument u and parameter v is expressed by the σ-function.

Another form of the integral of the third kind is obtained by integrating (10):

$$\log\frac{\sigma(u+v)}{\sigma(u)\sigma(v)}-u\,\zeta(v)=\frac{1}{2}\int\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\,du. \tag{13}$$

§ 50. Developments of the elliptic functions.

If in the developments of the theta quotients that were derived in § 26 and collected in Table II one sets $a=0$, then, if one first omits the formulae that contain $\vartheta_{11}(a)$ in the denominator, one obtains the partial fraction developments of twelve elliptic functions:

$$\begin{array}{lll}
\operatorname{sn}2Ku, & \operatorname{cn}2Ku, & \operatorname{dn}2Ku,\\
\dfrac{1}{\operatorname{sn}2Ku}, & \dfrac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku}, & \dfrac{\operatorname{dn}2Ku}{\operatorname{sn}2Ku},\\
\dfrac{1}{\operatorname{cn}2Ku}, & \dfrac{\operatorname{sn}2Ku}{\operatorname{cn}2Ku}, & \dfrac{\operatorname{dn}2Ku}{\operatorname{cn}2Ku},\\
\dfrac{1}{\operatorname{dn}2Ku}, & \dfrac{\operatorname{sn}2Ku}{\operatorname{dn}2Ku}, & \dfrac{\operatorname{cn}2Ku}{\operatorname{dn}2Ku}.
\end{array}$$

Thus there results from formula (2), Table II:

$$\frac{\vartheta_{11}'\vartheta_{10}(u)}{\pi\vartheta_{11}(u)\vartheta_{10}}=\operatorname{cotg}\pi u-2i\sum(-1)^{\frac{m}{2}}q^m\left(\frac{1}{e^{-2\pi iu}-q^m}-\frac{1}{e^{2\pi iu}-q^m}\right)$$

$$=\operatorname{cotg}\pi u+4\sin 2\pi u\sum\frac{(-1)^{\frac{m}{2}}q^m}{1-2q^m\cos 2\pi u+q^{2m}},$$

where m runs through the sequence of positive even numbers 2, 4, 6, 8, ...

But according to § 42, (4) we have

$$\frac{\vartheta_{10}(u)}{\vartheta_{11}(u)}=\frac{\vartheta_{00}}{\vartheta_{01}}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku},$$

and from this there results

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku}=\operatorname{cotg}\pi u+4\sin 2\pi u\sum\frac{(-1)^{\frac{m}{2}}q^m}{1-2q^m\cos\pi u+q^{2m}}. \tag{1}$$

If one applies the same procedure to formula (2) of Table III, one obtains:

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku}=\operatorname{cotg}\pi u+4\sum(-1)^{\frac{m'}{2}}q^{\frac{mm'}{2}}\sin\pi mu.$$

Here m and m' run, independently of one another, through the sequences of positive even numbers, and the summation with respect to m' can still be carried out:

$$\sum(-1)^{\frac{m'}{2}} q^{\frac{mm'}{2}} = -\frac{q^m}{1+q^m},$$

hence:

$$\frac{2K}{\pi}\frac{\operatorname{cn} 2Ku}{\operatorname{sn} 2Ku} = \operatorname{cotg}\pi u - 4\sum \frac{q^m}{1+q^m}\sin \pi m u. \tag{2}$$

But while the first two formulae converge for all values of u, the convergence of the third is restricted to the region in which the imaginary part of u is absolutely smaller than the imaginary part of ω. The formula is therefore in any case valid for real u.

In Table IV the developments for the 12 functions are collected.

Among the 16 formulae of Tables II, III there are four which contain the factor $\vartheta_{11}(a)$ in the denominator, and in which one therefore cannot simply set $a = 0$. These are the formulae (1), (5), (9), (13). If one develops these formulae in ascending powers of a, the developments on the right and on the left begin with a^{-1}, and if one sets the terms independent of a equal to one another on both sides, the developments sought result.

One may proceed, for example, in such a way that in formula (1) one changes a into $-a$ and takes the arithmetic mean of the two formulae. Then on the right the infinite term $1 : \operatorname{cotg}\pi a$ cancels out, and on the left one obtains:

$$\frac{\vartheta'_{11}[\vartheta_{11}(v+a) - \vartheta_{11}(v-a)]}{2\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \frac{\vartheta'_{11}(v)}{\pi\vartheta_{11}(v)} \text{ for } a = 0,$$

on the right-hand side of (1) in Table II there results:

$$\operatorname{cotg}\pi v - 4i\sum\left(\frac{q^m}{e^{-2\pi iv} - q^m} - \frac{q^m}{e^{2\pi iv} - q^m}\right)$$

$$= \operatorname{cotg}\pi v + 4\sum \frac{q^m \sin 2\pi v}{1 - 2q^m\cos 2\pi v + q^{2m}},$$

and on the right-hand side of formula (1) of Table III one obtains

$$\operatorname{cotg}\pi v + 4\sum \frac{q^{\frac{m}{2}}}{1-q^m}\sin m\pi v,$$

and if one further introduces the Jacobi H-function, and writes u for v, one obtains

$$\frac{2K}{\pi}\frac{d\log H(2Ku)}{d\,2Ku} = \operatorname{cotg}\pi u + 4\sin 2\pi u\sum\frac{q^m}{1-2q^m\cos 2\pi u + q^m} \tag{3}$$

$$= \operatorname{cotg}\pi u + 4\sum\frac{q^m}{1-q^m}\sin m\pi u.$$

On the left-hand side there stands a transcendent of the second kind, which by § 47, (12) can be expressed by means of the Z-function:

$$\frac{2K}{\pi}Z(2Ku) + \frac{2K}{\pi}\frac{\operatorname{cn} 2Ku\,\operatorname{dn} 2Ku}{\sin 2Ku},$$

for which one may also put:

$$\frac{2K}{\pi}Z(2Ku) + \frac{d\log\sin 2Ku}{\pi\,du}.$$

In Table V the four formulae which arise in this way are collected. If one subtracts formula (3) of this table from the three remaining ones, there result developments for the logarithmic derivatives of the three fundamental elliptic functions, of which we will cite one:

$$\frac{d\log \operatorname{sn} 2Ku}{\pi\, du} = \frac{d\log \sin \pi u}{\pi\, du} + 4\sin 2\pi u \sum_{h=1}^{\infty} \frac{(-1)^h q^h}{1 - 2q^h \cos 2\pi u + q^{2h}} \tag{4}$$

$$= \frac{d\log \sin \pi u}{\pi\, du} - 4\sum_{h=1}^{\infty} \frac{q^h \sin 2h\pi u}{1+q^h}.$$

Developments for the transcendents of the third kind result when one integrates the formulae of the fifth table between the limits $u+v$, $u-v$. In this way one obtains, e. g., from formula (3) of this table:

$$\frac{1}{2}\log \frac{\theta[2K(u-v)]}{\theta[2K(u+v)]} = -4\sum \frac{q^{\frac{m}{2}}}{m(1-q^m)} \sin m\pi u \sin m\pi v. \tag{5}$$

If one puts $u = 0$ in the formulae of Tables IV and V, the following developments result:

$$\frac{2K}{\pi} = 1 + 4\sum \frac{q^{\frac{m}{2}}}{1+q^m} = 1 + 4\sum \frac{(-1)^{\frac{n-1}{2}} q^n}{1-q^n},$$

$$\frac{2\varkappa K}{\pi} = 4\sum \frac{q^{\frac{n}{2}}}{1+q^n} = 4\sum \frac{(-1)^{\frac{n-1}{2}} q^{\frac{n}{2}}}{1-q^n},$$

$$\frac{2\varkappa' K}{\pi} = 1 + 4\sum \frac{(-1)^{\frac{m}{2}} q^{\frac{m}{2}}}{1+q^m} = 1 - 4\sum \frac{(-1)^{\frac{n-1}{2}} q^n}{1+q^n}.$$

Finally one obtains a development for the complete integral of the second kind E from the formula

$$Z(v) = E(v) - \frac{E}{K} v.$$

If in this, after the differentiation, one puts $v = 0$, it follows that:

$$Z'(0) = \frac{K-E}{K},$$

and hence, by Table V, formula (3):

$$K - E = \frac{2\pi^2}{K} \sum \frac{q^n}{(1-q^n)^2} = \frac{\pi^2}{K} \sum \frac{m q^{\frac{m}{2}}}{1-q^m}.$$

Fifth Section.
The Modular Functions.

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§ 51. The Elliptic Differential Equations.

The elliptic functions defined so far are functions of the two variables v, ω, and the moduli $\varkappa$, $\varkappa'$, and further $j(\omega)$, K, K' depend on the ratio of periods ω, which must have a positive imaginary part. Likewise the function $\wp(u)$ is a function of the three variables u, ω_1, ω_2, and the invariants g_2, g_3 depend on ω_1, ω_2. The problem of the inversion of the elliptic integrals, as we formulated it in § 14, presupposes however that in the elliptic functions $\varkappa^2$ (or,

in the Weierstrass normal form, g_2, g_3) have arbitrarily given values, and the complete solution of this problem therefore requires that not ω, but $\varkappa^2$ (respectively g_2, g_3) appear as the second independent variable, and that ω be expressed by these. The next considerations will be devoted to this problem.

We start from the problem of integrating a system of differential equations, which we call that of the elliptic differential equations:

$$\begin{aligned} \frac{dx}{dv} &= yz, \\ \frac{dy}{dv} &= -zx, \\ \frac{dz}{dv} &= -\varkappa^2 xy, \end{aligned} \tag{1}$$

under the subsidiary conditions:

$$\text{for } v = 0 \text{ we are to have } x = 0,\ y = 1,\ z = 1. \tag{2}$$

We saw in the previous section that, if

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4} \tag{3}$$

holds, this problem is solved by the elliptic functions, and indeed in such a way that x, y, z become single-valued functions of v in the whole v-plane, continuous except where they are infinite. But if $\varkappa^2$ is now given, the question arises whether for every value of $\varkappa^2$ there is a value of ω satisfying the condition (3), and whether there are several such values.

We prove first that the functions x, y, z are uniquely determined by the differential equations (1) together with the subsidiary conditions.

It follows first from (1) that $x^2 + y^2$, $\varkappa^2 x^2 + z^2$ are constant, and from the subsidiary conditions there follows:

$$y^2 = 1 - x^2, \qquad z^2 = 1 - \varkappa^2 x^2. \tag{4}$$

Suppose that a second system of functions x_1, y_1, z_1 satisfying the same conditions exists; then first

$$y_1^2 = 1 - x_1^2, \qquad z_1^2 = 1 - \varkappa^2 x_1^2,$$

and a simple differentiation shows that

$$\frac{x\, y_1 z_1 - x_1\, yz}{1 - \varkappa^2 x^2 x_1^2}$$

is constant. (The addition theorem of the elliptic integrals, from which one obtains the form of this expression, is confirmed directly by differentiation.) But from the subsidiary conditions it follows that this constant vanishes, and hence:

$$x\, y_1 z_1 = x_1\, yz,$$

from which one easily concludes, by squaring both sides:

$$x = x_1, \qquad y = y_1, \qquad z = z_1.$$

§ 52. The independent variable $\varkappa^2$. Linear differential equation for K.

We now show first that, and how, for an arbitrarily given value of $\varkappa^2$ one can find at least one value of ω satisfying the condition

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4} \tag{1}$$

This question is answered most completely by reduction to a linear differential equation of the second order.

Such a differential equation, however, is already contained in the formulae of § 23.

First of all it follows from § 23, (14):

$$d \log \varkappa^2 = i\pi\vartheta_{01}^4 \, d\omega,$$

or, in view of

$$\pi\vartheta_{00}^2 = 2K, \quad \pi\vartheta_{10}^2 = 2\varkappa K, \quad \pi\vartheta_{01}^2 = 2\varkappa' K \ [\S\,42,\ (3),\ (15)] : \tag{2}$$

$$\pi \, d(\varkappa^2) = 4i\varkappa^2\varkappa'^2 K^2 \, d\omega, \tag{3}$$

and from § 23, (19):

$$\frac{d}{d\omega}\frac{dK}{K^2 d\omega} = -\frac{4}{\pi^2}\varkappa^2\varkappa'^2 K^3.$$

If by means of (3) one introduces the variable $\varkappa^2$ in place of ω, there follows

$$\frac{d}{d(\varkappa^2)}\left(\varkappa^2\varkappa'^2\frac{dK}{d(\varkappa^2)}\right) = \frac{1}{4}K, \tag{4}$$

and this is the differential equation sought.

If we set further

$$-i\omega = \frac{K'}{K}, \tag{5}$$

$$-iK^2 d\omega = K\,dK' - K'\,dK, \tag{6}$$

then by (3) there results:

$$\varkappa^2\varkappa'^2\left(K\frac{dK'}{d(\varkappa^2)} - K'\frac{dK}{d(\varkappa^2)}\right) = -\frac{\pi}{4}, \tag{7}$$

and hence, by differentiating once again:

$$\frac{1}{K}\frac{d}{d(\varkappa^2)}\left(\varkappa^2\varkappa'^2\frac{dK}{d(\varkappa^2)}\right) = \frac{1}{K'}\frac{d}{d(\varkappa^2)}\left(\varkappa^2\varkappa'^2\frac{dK'}{d(\varkappa^2)}\right),$$

from which it is to be seen that K' is the second particular integral of the differential equation (4).

This differential equation can be integrated by hypergeometric series (Gauss, Disq. circa seriem infinitum Werke, Bd. III, S. 125).

If for the moment we set $\varkappa^2 = x$, $K = y$, then the differential equation (4) reads

$$x(1-x)\frac{d^2y}{dx^2} + (1-2x)\frac{dy}{dx} - \frac{1}{4}y = 0, \tag{8}$$

and it arises from the general Gauss differential equation

$$(x-x^2)\frac{d^2y}{dx^2} + [\gamma - (\alpha+\beta+1)x]\frac{dy}{dx} - \alpha\beta y = 0 \tag{9}$$

if one sets $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$. One of its particular integrals is therefore

$$F(\tfrac{1}{2},\tfrac{1}{2},1,x) = 1 + \sum_{\nu=1}^{\infty}\Big(\frac{1.3.5\ldots 2\nu-1}{2.4\ldots 2\nu}\Big)^2 x^\nu \tag{10}$$

$$= \sum_{\nu=0}^{\infty}\frac{\Pi(2\nu)^2}{\Pi(\nu)^4}\Big(\frac{x}{16}\Big)^\nu,$$

and this series is convergent as long as the absolute value of x is less than 1. As the second particular integral one may choose

$$F(\tfrac{1}{2},\tfrac{1}{2},1,1-x), \tag{11}$$

which, however, has a different domain of convergence. If we think of x as a complex variable represented in a plane, then (10) converges in a circle described about the origin with radius 1, and (11) in an equal circle described about the point 1, so that the common domain of convergence consists of a lune containing the part of the real axis that lies between 0 and 1. But one can also, and this is important for us, set up the second particular integral in a form which converges in the same circle as the series (10). Here it is to be observed that the second particular integral becomes infinite for $x = 0$.

This second particular integral is obtained by the following passage to the limit. If we denote the hypergeometric series after Gauss by

$$F(\alpha,\beta,\gamma,x) = 1 + \frac{\alpha.\beta}{1.\gamma}x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1.2.\gamma(\gamma+1)}x^2 + \cdots$$

$$= \frac{\Pi(\gamma-1)}{\Pi(\alpha-1)\Pi(\beta-1)}\sum_{\nu=0}^{\infty}\frac{\Pi(\alpha+\nu-1)\Pi(\beta+\nu-1)}{\Pi(\gamma+\nu-1)\Pi(\nu)}x^\nu,$$

then in general one obtains the two particular integrals of (9):

$$y_1 = F_1(x) = F(\alpha,\beta,\gamma,x),$$
$$y_2 = F_2(x) = x^{1-\gamma}(1-x)^{\gamma-\alpha-\beta}F(1-\alpha,1-\beta,2-\gamma,x).$$

But since for $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$ these become identical with one another, one first sets $\alpha = \beta = \frac{1}{2}$, $\gamma = 1+\varepsilon$, hence

$$y_1 = F(\tfrac{1}{2},\tfrac{1}{2},1+\varepsilon,x),$$
$$y_2 = \Big(\frac{1-x}{x}\Big)^\varepsilon F(\tfrac{1}{2},\tfrac{1}{2},1-\varepsilon,x).$$

But we have

$$F(\tfrac{1}{2},\tfrac{1}{2},1\pm\varepsilon,x)=\frac{\Pi(\pm\varepsilon)}{[\Pi(-\frac{1}{2})]^2}\sum_{\nu=0}^{\infty}\frac{\Pi(\nu-\frac{1}{2})^2}{\Pi(\nu\pm\varepsilon)\Pi(\nu)}x^\nu,$$

and, with the addition of a constant factor, one may also take the following as the second particular integral:

$$\frac{1}{2\varepsilon}\cdot\sum_{\nu=0}^{\infty}\frac{\Pi(\nu-\frac{1}{2})^2}{\Pi(\nu)}\Big[\frac{-1}{\Pi(\nu+\varepsilon)}+\Big(\frac{x}{1-x}\Big)^{-\varepsilon}\frac{1}{\Pi(\nu-\varepsilon)}\Big]x^\nu.$$

If we here develop in powers of ε and then set $\varepsilon=0$, it follows that:

$$\sum_{\nu=0}^{\infty}\frac{\Pi(\nu-\frac{1}{2})^2}{\Pi(\nu)^2}x^\nu\Big[\frac{\Pi'(\nu)}{\Pi(\nu)}-\frac{1}{2}\log\frac{x}{1-x}\Big],$$

for which, using the Gauss formula

$$\frac{\Pi(\nu-\frac{1}{2})\Pi(\nu)}{\Pi(2\nu)}=\sqrt{\pi}\,2^{-2\nu}$$

one may also write, on discarding a constant factor:

$$\sum_{\nu=0}^{\infty}\frac{\Pi(2\nu)^2}{\Pi(\nu)^4}\Big(\frac{x}{16}\Big)^\nu\Big[\frac{\Pi'(\nu)}{\Pi(\nu)}-\frac{1}{2}\log\frac{x}{1-x}\Big]. \tag{12}$$

But for a positive integer ν we have:

$$\frac{\Pi'(\nu)}{\Pi(\nu)}=\Pi'(0)+1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{\nu},$$

and if we therefore set, for brevity:

$$\begin{aligned}F(x)&=\sum_{\nu=0}^{\infty}\frac{\Pi(2\nu)^2}{\Pi(\nu)^4}\Big(\frac{x}{16}\Big)^\nu,\\ G(x)&=2\sum_{\nu=1}^{\infty}\frac{\Pi(2\nu)^2}{\Pi(\nu)^4}\Big(1+\frac{1}{2}+\frac{1}{3}+\cdots\frac{1}{\nu}\Big)\Big(\frac{x}{16}\Big)^\nu,\end{aligned} \tag{13}$$

then the two particular solutions of the differential equation (8) result, if for the second one takes a linear combination of (10) and (12):

$$y_1=F(x),\qquad y_2=\frac{1}{2}G(x)-\frac{1}{2}\log\frac{x}{16(1-x)}F(x). \tag{14}$$

But in order now to represent the particular integrals K and K' by these functions, we must investigate the behaviour of K,K' for $q=0$, for which at the same time $\varkappa^2=0$.

But for $q=0$ we have, by § 25 and § 42:

$$q^{-1}\varkappa^2=16,\qquad 2K=\pi,$$

thus:

$$\operatorname{Lim}\Big(\log\frac{\varkappa^2}{16}-i\pi\omega\Big)=0,$$

on the other hand we have $2iK'=\pi\vartheta_{00}^2\,\omega$ (§ 42), or:

$$2K'+\log\frac{\varkappa^2}{16}=\Big(\log\frac{\varkappa^2}{16}-i\pi\omega\Big)+i\pi\omega(1-\vartheta_{00}^2),$$

from which:

$$\text{Lim}_{\varkappa^2=0}\Big(K' + \frac{1}{2}\log\frac{\varkappa^2}{16}\Big) = 0. \tag{15}$$

Since K and K' are now particular integrals of (8) (for $x = \varkappa^2$), both have the form:

$$K = a_1 y_1 + a_2 y_2, \qquad K' = a_1' y_1 + a_2' y_2,$$

in which a_1, a_2, a_1', a_2' are constant. Since K remains finite for $x = 0$ and takes the value $\frac{1}{2}\pi$, we have $a_2 = 0$, $a_1 = \frac{1}{2}\pi$; and in order that $K' + \frac{1}{2}\log x$ remain finite and [by (15)] take the value $\log 4$, $a_2' = 1$ and $a_1' = 0$ must hold. Thus there results, if one puts $x = \varkappa^2$, $1 - x = \varkappa'^2$:

$$K = \frac{\pi}{2}F(\varkappa^2), \qquad K' = \frac{1}{2}G(\varkappa^2) - \frac{1}{2}\log\frac{\varkappa^2}{16\varkappa'^2}F(\varkappa^2) \tag{16}$$

and

$$q = e^{-\frac{\pi K'}{K}} = \frac{\varkappa^2}{16\varkappa'^2}\, e^{-\frac{G(\varkappa^2)}{F(\varkappa^2)}}, \tag{17}$$

in which, as already remarked, the series $G(\varkappa^2)$, $F(\varkappa^2)$ are convergent so long as the absolute value of $\varkappa^2$ is a proper fraction. In the differential equation (4) there lie, to be sure, the means of continuing the expressions found for K, K', q beyond this domain of convergence. One reaches this more simply, however, by the application of the Landen transformation.

We bound the plane of the complex values $\varkappa^2$ by laying two cuts along the real axis, from 0 to $-\infty$ and from 1 to ∞. In the plane so bounded all roots of $\varkappa^2, \varkappa'^2$ are then uniquely determined by the requirement that, for positive properly fractional $\varkappa^2$, they are to be real and positive. Now by the formulae of the Landen transformation [§ 44, (28)], if we denote by $\varkappa_1, \varkappa_1', K_1, K_1'$ the functions of ω that arise from $\varkappa, \varkappa', K, K'$ when ω is replaced by 2ω, we have:

$$\begin{aligned} \varkappa_1 &= \frac{1-\varkappa'}{1+\varkappa'}, \qquad \varkappa_1' = \frac{2\sqrt{\varkappa'}}{1+\varkappa'}, \\ 2K_1 &= (1+\varkappa')K, \qquad K_1' = (1+\varkappa')K'. \end{aligned} \tag{18}$$

If we therefore replace ω by 2ω in (17) and extract the square root, it follows that:

$$q = \frac{1}{4}\frac{\varkappa_1}{\varkappa_1'}\, e^{-\frac{1}{2}\frac{G(\varkappa_1{}^2)}{F(\varkappa_1{}^2)}}. \tag{19}$$

Here the domain of convergence now extends over the unit circle lying in the plane $\varkappa_1{}^2$, and this circle corresponds to the whole plane $\varkappa^2$. For the real part of $\varkappa'$ is positive in the whole plane of the $\varkappa^2$, with the exception of the two banks of the cut running from $+1$ to $+\infty$, on which $\varkappa'$ is purely imaginary. From this it follows that the absolute value of $\varkappa_1$ is less than 1, and that the periphery of the unit circle in the plane $\varkappa_1$ corresponds to the two banks of this cut.

But one can improve the convergence of these expressions still further by a renewed application of the Landen transformation. If we denote the result of a further doubling of ω by the index 2, then it follows that:

$$\begin{aligned} \sqrt{\varkappa_2} &= \frac{1-\sqrt{\varkappa'}}{1+\sqrt{\varkappa'}}, \\ 4K_2 &= (1+\sqrt{\varkappa'})^2 K, \quad K_2' = (1+\sqrt{\varkappa'})^2 K', \\ q &= \sqrt{\frac{\varkappa_2}{4\varkappa_2'}}\, e^{-\frac{1}{4}\frac{G(\varkappa_2{}^2)}{F(\varkappa_2{}^2)}}. \end{aligned} \tag{20}$$

in which the domain of convergence now covers the plane of the $\varkappa^2$ twice (with a branching at the point $\varkappa^2 = 1$).

This operation can be continued, by deriving from $\varkappa_2$, according to formula (17), a new quantity $\varkappa_3$ through doubling of ω, and from this in the same way $\varkappa_4$, and so on. One then obtains a sequence of expressions for q whose convergence becomes ever better. For an arbitrary ν we have:

$$q = \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu}}\, e^{-\frac{G(\varkappa_\nu{}^2)}{2^\nu F(\varkappa_\nu{}^2)}}. \tag{21}$$

The first equation (18)

$$\varkappa_1 = \frac{\varkappa^2}{(1+\varkappa')^2}$$

shows that, so long as the absolute value of $\varkappa^2$ is less than 1, and consequently the absolute value of $1+\varkappa'$ greater than 1, the absolute value of $\varkappa_1$ is less than that of $\varkappa^2$, and consequently

$\varkappa_\nu$ approaches the limit zero as ν increases infinitely. From this one obtains for q the representation:

$$q = \operatorname*{Lim}_{\nu=\infty} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu}} = \operatorname{Lim} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4}}, \tag{22}$$

which is suited to computation for real values of $\varkappa$.

From each of the formulae (21) one can derive a series development of q in ascending powers of $\varkappa_\nu$, whose convergence increases as ν increases. If in particular one applies formula (20), one obtains a development which was communicated by Weierstrass in the Monatsberichte der Berliner Akademie for the year 1883, and whose domain of convergence fills the plane $\varkappa^2$ twice.

The coefficients of this development are most simply computed from:

$$\sqrt{\varkappa_2} = \frac{\vartheta_{10}(0,4\omega)}{\vartheta_{00}(0,4\omega)} = \frac{2q + 2q^9 + 2q^{25} + \cdots}{1 + 2q^4 + 2q^{16} + \cdots}, \tag{23}$$

and one thus obtains, by the method of undetermined coefficients:

$$q = \frac{\sqrt{\varkappa_2}}{2} + 2\left(\frac{\sqrt{\varkappa_2}}{2}\right)^5 + 15\left(\frac{\sqrt{\varkappa_2}}{2}\right)^9 + 150\left(\frac{\sqrt{\varkappa_2}}{2}\right)^{13} + 1707\left(\frac{\sqrt{\varkappa_2}}{2}\right)^{17} + \cdots \tag{24}$$

In the same way, conversely, $\sqrt{\varkappa_2}$ can also be developed by (23) into a series proceeding in powers of q:

$$\tfrac{1}{2}\sqrt{\varkappa_2} = q - 2q^5 + 5q^9 - 10q^{13} + 18q^{17} \cdots, \tag{25}$$

which one also obtains by inversion of the series (24).

With this the question posed at the beginning of this section is completely answered.

As regards the application of the formulae developed here to the numerical computation of q, it may still be remarked that, when $\varkappa^2$ lies nearer to 1, one obtains a better converging procedure if one interchanges $\varkappa^2$ with $\varkappa'^2$. The calculation then yields at first not q, but

$$q' = e^{-\frac{\pi i}{\omega}},$$

from which, however, one obtains q from the formula:

$$\log q \log q' = \pi^2.$$

§ 53. The Solutions of the Equation $j(\omega) = j(\omega')$.

The results of § 51 suffice, to begin with, to establish the relations among the various values of ω which lead to the same value of $\varkappa^2$. Let then:

$$\frac{\vartheta_{10}^4(0,\omega')}{\vartheta_{00}^4(0,\omega')} = \frac{\vartheta_{10}^4(0,\omega)}{\vartheta_{00}^4(0,\omega)} = \varkappa^2. \tag{1}$$

If we set

$$v = \pi\vartheta_{00}^2(0,\omega)\, u = \pi\vartheta_{00}^2(0,\omega')\, u', \tag{2}$$

then (by § 42) both the functions

$$\frac{\vartheta_{00}(0,\omega)\vartheta_{11}(u,\omega)}{\vartheta_{10}(0,\omega)\vartheta_{01}(u,\omega)}, \quad \frac{\vartheta_{01}(0,\omega)\vartheta_{10}(u,\omega)}{\vartheta_{10}(0,\omega)\vartheta_{01}(u,\omega)}, \quad \frac{\vartheta_{01}(0,\omega)\vartheta_{00}(u,\omega)}{\vartheta_{00}(0,\omega)\vartheta_{01}(u,\omega)}, \tag{3}$$

as well as

$$\frac{\vartheta_{00}(0,\omega')\vartheta_{11}(u',\omega')}{\vartheta_{10}(0,\omega')\vartheta_{01}(u',\omega')}, \quad \frac{\vartheta_{01}(0,\omega')\vartheta_{10}(u',\omega')}{\vartheta_{10}(0,\omega')\vartheta_{01}(u',\omega')}, \quad \frac{\vartheta_{01}(0,\omega')\vartheta_{00}(u',\omega')}{\vartheta_{00}(0,\omega')\vartheta_{01}(u',\omega')}, \tag{4}$$

put for x, y, z, satisfy the differential equations (1) of the preceding section, and the corresponding ones among these functions are therefore identical. Now if $u' = \frac{1}{2}$, then $\vartheta_{10}(u',\omega')$ vanishes, and $\vartheta_{10}(u,\omega)$ must then vanish as well. Accordingly there follows from (2), with regard to § 21:

$$\vartheta_{00}^2(0,\omega') = \vartheta_{00}^2(0,\omega)(\alpha + \beta\omega), \tag{5}$$

in which α, β are integers which satisfy the condition

$$\alpha \equiv 1, \quad \beta \equiv 0 \pmod 2 \tag{6}$$

If $u' = \omega' : 2$, then $\vartheta_{01}(u',\omega')$ and $\vartheta_{01}(u,\omega)$ vanish, and therefore, as above,

$$\vartheta_{00}^2(0,\omega')\,\omega' = \vartheta_{00}^2(0,\omega)(\gamma + \delta\omega), \tag{7}$$

in which

$$\gamma \equiv 0, \quad \delta \equiv 1 \pmod 2. \tag{8}$$

From this it follows:

(9) $$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}.$$

But since one may conclude in the same way:

$$\vartheta_{00}^2(0,\omega) = \vartheta_{00}^2(0,\omega')(\alpha' + \beta'\omega'),$$
$$\vartheta_{00}^2(0,\omega)\,\omega = \vartheta_{00}^2(0,\omega')(\gamma' + \delta'\omega'),$$

in which $\alpha', \beta', \gamma', \delta'$ are likewise integers, the substitution of (5) in these equations gives:

$$1 = (\alpha + \beta\omega)(\alpha' + \beta'\omega'),$$
$$\omega = (\alpha + \beta\omega)(\gamma' + \delta'\omega').$$

If in these equations one expresses ω' in terms of ω according to (9), one obtains

$$1 = \alpha'(\alpha + \beta\omega) + \beta'(\gamma + \delta\omega),$$
$$\omega = \gamma'(\alpha + \beta\omega) + \delta'(\gamma + \delta\omega),$$

and since ω is not real, each of these equations splits into two others:

$$\alpha\alpha' + \gamma\beta' = 1, \quad \beta\alpha' + \delta\beta' = 0,$$
$$\alpha\gamma' + \gamma\delta' = 0, \quad \beta\gamma' + \delta\delta' = 1.$$

Consequently

$$(\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma') = 1,$$

and therefore, since ω, ω' must both have a positive imaginary part, so that by (9) the determinant $\alpha\delta - \beta\gamma$ must be positive:

$$\alpha\delta - \beta\gamma = 1,$$
$$\alpha = \delta', \quad \delta = \alpha', \quad \gamma = -\gamma', \quad \beta = -\beta'.$$

1. Thus ω, ω' are connected with one another by a linear transformation, and indeed [by (6), (8)] by one of the first class (§ 36). That conversely two such values ω, ω' also give the same value $\varkappa^2$ has already been shown above.

A similar theorem follows from this as an immediate consequence, for the invariant $j(\omega)$ (§ 46).

2. The equation:

(10) $$j(\omega') = j(\omega)$$

is then and **only** then satisfied, when ω' is connected with ω by some linear transformation

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

that is, when

(11) $$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \quad \alpha\delta - \beta\gamma = 1$$

holds.

For if we denote the values of $\varkappa^2, \varkappa'^2$ belonging to ω' by λ^2, λ'^2, then equation (10) can be written thus:

$$\frac{(1-\lambda^2\lambda'^2)^3}{\lambda^4\lambda'^4} = \frac{(1-\varkappa^2\varkappa'^2)^3}{\varkappa^4\varkappa'^4},$$

and with respect to λ^2 it is an equation of the sixth degree, whose roots

$$\varkappa^2, \quad \varkappa'^2, \quad \frac{1}{\varkappa^2}, \quad \frac{1}{\varkappa'^2}, \quad -\frac{\varkappa'^2}{\varkappa^2}, \quad -\frac{\varkappa^2}{\varkappa'^2}$$

are. Consequently (having regard to § 45) one of the following relations holds:

$$\lambda^2 = \varkappa^2(\omega), \quad \varkappa^2\left(-\frac{1}{\omega}\right), \quad \varkappa^2(\omega+1), \quad \varkappa^2\left(\frac{\omega-1}{\omega}\right),$$

$$\varkappa^2\left(-\frac{1}{\omega+1}\right), \quad \varkappa^2\left(\frac{\omega}{1-\omega}\right),$$

whereby the correctness of the second theorem follows from the first.

The variable ω is here always restricted to a positive imaginary part, and if we call two numbers ω, ω' that are connected by an equation (11) equivalent, then all numbers equivalent to ω have a positive imaginary part. We can also state our theorem 2. in this way:

3. The function $j(\omega)$ has one and the same value for all equivalent values ω and only for these[1].

To every value of ω there belongs a definite value of $j(\omega)$, and to every value of $j(\omega)$ a value of ω and the totality of the equivalent values ω'. If therefore a one-valued function $\Phi(\omega)$ of ω satisfies the condition:

$$\Phi\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \Phi(\omega),$$

then $\Phi(\omega)$ is a one-valued function of $j(\omega)$, and if we therefore assume that $\Phi(\omega)$ becomes infinite only for a finite number of values $j(\omega)$ and only in finite order, it follows that $\Phi(\omega)$ is a rational function of $j(\omega)$.

The assumption always holds, for example, when $\Phi(\omega)$ stands in an algebraic connection with $j(\omega)$.

§ 54. The Modular Functions.

Let $\psi(\omega)$ be a single-valued function of ω. If one applies to ω a linear transformation $S = \begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}$, then $\psi(\omega)$ passes into another function

$$\psi\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) \tag{1}$$

which we shall denote by $\psi \,|\, S$.

The function $\psi(\omega)$ may possibly remain unchanged under certain transformations S (e. g. always under the identity transformation). All transformations that leave a function $\psi(\omega)$ unchanged form a group $\mathfrak{G}$. For if

$$\psi \,|\, S = \psi, \qquad \psi \,|\, S_1 = \psi,$$

then also

$$\psi \,|\, SS_1 = \psi \,|\, S_1 = \psi.$$

[1] This theorem was first proved by Dedekind (Crelle, Bd. 83). Dedekind accordingly calls the function $\mathrm{val}(\omega) = 3^{-3}2^{-6}j(\omega)$ the valence of ω.

The group $\mathfrak{G}$ is a subgroup of the group $\mathfrak{L}$ of all linear transformations.

If $\mathfrak{G}$ is a subgroup of $\mathfrak{L}$ of finite index $(\mathfrak{L}, \mathfrak{G})$, and $\psi(\omega)$ a function of ω with the property that conversely

$$\psi(\omega') = \psi(\omega)$$

in turn entails a linear connection

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}$$

in which $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ is a transformation belonging to $\mathfrak{G}$, then $\psi(\omega)$ is called a modular function belonging to the group $\mathfrak{G}$.

We restrict ourselves to the consideration of those modular functions that stand in an algebraic connection with the modulus $\varkappa^2$, and hence also with $j(\omega)$.

Under this hypothesis the following general theorem can be stated:

If $\psi(\omega)$ belongs to the group $\mathfrak{G}$ and $\chi(\omega)$ remains unchanged under the transformations of $\mathfrak{G}$, then $\chi(\omega)$ is **rationally** expressible through $\psi(\omega)$.

For in the first place $\chi(\omega)$ is an algebraic function of $\psi(\omega)$, and $\psi(\omega)$, as an algebraic function of $j(\omega)$, can take every value. But to each value of $\psi(\omega)$ there belongs, by hypothesis, only one value of $\chi(\omega)$, and therefore $\chi(\omega)$, as a one-valued algebraic function of $\psi(\omega)$, is rational.

The study of the groups contained in $\mathfrak{L}$ and of the modular functions belonging to them forms the principal subject of the great work of Klein and Fricke: „Vorlesungen über die Theorie der elliptischen Modulfunktionen“ (2 Bde., Leipzig, Teubner, 1890, 1892). Here we consider only a few quite special cases of these groups and functions, upon which we have come in the course of our investigations.

A large class of subgroups of the group $\mathfrak{L}$ with finite index is formed by the so-called congruence groups, which, if m is any integer, are characterised by the four congruences

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} (\mathrm{mod}\ m)$$

The modular functions belonging to these groups are called congruence moduli of the mth level. For $m = 2$ this is the group $\mathfrak{A}$ (§ 36). We shall make use of the following theorems:

1. By § 52 the modulus $\varkappa^2$ is a modular function belonging to this group; and since by § 36 all these transformations can be composed from the two

$$\begin{pmatrix} 1, & 0 \\ 2, & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1, & 2 \\ 0, & 1 \end{pmatrix}$$

we can state the theorem:

Every modular function which remains unchanged under the two substitutions

$$(\omega,\ \omega + 2), \qquad \left(\omega,\ \frac{\omega}{1 + 2\omega}\right)$$

is a rational function of $\varkappa^2$.

2. The function $\varkappa^2\varkappa'^2$ belongs to the group $\mathfrak{A}'$ formed jointly from the first and second classes of § 36. This group can be derived by repetition of the two transformations

$$\begin{pmatrix} 1, & 0 \\ 2, & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

and we can therefore state the theorem:

Every modular function which remains unchanged under the two substitutions

$$(\omega,\ \omega+2), \qquad \left(\omega,\ -\frac{1}{\omega}\right)$$

is a rational function of $\varkappa^2\varkappa'^2$. From this there follows further:

3. Every modular function which remains unchanged under $(\omega, \omega+2)$ and changes its sign under $\left(\omega, -\frac{1}{\omega}\right)$ is the product of $(\varkappa'^2-\varkappa^2)$ with a rational function of $\varkappa^2\varkappa'^2$.

4. The invariant $j(\omega)$ belongs to the group $\mathfrak{L}$ (§ 53), and hence the theorem:

Every modular function that remains unchanged under the two substitutions

$$(\omega,\ \omega+1), \qquad \left(\omega,\ -\frac{1}{\omega}\right)$$

is a rational function of $j(\omega)$.

We also introduce two other modular functions.

Multiplying the two equations in § 42, (3), the definitions of the functions $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ [§ 34, (1), (10)] give:

$$f(\omega)=\frac{\sqrt[6]{2}}{\sqrt[12]{\varkappa\varkappa'}}, \qquad f_1(\omega)=\sqrt[6]{2}\ \sqrt[12]{\frac{\varkappa'^2}{\varkappa}}, \qquad f_2(\omega)=\sqrt[6]{2}\ \sqrt[12]{\frac{\varkappa^2}{\varkappa'}}, \tag{2}$$

$$\frac{f_1(\omega)}{f(\omega)}=\sqrt[4]{\varkappa'}, \qquad \frac{f_2(\omega)}{f(\omega)}=\sqrt[4]{\varkappa}. \tag{3}$$

On the basis of § 46, (15), (16) we define two functions $\gamma_2(\omega)$, $\gamma_3(\omega)$:

$$\begin{aligned}\gamma_2(\omega)&=\sqrt[3]{j(\omega)}=\sqrt[3]{2^8}\ \frac{1-\varkappa^2\varkappa'^2}{\sqrt[3]{\varkappa^4\varkappa'^4}},\\ \gamma_3(\omega)&=\sqrt{j(\omega)-27\cdot 64}=\frac{8(2+\varkappa^2\varkappa'^2)(\varkappa'^2-\varkappa^2)}{\varkappa^2\varkappa'^2},\end{aligned} \tag{4}$$

which, by (2) and (3), admit of the following representation as one-valued functions of ω:

$$\begin{aligned}\gamma_2(\omega)&=\frac{f(\omega)^{24}-16}{f(\omega)^8},\\ \gamma_3(\omega)&=\frac{[f(\omega)^{24}+8][f_1(\omega)^8-f_2(\omega)^8]}{f(\omega)^8},\end{aligned} \tag{5}$$

for which, by the fundamental formulae for the f-functions [§ 34, (11), (12)], one may also put:

$$\gamma_2(\omega)=f(\omega)^8f_1(\omega)^8+f(\omega)^8f_2(\omega)^8-f_1(\omega)^8f_2(\omega)^8, \tag{6}$$

$$\gamma_3(\omega)=\tfrac{1}{2}[f(\omega)^8+f_1(\omega)^8][f(\omega)^8+f_2(\omega)^8][f_1(\omega)^8-f_2(\omega)^8]. \tag{7}$$

Thus $x=f^8,\ -f_1^8,\ -f_2^8$ are the roots of the cubic equation

$$x^3-\gamma_2x-16=0, \tag{8}$$

and $4\gamma_3^2$ is the discriminant of this equation.

Also noteworthy are the differential equations:

$$d\varkappa^2=-d\varkappa'^2=\pi i\,\vartheta_{00}^4\,\varkappa^2\varkappa'^2\,d\omega\ [\S\,52,\ (3)], \tag{9}$$

(10) $$d\varkappa^2\varkappa'^2 = -\pi i\,\vartheta_{00}^4(\varkappa^2-\varkappa'^2)\varkappa^2\varkappa'^2\,d\omega,$$

(11) $$df(\omega)^8 = \frac{\pi i}{3}\vartheta_{00}^4[f_2(\omega)^8 - f_1(\omega)^8]\,d\omega,$$

(12) $$d\gamma_2(\omega) = -\frac{2\pi i}{3}\gamma_3(\omega)\eta(\omega)^4\,d\omega \text{ [by (5) and § 34, (10)]},$$

(13) $$dj(\omega) = -2\pi i\,\gamma_2(\omega)^2\gamma_3(\omega)\eta(\omega)^4\,d\omega.$$

One then obtains for the linear fundamental transformations, from § 34, (13), (14), (15):

(14) $$\begin{aligned} &\gamma_2(\omega+1) = e^{-\frac{2\pi i}{3}}\gamma_2(\omega), \quad &&\gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega),\\ &\gamma_3(\omega+1) = -\gamma_3(\omega), &&\gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega), \end{aligned}$$

and from § 40 one finds in general:

(15) $$\begin{aligned} &\gamma_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varrho\,\gamma_2(\omega),\\ &\gamma_3\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = (-1)^{\alpha\gamma+\beta\gamma+\beta\delta}\gamma_3(\omega), \end{aligned}$$

in which ϱ has the meaning it has in § 40, (3):

$$\varrho = e^{-\frac{2\pi i}{3}(\alpha\gamma+\beta\delta-\alpha\beta-\alpha^2\beta\delta)}.$$

These, then, are modular functions, and the groups to which they belong are characterised by the two congruences:

$$\begin{aligned} -\alpha\beta+\alpha\gamma+\beta\delta-\alpha^2\beta\gamma &\equiv 0 \pmod 3,\\ \alpha\gamma+\beta\gamma+\beta\delta &\equiv 0 \pmod 2. \end{aligned}$$

Accordingly we give our Theorem 4. the following supplement, which follows from the formulae (14).

5. A modular function which changes its sign under the two substitutions

$$(\omega,\omega+1), \qquad \left(\omega,-\frac{1}{\omega}\right)$$

is the product of $\gamma_3(\omega)$ with a rational function of $j(\omega)$.

6. A modular function which remains unchanged under the substitution $\left(\omega,-\frac{1}{\omega}\right)$ and which under $(\omega,\omega+1)$ takes on the factor $e^{\mp\frac{2\pi i}{3}}$ is the product of $\gamma_3(\omega)^{\pm 1}$ with a rational function $j(\omega)$.

7. A modular function which changes sign under the substitutions $\left(\omega,-\frac{1}{\omega}\right)$ and which under $(\omega,\omega+1)$ takes on the factor $-e^{\mp\frac{2\pi i}{3}}$ is the product of $\gamma_2(\omega)\gamma_3(\omega)^{\pm 1}$ with a rational function of $j(\omega)$.

We shall occupy ourselves more thoroughly with the modular functions in the second part.

§ 55. Representation of the elliptic functions by v and $\varkappa^2$.

If we conceive the inversion problem of the elliptic integrals, as in § 51, as the task of integrating the elliptic differential equations, then it demands the representation of the three functions $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ by the two independent variables v, $\varkappa^2$. This task is solved by § 42, but with respect to the variable $\varkappa^2$ only implicitly. One can, however, also find representations in which $\varkappa^2$ occurs explicitly, namely by series which proceed in powers of v and whose coefficients are rational functions of $\varkappa^2$. The coefficients of these series cannot, it is true, be represented by a general law; they can only be computed successively. These representations are owed chiefly to Weierstrass[9].

Since the σ-functions are throughout finite and continuous functions of u, they can be developed in unconditionally convergent power series in u, and if we accordingly develop the σ-functions occurring in § 45, (2) in this way, we obtain a solution of our task. We therefore put

$$
\begin{aligned}
\sigma(v,\, 2K,\, 2iK') &= \sum_{0,\infty}^{\nu} A_\nu \frac{v^{2\nu+1}}{(2\nu+1)!},\\
\sigma_{00}(v,\, 2K,\, 2iK') &= \sum_{0,\infty}^{\nu} B_\nu \frac{v^{2\nu}}{(2\nu)!},\\
\sigma_{01}(v,\, 2K,\, 2iK') &= \sum_{0,\infty}^{\nu} C_\nu \frac{v^{2\nu}}{(2\nu)!},\\
\sigma_{10}(v,\, 2K,\, 2iK') &= \sum_{0,\infty}^{\nu} D_\nu \frac{v^{2\nu}}{(2\nu)!},
\end{aligned}
\tag{1}
$$

and the A_ν, B_ν, C_ν, D_ν are the functions of $\varkappa^2$, or of ω, that are to be determined. If we regard them first as functions of ω, then the linear transformations [§ 45, (3)] give:

$$
\begin{aligned}
A_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu A_\nu(\omega),\\
B_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu B_\nu(\omega),\\
C_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu D_\nu(\omega),\\
D_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu C_\nu(\omega).
\end{aligned}
\tag{2}
$$

$$
\begin{aligned}
\varkappa'^{2\nu} A_\nu(\omega+1) &= A_\nu(\omega),\\
\varkappa'^{2\nu} B_\nu(\omega+1) &= C_\nu(\omega),\\
\varkappa'^{2\nu} C_\nu(\omega+1) &= B_\nu(\omega),\\
\varkappa'^{2\nu} D_\nu(\omega+1) &= D_\nu(\omega).
\end{aligned}
\tag{3}
$$

$$
\begin{aligned}
A_\nu(\omega+2) &= A_\nu(\omega),\\
B_\nu(\omega+2) &= B_\nu(\omega),\\
C_\nu(\omega+2) &= C_\nu(\omega),\\
D_\nu(\omega+2) &= D_\nu(\omega).
\end{aligned}
\tag{4}
$$

From this we infer the characteristic properties of these coefficients as functions of $\varkappa^2$. It is evident at once from the meaning of the coefficients that, for every value of $\varkappa^2$ with the possible exception of $\varkappa^2 = 0, 1, \infty$, the coefficients $A_\nu(\varkappa^2)$, $B_\nu(\varkappa^2)$, $C_\nu(\varkappa^2)$, $D_\nu(\varkappa^2)$ are finite and continuous functions of $\varkappa^2$.

[9] On Weierstrass's theory of the elliptic functions cf. especially the "Formeln und Lehrsätze zum Gebrauch der elliptischen Funktionen" edited by H. A. Schwarz. On the series developments cf. also Königsberger, "Vorlesungen über die Theorie der elliptischen Funktionen", 25. Vorlesung.

But these functions remain finite for $\varkappa^2 = 0$ as well, and their values can easily be found by letting q, and hence $\varkappa^2$, pass into zero in the representations of § 37.

From the developments in § 25 one then obtains, in view of the fact that by § 34 and § 42, for a vanishing q,

$$\eta(\omega) = q^{\frac{1}{12}}, \qquad 2K = \pi$$

holds:

$$\begin{aligned} e^{\frac{v^2}{6}} \sin v &= \sum_{0,\infty}^{\nu} A_\nu(0)\, \frac{v^{2\nu+1}}{(2\nu+1)!}, \\ e^{\frac{v^2}{6}} &= \sum_{0,\infty}^{\nu} B_\nu(0)\, \frac{v^{2\nu}}{(2\nu)!}, \\ e^{\frac{v^2}{6}} &= \sum_{0,\infty}^{\nu} C_\nu(0)\, \frac{v^{2\nu}}{(2\nu)!}, \\ e^{\frac{v^2}{6}} \cos v &= \sum_{0,\infty}^{\nu} D_\nu(0)\, \frac{v^{2\nu}}{(2\nu)!}. \end{aligned} \tag{5}$$

Formulae (2) now give:

$$\begin{aligned} A_\nu(\varkappa'^2) &= (-1)^\nu A_\nu(\varkappa^2), \\ B_\nu(\varkappa'^2) &= (-1)^\nu B_\nu(\varkappa^2), \\ C_\nu(\varkappa'^2) &= (-1)^\nu D_\nu(\varkappa^2), \\ D_\nu(\varkappa'^2) &= (-1)^\nu C_\nu(\varkappa^2), \end{aligned} \tag{6}$$

from which it follows that these functions remain finite for $\varkappa^2 = 1$ as well, namely:

$$\begin{aligned} A_\nu(1) &= (-1)^\nu A_\nu(0), \\ B_\nu(1) &= (-1)^\nu B_\nu(0), \\ C_\nu(1) &= (-1)^\nu D_\nu(0), \\ D_\nu(1) &= (-1)^\nu C_\nu(0). \end{aligned} \tag{7}$$

The system (3) can moreover be written thus:

$$\begin{aligned} \varkappa'^{2\nu} A_\nu\left(\frac{-\varkappa^2}{\varkappa'^2}\right) &= A_\nu(\varkappa^2), \\ \varkappa'^{2\nu} B_\nu\left(\frac{-\varkappa^2}{\varkappa'^2}\right) &= C_\nu(\varkappa^2), \\ \varkappa'^{2\nu} C_\nu\left(\frac{-\varkappa^2}{\varkappa'^2}\right) &= B_\nu(\varkappa^2), \\ \varkappa'^{2\nu} D_\nu\left(\frac{-\varkappa^2}{\varkappa'^2}\right) &= D_\nu(\varkappa^2), \end{aligned} \tag{8}$$

from which, for an infinite $\varkappa^2$:

$$\begin{aligned} A_\nu(\varkappa^2) &= \varkappa^{2\nu} A_\nu(0), \\ B_\nu(\varkappa^2) &= \varkappa^{2\nu} D_\nu(0), \\ C_\nu(\varkappa^2) &= \varkappa^{2\nu} B_\nu(0), \\ D_\nu(\varkappa^2) &= \varkappa^{2\nu} C_\nu(0), \end{aligned} \tag{9}$$

and from this it is concluded that the coefficients $A_\nu, B_\nu, C_\nu, D_\nu$ are integral rational functions of $\varkappa^2$ of degree ν, and from § 54, 2, 3, it follows moreover that A_ν, B_ν are, for even

ν, expressed rationally by $\varkappa^2\varkappa'^2$, while for odd ν they are equal to the product of $(\varkappa'^2-\varkappa^2)$ with a rational function of $\varkappa^2\varkappa'^2$.

From B_ν one finds C_ν by means of the formula:

$$C_\nu(\varkappa^2)=\varkappa'^{2\nu}B_\nu\left(\frac{-\varkappa^2}{\varkappa'^2}\right) \tag{10}$$

and D_ν by applying (6) and (8):

$$D_\nu(\varkappa^2)=(-1)^\nu C_\nu(\varkappa'^2)=(-1)^\nu\varkappa^{2\nu}B_\nu\left(\frac{-\varkappa'^2}{\varkappa^2}\right). \tag{11}$$

Since the coefficients $A_\nu(0),B_\nu(0),C_\nu(0),D_\nu(0)$ can easily be determined from (5), the system of formulae developed here yields the coefficients A_ν,B_ν,C_ν,D_ν without further calculation up to $\nu=3$ inclusive.

One finds:

$$\begin{aligned}
&A_0=1,\quad A_1=0,\qquad A_2=-\frac{2}{3}(1-\varkappa^2\varkappa'^2),\\
&B_0=1,\quad B_1=\frac{1}{3}(\varkappa'^2-\varkappa^2),\quad B_2=\frac{1}{3}(1+2\varkappa^2\varkappa'^2),\\
&C_0=1,\quad C_1=\frac{1}{3}(1+\varkappa^2),\quad C_2=\frac{1}{3}(\varkappa'^4-2\varkappa^2),\\
&D_0=1,\quad D_1=-\frac{1}{3}(1+\varkappa'^2),\quad D_2=\frac{1}{3}(\varkappa^4-2\varkappa'^2).\\
&A_3=-\frac{8}{9}(\varkappa'^2-\varkappa^2)(2+\varkappa^2\varkappa'^2),\\
&B_3=\frac{1}{9}(\varkappa'^2-\varkappa^2)(5-2\varkappa^2\varkappa'^2),\\
&C_3=\frac{1}{9}(1+\varkappa^2)(5\varkappa'^4+2\varkappa^2),\\
&D_3=-\frac{1}{9}(1+\varkappa'^2)(5\varkappa^4+2\varkappa'^2).
\end{aligned}$$

Further coefficients can be calculated in the same way, though more laboriously, by developing the expressions for the σ-functions in § 32 in powers of q and using the higher powers of q as the lowest one has been used here. Weierstrass makes use, for the recurrent calculation of the coefficients, of certain partial differential equations, similar to the one that we shall derive in the next section for the function σ.

§ 56. Power series for the Weierstrass functions $\wp(u),\sigma(u)$.

The function $\sigma(u)$, regarded as a function of u, can, as has already been remarked, be developed into a series in powers of u converging in the whole u-plane, and by § 35, (8), (12) this development has the following form:

$$\sigma(u)=u+\alpha u^3+\alpha_1u^5+\alpha_2u^7+\alpha_3u^9+\cdots \tag{1}$$

The function

$$\wp(u)=-\frac{d^2\log\sigma(u)}{du^2}=\frac{\sigma'(u)^2-\sigma\sigma''(u)}{\sigma(u)^2} \tag{2}$$

can likewise be developed in ascending powers of u, but no longer into a series that is always convergent; rather, this series has a circle of convergence. It has the form:

$$\wp(u) = \frac{1}{u^2} + a + a_1 u^2 + a_2 u^4 + a_3 u^6 + \cdots \tag{3}$$

and one can determine the coefficients $a, a_1, a_2, a_3, \ldots$ in turn, from the differential equation [§ 46, (18)]

$$\wp'(u)^2 = 4\wp(u)^3 - g_2\wp(u) - g_3 \tag{4}$$

as functions of g_2, g_3. First of all we have

$$\wp'(u) = -\frac{2}{u^3} + 2a_1 u + 4a_2 u^3 + 6a_3 u^5 + \cdots,$$

$$\wp'(u)^2 = \frac{4}{u^6} - \frac{8a_1}{u^2} - 16a_2 + (4a_1^2 - 24a_3)u^2 + \cdots,$$

and since $\wp'(u)^2$ contains no term in u^{-4}, while in $\wp(u)^3$ the term $3au^{-4}$ occurs, $a = 0$ must hold. Accordingly there results:

$$\begin{aligned} \wp(u)^3 &= \frac{1}{u^6} + \frac{3a_1}{u^2} + 3a_2 + 3(a_3 + a_1^2)u^2 + \cdots, \\ \wp(u) &= \frac{1}{u^2} \qquad\qquad\qquad + a_1 u^2 + \cdots \end{aligned}$$

And if one substitutes this into equation (4), it follows that

$$a_1 = \frac{g_2}{20}, \quad a_2 = \frac{g_3}{28}, \quad a_3 = \frac{g_2^2}{1200}. \tag{5}$$

By integration of (2) one finds:

$$\sigma'(u) = \left(\frac{1}{u} - \frac{a_1 u^3}{3} - \frac{a_2 u^5}{5} - \frac{a_3 u^7}{7} \cdots\right)\sigma(u),$$

and from this equation it follows:

$$\alpha = 0, \quad \alpha_1 = \frac{-g_2}{240}, \quad \alpha_2 = \frac{-g_3}{840}, \quad \alpha_3 = \frac{-g_3^2}{161\,280}. \tag{6}$$

That $\alpha = 0$ follows also from the $\sigma'''(u) = 0$ proved in § 35; there, however, this was proved by a rather cumbersome route. We thus have the following beginning of the development of $\sigma(u)$, where, for the sake of clarity, the numerical denominators are decomposed into their prime factors:

$$\sigma(u) = u - \frac{g_2}{2^4 \cdot 3 \cdot 5} u^5 - \frac{g_3}{2^3 \cdot 3 \cdot 5 \cdot 7} u^7 - \frac{g_3^2}{2^9 \cdot 3^2 \cdot 5 \cdot 7} u^9 - \cdots \tag{7}$$

For the recurrent calculation of the coefficients in the power series for $\sigma(u)$ there serves a partial differential equation which may be regarded as a transformation of the partial differential equation for the ϑ-functions. We derive it in the following way.

The function $\sigma(u)$ was determined by the two period equations § 35, (9)

$$
\begin{aligned}
&\sigma(u+\omega_1) = -e^{\eta_1(2u+\omega_1)}\sigma(u),\\
&\sigma(u+\omega_2) = -e^{\eta_2(2u+\omega_2)}\sigma(u),\\
&\eta_1\omega_2 - \eta_2\omega_1 = \pi i
\end{aligned}
\tag{8}
$$

up to a factor independent of u. But differentiation of the last equation (8) yields:

$$
\begin{aligned}
\frac{\partial\eta_1}{\partial\omega_1}\omega_2 - \frac{\partial\eta_2}{\partial\omega_1}\omega_1 - \eta_2 = 0,\\
\frac{\partial\eta_1}{\partial\omega_2}\omega_2 - \frac{\partial\eta_2}{\partial\omega_2}\omega_1 + \eta_1 = 0,
\end{aligned}
$$

hence:

$$
\omega_2\left(\eta_1\frac{\partial\eta_1}{\partial\omega_1} + \eta_2\frac{\partial\eta_1}{\partial\omega_2}\right) = \omega_1\left(\eta_1\frac{\partial\eta_2}{\partial\omega_1} + \eta_2\frac{\partial\eta_2}{\partial\omega_2}\right),
$$

and we therefore introduce a quantity r, which we define as follows:

$$
\begin{aligned}
\eta_1\frac{\partial\eta_1}{\partial\omega_1} + \eta_2\frac{\partial\eta_1}{\partial\omega_1} = -r\omega_1,\\
\eta_1\frac{\partial\eta_1}{\partial\omega_2} + \eta_2\frac{\partial\eta_2}{\partial\omega_2} = -r\omega_2.
\end{aligned}
\tag{9}
$$

Now one shows, by differentiating the period equations (8), that the function

$$
S(u) = \frac{\partial^2\sigma}{\partial u^2} + 4\eta_1\frac{\partial\sigma}{\partial\omega_1} + 4\eta_2\frac{\partial\sigma}{\partial\omega_2} + 4ru^2\sigma
$$

satisfies the period equations (8), and is therefore equal to $C\sigma(u)$, where C is independent of u. But the development of $\sigma(u)$ in powers of u begins with u^3, while $\sigma(u)$ begins with u, and consequently we must have $C = 0$.

By (7) the development of $\partial\sigma/\partial\omega_1$ and $\partial\sigma/\partial\omega_2$ begins only with u^5, and the first terms of $\partial^2\sigma/\partial u^2$ and $4ru^2\sigma$ are $-g_2u^3/12$, $4ru^3$, from which $4r = g_2/12$ follows; accordingly the differential equation for σ takes the form:

$$
\frac{\partial^2\sigma}{\partial u^2} + 4\eta_1\frac{\partial\sigma}{\partial\omega_1} + 4\eta_2\frac{\partial\sigma}{\partial\omega_2} + \frac{g_2}{12}u^2\sigma = 0,
\tag{10}
$$

and incidentally there follow from (9) the relations:

$$
\begin{aligned}
\eta_1\frac{\partial\eta_1}{\partial\omega_1} + \eta_2\frac{\partial\eta_1}{\partial\omega_1} = -\frac{g_2}{48}\omega_1,\\
\eta_1\frac{\partial\eta_1}{\partial\omega_2} + \eta_2\frac{\partial\eta_2}{\partial\omega_2} = -\frac{g_2}{48}\omega_2.
\end{aligned}
\tag{11}
$$

Now in the differential equation (10) the variables g_2, g_3 are to be introduced in place of the variables ω_1, ω_2, and if we accordingly put

$$
\begin{aligned}
\frac{\partial\sigma}{\partial\omega_1} = \frac{\partial\sigma}{\partial g_2}\frac{\partial g_2}{\partial\omega_1} + \frac{\partial\sigma}{\partial g_3}\frac{\partial g_3}{\partial\omega_1},\\
\frac{\partial\sigma}{\partial\omega_2} = \frac{\partial\sigma}{\partial g_2}\frac{\partial g_2}{\partial\omega_2} + \frac{\partial\sigma}{\partial g_3}\frac{\partial g_3}{\partial\omega_2}
\end{aligned}
$$

then (10) takes the form:

$$\frac{\partial^2\sigma}{\partial u^2} + h_2\frac{\partial\sigma}{\partial g_2} + h_3\frac{\partial\sigma}{\partial g_3} + \frac{g_2}{12}u^2\sigma = 0, \tag{12}$$

where

$$h_2 = 4\left(\eta_1\frac{\partial g_2}{\partial\omega_1} + \eta_2\frac{\partial g_2}{\partial\omega_2}\right),$$
$$h_3 = 4\left(\eta_1\frac{\partial g_3}{\partial\omega_1} + \eta_2\frac{\partial g_3}{\partial\omega_2}\right).$$

The quantities h_2, h_3, expressed through g_2, g_3, now follow from the differential equation itself, if one substitutes the development (7) for σ. Accordingly, up to the 7th power of u:

$$\begin{aligned}
\frac{\partial^2\sigma}{\partial u^2} &= -\frac{g_2}{12}u^3 - 12\,g_3\frac{u^5}{2^4\cdot 3\cdot 5} - \frac{3}{8}g_2^2\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},\\
\frac{\partial\sigma}{\partial g_2} &= -\frac{u^5}{2^4\cdot 3\cdot 5},\\
\frac{\partial\sigma}{\partial g_3} &= -\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},\\
\frac{g_2}{12}u^2\sigma &= \frac{g_2}{12}u^3 - \frac{7}{24}g_2^2\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},
\end{aligned}$$

and if we substitute this into (12) and set the coefficients of u^5 and u^7 equal to zero, we find:

$$h_2 = -12\,g_3, \qquad h_3 = -\tfrac{2}{3}g_2^2,$$

and we obtain the differential equation:

$$\frac{\partial^2\sigma}{\partial u^2} - 12\,g_3\frac{\partial\sigma}{\partial g_2} - \frac{2}{3}g_2^2\frac{\partial\sigma}{\partial g_3} + \frac{g_2}{12}u^2\sigma = 0. \tag{13}$$

If we set up the series development for σ with undetermined coefficients:

$$\sigma = \Sigma A_\nu\frac{u^{2\nu+1}}{(2\nu+1)!}, \tag{14}$$

then from (13) we obtain the recursion formula:

$$A_\nu = 12\,g_3\frac{\partial A_{\nu-1}}{\partial g_2} + \frac{2}{3}g_2^2\frac{\partial A_{\nu-1}}{\partial g_3} - \frac{(2\nu-1)(2\nu-2)}{12}g_2^2A_{\nu-2}, \tag{15}$$

from which it is to be concluded that the A_ν are integral rational functions of g_2 and g_3, whose coefficients are rational numbers that can have only powers of 2 and of 3 in the denominator. The series (14) therefore consists of terms of the form

$$a_{m,n}\left(\tfrac{1}{2}g_2\right)^m(2g_3)^n\frac{u^{2\nu+1}}{(2\nu+1)!},$$

and from the homogeneity of the function σ [§ 37, (8), § 46, (13)] it follows that:

$$2m + 3n = \nu, \tag{16}$$

for the coefficients $a_{m,n}$ there then results from (15):

$$(17)\qquad \begin{aligned} a_{m,n} = {} & 3(m+1)a_{m+1,\,n-1} + \frac{16}{3}(n+1)a_{m-2,\,n+1} \\ & - \frac{1}{3}(2m+3n-1)(4m+6n-1)a_{m-1,\,n}, \end{aligned}$$

where $a_{00} = 1$, and those a for which one of the two indices is negative are to be set equal to zero.

From this $a_{m,n}$ can be found, once the earlier $a_{m,n}$, that is those in which $2m+3n$ is smaller than ν, have already been calculated.

One sees from this that the $a_{m,n}$ contain only powers of 3 in the denominator. In the „Formeln und Lehrsätzen zum Gebrauch der elliptischen Funktionen“ edited by H. A. Schwarz, after lectures of Weierstrass (2. Ausgabe 1893), the $a_{m,n}$ have been calculated up to $\nu = 17$. It appears thereby that these coefficients are not only integers, but that they also contain as factors powers of 3 which increase with ν. I am unable to give a proof of this.

Sixth Section.

Multiplication and Division of the Elliptic Functions.

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§ 57. Multiplication of the Elliptic Functions.

By the multiplication of the elliptic functions one understands the representation of the functions $\operatorname{sn} nv$, $\operatorname{cn} nv$, $\operatorname{dn} nv$ for an integer n as rational functions of $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ — a problem which, as is evident from the addition theorem, can always be solved.

The form of the solution follows easily from the consideration of the ϑ-functions.

As is to be seen immediately from § 21, the functions $\vartheta_{g_1,g_2}(nu)$ are Θ-functions of order n^2, whose characteristic is equal to $(0,0)$ for even n and equal to (g_1,g_2) for odd n. Moreover $\vartheta_{11}(nu)$ is an odd function, while $\vartheta_{00}(nu)$, $\vartheta_{10}(nu)$, $\vartheta_{01}(nu)$ are even functions of u. These functions can therefore be represented rationally by the ϑ-functions, and the theorems of § 21 yield the form of these expressions. If we again denote by $F^{(\nu)}(x,y)$ an integral, rational, homogeneous function of order ν, we obtain:

for even n:

$$(1)\qquad \begin{aligned} \vartheta_{11}(nu) &= \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{11}(u)\,F^{\frac{n^2-4}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right], \\ \vartheta_{g_1,g_2}(nu) &= F^{\frac{n^2}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right], \quad (g_1,g_2) = (10),(01),(00), \end{aligned}$$

for odd n:

$$(2)\qquad \vartheta_{g_1,g_2}(nu) = \vartheta_{g_1,g_2}(u)\,F^{\frac{n^2-1}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right].$$

If we divide these formulae by $\vartheta_{01}(u)^{n^2}$, the right-hand sides can be represented as integral rational functions of the elliptic functions $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ (§ 42). If, then, we set

$$v = \pi\vartheta_{00}^2 u, \quad \operatorname{sn} v = x, \quad \operatorname{cn} v = y, \quad \operatorname{dn} v = z \tag{3}$$

we can write the formulae (1), (2), on determining a constant factor suitably, as follows:

I. for even n:

$$\begin{aligned}
&\frac{\vartheta_{01}^{n^2}\vartheta_{00}}{\vartheta_{10}}\,\frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} = xyz\,A(x^2), && A(0) = n,\\
&\frac{\vartheta_{01}^{n^2}}{\vartheta_{10}}\,\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} = B(x^2), && B(0) = 1,\\
&\frac{\vartheta_{01}^{n^2}}{\vartheta_{00}}\,\frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} = C(x^2), && C(0) = 1,\\
&\vartheta_{01}^{n^2-1}\,\frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} = D(x^2), && D(0) = 1.
\end{aligned}$$

II. for odd n:

$$\begin{aligned}
&\frac{\vartheta_{00}\vartheta_{01}^{n^2-1}}{\vartheta_{10}}\,\frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} = x\,A(x^2), && A(0) = n,\\
&\frac{\vartheta_{01}^{n^2}}{\vartheta_{10}}\,\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} = y\,B(x^2), && B(0) = 1,\\
&\frac{\vartheta_{01}^{n^2}}{\vartheta_{00}}\,\frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} = z\,C(x^2), && C(0) = 1,\\
&\vartheta_{01}^{n^2-1}\,\frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} = D(x^2), && D(0) = 1,
\end{aligned}$$

where A, B, C, D are integral rational functions of x^2, whose degrees follow from the formulae (1), (2) in this way:

$$\begin{array}{lcccc}
 & A(x^2), & B(x^2), & C(x^2), & D(x^2), \\
\text{Degree:} & \dfrac{n^2}{2} - 2, & \dfrac{n^2}{2}, & \dfrac{n^2}{2}, & \dfrac{n^2}{2} & n \text{ even},\\
 & \dfrac{n^2-1}{2}, & \dfrac{n^2-1}{2}, & \dfrac{n^2-1}{2}, & \dfrac{n^2-2}{2} & n \text{ odd}.
\end{array}$$

From the definitions I., II. one obtains immediately the values of $A(0)$, $B(0)$, $C(0)$, $D(0)$, i. e. the terms independent of n, as they are appended.

Various relations among these functions are obtained in the following way:

If, in order to carry out an example here in fuller detail, one replaces v by $v + K$, and hence u by $u + \frac{1}{2}$, then by § 44, (18) x, y, z pass over into y/z, $-\varkappa' x/z$, $\varkappa'/z$, and from § 21, (8) one obtains from the first formula II, for an odd n:

$$\frac{\vartheta_{00}\,\vartheta_{01}^{n^2-1}}{\vartheta_{10}}\,\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}}\left(\frac{\vartheta_{01}(u)}{\vartheta_{00}(u)}\right)^{n^2} = (-1)^{\frac{n-1}{2}}\,\frac{y}{z}\,A\left(\frac{y^2}{z^2}\right).$$

On the other hand, by the second equation II and by § 42 the left side is equal to

$$\frac{y}{z}\left(\frac{\sqrt{\varkappa'}}{z}\right)^{n^2-1} B(x^2),$$

and consequently:

$$B(x^2) = (-1)^{\frac{n-1}{2}} \left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} A\left(\frac{y^2}{z^2}\right).$$

If one puts here $x = 1$, $y = 0$, $z = \varkappa'$ or $x = 0$, $y = 1$, $z = 1$, it follows that:

$$B(1) = (-1)^{\frac{n-1}{2}}\, n\, \sqrt{\varkappa'}^{\,n^2-1}, \qquad A(1) = (-1)^{\frac{n-1}{2}}\, \sqrt{\varkappa'}^{\,n^2-1}.$$

A whole system of such formulae can be derived by making the following substitutions in I and II:

$$\begin{array}{cccc} u, & x, & y, & z, \\ u + \dfrac{1}{2}, & \dfrac{y}{z}, & \dfrac{-\varkappa' x}{z}, & \dfrac{\varkappa'}{z}, \\ u + \dfrac{\omega}{2}, & \dfrac{1}{\varkappa x}, & \dfrac{-iz}{\varkappa x}, & \dfrac{-iy}{x}, \\ u + \dfrac{1+\omega}{2}, & \dfrac{z}{\varkappa y}, & \dfrac{i\varkappa'}{\varkappa}\dfrac{1}{y}, & \dfrac{i\varkappa' x}{y}. \end{array}$$

In this way one obtains:

(4)

$$\begin{aligned} &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-4} A\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}-1} A(x^2), && A(1) = (-1)^{\frac{n}{2}-1}\, n\, \sqrt{\varkappa'}^{\,n^2-4}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} B\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}} B(x^2), && B(1) = (-1)^{\frac{n}{2}}\, \sqrt{\varkappa'}^{\,n^2}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} C\left(\frac{y^2}{z^2}\right) = C(x^2), && C(1) = \sqrt{\varkappa'}^{\,n^2}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} D\left(\frac{y^2}{z^2}\right) = D(x^2), && D(1) = \sqrt{\varkappa'}^{\,n^2}, \\ & && n \text{ even}. \end{aligned}$$

(5)

$$\begin{aligned} &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} A\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n-1}{2}} B(x^2), && A(1) = (-1)^{\frac{n-1}{2}}\, \sqrt{\varkappa'}^{\,n^2-1}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} B\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n-1}{2}} A(x^2), && B(1) = (-1)^{\frac{n-1}{2}}\, n\, \sqrt{\varkappa'}^{\,n^2-1}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} C\left(\frac{y^2}{z^2}\right) = D(x^2), && C(1) = \sqrt{\varkappa'}^{\,n^2-1}, \\ &\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} D\left(\frac{y^2}{z^2}\right) = C(x^2), && D(1) = \sqrt{\varkappa'}^{\,n^2-1}, \\ & && n \text{ odd}. \end{aligned}$$

(6)

$$\begin{aligned} &(\sqrt{\varkappa}\, x)^{n^2-4}\, A\left(\frac{1}{\varkappa^2 x^2}\right) = (-1)^{\frac{n}{2}-1} A(x^2), && A(\infty) = (-1)^{\frac{n}{2}-1}\, n\, (\sqrt{\varkappa})^{n^2-4}, \\ &(\sqrt{\varkappa}\, x)^{n^2}\, B\left(\frac{1}{\varkappa^2 x^2}\right) = B(x^2), && B(\infty) = \sqrt{\varkappa}^{\,-n^2}, \\ &(\sqrt{\varkappa}\, x)^{n^2}\, C\left(\frac{1}{\varkappa^2 x^2}\right) = C(x^2), && C(\infty) = \sqrt{\varkappa}^{\,-n^2}, \\ &(\sqrt{\varkappa}\, x)^{n^2}\, D\left(\frac{1}{\varkappa^2 x^2}\right) = (-1)^{\frac{n}{2}} D(x^2), && D(\infty) = (-1)^{\frac{n}{2}}\, \sqrt{\varkappa}^{\,-n^2}, \\ & && n \text{ even}. \end{aligned}$$

$$
\begin{aligned}
&(\sqrt{\varkappa}\,x)^{n^2-1}\,A\left(\frac{1}{\varkappa^2x^2}\right) = (-1)^{\frac{n-1}{2}}D(x^2), && A(\infty) = (-1)^{\frac{n-1}{2}}\,\sqrt{\varkappa}^{-(n^2-1)},\\
&(\sqrt{\varkappa}\,x)^{n^2-1}\,B\left(\frac{1}{\varkappa^2x^2}\right) = C(x^2), && B(\infty) = \sqrt{\varkappa}^{-(n^2-1)},\\
&(\sqrt{\varkappa}\,x)^{n^2-1}\,C\left(\frac{1}{\varkappa^2x^2}\right) = B(x^2), && C(\infty) = \sqrt{\varkappa}^{-(n^2-1)},\\
&(\sqrt{\varkappa}\,x)^{n^2-1}\,D\left(\frac{1}{\varkappa^2x^2}\right) = (-1)^{\frac{n-1}{2}}A(x^2), && D(\infty) = (-1)^{\frac{n-1}{2}}\,n\,\sqrt{\varkappa}^{n^2-1},\\
& && n \text{ odd.}
\end{aligned}
\tag{7}
$$

Here $A(\infty)$, $B(\infty)$, $C(\infty)$, $D(\infty)$ are in every case to be understood as the coefficients of the highest power of x in these functions. In (7) the last two formulae can be derived from the first by the interchange of x with $1/\varkappa x$.

The third system of formulae can be obtained either in the same way, or else by replacing x in the formulae (4), (5) by $1/\varkappa x$ and applying (6), (7).

$$
\begin{aligned}
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-4}A\left(\frac{z^2}{x^2y^2}\right) = A(x^2), && A\left(\frac{1}{x^2}\right) = n\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2}B\left(\frac{z^2}{x^2y^2}\right) = (-1)^{\frac{n}{2}}B(x^2), && B\left(\frac{1}{x^2}\right) = (-1)^{\frac{n}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2}C\left(\frac{z^2}{x^2y^2}\right) = C(x^2), && C\left(\frac{1}{x^2}\right) = \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{2}D\left(\frac{z^2}{x^2y^2}\right) = (-1)^{\frac{n}{2}}D(x^2), && D\left(\frac{1}{x^2}\right) = (-1)^{\frac{n}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4},\\
& && n \text{ even.}
\end{aligned}
\tag{8}
$$

$$
\begin{aligned}
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1}A\left(\frac{z^2}{x^2y^2}\right) = (-1)^{\frac{n-1}{2}}C(x^2), && A\left(\frac{1}{x^2}\right) = (-1)^{\frac{n-1}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1}B\left(\frac{z^2}{x^2y^2}\right) = D(x^2), && B\left(\frac{1}{x^2}\right) = \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1}C\left(\frac{z^2}{x^2y^2}\right) = (-1)^{\frac{n-1}{2}}A(x^2), && C\left(\frac{1}{x^2}\right) = (-1)^{\frac{n-1}{2}}\,n\,\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1},\\
&\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1}D\left(\frac{z^2}{x^2y^2}\right) = B(x^2), && D\left(\frac{1}{x^2}\right) = \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1},\\
& && n \text{ odd.}
\end{aligned}
\tag{9}
$$

By these formulae one can, in the case of an odd n, reduce the four polynomials $A(x^2)$, $B(x^2)$, $C(x^2)$, $D(x^2)$ to one of them, for example to $A(x^2)$.

If in each of the formulae I, II one divides the first three by the last, one obtains the multiplication formulae for the elliptic functions (§ 42).

III. For even n:	IV. For odd n:
$\operatorname{sn} nv = \dfrac{xyzA(x^2)}{D(x^2)}$,	$\operatorname{sn} nv = \dfrac{xA(x^2)}{D(x^2)}$,
$\operatorname{cn} nv = \dfrac{B(x^2)}{D(x^2)}$,	$\operatorname{cn} nv = \dfrac{yB(x^2)}{D(x^2)}$,
$\operatorname{dn} nv = \dfrac{C(x^2)}{D(x^2)}$,	$\operatorname{dn} nv = \dfrac{zC(x^2)}{D(x^2)}$.

The coefficients of A, B, C, D still depend on ω or on $\varkappa^2$. The nature of this dependence is disclosed by the recursion formulae, which one uses for the successive calculation of

A, B, C, D from the addition theorem. If we indicate the value of the multiplier n to which these functions belong by an index, we have first

$$A_1 = 1, \quad B_1 = 1, \quad C_1 = 1, \quad D_1 = 1; \tag{10}$$

further, from the addition formulae [§ 44, (16), for $u = v$]:

$$\begin{aligned} \operatorname{sn} 2v &= \frac{2 \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^4 v}, \\ \operatorname{cn} 2v &= \frac{\operatorname{cn}^2 v - \operatorname{sn}^2 v \operatorname{dn}^2 v}{1 - \varkappa^2 \operatorname{sn}^4 v}, \\ \operatorname{dn} 2v &= \frac{\operatorname{dn}^2 v - \varkappa^2 \operatorname{sn}^2 v \operatorname{cn}^2 v}{1 - \varkappa^2 \operatorname{sn}^4 v}. \end{aligned} \tag{11}$$

$$\begin{aligned} A_2 &= 2, \\ B_2 &= 1 - 2x^2 + \varkappa^2 x^4, \\ C_2 &= 1 - 2\varkappa^2 x^2 + \varkappa^2 x^4, \\ D_2 &= 1 - \varkappa^2 x^4. \end{aligned} \tag{12}$$

If we replace v in (11) by nv, it follows that:

$$\begin{aligned} A_{2n} &= 2A_n B_n C_n D_n, \\ B_{2n} &= B_n^2 D_n^2 - x^2 y^2 z^2 A_n^2 C_n^2, && n \text{ even}, \\ &= y^2 B_n^2 D_n^2 - x^2 z^2 A_n^2 C_n^2, && n \text{ odd}, \\ C_{2n} &= C_n^2 D_n^2 - \varkappa^2 x^2 y^2 z^2 A_n^2 B_n^2, && n \text{ even}, \\ &= z^2 C_n^2 D_n^2 - \varkappa^2 x^2 y^2 A_n^2 B_n^2, && n \text{ odd}, \\ D_{2n} &= D_n^4 - \varkappa^2 x^4 y^4 z^4 A_n^4, && n \text{ even}, \\ &= D_n^4 - \varkappa^2 x^4 A_n^4, && n \text{ odd}, \end{aligned} \tag{13}$$

and if in the addition formulae of § 44 one sets $u = nv$, $v = (n+1)v$:

$$\begin{aligned} A_{2n+1} &= y^2 z^2 A_n D_n B_{n+1} C_{n+1} + A_{n+1} D_{n+1} C_n B_n, && n \text{ even}, \\ &= A_n D_n B_{n+1} C_{n+1} + y^2 z^2 A_{n+1} D_{n+1} C_n B_n, && n \text{ odd}, \\ B_{2n+1} &= B_n B_{n+1} D_n D_{n+1} - x^2 z^2 A_n A_{n+1} C_n C_{n+1}, \\ C_{2n+1} &= C_n C_{n+1} D_n D_{n+1} - \varkappa^2 x^2 y^2 A_n A_{n+1} B_n B_{n+1}, \\ D_{2n+1} &= D_n^2 D_{n+1}^2 - \varkappa^2 x^4 y^2 z^2 A_n^2 A_{n+1}^2. \end{aligned} \tag{14}$$

From these formulae one concludes that the A, B, C, D are polynomials in $\varkappa^2$, and that their numerical coefficients are integers.

For by (10), (12) this property holds for $n = 1, 2$, and consequently by (13), (14) in general.

As to the degree with respect to $\varkappa^2$ it can further be concluded that the coefficients of $x^{2\nu}$ do not exceed the degree ν with respect to $\varkappa^2$. For if this rule is correct for n and $n+1$, its correctness follows for $2n$ and $2n+1$, and for $n = 1$, $n = 2$ it holds.

From the fundamental equations between the three elliptic functions:

$$1 = \operatorname{cn}^2 v + \operatorname{sn}^2 v = \operatorname{dn}^2 v + \varkappa^2 \operatorname{sn}^2 x$$

there follow further the relations:

$$(15)\qquad \begin{aligned} D^2 &= B^2 + x^2y^2z^2A^2 = C^2 + \varkappa^2x^2y^2z^2A^2, \quad n \text{ even},\\ D^2 &= y^2B^2 + x^2A^2 = z^2C^2 + \varkappa^2x^2A^2, \quad n \text{ odd}, \end{aligned}$$

with whose help one can, by (14), express B_{2n} and C_{2n} through A_n and D_n alone as follows:

$$(16)\qquad \begin{aligned} B_{2n} &= D_n^4 - 2x^2y^2z^2A_n^2D_n^2 + \varkappa^2x^4y^4z^4A_n^4 \\ C_{2n} &= D_n^4 - 2\varkappa^2x^2y^2z^2A_n^2D_n^2 + \varkappa^2x^4y^4z^4A_n^4 \qquad \} \ n \text{ even},\\ B_{2n} &= D_n^4 - 2x^2A_n^2D_n^2 + \varkappa^2x^4A_n^4 \\ C_{2n} &= D_n^4 - 2\varkappa^2x^2A_n^2D_n^2 + \varkappa^2x^4A_n^4 \qquad \} \ n \text{ odd}. \end{aligned}$$

§ 58. Multiplication of the Function $\wp(u)$.

The theory of multiplication takes a very elegant form for the Weierstrass function $\wp(u)$ with the two periods ω_1, ω_2.

Let us consider the doubly periodic function

$$(1)\qquad \wp(nu) - \wp(u),$$

in which n may be an arbitrary positive integer. This function becomes infinite for $u = 0$, and indeed in such a way that

$$(2)\qquad \wp(nu) - \wp(u) + \frac{n^2-1}{n^2}\,\frac{1}{u^2}$$

remains finite for $u = 0$. But besides this it becomes infinite for all values u^* of u which satisfy the condition

$$n\,u^* \equiv 0 \pmod{\omega_1, \omega_2}$$

and which are therefore of the form:

$$(3)\qquad u^* = \frac{\nu_1\omega_1 + \nu_2\omega_2}{n},$$

in which ν_1, ν_2 denote arbitrary integers. We obtain all incongruent values contained in (3) if we let ν_1 and ν_2 each run through a complete system of residues with respect to the modulus n.

The function (1) vanishes for all values u^0 of u different from zero which satisfy the condition

$$\pm n\,u^0 \equiv u^0 \pmod{\omega_1, \omega_2}$$

hence for

$$(4)\qquad u^0 = \frac{\nu_1\omega_1 + \nu_2\omega_2}{n \pm 1},$$

where again ν_1, ν_2 are arbitrary integers. The zero points are of the first order, the points of infinity of the second.

We therefore now introduce a function $\psi_n(u)$, which we define in the following way:

$$(5)\qquad \psi_n(u)^2 = n^2 \prod^{\nu_1,\nu_2} \left[\wp(u) - \wp\left(\frac{\nu_1\omega_1 + \nu_2\omega_2}{n}\right)\right],$$

where ν_1, ν_2 each run through a complete system of residues with respect to the modulus n, with the sole exclusion of the pair of values $0, 0$; and we add the further stipulation that $\psi_1 = 1$ shall hold.

If n is odd, then in the product (5) every linear factor occurs twice, since the two incongruent values

$$u = \pm\frac{\nu_1\omega_1 + \nu_2\omega_2}{n}$$

give the same value of $\wp(u)$. But if n is even, then again all linear factors occur twice in (5), with the exception of the three

$$\wp(u) - \wp\left(\frac{\omega_1}{2}\right), \quad \wp(u) - \wp\left(\frac{\omega_2}{2}\right), \quad \wp(u) - \wp\left(\frac{\omega_1+\omega_2}{2}\right),$$

which occur only once. If one now observes that, according to § 41,

$$\wp'(u)^2 = 4\left[\wp(u) - \wp\left(\frac{\omega_1}{2}\right)\right]\left[\wp(u) - \wp\left(\frac{\omega_2}{2}\right)\right]\left[\wp(u) - \wp\left(\frac{\omega_1+\omega_2}{2}\right)\right]$$

holds, then the result follows:

$$\begin{array}{ll} n \text{ odd:} & \psi_n = P_n, \\ n \text{ even:} & \psi_n = \wp'(u)\, P_n, \end{array} \tag{6}$$

where P_n denotes a polynomial in $\wp(u)$ which for odd n is of degree $\frac{1}{2}(n^2-1)$ and for even n of degree $\frac{1}{2}(n^2-4)$, and in which the term of highest order is respectively equal to

$$n\,\wp(u)^{\frac{n^2-1}{2}}, \qquad -\frac{n}{2}\,\wp(u)^{\frac{n^2-4}{2}}$$

is, by which at the same time the sign still left undetermined by (5) is determined. With this determination of the signs, the initial term in the development of $\psi_n(u)$ in ascending powers of u is, in both cases (§ 56):

$$\frac{n}{u^{n^2-1}}. \tag{7}$$

If one now considers that the points of infinity of the function (1) coincide with the zero points of $\psi_n(u)$, and the zero points of (1) with the zero points of $\psi_{n\pm1}(u)$, so that accordingly

$$\frac{\psi_n(u)^2\,[\wp(nu) - \wp(u)]}{\psi_{n+1}(u)\,\psi_{n-1}(u)} \tag{8}$$

is an everywhere finite doubly periodic function and therefore a constant, whose value is found from $u = 0$ to be -1, then it follows:

$$\wp(nu) = \wp(u) - \frac{\psi_{n+1}(u)\,\psi_{n-1}(u)}{\psi_n(u)^2}, \tag{9}$$

from which the two following formulae result:

$$\begin{aligned} \wp(nu) &= \wp(u) - \frac{P_{n+1}\,P_{n-1}}{\wp'(u)^2\,P_n^2}, && n \text{ even}, \\ &= \wp(u) - \frac{\wp'(u)^2\,P_{n+1}\,P_{n-1}}{P_n^2}, && n \text{ odd}. \end{aligned} \tag{10}$$

We determine first of all the function P_n in the first cases $n = 1, 2, 3, 4$. For this we form, according to the addition theorem of the $\wp$-function [§ 49, (11)]:

$$4\big[\wp(u) + \wp(v) + \wp(u+v)\big] = \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}\right)^2,$$

by setting $u = v$ and determining the limiting value on the right by differentiation:

$$\wp(2u) + 2\wp(u) = \frac{1}{4}\left(\frac{\wp''(u)}{\wp'(u)}\right)^2 = \frac{\frac{1}{4}\left(6\wp(u)^2 - \frac{1}{2}g_2\right)^2}{4\wp(u)^3 - g_2\wp(u) - g_3},$$

or

$$\wp(2u) = \wp(u) - \frac{3\wp(u)^4 - \frac{3}{2}g_2\wp(u)^2 - 3g_3\wp(u) - \frac{g_2^2}{16}}{\wp'(u)^2}, \tag{11}$$

so that

$$\psi_1 = 1, \quad \psi_2 = -\wp'(u), \quad \psi_3 = P_3,$$

$$P_1 = 1, \quad P_2 = -1, \quad P_3 = 3\wp(u)^4 - \frac{3}{2}g_2\wp(u)^2 - 3g_3\wp(u) - \frac{g_2^2}{16} \tag{12}$$

becomes. We obtain further, from the addition formula (§ 49):

$$\wp(v+u) - \wp(v-u) = -\frac{\wp'(u)\,\wp'(v)}{[\wp(v) - \wp(u)]^2},$$

on setting $v = 2u$, by (9):

$$\wp(3u) - \wp(u) = -\frac{\wp'(u)\,\wp'(2u)}{[\wp(2u) - \wp(u)]^2} = -\frac{\wp'(u)^5\,\wp'(2u)}{\psi_3(u)^2} = -\frac{\psi_2(u)\,\psi_4(u)}{\psi_3(u)^2},$$

hence:

$$\psi_4(u) = \wp'(u)\,P_4 = -\wp'(u)^4\,\wp'(2u),$$

and if one forms the value $\wp'(2u)$ from (11):

$$P_4 = -2\wp(u)^6 + \frac{5}{2}g_2\wp(u)^4 + 10g_3\,p(u)^3 + \frac{5}{8}g_2^2\wp(u)^2 + \frac{1}{2}g_2g_3\wp(u) + g_3^2 - \frac{g_2^3}{32}. \tag{13}$$

For larger values of n we derive recursion formulae. To this end we apply formula (9) to two different numbers m, n and obtain:

$$\wp(u) - \wp(mu) = \frac{\psi_{m+1}(u)\,\psi_{m-1}(u)}{\psi_m(u)^2},$$

$$\wp(u) - \wp(nu) = \frac{\psi_{n+1}(u)\,\psi_{n-1}(u)}{\psi_n(u)^2},$$

from which it is to be seen that the right-hand sides become equal to one another for those values u^0 of u which are different from zero and satisfy the condition

$$m\,u^0 \equiv \pm n\,u^0 \pmod{\omega_1, \omega_2},$$

or

$$(m \pm n)\,u^0 \equiv 0 \pmod{\omega_1, \omega_2};$$

but for these same values u^0 the functions

$$\psi_{m\pm n}(u),$$

also vanish, so that the two functions:

$$\psi_{m+1}(u)\,\psi_{m-1}(u)\,\psi_n(u)^2 - \psi_{n+1}(u)\,\psi_{n-1}(u)\,\psi_m(u)^2,$$

$$\psi_{m+n}(u)\,\psi_{m-n}(u),$$

which by (6) are polynomials in $\wp(u)$, vanish for the same finite values of $\wp(u)$.

They therefore differ from one another only by a constant factor, which by (7) turns out to be equal to 1. We therefore have

$$\begin{aligned}(14)\qquad &\psi_{m+n}(u)\,\psi_{m+n}(u)\\ &= \psi_{m+1}(u)\,\psi_{m-1}(u)\,\psi_n(u)^2 - \psi_{n+1}(u)\,\psi_{n-1}(u)\,\psi_m(u)^2.\end{aligned}$$

We apply this formula to two special cases, by putting $n+1$, n or $n+1$, $n-1$ in place of m, n, and so obtain:

$$\begin{aligned}\psi_{2n+1}(u) &= \psi_{n+2}(u)\,\psi_n(u)^3 - \psi_{n+1}(u)^3\,\psi_{n-1}(u)\\ \wp'(u)\,\psi_{2n}(u) &= -\psi_n(u)\,[\psi_{n+2}(u)\,\psi_{n-1}(u)^2 - \psi_{n+1}(u)^2\,\psi_{n-2}(u)],\end{aligned}$$

and from this one obtains, by (6), the recursion formulae for P_n:

$$\begin{aligned}(15)\qquad P_{2n+1} &= \wp'(u)^4\,P_{n+2}\,P_n^3 - P_{n+1}^3\,P_{n-1}, \quad n \text{ even},\\ &= P_{n+2}\,P_n^3 - \wp'(u)^4\,P_{n+1}^3\,P_{n-1}, \quad n \text{ odd}.\end{aligned}$$

$$(16)\qquad P_{2n} = -P_n\,(P_{n+2}\,P_{n-1}^2 - P_{n+1}^2\,P_{n-2}).$$

Since in this way the functions P_n can all be calculated from P_1, P_2, P_3, P_4, it follows that the coefficients of P_n are rationally composed of g_2, g_3 and rational numbers.

Our further considerations on the division, however, we attach to the multiplication formulae of the functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$, whose advantages will come fully into their own only in the fourth part.

§ 59. Division by 2.

Turning first to the consideration of the simplest case, we put v in place of $2v$ in the equations (5) of § 57. Then, if we set

$$(1)\qquad x = \operatorname{sn}\frac{v}{2}, \qquad y = \operatorname{cn}\frac{v}{2}, \qquad z = \operatorname{dn}\frac{v}{2}$$

we have:

$$\begin{aligned}(2)\qquad \operatorname{sn} v &= \frac{2\,x\,y\,z}{1-\varkappa^2\,x^4},\\ \operatorname{cn} v &= \frac{y^2 - x^2\,z^2}{1-\varkappa^2\,x^4},\\ \operatorname{dn} v &= \frac{z^2 - \varkappa^2\,x^2\,y^2}{1-\varkappa^2\,x^4}.\end{aligned}$$

The last two of these equations are quadratic with respect to x^2, but they have only one common root, for the roots of the second of the equations (2) are

$\operatorname{sn}^2\dfrac{v}{2}, \quad \operatorname{sn}^2\left(\dfrac{v}{2}+K+iK'\right)$,

and those of the third

$\operatorname{sn}^2\dfrac{v}{2}, \quad \operatorname{sn}^2\left(\dfrac{v}{2}+K\right)$.

One now finds easily from (2):

$$1+\operatorname{cn} v = \frac{2y^2}{1-\varkappa^2x^4}, \qquad 1+\operatorname{dn} v = \frac{2z^2}{1-\varkappa^2x^4},$$

$$1-\operatorname{cn} v=\frac{2x^2z^2}{1-\varkappa^2x^4},\qquad 1-\operatorname{dn} v=\frac{2\varkappa^2x^2y^2}{1-\varkappa^2x^4},$$
$$\operatorname{dn} v+\operatorname{cn} v=\frac{2y^2z^2}{1-\varkappa^2x^4},$$

hence

$$(3)\qquad \begin{aligned} x&=\sqrt{\frac{1-\operatorname{cn} v}{1+\operatorname{dn} v}}=\frac{1}{\varkappa}\sqrt{\frac{1-\operatorname{dn} v}{1+\operatorname{cn} v}},\\ y&=\sqrt{\frac{\operatorname{dn} v+\operatorname{cn} v}{1+\operatorname{dn} v}},\qquad z=\sqrt{\frac{\operatorname{dn} v+\operatorname{cn} v}{1+\operatorname{cn} v}};\end{aligned}$$

between the signs of these three quantities there subsists, by the first equation (2), a further relation, so that one obtains only four different systems of values, which have the following meaning:

$$\begin{array}{ccc}
\operatorname{sn}\left(\frac{v}{2}\right), & \operatorname{cn}\left(\frac{v}{2}\right), & \operatorname{dn}\left(\frac{v}{2}\right),\\
\operatorname{sn}\left(\frac{v}{2}+2K\right), & \operatorname{cn}\left(\frac{v}{2}+2K\right), & \operatorname{dn}\left(\frac{v}{2}+2K\right),\\
\operatorname{sn}\left(\frac{v}{2}+2iK'\right), & \operatorname{cn}\left(\frac{v}{2}+2iK'\right), & \operatorname{dn}\left(\frac{v}{2}+2iK'\right),\\
\operatorname{sn}\left(\frac{v}{2}+2K+2iK'\right), & \operatorname{cn}\left(\frac{v}{2}+2K+2iK'\right), & \operatorname{dn}\left(\frac{v}{2}+2K+2iK'\right).
\end{array}$$

We further state the special cases which follow from (3):

$$\operatorname{sn}\frac{K}{2}=\frac{1}{\sqrt{1+\varkappa'}},\qquad \operatorname{sn}\frac{iK'}{2}=\frac{i}{\sqrt{\varkappa}},\qquad \operatorname{sn}\frac{K+iK'}{2}=\sqrt{\frac{\varkappa+i\varkappa'}{\varkappa}},$$
$$\operatorname{cn}\frac{K}{2}=\sqrt{\frac{\varkappa'}{1+\varkappa'}},\qquad \operatorname{cn}\frac{iK'}{2}=\sqrt{\frac{1+\varkappa}{\varkappa}},\qquad \operatorname{cn}\frac{K+iK'}{2}=\sqrt{\frac{-i\varkappa'}{\varkappa}},$$
$$\operatorname{dn}\frac{K}{2}=\sqrt{\varkappa'},\qquad \operatorname{dn}\frac{iK'}{2}=\sqrt{1+\varkappa},\qquad \operatorname{dn}\frac{K+iK'}{2}=\sqrt{\frac{\varkappa}{\varkappa+i\varkappa'}}.$$

From this it now follows that the division by 2, and hence also by every power of 2, can be carried out by a chain of square roots. If we therefore assume the problem of the division by an odd number to be solved, then the division by an even number is reduced to square roots. In what follows we shall occupy ourselves exclusively with the division by odd numbers.

§ 60. The Division by an Odd Number.

If we put, when n is an odd number,

$$(1)\qquad x=\operatorname{sn}\frac{v}{n},\qquad y=\operatorname{cn}\frac{v}{n},\qquad z=\operatorname{dn}\frac{v}{n},$$

then from the equations IV, § 57 we obtain:

$$(2)\qquad \begin{aligned} D(x^2)\operatorname{sn} v-x\,A(x^2)&=0,\\ D(x^2)\operatorname{cn} v-y\,B(x^2)&=0,\\ D(x^2)\operatorname{dn} v-z\,C(x^2)&=0.\end{aligned}$$

The first of these equations is of degree n^2 with respect to the unknown x; by the second and the third, y and z are expressed rationally in terms of x (and of $\operatorname{cn} v$, $\operatorname{dn} v$). There result, then, n^2 systems of values for the three unknowns x, y, z, whose meaning is:

$$
\begin{aligned}
x_{\mu,\mu'} &= \operatorname{sn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right), \\
y_{\mu,\mu'} &= \operatorname{cn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right), \\
z_{\mu,\mu'} &= \operatorname{dn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right),
\end{aligned} \tag{3}
$$

in which μ, μ' each run through a complete system of residues (mod n).

The first equation (2):

$$
D(x^2)\operatorname{sn} v - x\,A(x^2) = 0, \tag{4}
$$

of degree n^2, is called the general division equation. It is irreducible in the sense that it is not decomposable into factors which are rational with respect to $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ and have arbitrary coefficients independent of v.

For, in the first place, the n^2 values $x_{\mu,\mu'}$ are all different from one another, and if any rational equation:

$$
F\left(\operatorname{sn}\frac{v}{n},\ \operatorname{sn} v,\ \operatorname{cn} v,\ \operatorname{dn} v\right) = 0
$$

holds, then in it the variable v may be replaced by $v + 4\mu K + 4\mu' iK'$. On account of the periodicity of $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ it follows from this, however, that the equation

$$
F(x,\ \operatorname{sn} v,\ \operatorname{cn} v,\ \operatorname{dn} v) = 0
$$

is satisfied for all roots of equation (4). In order to determine its Galois group, we first fix the domain of rationality. It is to comprise the following quantities:

1. rational numbers,
2. rational functions of $\varkappa^2$,
3. the three functions $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$,
4. the quantities $\operatorname{sn}\left(\dfrac{4\mu K' + 4\mu' iK'}{n}\right)$.

From (2) it then follows that

$$
\operatorname{cn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right), \qquad \operatorname{dn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right)
$$

also belong to the domain of rationality.

We now remark that, by the addition theorem, each of the quantities $x_{\mu,\mu'}$ is rationally expressible through every other one among them. If in particular

$$
x_{\mu,\mu'} = F_{\mu,\mu'}(x_{0,0}) \tag{5}
$$

holds, then

$$
x_{\mu+\nu,\mu'+\nu'} = F_{\mu,\mu'}(x_{\nu,\nu'}) = F_{\mu+\nu,\mu'+\nu'}(x_{0,0}). \tag{6}
$$

From this it follows that the Galois group of our equation consists of the n^2 permutations $S_{\nu,\nu'}$ which one obtains by increasing, in the roots $x_{\mu,\mu'}$, the indices μ, μ' all by the same pair of numbers ν, ν' (where each index is to be taken according to the modulus n).

For by (5) every rational function of the roots $x_{\mu,\mu'}$ can be expressed rationally through $x_{0,0}$ in the form

$$\Phi(x_{0,0})$$

and therefore passes over, by (6), under the substitution $S_{\nu,\nu'}$ into $\Phi(x_{\nu,\nu'})$; if it thus remains unchanged under this substitution, then it is rational. Conversely it follows from the irreducibility that every rational equation between the roots, since it can be put into the form:

$$\psi(x_{0,0}) = 0,$$

and thus

$$\psi(x_{\nu,\nu'}) = 0$$

has as a consequence, admits the substitution $S_{\nu,\nu'}$. But the group of the $S_{\nu,\nu'}$ is, on account of

$$S_{\nu,\nu'} S_{\mu,\mu'} = S_{\nu+\mu,\nu'+\mu'}$$

an Abelian one, which by the formula

$$S_{\nu,\nu'} = S_{1,0}^{\nu} S_{0,1}^{\nu'}$$

is representable by the basis $S_{1,0}$, $S_{0,1}$; and so the general division equation is an Abelian equation and hence algebraically solvable (Bd. I, § 169 f.).

§ 61. The Division of the Periods.

The problem that still remains to be solved now consists in determining by algebraic means the quantities

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' i K'}{n}\right) \tag{1}$$

One obtains all values of this quantity by letting μ, μ', independently of one another, each run through a complete system of residues with respect to the modulus n. But these values are also all different from one another, for $\operatorname{sn} v$ can be $= \operatorname{sn} v'$ only if v' is congruent to v or congruent to $2K - v$ modulo $4K$, $2iK'$, and neither of these can occur for two different ones among the arguments of (1).

The n^2 quantities (1) are the roots of the equation

$$xA(x^2) = 0, \tag{2}$$

and by the formulae

$$y = \frac{D(x^2)}{B(x^2)}, \qquad z = \frac{D(x^2)}{C(x^2)} \tag{3}$$

one can also express

$$y_{\mu,\mu'} = \operatorname{cn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right),$$
$$z_{\mu,\mu'} = \operatorname{dn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right)$$

rationally in terms of $x_{\mu,\mu'}$, if the domain of rationality consists of rational numbers and rational functions of x^2.

From the roots x^2 of the equation $A = 0$ the roots of $B = 0$, $C = 0$, $D = 0$ can be derived rationally by § 60, (2) and § 44, (18), (19), (20); for if x^2 is a root of $A = 0$, then

$$\frac{1 - x^2}{1 - \varkappa^2 x^2}, \qquad \frac{1 - \varkappa^2 x^2}{\varkappa^2(1 - x^2)}, \qquad \frac{1}{\varkappa^2 x^2}$$

the roots of $B = 0$, $C = 0$, $D = 0$. The meanings of these latter roots, however, are

$$\operatorname{sn}\left(\frac{(4\mu + 1)K + 4\mu' iK'}{n}\right)^2,$$
$$\operatorname{sn}\left(\frac{(4\mu + 1)K + (4\mu' + 1)iK'}{n}\right)^2,$$
$$\operatorname{sn}\left(\frac{4\mu K + (4\mu' + 1)iK'}{n}\right)^2,$$

and accordingly, by the solution of the equation $A = 0$, all quantities of the form

$$\operatorname{sn}\left(\frac{\mu K + \mu' iK'}{n}\right)^2,$$

in which μ, μ' are arbitrary integers, are rationally determined.

§ 62. The Abelian Relations.

For a closer investigation of the algebraic nature of the equation of the division of the periods, a system of relations among its roots is of great importance, and we now pass on to the derivation of that system [1]).

We consider the sum:

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(u + \frac{2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(u + \frac{2\nu}{n}\right)}, \tag{1}$$

taken over ν through a complete system of residues for the (odd) modulus n. ν' is an arbitrary integer, and g_1, g_2 is one of the three even characteristics $(0,0)$, $(0,1)$, $(1,0)$. This sum is independent of the particular system of residues that has been taken for ν.

[1]) These relations are due to Abel (Oeuvres complètes, Bd. I, S. 523; Bd. II, S. 251 der neuen Ausgabe). The proof of these relations given above follows Hermite (Crelles Journal, Bd. 32, S. 283). Also to be mentioned is Sylow, Christiania Videnskabsselskabs Forhandlinger 1864 und 1871. Kronecker, Berichte der Berliner Akademie, 19. Juli 1875. Engel, Berichte der Sächsischen Gesellschaft der Wissenschaften, 31. Juli 1884.

The common denominator of the fractions occurring in (1):

$$\prod^{\nu} \vartheta_{g_1,g_2}\left(u + \frac{2\nu}{n}\right) = \pm \prod^{\nu} \vartheta_{g_1,g_2}\left(u + \frac{\nu}{n}\right)$$

is, by the last formulae of § 33, equal, apart from a constant factor, to

$$\vartheta_{g_1,g_2}(nu, n\omega),$$

and if we therefore set the sum (1) equal to

$$\frac{\Phi(u)}{\vartheta_{g_1,g_2}(nu, n\omega)} \tag{2}$$

then $\Phi(u)$ is an entire transcendental function of u. Now we have

$$\frac{\vartheta_{11}\left(u+\frac{2\nu+1}{u}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu+1}{u}\right)} = (-1)^{g_1+1}\frac{\vartheta_{11}\left(u+\frac{2\nu+n+1}{n}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu+n+1}{n}\right)}$$

and ν runs, simultaneously with $\nu+\frac{1}{2}(n+1)$, through a complete system of residues with respect to n. From this it follows by (1) and (2) that

$$\Phi\left(u+\frac{1}{n}\right) = -e^{-\frac{4\nu'\pi i}{n}}\Phi(u) \tag{3}$$

and similarly

$$\Phi(u+\omega) = -e^{-n\pi i(2u+\omega)}\Phi(u). \tag{4}$$

From this it follows that the function $\Phi(u)$ is a t-function, that is, a theta-function, with the periods $1/n$, ω, one that is determined by the conditions (3), (4) up to a factor independent of u and can be expressed by a ϑ-function (§ 19). One sees easily that the conditions (3), (4) are satisfied by the function

$$e^{-4\pi i\nu' u}\vartheta_{11}(nu - 4\nu'\omega, n\omega)$$

so that there results:

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}}\frac{\vartheta_{11}\left(u+\frac{2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu}{n}\right)} = Ce^{-4\pi i\nu' u}\frac{\vartheta_{11}(nu-2\nu'\omega, n\omega)}{\vartheta_{g_1,g_2}(nu, n\omega)}. \tag{5}$$

The determination of the constant C can easily be carried out if one multiplies on the right and on the left by

$$\vartheta_{g_1,g_2}(u)$$

and then

$$u = \frac{(1-g_2)+(1-g_1)\omega}{2}$$

sets. Formula (2) of § 21 then gives:

$$C = (-1)^{(g_1+1)(g_2+1)\frac{n-1}{2}} n\frac{\vartheta_{g_1,g_2}(0,\omega)}{\vartheta_{g_1,g_2}(2\nu'\omega, n\omega)}\frac{\vartheta'_{11}(0, n\omega)}{\vartheta'_{11}(0,\omega)}. \tag{6}$$

One obtains the Abelian relations quite simply by setting $nu = 2\nu'\omega$ in (5), whereby the right-hand side vanishes:

$$\sum^{\nu} e^{\frac{2\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2\nu'\omega+2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(\frac{2\nu'\omega+2\nu}{n}\right)} = 0. \tag{7}$$

If one applies to the ϑ-functions occurring here a linear transformation

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1$$

then by § 39 one obtains the general formula:

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2(\nu\alpha+\nu'\gamma)+2(\nu\beta+\nu'\delta)\omega}{n}\right)}{\vartheta_{g_1,g_2}\left(\frac{2(\nu\alpha+\nu'\gamma)+2(\nu\beta+\nu'\delta)\omega}{n}\right)} = 0. \tag{8}$$

These equations now pass over, when one sets $g_1, g_2 = 0, 1$ and introduces the notation of § 42, directly into relations between the roots $x_{\nu,\nu'}$ (§ 61):

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu\alpha+\nu'\gamma,\ \nu\beta+\nu'\delta} = 0, \tag{9}$$

and to the two special cases:

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

correspondingly:

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu,\nu'} = 0, \tag{10}$$

$$\sum^{\nu'} e^{-\frac{8\nu\nu'\pi i}{n}} x_{\nu,\nu'} = 0. \tag{11}$$

By (5), (6) one can also determine the value of these sums, if in them the nth root of unity

$$e^{\frac{8\pi i}{n}}$$

is replaced by another, by replacing ν' in (5) by another symbol, m, and then setting $nu = 2\nu'\omega$. If we restrict ourselves to the case $(g_1, g_2) = (0, 1)$ and further multiply (5) by $\vartheta_{00} : \vartheta_{10}$, there follows:

$$\sum^{\nu} e^{\frac{8m\nu\pi i}{n}} x_{\nu,\nu'} = (-1)^{\frac{n-1}{2}} n\, e^{\frac{-4\pi i m\nu'\omega}{n}} \frac{\vartheta_{00}\,\vartheta_{01}\,\vartheta'_{11}(0, n\omega)\,\vartheta_{11}[2(\nu'-m)\omega, n\omega]}{\vartheta_{10}\,\vartheta'_{11}\,\vartheta_{01}(2m\omega, n\omega)\,\vartheta_{01}(2\nu'\omega, n\omega)}, \tag{12}$$

where the left side vanishes if and only if $m \equiv \nu' \pmod{n}$.

§ 63. The Galois Group of the Division Equation.

In the thirteenth chapter of the first volume it was shown how the algebraic nature of an equation finds its simplest expression in the Galois group of the equation.

Let the definition of the Galois group be briefly recalled here.

Once the domain of rationality Ω has been fixed, the Galois group of an equation $F(x) = 0$, of which it is assumed only that it has no repeated roots, is defined as the group G of the permutations among the roots of this equation to which the twofold property belongs:

a) Every rational equation in Ω which subsists among the roots of $F(x)$ remains correct when the roots are subjected to any permutation of this group.

b) Every rational function in Ω of the roots of $F(x)$ which admits all the permutations of the group is contained in Ω.

Now, as we seek to determine the group G of the division equation, we assume as domain of rationality in the first place the totality of all rational numbers and rational functions of $\varkappa^2$.

By the addition and multiplication theorem (cf. § 57, 61), if f and φ_m denote rational functions which are independent of μ, μ', ν, ν', then

$$x_{\mu+\nu,\,\mu'+\nu'} = f(x_{\mu,\mu'}, x_{\nu,\nu'}), \tag{1}$$

$$x_{m\mu,\,m\mu'} = \varphi_m(x_{\mu,\mu'}). \tag{2}$$

Now if S is any substitution of the group $\mathfrak{G}$ through which, when a, b, c, ∂ denote integers taken with respect to the modulus n, the passage

$$x_{1,0}, x_{0,1} \quad \text{into} \quad x_{\partial,-c}, x_{-b,a}$$

takes place, then we can apply S to each of the formulae (1), (2). But by (2):

$$x_{\mu,0} = \varphi_\mu(x_{1,0}),$$

from which it is to be concluded that through the substitution S the passage

$$x_{\mu,0} \quad \text{into} \quad \varphi_\mu(x_{\partial,-c}) = x_{\partial\mu,\,-c\mu}$$

takes place. Likewise one finds that through S the passage

$$x_{0,\mu'} \quad \text{into} \quad \varphi_{\mu'}(x_{-b,a}) = x_{-b\mu',\,a\mu'}$$

takes place.

If we apply this to the equation resulting from (1)

$$x_{\mu,\mu'} = f(x_{\mu,0}, x_{0,\mu'})$$

then it follows that through S the passage

$$x_{\mu,\mu'} \quad \text{into} \quad x_{\partial\mu-b\mu',\,-c\mu+a\mu'} \tag{3}$$

takes place.

From this we conclude first of all that the numbers a, b, c, ∂ must be so constituted that their determinant

$$m = a\partial - bc \tag{4}$$

has no divisor in common with n. For if such a divisor were present, then μ, μ' could be so determined, without both being divisible by n, that

$$\partial\mu - b\mu' \equiv 0, \quad -c\mu + a\mu' \equiv 0 \pmod{n},$$

and several roots $x_{\mu,\mu'}$ distinct from one another would pass through S into one and the same root, which is impossible. For, to begin with, neither a and b nor c and ∂ can have a divisor in common with n, since otherwise, if $a\mu'$, $b\mu'$ or $c\mu$, $\partial\mu$ are taken divisible by n, all the $x_{\mu',0}$ or all the $x_{0,\mu}$ would pass into $x_{0,0}$. If one then puts $m\mu = n(ha + h'b)$, $m\mu' = n(hc + h'\partial)$, and determines h and h' so that $ha + h'b$ has no divisors in common with n, then, if n and m have a common divisor, μ, μ' need not both be divisible by n, and not only $x_{0,0}$ but also $x_{\mu,\mu'}$ passes into $x_{0,0}$ by (3).

If we denote the permutation (3) briefly by

$$(x_{\mu,\mu'}, x_{\nu,\nu'}),$$

then

$$\begin{matrix} \nu \equiv \partial\mu - b\mu' \\ \nu' \equiv -c\mu + a\mu' \end{matrix} \pmod{n},$$

and from this, by solving, using the notation of § 28, (4):

$$m(\mu, \mu') \equiv \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} (\nu, \nu'). \tag{5}$$

Therefore

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} \tag{6}$$

is a suitable symbol for the permutation S, and from (5) one sees that two such permutations:

$$S = \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}, \qquad S' = \begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix}$$

compose exactly according to the rule given in § 28:

$$SS' = S'' = \begin{pmatrix} aa' + bc', & ab' + b\partial' \\ ca' + \partial c', & cb' + \partial\partial' \end{pmatrix} \tag{7}$$

so that the permutation S'' of the roots arises when first the permutation S, and then the permutation S', is carried out among the roots of the division equation. It must always be observed, however, that here only the residues of the numbers a, b, c, ∂ taken with respect to the modulus n come into consideration, so that the number of permutations S is always finite[10].

[10] If one wished to choose the notation $S = \begin{pmatrix} \partial, & -b \\ -c, & a \end{pmatrix}$, which at first sight seems to lie nearer, then the composition would have to be denoted by the formula $S'S = S''$, that is, in the reverse sense from the usual composition of permutations .

1. The totality of all substitutions $\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$ forms a group, which we denote by $\mathfrak{A}$, and in it, as we have proved, the group $\mathfrak{G}$ of the division equation is contained.

In $\mathfrak{A}$ there is contained a subgroup $\mathfrak{B}$ which consists of all substitutions (5)

$$T = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \tag{8}$$

for which the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n} \tag{9}$$

holds.

Every substitution S can be derived by composition of

$$\begin{pmatrix} m, & 0 \\ 0, & 1 \end{pmatrix}$$

with a substitution T; in order that

$$S = \begin{pmatrix} m, & 0 \\ 0, & 1 \end{pmatrix} T$$

hold, one has simply to determine $\alpha, \beta, \gamma, \delta$ from the congruences

$$a \equiv \alpha m, \quad b \equiv \beta m, \quad c \equiv \gamma, \quad d \equiv \delta \pmod{n}$$

We now consider

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right) = \frac{\vartheta_{00}\,\vartheta_{11}\left(\dfrac{2\mu + 2\mu'\omega}{n}\right)}{\vartheta_{10}\,\vartheta_{01}\left(\dfrac{2\mu + 2\mu'\omega}{n}\right)}, \tag{10}$$

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}$$

as functions of ω, and apply to them a linear transformation

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} \pmod{8}$$

in which we replace ω by

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega} \tag{11}$$

Then, by the formulae (3), (5) and (6), § 39, it follows that

$$x_{\mu,\mu'} \quad \text{into} \quad x_{\alpha\mu+\gamma\mu',\,\beta\mu+\delta\mu'}$$

passes over; that is, the $x_{\mu,\mu'}$ undergo a substitution which, by (3), (6), is to be denoted by

$$T = \begin{pmatrix} \delta, & -\gamma \\ -\beta, & \alpha \end{pmatrix}$$

while $\varkappa^2$ remains unchanged.

Conversely, every one of the substitutions T can arise in this way as well; for if any substitution T whatever is given, then by adding suitable multiples of the (odd) number n to the numbers $\alpha, \beta, \gamma, \delta$ the conditions

$$\alpha \equiv 1, \quad \delta \equiv 1, \quad \beta \equiv 0, \quad \gamma \equiv 0 \pmod 8$$

can always be satisfied.

Any rational equation between the roots $x_{\mu,\mu'}$ and $\varkappa^2$ passes by (10) into an identity, even when arbitrary constant numerical coefficients, i. e. coefficients independent of ω, occur in it, and one may therefore substitute for ω

$$\frac{\gamma + \delta\omega}{\alpha + \beta\omega}$$

i. e. one may apply every substitution T to the rational equation between the $x_{\mu,\mu'}$. From this it follows that the whole group $\mathfrak{B}$ is contained in $\mathfrak{G}$ (Bd. I, § 156), and indeed in the group $\mathfrak{G}'$, which arises from the group $\mathfrak{G}$ of the division equation by the adjunction of arbitrary numbers.

If we extend the domain of rationality of the rational numbers by the adjunction of nth roots of unity, then the Abelian relations of the preceding section also belong among the rational equations. To these relations, say to

$$\sum_{\nu}^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu,\nu'} = 0$$

however, none of the substitutions

$$M = \begin{pmatrix} m, & 0 \\ 0, & 1 \end{pmatrix},$$

in which m differs from 1, is applicable; for by this substitution, if $mm' \equiv 1 \pmod n$, there passes

$$\sum_{\nu}^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu,\nu'} \text{ into } \sum_{\nu}^{\nu} e^{\frac{8\nu m'\nu'\pi i}{n}} x_{\nu,\nu'}$$

which by (12), § 62 is different from zero unless m' and hence also $m \equiv 1 \pmod n$. From this follows the theorem:

2. After adjunction of the nth roots of unity (and of arbitrary other constants), $\mathfrak{B}$ is the Galois group of the division equation. $\mathfrak{B}$ is also called the monodromy group of the division equation.

3. We now prove further that in the original domain of rationality, that is, without adjunction of the nth roots of unity or of other irrational numbers, the group of the division equation is identical with the group $\mathfrak{A}$.

It is shown in Bd. I, § 174 that the primitive nth roots of unity ϱ are roots of an irreducible equation

(12) $$\Phi(\varrho) = 0$$

are, and this equation cannot split even when the domain of rationality of the rational numbers is extended by adjoining the independent variable $\varkappa^2$.

The degree of this equation is $\varphi(n)$, i. e. equal to the number of the numbers m incongruent modulo n that are relatively prime to n.

Now let

$$F'(r) = 0 \tag{13}$$

be a Galois resolvent of degree μ of the division equation in the original domain of rationality (so that all roots $x_{\nu,\nu'}$ can be represented rationally by r, Bd. I, § 152). This equation must split after the adjunction of a root of (12); for if it did not, then every rational relation

$$\psi(r, \varrho) = 0 \tag{14}$$

would be satisfied for all roots of (13), and $\psi(t, \varrho)$ would be divisible by $F(t)$ (for a variable t). Hence (14) would still continue to hold if ϱ were replaced by another root ϱ^m of (12). But by § 62, (12) this is not the case when one puts one of the Abelian relations in place of (14).

Now let, after the adjunction of ϱ,

$$F(r, \varrho) = 0 \tag{15}$$

be the Galois resolvent of degree ν of the division equation, so that, by 2., ν is equal to the order of the group $\mathfrak{B}$. Then $F(t)$ is algebraically divisible by $F(t, \varrho)$, and hence, because of the irreducibility of (12), also by each of the functions $F(t, \varrho^m)$. Since the functions $F(t, \varrho^m)$ are irreducible, two of them can have a common divisor only when they are entirely identical. But if

$$F(t, \varrho) = F(t, \varrho^m)$$

were the case, then from every equation of the form (14) it would follow that $\psi(t, \varrho)$ is divisible by $F(t, \varrho) = F(t, \varrho^m)$, and it would follow that in (14) the interchange (ϱ, ϱ^m) is admitted, which again is not the case for the Abelian relations. Consequently the $\varphi(n)$ functions $F(t, \varrho^m)$ are all different from one another, and $F(t)$ is divisible by their product. The degree μ of $F(t)$ is therefore at least $= \nu\varphi(n)$. But it also cannot be higher than $\nu\varphi(n)$, since $\nu\varphi(n)$ is the order of the group $\mathfrak{A}$, and the

group $\mathfrak{G}$ of order μ is certainly contained in $\mathfrak{A}$. From this there results

$$\mu = \nu\varphi(n) \tag{16}$$

and at the same time the identity of $\mathfrak{G}$ with $\mathfrak{A}$.

But it also follows from this that $F(t)$ is equal to the product of all the factors $F(t, \varrho^m)$, so that

$$F(t) = \overset{m}{\Pi} F(t, \varrho^m), \tag{17}$$

and, since $F(t) = 0$ has no repeated roots, that $F(r, \varrho)$ does indeed vanish, but that none of the other factors $F(r, \varrho^m)$ does. The equations

$$F(r, t) = 0, \quad \Phi(t) = 0 \tag{18}$$

have therefore only the single root $t = \varrho$ in common with one another, and by seeking out their greatest common divisor one finds ϱ expressed rationally by r, that is, by the roots of the division equation.

These expressions for ϱ do not change their value when a substitution of the group $\mathfrak{B}$ is applied to r, while [according to (12)] ϱ passes over into a power of ϱ under a substitution of the group $\mathfrak{A}$.

The Abelian relations show that under the substitution contained in $\mathfrak{A}$

$$\begin{pmatrix} m, & 0 \\ 0, & 1 \end{pmatrix} \tag{19}$$

ϱ passes over into ϱ^m; for if we set

$$e^{\frac{8\pi i}{n}} = \varrho^{\lambda},$$

then one of the Abelian relations reads

$$\sum^{\nu} \varrho^{\lambda\nu\nu'} x_{\nu,\nu'} = 0, \tag{20}$$

to which, when the expression in terms of r is put in place of ϱ, all substitutions of $\mathfrak{A}$, and hence (19) as well, are applicable. But by § 62, (12), equation (20) remains correct under this substitution only if ϱ passes over into ϱ^m.

Since ϱ is not changed by the substitutions in $\mathfrak{B}$, and since the group $\mathfrak{A}$ arises from $\mathfrak{B}$ by composition with (19), it follows that under any substitution in $\mathfrak{A}$

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$$

ϱ passes into ϱ^m if m is congruent to the determinant $(a\partial - bc)$ with respect to the modulus n.

4. We wish finally to determine the number ν, that is, the order of the group $\mathfrak{B}$.

$\varphi(n)$ has, as is well known, the expression (Bd. I, § 140)

$$\varphi(n) = n\Pi\left(1 - \frac{1}{p}\right), \tag{21}$$

where the product sign Π extends over all the primes p dividing n that are distinct from one another.

The number ν is equal to the number of systems of numbers $\alpha, \beta, \gamma, \delta$ incongruent with respect to n which satisfy the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n} \tag{22}$$

We ask first for the number of pairs α, β which have no common divisor with n, and denote it by $\chi(n)$.

If $n = n'n''$ and n' is relatively prime to n'', then out of every combination of a pair of numbers α', β' belonging to n' with a pair of numbers α'', β'' belonging to n'' one can derive a pair belonging to n,

$$\alpha = n''\alpha' + n'\alpha'', \quad \beta = n''\beta' + n'\beta''$$

and one obtains in this way all the pairs of numbers α, β, and each only once. From this it follows:

$$\chi(n) = \chi(n')\chi(n''). \tag{23}$$

It remains therefore to determine $\chi(p^\pi)$, that is, $\chi(n)$ for a prime power $n = p^\pi$.

If we first put for α, β all the numbers distinct modulo p^π, this number is $p^{2\pi}$. From these, however, all those pairs are to be omitted in which α and β are divisible by p, whose number amounts to $p^{2\pi-2}$, so that

$$\chi(p^\pi) = p^{2\pi} - p^{2\pi-2},$$

and consequently, by (23), there follows in general

$$\chi(n) = n^2 \Pi \left(1 - \frac{1}{p^2}\right) \tag{24}$$

where p runs through the primes dividing n.

Every pair of numbers α, β can be changed, by increase through multiples of n, into one whose numbers are relatively prime to one another, and then γ, δ can be so determined that

$$\alpha\delta - \beta\gamma = 1$$

in it γ, δ can be replaced by $\gamma + h\alpha$, $\delta + h\beta$, and by letting h run through a complete system of residues modulo n, one recognises that to every pair of numbers α, β there belong n pairs of numbers γ, δ satisfying condition (23). Accordingly the order of the group $\mathfrak{B}$ is:

$$\nu = n^3 \Pi \left(1 - \frac{1}{p^2}\right),$$

or, if we further introduce the numerical function

$$\psi(n) = n \Pi \left(1 + \frac{1}{p}\right) \tag{25}$$

then:

$$\nu = n\varphi(n)\psi(n) \tag{26}$$

and the order of the group $\mathfrak{A}$:

$$\mu = n\varphi(n)^2\psi(n). \tag{27}$$

§64. The irreducible factors of the division equation.

If

$$\nu = \partial\mu - b\mu'$$
$$\nu' = -c\mu + a\mu'$$

and $a\partial - bc$ is relatively prime to n, then the greatest common divisor of μ, μ', n is at the same time the greatest common divisor of ν, ν', n. The roots $x_{\mu,\mu'}$ of the division equation therefore split,

according to the greatest common divisor of μ, μ', n, into systems which under the substitutions of the group $\mathfrak{A}$ always pass only into one another, that is, the group $\mathfrak{G}$ is intransitive, and the systems of intransitivity are the systems $x_{\mu,\mu'}$ in which μ, μ', n have one and the same greatest common divisor. The division equation is therefore reducible (except when n is a prime) and the roots of one of these systems of intransitivity satisfy a rational equation (Bd. I, § 157).

According to the preceding section there are $\varphi(n)\psi(n)$ pairs of numbers μ, μ' whose greatest common divisor with n is equal to 1, and the roots corresponding to these pairs of numbers $x_{\mu,\mu'}$ therefore satisfy a rational equation of degree $\varphi(n)\psi(n)$. This equation we call the proper division equation for the divisor n, because only when μ, μ', n are without a common divisor is $x_{\mu,\mu'}$ not at the same time a root of a division

equation for a smaller divisor. In contrast to this we call the equation whose roots are all the $x_{\mu,\mu'}$ the general division equation for the divisor n.

By any substitution of the group $\mathfrak{A}$

$$S = \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$$

$(1, 0)$, $(0, 1)$ pass over into $(\partial, -c)$, $(-b, a)$, and so if S is not the identical substitution, then certainly at least one of the two roots $x_{1,0}$, $x_{0,1}$ is altered by S. From this it results that the Galois group of the proper division equation is exactly the same as that of the general one, only applied to the case in which μ, μ', n have no common divisor, namely, according as the nth roots of unity are adjoined or not, $\mathfrak{B}$ or $\mathfrak{A}$.

From this it follows further that the proper division equation is irreducible, even after the adjunction of arbitrary constants.

For if γ, δ are any two numbers having no common divisor with n, then one can determine α, β in accordance with the congruence

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n}$$

and apply the substitution contained in $\mathfrak{B}$

$$T = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

to every rational equation between the roots $x_{\mu,\mu'}$. If then $x_{1,0}$ satisfies a rational equation (with arbitrary constant coefficients)

$$\Phi(x_{1,0}) = 0$$

then every other root $x_{\delta,-\gamma}$ of the proper division equation satisfies the same equation, from which the irreducibility of the latter follows.

§ 65. Reduction of the division equation to transformation equations.

The roots of the proper division equation can be arranged in rows in the following way. Choose at will one of the roots:

$$x_{\mu_1,\mu_1'} = \operatorname{sn}\left(\frac{4\mu_1 K + 4\mu_1' i K'}{n}\right) = \operatorname{sn} \Omega_1.$$

Among the roots of the division equation there occur also the $\varphi(n)$ quantities

$$(R_1) \qquad \operatorname{sn} h\,\Omega_1$$

which, when h runs through a complete system of numbers incongruent and coprime to n, are all different from one another. The system (R_1) we shall call the first row of the roots.

If now $\operatorname{sn}\Omega_2$ is a root not contained in (R_1), then the $\varphi(n)$ quantities

$$(R_2) \qquad \operatorname{sn} h\,\Omega_2,$$

which are different both among themselves and from the roots (R_1), form a second row; and in this way one can proceed until all the $\varphi(n)\psi(n)$ roots of the proper division equation are distributed into $\psi(n)$ rows of $\varphi(n)$ terms each.

By the multiplication theorem, every other root of the same row can be expressed rationally by means of one root, in the form:

$$(1) \qquad \operatorname{sn} h\,\Omega = f_h(\operatorname{sn}\Omega),$$

where f_h is a rational function depending only on the multiplier h, not on the choice of Ω.

If the two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ belong to the same row, then the number h must admit of being so determined that

$$(2) \qquad h\mu \equiv \nu, \quad h\mu' \equiv \nu' \pmod{n},$$

and conversely, if this is the case, then the two roots belong to the same row. From (2), however, there follows the congruence

$$(3) \qquad \mu\nu' - \nu\mu' \equiv 0 \pmod{n},$$

and from this, conversely, the congruences (2) can be derived again.

For since μ, μ', n are relatively prime, one can determine the numbers α, β so that

$$\alpha\mu - \beta\mu' \equiv 1 \pmod{n}$$

holds, and from this there follows by means of (3):

$$\nu \equiv (\alpha\nu - \beta\nu')\mu, \quad \nu' \equiv (\alpha\nu - \beta\nu')\mu' \pmod{n},$$

hence, if $\alpha\nu - \beta\nu' = h$ is put, the congruences (2).

The congruence (3) is therefore the necessary and sufficient condition that the two roots of the proper division equation $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ belong to the same row.

From this it follows that the division into rows is entirely independent of the arbitrariness in the assumption about the roots $\operatorname{sn}\Omega_1, \operatorname{sn}\Omega_2 \ldots$

It follows further, however, that under the substitutions of the group $\mathfrak{A}$ the rows are not torn apart, but are only interchanged with one another. The group of the division equation is therefore imprimitive, and the individual rows are the systems of imprimitivity (Bd. I, § 158).

For if one applies to (μ, μ') and (ν, ν') simultaneously the substitution

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$$

the congruence (3) remains preserved.

If one selects, among the substitutions of the group $\mathfrak{A}$, those substitutions which interchange the roots of one row only among themselves, one obtains a group, and indeed a subgroup of $\mathfrak{A}$. To

every row there thus belongs such a subgroup of $\mathfrak{A}$. If we denote by $R_{\mu,\mu'}$ the row in which the root $x_{\mu,\mu'}$ occurs, and by $\mathfrak{A}_{\mu,\mu'}$ the subgroup of $\mathfrak{A}$ belonging to this row, then by (3) [cf. § 63, (3)] the substitutions

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$$

of $\mathfrak{A}_{\mu,\mu'}$ are characterized by the congruence

$$(\partial\mu - b\mu')\mu' + (c\mu - a\mu')\mu \equiv 0 \pmod{n} \tag{4}$$

In the same way one obtains a group $\mathfrak{B}_{\mu,\mu'}$ if one seeks out the substitutions in $\mathfrak{B}$ which interchange the roots of the row $R_{\mu,\mu'}$ only among themselves; their substitutions:

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

are characterized by the two congruences:

$$\begin{aligned} (\delta\mu - \beta\mu')\mu' + (\gamma\mu - \alpha\mu')\mu &\equiv 0 \\ \alpha\delta - \beta\gamma &\equiv 1 \end{aligned} \pmod{n} \tag{5}$$

For example, the groups $\mathfrak{A}_{1,0}$, $\mathfrak{B}_{1,0}$ consist of the substitutions

$$\begin{pmatrix} a, & b \\ 0, & \partial \end{pmatrix}, \quad \begin{pmatrix} \alpha, & \beta \\ 0, & \delta \end{pmatrix}, \quad \alpha\delta \equiv 1 \pmod{n}.$$

The groups $\mathfrak{A}_{\mu,\mu'}$ are conjugate subgroups of $\mathfrak{A}$ (Bd. II, § 3); for if $x_{1,0}$ passes by S into $x_{\mu,\mu'}$, and hence $R_{1,0}$ into $R_{\mu,\mu'}$, then by the group

$$S^{-1}\mathfrak{A}_{1,0}S$$

the roots of $R_{\mu,\mu'}$ are permuted only among themselves, and therefore

$$S^{-1}\mathfrak{A}_{1,0}S = \mathfrak{A}_{\mu,\mu'}. \tag{6}$$

Likewise, if T is a substitution in $\mathfrak{B}$ by which $x_{1,0}$ passes into $x_{\mu,\mu'}$:

$$T^{-1}\mathfrak{B}_{1,0}T = \mathfrak{B}_{\mu,\mu'}. \tag{7}$$

The largest common subgroup $\mathfrak{A}_0$ of all the groups $\mathfrak{A}_{\mu,\mu'}$ is determined, according to (4), by

$$b \equiv 0, \quad c \equiv 0, \quad a \equiv \partial \pmod{n}$$

and consists therefore of the substitutions:

$$\begin{pmatrix} a, & 0 \\ 0, & a \end{pmatrix},$$

in which a is an arbitrary number coprime to n. $\mathfrak{A}_0$ is identical with the largest common subgroup of three of the groups $\mathfrak{A}_{\mu,\mu'}$ that belong to different rows. For $\mathfrak{A}_0$ is then the largest common subgroup of $\mathfrak{A}_{\mu,\mu'}$, $\mathfrak{A}_{\nu,\nu'}$, $\mathfrak{A}_{\varrho,\varrho'}$ if the three congruences

$$\begin{aligned} c\mu^2 - (\partial - a)\mu\mu' - b\mu'^2 &\equiv 0 \\ c\nu^2 - (\partial - a)\nu\nu' - b\nu'^2 &\equiv 0 \\ c\varrho^2 - (\partial - a)\varrho\varrho' - b\varrho'^2 &\equiv 0 \end{aligned} \pmod{n}$$

are satisfied only under the assumption

$$\partial \equiv a, \quad b \equiv 0, \quad c \equiv 0 \pmod{n}$$

This takes place when the determinant

$$\begin{vmatrix} \mu^2, & \mu\mu', & \mu'^2 \\ \nu^2, & \nu\nu', & \nu'^2 \\ \varrho^2, & \varrho\varrho', & \varrho'^2 \end{vmatrix} = -(\nu\varrho' - \varrho\nu')(\varrho\mu' - \mu\varrho')(\mu\nu' - \nu\mu')$$

is relatively prime to n.

Likewise, by (5), the largest common subgroup $\mathfrak{B}_0$ of all the $\mathfrak{B}_{\mu,\mu'}$ is the totality of the substitutions:

$$\begin{pmatrix} \alpha, & 0 \\ 0, & \alpha \end{pmatrix} \tag{8}$$

with the condition:

$$\alpha^2 \equiv 1 \pmod{n}. \tag{9}$$

The congruence (9) possesses, if k is the number of the mutually distinct primes dividing n, 2^k incongruent solutions, and this is therefore the order of the group $\mathfrak{B}_0$[11].

A rational function ξ of the roots of one row, say of the row $R_{1,0}$, can by (1) be represented rationally by one of these roots. If this function has the property of remaining unchanged when this one root is replaced by another of the same row, that is, if for example ξ is a symmetric function of the roots of one row, then ξ takes, through application of the substitutions of $\mathfrak{A}$ (or also of $\mathfrak{B}$), only

$$\nu = \psi(n) \tag{10}$$

different values

$$\xi_1, \xi_2, \ldots, \xi_\nu, \tag{11}$$

and if these values are all different from one another, then ξ belongs to the group $\mathfrak{A}_{1,0}$, and all other symmetric functions of the roots of $R_{1,0}$ can be represented rationally by ξ (Bd. I, § 162).

The ν quantities (11) are the roots of an irreducible rational equation of degree ν, which we call a transformation equation.

The transformation equation itself has rational numerical coefficients, provided that only rational numerical coefficients occur in ξ. But it remains irreducible even when nth roots of unity, or indeed any constants whatever, are adjoined, as follows from the fact that by substitutions of the group $\mathfrak{B}$ every root of the division equation can be carried over into every other, and hence also every row into every other row.

By the adjunction of a root ξ of the transformation equation, say of the one belonging to the group $\mathfrak{A}_{1,0}$, the group of the division equation reduces to $\mathfrak{A}_{1,0}$; but this latter group is no longer transitive, and instead permutes among themselves the $\varphi(n)$ roots $x_{h,0}$, in which h runs through a complete system of incongruent numbers coprime to n. The division equation therefore becomes reducible and has a factor of degree $\varphi(n)$,

whose roots are the $x_{h,0}$. The influence of a substitution of the group $\mathfrak{A}_{1,0}$

$$\begin{pmatrix} a, & b \\ 0, & \partial \end{pmatrix}$$

[11] Cf. Dirichlet-Dedekind, Vorlesungen über Zahlentheorie, 4. Aufl., § 37.

upon $x_{h,0}$ consists in this, that $x_{h,0}$ passes over into $x_{\partial h,0}$, and the group of this equation of degree $\varphi(n)$ therefore consists of the permutations

$$(x_{h,0},\ x_{\partial h,0}),$$

in which ∂ may be any number whatever coprime to n. This group is an Abelian one, and therefore the roots $x_{h,0}$ are, after the adjunction of ξ, to be determined algebraically by radicals.

If, however, we adjoin not merely one, but all the roots ξ of the transformation equation, or, what comes to the same thing, the three belonging to $R_{1,0}$, $R_{0,1}$, $R_{1,1}$, then the group of the division equation reduces to $\mathfrak{A}_0$, and if we adjoin in addition nth roots of unity, to $\mathfrak{B}_0$. The group $\mathfrak{B}_0$ is an Abelian one of order 2^k which contains only elements of order $= 2$. In consequence the division equation is solvable by square roots.

In order to find the form of this solution, let one put

$$n = p^{\pi} p'^{\pi'} \dots,$$

in which $p, p', \dots$ are primes different from one another, and $\pi, \pi', \dots$ are positive exponents. Let the numbers $c, c', \dots$ be determined from the congruences

$$\begin{aligned} c &\equiv -1 \pmod{p^{\pi}}, \qquad c' \equiv -1 \pmod{p'^{\pi'}} \dots \\ c &\equiv +1 \pmod{np^{-\pi}}, \quad c' \equiv +1 \pmod{np'^{-\pi'}} \dots \end{aligned}$$

then one obtains every solution α of the congruence

$$\alpha^2 \equiv 1 \pmod{n},$$

and each only once, in the form

$$\alpha \equiv c^{\varepsilon} c'^{\varepsilon'} \dots \pmod{n}, \tag{12}$$

when $\varepsilon, \varepsilon', \dots$ take the values 0, 1.

Now if Ω is any one of the values

$$\frac{4\mu K + 4\mu' i K'}{n},$$

and thus $\operatorname{sn}\Omega$ any root of the proper division equation, then the sum, to be extended over all α,

$$\Sigma(\pm 1)^{\varepsilon}(\pm 1)^{\varepsilon'} \dots \operatorname{sn}\alpha\Omega = \psi(\Omega), \tag{13}$$

in which the k signs of ± 1 may be chosen arbitrarily, has the property that for every α determined according to (12)

$$\psi(\alpha\Omega) = (\pm 1)^{\varepsilon}(\pm 1)^{\varepsilon'} \dots \psi(\Omega) \tag{14}$$

holds; and the square of $\psi(\Omega)$ remains unchanged under the substitutions of the group $\mathfrak{B}_0$, and is therefore rationally expressible by nth roots of unity and the roots of a transformation equation. $\psi(\Omega)$ is therefore the square root $\sqrt{A}$ of such an expression A. But the number of the expressions (13) amounts to 2^k, and by addition of all the equations so formed we obtain

$$2^k \operatorname{sn}\Omega = \Sigma\sqrt{A}. \tag{15}$$

It is still to be remarked that of the quantities ψ, and hence also of the A, half vanish. For we have

$$-1 \equiv cc' \dots \pmod{n},$$

and if we therefore set $\alpha \equiv -1$ in (14), it follows that:

$$\psi(-\Omega) = (\pm 1)(\pm 1)\ldots\psi(\Omega);$$

but on the other hand

$$\psi(-\Omega) = -\psi(\Omega),$$

and consequently $\psi(\Omega)$ always vanishes when among the k quantities ± 1 occurring in (14) an even number of minus signs occurs.

The roots $x_{\mu,\mu'}$ can therefore be expressed linearly by 2^{k-1} square roots.

If n is a prime or a power of a prime, then the squares of the roots of the division equation are rationally expressible by the roots of the transformation equation[12].

If n is a composite number, the classification of the roots $x_{\mu,\mu'}$ into rows can be carried still further. If p is a prime divisor of

n, then one takes two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ into the same row or into different rows, according as

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{p}$$

or not; if accordingly $x_{\nu,\nu'}$ and $x_{\nu_1,\nu_1'}$ belong to one row with $x_{\mu,\mu'}$, then they belong to the same row with one another as well. The number of the rows so formed is, as is easily shown, $p+1$, and in this way one obtains, as the first resolvent of the division equation, a transformation equation belonging to the divisor p. We shall show later, by another route, how the transformation equations for composite divisors can be reduced to those for prime divisors.

Seventh Section.

Theory of the Transformation Equations.

---

§ 66. Formation of Transformation Equations.

Now that the division equations have been reduced to transformation equations, we proceed to a more exact study of these latter equations.

We assume n to be odd, understand by p the primes that divide n, and put

$$\nu = \psi(n) = n\,\Pi\left(1 + \frac{1}{p}\right), \tag{1}$$

and denote the ν rows of the roots of the division equation by $R_1, R_2, \ldots, R_\nu$.

From each of these rows we take, for

$$\Omega_{\mu,\mu'} = \frac{4\mu K + 4\mu' i K'}{n} \tag{2}$$

a representative $\Omega_1, \Omega_2, \ldots, \Omega_\nu$, and we obtain the $\varphi(n)$ roots of one row if in

$$\operatorname{sn} m\,\Omega$$

[12]Cf. on this theorem Kronecker, Monatsbericht der Berliner Akademie, 19. Juli 1875. There Kronecker draws attention to an oversight which occurs in a memoir of Jacobi bearing on this subject (Crelle, Bd. 47 and Bd. 50).

we let m run through a complete system of incongruent numbers coprime to n.

The simplest expressions that can be introduced as roots of transformation equations are the products

$$\prod_{1,\,n-1}^{n} \Phi(\operatorname{sn} h\,\Omega), \tag{3}$$

where Φ is an arbitrary rational function, and h runs through the sequence of numbers $1, 2, \ldots, n-1$.

Every such function is rationally expressible by $\operatorname{sn}\Omega$, and evidently remains unchanged when Ω is replaced by any $m\,\Omega$;

one therefore only has to ensure that the ν values of (3), which correspond to the ν rows, are distinct from one another, so that (3) becomes the root of an (irreducible) transformation equation[1].

We assume that the function $\Phi(x)$ is either an even or an odd function of x, and accordingly make the following distinction.

If $\Phi(x)$ is an even function, then among the factors of the product (3) two at a time, namely

$$\Phi(\operatorname{sn} h\,\Omega), \quad \Phi[\operatorname{sn}(n-h)\,\Omega] \tag{4}$$

are equal to one another. If we therefore set

$$\Pi(\Omega) = \prod_{1,\,\frac{n-1}{2}}^{h} \Phi(\operatorname{sn} h\,\Omega), \tag{5}$$

then, if m is any number coprime to n,

$$\Pi(m\,\Omega) = \Pi(\Omega) \tag{6}$$

[because among the $\frac{1}{2}(n-1)$ numbers $h\,m$ no two have a sum or a difference divisible by n, so that, apart from sign, the $\operatorname{sn} h\,\Omega$ are the same as the $\operatorname{sn} h\,m\,\Omega$]. Thus $\Pi(\Omega)$ is the root of a transformation equation. This class of transformation equations we call modular equations.

If $\Phi(x)$ is an odd function, then the two quantities (4) are opposite to one another and (6) is no longer generally correct; instead we have

$$\Pi(m\,\Omega) = \pm\,\Pi(\Omega). \tag{7}$$

Hence, at least in general, it is no longer $\Pi(\Omega)$, but only $\Pi(\Omega)^2$, that is a root of a transformation equation. This kind of transformation equations we call multiplier equations. In order to learn the cases in which $\Pi(\Omega)$ itself is a root of a transformation, the sign in (7) must be determined.

This is achieved by means of a theorem of number theory which is derived in Bd. I, § 145.

The theorem runs:

[1] Expressions like (3) are still roots of transformation equations even when h takes only values coprime to n. Such a transformation equation has hitherto been little used. We shall become acquainted with an example further on.

If m, n are any two coprime numbers, the latter odd, and if μ further denotes the number of those among the numbers

$$m,\ 2m,\ 3m \cdots \frac{n-1}{2}\, m,$$

whose absolutely least residues (mod n) are negative, then

$$(-1)^{\mu} = \left(\frac{m}{n}\right), \tag{8}$$

where $\left(\frac{m}{n}\right)$ is the Legendre-Jacobi symbol from the theory of quadratic residues[1].

This theorem now leads immediately to the determination of the sign in (7). For if the function $\Phi(x)$ in (5) is odd, then in the passage from Ω to $m\,\Omega$ exactly μ factors in (5) change their sign, and we conclude

$$\Pi(m\,\Omega) = \left(\frac{m}{n}\right)\Pi(\Omega). \tag{9}$$

If n is not a square number, then m can always be assumed such that $\left(\frac{m}{n}\right) = -1$. For if $n = p^{\lambda}n'$, with λ odd, n' not divisible by p, and β a quadratic non-residue of p, then one need only determine m from the congruences

$$m \equiv \beta \pmod{p^{\lambda}}, \quad m \equiv 1 \pmod{n'}$$

in order to find such a number m. And then $\Pi(m\,\Omega) = -\Pi(\Omega)$, and not $\Pi(\Omega)$, but only $\Pi(\Omega)^2$ is a root of a transformation equation. But if n is a square number, then $\Pi(m\,\Omega) = \Pi(\Omega)$ for every m, and consequently $\Pi(\Omega)$ itself is a root of a transformation equation. Hence the theorem follows:

> For an odd function Φ it is $\Pi(\Omega)^2$, and only when n is a square number $\Pi(\Omega)$ itself, that is a root of a transformation equation[2].

[1] Cf. on this theorem: Schering and Kronecker in the Monatsbericht der Berliner Akademie of 22 June 1876. Schering, Acta mathematica I.

[2] This simplification of the multiplier equation in the case where n is a square was discovered by Joubert, but was proved by a method entirely different from ours. "Sur les équations, qui se rencontrent dans la théorie de la transformation des fonctions elliptiques." Paris 1876.

§67. Special Transformation Equations.

The roots of the transformation equations are now to be represented by ϑ-functions. For this purpose we set:

$$\begin{aligned} \Omega &= \frac{4\mu K + 4\mu' i K'}{n}, \quad iK' = K\omega, \\ \varpi &= \frac{\mu + \mu'\omega}{n}, \quad \Omega = 4K\varpi \end{aligned} \tag{1}$$

and consider the products

$$\prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{00}(2h\varpi), \quad \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{01}(2h\varpi), \quad \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{10}(2h\varpi), \quad \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{11}(2h\varpi).$$

The following notation is recommended:

$$
\begin{aligned}
P_{00}\,\vartheta_{00}^{\frac{n-1}{2}} &= e^{\frac{\pi i}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{00}(2h\varpi),\\
P_{10}\,\vartheta_{10}^{\frac{n-1}{2}} &= e^{\frac{\pi i}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{10}(2h\varpi),\\
P_{01}\,\vartheta_{01}^{\frac{n-1}{2}} &= e^{\frac{\pi i}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{01}(2h\varpi).
\end{aligned}
\tag{2}
$$

Formula § 39, (9) is now to be applied, but for the present purpose it is presented somewhat differently.

If one sets $v' = 2h/n$, $v = 2h(\alpha+\beta\omega)/n$ in § 39, (8) and again takes the product over $h = 1, 2, \ldots (n-1)/2$, then, with the application of § 32, (24), it follows that:

$$
\begin{aligned}
&2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{00}\Big(\frac{2h(\alpha+\beta\omega)}{n}\Big)\,\vartheta_{01}\Big(\frac{2h(\alpha+\beta\omega)}{n}\Big)\,\vartheta_{10}\Big(\frac{2h(\alpha+\beta\omega)}{n}\Big)\\
&= (-1)^{\frac{n^2-1}{8}} e^{-\frac{\pi i}{2}\frac{n^2-1}{n}\beta(\alpha+\beta\omega)} \vartheta_{00}^{\frac{n-1}{2}}\,\vartheta_{10}^{\frac{n-1}{2}}\,\vartheta_{01}^{\frac{n-1}{2}}\,1).
\end{aligned}
$$

Accordingly, by multiplication of the three equations (2), one obtains:

$$
(-1)^{\frac{n^2-1}{8}} 2^{\frac{n-1}{2}} P_{00}P_{10}P_{01} = 1. \tag{3}
$$

[1] In § 34, (9) of the first edition the formula in question is given at once in this form, which can also be derived, though somewhat more circumstantially, from § 39, (9).

In addition, we set:

$$
P_{11}\,\eta(\omega)^{\frac{n-1}{2}} = e^{\frac{\pi i}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{1,\,\frac{n-1}{2}}^{h} \vartheta_{11}(2h\varpi). \tag{4}
$$

These quantities P can be expressed by the roots of the division equation if, applying (3), one forms the quotients

$$
\frac{P_{00}^2}{P_{01}P_{10}},\quad \frac{P_{10}^2}{P_{01}P_{00}},\quad \frac{P_{01}^2}{P_{10}P_{00}},\quad \frac{P_{11}^3}{P_{00}P_{01}P_{10}}
$$

and by means of the formulae of § 42 introduces elliptic functions.

One obtains in this way

$$
\begin{aligned}
(-1)^{\frac{n^2-1}{8}}\,2^{\frac{n-1}{2}}\,P_{00}^3 &= \prod_{1,\,\frac{n-1}{2}}^{h} \frac{\operatorname{dn}^2 h\Omega}{\operatorname{cn} h\Omega},\\
(-1)^{\frac{n^2-1}{8}}\,2^{\frac{n-1}{2}}\,P_{10}^3 &= \prod_{1,\,\frac{n-1}{2}}^{h} \frac{\operatorname{cn}^2 h\Omega}{\operatorname{dn} h\Omega},\\
(-1)^{\frac{n^2-1}{8}}\,2^{\frac{n-1}{2}}\,P_{01}^3 &= \prod_{1,\,\frac{n-1}{2}}^{h} \frac{1}{\operatorname{cn} h\Omega\,\operatorname{dn} h\Omega},
\end{aligned}
\tag{5}
$$

$$
(-1)^{\frac{n^2-1}{8}} P_{11}^3 = (\varkappa\varkappa')^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{h} \frac{\operatorname{sn}^3 h\Omega}{\operatorname{cn} h\Omega\,\operatorname{dn} h\Omega}. \tag{6}
$$

These expressions now show that P_{00}^3, P_{10}^3, P_{01}^3 are the roots of transformation equations (modular equations). P_{11}^6 is the root of a multiplier equation, and if n is a square number, then P_{11}^3 too is the root of a multiplier equation.

One can combine the functions P with one another in manifold ways in order to obtain new transformation equations. We adduce the following:

$$\begin{aligned}\frac{P_{10}}{P_{00}} &= \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{cn} h\Omega}{\operatorname{dn} h\Omega},\\ \frac{P_{01}}{P_{00}} &= \prod_{1,\frac{n-1}{2}}^{h} \frac{1}{\operatorname{dn} h\Omega},\end{aligned} \tag{7}$$

$$\begin{aligned}(-1)^{\frac{n^2-1}{8}} \frac{2^{\frac{n-1}{3}} P_{00}^2 P_{11}}{\sqrt{\varkappa\varkappa'}^{\frac{n-1}{3}}} &= \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{sn} h\Omega \operatorname{dn} h\Omega}{\operatorname{cn} h\Omega},\\ (-1)^{\frac{n^2-1}{8}} \frac{2^{\frac{n-1}{3}} P_{10}^2 P_{11}}{\sqrt{\varkappa\varkappa'}^{\frac{n-1}{3}}} &= \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{sn} h\Omega \operatorname{cn} h\Omega}{\operatorname{dn} k\Omega},\\ (-1)^{\frac{n^2-1}{8}} \frac{2^{\frac{n-1}{3}} P_{01}^2 P_{11}}{\sqrt{\varkappa\varkappa'}^{\frac{n-1}{3}}} &= \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{sn} h\Omega}{\operatorname{cn} h\Omega \operatorname{dn} h\Omega}.\end{aligned} \tag{8}$$

One further important result, however, is obtained by applying the multiplication theorem. If in the last formula II of § 57 one sets $u = 2h\varpi$, one finds

$$e^{4\pi i\omega h^2\mu'^2}\vartheta_{01}(2h\varpi)^{n^2} = \frac{\vartheta_{01}^{n^2}}{D(\operatorname{sn}^2 h\Omega)}, \tag{9}$$

in which, as we recall, D is a polynomial whose coefficients are formed rationally from $\varkappa^2$ and integers.

If we take the product of these expressions, for $h = 1, 2, \ldots, \frac{n-1}{2}$, then from the last equation (2) it follows (because, as is well known, $\Sigma h^2 = \frac{n(n^2-1)}{24}$):

$$P_{01}^{n^2} = \prod_{1,\frac{n-1}{2}}^{h} \frac{1}{D(\operatorname{sn}^2 h\Omega)}. \tag{10}$$

We conclude from this that $P_{01}^{n^2}$ too is a root of a transformation equation. This is nothing new when n is divisible by 3, but it is new when n is not divisible by 3, so that $n^2 - 1$ is divisible by 3. For in that case we obtain from (10), with the use of (5), (7), (8):

$$\begin{aligned}P_{01} &= 2^{\frac{(n-1)(n^2-1)}{6}} \prod_{1,\frac{n-1}{2}}^{h} \frac{(\operatorname{cn} h\Omega)^{\frac{n^2-1}{3}} (\operatorname{dn} h\Omega)^{\frac{n^2-1}{3}}}{D(\operatorname{sn}^2 h\Omega)},\\ P_{00} &= 2^{\frac{(n-1)(n^2-1)}{6}} \prod_{1,\frac{n-1}{2}}^{h} \frac{(\operatorname{cn} h\Omega)^{\frac{n^2-1}{3}} (\operatorname{dn} h\Omega)^{\frac{n^2+2}{3}}}{D(\operatorname{sn}^2 h\Omega)},\\ P_{10} &= 2^{\frac{(n-1)(n^2-1)}{6}} \prod_{1,\frac{n-1}{2}}^{h} \frac{(\operatorname{cn} h\Omega)^{\frac{n^2+2}{3}} (\operatorname{dn} h\Omega)^{\frac{n^2-1}{3}}}{D(\operatorname{sn}^2 h\Omega)},\end{aligned} \tag{11}$$

$$(-1)^{\frac{n^2-1}{8}} 2^{\frac{n^2(n-1)}{3}} P_{11} = \sqrt{\varkappa\varkappa'}^{\frac{n-1}{3}} \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{sn} h\Omega\, D(\operatorname{sn}^2 h\Omega)^2}{(\operatorname{cn} h\Omega)^{\frac{2n^2+1}{3}} (\operatorname{dn} h\Omega)^{\frac{2n^2+1}{3}}}. \tag{12}$$

If the domain of rationality is to be maintained as that of the rational numbers and rational functions of $\varkappa^2$, then the last equation is better written in the form [cf. § 54, (4)]:

$$P_{11}\gamma_2(\omega)^{\frac{n-1}{2}} = (-1)^{\frac{n^2-1}{8}}\, 2^{-\frac{(n^2-4)(n-1)}{3}} (\varkappa\varkappa')^{-\frac{n-1}{2}} (1-\varkappa^2\varkappa'^2)^{\frac{n-1}{2}} \times \prod_{1,\frac{n-1}{2}}^{h} \frac{\operatorname{sn} h\Omega\, D(\operatorname{sn}^2 h\Omega)^2}{(\operatorname{cn} h\Omega)^{\frac{2n^2+1}{3}} (\operatorname{dn} h\Omega)^{\frac{2n^2+1}{3}}}, \tag{13}$$

and from this one infers the following theorem:

If n is not divisible by 3, then P_{00}, P_{01}, P_{10}, $P_{11}^2\gamma_2(\omega)^{n-1}$, and, when n is a square, also $P_{11}\gamma_2(\omega)^{\frac{n-1}{2}}$, are roots of transformation equations.

§ 68. Second Representation of the Roots of the Transformation Equations.

If we first fix our attention on one of the rows, namely the one to which the root $x_{1,0}$ belongs, and accordingly put $\mu = 1$, $\mu' = 0$, then we may apply to (2), (3) of the preceding section the formulae (20), (21), (10), § 34. If there, when ν is odd, one puts $n - \nu$ in place of ν and uses the formula

$$\vartheta_{g_1 g_2}\left(\frac{n-\nu}{n}\right) = (-1)^{g_1(g_2+1)}\vartheta_{g_1 g_2}\left(\frac{\nu}{n}\right),$$

then only the even ν occur in these formulae, and if one accordingly puts $\nu = 2h$, then the formulae (20), (21) there may also be written thus:

$$\begin{aligned}
\sqrt{n}\,\eta(n\omega)\eta(\omega)^{\frac{n-3}{2}} &= \prod_{1,\frac{n-1}{2}}^{h} \vartheta_{11}\left(\frac{2h}{n}\right),\\
f(n\omega)\eta(\omega)^{\frac{n-1}{2}} &= f(\omega) \prod_{1,\frac{n-1}{2}}^{h} \vartheta_{00}\left(\frac{2h}{n}\right),\\
f_1(n\omega)\eta(\omega)^{\frac{n-1}{2}} &= f_1(\omega) \prod_{1,\frac{n-1}{2}}^{h} \vartheta_{01}\left(\frac{2h}{n}\right),\\
f_2(n\omega)\eta(\omega)^{\frac{n-1}{2}} &= \left(\frac{2}{n}\right) f_2(\omega) \prod_{1,\frac{n-1}{2}}^{h} \vartheta_{10}\left(\frac{2h}{n}\right),
\end{aligned}$$

in which $\left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}$, and if we accordingly denote by P^0 the functions P that correspond to the case $\mu = 1$, $\mu' = 0$, then from § 67, (2) and (4), with regard to § 34, (10), there follows:

$$\begin{aligned}
P_{00}^0 &= \frac{f(n\omega)}{f(\omega)^n},\\
P_{01}^0 &= \frac{f_1(n\omega)}{f_1(\omega)^n},\\
P_{10}^0 &= \left(\frac{2}{n}\right)\frac{f_2(n\omega)}{f_2(\omega)^n},
\end{aligned} \tag{1}$$

$$P_{11}^0 = \sqrt{n}\,\frac{\eta(n\omega)}{\eta(\omega)}. \tag{2}$$

In order to represent all the roots of the transformation equations by such formulae, we determine a linear transformation as follows:

$$\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix} \equiv \begin{pmatrix}1, & 0\\ 0, & 1\end{pmatrix} \pmod{16}, \tag{3}$$

$$\alpha \equiv \mu, \qquad \beta \equiv \mu' \pmod{n}, \tag{4}$$

and replace ω in (1), (2) by

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}; \tag{5}$$

the formulae (3) to (6) of § 39 and (19) of § 38 may then be applied to the left-hand side, giving:

$$\begin{aligned} P_{00} &= \frac{f(n\omega')}{f(\omega')^n}, \\ P_{01} &= \frac{f_1(n\omega')}{f_1(\omega')^n}, \\ P_{10} &= \left(\frac{2}{n}\right)\frac{f_2(n\omega')}{f_2(\omega')^n}, \\ P_{11} &= \varepsilon^{1-n}\sqrt{n}\,\frac{\eta(n\omega')}{\eta(\omega')}, \end{aligned} \tag{6}$$

in which ε is the 24th root of unity exactly defined in § 33, (15), (18).

Now two transformations

$$\begin{pmatrix} \alpha'; & \beta' \\ \gamma', & \delta' \end{pmatrix}, \quad \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}, \quad a\partial = n,$$

of which the first is linear and the other is of the nth order, can be determined in such a way that

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}. \tag{7}$$

For if we write the relation (7) thus:

$$\begin{pmatrix} \delta', & -\beta' \\ -\gamma', & \alpha' \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} = \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \begin{pmatrix} \delta, & -\beta \\ -\gamma, & \alpha \end{pmatrix},$$

then one derives from it simply:

$$\begin{aligned} \delta' = a\delta, &\quad \gamma' = \partial\gamma - c\delta, \\ \partial\beta' = \beta, &\quad n\alpha' = \partial\alpha - c\beta, \quad a\alpha' = \alpha - c\beta'. \end{aligned} \tag{8}$$

By this ∂ is first of all determined as the greatest common divisor of β and n, or [because of (4)] of μ' and n. For if ∂ were not the greatest common divisor of $n = \partial a$ and $\beta = \partial\beta'$, then a and β' would have to have a common divisor, and consequently so would β and $\alpha = \alpha a' + c\beta$, which cannot be. At the same time $a = n/\partial$ is thereby determined, and c must satisfy the congruence

$$\partial\alpha - c\beta \equiv 0 \pmod{n} \tag{9}$$

by which, however, c is determined not absolutely, but only to the modulus a. One can therefore attach to c the further condition, e. g., that it shall be divisible by some power of 2, or, if a is not divisible by 3, by some power of 3, as we shall sometimes do later.

The three numbers a, ∂, c can have no common divisor, for such a one would [by (8)] also be a divisor of α and β, which is impossible. If we therefore denote the greatest common divisor of a and ∂ by e, then c must be relatively prime to e.

The numbers a, ∂, c, the last modulo a, are, on account of (4) and (8), completely determined by the two numbers μ, μ', and do not change when μ, μ' are multiplied by a common factor coprime to n, i. e. are replaced by another pair of the same row. If one decomposes n in any way into two factors a, ∂, then for c one can still take

$$\frac{a}{e}\varphi(e)$$

values to the modulus a that are distinct and coprime to e. Each of these assumptions leads to a pair of numbers μ, μ' through the congruences

$$\begin{aligned} \mu &\equiv \alpha = a\alpha' + c\beta' \\ \mu' &\equiv \beta = \partial\beta' \end{aligned} \pmod{n}, \tag{10}$$

where β' denotes an arbitrary number coprime to a, and α' is so determined that α, β become relatively prime; and it is also easy to see [by § 65, (3)] that, so long as a, ∂, c remain the same, the pairs of numbers μ, μ' determined by (10) belong to the same row.

We call the numbers a, c, ∂ (as in § 27) the transformation numbers, and n the degree of transformation.

The system of numbers a, c, ∂ is therefore completely characteristic for a row, and the number of rows is equal to the number of these systems of numbers, from which there follows for the number $\psi(n)$, (§ 63):

$$\psi(n) = \Sigma\frac{a}{e}\varphi(e), \tag{11}$$

when the sum is extended over all divisors a of n.

It is of interest to prove this relation directly as well, in which case the restriction to an odd n can be dropped. If we now regard $\psi(n)$ as a symbol for the sum (11), then, when n', n'' are relatively prime, we obtain first

$$\psi(n')\psi(n'') = \psi(n'n''),$$

and it thus only remains to determine the sum $\psi(n)$ for the case in which $n = p^\pi$ is a power of a prime. In this case e is equal to the smaller of the two divisors a, ∂, and when $a = \partial$, $e = a$. If, then, we separate off the first and last term of the sum $\psi(n)$, we obtain

$$\psi(p^\pi) = 1 + p^\pi + \sum_{1<a\leqq\sqrt{n}} \varphi(a) + \sum_{\sqrt{n}<a<n} \frac{a}{\partial}\varphi(\partial).$$

But we have

$$\varphi(a) = a\frac{p-1}{p}, \qquad \varphi(\partial) = \partial\frac{p-1}{p},$$

hence

$$\begin{aligned} \psi(p^\pi) &= 1 + p^\pi + \frac{p-1}{p}\sum_{1<a<n} a \\ &= 1 + p^\pi + (p-1)(1 + p + p^2 + \cdots p^{\pi-2}) = p^\pi\Big(1 + \frac{1}{p}\Big), \end{aligned}$$

from which, for $\psi(n)$, the expression

$$\psi(n) = n\,\Pi\Big(1 + \frac{1}{p}\Big)$$

results, as in § 63[13].

[13] Cf. Dedekind: Über die elliptischen Modulfunktionen. Journal f. Mathematik, Bd. 83.

By (7) we must now put in the formulae (6) for $n\omega'$:

$$\frac{\gamma' + \delta' \dfrac{c + \partial\omega}{a}}{\alpha' + \beta' \dfrac{c + \partial\omega}{a}},$$

and the linear transformations of the f-functions [§ 40, (4), (8), (11)] can be applied. In doing so we take

$$c \equiv 0 \pmod{16}, \tag{12}$$

so that, by (3) and (8):

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \equiv \begin{pmatrix} a^3, & 0 \\ 0, & a \end{pmatrix} \pmod{16}.$$

If we denote by ϱ the third root of unity

$$\varrho = e^{-\frac{2\pi i}{3}[\alpha'(\gamma'-\beta') - (\alpha'^2-1)\beta'\delta']}\, e^{\frac{2n\pi i}{3}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}, \tag{13}$$

then we obtain

$$\begin{aligned} P_{00} &= \varrho \frac{f\left(\dfrac{c+\partial\omega}{a}\right)}{f(\omega)^n}, \\ P_{01} &= \varrho \left(\frac{2}{a}\right) \frac{f_1\left(\dfrac{c+\partial\omega}{a}\right)}{f_1(\omega)^n}, \\ P_{10} &= \varrho \left(\frac{2}{\partial}\right) \frac{f_2\left(\dfrac{c+\partial\omega}{a}\right)}{f_2(\omega)^n}. \end{aligned} \tag{14}$$

The quotient

$$\frac{P_{10}}{P_{00}} = \left(\frac{2}{\partial}\right) \frac{f_2\left(\dfrac{c+\partial\omega}{a}\right)}{f\left(\dfrac{c+\partial\omega}{a}\right)} \left(\frac{f\omega}{f_2\omega}\right)^n \tag{15}$$

gives, by § 54, (3), if one puts

$$u(\omega) = \sqrt[4]{k}$$

the quantities

$$\left(\frac{2}{\partial}\right) \frac{u\left(\dfrac{c+\partial\omega}{a}\right)}{u(\omega)^n} \tag{16}$$

as roots of a modular equation, and this is the Jacobi modular equation[14].

[14] If M is the Jacobi multiplier, then

$$\frac{(kk')^{\frac{n-1}{3}}}{M} = 4^{\frac{n-1}{3}} P_{11}^2 P_{00}^4.$$

If n is not divisible by 3, we assume

$$c \equiv 0 \pmod 3 \tag{17}$$

whereby ϱ takes the value 1.

In the same way one can apply the transformation of the η-function [§ 38, (15), (18), (19)] to the last of the equations (6), and one obtains:

$$P_{11} = \left(\frac{\beta}{\alpha}\right)\left(\frac{\beta'}{\alpha'}\right)\varrho\, i^{\frac{a-1}{2}}\sqrt{\partial}\,\frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)} \tag{18}$$

(where it would already suffice if c is divisible by 8). If n is not divisible by 3 and c is divisible by 3, then here too one must set $\varrho = 1$. The determination of the sign in (18) is of interest to us only in the case where n is a square. But we have [by (8)]

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\beta'}{\alpha'}\right) = \left(\frac{\partial}{\alpha}\right)\left(\frac{\beta'}{\alpha\alpha'}\right) = \left(\frac{\partial}{\alpha}\right)\left(\frac{\beta'}{a}\right) = \left(\frac{\alpha}{\partial}\right)\left(\frac{\beta'}{a}\right)$$

[the latter by the law of reciprocity of the quadratic residues, because $\alpha \equiv 1 \pmod 4$]. If now n is a square, then $a:e$, $\partial:e$ are squares as well, and there results:

$$\left(\frac{\alpha}{\partial}\right)\left(\frac{\beta'}{a}\right) = \left(\frac{\alpha\beta'}{e}\right) = \left(\frac{c}{e}\right),$$

thus in this case

$$P_{11} = \varrho\, i^{\frac{a-1}{2}}\left(\frac{c}{e}\right)\sqrt{\partial}\,\frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}.$$

The formulae (10) set up for the characterization of a row are a special case of a more general system, which one obtains by applying a linear transformation to α', β' in (10). One then obtains the following:

If a, b, c, ∂ are four integers without a common divisor satisfying the condition:

$$a\partial - bc = n$$

then one obtains the pairs of numbers μ, μ' corresponding to one row by letting, in

$$\begin{aligned} \mu &\equiv a\alpha' + c\beta' \\ \mu' &\equiv b\alpha' + \partial\beta' \pmod n \end{aligned} \tag{19}$$

α', β' run through all those values, and only those, for which μ, μ' have no common divisor with n.

That two pairs of values μ, μ' corresponding to the congruences (19) really do belong to the same row follows immediately from § 65 (3), and it is likewise self-evident that all pairs of numbers of one row are contained in this form, since α', β' may be replaced by $h\alpha'$, $h\beta'$ when h is relatively prime to n. That one obtains all the rows in this way is shown by the formulae (10).

§ 69. The Invariant Equation.

Among the transformation equations, one is of particular interest: the equation whose roots are the $\psi(n)$ quantities

$$j\left(\frac{c+\partial\omega}{a}\right) \tag{1}$$

or, by the determination (19), § 68, the quantities identical with them

$$j\left(\frac{c+\partial\omega}{a+b\omega}\right) \tag{2}$$

where $j(\omega)$ is the invariant defined in § 46.

This equation is called the invariant equation.

The function $j(\omega)$ can, by § 54, be represented rationally by $f(\omega)^{24}$, namely

$$j(\omega) = \frac{[f(\omega)^{24} - 16]^3}{f(\omega)^{24}}, \tag{3}$$

so that, by the results of the preceding section, the quantities (1) are the roots of an equation whose coefficients are rational functions of $\varkappa^2$.

But if for $f(\omega)$ in (3) one substitutes one of the developments found earlier, e.g. § 24, (11):

$$f(\omega) = q^{-\frac{1}{24}} \prod_{1,}^{\nu} (1 + q^{2\nu-1}),$$

one recognises that $j(\omega)$ can be developed into a series progressing according to powers of q^2, which has the form

$$j(\omega) = q^{-2} + a_1 + a_2 q^2 + a_3 q^4 + \cdots, \tag{4}$$

in which the a_1, a_2, $a_3 \ldots$ are integers which can be calculated successively (there results, for example, $a_1 = 744$, $a_2 = 196\,884$). The developments of the quantities (1) therefore begin with

$$e^{-\frac{2\pi i c}{a}} e^{-\frac{2\pi i \partial}{a}\omega} \tag{5}$$

and are therefore all different from one another. The invariant equation is therefore, by § 65, irreducible.

There is, however, a second way of arriving at this equation, and at the transformation equations generally, which we shall now first become acquainted with in the case of the invariant equation. This derivation rests upon the theorems of § 54 on modular functions and holds also for an even n. Here it is Theorem 4, § 54, that comes into application:

I. Every modular function which under the two transformations

$$\begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \tag{6}$$

$$\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \tag{7}$$

remains unchanged is a rational function of $j(\omega)$.

We first show that by application of the substitutions (6), (7) the ν quantities (1) are permuted among one another. We have the composition

$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \lambda, & 1 \end{pmatrix} \begin{pmatrix} a, & 0 \\ c_1, & \partial \end{pmatrix}, \tag{8}$$

when

$$c_1 = c + \partial - \lambda a \tag{9}$$

is set. Since $j(\omega)$ remains unchanged under every linear transformation, transformation (6) acts as follows:

$$j\left(\frac{c+\partial\omega}{a}\right) \text{ goes to } j\left(\frac{c_1+\partial\omega}{a}\right)$$

Further we determine a_2, ∂_2, c_2 so that

$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}\begin{pmatrix} a_2, & 0 \\ c_2, & \partial_2 \end{pmatrix}, \tag{10}$$
$$\alpha\delta - \beta\gamma = 1.$$

For this it is required that

$$\begin{aligned} \alpha a_2 + \beta c_2 &= 0 \\ \gamma a_2 + \delta c_2 &= -\partial, \end{aligned} \tag{11}$$

$$\begin{aligned} \beta\partial_2 &= a \\ \delta\partial_2 &= c. \end{aligned} \tag{12}$$

Thus ∂_2 is determined as the greatest common divisor of a and c, and with it at the same time, because of $a_2\partial_2 = n$, also a_2.

By the two equations (12) one now knows β, δ as relatively prime numbers, and can determine α, γ from the Diophantine equation

$$\alpha\delta - \beta\gamma = 1$$

When this has been done, there follows from the two equations (11)

$$\begin{aligned} a_2 &= \partial\beta, \\ c_2 &= -\partial\alpha, \end{aligned} \tag{13}$$

whereby c_2 also is determined, and it appears at the same time that ∂ is the greatest common divisor of a_2, c_2. If one replaces α, γ by another solution $\alpha + h\beta$, $\gamma + h\delta$, then only c_2 is altered, by a multiple of a_2.

Through the transformation (7), therefore,

$$j\left(\frac{c+\partial\omega}{a}\right) \text{ is carried to } j\left(\frac{c_2+\partial_2\omega}{a_2}\right)$$

Let us now form a symmetric function of all the quantities (1), say, for an indeterminate x, the product

$$\Pi\left[x - j\left(\frac{c+\partial\omega}{a}\right)\right],$$

then this function is not altered by the transformations (6), (7), and is therefore, by the theorem mentioned above, a rational function of $j(\omega)$. Moreover it is a polynomial in x of degree ν with leading term x^ν, and we denote it by $F_n[x, j(\omega)]$. The equation

$$F_n[x, j(\omega)] = 0 \tag{14}$$

has the quantities $j\left(\frac{c+\partial\omega}{a}\right)$ as roots, and is the invariant equation, whose most important properties we now propose to derive.

1. The invariant equation is irreducible when, as domain of rationality, the totality of all rational functions of

$j(\omega)$ is considered with arbitrary numerical coefficients. For if there exists any rational equation

$$\Phi[j(n\omega),\ j(\omega)] = 0, \tag{15}$$

then any arbitrary linear transformation

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

may be applied to it, and by § 29, (5), if

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$$

is an arbitrary transformation of order n, the linear transformations

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix}$$

can always be determined so that

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}. \tag{16}$$

But from this it follows that equation (9) is satisfied by each of the quantities

$$j\left(\frac{c+\partial\omega}{a+b\omega}\right)$$

and hence also by each of the quantities (1), from which [because of the distinctness of the quantities (1)] the irreducibility of (14) follows.

2. The function

$$F_n[x, j(\omega)] = \Pi\left[x - j\left(\frac{c+\partial\omega}{a}\right)\right] \tag{17}$$

has a finite value for every finite value of ω with positive imaginary part, and hence also for every finite value of $j(\omega)$, and is accordingly a polynomial in $j(\omega)$.

If we next seek, by (3), the initial term of the development of the function (17) in powers of q, we obtain

$$(-1)^\nu e^{-2\pi i \Sigma \frac{c}{a}} e^{-2\pi i\omega \Sigma \frac{\partial}{e}\varphi(e)} = Cq^{-2\nu},$$

where C is a finite constant. Therefore

$$j(\omega)^{-\nu} F_n[x, j(\omega)]$$

is finite for an infinite $j(\omega)$, that is, the degree of $F_n[x, j(\omega)]$ with respect to $j(\omega)$ is likewise the νth.

3. We have

$$F_n[j(n\omega), j(\omega)] = 0$$

and if we replace $n\omega$ by ω:

$$F_n\Big[j(\omega), j\Big(\frac{\omega}{n}\Big)\Big] = 0.$$

Since now $j\left(\frac{\omega}{n}\right)$ is likewise a root of the invariant equation, it follows that the two equations of degree ν

$$F_n[x, j(\omega)] = 0, \quad F_n[j(\omega), x] = 0$$

have one root in common, and consequently, because of the irreducibility of the former and the equality of the degree, all of them. Therefore

$$F_n(x, y) = CF_n(y, x)$$

for indeterminate values of the variables x, y and a constant C. Hence also, by interchange of x with y:

$$F_n(y,x) = CF_n(x,y),$$

and so $C^2 = 1$ or

$$F_n(x,y) = \pm F_n(y,x).$$

If we now set $x = y$, it follows that:

$$F_n(y,y) = \pm F_n(y,y),$$

and so, if the lower sign holds,

$$F_n(y,y) = 0.$$

But this is possible only when $n = 1$ [where $F_1(x,y) = x - y$], since otherwise $F_n(x,y)$ would have to be divisible by $x - y$, which contradicts the irreducibility. Therefore we always have, as soon as $n > 1$:

$$F_n(x,y) = F_n(y,x). \tag{18}$$

4. Let

$$n = n'n'', \quad \nu' = \psi(n'), \quad \nu'' = \psi(n''),$$

and let n', n'' be without common divisor, and let $x_1, x_2, \ldots x_{\nu''}$ be the roots of the equation

$$F_{n''}[x, j(\omega)] = 0. \tag{19}$$

The product

$$F_{n'}(x,x_1)\,F_{n'}(x,x_2)\ldots F_{n'}(x,x_{\nu''}), \tag{20}$$

whose degree with respect to x is equal to

$$\nu = \psi(n) = \psi(n')\psi(n'')$$

depends, as a symmetric function of the roots of (19), rationally on $j(\omega)$, and vanishes for

$$x = j\left(\frac{\omega}{n'n''}\right),$$

i.e. for a root of the equation $F_n[x, j(\omega)] = 0$. In view of the irreducibility of the latter equation, and of the equality of the degree and of the coefficient of x^ν, we therefore have

$$F_n[x, j(\omega)] = F_{n'}(x,x_1)\,F_{n'}(x,x_2)\ldots F_{n'}(x,x_{\nu''}). \tag{21}$$

5. If $n = p^\pi$ is a power of a prime, then the degree of the function $F_n[x, j(\omega)]$ is equal to $p^{\pi-1}(p+1)$, and the equation

$$F_{p^{\pi-1}}[x, j(\omega)] = 0 \tag{22}$$

is of degree

$$\nu' = p^{\pi-2}(p+1).$$

We denote its roots by $x_1, x_2, \ldots x_{\nu'}$. The product

$$P = F_p(x,x_1)\,F_p(x,x_2)\ldots F_p(x,x_{\nu'})$$

is, with respect to x, of degree $\nu'(p+1) = p^{\pi-2}(p+1)^2$; it depends symmetrically on the roots of (22), and hence rationally on $j(\omega)$, and vanishes for

$$x = j\left(\frac{\omega}{p^\pi}\right).$$

Therefore P is divisible by $F_n[x, j(\omega)]$. But the degree of P is higher than that of F_n. If we take

$$x_1 = j\left(\frac{p^{\pi-2}c + \omega}{p^{\pi-1}}\right),$$

where c may have any of the values $0, 1, 2, \ldots p-1$, then $F_p(x, x_1)$ vanishes for

$$x = j\left(\frac{\omega}{p^{\pi-2}}\right),$$

i.e. p of the factors of P have a definite factor in common with $F_{p^{\pi-2}}[x, j(\omega)]$; from this it follows that P is divisible by

$$\{F_{p^{\pi-2}}[x, j(\omega)]\}^p$$

and hence, as the comparison of the degrees and of the highest terms shows:

$$F_{p^\pi}[x, j(\omega)] = \frac{F_p(x, x_1)\, F_p(x, x_2) \ldots F_p(x, x_{\nu'})}{\{F_{p^{\pi-2}}[x, j(\omega)]\}^p}. \tag{23}$$

This formula is subject to an exception for $\pi = 2$, because in this case the degree of the function on the right-hand side is still too high. For this case, however, each of the $p+1$ factors of the product P has the divisor $x - j(\omega)$, precisely because every x_i is a root of the equation $F_p[x, j(\omega)]$, and in place of (23) there appears the formula

$$F_{p^2}[x, j(\omega)] = \frac{F_p(x, x_1)\, F_p(x, x_2) \ldots F_p(x, x_{p+1})}{[x - j(\omega)]^{p+1}}. \tag{24}$$

In this way the solution of the invariant equation $F_n = 0$ is reduced to the successive solution of those cases in which n is a prime.

6. In the derivation of the transformation equations from the division equations it is settled from the outset that the numerical coefficients in these equations are rational numbers; the second derivation, however, teaches us nothing at first about the numerical coefficients. But we can prove afterwards that these coefficients are not only rational numbers but integers, and we arrive at the same time at an important theorem on the divisibility of these coefficients.

If we can prove that, when p is a prime, the coefficients in $F_p(x, y)$ are integers, then the same follows from 4. and 5. for every composite n.

By (4), $j(\omega)$ can be developed in a series of the form

$$j(\omega) = q^{-2} \sum_{0,\infty}^{h} a_h q^{2h}, \tag{25}$$

whose coefficients a_h are integers, and indeed $a_0 = 1$. If from this we form, when p is a prime, the pth power, and observe Fermat's theorem, valid for every integer:

$$a^p \equiv a \pmod{p},$$

then it follows

$$j(\omega)^p = q^{-2p} \sum_{0,\infty}^{h} a_h q^{2hp} + p\, q^{-2(p-1)} \sum_{0,\infty}^{h} b_h q^{2h}, \tag{26}$$

in which b_h are likewise integers. On the other hand, if one replaces ω by $p\omega$ in (25), then:

$$j(p\omega) = q^{-2p} \sum_{0,\infty}^{h} a_h q^{2hp} \tag{27}$$

and from this, if one puts

$$j(\omega) = u, \quad j(p\omega) = v$$

one obtains:

$$u^p - v = p\, q^{-2(p-1)} \sum_{0,\infty}^{h} b_h q^{2h},$$

for which we may also write:

$$(u^p - v)(u - v^p) = p\, q^{-2(p^2+p-1)} \sum_{0,\infty}^{h} c_h q^{2h}, \tag{28}$$

if c_h denotes a third system of integers. Now in $F_p(x, y)$ there occur the terms $x^{p+1} + y^{p+1}$, the highest with respect to x, y, and we accordingly put

$$F_p(x, y) = (x^p - y)(x - y^p) - \sum_{0,p}^{h,k} c_{h,k} x^h y^k, \tag{29}$$

where $c_{h,k}$ are the coefficients to be determined, which by 3. satisfy the condition

$$c_{h,k} = c_{k,h}.$$

To determine them, we set in (29)

$$x = u, \quad y = v,$$

whereby F_p vanishes, and obtain

$$(u^p - v)(u - v^p) = \sum^{h,k} c_{h,k} u^h v^k. \tag{30}$$

From this it follows first of all that

$$c_{p,p} = 0$$

must hold, since by (28), in the development (of the left side) in powers of q, the power $q^{-2(p^2+p)}$ cannot occur, and we can now put (29) into the form:

$$(u^p - v)(u - v^p) = \sum_{0,p}^{h} \sum_{0,h-1}^{k} c_{h,k}(u^h v^k + u^k v^h) + \sum_{0,p-1}^{h} c_{h,h} u^h v^h. \tag{31}$$

In this there are to be inserted the developments, resulting from (25) and (27), of

$$u^h v^k + u^k v^h, \quad u^h v^h$$

in powers of q, whose initial terms

$$q^{-2(hp+k)}, \quad q^{-2h(p+1)}$$

have the coefficients 1. On the right side of (31) no two terms occur whose development begins with the same power of q; for from

$$hp + k = h'p + k'$$

it follows that $k \equiv k' \pmod{p}$, and hence, since k and $k' < p$, that $k = k'$, $h = h'$. If one therefore arranges the series which represent the two sides of (31) according to ascending powers of q, and then sets equal to one another the coefficients of equally high powers, one obtains a sequence of linear equations for the unknowns $c_{h,k}$, of which each succeeding one contains only one new unknown, and this with the coefficient 1. The known terms of these equations which arise from

the left side are, by (28), all integers divisible by p, and hence the $c_{h,k}$ likewise have integer values divisible by p. Accordingly we have

$$F_p(x,y) = (x^p - y)(x - y^p) - p\sum_{0,p}^{h,k} a_{h,k}x^h y^k, \tag{32}$$

in which the $a_{h,k}$ are integers, and the conditions

$$a_{h,k} = a_{k,h}, \quad a_{p,p} = 0$$

are satisfied.

§ 70. Transformation Equations of the First Level.

The invariant equation is of great theoretical importance, partly because of its general validity (for even n as well), partly because of the ease with which the linear transformations can be applied to $j(\omega)$. The actual computation of this equation, however, turns out to be so complicated, and the numerical coefficients are so large, that the computation has so far been carried out only in the simplest case $p = 2$. On the other hand, by making use of other modular functions, one can obtain far simpler transformation equations.

Concerning the principle to be applied here we remark the following:

If there is given any system of ν functions of ω, corresponding to the ν systems of transformation numbers a, c, ∂, which we shall denote by

$$\Phi_{a,c,\partial} \tag{1}$$

and which are permuted among one another isomorphically with the functions

$$j\left(\frac{c+\partial\omega}{a}\right) \tag{2}$$

by the substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \quad (\omega, \omega+1) \tag{3}$$

then (1) is expressible rationally through (2) and through $j(\omega)$; for the function

$$F_n[x, j(\omega)] \sum^{a,c,\partial} \frac{\Phi_{a,c,\partial}}{x - j\left(\frac{c+\partial\omega}{a}\right)} = \Psi[x, j(\omega)] \tag{4}$$

remains unaltered under the substitutions (3), and is therefore (for an indeterminate x) a rational function of $j(\omega)$, and moreover a polynomial in x of degree at most $\nu - 1$; $F_n[x, j(\omega)]$ has the same meaning as in (17) of the preceding section, and $\nu = \psi(n)$ is the degree of this function with respect to x. From (4), however, on setting

$$x = j\left(\frac{c+\partial\omega}{a}\right)$$

one obtains:

$$\Phi_{a,c,\partial} = \frac{\Psi\left[j\left(\frac{c+\partial\omega}{a}\right), j(\omega)\right]}{F'\left[j\left(\frac{c+\partial\omega}{a}\right)\right]}. \tag{5}$$

Thus $\Phi_{a,c,\partial}$ belongs to the algebraic field which consists of the rational functions of $j\left(\frac{c+\partial\omega}{a}\right)$ and $j(\omega)$, and we can apply the theorems of Bd. I, § 151. From these it follows that the ν quantities $\Phi_{a,c,\partial}$ are the roots of an equation of degree ν whose coefficients depend rationally on $j(\omega)$. If the ν quantities $\Phi_{a,c,\partial}$ are different from one another, then this equation is irreducible. Such an equation we call a transformation equation of the first level belonging to the transformation degree n[15]. Every other quantity of the field can be rationally expressed by such a $\Phi_{a,c,\partial}$ and by $j(\omega)$.

If the functions $\Phi_{a,c,\partial}$ have the property of remaining finite for every finite value of ω with positive imaginary part, and hence for every finite $j(\omega)$, then the function $\Psi[x, j(\omega)]$ in (4) is also a polynomial in $j(\omega)$. Formula (5) could therefore fail only for those particular values of ω for which two roots of the invariant equation become equal to one another.

If the ν quantities $\Phi_{a,c,\partial}$ are not all different from one another, then they split into groups of equal size, each group consisting of equal quantities, and the equation of degree ν to be derived from (5) is a power of an irreducible equation, which we shall occasionally also designate as a transformation equation (Bd. I, § 151, 2).

If the quantities $\Phi_{a,c,\partial}$ are so chosen that they become infinite for no finite ω with positive imaginary part, then they remain finite for every finite $j(\omega)$, from which it follows that the coefficients in the function of degree ν

$$\Pi(x - \Phi_{a,c,\partial}) \tag{6}$$

are polynomials in $j(\omega)$, i.e. the $\Phi_{a,c,\partial}$ are integral algebraic functions of $j(\omega)$.

If the functions $\Phi_{a,c,\partial}$ are so arranged, say by a suitable determination of the constants that are still at our disposal in them, that they remain finite for an infinite imaginary ω as well, i.e. for $q = 0$, then these functions are finite for an infinite $j(\omega)$ as well, and the coefficients in (6) are constant. But this is possible only if the $\Phi_{a,c,\partial}$ are all equal to one and the same constant C. If one knows $\Phi_{a,c,\partial}$ as a rational function of another quantity $\Psi_{a,c,\partial}$, then $\Phi_{a,c,\partial} = C$ is either an identity or a transformation equation for $\Psi_{a,c,\partial}$.

§ 71. The Transformation Equations for γ_2 and γ_3.

Transformation equations of the first level are obtained first of all from the consideration of the functions § 54, (4), (5):

$$\gamma_2(\omega) = \sqrt[3]{j(\omega)}, \quad \gamma_3(\omega) = \sqrt{j(\omega) - 1728}.$$

If n is not divisible by 3, we can choose the numbers a, c, ∂ in such a way that c always becomes divisible by 3.

Now, by § 54, (15), a linear transformation $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ has the following effect on the function $\gamma_2(\omega)$:

$$\gamma_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = e^{-\frac{2\pi i}{3}(\alpha\gamma + \beta\delta - \alpha\beta - \alpha^2\beta\delta)}\gamma_2(\omega).$$

In the compositions (8), (10) of § 69 we obtain

$$\lambda \equiv n, \quad \delta \equiv 0 \pmod 3,$$

and then α can be further determined in such a way that α also, and consequently c_2, become divisible by 3.

[15] In the first edition I called these equations "invariant transformation equations". This expression has been objected to by Klein (Vorlesungen über ausgewählte Kapitel der Zahlentheorie, autographiertes Heft, Göttingen 1897). I adopt Klein's proposal, in that I now call these equations "transformation equations of the first level", without entering here more closely into the justification of this expression.

From this it follows that by the two substitutions

$$(\omega, \omega+1), \quad \left(\omega, -\frac{1}{\omega}\right)$$

the ν functions

$$\gamma_2\left(\frac{c+\partial\omega}{a}\right)\gamma_2(\omega)^{-n} \tag{1}$$

are only permuted among one another, and hence are the roots of a transformation equation of the first level.

If, secondly, n is an odd number, let c be assumed even.

For the function $\gamma_3(\omega)$ we have [by § 54, (15)]:

$$\gamma_3\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = (-1)^{\alpha\gamma+\beta\gamma+\beta\delta}\gamma_3(\omega),$$

and in the compositions (8), (10), § 69 we have

$$\lambda \equiv 1, \quad \alpha \equiv \delta \equiv 0 \pmod 2.$$

The ν quantities

$$\gamma_3\left(\frac{c+\partial\omega}{a}\right)\gamma_3(\omega) \tag{2}$$

are therefore permuted among themselves, and are accordingly likewise the roots of a transformation equation of the first level.

In order to form the former of these equations for the simplest case $n = 2$, note the relation [§ 54, (5)]:

$$\gamma_2(\omega) = \frac{f(\omega)^{24}-16}{f(\omega)^8} = \frac{f_1(\omega)^{24}+16}{f_1(\omega)^8} = \frac{f_2(\omega)^{24}+16}{f_2(\omega)^8}, \tag{3}$$

from which, in virtue of

$$f_1(2\omega)f_2(\omega) = \sqrt{2} \qquad [\S\ 34, (16)]$$

there follows:

$$\begin{aligned} \gamma_2(2\omega) &= \frac{2^8 + f_2(\omega)^{24}}{f_2(\omega)^{16}} \\ \gamma_2\left(\frac{\omega}{2}\right) &= \frac{2^8 + f_1(\omega)^{24}}{f_1(\omega)^{16}} \\ \gamma_2\left(\frac{\omega+3}{2}\right) &= \frac{2^8 - f(\omega)^{24}}{f(\omega)^{16}}. \qquad [\S\ 34, (13)] \end{aligned} \tag{4}$$

If we denote these three quantities by x, x_0, x_1, then from the relations (6), (8), § 54 there results:

$$\begin{aligned} x + x_0 + x_1 &= \gamma_2(\omega)^2 \\ xx_0 + xx_1 + x_0x_1 &= 495\,\gamma_2(\omega) \\ xx_0x_1 &= -j(\omega) + 2^4 \cdot 3^3 \cdot 5^3, \end{aligned}$$

so that x, x_0, x_1 are the roots of the equation

$$x^3 - \gamma_2(\omega)^2x^2 + 5 \cdot 9 \cdot 11 \cdot \gamma_2(\omega)x + j(\omega) - 2^4 \cdot 3^3 \cdot 5^3 = 0. \tag{5}$$

The equation whose roots are the cubes of the roots of (5) is the invariant equation for $n = 2$, and it can be computed from this without difficulty.

§ 72. Multiplier Equations of the First Level.

We shall consider here those multiplier equations whose roots are the various powers of P_{11} multiplied by powers of $\gamma_2(\omega)$ and $\gamma_3(\omega)$, and whose coefficients

depend rationally on $j(\omega)$. These equations we call multiplier equations of the first level[16].

We consider the quantities [§ 68, (16)]:

$$P_{c,\mathfrak{d},a} = i^{\frac{a-1}{2}} \left(\frac{c}{e}\right) \sqrt{\mathfrak{d}}\, \frac{\eta\left(\dfrac{c+\mathfrak{d}\,\omega}{a}\right)}{\eta(\omega)}, \tag{1}$$

and the influence which the transformations

$$\begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \quad \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

exert upon them. This influence is determined by the formulae (8) to (13), § 69, according to which [because $c_1 \equiv c \pmod{e}$]

$$P_{c,\mathfrak{d},a} \text{ under } (\omega, \omega+1) \text{ is carried to } e^{\frac{\pi i(\lambda-1)}{12}} P_{c_1,\mathfrak{d},a}$$

$$\text{under } \left(\omega, -\frac{1}{\omega}\right), \text{ the image is } \sqrt{\frac{\mathfrak{d}}{\mathfrak{d}_2}}\, \frac{i^{\frac{a-a_2}{2}}}{\sqrt{-i\omega}} \left(\frac{c}{e}\right) \left(\frac{c_2}{e_2}\right) E\begin{pmatrix} \alpha, & \beta\ \dfrac{c_2+\mathfrak{d}_2\,\omega}{a_2} \\ \gamma, & \delta' \end{pmatrix} P_{c_2,\mathfrak{d}_2,a_2} \tag{2}$$

Here E has the meaning given in § 38, (15). But we have [§ 69, (11), (12)]

$$\sqrt{\mathfrak{d}}\, \sqrt{\alpha + \beta\, \frac{c_2+\mathfrak{d}_2\,\omega}{a_2}} = \sqrt{\mathfrak{d}_2\,\omega},$$

and consequently, if we set

$$E\begin{pmatrix} \alpha, & \beta\ \dfrac{c_2+\mathfrak{d}_2\,\omega}{a_2} \\ \gamma, & \delta' \end{pmatrix} = r\,\sqrt{-i\omega}\,\sqrt{\frac{\mathfrak{d}_2}{\mathfrak{d}}}$$

then

$$r = \left(\frac{\alpha}{\beta}\right) i^{\frac{1-\beta}{2}} e^{\frac{\pi i}{12}\left[\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma\right]}$$

is a 24th root of unity. Under the substitution

$$\left(\omega, -\frac{1}{\omega}\right)$$

$$P_{c,\mathfrak{d},a} \text{ is carried to } r\, i^{\frac{a-a_2}{2}} \left(\frac{c}{e}\right) \left(\frac{c_2}{e_2}\right) P_{c_2,\mathfrak{d}_2,a_2}$$

From this there follows, as above, the conclusion:

1. For every n the quantities

$$P^{24}_{c,\mathfrak{d},a}$$

[16] These equations have been investigated in particular detail by Kiepert, first in several memoirs in Crelles Journal, Bd. 87, 88, 95, and most fully in the memoirs in the Mathematischen Annalen, Bd. 26, 33. Cf. also F. Klein, Mathematische Annalen, Bd. 14, 15 and the autographed lectures mentioned above.

are the roots of a transformation equation.

If n is not divisible by 3, one may assume the c, c_2, c_1 to be divisible by 3. Then

$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \delta \equiv 0 \pmod 3.$$

Thus, as follows from § 38, (15), r is an eighth root of unity; if we therefore also observe

$$\gamma_2(\omega+1) = e^{-\frac{2\pi i}{3}}\gamma_2(\omega), \quad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega), \quad [\S\,54,\ (14)]$$

we have the theorem:

2. If n is not divisible by 3 and c is divisible by 3, then the quantities

$$P^8_{c,\mathfrak{d},a}\,\gamma_2(\omega)^{n-1}$$

are roots of a transformation equation.

To make the application to the simplest case $n = 2$, we set

$$x = 16\,\frac{\eta(2\omega)^8}{\eta(\omega)^8}, \quad x_0 = \frac{\eta\left(\frac{\omega}{2}\right)^8}{\eta(\omega)^8}, \quad x_1 = \frac{\eta\left(\frac{\omega+3}{2}\right)^8}{\eta(\omega)^8}, \tag{3}$$

and then x, x_0, x_1 are nothing other than our functions

$$f_2(\omega)^8, \quad f_1(\omega)^8, \quad -f(\omega)^8. \qquad [\S\,34,\ (9)]$$

These are, as we have already seen earlier [§ 54, (8)], the roots of the cubic equation

$$x^3 - x\,\gamma_2(\omega) + 16 = 0. \tag{4}$$

The circumstance that $f_2^8, f_1^8, -f^8$ are themselves roots of a transformation equation for the transformation degree 2 explains the phenomenon that on the adjunction of these quantities, and hence also on the adjunction of x^2, the transformation equations belonging to an even n become reducible.

If n is an odd number, one takes c to be divisible by 4; then it follows that [§ 69, (9), (11), (12)]

$$\lambda \equiv n \pmod 4,$$

$$\alpha \equiv 0, \quad \delta \equiv 0, \quad \beta \equiv a\,\mathfrak{d}_2, \quad \gamma \equiv -\mathfrak{d}\,a_2 \pmod 4,$$

and hence there results

$$r^6 = (-1)^{\frac{\beta-1}{2}} = (-1)^{\frac{n-1}{2}}(-1)^{\frac{a-a_2}{2}}$$

and from this there follows, having regard to

$$\gamma_3(\omega+1) = -\gamma_3(\omega), \quad \gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega): \qquad [\S\,54,\ (14)]$$

3. If n is odd and c divisible by 4, then the quantities

$$P^6_{c,\mathfrak{d},a}\,\gamma_3(\omega)^{\frac{n-1}{2}}$$

are roots of a multiplier equation. If $n \equiv 1 \pmod 4$, the same holds of $P^6_{c,\mathfrak{d},a}$.

As an example we choose $n = 3$ and set

(5)
$$x = 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6, \quad x_0 = -\left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad x_1 = -\left(\frac{\eta\left(\frac{4+\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad x_2 = -\left(\frac{\eta\left(\frac{8+\omega}{3}\right)}{\eta(\omega)}\right)^6.$$

The coefficients of the equation which has these quantities for its roots are rational functions of $\gamma_3(\omega)$, and since none of the quantities (5) becomes infinite for a finite value of γ_3, they are polynomials in γ_3. According to 3. we may therefore set this equation up in the form:

$$x^4 + A_1\gamma_3 x^3 + A_2 x^2 + A_3\gamma_3 x + A_4 = 0,$$

in which A_1, A_2, A_3, A_4 are polynomials in $j(\omega)$. First of all one obtains A_4 as the product $xx_0x_1x_2$, which can vanish for no value of $j(\omega)$ and must therefore be constant. From the beginnings of the development [§ 24]

(6)
$$\begin{aligned} x &= 27\,q\ldots, \\ x_0 &= -q^{-\frac{1}{3}}\ldots, \\ x_1 &= -e^{\frac{2\pi i}{3}}q^{-\frac{1}{3}}\ldots, \\ x_2 &= -e^{-\frac{2\pi i}{3}}q^{-\frac{1}{3}}\ldots, \\ \gamma_3 &= q^{-1}\ldots \end{aligned}$$

one therefore finds
$$A_4 = -27.$$

Further we have
$$-A_1\gamma_3 = x + x_0 + x_1 + x_2.$$

Since the right-hand side does not become infinite even in the first order for an infinite γ_3, that is, for $q = 0$, A_1 must vanish.

From
$$-A_3\gamma_3 = xx_0x_1x_2\left(\frac{1}{x} + \frac{1}{x_0} + \frac{1}{x_1} + \frac{1}{x_2}\right)$$
there results by (6) the constant value 1 for A_3. But in order to determine A_2, we must go one term further in the development. In
$$x_0^4 + \gamma_3 x_0 - 27 = -A_2 x_0^2$$
we substitute for x_0 the development
$$x_0 = -q^{-\frac{1}{3}} + 6\,q^{\frac{1}{3}} + \cdots,$$
and find $A_2 = 18$, so that the equation sought for reads:

(7)
$$x^4 + 18\,x^2 + \gamma_3\,x - 27 = 0.$$

If n is odd and not divisible by 3, let c be assumed divisible by 12.

It is then
$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \delta \equiv 0, \quad \beta \equiv a\,\mathfrak{d}_2, \quad \gamma \equiv -\mathfrak{d}\,a_2 \pmod{12}$$
and there results
$$r^2 = (-1)^{\frac{\beta-1}{2}} = (-1)^{\frac{n-1}{2}}(-1)^{\frac{a-a'}{2}},$$

hence the theorem

4. If n is relatively prime to 6 and c divisible by 12, then the quantities

$$P^2_{c,\mathfrak{d},a}\,\gamma_2(\omega)^{n-1}\,\gamma_3(\omega)^{\frac{n-1}{2}}$$

are roots of a multiplier equation of the first level.

Since the functions P^2 become infinite or zero for no finite value of $j(\omega)$, we conclude that the coefficients of the equation whose roots are the P^2 are polynomials in γ_2, γ_3, and that the last coefficient is a constant.

If n is a prime p, the roots of this equation are

$$x = p\left(\frac{\eta(p\omega)}{\eta(\omega)}\right)^2, \quad x_h = (-1)^{\frac{p-1}{2}}\left(\frac{\eta\left(\frac{12h+\omega}{p}\right)}{\eta(\omega)}\right)^2 \qquad h = 0, 1, \ldots, p-1$$

and the initial terms of the developments are the following:

$$x = p\,q^{\frac{p-1}{6}}\ldots, \quad x_h = (-1)^{\frac{p-1}{2}} e^{\frac{2\pi i h}{p}} q^{-\frac{p-1}{6p}}\ldots,$$

whereby the value $(-1)^{\frac{p-1}{2}}p$ results for the last coefficient. From the fact that the first term in the development of

$j(\omega)$ in powers of q has the coefficient 1, one concludes easily that the numerical coefficients in these equations are integers.

For $p = 5$ the equation in question has the form:

$$x^6 + A_1\gamma_2^2 x^5 + A_2\gamma_2 x^4 + A_3 x^3 + A_4\gamma_2 x + 5 = 0,$$

in which A_1, A_2, A_3, A_4 are polynomials in $j(\omega)$ and, as is easily seen, constants.

From the initial terms there follows at once

$$A_1 = 0, \quad A_2 = 0, \quad A_4 = -1.$$

But in order to determine A_3, one carries the development of x_0 as far as the next following term:

$$x_0 = q^{-\frac{7}{15}}\left(1 - 2q^{\frac{2}{5}}\ldots\right),$$

from which one obtains $A_3 = 10$; thus for $n = 5$ the multiplier equation reads

$$x^6 + 10x^3 - \gamma_2 x + 5 = 0. \tag{8}$$

In the same way one computes the equations for $p = 7$, $p = 11$, by using the developments of x_0:

$$p = 7: \; x_0 = -q^{-\frac{1}{7}}\left(1 - 2q^{\frac{2}{7}} - q^{\frac{4}{7}} + 2q^{\frac{6}{7}}\ldots\right),$$

$$p = 11: \; x_0 = -q^{-\frac{5}{33}}\left(1 - 2q^{\frac{2}{11}} - q^{\frac{4}{11}} + 2q^{\frac{6}{11}} + q^{\frac{8}{11}} + 2q^{\frac{10}{11}} + \ldots\right),$$

while of $\gamma_2(\omega)$, $\gamma_3(\omega)$ only the first terms are ever needed. Thus one finds

$$p = 7: \quad x^8 + 7\cdot 2x^6 + 7\cdot 9x^4 + 7\cdot 10x^2 + \gamma_3 x - 7 = 0, \tag{9}$$

$$p = 11: \quad x^{12} - 11\cdot 90x^6 + 11\cdot 40\gamma_2 x^4 + 11\cdot 15\gamma_3 x^3 + 11\cdot 2\gamma_2^2 x^2 + \gamma_2\gamma_3 x - 11 = 0. \tag{10}$$

If n is an odd square number, we assume c to be divisible by 8. Then $a : e$ and $\partial : e$ are likewise square numbers. Further we have

$$\alpha \equiv \delta \equiv 0 \pmod 8, \quad \lambda \equiv 1 \pmod 8 \tag{11}$$

(12)
$$\left.\begin{array}{lll} a=\beta\partial_2, & a_2=\beta\partial, & a\equiv\partial\equiv e\equiv 1 \\ c=\delta\partial_2, & c_2=-\alpha\partial, & a_2\equiv\partial_2\equiv e_2\equiv 1 \end{array}\right\} \pmod 8.$$

The root of unity r in (2) takes the value

(13)
$$r=\left(\frac{\alpha}{\beta}\right) i^{\frac{1-\beta}{2}} e^{\frac{2\pi i}{3}\frac{\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma}{8}}.$$

But by (12) β is divisible both by e and by e_2, and $\beta e e_2$ is a square; hence

$$\left(\frac{\alpha}{\beta}\right)=\left(\frac{\alpha}{ee_2}\right)=\left(\frac{\alpha}{e}\right)\left(\frac{\alpha}{e_2}\right)=\left(\frac{\delta}{e}\right)\left(\frac{\alpha}{e_2}\right) \text{ (since } \alpha\delta\equiv 1 \bmod e);$$

furthermore, by (12),

$$\left(\frac{\delta}{e}\right)=\left(\frac{c}{e}\right)\left(\frac{\partial_2}{e}\right), \quad \left(\frac{\alpha}{e_2}\right)=\left(\frac{c_2}{e_2}\right)\left(\frac{-\partial}{e_2}\right),$$

and

$$\left(\frac{\partial_2}{e}\right)=\left(\frac{e_2}{e}\right), \quad \left(\frac{-\partial}{e_2}\right)=\left(\frac{-e}{e_2}\right), \quad \left(\frac{e_2}{e}\right)\left(\frac{-e}{e_2}\right)=(-1)^{\frac{(e+1)(e_2-1)}{4}},$$

hence

$$\left(\frac{\alpha}{\beta}\right)=(-1)^{\frac{(e+1)(e_2-1)}{4}}\left(\frac{c}{e}\right)\left(\frac{c_2}{e_2}\right)$$

$$i^{\frac{1-\beta}{2}}=i^{\frac{1-ee_2}{2}}; \quad i^{\frac{a-a_2}{2}}=i^{\frac{e-e_2}{2}}.$$

Thus, under the transformations (2),

(14)
$$P_{c,\partial,a} \text{ is carried to } e^{\frac{2\pi i}{3}\frac{\lambda-1}{8}}P_{c_1,\partial,a} \text{ and to } e^{\frac{2\pi i}{3}\frac{\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma}{8}}P_{c_2,\partial_2,a_2}.$$

If n is moreover not divisible by 3, then let c be assumed divisible by 3, whereby

$$\lambda\equiv 1, \quad \alpha\equiv\delta\equiv 0 \pmod 3$$

result, and the third roots of unity occurring in (14) receive the value 1.

From this we obtain the following theorems:

5. If n is an odd square number, c divisible by 8, then the quantities

$$P^3_{c,\partial,a},$$

and if n is an odd square number not divisible by 3, c divisible by 24, then the quantities

$$P_{c,\partial,a}$$

are roots of multiplier equations of the first level.

The first examples are $n=9$, $n=25$.

These equations can be formed in various ways. Here, following Kiepert[1]), we take the following course: by § 69, we express the roots of a transformation equation belonging to degree p rationally in terms of those of the corresponding equation for degree p^2, and substitute this expression into the transformation equation belonging to p.

According to 3., formula (7), the equation

(15)
$$x^4+18x^2+\gamma_3(\omega)x-27=0$$

[1]) Zur Transformationstheorie der elliptischen Funktionen. Crelles Journal, Bd. 87, 88, 95.

has the following two functions as roots:

$$-\left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6 \tag{16}$$

Let the first of these values be x. Under the substitution $(\omega, \frac{\omega}{3})$, the second becomes $\frac{-27}{x}$; consequently x also satisfies the equation

$$27^3 + 18 \cdot 27x^2 - \gamma_3\left(\frac{\omega}{3}\right)x^3 - x^4 = 0 \tag{17}$$

If we now set

$$y = \left(\frac{\eta\left(\frac{\omega}{9}\right)}{\eta(\omega)}\right)^3, \tag{18}$$

then, under the substitution $(\omega, \frac{\omega}{3})$, x becomes $\frac{y^2}{x}$, so that one also obtains the equation:

$$y^8 + 18y^4x^2 + \gamma_3\left(\frac{\omega}{3}\right)y^2x^3 - 27x^4 = 0, \tag{19}$$

and by elimination of $\gamma_3\left(\frac{\omega}{3}\right)$ from (17) and (19) one obtains, after cancellation of the factor $y^2 + 27$,

$$x^4 - 18x^2y^2 - y^2(y^4 - 27y^2 + 27^2) = 0. \tag{20}$$

If this quadratic equation is solved for x^2, it follows that:

$$x^2 = y^3 + 9y^2 + 27y = (y+3)^3 - 27, \tag{21}$$

where the sign is decided by substituting the initial terms of the developments, most simply, no doubt, since this equation (by § 69) is also satisfied for

$$x = 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6, \quad y = 27\left(\frac{\eta(9\omega)}{\eta(\omega)}\right)^3$$

when one sets

$$x = 27q + \cdots, \quad y = 27q^2 + \cdots$$

in it.

If one isolates the term linear in x in (15), squares, and substitutes (21), one obtains the multiplier equation sought for transformation degree 9. It takes a simpler form if one sets

$$(y+3)^3 = t \tag{22}$$

then it becomes:

$$[j(\omega) - 27 \cdot 64](t - 27) = (t^2 - 36t + 27 \cdot 8)^2 \tag{23}$$

or

$$j(\omega)(t - 27) = t(t - 24)^3. \tag{24}$$

One can proceed in quite a similar way for transformation degree 25.

If we set

$$x = \left(\frac{\eta\left(\frac{\omega}{5}\right)}{\eta(\omega)}\right)^2, \quad y = \frac{\eta\left(\frac{\omega}{25}\right)}{\eta(\omega)} \tag{25}$$

then we have first of all, by 4. (8):

$$x^6 + 10x^3 - \gamma_2(\omega)x + 5 = 0, \tag{26}$$

from which, as above, the two equations

$$5^5 + 10 \cdot 5^2 x^3 - \gamma_2\left(\frac{\omega}{5}\right) x^5 + x^6 = 0,$$

$$y^{12} + 10y^6 x^3 - \gamma_2\left(\frac{\omega}{5}\right) y^2 x^5 + 5x^6 = 0,$$

and, by elimination of $\gamma_2\left(\frac{\omega}{5}\right)$,

$$x^6 - 10x^3y^2(5 + y^2) = y^2(y^8 + 5y^6 + 5^2y^4 + 5^3y^2 + 5^4).$$

This equation, solved for x^3, gives:

$$x^3 = y^5 + 5y^4 + 15y^3 + 25y^2 + 25y, \tag{27}$$

and if for brevity we denote the right-hand side of this equation by $\chi(y)$, then by (26)

$$j(\omega)\chi(y) = [\chi(y)^2 + 10\chi(y) + 5]^3. \tag{28}$$

This is the equation of degree 30 for y that was sought.

§ 73. The Schlaefli Modular Equations.

One arrives at simpler transformation equations by adjoining to the domain of rationality, which up to now consisted of the rational functions of $j(\omega)$, the quantity $f(\omega)^{24}$. To this

domain of rationality belong the functions of ω which, under the two substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \quad (\omega, \omega + 2) \tag{1}$$

remain unchanged (§ 54, 2). If, then, a system of ν functions $\Phi_{a,c,\partial}$ is permuted by the substitutions (1) only among themselves, these functions are the roots of a transformation equation whose coefficients depend rationally on $f(\omega)^{24}$. To these belong (as we have already shown earlier by another route), for an odd n, which we shall now always assume, certain powers of the quantities

$$f\left(\frac{c + \partial\omega}{a}\right),$$

in which, as may be remarked once and for all, c is assumed divisible by 16, and, when n is not divisible by 3, divisible by 48.

The somewhat extended principles of § 70 lead comparatively simply to the calculation of these equations.

An extension is, however, necessary for the following reason:

In the considerations hitherto we were able to draw the conclusion: if a rational function of $j(\omega)$ remains finite for every finite ω with positive imaginary part, then it is a polynomial in $j(\omega)$, because to every finite $j(\omega)$ there belongs also a finite, non-real ω (§ 52). In the case of the rational functions of $f(\omega)$, however, from the finiteness for every finite imaginary ω we can only conclude that they are Laurent polynomials in $f(\omega)$, allowing negative powers represented by $1 : f(\omega)$, because only to every finite $f(\omega)$, with the exception of $f(\omega) = 0$, does there belong a finite imaginary ω.

But to the substitution

(2) $$\left(\omega, \frac{\omega-1}{\omega+1}\right)$$

there corresponds the interchange

$$\left(f(\omega), \frac{\sqrt{2}}{f(\omega)}\right), \qquad [\S\,34, (18)]$$

and if, then, the functions $\Phi_{a,c,\partial}$ are so chosen that they are permuted only among one another by the substitution (2) as well, then the coefficients of the equation of which they are the roots will depend rationally on

$$f(\omega) + \frac{\sqrt{2}}{f(\omega)}$$

depend. They are polynomials in this combination if the $\Phi_{a,c,\partial}$ remain finite for every finite, non-real ω, and they are constant if all the $\Phi_{a,c,\partial}$ remain finite for $q = 0$ as well, that is, for $f(\omega) = 0$ and $f(\omega) = \infty$. In this case all the $\Phi_{a,c,\partial}$ are equal to one and the same constant (§ 70).

From this the theorem follows:

If one forms polynomials $\Phi_{a,c,\partial}$ in

$$f\left(\frac{c+\partial\omega}{a}\right), \quad f(\omega), \quad \frac{1}{f\left(\frac{c+\partial\omega}{a}\right)}, \quad \frac{1}{f(\omega)},$$

which have the property:

1. by the substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \quad (\omega, \omega+2), \quad \left(\omega, \frac{\omega-1}{\omega+1}\right)$$

of being only permuted among one another,

2. of not becoming infinite for $q = 0$, then

$$\Phi_{a,c,\partial} = \text{constans}$$

is a transformation equation.

In order to satisfy these conditions, the influence of the substitutions (1) upon the functions f must first be investigated. This follows from the compositions § 69, (8) to (13) and from the transformation formulae for the f-functions § 40.

For brevity we introduce the notation:

(3)
$$\begin{aligned} f(\omega) &= u, & f\left(\frac{c+\partial\omega}{a}\right) &= v, \\ f_1(\omega) &= u_1, & \left(\frac{2}{a}\right) f_1\left(\frac{c+\partial\omega}{a}\right) &= v_1, \\ f_2(\omega) &= u_2, & \left(\frac{2}{\partial}\right) f_2\left(\frac{c+\partial\omega}{a}\right) &= v_2. \end{aligned}$$

There then result the following corresponding changes, if in the notation no account is taken of the conversion of the numbers a, c, ∂ into a, c_1, ∂ or a_2, c_2, ∂_2, as it has just been characterised by the cited formulae of § 69:

(4)
$$\begin{array}{cccc}
\omega, & u, & u_1, & u_2, \\
-\dfrac{1}{\omega}, & u, & u_2, & u_1, \\
\omega+1, & e^{-\frac{\pi i}{24}}u_1, & e^{-\frac{\pi i}{24}}u, & e^{\frac{\pi i}{12}}u_2, \\
\omega+2, & e^{-\frac{\pi i}{12}}u, & e^{-\frac{\pi i}{12}}u_1, & e^{\frac{\pi i}{6}}u_2, \\
\hline
\omega, & v, & v_1, & v_2, \\
-\dfrac{1}{\omega}, & \varrho v, & \varrho v_2, & \varrho v_1, \\
\omega+1, & \sigma e^{-\frac{n\pi i}{24}}v_1, & \sigma e^{-\frac{n\pi i}{24}}v, & \sigma e^{\frac{n\pi i}{12}}v_2, \\
\omega+2, & \sigma^2 e^{-\frac{n\pi i}{12}}v, & \sigma^2 e^{-\frac{n\pi i}{12}}v_1, & \sigma^2 e^{\frac{n\pi i}{6}}v_2,
\end{array}$$

where

$$\varrho = e^{-\frac{2\pi i}{3}\left[\alpha(\gamma-\beta)+(\alpha^2-1)\beta^3\delta\right]}, \qquad \sigma = \left(\frac{2}{a}\right)e^{\frac{(n-\lambda)\pi i}{24}}$$

are two third roots of unity, which, if n is not divisible by 3, have the value 1.

In order to determine further the effect of the substitution (2), which arises from the transformation of the second degree

$$\begin{pmatrix} 1, & 1 \\ -1, & 1 \end{pmatrix}$$

upon the functions u, v, we must determine the transformations of the first and nth order

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix}$$

in such a way that

(5)
$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}\begin{pmatrix} 1, & 1 \\ -1, & 1 \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}\begin{pmatrix} 1, & 1 \\ -1, & 1 \end{pmatrix}\begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix}$$

holds. This approach leads to the equations

(6)
$$\begin{aligned} a &= a'(\alpha-\beta)+c'(\alpha+\beta), & a &= \partial'(\alpha+\beta), \\ c-\partial &= a'(\gamma-\delta)+c'(\gamma+\delta), & c+\partial &= \partial'(\gamma+\delta). \end{aligned}$$

With this, since $\alpha+\beta$ and $\gamma+\delta$ are relatively prime in consequence of $\alpha\delta-\beta\gamma=1$, ∂' is first determined as the greatest common divisor of a and $c+\partial$, and a' results from $n=a'\partial'$. Then, by equations (6):

$$\alpha+\beta = \frac{a}{\partial'}, \qquad \gamma+\delta = \frac{c+\partial}{\beta'},$$

and δ and β can be so determined that

(7)
$$\delta(\alpha+\beta)-\beta(\gamma+\delta) = \alpha\delta-\beta\gamma = 1.$$

From the equations (6) for a and $c-\partial$ there is then found

(8) $$c' + a' = a\delta - (c - \partial)\beta.$$

Since $\alpha + \beta$ and $\gamma + \delta$ are both odd, δ and β cannot, by (7), both be odd; consequently c' turns out even by (8). If one replaces δ, β, which by (7) is permitted, by $\delta + h(\gamma + \delta)$, $\beta + h(\alpha + \beta)$, for an arbitrary h, then c' changes by $2ha'$ according to (6) and (8), and h can be so disposed of that c' becomes divisible by 16 or 48.

Then, by (6), $\alpha + \beta + \gamma - \delta$ is even, and therefore formula (12), § 40 is to be applied. In it the following is now to be taken into account

$$\begin{matrix} a'(\alpha - \beta) \equiv a, & \partial(\alpha - \beta) \equiv \partial' \\ \partial'(\alpha + \beta) \equiv a, & a'(\gamma - \delta) \equiv -\partial \end{matrix} \pmod{16},$$

from which it follows:

$$n(\alpha - \beta)(\alpha + \beta + \gamma - \delta) \equiv a^2 - \partial'^2 \pmod{16},$$

hence

$$\left(\frac{2}{\alpha - \beta}\right) e^{-\frac{3\pi i}{8}(\alpha-\beta)(\alpha+\beta+\gamma-\delta)} = \left(\frac{2}{a}\right)\left(\frac{2}{a'}\right)\left(\frac{2}{a}\right)\left(\frac{2}{\partial'}\right) = \left(\frac{2}{n}\right)$$

and

$$\varrho = e^{-\frac{2\pi i}{3}\left[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta\right]}.$$

If n is not divisible by 3, then

$$\begin{matrix} 2\alpha \equiv a(a' + \partial'), & 2\delta \equiv \partial(a' + \partial'), \\ 2\beta \equiv a(\partial' - a'), & 2\gamma \equiv \partial(\partial' - a') \end{matrix} \pmod{3},$$

hence either α and δ or β and γ are divisible by 3, and therefore $\varrho = 1$.

From this there result the following interchanges belonging together (where the change of a, c, ∂ into a', c', ∂' is determined by (6)):

(9)
$$\begin{matrix} \omega, & u, & v \\ \dfrac{\omega - 1}{\omega + 1}, & \dfrac{\sqrt{2}}{u}, & \varrho\left(\dfrac{2}{n}\right)\dfrac{\sqrt{2}}{v}. \end{matrix}$$

We apply the interchanges (4), (5), (9) to the following functions:

(10)
$$\begin{aligned} A &= \left(\frac{u}{v}\right)^r + \left(\frac{v}{u}\right)^r \\ B &= (uv)^s + \left(\frac{2}{n}\right)^{r+s} \frac{2^s}{(uv)^s}, \end{aligned}$$

in which r, s are two integers that satisfy the conditions

(11) $$(n - 1)r \equiv 0, \quad (n + 1)s \equiv 0 \pmod{12}$$

and which therefore, if n is divisible by 3, are both divisible by 3, and where

$$\left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}$$

holds.

There then result the following mutually corresponding permutations:

$$
\begin{array}{lll}
\omega, & A, & B, \\
-\dfrac{1}{\omega}, & A, & B, \\
\omega+2, & (-1)^{\frac{(n-1)r}{12}} A, & (-1)^{\frac{(n+1)s}{12}} B, \\
\dfrac{\omega-1}{\omega+1}, & \left(\dfrac{2}{n}\right)^r A, & \left(\dfrac{2}{n}\right)^r B.
\end{array} \tag{12}
$$

If we now form a polynomial in A and B with numerical coefficients

$$\Phi_{a,c,\partial} = \Sigma\, M_{h,k}\, A^h B^k, \tag{13}$$

in which, should changes of sign occur in (12), the exponents h, k are to be so arranged that in all terms of (13) the same alterations of sign take place, then the system of functions $\Phi_{a,c,\partial}$, or at least $\Phi^2_{a,c,\partial}$, will satisfy requirement 1. of the theorem set up above; and in order to satisfy requirement 2. as well, and so to obtain a transformation equation, we have to determine the coefficients $M_{h,k}$ in such a way that all the ν values of $\Phi_{a,c,\partial}$ remain finite for $q = 0$. This problem is simplified essentially if n has no square divisor, and still more if n is a prime.

For if n has no square divisor, one can derive all the values $\Phi_{a,c,\partial}$ from $\Phi_{a,0,\partial}$ by increasing ω by certain integers; if therefore, as is our requirement, no negative powers occur in the expansion of $\Phi_{a,0,\partial}$ in ascending powers of q, then the same holds of all the $\Phi_{a,c,\partial}$.

But if n is a prime, it is enough to show that $\Phi_{1,0,n}$ contains no negative powers of q, since the same follows for $\Phi_{n,0,1}$ by the interchange of ω with $\omega : n$.

The constant value which the function $\Phi_{a,c,\partial}$ takes is determined from one term of the expansion.

In carrying out these calculations the expansions § 24, (11), are of use,

$$
\begin{aligned}
A &= q^{-\frac{n-1}{24}r} \prod_{1,\infty}^{h} \left(\frac{1+q^{n(2h-1)}}{1+q^{2h-1}}\right)^r \\
&\quad + q^{\frac{n-1}{24}r} \prod_{1,\infty}^{h} \left(\frac{1+q^{2h-1}}{1+q^{n(2h-1)}}\right)^r \\
B &= q^{-\frac{n+1}{24}s} \prod_{1,\infty}^{h} (1+q^{2h-1})^s (1+q^{n(2h-1)})^s \\
&\quad + \left(\frac{2}{n}\right)^{r+s} q^{\frac{n+1}{24}s} 2^s \prod_{1,\infty}^{h} \frac{1}{(1+q^{2h-1})^s (1+q^{n(2h-1)})^s}.
\end{aligned} \tag{14}
$$

The formulae do not always come out simplest when r, s are assumed as small as possible; rather it proves most expedient to add the further condition

$$\frac{(n-1)r}{12} \equiv \frac{(n+1)s}{12} \pmod 2 \tag{15}$$

as well.

In this way we obtain, for the first seven odd primes, the following determination of A and B:

$$\begin{aligned}
n=3,\quad & A=\left(\frac{u}{v}\right)^6+\left(\frac{v}{u}\right)^6, && B=(uv)^3-\frac{8}{(uv)^3},\\
n=5,\quad & A=\left(\frac{u}{v}\right)^3+\left(\frac{v}{u}\right)^3, && B=(uv)^2-\frac{4}{(uv)^2},\\
n=7,\quad & A=\left(\frac{u}{v}\right)^4+\left(\frac{v}{u}\right)^4, && B=(uv)^3+\frac{8}{(uv)^3},\\
n=11,\quad & A=\left(\frac{u}{v}\right)^6+\left(\frac{v}{u}\right)^6, && B=uv-\frac{2}{vu},\\
n=13,\quad & A=\frac{u}{v}+\frac{v}{u}, && B=(uv)^6-\frac{64}{(uv)^6},\\
n=17,\quad & A=\left(\frac{u}{v}\right)^3+\left(\frac{v}{u}\right)^3, && B=(uv)^4+\frac{16}{(uv)^4},\\
n=19,\quad & A=\left(\frac{u}{v}\right)^2+\left(\frac{v}{u}\right)^2, && B=(uv)^3-\frac{8}{(uv)^3}.
\end{aligned}$$

The expansions in powers of q for A and B result from (14), so far as they are needed for the calculation, as follows:

$$\begin{aligned}
n=3,\quad & A=q^{-\frac{1}{2}}(1-5q\dots),\\
& B=q^{-\frac{1}{2}}(1-5q\dots),\\
n=5,\quad & A=q^{-\frac{1}{2}}(1-2q\dots),\\
& B=q^{-\frac{1}{2}}(1-2q\dots),\\
n=7,\quad & A=q^{-1}(1-4q\dots),\\
& B=q^{-1}(1+3q\dots),\\
n=11,\quad & A=q^{-\frac{5}{2}}(1-6q+21q^2\dots),\\
& B=q^{-\frac{1}{2}}(1-q+2q^2\dots),\\
n=13,\quad & A=q^{-\frac{1}{2}}(1+2q^2-2q^3\dots),\\
& B=q^{-\frac{7}{2}}(1+6q+15q^2+26q^3\dots),\\
n=17,\quad & A=q^{-2}(1-3q+6q^2-13q^3+25q^4-39q^5+76q^6\dots),\\
& B=q^{-3}(1+4q+6q^2+8q^3+17q^4+28q^5+54q^6\dots),\\
n=19,\quad & A=q^{-\frac{3}{2}}(1-2q+3q^2-5q^3+11q^4-13q^5+24q^6-28q^7\dots),\\
& B=q^{-\frac{5}{2}}(1+3q+3q^2+4q^3+9q^4+4q^5+39q^6-27q^7\dots),
\end{aligned}$$

and from these one obtains, by elimination of the negative powers, the required equations between A and B:

I.
$$\begin{aligned}
n=3,\quad & A-B=0,\\
n=5,\quad & A-B=0,\\
n=7,\quad & A-B+7=0,\\
n=11,\quad & A-B^5+B^3+2B=0,\\
n=13,\quad & A^7+6A^5+A^3-20A-B=0,\\
n=17,\quad & A^3-B^2+17AB-34A^2+34B+116A\\
& \qquad +440=0,\\
n=19,\quad & A^5-B^3+19AB^2-95A^2B+109A^3\\
& \qquad +128B-128A=0\ ^1).
\end{aligned}$$

[1]) These equations were first set up by Schlaefli (Journal für Mathematik, Bd. 72).

From this system of equations one derives a second and a third for the functions f_1, f_2, by replacing ω by $\omega+1$ and thereupon ω by $-1:\omega$. These two systems have the same form; only that in the one case

$$u_1 = f_1(\omega), \qquad v_1 = \left(\frac{2}{a}\right) f_1\left(\frac{c+\partial\omega}{a}\right),$$

and in the other case

$$u_1 = f_2(\omega), \qquad v_1 = \left(\frac{2}{\partial}\right) f_2\left(\frac{c+\partial\omega}{a}\right)$$

is to be set. From the interchanges (4) there results in this way:

II. $n=3$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^6 - \left(\frac{v_1}{u_1}\right)^6, \quad B_1 = (u_1v_1)^3 + \frac{8}{(u_1v_1)^3},$$
$$A_1 + B_1 = 0,$$

$n=5$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^3 - \left(\frac{v_1}{u_1}\right)^3, \quad B_1 = (u_1v_1)^2 + \frac{4}{(u_1v_1)^2},$$
$$A_1 + B_1 = 0,$$

$n=7$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^4 + \left(\frac{v_1}{u_1}\right)^4, \quad B_1 = (u_1v_1)^3 + \frac{8}{(u_1v_1)^3},$$
$$A_1 - B_1 - 7 = 0,$$

$n=11$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^6 - \left(\frac{v_1}{u_1}\right)^6, \quad B_1 = u_1v_1 + \frac{2}{u_1v_1},$$
$$A_1 + B_1^5 + B_1^3 - 2B_1 = 0,$$

$n=13$,
$$A_1 = \frac{u_1}{v_1} - \frac{v_1}{u_1}, \qquad B_1 = (u_1v_1)^6 + \frac{64}{(u_1v_1)^6},$$
$$A_1^7 - 6A_1^5 + A_1^3 + 20A_1 + B_1 = 0,$$

$n=17$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^3 + \left(\frac{v_1}{u_1}\right)^3, \quad B_1 = (u_1v_1)^4 + \frac{16}{(u_1v_1)^4},$$
$$A_1^3 - B_1^2 - 17A_1B_1 - 34A_1^2 - 34B_1 + 116A_1 + 440 = 0,$$

$n=19$,
$$A_1 = \left(\frac{u_1}{v_1}\right)^2 - \left(\frac{v_1}{u_1}\right)^2, \quad B_1 = (u_1v_1)^3 + \frac{8}{(u_1v_1)^3},$$
$$A_1^5 + B_1^3 - 19A_1B_1^2 + 95A_1^2B_1 - 109A_1^3 + 128B_1 - 128A_1 = 0.$$

§ 74. The Form of the Schlaefli Modular Equations for a Prime Degree.

The form which we have found in the preceding section for the relations subsisting between

(1) $$u = f(\omega), \qquad v = f\Big(\frac{c+\partial\omega}{a}\Big)$$

can easily be brought under a general law, at least when the degree of transformation is a prime p. Since much depends on this form in what follows, we enter here somewhat more closely into it.

The determination of the numbers r, s by (11) and (15) of the preceding section depends on the behaviour of p with respect to the modulus 24, and since we may exclude the case $p=3$, we have

the following cases:

$$
\begin{array}{lll}
p \equiv 1 \pmod{24} & r = 1 & s = 12 \\
p \equiv 5 & r = 3 & s = 2 \\
p \equiv 7 & r = 4 & s = 3 \\
p \equiv 11 & r = 6 & s = 1 \\
p \equiv 13 & r = 1 & s = 6 \\
p \equiv 17 & r = 3 & s = 4 \\
p \equiv 19 & r = 2 & s = 3 \\
p \equiv 23 & r = 12 & s = 1,
\end{array} \tag{2}
$$

so that $r + s$ is always odd, and $p + 1$ is divisible by $2r$, $p - 1$ by $2s$. Accordingly we obtain

$$
\begin{aligned}
A &= \left(\frac{u}{v}\right)^r + \left(\frac{v}{u}\right)^r, \\
B &= (uv)^s + \left(\frac{2}{p}\right)\frac{2^s}{(uv)^s}.
\end{aligned} \tag{3}
$$

Now we know that the $p + 1$ quantities v are the roots of an irreducible transformation equation

$$\Phi_p(v, u) = v^{p+1} + U_1 v^p + \cdots U_{p+1} = 0 \tag{4}$$

in which the coefficients $U_1 \ldots, U_{p+1}$ are rational functions of u.

We conclude at once that they are polynomials in u. For in the first place, all the roots of (4) become infinite for $u = \infty$, as one recognises on letting $\omega = i\infty$, and therefore $q = 0$. In the second place, by (9) of the preceding

section, the interchange $\left(u, \frac{\sqrt{2}}{u}\right)$ carries the totality of the roots v over into

$$\left(\frac{2}{p}\right)\frac{\sqrt{2}}{v}$$

From this it follows that for $u = 0$ all the roots v pass over into zero, so that none of them becomes infinite; and from this we must conclude that the $U_1, U_2, \ldots U_{p+1}$ are not only polynomials in u, but that each of them must also contain the factor u.

We now conclude first of all, exactly as with the invariant equation (§ 69, 2., 3. and 6.), that

$$\Phi_p(v, u) = \Phi_p(u, v), \tag{5}$$

and that

$$\Phi_p(v, u) = (v^p - u)(v - u^p) + p \sum_{1,p-1}^{h,k} c_{h,k} u^h v^k, \tag{6}$$

where $c_{h,k} = c_{k,h}$ are integers. We wish to represent this function in the form

$$\Phi_p(v, u) = v^{p+1} + u^{p+1} + \sum_{1,p}^{h,k} a_{h,k} u^h v^k, \tag{7}$$

where accordingly $a_{h,k} = a_{k,h}$ are likewise integers, upon whose divisibility by p nothing further now depends, and in particular $a_{p,p} = -1$. If in the equation

$$\Phi_p[f(p\,\omega), f(\omega)] = 0$$

we replace ω by $\omega + 2$, then, on account of the irreducibility, it results from the relation

$$f(\omega + 2) = e^{-\frac{\pi i}{12}} f(\omega),$$

that in (7) not all terms occur, but only those in which the exponents h, k satisfy the congruence

$$hp + k \equiv p + 1 \pmod{24} \tag{8}$$

Now we know yet a further property of the functions $\Phi_p(v, u)$, one which follows from the interchange (9) of the preceding section already used, where $\varrho = 1$ is to be put, and which, if for brevity we set

$$\varepsilon = \Big(\frac{2}{p}\Big) = (-1)^{\frac{p^2-1}{8}}$$

set, it can be represented as follows:

$$\Phi_p(v, u) = \Big(\frac{uv}{\sqrt{2}}\Big)^{p+1} \Phi_p\Big(\frac{\varepsilon\sqrt{2}}{v}, \frac{\sqrt{2}}{u}\Big). \tag{9}$$

From this one infers the relations

$$\varepsilon^h 2^{\frac{h+k-p-1}{2}} a_{p+1-h,\, p+1-k} = a_{h,k}, \qquad a_{h,k} = a_{k,h}. \tag{10}$$

If, therefore, we collect together in (7) the terms with equal coefficients $a_{h,k}$, we may restrict ourselves to the assumption that $h \geqq k$, $h + k \geqq p + 1$; and, apart from the two terms v^{p+1}, u^{p+1}, Φ_p presents itself as an aggregate of terms of the two following forms (if it is further taken into account that, on account of (8), $\varepsilon^h = \varepsilon^k$):

$$\begin{gathered} v^h u^k + v^k u^h + \varepsilon^h 2^{\frac{h+k-p-1}{2}} (u^{p+1-h} v^{p+1-k} + u^{p+1-k} v^{p+1-h}) = \\ \sum_{h>k} (uv)^{\frac{p+1}{2}} \left[\Big(\frac{v}{u}\Big)^{\frac{h-k}{2}} + \Big(\frac{u}{v}\Big)^{\frac{h-k}{2}} \right] \left[(uv)^{\frac{h+k-p-1}{2}} + \frac{\varepsilon^h 2^{\frac{h+k-p-1}{2}}}{(uv)^{\frac{h+k-p-1}{2}}} \right]. \end{gathered} \tag{11}$$

$$\begin{gathered} v^h u^h + \varepsilon^h 2^{h-\frac{p+1}{2}} (uv)^{p+1-h} \\ = (uv)^{\frac{p+1}{2}} \left[(uv)^{h-\frac{p+1}{2}} + \frac{\varepsilon^h 2^{h-\frac{p+1}{2}}}{(uv)^{h-\frac{p+1}{2}}} \right]. \end{gathered} \tag{12}$$

The coefficients of these terms in Φ_p are integers. Now from (8) there results:

$$\begin{gathered} (h-k) \equiv (h-1)(p+1) \\ h + k - p - 1 \equiv -h(p-1) \end{gathered} \pmod{24},$$

and therefore $h - k$ is divisible by $2r$, and $h + k - p - 1$ (in which h may also be $= k$) by $2s$ [§ 74, (2)].

If we therefore set

$$\frac{h-k}{2} = r\alpha, \qquad \frac{h+k-p-1}{2} = s\beta,$$

then α and β are positive integers which lie between the limits

$$0 \leqq \alpha < \frac{p+1}{2r}, \qquad 0 \leqq \beta \leqq \frac{p+1}{2s}, \tag{13}$$

and moreover the congruence $-h(p-1) \equiv 2s\beta \pmod{24}$, which follows from (8), shows that at least in the cases where $\varepsilon = -1$, $h \equiv \beta \pmod 2$ [§ 74, (2)], and hence always $\varepsilon^h = \varepsilon^\beta$.

Accordingly (11) becomes

$$(uv)^{\frac{p+1}{2}} \left[\Big(\frac{v}{u}\Big)^{r\alpha} + \Big(\frac{u}{v}\Big)^{r\alpha} \right] \left[(uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}} \right],$$

and (12)

$$(uv)^{\frac{p+1}{2}}\left[(uv)^{s\beta}+\frac{\varepsilon^{\beta}2^{s\beta}}{(uv)^{s\beta}}\right].$$

Now the well-known formulae hold, when x, γ are arbitrary quantities and

$$x+\frac{\gamma}{x}=y$$

is put,

$$x^2+\frac{\gamma^2}{x^2}=y^2-2\gamma, \qquad x^3+\frac{\gamma^3}{x^3}=y^3-3\gamma y,\ \ldots,$$

$$\Big(x^n+\frac{\gamma^n}{x^n}\Big)\Big(x+\frac{\gamma}{x}\Big)=x^{n+1}+\frac{\gamma^{n+1}}{x^{n+1}}+\gamma\Big(x^{n-1}+\frac{\gamma^{n-1}}{x^{n-1}}\Big),$$

from which one recognises, by the inference from n to $n+1$, that

$$x^n+\frac{\gamma^n}{x^n}$$

can be represented, for every arbitrary n, as a polynomial of degree n in y which, when γ is an integer, has integer coefficients, its leading coefficient being $=1$.

The two quantities

$$\Big(\frac{v}{u}\Big)^{r\alpha}+\Big(\frac{u}{v}\Big)^{r\alpha} \qquad \text{and} \qquad (uv)^{s\beta}+\frac{\varepsilon^{\beta}2^{s\beta}}{(uv)^{s\beta}}$$

can therefore be represented in this way as polynomials in A and B of degrees α and β, and we find, if we further separate off the term corresponding to the value $\beta=\dfrac{p-1}{2s}$, which has the coefficient -1, and denote integer coefficients by $c_{\alpha,\beta}$:

$$\frac{\Phi_p(u,v)}{(uv)^{\frac{p+1}{2}}}=A^{\frac{p+1}{2r}}-B^{\frac{p-1}{2s}}+\sum^{\alpha,\beta}c_{\alpha,\beta}A^{\alpha}B^{\beta}, \tag{14}$$

in which α and β are bound to the limits

$$0\leqq\alpha<\frac{p+1}{2r}, \qquad 0\leqq\beta<\frac{p-1}{2s}$$

This is the form which we gave to the modular equations up to $p=19$ in the preceding section.

It should still be mentioned that the discussion carried out in § 69, 4., 5. for the invariant equation in the case of a composite degree of transformation holds unchanged for the Schlaefli modular equations as well, from which we may conclude that all these equations have rational numerical coefficients.

§ 75. The Irrational Forms of the Modular Equations.

By applying the same procedure, far simpler forms can be given to the transformation equations. The forms of equation with which we shall now occupy ourselves contain the three functions f, f_1, f_2 at once; but since, by § 34, (11), (12), two of these functions can be expressed by the third, two of them can be eliminated, and in this way one can arrive at the equations of the preceding section. If one substitutes for f_1, f_2 the expressions in terms of f, or for the three functions f, f_1, f_2 the expressions in terms of k^2, then radical signs occur, and this explains the name of these equations.

We set, as above,

$$
\begin{aligned}
u &= f(\omega), & v &= f\Big(\frac{c+\partial\,\omega}{a}\Big),\\
u_1 &= f_1(\omega), & v_1 &= \Big(\frac{2}{a}\Big) f_1\Big(\frac{c+\partial\,\omega}{a}\Big),\\
u_2 &= f_2(\omega), & v_2 &= \Big(\frac{2}{\partial}\Big) f_2\Big(\frac{c+\partial\,\omega}{a}\Big),\\
u\,u_1\,u_2 &= \sqrt{2}, & v\,v_1\,v_2 &= \Big(\frac{2}{n}\Big)\sqrt{2},
\end{aligned}
\tag{1}
$$

and obtain the following corresponding permutations:

$$
\begin{array}{llll}
\omega, & uv, & u_1v_1, & u_2v_2,\\
-\dfrac{1}{\omega}, & \varrho\,uv, & \varrho\,u_2v_2, & \varrho\,u_1v_1,\\
\omega+1, & e^{-\frac{(n+1)\pi i}{24}}\sigma\,u_1v_1, & e^{-\frac{(n+1)\pi i}{24}}\sigma\,uv, & e^{\frac{(n+1)\pi i}{12}}\sigma\,u_2v_2,
\end{array}
\tag{2}
$$

where ϱ, σ are the third roots of unity defined above [§ 73, (4)].

We distinguish three different cases according to the behaviour of n with respect to the modulus 8.

1. $n+1 \equiv 0$, (mod 8).

We set

$$
\begin{aligned}
2A &= uv + (-1)^{\frac{n+1}{8}}(u_1v_1 + u_2v_2),\\
B &= uvu_1v_1 + uvu_2v_2 + (-1)^{\frac{n+1}{8}}u_1v_1u_2v_2\\
&= \frac{2}{u_1v_1} + \frac{2}{u_2v_2} + (-1)^{\frac{n+1}{8}}\frac{2}{uv},
\end{aligned}
\tag{3}
$$

so that the interchanges that belong together come out as

$$
\begin{array}{lll}
\omega, & A, & B,\\
-\dfrac{1}{\omega}, & \varrho\,A, & \varrho^2B,\\
\omega+1, & e^{\frac{\pi i(n+1)}{12}}\sigma\,A, & e^{-\frac{\pi i(n+1)}{12}}\sigma^2B.
\end{array}
\tag{4}
$$

A product of the form A^hB^k therefore takes on, through the two interchanges

$$\Big(\omega, -\frac{1}{\omega}\Big), \quad (\omega, \omega+1)$$

the factors

$$\varrho^{h-k}, \quad e^{(h-k)\frac{(n+1)\pi i}{12}}\sigma^{h-k},$$

which are $= 1$ when $n+1$ is divisible by 3. We therefore now form, with numerical coefficients $M_{h,k}$, functions of the form

$$\Phi_{a,c,\partial} = \Sigma\, M_{h,k}A^hB^k, \tag{5}$$

in which, if $n \equiv 0$ or $\equiv 1$ (mod 3), h and k may take only such (integral) values whose differences $h-k$ leave the same residue on division by 3, while if $n \equiv -1$ (mod 3) they are not subject to this restriction. Such a function itself, or at least its third power, therefore satisfies a transformation equation of the first level. It remains finite, moreover, for all finite values of $j(\omega)$, and is consequently an integral algebraic function of $j(\omega)$. [The transformation of the second order $\begin{pmatrix} 1, & 1\\ -1, & 1\end{pmatrix}$ need not be brought in here, since the totality of the $\Phi_{a,c,\partial}$ remains unchanged already under the substitution $(\omega, \omega+1)$, and not only under $(\omega, \omega+2)$. Cf. the remark at the beginning of § 73.]

If, then, we determine the constants in $\Phi_{a,c,\partial}$ so that no negative powers of q occur in the expansions of these functions, then all the $\Phi_{a,c,\partial}$ must be equal to one and the same constant.

To attain this goal it suffices here too that, when n has no square divisor, $\Phi_{a,0,\partial}$, and when n is a prime, $\Phi_{1,0,n}$ remain finite for $q = 0$.

In these calculations we make use of the expansions:

$$\begin{aligned} uv &= q^{-\frac{n+1}{24}} \Pi(1+q^{n(2h+1)})(1+q^{2h+1}), \\ u_1v_1 &= q^{-\frac{n+1}{24}} \Pi(1-q^{n(2h+1)})(1-q^{2h+1}), \\ u_2v_2 &= 2q^{\frac{n+1}{12}} \Pi(1+q^{2hn})(1+q^{2h}), \end{aligned} \tag{6}$$

from which, by expansion in powers of q:

$$\begin{aligned} uv &= q^{-\frac{n+1}{24}}(1+q+q^3+q^4+q^5+q^6+q^7+2q^8\dots), \\ u_1v_1 &= q^{-\frac{n+1}{24}}(1-q-q^3+q^4-q^5+q^6-q^7+2q^8\dots), \\ u_2v_2 &= 2q^{\frac{n+1}{12}}(1+q^2+q^4+2q^6+2q^8+\cdots). \end{aligned} \tag{7}$$

The latter formulae are correct for $n > 7$ (for $n = 3, 5, 7$ the terms from q^3, q^5, q^7 on are to be modified). For $n = 7$, $n = 23$ these expressions show that A itself remains finite for $q = 0$, from which the equations follow for these cases:

$$\begin{aligned} n &= 7, \quad A = 0, \\ n &= 23, \quad A = 1. \end{aligned} \tag{8}$$

By a simple calculation one finds further still:

$$\begin{aligned} n &= 31, & (A^2-B)^2 - A &= 0, \\ n &= 47, & A^2 - A - B &= 2, \\ n &= 71, & A^3 - 4A^2 + 2A - B &= 1. \end{aligned} \tag{9}$$

The calculation turns out not quite so simple for a composite n. Thus for $n = 15$, for example, one must take account of the condition of finiteness for $q = 0$ not only for $\Phi_{1,0,15}$, but also for $\Phi_{3,0,5}$. These two, however, suffice. One obtains in this way:

$$n = 15, \quad A^3 - AB + 1 = 0. \tag{10}$$

2. If $n \equiv 3 \pmod 8$, the same conclusions are to be drawn, if we set:

$$\begin{aligned} 4A &= u^2v^2 - u_1^2v_1^2 - u_2^2v_2^2, \\ B &= u^2v^2u_1^2v_1^2 + u^2v^2u_2^2v_2^2 - u_1^2v_1^2u_2^2v_2^2, \end{aligned} \tag{11}$$

for which one obtains from (2) the corresponding permutations:

$$\begin{array}{lll} \omega, & A, & B, \\ -\dfrac{1}{\omega}, & \varrho^2 A, & \varrho B, \\ \omega+1, & e^{\frac{(n+1)\pi i}{6}}\sigma^2 A, & e^{-\frac{(n+1)\pi i}{6}}\sigma B, \end{array} \tag{12}$$

and from this one derives in the same way the equations:

$$\begin{aligned} n &= 3, & A &= 0, \\ n &= 11, & A &= 1, \\ n &= 19, & A^5 - 7A^2 - B &= 0. \end{aligned} \tag{13}$$

3. If $n \equiv 1 \pmod 4$, one can proceed in the same way with the functions

$$\begin{aligned} 8A &= u^4v^4 - u_1^4v_1^4 - u_2^4v_2^4, \\ B &= u^4v^4u_1^4v_1^4 + u^4v^4u_2^4v_2^4 - u_1^4v_1^4u_2^4v_2^4, \end{aligned} \tag{14}$$

for which one obtains the corresponding permutations:

$$\begin{array}{lll} \omega, & A, & B, \\ -\dfrac{1}{\omega}, & \varrho\, A, & \varrho^2 B, \\ \omega+1, & e^{\frac{(n+1)\pi i}{3}}\sigma\, A, & e^{-\frac{(n+1)\pi i}{3}}\sigma^2 B. \end{array} \tag{15}$$

Only the first case $n = 5$ leads here to a simple result:

$$n = 5, \quad A = 1. \tag{16}$$

For $n = 5$ a still simpler form of the transformation equation can be obtained.

For if we apply to the three functions

$$w = \frac{u_1^2v_2^2 + u_2^2v_1^2}{uv}, \quad w_1 = \frac{u_2^2v^2 - v_2^2u^2}{u_1v_1}, \quad w_2 = \frac{u_1^2v^2 - v_1^2u}{u_2v_2} \tag{17}$$

the substitutions $\left(\omega, -\frac{1}{\omega}\right)$, $(\omega, \omega+1)$, then under the assumption $n \equiv 5 \pmod 8$ the corresponding permutations result:

$$\begin{array}{llll} \omega, & w, & w_1, & w_2, \\ -\dfrac{1}{\omega}, & w, & w_2, & w_1, \\ \omega+1, & e^{-\frac{\pi i}{24}(n-5)}w_1, & e^{-\frac{\pi i}{24}(n-5)}w, & e^{\frac{\pi i}{12}(n-5)}w_2. \end{array}$$

If we therefore set

$$\begin{aligned} A &= w^2 + w_1^2 + w_2^2, \\ B &= w^2w_1^2 + w^2w_2^2 + w_1^2w_2^2, \\ C &= w\, w_1\, w_2, \end{aligned}$$

we obtain

$$\begin{array}{llll} \omega, & A, & B, & C, \\ -\dfrac{1}{\omega}, & A, & B, & C, \\ \omega+1, & e^{-\frac{\pi i}{12}(n-5)}A, & e^{-\frac{\pi i}{6}(n-5)}B, & C, \end{array}$$

so that two of these functions can be used just as A and B were used above. For $n = 5$ it appears, however, that in the expansions of w, w_1, w_2 in powers of q no negative powers occur, and that therefore A, B, C, and hence also w, w_1, w_2 themselves, are constant.

The modular equation for $n = 5$ can therefore be set up in each of the three forms:

$$\begin{aligned} u_1^2v_2^2 + u_2^2v_1^2 &= 2uv, \\ u_2^2v^2 - v_2^2u^2 &= 2u_1v_1, \\ u_1^2v^2 - v_1^2u^2 &= 2u_2v_2. \end{aligned} \tag{18}$$

These equations can also be derived from the one given by Jacobi (Fund. art. 30, gesammelte Werke, Bd. 1, S. 123).

We close these considerations by deriving, in the simplest cases, the Jacobi modular equations from these irrational forms.

We set [cf. §54, (3)]:

$$\frac{u_2}{u} = x = \sqrt[4]{\varkappa}, \qquad \frac{u_1}{u} = x' = \sqrt[4]{\varkappa'},$$
$$\frac{v_2}{v} = y = \sqrt[4]{\lambda}, \qquad \frac{v_1}{v} = y' = \sqrt[4]{\lambda'},$$

and eliminate, by means of the relations

$$\begin{aligned} x^8 + x'^8 = 1, \quad xx' = \frac{\sqrt{2}}{u^3} \\ y^8 + y'^8 = 1, \quad yy' = \frac{\sqrt{2}}{v^3} \end{aligned} \tag{19}$$

the quantities u and v.

For $n = 3$ one obtains from (13):

$$x^2y^2 + x'^2y'^2 = 1, \tag{20}$$

a form of the modular equation which goes back to Legendre.

From this, by substituting for x', y' the values from (19),

$$x^8 + y^8 = 4x^2y^2 - 6x^4y^4 + 4x^6y^6,$$

or

$$(x^4 - y^4)^2 = 4(xy - x^3y^3)^2$$

and by extracting the root from this and determining the sign by $q = 0$, one finds:

$$x^4 - y^4 = 2xy - 2x^3y^3, \qquad (n = 3), \tag{21}$$

which, apart from the sign of $\sqrt[4]{\lambda}$ that is not further explained there, agrees with §10, (12).

For $n = 5$ there follows from the second equation (18):

$$(x^2 - y^2)^3 = 4xy\, x'^4y'^4,$$

and from this by squaring

$$(x^2 - y^2)^6 = 16x^2y^2(1 - x^8 - y^8 + x^8y^8),$$

which is easily brought into the form

$$(x^2 - y^2)^2\left[(x^2 - y^2)^2 + 8x^2y^2\right]^2 = 16x^2y^2(1 - x^4y^4)^2.$$

If one extracts the root from this, it follows, as above:

$$x^6 - y^6 - 4xy(1 - x^4y^4) + 5x^2y^2(x^2 - y^2) = 0 \quad (n = 5). \tag{22}$$

For $n = 7$ one obtains, without extraction of roots, from (8):

$$xy + x'y' = 1, \tag{23}$$

and from this:

$$\begin{aligned} x^8 + y^8 - 8xy(1 + x^6y^6) + 28x^2y^2(1 + x^4y^4) \\ -56x^3y^3(1 + x^2y^2) + 70x^4y^4 = 0 \qquad (n = 7). \end{aligned} \tag{24}$$

§ 76. Composite Degrees of Transformation.

If the degree of transformation is a composite number, one can set up still simpler transformation equations than those which are obtained by the method of the preceding section.

We carry out these considerations here only in the simplest cases.

Let the odd degree of transformation n be decomposed into two factors

$$n = n'n'', \tag{1}$$

which are relatively prime to each other.

To each transformation of degree n of the form

$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \tag{2}$$

there can then be assigned one and only one transformation of each of the degrees n', n'':

$$\begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'', & 0 \\ c'', & \partial'' \end{pmatrix} \tag{3}$$

and these are determined by the following conditions:

$$\begin{gathered} a = a'a'', \qquad \partial = \partial'\partial'', \\ \partial''c' \equiv c \pmod{a'}, \quad \partial'c'' \equiv c \pmod{a''}, \end{gathered} \tag{4}$$

and conversely, from every pair of transformations of the form (3) there follows, by (4), one and only one transformation (2). For by (4) the quantities ∂', ∂'' are first determined as the greatest common divisors of ∂ with n' and n'', and thereupon c' is determined with respect to the modulus a', and c'' with respect to the modulus a'', from the two last congruences (4).

What now matters above all is to show that the congruences (4) are preserved when the transformations (2) and (3), in accordance with §69, (8) to (12), by means of the two linear transformations

$$(\omega, \omega + 1), \quad \left(\omega, -\frac{1}{\omega}\right)$$

are transformed.

Following the formulae mentioned, we set:

$$\begin{gathered} \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \lambda, & 1 \end{pmatrix} \begin{pmatrix} a, & 0 \\ c_1, & \partial \end{pmatrix} \\ \begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \lambda', & 1 \end{pmatrix} \begin{pmatrix} a', & 0 \\ c'_1, & \partial' \end{pmatrix} \\ \begin{pmatrix} a'', & 0 \\ c'', & \partial'' \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \lambda'', & 1 \end{pmatrix} \begin{pmatrix} a'', & 0 \\ c''_2, & \partial'' \end{pmatrix}. \end{gathered} \tag{5}$$

$$\begin{gathered} \begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} a_2, & 0 \\ c_2, & \partial_2 \end{pmatrix} \\ \begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} a'_2, & 0 \\ c'_2, & \partial'_2 \end{pmatrix} \\ \begin{pmatrix} a'', & 0 \\ c'', & \partial'' \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{pmatrix} \begin{pmatrix} a''_2, & 0 \\ c''_2, & \partial''_2 \end{pmatrix}. \end{gathered} \tag{6}$$

It follows first of all from §69, (9):

$$c_1 \equiv c + \partial \pmod{a}, \quad c'_1 \equiv c' + \partial' \pmod{a'}, \quad c''_1 \equiv c'' + \partial'' \pmod{a''},$$

from which, by (4),

$$\partial'' c_1' \equiv \partial'' c' + \partial \equiv c_1 \pmod{a'}$$
$$\partial' c_1'' \equiv \partial' c'' + \partial \equiv c_1 \pmod{a''}$$

in agreement with the congruences (4).

For the composition (6) it follows, by §69, (12), that ∂_2, ∂_2', ∂_2'' are the greatest common divisors of a, c; a', c'; a'', c''; but since ∂'' is relatively prime to a', and $\partial'' c' \equiv c \pmod{a'}$, ∂_2' is also the greatest common divisor of a' and c, and for the same reasons ∂_2'' the greatest common divisor of a'' and c. From this one concludes, since a', a'' are relatively prime:

$$\partial_2 = \partial_2' \partial_2'', \qquad a_2 = a_2' a_2''.$$

Further, by §69, (11), (12), (13):

$$\partial_2 = \alpha c - \gamma a, \qquad c_2 = -\partial\alpha,$$
$$\partial_2' = \alpha' c' - \gamma' a', \quad c_2' = -\partial'\alpha',$$

hence, by (4):

$$\partial_2 c_2' - c_2 \partial_2' \equiv \alpha\alpha'(\partial c' - c\partial') \equiv 0 \pmod{n'},$$

and consequently also

$$\partial_2'' c_2' - c_2 \equiv 0 \pmod{a_2'},$$

in agreement with (4), and in the same way it follows that:

$$\partial_2' c_2'' - c_2 \equiv 0 \pmod{a_2''},$$

whereby the proof is furnished that the belonging together of the transformations expressed by (4)

$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}, \quad \begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'', & 0 \\ c'', & \partial'' \end{pmatrix}$$

is not disturbed by the application of any linear transformation to ω. Since we here assume n to be odd, we may always take c, c', c'' to be divisible by 16; and if n, and consequently also n', n'', are indivisible by 3, then c, c', c'' may also be assumed divisible by 3. But if n is divisible by 3, then of the two factors n', n'' one, say n'', will be divisible by 3, and the other, n', not. It may then still be assumed that c' is divisible by 3, but not c and c''. In this case the dependence of c'' upon c shall be determined more closely by the congruence

$$\partial' c'' \equiv c \pmod{3a''}. \tag{7}$$

A solution of this congruence can always be derived from a solution of the congruence (4) $\partial' c'' \equiv c \pmod{a''}$ by adding to c'' a multiple of a''.

By § 69, (9) to (13), the congruence (7) then has as a consequence:

$$\lambda \equiv n'\lambda'' \pmod 3, \tag{8}$$

$$\begin{matrix} \alpha \equiv \alpha''\partial'\partial_2', & \beta \equiv \beta'' a'\partial_2' \\ \gamma \equiv \gamma'' a'\partial_2', & \delta \equiv \delta''\partial'\partial_2' \end{matrix} \pmod 3. \tag{9}$$

Now let $v', v_1', v_2'; v'', v_1'', v_2''$ have for the numbers n', n'' the same meaning that was given to v, v_1, v_2 in § 73, (3) for the number n, namely:

$$v' = f\left(\frac{c' + \partial'\omega}{a'}\right), \qquad v'' = f\left(\frac{c'' + \partial''\omega}{a''}\right),$$

$$
(10)\qquad
\begin{aligned}
v_1' &= \left(\frac{2}{a'}\right) f_1\left(\frac{c'+\partial'\omega}{a'}\right), \quad v_1'' = \left(\frac{2}{a''}\right) f_1\left(\frac{c''+\partial''\omega}{a''}\right),\\
v_2' &= \left(\frac{2}{\partial'}\right) f_2\left(\frac{c'+\partial'\omega}{a'}\right), \quad v_2'' = \left(\frac{2}{\partial''}\right) f_2\left(\frac{c''+\partial''\omega}{a''}\right).
\end{aligned}
$$

We apply the table of permutations (4), § 73 to these functions. Since n' is under all circumstances indivisible by 3, the cubic roots of unity $\varrho', \sigma' = 1$ are both to be set equal to 1, while, by the congruences (8), (9):

$$
(11)\qquad \varrho'' = \varrho^{n'}, \quad \sigma'' = \sigma^{n'}
$$

results. We accordingly obtain the following permutations that belong together:

$$
(12)\qquad
\begin{array}{llll}
\omega, & v', & v_1', & v_2',\\
-\dfrac{1}{\omega}, & v', & v_2', & v_1',\\
\omega+1, & e^{-\frac{n'\pi i}{24}} v_1', & e^{-\frac{n'\pi i}{24}} v', & e^{\frac{n'\pi i}{12}} v_2',\\
\hline
\omega, & v'', & v_1'', & v_2'',\\
-\dfrac{1}{\omega}, & \varrho^{n'} v'', & \varrho^{n'} v_2'', & \varrho^{n'} v_1'',\\
\omega+1, & \sigma^{n'} e^{-\frac{n''\pi i}{24}} v_1'', & \sigma^{n'} e^{-\frac{n''\pi i}{24}} v'', & \sigma^{n'} e^{\frac{n''\pi i}{12}} v_2'',
\end{array}
$$

which hold together with the permutations (4), § 73.

We now distinguish two cases.

1. If

$$
(13)\qquad (n'+1)(n''+1) = 8\mu \equiv 0 \pmod 8
$$

is, we set

$$
(14)\qquad U = uvv'v'', \quad U_1 = u_1 v_1 v_1' v_1'', \quad U_2 = u_2 v_2 v_2' v_2'',
$$

$$
(15)\qquad
\begin{aligned}
2A &= U + (-1)^\mu (U_1 + U_2),\\
B &= UU_1 + UU_2 + (-1)^\mu U_1 U_2,\\
&= \frac{4}{U_1} + \frac{4}{U_2} + (-1)^\mu \frac{4}{U}.
\end{aligned}
$$

From (12) and § 73, (4) we then obtain the following associated permutations:

$$
(16)\qquad
\begin{array}{lll}
\omega, & A, & B,\\
-\dfrac{1}{\omega}, & \varrho^{n'+1} A, & \varrho^{-(n'+1)} B,\\
\omega+1, & \sigma^{n'+1} e^{\frac{2\mu\pi i}{3}} A, & \sigma^{-(n'+1)} e^{-\frac{2\mu\pi i}{3}} B.
\end{array}
$$

We apply our principle for the derivation of modular equations to these functions, and remark in addition the following:

The third roots of unity occurring in (16) are $= 1$ if either n is indivisible by 3 and μ is divisible by 3, or n'' is divisible by 3 and n' leaves the residue 2. In these cases every rational function of A and B is a root of a transformation equation. In the other cases this property belongs to the cube of such a rational function of A and B, for which the differences of the exponents of all the terms are congruent to one another with respect to the modulus 3.

If these rational functions are p o l y n o m i a l s and if moreover all their values for $q = 0$ are finite, then they must be equal to a constant, and thereby we obtain transformation equations.

If n', n'' are primes, then here too it suffices (cf. § 72) that the negative powers of q in the expansion of such a function in ascending powers of q drop out in the one principal case, namely in that where

$$\begin{pmatrix} a, & 0 \\ c, & \partial \end{pmatrix}, \quad \begin{pmatrix} a', & 0 \\ c', & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'', & 0 \\ c'', & \partial'' \end{pmatrix}$$

are equal to

$$\begin{pmatrix} 1, & 0 \\ 0, & n \end{pmatrix}, \quad \begin{pmatrix} 1, & 0 \\ 0, & n' \end{pmatrix}, \quad \begin{pmatrix} 1, & 0 \\ 0, & n'' \end{pmatrix},$$

thus

$$U = f(\omega) f(n'\omega) f(n''\omega) f(n'n''\omega);$$

for from the expansion in this one case one can derive the expansions for the remaining cases, by replacing ω by

$$\frac{\omega}{n}, \quad \frac{\omega}{n'}, \quad \frac{\omega}{n''} \tag{17}$$

and then adding integers to ω, whereby no negative powers of q can be newly introduced.

For carrying out the calculation one makes use of the expansions

$$\begin{aligned} U &= q^{-\frac{(n'+1)(n''+1)}{24}} \times \Pi(1+q^{2h-1})(1+q^{(2h-1)n'})(1+q^{(2h-1)n''})(1+q^{(2h-1)n}) \\ U_1 &= q^{-\frac{(n'+1)(n''+1)}{24}} \times \Pi(1-q^{2h-1})(1-q^{(2h-1)n'})(1-q^{(2h-1)n''})(1-q^{(2h-1)n}) \\ U_2 &= 4q^{\frac{(n'+1)(n''+1)}{12}} \times \Pi(1+q^{2h})(1+q^{2hn'})(1+q^{2hn''})(1+q^{2hn}), \end{aligned} \tag{18}$$

which in the individual cases furnish the power-series expansions of A, B, from which the negative powers of q are to be eliminated. In this way one computes very simply the following equations:

$$\begin{aligned} &n = 15, \quad A = 1, \\ &n = 21, \quad (A^2 - B)^2 - A = 0, \\ &n = 33, \quad A^2 - B - A = 4, \\ &n = 35, \quad A^2 - B - A = 2, \\ &n = 55, \quad A^3 - B - 4A^2 - A + 4 = 0. \end{aligned} \tag{19}$$

2. If

$$(n'-1)(n''-1) = 8\mu \equiv 0 \pmod 8,$$

then we set

$$\begin{aligned} A &= \frac{uv}{v'v''} + (-1)^\mu \left(\frac{u_1 v_1}{v_1' v_1''} + \frac{u_2 v_2}{v_2' v_2''} \right), \\ B &= \frac{v'v''}{uv} + (-1)^\mu \left(\frac{v_1' v_1''}{u_1 v_1} + \frac{v_2' v_2''}{u_2 v_2} \right), \end{aligned} \tag{20}$$

and obtain, according to (12), the permutations belonging together:

$$\begin{array}{lll} \omega, & A, & B, \\ -\dfrac{1}{\omega}, & \varrho^{1-n'} A, & \varrho^{n'-1} B, \\ \omega + 1, & \sigma^{1-n'} e^{\frac{2\mu\pi i}{3}} A, & \sigma^{n'-1} e^{-\frac{2\mu\pi i}{3}} B. \end{array} \tag{21}$$

We can therefore apply the same procedure as above if we add the further restriction that only symmetric functions of A and B, that is, rational functions of AB, $A + B$, are used, because only under this assumption do all the others follow from one of the values of such a function by the permutations (17). The following examples are easily computed:

$$\begin{aligned} &n = 15, \quad AB + 1 = 0, \\ &n = 35, \quad 2(A+B) - AB = 5 \\ &n = 39, \quad 2(A+B) - AB = 3. \end{aligned} \tag{22}$$

The computation presents no insurmountable difficulties even in still more complicated cases. Thus in the memoir Acta mathematica Bd. II, S. 359 I have communicated the following formula for the case $n = 105$:

$$A = \sum \frac{f(3\omega)f(5\omega)f(7\omega)f(105\omega)}{f(\omega)f(15\omega)f(21\omega)f(35\omega)},$$
$$B = \sum \frac{f(\omega)f(15\omega)f(21\omega)f(35\omega)}{f(3\omega)f(5\omega)f(7\omega)f(105\omega)},$$
$$A^2 + B^2 - 4(A+B)^3 + 10AB(A+B) + 4(A+B)^2 + 10AB + 14(A+B) + 5 = 0,$$

if the sums Σ in A and B refer to the three functions f, f_1, f_2.

§ 77. Geometric interpretation of the irrational modular equations as modular correspondences.

We make the content of the irrational forms of the modular equations more intuitive by means of a geometric interpretation. Let us first consider the case $n \equiv -1 \pmod 8$ and put

$$x = f(\omega), \quad y = f_1(\omega), \quad z = f_2(\omega), \tag{1}$$

so that, if we regard x, y, z as Cartesian coordinates of a point P, a space curve is thereby represented, which we can also express by the two equations:

$$\begin{gathered} x^8 - y^8 - z^8 = 0 \\ xyz = \sqrt{2} \end{gathered} \tag{2}$$

and which we shall call the fundamental curve. This curve is therefore of the 24th order. One can map it onto

a plane curve of the third order, if one sets $x^8 = X$, $y^8 = Y$:

$$XY(X - Y) = 16. \tag{3}$$

If we set [§ 75, (1)]

$$\xi = f\left(\frac{c+\partial\omega}{a}\right), \ \eta = \left(\frac{2}{a}\right) f_1\left(\frac{c+\partial\omega}{a}\right), \ \zeta = \left(\frac{2}{\partial}\right) f_2\left(\frac{c+\partial\omega}{a}\right), \tag{4}$$

then the quantities ξ, η, ζ likewise satisfy the equations (2), since

$$\left(\frac{2}{a}\right)\left(\frac{2}{\partial}\right) = \left(\frac{2}{n}\right) = +1$$

holds, and the point Π, whose coordinates are ξ, η, ζ, therefore also lies on the fundamental curve.

An equation

$$\Phi(x, y, z, \xi, \eta, \zeta) = 0$$

denotes, when the point P is held fixed, a surface on which Π is to lie, and the intersection of this surface with the fundamental curve gives a certain number of points Π that correspond to the point P. In the same way, to a fixed point Π there corresponds a certain number of points P, and this assignment is called a correspondence on the fundamental curve. If the point P coincides with one of the points Π corresponding to it, we obtain a double point or coincidence of the correspondence.

If equation (5) is not altered by the interchange of P with Π, the correspondence is a mutual one. Such correspondences are given by equations (8), (9), (10), § 75.

By § 75, (3) we have then to set

$$\begin{aligned} 2A &= x\xi + (-1)^{\frac{n+1}{8}}(y\eta + z\zeta) \\ B &= x\xi y\eta + x\xi z\zeta + (-1)^{\frac{n+1}{8}} y\eta z\zeta \end{aligned} \tag{5}$$

A is of the first, B of the second degree with respect to the coordinates of each of the two points P and Π.

We wish to determine the degree of these correspondences, that is, the number of points Π corresponding to a point P. In doing so we have to multiply the degree of the equation $\Phi = 0$ with respect to ξ, η, ζ by the degree of the fundamental curve, that is, by 24, and there results:

$$\begin{array}{lll} \text{for } n = 7, & m = 24, \\ \text{`} \quad n = 23, & m = 24, \\ \text{`} \quad n = 31, & m = 96, \\ \text{`} \quad n = 47, & m = 48, \\ \text{`} \quad n = 71, & m = 72, \\ \text{`} \quad n = 15, & m = 72. \end{array}$$

In the cases 23, 47, 71 the degree of the correspondence is therefore equal to the degree of the corresponding modular equation, that is, equal to the number of the transformations in question; in the cases $n = 7, 31, 15$ the degree is three times the degree of the modular equations.

What do these supernumerary points signify? They are explained by the fact that in these cases, in which $n \equiv 0, 1 \pmod 3$, the exponents $h + 2k$ in the function $\Phi_{a,c,\partial}$ [§ 75, (5)], which by the above equation (5) is equal to a constant, leave the same residue with respect to the modulus 3 in all terms, and that accordingly the equation $\Phi = 0$, and likewise the equations (2), remain satisfied if ξ, η, ζ are multiplied by an arbitrary third root of unity. Any three points of the correspondence therefore give the same transformation.

We could also, if x, y, z are determined by (1), have put, in place of (4),

$$\xi = f\left(\frac{c+\partial\omega}{a}\right),\ \eta = \left(\frac{2}{\partial}\right) f_2\left(\frac{c+\partial\omega}{a}\right),\ \zeta = \left(\frac{2}{a}\right) f_1\left(\frac{c+\partial\omega}{a}\right) \tag{6}$$

and consequently for A, B:

$$\begin{aligned} 2A &= x\xi + (-1)^{\frac{n+1}{8}}(y\zeta + z\xi), \\ B &= x\xi y\zeta + x\xi z\eta + (-1)^{\frac{n+1}{8}} y\eta z\zeta \end{aligned} \tag{7}$$

The fundamental curve and the degree of the correspondence would then have remained the same, but the coincidences would have acquired a different and indeed simpler meaning. In both cases the coincidences belong to the singular values of the modular functions, in which ω is an imaginary quadratic irrationality, as we shall see more precisely in what follows.

In the case $n \equiv 3 \pmod 8$ we set

$$x = f^2(\omega),\ y = f_1^2(\omega),\ z = f_2^2(\omega),$$

$$\xi = f^2\left(\frac{c+\partial\omega}{a}\right),\ \eta = f_1^2\left(\frac{c+\partial\omega}{a}\right),\ \varrho = f_2^2\left(\frac{c+\partial\omega}{a}\right), \tag{8}$$

and obtain a fundamental curve

$$x^4 - y^4 - z^4 = 0, \qquad x\,y\,z = 2, \tag{9}$$

which is only of degree 12. We then have further, according to § 75:

$$4A = x\xi - y\eta - z\varrho, \qquad B = x\xi y\eta + x\xi z\varrho - y\eta z\varrho, \tag{10}$$

and for the degree m of the correspondence one obtains, according to § 75 (13):

$$n = 3, \quad m = 12, \qquad n = 11, \quad m = 12, \qquad n = 19, \quad m = 60,$$

and so again, as in the previous case, when $n \equiv 0, 1 \pmod 3$, a number three times too large.

If finally $n \equiv 1 \pmod 4$, then one sets

$$x = f^4(\omega),\ y = f_1^4(\omega),\ z = f_2^4(\omega)$$

and obtains a fundamental curve of the sixth degree.

For $n = 5$ the correct number $m = 6$ results.

Eighth Chapter.
The Group of the Transformation Equations and the Equation of Degree 5.

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§ 78.
The Galois Group of the Transformation Equations for a Prime Degree.

A more thorough algebraic study of the transformation equations requires a knowledge of their Galois group. Since we have derived the transformation equations from the division equations, whose group is known to us (§ 63), we can form the group of the transformation equations as well.

The transformation equations arose (§ 65) from the fact that the roots of the division equations admitted of a partition into rows which are not torn apart by the permutations of the group of the division equation, but are only interchanged among one another.

To each of these rows there corresponds a definite root of a transformation equation, and the group of the latter therefore consists of the totality of the permutations that are produced among the rows by the group of the division equation.

If the degree of transformation n is an odd prime p, this group admits a very elegant representation which is suited to further investigations, and we now hold fast to this assumption.

In § 65, (3) the necessary and sufficient condition was already determined for two roots of the division equation

$$x_{\mu,\mu'}, \quad x_{\nu,\nu'} \tag{1}$$

to belong to the same row R, namely the congruence

(2) $$\mu\,\nu' - \nu\,\mu' \equiv 0 \pmod{p}.$$

If we now, as is customary in number theory (Gauss, Disquisitiones arithmeticae, art. 31), understand by the symbol

$$\frac{a}{b} \pmod{p}$$

an integer [or class of numbers (mod p)] which, multiplied by b, leaves the residue a on division by p, then, provided that μ' and ν' are not divisible by p, we can also write the congruence (2) that defines a row as follows:

(3) $$\frac{\mu}{\mu'} \equiv \frac{\nu}{\nu'} \pmod{p},$$

and we are thus naturally led to denote this row R by the value of the ratio

(4) $$\frac{\mu}{\mu'} \equiv z \pmod{p},$$

which can be congruent to any one of the numbers $0, 1, \ldots, p-1$ and which is invariable for a whole row. At first one row here remains undesignated, namely that in which μ', and consequently all ν', are divisible by p; but this row too is subsumed very well under the general designation if, in the case $\mu' \equiv 0 \pmod{p}$, we put

(5) $$\frac{\mu}{\mu'} \equiv \infty \pmod{p}$$

so that we thus obtain a further $(p+1)$th row R_∞. The totality of the rows, whose number amounts to $p+1$, is accordingly to be denoted by

(6) $$R_\infty, R_0, R_1, \ldots, R_{p-1}.$$

Correspondingly, for the associated roots of a transformation equation the designations

(7) $$v_\infty, v_0, v_1, \ldots, v_{p-1}$$

will have to be used.

If, for example, we take the invariant equation (§ 69) as our basis, then [according to the determinations of § 68, (10) concerning the numbers a, c, ∂] we must put

(8) $$v_\infty = j(p\,\omega), \quad v_z = j\left(\frac{z+\omega}{p}\right)$$

and likewise, if any other transformation equation is chosen.

By this mode of notation we are in a position to derive the group of the transformation equation at once from that of the division equation.

We first fix as domain of rationality the totality of the rational functions of $\varkappa^2$ with rational numerical coefficients.

By § 63, 3. the group of the division equation consists, in this domain of rationality, of all substitutions by which μ, μ' are carried over into

$$\begin{matrix} \partial\mu - b\mu' \\ -c\mu + a\mu' \end{matrix}$$

where a, b, c, ∂ are arbitrary integers taken according to the modulus p, whose determinant

(9) $$\Delta = a\partial - bc$$

is not divisible by p; and the number of all these substitutions amounts to

$$p(p-1)(p^2-1) \qquad [\S\ 63,\ (27)] \tag{10}$$

But from this there results, by the mode of notation (4), (5), the group of the transformation equation as consisting of all the permutations which the symbol

$$\left(z, \frac{\partial z - b}{-cz + a}\right) \pmod p \tag{11}$$

expresses.

We denote a substitution (z, z'), when

$$z \equiv \frac{c + \partial z'}{a + bz'} \pmod p \tag{12}$$

holds, similarly as before by

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}, \tag{13}$$

and obtain for the composition of two such substitutions, when

$$z' \equiv \frac{c' + \partial' z''}{a' + b' z''} \pmod p$$

holds, the rule:

$$(z, z')(z' z'') \equiv \begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} \begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix} \equiv \begin{pmatrix} aa' + bc', & ab' + b\partial' \\ ca' + \partial c', & cb' + \partial\partial' \end{pmatrix} \pmod p, \tag{14}$$

in agreement with the rule for the composition of two transformations in § 28. The substitution (11) is accordingly to be denoted by

$$\begin{pmatrix} \partial, & c \\ b, & a \end{pmatrix}.$$

The group formed by all substitutions of this form [according to (14)] we denote by $\mathfrak{L}$ (group of linear substitutions).

The functions (12) of z', and therefore also the substitutions (13), remain unchanged if the four numbers a, b, c, ∂ are multiplied by one and the same factor not divisible by p, and from this there follows, by (10), the number of these functions, or the degree of the group $\mathfrak{L}$

$$p(p^2 - 1), \tag{15}$$

which one also finds easily by direct enumeration.

If in one of the substitutions (13) the four numbers a, b, c, ∂ are multiplied by a common factor indivisible by p, then the determinant Δ is multiplied by the square of this factor. Hence it is not the determinant Δ, but rather its quadratic character, that is, the value of the symbol

$$\left(\frac{\Delta}{p}\right)$$

that is preserved under this multiplication.

If we compose two of the substitutions (13), their determinants are multiplied together, and from this it follows that all substitutions of the group $\mathfrak{L}$ in which Δ is a quadratic residue of p form a group among themselves, which we shall denote by $\mathfrak{L}_0$.

The group $\mathfrak{L}_0$ is a (proper) divisor of the group $\mathfrak{L}$ of index 2.

If one composes the substitutions of $\mathfrak{L}_0$ with any substitution $(z, \beta z) = \begin{pmatrix} 1, & 0 \\ 0, & \beta^{-1} \end{pmatrix}$, where β, and therefore also β^{-1}, is a quadratic non-residue of p, then one obtains the whole group $\mathfrak{L}$.

If Δ is a quadratic residue of p, then one can choose a common factor by which to multiply a, b, c, ∂ in such a way that $\Delta \equiv 1 \pmod{p}$, so that we can also represent the group $\mathfrak{L}_0$ by

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma \equiv 1 \pmod{p}. \tag{16}$$

In one of these substitutions the numbers α, β, γ, δ are determined, with respect to the modulus p, up to the common sign. The degree of the group $\mathfrak{L}_0$ is

$$\frac{1}{2}p(p^2 - 1). \tag{17}$$

One arrives at the form (16) directly, however, if one takes as the group of the division equation not the group $\mathfrak{A}$ but the group $\mathfrak{B}$ of § 63, that is, if one adjoins the pth roots of unity to the domain of rationality; from which the theorem flows:

The group $\mathfrak{L}_0$ is the group of the transformation equation when the pth roots of unity are adjoined to the domain of rationality.

For the reduction of the group $\mathfrak{L}$ to the group $\mathfrak{L}_0$, however, the adjunction of a two-valued function already suffices, and with the adjunction of the pth roots of unity too much has therefore been done. To recognise which irrationality it is necessary to adjoin, the theorems of §§ 63, 65 serve.

In § 63, 3. we have seen that the pth root of unity ϱ is rationally representable by the roots of the division equation, and that by a substitution of the group $\mathfrak{A}$ whose determinant is congruent to m, ϱ passes over into ϱ^m; further, we proved in § 65 that by the adjunction of all the roots of a transformation equation the group of the division equation is reduced to the group $\mathfrak{A}_0$, which consists of all substitutions of the form

$$\begin{pmatrix} a, & 0 \\ 0, & a \end{pmatrix} \tag{18}$$

where a is an arbitrary number not divisible by p. The determinants of the substitutions of $\mathfrak{A}_0$ are therefore squares, and are consequently each congruent modulo p to one of the $\frac{1}{2}(p-1)$ quadratic residues of p.

The sum

$$A = \overset{a}{\Sigma}\varrho^a, \tag{19}$$

in which all the quadratic residues of p are to be put for a, therefore remains unchanged by the substitutions

of $\mathfrak{A}_0$ and is consequently expressible rationally through the roots of the transformation equation. The sum A remains unchanged under the substitutions of the group $\mathfrak{B}$, and hence also under $\mathfrak{L}_0$, while under the substitutions of $\mathfrak{A}$, and therefore also of $\mathfrak{L}$, it takes two different values, namely, when b runs through the sequence of non-residues,

$$A = \overset{a}{\Sigma}\varrho^a, \qquad B = \overset{b}{\Sigma}\varrho^b.$$

A is therefore a function belonging to the group $\mathfrak{L}_0$, and by its adjunction $\mathfrak{L}$ is reduced to $\mathfrak{L}_0$.

The values of the sums A, B are, however, known (Bd. I, § 179):

$$A = \frac{-1 \pm \sqrt{(-1)^{\frac{p-1}{2}} p}}{2}, \qquad B = \frac{-1 \mp \sqrt{(-1)^{\frac{p-1}{2}} p}}{2}.$$

The sign of the root depends on the choice of the root ϱ and can be determined, but does not come into consideration here. We have therefore the theorem:

The group of the transformation equation is $\mathfrak{L}_0$, when $\sqrt{(-1)^{\frac{p-1}{2}}\, p}$ is adjoined to the domain of rationality.

The group of the invariant equation can also be derived, without the division of the elliptic functions, in the following simple way; in doing so one obtains, however, only the monodromy group, i. e. the group in the field of the rational functions of $j(\omega)$, without regard to the constants to be adjoined. We restrict ourselves to the case of a prime degree p of the transformation, and set

$$v_c = j\left(\frac{c+\omega}{p}\right), \qquad v_\infty = j(p\omega), \qquad u = j(\omega), \tag{20}$$

where c is to be taken with respect to the modulus p.

If one applies to ω a linear substitution $S = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$, then $j(\omega)$ remains unchanged, and v_c passes into $v_{c'}$. To find c', one has to seek a second linear substitution $S' = \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix}$, so that

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} p, & 0 \\ c', & 1 \end{pmatrix} = \begin{pmatrix} p, & 0 \\ c, & 1 \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

or

$$\begin{aligned} p\alpha' + c'\beta' &= p\alpha, & \beta' &= p\beta \\ p\gamma' + c'\delta' &= c\alpha + \gamma, & \delta' &= c\beta + \delta. \end{aligned}$$

From this there follows, by the third equation: $c'\delta' \equiv c\alpha + \gamma$, and with the help of the fourth:

$$c' \equiv \frac{c\alpha + \gamma}{c\beta + \delta} \pmod{p}.$$

If $c\beta + \delta \equiv 0$, then $c' \equiv \infty$, and if $c \equiv \infty$, then $c' \equiv \alpha/\beta$.

In the application to the determination of c, c' one may replace the α, β, γ, δ by congruent numbers a, b, c, d whose determinant $ad - bc$ is a quadratic residue of c, and it can also be shown that in this way every substitution of the group $\mathfrak{L}_0$ can be obtained. Every function of v_c, then, which remains unchanged under the substitutions of the group $\mathfrak{L}_0$, remains unchanged also when a linear substitution is applied to ω, and is consequently a rational function of $j(\omega)$. But of the nature of the coefficients in this function this consideration teaches us nothing, and therefore $\mathfrak{L}_0$ is recognised only as the monodromy group of the invariant equation.

§ 79. Investigation of the Group $\mathfrak{L}_0$[1]).

In the 10th chapter of Band II we investigated the congruence group and its divisors quite generally. We carry this investigation through here once more in a special form, so far as it bears on the transformation problem.

The substitutions contained in $\mathfrak{L}_0$

$$\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix} = \left(z, \frac{-c + az}{\partial - bz}\right), \tag{1}$$

in which

$$\Delta = a\partial - bc \tag{2}$$

is a quadratic residue of p, can, as already remarked above, be brought into the form:

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \left(z, \frac{-\gamma + \alpha z}{\delta - \beta z}\right), \tag{3}$$

in which

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{p}, \tag{4}$$

and if by $\sqrt{\Delta} \pmod{p}$ we understand an integer whose square is congruent with Δ with respect to the modulus p, we have only to set:

$$\begin{matrix} a \equiv \alpha\sqrt{\Delta}, & b \equiv \beta\sqrt{\Delta} \\ c \equiv \gamma\sqrt{\Delta}, & \partial \equiv \delta\sqrt{\Delta} \end{matrix} \pmod{p}$$

In the form (3) the signs of α, β, γ, δ can still be changed simultaneously.

We ask for those elements z which remain unchanged under a substitution of the form (3). These are determined by the congruence

$$z \equiv \frac{\gamma + \delta z}{\alpha + \beta z} \pmod{p},$$

or

$$\beta z^2 + (\alpha - \delta)z - \gamma \equiv 0 \pmod{p}, \tag{5}$$

a congruence which, if it is not identical, has at most two incongruent roots. To obtain these we write the congruence (5), first under the assumption that β is not divisible by p, thus:

$$\left(\beta z + \frac{\alpha - \delta}{2}\right)^2 \equiv \left(\frac{\alpha - \delta}{2}\right)^2 + \beta\gamma \pmod{p},$$

or with the help of (4):

$$\left(\beta z + \frac{\alpha - \delta}{2}\right)^2 \equiv \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \pmod{p}. \tag{6}$$

Accordingly three different cases are to be distinguished:

$$\text{I.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \equiv 0 \pmod{p};$$

then the congruence (5) has one root; there is one and only one element z that remains unchanged.

$$\text{II.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \text{ quadratic residue of } p;$$

the congruence (5) has two different roots; there are two elements z that remain unchanged.

$$\text{III.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \text{ quadratic non-residue of } p;$$

the congruence (5) has no root at all, and all elements z are displaced.

If $\beta \equiv 0$, then (5) always has the one root $z = \infty$; if moreover $\delta \equiv \alpha$, and γ not $\equiv 0 \pmod{p}$, then there is only this one; in this case, however,

$$\alpha\delta \equiv 1, \qquad \alpha \equiv \delta, \qquad \left(\frac{\alpha + \delta}{2}\right)^2 \equiv 1 \pmod{p},$$

and condition I. is satisfied. But if at the same time $\gamma \equiv 0 \pmod{p}$, then the substitution is the identical one.

[1] On the group of linear substitutions $\mathfrak{L}_0$ cf.: Galois, Liouvilles Journal, Bd. XI. Serret, Algèbre supérieure, Section IV, Chapitre IV. C. Jordan, Traité des Substitutions. Gierster, Gruppe der Modulargleichungen. Mathematische Annalen, Bd. 18.

If $\beta \equiv 0$, but $\alpha - \delta$ not $\equiv 0 \pmod{p}$, then (5) has yet a second root; since now $\alpha\delta \equiv 1 \pmod{p}$, we have:

$$\left(\frac{\alpha+\delta}{2}\right)^2 - 1 \equiv \left(\frac{\alpha-\delta}{2}\right)^2 \pmod{p},$$

hence a quadratic residue, and condition II. is satisfied. We summarise these theorems as follows:

In case I. one element or all elements remain unchanged, in case II. two elements remain unchanged and in case III. all elements are changed.

When we repeat one and the same substitution

$$A = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \tag{7}$$

several times, the substitutions

$$A,\ A^2,\ A^3 \ldots,$$

arise, in whose sequence the identical substitution $\begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$ must once occur. If n is the least positive number for which

$$A^n = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$$

holds, then n is called the degree of A (Bd. II, § 2).

If we set, for an arbitrary m,

$$A^m = \begin{pmatrix} \alpha_m, & \beta_m \\ \gamma_m, & \delta_m \end{pmatrix},$$

then for the computation of the numbers α_m, β_m, γ_m, δ_m we obtain the following system of recurrent formulae:

$$\begin{aligned} \alpha_{m+1} &= \alpha\alpha_m + \beta\gamma_m, \quad \beta_{m+1} = \alpha\beta_m + \beta\delta_m, \\ \gamma'_{m+1} &= \gamma\alpha_m + \delta\gamma_m, \quad \delta_{m+1} = \gamma\beta_m + \delta\delta_m. \end{aligned} \tag{8}$$

From these the numbers α_m, β_m, γ_m, δ_m can be computed with particular ease in case I.

In this case, since the signs of α, β, γ, δ may all be reversed at the same time, we may assume:

$$\alpha + \delta \equiv 2, \quad \alpha\delta - \beta\gamma \equiv 1 \pmod{p},$$

and thus we obtain from (8):

$$\begin{aligned} \alpha_2 &\equiv -1 + 2\alpha, \quad \beta_2 \equiv 2\beta \\ \delta_2 &\equiv -1 + 2\delta, \quad \gamma_2 \equiv 2\gamma \\ \alpha_3 &\equiv -2 + 3\alpha, \quad \beta_3 \equiv 3\beta \\ \delta_3 &\equiv -2 + 3\delta, \quad \gamma_3 \equiv 3\gamma \end{aligned} \pmod{p},$$

from which, by the inference from m to $m+1$, it follows that:

$$\begin{aligned} \alpha_m &\equiv -(m-1) + m\alpha, \quad \beta_m \equiv m\beta \\ \delta_m &\equiv -(m-1) + m\delta, \quad \gamma_m \equiv m\gamma \end{aligned} \pmod{p}. \tag{9}$$

From this one sees that all the A^m again belong to Case I., and that p is the degree of A.

The analogous consideration of the two other cases has been carried out by Serret, but appears dispensable for our purpose.

On the other hand, we wish to add here the further theorem, that the whole group $\mathfrak{L}_0$ can be composed out of the two following special substitutions

$$A = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}. \tag{10}$$

The proof follows from § 30, if one observes that for every system of numbers α, β, γ, δ satisfying the condition $\alpha\delta - \beta\gamma \equiv 1 \pmod{p}$ the system of numbers α', β', γ', δ' can be chosen so that

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \equiv \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \pmod{p} \text{ and } \alpha'\delta' - \beta'\gamma' = 1$$

holds. All substitutions of $\mathfrak{L}_0$ are therefore contained among those considered in § 30. But here we can also easily exhibit the actual composition of an arbitrary substitution in $\mathfrak{L}_0$ out of A and B, and so carry out the proof independently of § 30.

For if in the first place c is an arbitrary number not divisible by p, then actual calculation easily gives

$$A^c = \begin{pmatrix} 1, & 0 \\ c, & 1 \end{pmatrix}, \quad C = \begin{pmatrix} c, & 0 \\ 0, & c^{-1} \end{pmatrix} = A^{-c^{-1}} B A^{-c} B A^{-c^{-1}} B, \tag{11}$$

and then, if $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ is an arbitrary substitution in $\mathfrak{L}_0$ and β is different from 0:

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \delta\beta^{-1}, & 1 \end{pmatrix} \begin{pmatrix} \beta, & 0 \\ 0, & \beta^{-1} \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ \alpha\beta^{-1}, & 1 \end{pmatrix}, \tag{12}$$

and if $\beta = 0$:

$$\begin{pmatrix} \alpha, & 0 \\ \gamma, & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \alpha, & 0 \\ 0, & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1, & 0 \\ \gamma\alpha, & 1 \end{pmatrix}. \tag{13}$$

From this theorem it is to be concluded that a group contained in $\mathfrak{L}_0$ which contains the two substitutions A, B must necessarily be identical with $\mathfrak{L}_0$.

§ 80. Normal divisors of the group $\mathfrak{L}_0$.

In order to learn the reductions of the transformation equation that may perhaps be possible, it is above all necessary to investigate the divisors of the group $\mathfrak{L}_0$. We ask first after the existence of a normal divisor $\mathfrak{R}$ of $\mathfrak{L}_0$ (Bd. II, § 3)[17].

If

$$S = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \tag{1}$$

is any substitution in $\mathfrak{R}$ different from the identical substitution, then, by the nature of the normal divisor, every substitution

$$U = TST^{-1} \tag{2}$$

is likewise contained in $\mathfrak{R}$, if

$$T = \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \tag{3}$$

[17] After Galois: „Proper divisors“.

is an arbitrary substitution in $\mathfrak{L}_0$. We set ourselves the task of determining T and ξ in such a way that

$$U = \begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix} \tag{4}$$

results. This condition can also be written thus:

$$\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix} \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix},$$

and leads to the congruences:

$$\begin{array}{ll} 1. & \alpha'\alpha + \beta'\gamma \equiv \gamma' \\ 2. & \alpha'\beta + \beta'\delta \equiv \delta' \\ 3. & \gamma'\alpha + \delta'\gamma \equiv -\alpha' + \gamma'\xi \\ 4. & \gamma'\beta + \delta'\delta \equiv -\beta' + \delta'\xi \end{array} \pmod{p}, \tag{5}$$

and from 1. and 2. there follows further:

$$\begin{array}{ll} 5. & \alpha' \equiv \gamma'\delta - \delta'\gamma \\ 6. & \beta' \equiv -\gamma'\beta + \delta'\alpha \end{array} \pmod{p}. \tag{6}$$

If these values are substituted in (5) 3. and 4., it follows that:

$$\begin{array}{l} \gamma'(\alpha + \delta - \xi) \equiv 0 \\ \delta'(\alpha + \delta - \xi) \equiv 0 \end{array} \pmod{p},$$

from which, since γ', δ' cannot both be divisible by p, it follows that:

$$\xi \equiv \alpha + \delta \pmod{p}; \tag{7}$$

if ξ is determined in this way, then in (5) the congruences 3., 4. follow from 1., 2. But if one substitutes γ', δ' from 1., 2. in

$$\alpha'\delta' - \beta'\gamma' \equiv 1 \pmod{p} \tag{8}$$

then one obtains

$$\alpha'^2\beta + \alpha'\beta'(\delta - \alpha) - \beta'^2\gamma \equiv 1 \pmod{p}. \tag{9}$$

If α', β' are determined from this, then all the requirements stated in (5), and hence also in (2), (4), are satisfied.

It therefore remains only to show that the congruence (9) is always solvable.

This possibility is evident if β or $-\gamma$ is a quadratic residue of p; for then

$$\begin{array}{lll} & \alpha' \equiv \sqrt{\beta^{-1}}, & \beta' \equiv 0 \\ \text{or} & \alpha' \equiv 0, & \beta' \equiv \sqrt{-\gamma^{-1}} \end{array} \pmod{p}$$

satisfies the requirement laid down.

In what follows we must now distinguish whether the substitution S belongs to class I, II, III of the preceding section.

If first

$$\text{I.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 \equiv 1 \pmod{p},$$

then, if we put S^m in place of S, by § 79, (9), β passes into $m\beta$, γ into $m\gamma$, and m can always be so determined that $m\beta$ or $-m\gamma$ becomes a quadratic residue of p.

Here belongs also the case that $\beta \equiv 0$ or $\gamma \equiv 0$ and $\alpha \equiv \delta \pmod p$.

If, on the other hand, $\beta \equiv 0$ and $\alpha - \delta$ is not divisible by p, then one may choose β' in (9) arbitrarily and then determine α' from a congruence of the first degree.

If β is not divisible by p, then (9) may be put into the form

$$\left[\alpha'\beta + \beta'\left(\frac{\delta-\alpha}{2}\right)\right]^2 \equiv \beta + \beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \pmod p, \tag{10}$$

and if now

II. $\left(\frac{\alpha+\delta}{2}\right)^2 - 1$ is a quadratic residue of p, then determine

β' from the congruence:

$$\beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \equiv \left(\frac{\beta-1}{2}\right)^2,$$

whereby (10) passes into

$$\alpha'\beta + \beta'\left(\frac{\delta-\alpha}{2}\right) \equiv \pm\frac{\beta+1}{2} \pmod p,$$

and from this α' can be determined, whichever sign be chosen.

If

III. $\left(\frac{\alpha+\delta}{2}\right)^2 - 1$ is a quadratic non-residue of p and at the

same time β is a quadratic non-residue, then a non-residue ν can always be determined such that $\beta + \nu$ becomes a quadratic residue; for if one lets ν run through the sequence of non-residues in $\beta + \nu$, then the results cannot be nothing but non-residues, since among them β too would have to occur.

Then β' can be determined so that

$$\beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \equiv \nu \pmod p,$$

and then α' from

$$\alpha'\beta + \beta'\left(\frac{\delta-\alpha}{2}\right) \equiv \pm\sqrt{\beta+\nu} \pmod p.$$

From this it follows:

In a normal divisor $\mathfrak{R}$ of the group $\mathfrak{L}_0$ a substitution U of the form

$$\begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix} \tag{11}$$

must certainly occur.

Let ξ first be different from zero $(\bmod p)$.

By the concept of the normal divisor, $\mathfrak{N}$ also contains the substitution

$$\begin{pmatrix} 0, & -1 \\ 1, & 0 \end{pmatrix}\begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix}\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} \xi, & 1 \\ -1, & 0 \end{pmatrix} \tag{12}$$

and consequently also

$$\begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix}\begin{pmatrix} \xi, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 2\xi, & 1 \end{pmatrix} \equiv A^{2\xi} \text{ [§ 79, (11)]}, \tag{13}$$

hence also A and all its powers. Therefore $\mathfrak{N}$ also contains

$$\begin{pmatrix} 1, & 0 \\ \eta, & 1 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & \xi \end{pmatrix} = \begin{pmatrix} 0, & 1 \\ -1, & \xi + \eta \end{pmatrix} \tag{14}$$

for an arbitrary η, that is, the substitution (11) for every arbitrary ξ, and consequently also

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}.$$

Accordingly, by the theorem of the preceding section, the group $\mathfrak{N}$ is identical with $\mathfrak{L}_0$.

It still remains to discuss the possibility that in (11) $\xi \equiv 0$.

In this case, then, the group $\mathfrak{N}$ contains the substitution

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix},$$

and consequently also, for an arbitrary α not divisible by p,

$$V = \begin{pmatrix} 0, & \alpha \\ -\alpha^{-1}, & 0 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \begin{pmatrix} 0, & -\alpha \\ \alpha^{-1}, & 0 \end{pmatrix} \begin{pmatrix} 0, & -1 \\ 1, & 0 \end{pmatrix} = \begin{pmatrix} \alpha^2, & 0 \\ 0, & \alpha^{-2} \end{pmatrix}.$$

From this one derives further, as contained in $\mathfrak{N}$:

$$W = \begin{pmatrix} 1, & 0 \\ -1, & 1 \end{pmatrix} \begin{pmatrix} \alpha^{-2}, & 0 \\ 0, & \alpha^2 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} \begin{pmatrix} \alpha^2, & 0 \\ 0, & \alpha^{-2} \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ \alpha^4 - 1, & 1 \end{pmatrix}.$$

If now p is greater than 5, then α can be so assumed that $\alpha^4 - 1$ is not divisible by p; then $\mathfrak{N}$ also contains the substitution A and its powers, and consequently $\mathfrak{N}$ is identical with $\mathfrak{L}_0$.

If $p = 5$, then $\alpha^4 - 1$ is always divisible by p, so this conclusion is not applicable.

In this case, however, it follows from the concept of the normal divisor, as contained in $\mathfrak{N}$:

$$\begin{pmatrix} 1, & 1 \\ 2, & -2 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \begin{pmatrix} -2, & -1 \\ -2, & 1 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} -2, & 0 \\ 0, & 2 \end{pmatrix},$$

$$\begin{pmatrix} -2, & 0 \\ 0, & 2 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ -1, & 1 \end{pmatrix} \begin{pmatrix} 2, & 0 \\ 0, & -2 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} = \begin{pmatrix} 1, & 0 \\ 2, & 1 \end{pmatrix},$$

i. e. A^2, and hence all powers of A, from which it is to be concluded, as above, that $\mathfrak{N}$ is identical with $\mathfrak{L}_0$. We have therefore the theorem:

If $p > 3$, then the group $\mathfrak{L}_0$ has no normal divisor.

In the case $p = 3$, $\mathfrak{N}$ likewise contains

$$\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix},$$

and therefore also

$$\begin{pmatrix} 1, & 0 \\ \pm 1, & 1 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ \mp 1, & 1 \end{pmatrix} = \begin{pmatrix} \mp 1, & 1 \\ 1, & \pm 1 \end{pmatrix},$$

and in fact the four substitutions

$$\begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}, \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \begin{pmatrix} -1, & 1 \\ 1, & 1 \end{pmatrix}, \begin{pmatrix} 1, & 1 \\ 1, & -1 \end{pmatrix} \tag{15}$$

do form a normal divisor of $\mathfrak{L}_0$ of index 3.

In the case $p = 3$ the group $\mathfrak{L}_0$ is isomorphic with the alternating group of the permutations of four elements. This is the group of an arbitrary equation of the fourth degree, when the square root of the discriminant is adjoined to the domain of rationality. The divisor (15) of this group, of index 3, then furnishes the cubic resolvent of the biquadratic equation that is known in algebra.

A function belonging to the group (15), formed from the roots v_∞, v_0, v_1, v_2 of the transformation equation for the third degree of transformation, is for example

$$v_\infty v_0 + v_1 v_2$$

or

$$(v_\infty - v_0)(v_1 - v_2).$$

The difference between these two functions is this, that the first remains unchanged under the substitution of determinant -1

$$\begin{pmatrix} -1, & 0 \\ 0, & 1 \end{pmatrix}$$

while the second thereby changes its sign. The first will therefore lead to an equation of the third degree with rational coefficients, while in the cubic equation for the second $\sqrt{-3}$ still occurs (§ 78).

§ 81. Non-normal divisors of $\mathfrak{L}_0$.

We now inquire, leaving aside the case $p = 3$, after the non-normal divisors of the group $\mathfrak{L}_0$ whose index is smaller than $p + 1$; upon the existence of such divisors depends the possibility of forming resolvents of the transformation equation whose degree is lower than $p + 1$.

First of all it can be shown that the index of a divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ can never be smaller than p.

Let

$$\mathfrak{K} = S_0,\ S_1,\ \ldots,\ S_{\nu-1} \tag{1}$$

be any divisor whatever of $\mathfrak{L}_0$ of degree ν, and, if it is possible, let

$$T = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \tag{2}$$

be a substitution of degree p which occurs in $\mathfrak{L}_0$ but not in $\mathfrak{K}$. Then too, since p is a prime, no power of T lower than the pth can occur in $\mathfrak{K}$. The subsidiary groups

$$\mathfrak{K},\ T\mathfrak{K},\ T^2\mathfrak{K},\ \ldots,\ T^{p-1}\mathfrak{K} \tag{3}$$

contain none but mutually distinct elements $T^i S_k$, and accordingly νp is at most equal to the degree of the group $\mathfrak{L}_0$, that is:

$$\nu p \leqq p\,\frac{p^2 - 1}{2}.$$

Thus the index of $\mathfrak{K}$, that is the quotient

$$p\,\frac{p^2 - 1}{2} : \nu$$

is equal to p or greater than p.

A divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ whose index is smaller than p must therefore contain all substitutions of degree p, and hence also (§ 79, I.) all substitutions T in which

$$\alpha + \delta \equiv 2 \pmod{p} \tag{4}$$

holds. Accordingly such a group $\mathfrak{K}$ contains in the first place the substitution

$$A = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix} \tag{5}$$

and their powers; likewise the substitutions

$$\begin{pmatrix} 1, & 2 \\ 0, & 1 \end{pmatrix}, \quad \begin{pmatrix} 2, & 1 \\ -1, & 0 \end{pmatrix},$$

and consequently also

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} 1, & 2 \\ 0, & 1 \end{pmatrix} \begin{pmatrix} 2, & 1 \\ -1, & 0 \end{pmatrix}. \tag{6}$$

But if the substitutions A, B are contained in $\mathfrak{K}$, then by § 79 $\mathfrak{K}$ must be identical with $\mathfrak{L}_0$. We have therefore the theorem:

The index of a divisor of $\mathfrak{L}_0$ cannot be smaller than p.

Our problem is thus restricted to the question of the divisors of $\mathfrak{L}_0$ of index p, or of degree $\frac{p^2-1}{2}$.

Let $\mathfrak{K}$ be such a divisor, and let $S_0, S_1, \ldots, S_{\nu-1}$ be its elements, so that the number ν has the value

$$\nu = \frac{p^2-1}{2} \tag{7}$$

.

Since the degree of every element of a group is always a divisor of the degree of the group, and since ν is not divisible by p, no element of degree p can occur in $\mathfrak{K}$; if we therefore take for T the substitution of degree p:

$$A = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \tag{8}$$

then the group $\mathfrak{L}_0$ can be decomposed into the p subsidiary groups:

$$\mathfrak{L}_0 = \mathfrak{K},\ A\mathfrak{K},\ A^2\mathfrak{K}\ \ldots,\ A^{p-1}\mathfrak{K}. \tag{9}$$

Of the same degree as $\mathfrak{K}$, and hence also of the same index, are all its conjugate divisors

$$T\mathfrak{K}T^{-1}.$$

Now let g be a primitive root of the prime p, and

$$C = \begin{pmatrix} g, & 0 \\ 0, & g^{-1} \end{pmatrix}, \tag{10}$$

a substitution which is evidently of degree $\frac{p-1}{2}$. Like every substitution of $\mathfrak{L}_0$, it must occur in one of the rows (9); that is, there is a substitution S in $\mathfrak{K}$ and an exponent λ for which

$$C = A^\lambda S, \quad S = A^{-\lambda} C. \tag{11}$$

But γ can be further determined so that

$$A^{\gamma-\lambda} C A^{-\gamma} = C, \tag{12}$$

and for this one obtains the condition

$$\gamma(g - g^{-1}) \equiv \lambda g \pmod{p},$$

which, since $p > 3$, can always be satisfied; but then, on account of (11),

$$C = A^{\gamma} S A^{-\gamma}.$$

Thus C occurs in the divisor $A^{\gamma}\mathfrak{K}A^{-\gamma}$ conjugate to $\mathfrak{K}$, and we may therefore, by putting this group in place of $\mathfrak{K}$, assume without loss of generality that $\mathfrak{K}$ itself contains the substitution C.

Now the substitution

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}$$

may either occur in $\mathfrak{K}$ or not occur. In the first case $\mathfrak{K}$ also contains all the substitutions composed of B and C, which are all of one of the two forms:

$$\begin{pmatrix} g^r, & 0 \\ 0, & g^{-r} \end{pmatrix}, \qquad r = 0, 1, \ldots \frac{p-3}{2}, \tag{13}$$

$$\begin{pmatrix} 0, & g^{-r} \\ -g^r, & 0 \end{pmatrix}, \tag{14}$$

and whose number amounts to $p - 1$. In the first case, then, $\mathfrak{K}$ contains all substitutions of the form (13), (14), in the second all substitutions of the form (13) and none of the form (14).

Now let

$$V = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \tag{15}$$

be a substitution contained in $\mathfrak{K}$ which is contained neither in the form (13) nor in the form (14), and for which therefore neither α and δ, nor β and γ, are simultaneously congruent to 0. Then all the substitutions of the form

$$C^r V C^s = \begin{pmatrix} \alpha g^{r+s}, & \beta g^{r-s} \\ \gamma g^{-r+s}, & \delta g^{-r-s} \end{pmatrix}, \tag{16}$$

when r, s both run through the sequence of numbers:

$$0, 1, 2, \ldots \frac{p-3}{2}$$

are contained in $\mathfrak{K}$ and are all different from one another. For if two among the substitutions (16) are equal to one another, then we must have

there would also be one among them which, without r, s vanishing, becomes identical with V. But this requires

$$\begin{aligned} \alpha &\equiv \pm\alpha g^{r+s}, & \beta &\equiv \pm\beta g^{r-s} \\ \gamma &\equiv \pm\gamma g^{-r+s}, & \delta &\equiv \pm\delta g^{-r-s} \pmod{p}, \end{aligned}$$

where in all four formulae the upper or the lower signs hold, and hence, since neither α and δ nor β and γ vanish simultaneously,

$$r + s \equiv 0, \qquad r - s \equiv 0 \qquad (\text{mod } p-1),$$

or

$$r + s \equiv \frac{p-1}{2}, \qquad r - s \equiv \frac{p-1}{2} \qquad (\text{mod } p-1),$$

and so in both cases

$$r \equiv 0, \qquad s \equiv 0 \ \left(\text{mod } \frac{p-1}{2}\right), \qquad \text{q. e. d.}$$

In the form (16) there are therefore contained $\left(\frac{p-1}{2}\right)^2$ different substitutions. If the group $\mathfrak{K}$ is not yet exhausted by them, one chooses a substitution V' not contained in (13), (14), (16) and forms in the same way the sequence

(17) $$C^r V' C^s,$$

whose substitutions are different both among themselves and from those contained in (13), (14), (16); and one proceeds in this way until the whole group $\mathfrak{K}$ is exhausted.

If we denote by q the number of the sequences (16), (17), ... so formed, then the degree of the group $\mathfrak{K}$ comes out:

1. if B does not occur in $\mathfrak{K}$:

$$= \frac{p-1}{2} + q\left(\frac{p-1}{2}\right)^2;$$

2. if B occurs in $\mathfrak{K}$:

$$= p - 1 + q\left(\frac{p-1}{2}\right)^2,$$

and by the requirement of our problem this number is to be

$$= \frac{(p-1)(p+1)}{2}$$

From this, however, it follows that case 1. is impossible, for in this case we should necessarily have

$$q\left(\frac{p-1}{2}\right) = p,$$

so that

$$q = p, \qquad p = 3$$

would have to hold, which we have excluded.

In case 2., however, it follows for every arbitrary p:

$$q = 2.$$

Hence the substitution B must in any case occur in $\mathfrak{K}$, and the group sought is exhausted by (13), (14), (16), (17).

We show first that in the two substitutions V, V' of the group $\mathfrak{K}$ none of the numbers $\alpha, \beta, \gamma, \delta, \alpha', \beta', \gamma', \delta'$ can be congruent to 0.

For, as we have already seen, there does not occur in $\mathfrak{K}$

$$\begin{pmatrix} 1, & 0 \\ \gamma, & 1 \end{pmatrix},$$

if γ is different from 0, and hence neither

$$\begin{pmatrix} \alpha, & 0 \\ 0, & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1, & 0 \\ \alpha\gamma, & 1 \end{pmatrix} = \begin{pmatrix} \alpha, & 0 \\ \gamma, & \alpha^{-1} \end{pmatrix},$$

where α is arbitrary; and consequently neither

$$\begin{pmatrix} \alpha, & 0 \\ \gamma, & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} 0, & \alpha \\ -\alpha^{-1}, & \gamma \end{pmatrix}$$
$$\begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \begin{pmatrix} \alpha, & 0 \\ \gamma, & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \gamma, & \alpha^{-1} \\ -\alpha, & 0 \end{pmatrix}$$
$$\begin{pmatrix} 0, & -1 \\ 1, & 0 \end{pmatrix} \begin{pmatrix} 0, & \alpha \\ -\alpha^{-1}, & -\gamma \end{pmatrix} = \begin{pmatrix} \alpha^{-1}, & \gamma \\ 0, & \alpha \end{pmatrix},$$

in which are contained all substitutions $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ in which one of the four numbers $\alpha, \beta, \gamma, \delta$ is congruent with 0.

We now denote the substitutions of the system (16) by

$$W = \begin{pmatrix} \alpha g^{r+s}, & \beta g^{r-s} \\ \gamma g^{-r+s}, & \delta g^{-r-s} \end{pmatrix},$$

and set

$$\xi \equiv \alpha g^{r+s}, \qquad \varrho \equiv \alpha^{-1}\beta g^{-2s} \tag{18}$$

$$\alpha\delta \equiv m, \qquad \beta\gamma \equiv m - 1 \pmod{p}. \tag{19}$$

m does not change when V is replaced by an arbitrary substitution of the system (16). The number m' shall have the corresponding meaning for the system (17); in each of these two systems ξ can assume any of the values

$$1, 2, \ldots p - 1 \tag{20}$$

assume, and ϱ runs, according as $\alpha^{-1}\beta$ is a quadratic residue or a non-residue, in one system through the sequence of the residues or of the non-residues. The change of the sign of ξ gives no new substitution W. Accordingly we can represent W in this way:

$$W = \begin{pmatrix} \xi, & \xi\varrho \\ \xi^{-1}\varrho^{-1}(m-1), & \xi^{-1}m \end{pmatrix}. \tag{21}$$

Now since the square of W

$$W^2 = \begin{pmatrix} \xi^2 + m - 1, & (\xi^2 + m)\varrho \\ \xi^{-2}\varrho^{-1}(m-1)(\xi^2+m), & [(m-1)\xi^2 + m^2]\xi^{-2} \end{pmatrix}$$

must belong to $\mathfrak{K}$, it is contained under one of the systems (13), (14), (16), (17), from which the following four possibilities result:

(22)
$$\begin{aligned} &1.\quad \xi^2 + m \equiv 0, \\ &2.\quad \xi^2 + m - 1 \equiv 0, \quad (m-1)\xi^2 + m^2 \equiv 0, \\ &3.\quad (\xi^2 + m - 1)[(m-1)\xi^2 + m^2] \equiv m\xi^2, \\ &4.\quad (\xi^2 + m - 1)[(m-1)\xi^2 + m^2] \equiv m'\xi^2, \end{aligned}$$

and each of the $p-1$ values (20), put for ξ, must satisfy one of these four cases. If one of the congruences (22), 2. is fulfilled, then the other must follow from it, which is possible only if

(23)
$$2m \equiv 1, \qquad \xi^2 \equiv m \pmod{p}.$$

Now a congruence with respect to a prime modulus has at most as many incongruent roots as the degree of the congruence amounts to; 1. and 2. can therefore be satisfied for at most two values each, 3. and 4. for at most four values each of ξ; there are therefore in all at most twelve values of ξ which satisfy one of the four congruences (22).

From this it follows:

$$p - 1 \leqq 12, \qquad p \leqq 13.$$

But if $p = 13$, then each of the congruences (22) must have the maximum number of roots; 2. must therefore also be satisfied for two values of ξ; then by (23) we should have $m \equiv 7$, $\xi^2 \equiv 7 \pmod{13}$, which is impossible, since 7 is a quadratic non-residue of 13.

A divisor of $\mathfrak{L}_0$ of index p therefore does not exist when $p > 11$.

§ 82. Divisors of $\mathfrak{L}_0$ of index p for $p = 5, 7, 11$.

There remains the possibility that for $p = 5, 7, 11$ divisors of $\mathfrak{L}_0$ of index p exist.

1. $p = 5$. We investigate first the case $p = 5$. Since m and $m - 1$ cannot be congruent to 0, there remain for m the assumptions

$$m \equiv 2, 3, 4 \pmod{p}.$$

If, however, we replace the substitution W § 81, (21) by

(1)
$$WB = \begin{pmatrix} -\xi\varrho, & \xi \\ -\xi^{-1}m, & \xi^{-1}\varrho^{-1}(m-1) \end{pmatrix},$$

whereby m passes into $1 - m$, then the case $m \equiv 4$ is brought back to the case $m \equiv 2$.

For $m \equiv 2$, however, (23), § 81 is not satisfied, and of the congruences (22) none is possible, since ± 2 are non-residues of 5 and the left sides of (22), 3., 4. reduce for $m = 2$ to $\xi^4 - 1$, which for every ξ different from zero is divisible by 5. It therefore only remains that

(2)
$$m \equiv 3, \quad m' \equiv 3 \pmod{5}$$

and the two systems W, W' [§ 81, (16), (17)] can differ from one another only in this, that $\alpha^{-1}\beta$, and hence also ϱ, is in the one a quadratic residue, in the other a quadratic non-residue of 5. The group sought therefore consists, if it exists, of the 12 substitutions contained in the three forms

$$\begin{pmatrix} \xi, & 0 \\ 0, & \xi^{-1} \end{pmatrix}, \quad \begin{pmatrix} 0, & \xi \\ -\xi^{-1}, & 0 \end{pmatrix}, \quad \begin{pmatrix} \xi, & \eta \\ 2\eta^{-1}, & 3\xi^{-1} \end{pmatrix} \tag{3}$$

in which

$$\xi \equiv 1, 2, \quad \eta \equiv \pm 1, \pm 2 \pmod 5$$

is to be set.

2. $p = 7, 11$.

Since -1 is to be found among the quadratic non-residues for $p = 7, 11$, the composed substitution WB, (1), belongs to the W' [because the quadratic character of the quantity denoted by ϱ in (18), § 81 is the opposite one in W and in WB]. Accordingly, by (1) and § 81, (19), (21), $\beta'\gamma' \equiv -m \equiv m' - 1$, hence:

$$m + m' \equiv 1 \pmod p. \tag{4}$$

For $p = 7$, since the interchange of m with m' yields nothing new, the following three possibilities remain: 1. $m \equiv 2$, $m' \equiv 6$. 2. $m \equiv 3$, $m' \equiv 5$. 3. $m \equiv 4$, $m' \equiv 4$.

In case 1., of the congruences § 81, (22), numbers 1. and 2. are impossible; hence for every ξ one of the two congruences 3., 4. would have to be satisfied, which here read:

$$(\xi^2 + 1)(\xi^2 + 4) \equiv 2\xi^2, 6\xi^2 \pmod 7,$$

none of which is satisfied for $\xi \equiv 1$.

In case 3., (22), 1. cannot be satisfied, and (22), 2. is satisfied for $\xi^2 \equiv 4$; hence for $\xi^2 \equiv 1, 2$ the congruence (22) 3. or 4.:

$$(\xi^2 + 3)(3\xi^2 + 2) \equiv 4\xi^2 \pmod 7$$

must be satisfied, which again is not the case. Thus for $p = 7$ there remains only:

$$m \equiv 3, \quad m' \equiv 5 \pmod 7. \tag{5}$$

For $p = 11$ the possibility that the congruences (22), 2. be satisfied is to be excluded from the outset, because in that case, in consequence of (4) and § 81, (23):

$$m \equiv m'$$

would have to hold; then (22), 3., 4. would not be distinct, and the congruences (22) could together have at most eight roots and not ten, as they nevertheless must.

Accordingly the congruences (22), 1., 3., 4 must each have the maximal number of roots, and in order that 1. be satisfied, m, and for the same reasons m', must be a quadratic non-residue of 11. But this condition, and at the same time the condition (4), is satisfied only by the two numbers

$$m, m' \equiv 2, -1 \pmod{11}. \tag{6}$$

The groups sought therefore consist, in these two cases, if they exist, of the following substitutions:

$p = 7$

$$\begin{pmatrix} \xi, & 0 \\ 0, & \xi^{-1} \end{pmatrix}, \begin{pmatrix} 0, & \xi \\ \xi^{-1}, & 0 \end{pmatrix}, \begin{pmatrix} \xi & \xi\varrho \\ 2\xi^{-1}\varrho^{-1}, & 3\xi^{-1} \end{pmatrix}, \begin{pmatrix} -\xi\varrho, & \xi \\ -3\xi^{-1}, & 2\xi^{-1}\varrho^{-1} \end{pmatrix} \tag{7}$$

$$\xi \equiv 1, 2, 3 \pmod 7,$$

$p = 11$

$$(8) \qquad \begin{pmatrix} \xi, & 0 \\ 0, & \xi^{-1} \end{pmatrix}, \begin{pmatrix} 0, & \xi \\ -\xi^{-1}, & 0 \end{pmatrix}, \begin{pmatrix} \xi, & \xi\varrho \\ \xi^{-1}\varrho^{-1}, & 2\xi^{-1} \end{pmatrix}, \begin{pmatrix} -\xi\varrho, & \xi \\ -2\xi^{-1}, & \xi^{-1}\varrho^{-1} \end{pmatrix}$$

$$\xi \equiv 1, 2, 3, 4, 5 \pmod{11};$$

ϱ runs in (7) and (8) through either the sequence of the quadratic residues or that of the non-residues, so that for $p = 7$ or 11 one obtains in each case two groups of index p.

That the systems of substitutions collected in (3), (7), (8) really constitute groups can be shown by direct composition in various ways. One reaches this proof more simply, however, by forming functions of the v_z, § 78, (7), which remain unchanged under the substitutions of these systems, and which under the substitutions of $\mathfrak{L}_0$ take only p different values in all.

The systems (3), (7), (8) can, as follows from their mode of origin, be composed by repeated application of B, C, U, U', if U, U' are any two special substitutions W, W'. For $p = 5$ the substitution U^2, and for $p = 7, 11$ the substitution UB, belongs to the W', so that U' can then be omitted. If we choose U quite arbitrarily, and take for the primitive root g in C, for $p = 5, 7, 11$ respectively, $2, -2, 2$, then we can compose the group (3) from:

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2, & 0 \\ 0, & 3 \end{pmatrix}, \quad U = \begin{pmatrix} 1, & 1 \\ 2, & 3 \end{pmatrix} \quad (p = 5),$$

the group (7) from:

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2, & 0 \\ 0, & 4 \end{pmatrix}, \quad U = \begin{pmatrix} 1, & \pm 1 \\ \pm 2, & 3 \end{pmatrix} \quad (p = 7),$$

the group (8) from:

$$B = \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2, & 0 \\ 0, & 6 \end{pmatrix}, \quad U = \begin{pmatrix} 1, & \pm 1 \\ \pm 1, & 2 \end{pmatrix} \quad (p = 11),$$

where in the two last cases the two different groups mentioned above arise from the double signs.

We now collect the permutations of the indices z that correspond to these substitutions [§ 79, (3)].

$p = 5$:

$$\begin{array}{cccccccc} z = \infty, & 0, & 1, & 2, & 3, & 4 & \\ 0, & \infty, & 4, & 2, & 3, & 1 & (B) \\ \infty, & 0, & 4, & 3, & 2, & 1 & (C) \\ 4, & 1, & 2, & 0, & \infty, & 3 & (U). \end{array}$$

Thus by B, C, U the index pairs

$$(\infty, 0), \quad (1, 4), \quad (2, 3)$$

not torn apart, but only interchanged among one another. Moreover, in two of these pairs the elements are interchanged each time. If we therefore form a function like

$$(9) \qquad V = (v_\infty - v_0)(v_1 - v_4)(v_2 - v_3),$$

then it remains unchanged under B, C, U, and hence under the whole system (3), while by repeated application of the cyclic permutation $(0, 1, 2, 3, 4)$, that is, of the substitution

$$A = \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}$$

it takes five different values. But the whole group $\mathfrak{L}_0$ can be composed from A and B [§ 79, (10)], and V therefore takes no more than five values under the application of $\mathfrak{L}_0$.

The five values of V are the roots of an equation of the fifth degree.

For $p = 7$ we have

$$B = \left(z, -\frac{1}{z}\right), \quad C = (z, 4z), \quad U = \left(z, \frac{\mp 2 + z}{3 \mp z}\right).$$

For a better survey we also represent B, C, U by the cyclic permutations which they produce among the eight values of z:

$$\begin{aligned} B &= (0, \infty)(1, -1)(2, 3)(-2, -3), \\ C &= (1, -3, 2)(-1, 3, -2), \\ U &= (\infty, \mp 1, \pm 1, \pm 3)(0, \mp 3, \mp 2, \pm 2), \end{aligned}$$

and from this one obtains the seven-valued function

$$V = (v_\infty - v_0)(v_{\mp 1} - v_{\mp 3})(v_{\pm 1} - v_{\mp 2})(v_{\pm 3} - v_{\pm 2}), \tag{10}$$

in which the one or the other sign may be taken.

For $p = 11$ we have

$$B = \left(z, -\frac{1}{z}\right), \quad C = (z, 4z), \quad U = \left(z, \frac{\mp 1 + z}{2 \mp z}\right),$$

or, represented by the cycles:

$$\begin{aligned} B &= (0, \infty)(1, -1)(2, 5)(-2, -5)(3, -4)(-3, 4), \\ C &= (1, 4, 5, -2, 3)(-1, -4, -5, 2, -3), \\ U &= (\infty, \mp 1, \pm 3, \mp 2, \pm 2)(0, \pm 5, \mp 5, \mp 4, \pm 1), \end{aligned}$$

from which the two 11-valued functions

$$V = (v_\infty - v_0)(v_{\mp 1} - v_{\pm 5})(v_{\pm 3} - v_{\mp 5})(v_{\mp 2} - v_{\mp 4})(v_{\pm 2} - v_{\pm 1})(v_{\pm 4} - v_{\mp 3}). \tag{11}$$

The following remark is still of interest here. The resolvents of degree p, whose roots are the quantities (9), (10), (11), still contain $\sqrt{\pm p}$ in their coefficients, according to § 78.

But the group $\mathfrak{L}_0$ is extended to the group $\mathfrak{L}$ if we add a linear substitution of the form § 78, (13) in which $ad - bc$ is a quadratic non-residue, thus for instance:

$$\begin{aligned} &\text{for } p = 5 && (z,\ 2z) \\ &\text{for } p = 7,\ 11 && (z,\ -z). \end{aligned}$$

By applying this substitution, however, the value (9)

$$V = (v_\infty - v_0)(v_1 - v_4)(v_2 - v_3)$$

passes into its opposite, and from this it follows that, when $p = 5$, V^2 satisfies an equation of degree 5 with rational numerical coefficients. From this one concludes that in the equation for V the coefficients of the odd powers have the factor $\sqrt{5}$, while the others are rational; or that for the unknown $\sqrt{5}\,V$ one obtains an equation with rational coefficients.

For $p = 7$, 11 the expressions corresponding to the two signs in (10) or (11) pass into one another through the interchange $(z, -z)$. The resolvents of degree 7 or 11, which one of the two expressions (10) or (11) respectively satisfies, therefore pass into one another through a change of the sign of $\sqrt{-7}$, $\sqrt{-11}$.

§ 83. Various Resolvents of Degree 5 for the 5th Degree of Transformation.

In the formation of the resolvents of degree p we restrict ourselves here to the case $p = 5$.

These resolvents can be given very manifold forms; for one may not only choose, in the function V, (9) of the preceding section, the roots of an arbitrary transformation equation for the quantities v, but may also replace V by various other functions, for example by

$$(v_\infty + v_0)(v_1 + v_4)(v_2 + v_3),$$

or by

$$v_\infty v_0 + v_1 v_4 + v_2 v_3.$$

We wish first to apply the transformation equation § 72, (8)', in which now, in agreement with the preceding section, v is put for x:

$$v^6 + 10\,v^3 - \gamma_2 v + 5 = 0 \tag{1}$$

Here, if

$$c \equiv 0 \pmod{12}, \quad z \equiv c \pmod 5$$

then

$$v_\infty = 5\left(\frac{\eta(5\omega)}{\eta\,\omega}\right)^2, \quad v_z = \left(\frac{\eta\left(\dfrac{\omega + c}{5}\right)}{\eta(\omega)}\right)^2. \tag{2}$$

If we then take

$$\begin{aligned} \sqrt{5}\,w_z &= (v_\infty - v_z)(v_{z+1} - v_{z-1})(v_{z-2} - v_{z+2}) \\ &= (v_\infty - v_c)(v_{c+12} - v_{c-12})(v_{c+24} - v_{c-24}), \\ z &= 0, 1, 2, 3, 4, \quad c = 0, \pm 12, \pm 24, \end{aligned} \tag{3}$$

then, by the concluding remark of the preceding section, the w_0, w_1, w_2, w_3, w_4 are the roots of an equation of degree 5, whose coefficients are formed rationally from γ_2 and rational numbers.

If, however, we observe the relations [§ 54, (14)]:

$$\gamma_2(\omega + 1) = e^{-\frac{2\pi i}{3}}\gamma_2(\omega), \quad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega),$$

it follows from the form of (1) that under the interchange $(\omega, \omega + 1)$ the roots v, apart from a rearrangement, take on the factor $e^{\frac{2\pi i}{3}}$, while under $\left(\omega, -\dfrac{1}{\omega}\right)$ they remain unchanged.

The w_z are therefore interchanged only among themselves, and their symmetric functions remain unchanged and depend therefore rationally only on the invariant $j(\omega)$. The coefficients in the equation for w can moreover become infinite for no finite $j(\omega)$, and they are accordingly polynomials in $j(\omega)$.

Further information about these coefficients results from the expansion in ascending powers of q. By (2) the expansions of the v have the beginnings

$$v_\infty = 5\,q^{\frac{2}{3}} + \cdots, \quad v_z = e^{\frac{c}{12}\frac{2\pi i}{5}} q^{-\frac{2}{15}} + \cdots,$$

and from this it follows, with the use of the well-known equation

$$4 \sin\frac{2\pi}{5} \cdot \sin\frac{4\pi}{5} = \sqrt{5}$$

the beginning of the expansion for w_z:

$$w_z = q^{-\frac{2}{5}} e^{\frac{c}{2}\frac{\pi i}{5}} + \cdots \tag{4}$$

From this it follows that the power sums of the w_z, up to the fourth inclusive, are constants, since they cannot contain even the first power of $j(\omega)$, and that the product of the w_z must be a linear function of $j(w)$, in which $j(\omega)$ has the coefficient 1.

The equation of the w therefore has the form:

$$w^5 + b_1 w^4 + b_2 w^3 + b_3 w^2 + b_4 w + b_5 = j(\omega), \tag{5}$$

where the b are rational numbers. These coefficients can be calculated by carrying the expansion (4) further. Here we take another route, which, strictly speaking, belongs to the complex multiplication treated in the next part, but which, being so simple, may nevertheless quite well find its place here.

If we set $\omega = i = \sqrt{-1}$, then

$$\omega = -\frac{1}{\omega}$$

and consequently [§ 34, (11), (12), (14)]:

$$f_1(i) = f_2(i) = \sqrt[8]{2}, \quad f(i) = \sqrt[4]{2},$$

hence [§ 54, (5)]:

$$\gamma_3(i) = 0, \quad \gamma_2(i) = 12, \quad j(i) = 12^3. \tag{6}$$

But one finds further, from the transformation formulae for $\eta(\omega)$ [§ 34, (4), (5)]:

$$\eta(\omega + 1) = e^{\frac{\pi i}{12}} \eta(\omega), \quad \eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\, \eta(\omega),$$

if the decomposition

$$5 = (2 + i)(2 - i)$$

is taken into account:

$$\eta\left(\frac{12+i}{5}\right) = e^{\frac{\pi i}{6}} \eta\left(\frac{2+i}{5}\right) = e^{\frac{\pi i}{6}} \eta\left(\frac{1}{2-i}\right) = \sqrt{1+2i}\, \eta(i),$$

hence

$$v_2 = 1 + 2i, \quad v_3 = 1 - 2i,$$

and now, since for $\gamma_2 = 12$ two roots of equation (1) are known, one can easily decompose the left part of this equation into factors for this particular value of γ_2:

$$v^6 + 10v^3 - 12v + 5 = (v^2 + 2v + 5)(v^2 + v - 1)^2,$$

from which it is to be concluded:

$$\begin{aligned} v_\infty = v_0 &= \frac{-1 + \sqrt{5}}{2}, \\ v_1 = v_4 &= \frac{-1 - \sqrt{5}}{2}. \end{aligned} \tag{7}$$

(v_0, v_∞ must be positive, since q is real in them.)

From (3) there then results:

$$\begin{aligned} &w_0 = 0, \quad w_1 = w_3 = -5 - 2i\sqrt{5}, \\ &w_2 = w_4 = -5 + 2i\sqrt{5}. \end{aligned} \tag{8}$$

These must be the roots of (5) for $j = 12^3$, and from this one obtains the resolvent sought in the form:

$$w(w^2 + 10w + 45)^2 = j(\omega) - 12^3 = \gamma_3^2. \tag{9}$$

This equation can also be brought into the form:

$$(w + 3)^3(w^2 + 11w + 64) = j(\omega) = \gamma_2^3, \tag{10}$$

which one also reaches directly if, in place of setting $\omega = i$,

$$\omega = \frac{-1 + i\sqrt{3}}{2}$$

is assumed.

The resolvent found simplifies still further if [according to (9)]:

$$z = \sqrt{w} = \frac{\gamma_3}{w^2 + 10w + 45}$$

is set:

$$z^5 + 10z^3 + 45z = \gamma_3, \tag{11}$$

or if [according to (10)]

$$w^2 + 11w + 64 = y^3, \quad y = \frac{\gamma_2}{w + 3} \tag{12}$$

is set:

$$y^5 - 40y^2 - 5\gamma_2 y - \gamma_2^2 = 0, \tag{13}$$

an equation which also results if one makes the assumption from the outset:

$$2y = v_\infty v_0 + v_1 v_4 + v_2 v_3.$$

In equation (13), as one sees, the third and the fourth power of the unknown are missing. The same property belongs also, as is easily shown, to the equation whose roots

$$x = y(\lambda + \mu z) = \frac{(\lambda + \mu z)\gamma_2}{z^2 + 3} \tag{14}$$

[18]

are, when λ, μ denote arbitrary parameters which are the same for all five values x.

If one puts this equation into the form

$$x^5 - 5ax^3 - 5b\gamma_2 x - bc\gamma_2^2 = 0, \tag{15}$$

then, after some calculations making use of the particular values of x for $\gamma_3 = 0$ and $\gamma_3 = \infty$, the following expressions result for a, b, c:

$$\begin{aligned} a &= 8\lambda^3 - 72\lambda\mu^2 + \gamma_3(\lambda^2\mu - \mu^3), \\ b &= \lambda^4 + 18\lambda^2\mu^2 - 27\mu^4 + \gamma_3\lambda\mu^3, \\ c &= \lambda^5 + 10\lambda^3\mu^2 + 45\lambda\mu^3 + \gamma_3\mu^5. \end{aligned} \tag{16}$$

[18] Cf. Kiepert, Auflösung der Gleichung 5ten Grades. Crelles Journal, Bd. 87, S. 114.

One more resolvent of degree 5 is to be formed, making use of the function $f(\omega)$.

We put:

$$u = f(\omega), \quad v_\infty = f(5\omega), \quad v_z = f\left(\frac{\omega + c}{5}\right) \tag{17}$$
$$c \equiv 0 \pmod{48}, \quad z \equiv c \pmod 5,$$

and by § 73 we have between u and v the equation of degree 6:

$$u^6 + v^6 - u^5 v^5 + 4uv = 0. \tag{18}$$

We investigate the influence which the three permutations

$$(\omega, \omega + 2), \quad \left(\omega, -\frac{1}{\omega}\right), \quad \left(\omega, \frac{\omega - 1}{\omega + 1}\right)$$

have on the quantities v_z. By these permutations c passes over into c_1, c_2, c', which by § 69, (9), (11) and § 73, (6) are determined by the congruences

$$c_1 \equiv c + 2, \quad c_2 \equiv -\frac{1}{c}, \quad c' \equiv \frac{c-1}{c+1} \pmod 5 \tag{19}$$

Apart from this change of the index, by § 73, (4) and (9) the v pass over into

$$e^{-\frac{5\pi i}{12}} v, \quad v, \quad -\frac{\sqrt{2}}{v}. \tag{20}$$

Thus, if we set $e^{-\frac{5\pi i}{12}} = \varepsilon$ for brevity, we have the following permutations belonging together:

$$\begin{array}{cccccccc}
\omega, & u, & v_\infty, & v_0, & v_1, & v_2, & v_3, & v_4, \\
\omega + 2, & e^{-\frac{\pi i}{12}} u, & \varepsilon v_\infty, & \varepsilon v_2, & \varepsilon v_3, & \varepsilon v_4, & \varepsilon v_0, & \varepsilon v_1, \\
-\dfrac{1}{\omega}, & u, & v_0, & v_\infty, & v_4, & v_2, & v_3, & v_1, \\
\dfrac{\omega - 1}{\omega + 1}, & \dfrac{\sqrt{2}}{u}, & -\dfrac{\sqrt{2}}{v_1}, & -\dfrac{\sqrt{2}}{v_4}, & -\dfrac{\sqrt{2}}{v_0}, & -\dfrac{\sqrt{2}}{v_2}, & -\dfrac{\sqrt{2}}{v_3}, & -\dfrac{\sqrt{2}}{v_\infty}.
\end{array} \tag{21}$$

We now introduce the five quantities w_z by the equation:

$$w_z = \frac{(v_\infty - v_z)(v_{z+1} - v_{z-1})(v_{z+2} - v_{z-2})}{\sqrt{5}\, u^3}, \tag{22}$$

that is:

$$w_0 = \frac{(v_\infty - v_0)(v_1 - v_4)(v_2 - v_3)}{\sqrt{5}\, u^3},$$
$$w_1 = \frac{(v_\infty - v_1)(v_2 - v_0)(v_3 - v_4)}{\sqrt{5}\, u^3},$$
$$w_2 = \frac{(v_\infty - v_2)(v_3 - v_1)(v_4 - v_0)}{\sqrt{5}\, u^3},$$
$$w_3 = \frac{(v_\infty - v_3)(v_4 - v_2)(v_0 - v_1)}{\sqrt{5}\, u^3},$$
$$w_4 = \frac{(v_\infty - v_4)(v_0 - v_3)(v_1 - v_2)}{\sqrt{5}\, u^3},$$

so that, if one makes use in the last row of the fact that by (18) the product of the six quantities v_z has the value u^6, there follow from (21) these permutations of the w belonging together:

$$(23)\qquad \begin{array}{cccccc} \omega, & w_0, & w_1, & w_2, & w_3, & w_4, \\ \omega+2, & -w_2, & -w_3, & -w_4, & -w_0, & -w_1, \\ -\dfrac{1}{\omega}, & w_0, & w_2, & w_1, & w_4, & w_3, \\ \dfrac{\omega-1}{\omega+1}, & -w_0, & -w_3, & -w_4, & -w_2, & -w_1. \end{array}$$

The functions w cannot become infinite for any value of u different from 0 and ∞. Let us therefore assume the equation whose roots are the five quantities w_z in the form:

$$w^5 + A_1 w^4 + A_2 w^3 + A_3 w^2 + A_4 w + A_5 = 0,$$

then, by the principles of § 73, $A_1^2, A_2, A_3^2, A_4, A_5^2$ are polynomials in

$$(24)\qquad u^{24} + \frac{2^{12}}{u^{24}}.$$

The expansion of (24) in ascending powers of q begins with q^{-1}, while the beginning of the expansion of w_z is

$$q^{-\frac{1}{10}} e^{-\frac{c\pi i}{60}}$$

The quantities A_1^2, A_2, A_3^3, A_4 can therefore not contain even the first power of (24), and are accordingly constant, while A_5^2 has the form:

$$A_5^2 = u^{24} + \frac{2^{12}}{u^{24}} + C,$$

in which C is a constant. The values of the constants can be determined if one knows the values of the w_z for $\omega = i$. But for $\omega = i$ we have [cf. above (6), (7)]:

$$\begin{aligned} u &= f(i) = \sqrt[4]{2}, \\ v_3 &= f\left(\frac{i+48}{5}\right) = f\left(\frac{i-2}{5}+10\right) = e^{-\frac{10\pi i}{24}} f\left(\frac{i-2}{5}\right), \\ &= e^{-\frac{10\pi i}{24}} f(i+2) = -i\sqrt[4]{2}, \\ v_2 &= i\sqrt[4]{2} \end{aligned}$$

and hence, by (18):

$$v_\infty = v_0 = \sqrt[4]{2}\,\frac{1+\sqrt{5}}{2},$$
$$v_1 = v_4 = \sqrt[4]{2}\,\frac{1+\sqrt{5}}{2}.$$

Consequently:

$$w_0 = 0, \quad w_1 = w_2 = i\sqrt{5}, \quad w_3 = w_4 = -i\sqrt{5}.$$

Accordingly we obtain first of all $C = -2^7$, and so

$$A_5^2 = \left(u^{12} - \frac{64}{u^{12}}\right)^2,$$

and from this, by comparison of the beginnings of the expansion,

$$A_5 = -u^{12} + \frac{64}{u^{12}},$$

and the equation for w takes the form:

$$w(w^2+5)^2 = u^{12} - \frac{64}{u^{12}}. \tag{25}$$

But one can give it yet another, very remarkable form.

By the relations that subsist between the functions $f(\omega), f_1(\omega), f_2(\omega)$ [§ 34, (11), (12)] we have:

$$\frac{f(\omega)^{24} - 64}{f(\omega)^{12}} = \frac{[f_1(\omega)^8 - f_2(\omega)^8]^2}{f(\omega)^4}.$$

If, therefore, for w we introduce the new unknown

$$y = \sqrt{w} = \frac{f_1(\omega)^8 - f_2(\omega)^8}{f(\omega)^2(w^2+5)} \tag{26}$$

then we obtain the form:

$$y^5 + 5y = \frac{f_1(\omega)^8 - f_2(\omega)^8}{f(\omega)^2}. \tag{27}$$

On these formulae rests the solution of the equation of the fifth degree by elliptic functions created by Hermite and Kronecker, to which we have referred in the 14th chapter of Band II.

By Bd. I, § 60 the general equation of the fifth degree can be reduced to the form:

$$z^5 + 5z = a, \tag{28}$$

and this equation becomes identical with (27) if one sets:

$$f_1(\omega)^8 - f_2(\omega)^8 = af^2(\omega).$$

If one adds to this the equations:

$$f_1^8 = f_2^8 = f^8, \quad ff_1f_2 = \sqrt{2},$$

one obtains

$$f^{24} - a^2f^{12} - 64 = 0. \tag{29}$$

By this quadratic equation f^{12} is determined as a function of a.

The roots of equation (28) are then represented by the square roots of the expressions (22). The double sign of these square roots is explained by the fact that two opposite values of a lead to the same equation (29). But the roots can also be determined uniquely by formula (26):

$$z = \frac{a}{w^2+5}. \tag{30}$$

In the paper of Kiepert mentioned above the general equation of the fifth degree is transformed into the form (15), for which only the solution of a quadratic equation is required. If one regards a, b, c therein as arbitrarily given, then the equations (16) serve for the determination of λ, μ, γ_3. This likewise is done with the help of a quadratic equation, though this is not to be seen at first glance. (Cf. F. Klein, Vorlesungen über das Ikosaëder, S. 191 f.)

We shall return further on to the formation of resolvents for the seventh transformation degree, when we have at our disposal the means which complex multiplication offers us.

SECOND BOOK.

QUADRATIC FIELDS.

Ninth Chapter.
Discriminant.

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§ 84. Definition of the Discriminants.

In the theory of quadratic fields there occur as discriminants certain rational integers (positive or negative) which, if they are even, are divisible by 4, and which, if they are odd, leave the residue 1 on division by 4.

Kronecker has called such numbers „numbers of discriminant form“. We shall here call them briefly discriminants, and shall not take offence at the fact that this word has still various other meanings in algebra. We therefore define:

1. An integer, positive or negative and different from zero, which satisfies one of the two congruences

$$D \equiv 0, \quad D \equiv 1 \pmod 4 \tag{1}$$

is called a discriminant.

By this definition every square number is a discriminant. But since these occupy a certain exceptional position, we shall call them improper discriminants whenever the distinction is necessary.

2. The product of two, and consequently of arbitrarily many, discriminants is again a discriminant.

3. A discriminant which contains (apart from 1) no square divisor after the separation of which a discriminant remains is called a fundamental discriminant.

In what follows, fundamental discriminants shall be denoted by Δ, to distinguish them from others. If D is not a fundamental discriminant, then there is one and only one square number Q^2 and one fundamental discriminant Δ such that

$$D = \Delta Q^2 \tag{2}$$

holds; Δ is then called the fundamental discriminant associated with D.

A fundamental discriminant is divisible by no odd square number. If, however, Δ is divisible by 4, then

$$\Delta \equiv 8,\ 12 \pmod{16} \tag{3}$$

must hold. For were $\Delta \equiv 0, 4 \pmod{16}$, then after cancellation of the factor 4 a discriminant would remain.

In order, then, to find the associated fundamental discriminant for a given D, one has first to separate off the greatest odd square number, and then in addition so high a power of 4 that Δ becomes either odd and $\equiv 1 \pmod 4$, or $\equiv 8$, or $\equiv 12 \pmod{16}$.

4. A proper discriminant which is not decomposable into factors that are themselves again discriminants is called a prime discriminant.

If p is a natural odd prime, and if the sign $\pm$ is so determined that $\pm p \equiv 1 \pmod 4$, then the prime discriminants are the following:

$$\pm p, \quad -4, \quad +8, \quad -8. \tag{4}$$

5. Every fundamental discriminant can be decomposed in one and only one way into prime discriminants.

For if one first separates off from Δ all the factors $\pm p$, there remains only one of the numbers 1, -4, $+8$, -8.

§ 85. The extended Legendre-Jacobi symbol.

The Legendre symbol $\left(\frac{m}{p}\right)$ has, when m is an arbitrary positive or negative number different from zero and p is a natural prime not dividing m, the value $+1$ if m is a quadratic residue, and the value -1 if m is a quadratic non-residue of p (Bd. I, § 145).

The meaning of the symbol has been extended by Jacobi in such a way that a composite number may also be put for p. If, however, m is a discriminant, then a still further reaching generalization is sometimes to be recommended, and this we must now set forth[19].

If D is a discriminant and n a rational integer, then we define a symbol (D, n) by the following stipulations:

$$(D, 0) = 0. \tag{1}$$

$$(D, 1) = 1. \tag{2}$$

$$\begin{aligned} (D, -1) &= +1, \quad D > 0. \\ &= -1, \quad D < 0. \end{aligned} \tag{3}$$

$$(D, p) = 0, \tag{4}$$

if p is a prime divisor of D.

$$(D, 2) = (-1)^{\frac{D^2-1}{8}}, \quad D \text{ odd}. \tag{5}$$

$$(D, p) = \left(\frac{D}{p}\right), \tag{6}$$

[19] Kronecker, Berliner Sitzungsberichte, 30. Juli 1885. Dirichlet-Dedekind, Vorlesungen über Zahlentheorie, 4. Aufl., § 186. H. Weber, Göttinger Nachrichten, Januar 1893.

if p is an odd prime that does not divide D.

If one decomposes $\pm n$ into its prime factors

$$n = \pm p,\ p',\ p'',\ \ldots,$$

then

$$(D, n) = (D, \pm 1)(D, p)(D, p')(D, p'')\ldots \tag{7}$$

By (1) to (7) the symbol (D, n) is evidently defined without contradiction for all numbers n, and can have only one of the values -1, 0, $+1$. It is always $= 0$ then and only then, when D and n are not relatively prime to one another. From (7) there follows further:

$$(D, n)(D, n') = (D, nn'), \tag{8}$$

and if Q^2 is a square number that has no common divisor with n,

$$(Q^2 D, n) = (D, n). \tag{9}$$

The formulae relating to the law of reciprocity take a particular form in this notation. We consider first the prime discriminants.

If $\pm p \equiv 1 \pmod 4$, then by (3), (5), (6):

$$(\pm p, -1) = (-1)^{\frac{p-1}{4}} = \left(\frac{-1}{p}\right),$$

$$(\pm p, 2) = (-1)^{\frac{p^2-1}{8}} = \left(\frac{2}{p}\right),$$

$$(\pm p, n) = \left(\frac{\pm p}{n}\right) = \left(\frac{n}{p}\right),$$

when n is an odd prime (by the law of reciprocity and its supplementary theorems [Bd. I, § 145, 9.]), and these formulae can be summarised in

$$(\pm p, n) = \left(\frac{n}{p}\right). \tag{10}$$

But by (7) this formula holds for every arbitrary n that is not divisible by p.

In the same way one finds, for an odd n:

$$\begin{aligned} (-4, n) &= (-1)^{\frac{n-1}{2}}, \\ (8, n) &= (-1)^{\frac{n^2-1}{8}}, \\ (-8, n) &= (-1)^{\frac{n-1}{2}+\frac{n^2-1}{8}}. \end{aligned} \tag{11}$$

Further, in general, if D, D' are two discriminants:

$$(D, n)(D', n) = (DD', n). \tag{12}$$

This formula holds in the first place when DD' and n are not relatively prime, because then both sides are $= 0$. It holds further for $n = -1$ [by (3)], for $n = 2$ [by (5) and the congruence $\frac{D^2-1}{8} + \frac{D'^2-1}{8} \equiv \frac{D^2 D'^2 - 1}{8} \pmod 2$] and for an odd prime n [by (6)]. Hence by (7) it holds in general.

We can now summarise the law of reciprocity together with its supplementary theorems for discriminants in the following way:

$$(D, D') = \pm(D', D), \tag{13}$$

in which the upper sign holds when at least one of the two discriminants D, D' is positive, the lower when both are negative.

In order to derive this formula from the usual form of the law of reciprocity, we assume first that the two formulae hold:

$$(D, D') = \pm(D', D),$$
$$(D_1, D') = \pm(D', D_1).$$

Then from (8) and (12) it follows that

$$(DD_1, D') = \pm(D', DD_1),$$

in which again the lower sign holds only when D' and DD_1 are both negative. By a repeated application there results

$$(DD_1, D'D_1') = \pm(D'D_1', DD_1)$$

and consequently formula (12) is proved if it holds for any two prime discriminants.

But by (10), if p and p' are odd primes:

$$(\pm p, \pm p') = \pm\left(\frac{p'}{p}\right), \quad (\pm p', \pm p) = \pm\left(\frac{p'}{p}\right),$$

in which the negative sign stands on the right only when both signs on the left are negative, that is, when p as well as $p' \equiv 3 \pmod 4$. In this case, and only in this case, $\left(\frac{p'}{p}\right) = -\left(\frac{p}{p'}\right)$, otherwise $\left(\frac{p'}{p}\right) = \left(\frac{p}{p'}\right)$. Thus formula (13) proves to be correct for this case.

Further, by (11) and (3), (5)

$$(-4, \pm p) = +1, \quad (\pm p, -4) = \pm 1,$$

$$(p, \pm 8) = (p, \pm 2) = (p, 2) = (-1)^{\frac{p^2-1}{8}},$$
$$(\pm 8, p) = \left(\frac{\pm 8}{p}\right) = (-1)^{\frac{p^2-1}{8}},$$
$$[p \equiv 1 \pmod 4],$$
$$(-p, \pm 8) = (-p, \pm 2) = \pm(-p, 2) = -1,$$
$$(\pm 8, -p) = \pm\left(\frac{\pm 8}{p}\right) = -1,$$
$$[p \equiv 3 \pmod 8],$$

from which formula (13) follows for these cases, and it is thereby proved in general.

From (13) it follows further, since every square number is a positive discriminant, that if m is relatively prime to D,

$$(D, m^2) = 1 \tag{14}$$

and consequently, by (8):

$$(D, m^2 n) = (D, n), \tag{15}$$

if m and D have no common divisor.

We have further the theorem:

$$(D, n) = (D, n'), \tag{16}$$

if $n \equiv n' \pmod{D}$.

The theorem is correct when n and D have a common divisor, and by (12) it is proved in general if it holds for every prime discriminant D. For these, however, it follows at once from (10), (11).

Let us assume that in (16) n, n' are themselves discriminants:

$$n = D_1, \quad n' = D_2, \quad D_1 \equiv D_2 \pmod{D},$$

then

$$(D, D_1) = (D, D_2)$$

and from (13) it follows that

$$(D, D_1) = \pm(D_1, D)$$
$$(D, D_2) = \pm(D_2, D),$$

in which the upper signs hold when D is positive, the lower ones when D, D_1, D_2 are all three negative, and different signs when D is negative and D_1, D_2 are of different signs. Accordingly, by (16):

$$(D_1, D) = \pm(D_2, D), \tag{17}$$

in which the lower sign holds only when the two numbers D and $D_1 D_2$ are both negative.

If D_1, D_2 are odd, we may set $D = 4m$, where m is an arbitrary integer, and we obtain from (17)

$$(D_1, m) = \pm(D_2, m), \tag{18}$$

provided that

$$D_1 \equiv D_2 \pmod{4m} \tag{19}$$

holds, the lower sign holding only when m and $D_1 D_2$ are both negative.

This theorem has so far been proved under the assumption that D_1 and D_2 are odd. But it holds in general. For the congruence (19) requires in the first place that D_1 and D_2 be at the same time even or at the same time odd. If they are both even, and m is then even, (18) is correct, because both sides vanish. But if m is odd and D_1, D_2 are even, then in (17) let us set $D = \pm m \equiv 1 \pmod{4}$, where m is a positive odd number. Then from $D_1 \equiv D_2 \pmod{D}$ the congruence (19) follows, and by (17) we have

$$(D_1, m) = (D_2, m),$$

$$\begin{aligned} (D_1, m) &= (D_2, m), \\ (D_1, -m) &= \pm(D_2, -m), \\ (D_1, -1) &= \pm(D_2, -1), \end{aligned} \tag{20}$$

and consequently again

$$(D_1, m) = (D_2, m).$$

This formula therefore holds for every positive odd m, and by (20) then (18) is proved for a negative m as well. The formulae (16) and (18) can be used for the calculation of the symbol (D, n)

according to the algorithm of the greatest common divisor. For from the congruences

$$\begin{aligned} D &\equiv D' \pmod{4n}, \\ n &\equiv n' \pmod{D'}, \\ D' &\equiv D'' \pmod{4n'}, \\ n' &\equiv n'' \pmod{D''}, \\ &\cdots\cdots\cdots\cdots \end{aligned}$$

one can determine the sequence of numbers

$$D', \quad 2n', \quad D'', \quad 2n'', \quad \dots$$

in such a way that each following one is smaller in absolute value than the preceding one, so long as none of the numbers n', n'', n''', ... has become equal to ± 1.

If D is not a square number, then a number β can always be determined so that

$$(D, \beta) = -1. \tag{21}$$

For if Δ is the fundamental discriminant associated with D and β is relatively prime to D, then (by 9)

$$(D, \beta) = (\Delta, \beta).$$

If $\Delta = \delta\Delta'$ and δ is a prime discriminant, and therefore relatively prime to Δ', then β_0 can be assumed in such a way that

$$(\delta, \beta_0) = -1$$

holds [by (10) and (11)], and β can be determined from the two congruences

$$\begin{aligned} \beta &\equiv \beta_0 \pmod{\delta}, \\ &\equiv 1 \pmod{\Delta'} \end{aligned}$$

Then one obtains, by (12),

$$(D, \beta) = (\delta, \beta_0) = -1.$$

If one lets s run through a complete system of residues with respect to the modulus D and sets

$$S = \overset{s}{\Sigma}(D, s),$$

then multiplication by (D, β) gives

$$-S = \overset{s}{\Sigma}(D, \beta s),$$

and since βs runs through a system of residues at the same time as s, it follows that

$$S = 0. \tag{22}$$

From this it follows:

In a complete system of numbers s incongruent and coprime to D there are just as many numbers α for which $(D, \alpha) = +1$ as there are numbers β for which $(D, \beta) = -1$, and we have

$$\begin{aligned} (D, n) &= +1, \quad \text{if } n \equiv \alpha \\ &= -1, \quad \text{"} \quad n \equiv \beta \pmod{D}. \end{aligned}$$

§ 86. The Gauss Sums.

In the first volume we became acquainted with the Gauss sums from the theory of the division of the circle. These expressions admit of generalisation, and through the use of the symbol (D, n) they take on a simple shape.

We had [Bd. I, § 179, (16)], when n denotes an odd prime, s a number not divisible by n, and k runs through a system of residues with respect to the modulus n:

$$\sum \left(\frac{k}{n}\right) e^{\frac{2\pi i s k}{n}} = \left(\frac{s}{n}\right) i^{\left(\frac{n-1}{2}\right)^2} \sqrt{n}. \tag{1}$$

If we put $\pm n = \Delta \equiv 1 \pmod 4$, then Δ is a prime discriminant, and by § 85, (10) we have

$$\left(\frac{k}{n}\right) = (\Delta, k), \quad \left(\frac{\mp s}{n}\right) = (\Delta, \mp s) = (\Delta, s)$$

[the latter by § 85, (3), since the upper signs hold for positive and the lower for negative Δ], further:

$$i^{\left(\frac{n-1}{2}\right)^2}\sqrt{n} = \sqrt{\Delta},$$

where $\sqrt{\Delta}$ is positive real or positive imaginary according as $n \equiv 1$ or $\equiv 3 \pmod 4$, and we can write (1) in the form:

$$\overset{k}{\Sigma}(\Delta, k)e^{-\frac{2\pi i k s}{\Delta}} = (\Delta, s)\sqrt{\Delta}, \tag{2}$$

provided that in the case where Δ is positive we replace s by $-s$. This formula has so far been proved only for an odd prime discriminant. But it holds also, as direct calculation shows, for

$$\Delta = -4, \quad 8, \quad -8,$$

and therefore for every prime discriminant.

If we assume the correctness of formula (2) for $\Delta = \Delta'$ and for $\Delta = \Delta''$, then, provided Δ' and Δ'' have no common divisor, its correctness follows for $\Delta = \Delta'\Delta''$.

For we have

$$\sqrt{\Delta'}\sqrt{\Delta''} = \pm\sqrt{\Delta'\Delta''}, \tag{3}$$

when the upper sign holds, provided that at least one of the two fundamental discriminants Δ', Δ'' is positive, and the lower one when both are negative.

If we set

$$k = k'\Delta'' + k''\Delta',$$

and let k', k'' run through systems of residues with respect to Δ', Δ'', then k runs through a system of residues with respect to $\Delta = \Delta'\Delta''$.

The multiplication of the two sums

$$\overset{k'}{\Sigma}(\Delta', k')\, e^{-\frac{2s\pi i k'}{\Delta'}}, \quad \overset{k''}{\Sigma}(\Delta'', k'')\, e^{-\frac{2s\pi i k''}{\Delta''}} \tag{4}$$

gives

$$\overset{k'k''}{\Sigma}(\Delta', k')(\Delta'', k'')\, e^{-\frac{2s\pi i k}{\Delta}}. \tag{5}$$

But by § 85, (16) we have

$$(\Delta', k) = (\Delta', \Delta'')(\Delta', k'),$$
$$(\Delta'', k) = (\Delta'', \Delta')(\Delta'', k''),$$

and consequently, by § 85, (13),

$$(\Delta, k) = \pm(\Delta', k')(\Delta'', k''),$$

and accordingly the general validity of formula (2) for every fundamental discriminant Δ follows from (3), (4) and (5).

Tenth Chapter.

Algebraic Numbers and Forms.

§87. Ideals and Forms in Algebraic Fields.

The interest which the theory of elliptic functions has for the algebraist springs from its application to the theory of quadratic irrational numbers, to which these functions stand in a relation analogous to that of the roots of unity to the rational numbers. They offer us the first and hitherto only example, more exactly known in its laws, of a region of algebraic numbers going beyond the cyclotomic numbers. In order to be able to present the theory of these numbers more fully, we must first premise some theorems from the theory of forms and of algebraic fields, and especially of quadratic forms.

In Bd. II, § 163, 169 a connection was found between the ideals of an algebraic field Ω of the nth degree and certain homogeneous functions of the nth degree of n variables. Before we apply this to quadratic fields, let the general theorems be briefly recalled.

Let $\mathfrak{a}$ be an ideal of a field Ω and let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be a basis of this ideal. Further, let $\omega_1, \omega_2, \ldots, \omega_n$ be a minimal basis of Ω. Then

$$(\alpha_1, \alpha_2, \ldots, \alpha_n) = (A)(\omega_1, \omega_2, \ldots, \omega_n), \tag{1}$$

if (A) denotes a linear substitution with integral coefficients. In order to pass over to another basis $\alpha'_1, \alpha'_2, \ldots, \alpha'_n$ of $\mathfrak{a}$, one makes a linear substitution L with the determinant ± 1,

$$(\alpha'_1, \alpha'_2, \ldots, \alpha'_n) = (L)(\alpha_1, \alpha_2, \ldots, \alpha_n), \tag{2}$$

from which there results

$$(\alpha'_1, \alpha'_2, \ldots, \alpha'_n) = (L)(A)(\omega_1, \omega, \ldots, \omega_n)^{1)}. \tag{3}$$

---

[1]) On linear substitutions and their composition cf. the sixth chapter of the second volume.

By the norm $N(\mathfrak{a})$ of the ideal $\mathfrak{a}$ one understands the absolute value of the determinant A of the substitution (A):

$$A = \Sigma \pm a_{1,1}\, a_{2,2} \ldots a_{n,n},$$

and by (3) the norm is independent of the choice of the basis.

1. We shall call the basis (α_ν) positive if the determinant A is positive, and so equal to the norm of $\mathfrak{a}$. If (α_ν) is positive, then (α'_ν) is always positive when the substitution determinant $L = +1$.

From every basis one can derive a positive basis by a substitution with determinant -1, for example by an interchange of two of the α_ν, and in what follows we shall for the most part use only positive bases.

The basis determined in Bd. II, § 163, (3) is positive.

If now $t_1, t_2, \ldots, t_n$ denotes a system of independent variables and

$$\lambda = \sum_\nu \alpha_\nu t_\nu \tag{4}$$

is a basis form with positive basis, then there results (Bd. II, § 164)

$$N(\lambda) = N(\mathfrak{a})\, T, \tag{5}$$

in which

$$T = \Sigma \pm t_{1,1}\, t_{2,2} \ldots t_{n,n} \tag{6}$$

is a primitive homogeneous form with integral coefficients, of degree n, in the variables t_ν. The $t_{r,s}$ are linear functions of the t_ν.

If the substitution (2) is applied to (4), there results

$$\lambda = \lambda' = \sum \alpha'_\nu t'_\nu, \tag{7}$$

if

$$(t_1, t_2, \ldots, t_n) = L'(t'_1, t'_2, \ldots, t'_n), \tag{8}$$

in which L' is the transposed substitution of L; and if (α'_ν) is likewise a positive basis, then L' has the determinant $+1$, and formula (5) gives

$$N(\lambda) = N(\mathfrak{a})\, T', \tag{9}$$

in which T' is a homogeneous function of degree n in the variables t' which arises from T by the substitution (8).

Two forms which arise from one another by a linear substitution with integral coefficients of determinant $+1$ are called equivalent. Sometimes the forms are also called equivalent when the substitution determinant is -1, but then with the addition "improperly". It is then proved by (8):

2. If we call the form T the form belonging to the basis (α_ν) of $\mathfrak{a}$, then the forms belonging to different positive bases of the same ideal are equivalent.

Let us denote the conjugate values of α_s, ω_s by

$$\alpha_{s,1},\ \alpha_{s,2},\ \ldots,\ \alpha_{s,n};\quad \omega_{s,1},\ \omega_{s,2},\ \ldots,\ \omega_{s,n}$$

and set

$$\alpha_{s,r} = \overset{\nu}{\Sigma}\, a_{s,\nu}\, \omega_{\nu,r}, \tag{10}$$

then there results

$$(\Sigma \pm \alpha_{1,1}\, \alpha_{2,2}\, \ldots\, \alpha_{n,n})^2 = N(\mathfrak{a})^2 \Delta, \tag{11}$$

where Δ is the fundamental number of the field Ω, namely the determinant square:

$$\Delta = (\Sigma \pm \omega_{1,1}\, \omega_{2,2}\, \ldots\, \omega_{n,n})^2$$

(Bd. II, § 162).

If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the conjugate values of λ, then by (4)

$$\lambda_r = \overset{\nu}{\Sigma}\, \alpha_{\nu,r}\, t_\nu \tag{12}$$

is a linear substitution for the variables t_ν with the substitution determinant

$$r = N(\mathfrak{a})\, \sqrt{\Delta}, \tag{13}$$

and through this the form T passes, by (5), into the product

$$T' = [N(\mathfrak{a})]^{-1}\,\lambda_1\,\lambda_2\,\ldots\,\lambda_n \tag{14}$$

since $N(\lambda)$ is the product of the conjugate values of λ (Bd. II, § 151). We can apply invariant theory (Bd. I, § 65) to this, and if we form the Hesse determinant:

$$H = \Sigma \pm \frac{\partial^2 T}{\partial t_1 \partial t_1}\,\frac{\partial^2 T}{\partial t_2 \partial t_2}\cdots\frac{\partial^2 T}{\partial t_n \partial t_n}, \tag{15}$$

then

$$H' = r^{-2}\,H. \tag{16}$$

If, however, we form H' by (14) in the variables λ_ν, and observe that an n-rowed determinant in which the diagonal terms are $= 0$ and all the other terms are $= 1$ has the value $(-1)^{n-1}(n-1)$[20], then it follows:

$$\begin{aligned} H' &= (-1)^{n-1}(n-1)\,N(\mathfrak{a})^{-n}(\lambda_1\lambda_2\ldots\lambda_n)^{n-2}, \\ &= (-1)^{n-1}(n-1)\,N(\mathfrak{a})^{-2}\,T^{n-2} \end{aligned}$$

and consequently, by (13) and (16),

$$H = (-1)^{n-1}(n-1)\,T^{n-2}\Delta. \tag{17}$$

§ 88. Ideal Classes and Classes of Forms.

In Bd. II, § 170 we explained the equivalence of ideals and the ideal classes. According to this, two ideals $\mathfrak{a}$, $\mathfrak{b}$ were equivalent. If there is an integral or fractional number of the field Ω which is the quotient of $\mathfrak{b}$ and $\mathfrak{a}$, that is:

$$\eta\,\mathfrak{a} = \mathfrak{b}, \tag{1}$$

and it has been shown that the ideals are thereby divided into classes, and that the number of these classes is finite. In some cases it is expedient to take the concept of equivalence somewhat more narrowly and to call $\mathfrak{a}$ and $\mathfrak{b}$ equivalent only if there is a number η with positive norm that satisfies condition (1). This difference naturally does not come into consideration when the norms of all numbers are positive, as for example in the imaginary quadratic field. Neither does it come into consideration when there are units with the norm -1, because, if ε is such a unit, either $N(\eta)$ or $N(\varepsilon\eta)$ is positive, and ε and $\varepsilon\eta$ satisfy condition (1) at the same time.

But if there are no units whose norm $= -1$, while there are other numbers in Ω with negative norm, then under the narrower definition of equivalence every ideal class divides once again into two classes. We wish to hold here to the narrower concept of equivalence, which comes into force, for example, for some real quadratic fields[21].

In (1) η may be fractional. But if α is an integer divisible by $\mathfrak{a}$, then $\eta\alpha$ is an integer divisible by $\mathfrak{b}$, and from this it follows: If

$$(\alpha_1,\ \alpha_2,\ \ldots,\ \alpha_n) \tag{2}$$

[20] One can easily prove this theorem by extending the determinant through the addition of an $(n+1)$th row and an $(n+1)$th column, in which the added row contains only ones, and the column contains zeros with the exception of the last element.

[21] Namely, when the Pell equation $t^2 - D\,u^2 = -4$ has no solution.

is a basis of $\mathfrak{a}$, then

$$(\beta_1,\ \beta_2,\ \ldots,\ \beta_n) = (\alpha_1\eta,\ \alpha_2\eta,\ \ldots,\ \alpha_n\eta)$$

is a basis of $\mathfrak{b}$, and if the first basis is positive, then (under the narrower concept of equivalence) so is the second.

For the necessary and sufficient condition that (2) be a basis of $\mathfrak{a}$ consists in this, that every integer α divisible by $\mathfrak{a}$, and only such, is contained in the form:

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \cdots + x_n\alpha_n \tag{3}$$

with integer $x_1, x_2, \ldots, x_n$. Consequently all numbers

$$\beta = x_1\alpha_1\eta + x_2\alpha_2\eta + \cdots + x_n\alpha_n\eta \tag{4}$$

are divisible by $\mathfrak{b}$. If, conversely, β is an integer divisible by $\mathfrak{b}$, then β/η is divisible by $\mathfrak{a}$ and consequently representable in the form (3). Hence β is representable by (4).

Further, in the notation of (9) and (10), § 87, if α is a positive basis:

$$\begin{aligned}
\Sigma \pm \alpha_{1,1}\alpha_{2,2}\ldots\alpha_{n,n} &= N(\mathfrak{a})\,\Sigma \pm \omega_{1,1}\omega_{2,2}\ldots\omega_{n,n},\\
\Sigma \pm \beta_{1,1}\beta_{2,2}\ldots\beta_{n,n} &= N(\eta)\,\Sigma \pm \alpha_{1,1}\alpha_{2,2}\ldots\alpha_{n,n},\\
&= N(\eta)N(\mathfrak{a})\,\Sigma \pm \omega_{1,1}\omega_{2,2}\ldots\omega_{n,n},\\
&= \pm N(\mathfrak{b})\,\Sigma \pm \omega_{1,1}\omega_{2,2}\ldots\omega_{n,n},
\end{aligned}$$

and since by (1)

$$N(\mathfrak{b}) = N(\eta)N(\mathfrak{a}) \tag{5}$$

holds, the positive sign must stand in $\pm N(\mathfrak{b})$.

If λ is a basis form of $\mathfrak{a}$, then $\eta\lambda$ is a basis form of $\mathfrak{b}$, and

$$\begin{aligned}
N(\lambda) &= N(\mathfrak{a})T,\\
N(\eta\lambda) &= N(\mathfrak{b})T,
\end{aligned} \tag{6}$$

and thus the same form T belongs to the two ideals $\mathfrak{a}$, $\mathfrak{b}$. If we unite equivalent forms T into one class, then by 2. we can say:

3. To every ideal class there belongs a definite class of forms.

In answering the question whether different ideal classes can belong to one and the same class of forms, we make the assumption that Ω is a normal field: let us assume, then, that equivalent forms T and T' belong to two ideals $\mathfrak{a}$ and $\mathfrak{a}'$; then, if λ is a basis form of $\mathfrak{a}$,

$$N(\lambda) = N(\mathfrak{a})T, \tag{7}$$

and if we carry T' over into T by a linear substitution, and understand by λ' a basis form of $\mathfrak{a}'$ (with the same variables t as λ), then

$$N(\lambda') = N(\mathfrak{a}')T. \tag{8}$$

The linear factors of $N(\lambda)$ and $N(\lambda')$ must therefore agree with one another, apart from a constant factor, because a polynomial in arbitrary variables can be decomposed into irreducible (here linear) factors in only one way (Bd. I, § 20, Bd. II, § 152); and since all these factors belong to the same field Ω, there is among the conjugate factors λ' one that satisfies the condition:

$$\lambda = \eta\lambda',$$

in which η is a number in Ω. The functionals λ and λ', and accordingly the corresponding ideals, are therefore equivalent.

4. If in a normal field the same class of forms belongs to the two ideals $\mathfrak{a}$ and $\mathfrak{a}'$, then among the ideals conjugate to $\mathfrak{a}'$ there is one that is equivalent to $\mathfrak{a}$.

§ 89. Composition of forms and multiplication of ideals.

Let $\mathfrak{a}$ and $\mathfrak{b}$ now be two ideals in Ω, and let

$$\mathfrak{c} = \mathfrak{a}\mathfrak{b} \tag{1}$$

be the product formed from them both. Let further

$$\begin{aligned} \mathfrak{a} &= (\alpha_1,\ \alpha_2,\ \ldots,\ \alpha_n), \\ \mathfrak{b} &= (\beta_1,\ \beta_2,\ \ldots,\ \beta_n), \\ \mathfrak{c} &= (\gamma_1,\ \gamma_2,\ \ldots,\ \gamma_n) \end{aligned} \tag{2}$$

be bases of these ideals. Since every number $\alpha_h\beta_k$ belongs to $\mathfrak{c}$, there is a system of rational integers $c_{hk}^{(r)}$ such that

$$\alpha_h\beta_k = \sum^{r} c_{h,k}^{(r)}\,\gamma_r \tag{3}$$

holds; and since every number in $\mathfrak{c}$ arises by the multiplication of one number from $\mathfrak{a}$ and one from $\mathfrak{b}$ and the addition of these products (Bd. II, § 169), conversely also

$$\gamma_s = \sum^{h,k} a_{h,k}^{(s)}\,\alpha_h\beta_k, \tag{4}$$

in which the $a_{h,k}^{(s)}$ are likewise rational integers.

If one substitutes (3) into (4), it follows that

$$\gamma_s = \sum^{r}\gamma_r \sum^{h,k} a_{h,k}^{(s)}\,c_{h,k}^{(r)}, \tag{5}$$

and from this

$$\sum^{h,k} a_{h,k}^{(s)}\,c_{h,k}^{(r)} = (r,s), \tag{6}$$

in which

$$(r,r) = 1, \quad (r,s) = 0, \quad r \neq s.$$

If, then, we denote variables by x_r and set

$$\sum^{r} x_r\,c_{h,k}^{(r)} = y_{h,k}, \tag{7}$$

it follows that:

$$x_s = \sum^{h,k} a_{h,k}^{(s)}\,y_{h,k}. \tag{8}$$

Now let u_r, v_r, t_r denote three systems of variables, and let

$$\begin{aligned} \mu &= \Sigma\,\alpha_r\,u_r, \\ \nu &= \Sigma\,\beta_r\,v_r, \\ \lambda &= \Sigma\,\gamma_r\,t_r \end{aligned} \tag{9}$$

be basis forms of $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$, and further let U, V, T be the forms belonging to $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$, written in the variables u, v, t. Then

$$\begin{aligned} N(\mu) &= N(\mathfrak{a})\,U, \\ N(\nu) &= N(\mathfrak{b})\,V, \\ N(\lambda) &= N(\mathfrak{c})\,T. \end{aligned} \tag{10}$$

From (3) there follows therefore, by multiplication with $u_h v_k$ and summation according to (9):

$$\mu\nu = \overset{r}{\Sigma}\gamma_r \overset{h,k}{\Sigma} c_{h,k}^{(r)}\, u_h v_k, \tag{11}$$

hence, if one sets

$$t_r = \overset{h,k}{\Sigma} c_{h,k}^{(r)}\, u_h v_k \tag{12}$$

then:

$$\mu\nu = \lambda, \tag{13}$$

and from this, since $N(\mathfrak{a})N(\mathfrak{b}) = N(\mathfrak{c})$, there follows by (10)

$$T = UV. \tag{14}$$

In this, however, the t_r are no longer independent variables; rather, they arise from u_h, v_k through the bilinear substitution (12).

5. The form T passes, through the bilinear substitution (12), into the product of the two forms U, V.

But this substitution has still one essential peculiarity. We call the functions t_r determined by (12) linearly independent with respect to a prime modulus p if from the congruence

$$\Sigma\, x_r t_r \equiv 0 \pmod{p}, \tag{15}$$

in which the x_r are rational integers, it follows that all these numbers are divisible by p. If this holds for every arbitrary prime p, the t_r are called simply linearly independent.

By (12) and (7) the congruence (15) is equivalent to the n^2 congruences

$$y_{h,k} = \overset{r}{\Sigma}\, x_r\, c_{h,k}^{(r)} \equiv 0 \pmod{p}$$

and from (6) there then follows:

$$x_s \equiv 0 \pmod{p},$$

and this says that the substitutions (17) are linearly independent for every modulus p.

According to Gauss (Disq. aritm. art. 235) a form T is said to be compounded or composed of the two forms U, V when T passes over, by a bilinear substitution for the variable t whose equations are linearly independent for every modulus, into the product UV.

Accordingly we have the theorem:

6. The form that belongs to the product of two ideals $\mathfrak{a}, \mathfrak{b}$ is compounded from the forms of the ideals $\mathfrak{a}$ and $\mathfrak{b}$[22].

Eleventh Chapter.

Ideals in Quadratic Fields.

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[22]The theorems that Gauss proves at the passage mentioned by a very lengthy calculation for binary quadratic forms are here derived in the greatest generality by simple considerations. Cf. Dirichlet-Dedekind, Vorlesungen über Zahlentheorie, §182. For binary quadratic forms: Dedekind, Crelles Journal, Bd. 129; H. Weber, Göttinger Nachrichten 1907.

§ 90. Discriminant of the Quadratic Field.

A quadratic field arises when a square root $\sqrt{d}$ is adjoined to the field of rational numbers; here d may be assumed to be an integer without a square divisor. The field is real or imaginary according as d is positive or negative. If we denote the field of rational numbers, the absolute domain of rationality, by $\mathfrak{R}$, and by $\mathfrak{R}(x, x', \dots)$ the field which arises from $\mathfrak{R}$ by the adjunction of $x, x', \dots$, then the quadratic field Ω is to be denoted thus:

$$\Omega = \mathfrak{R}(\sqrt{d}). \tag{1}$$

Every number of the field Ω can be put into the form:

$$\omega = \frac{x + y\sqrt{d}}{2}, \tag{2}$$

in which x, y are rational integral or fractional numbers.

The number conjugate to ω

$$\omega' = \frac{x - y\sqrt{d}}{2} \tag{3}$$

is likewise contained in the field Ω, and consequently Ω is a normal field.

In order that ω be an integer, it is necessary that

$$\omega + \omega' = x, \quad (\omega - \omega')^2 = y^2 d$$

be rational integers, and since d is to have no square divisor, x and y must be integers. But in order that ω should really be an integer, the norm $\frac{1}{4}(x^2 - y^2 d)$ must also be a rational integer, i. e.

$$x^2 - y^2 d \equiv 0 \pmod 4 \tag{4}$$

must hold. If $d \equiv 2$ or $\equiv 3 \pmod 4$, this condition can only be satisfied if x and y are even numbers, and hence, if we replace x, y by $2x, 2y$, it follows:

1. If $d \equiv 2, 3 \pmod 4$, then all and only those numbers of the field Ω are integral which are contained in the form

$$x + y\sqrt{d}$$

with rational integers x, y.

But if $d \equiv 1 \pmod 4$, then (3) is satisfied when x and y are both even or both odd, that is, when x has the form $2x_1 - y$. If one then again replaces x_1 by x, it follows:

2. If $d \equiv 1 \pmod 4$, then all and only those numbers in Ω are integral which are contained in the form

$$x + y\,\frac{-1 + \sqrt{d}}{2}$$

with rational integers x, y.

Thus in case 1. $(1, \sqrt{d})$, in case 2. $\left(1, \dfrac{-1 + \sqrt{d}}{2}\right)$ is a minimal basis of the field Ω. The fundamental number or discriminant of the field Ω is therefore, in case 1.:

$$\Delta = \begin{vmatrix} 1, & \sqrt{d} \\ 1, & -\sqrt{d} \end{vmatrix}^2 = 4d, \tag{5}$$

in case 2.:

$$\Delta = \begin{vmatrix} 1, & \dfrac{-1 + \sqrt{d}}{2} \\ 1, & \dfrac{-1 - \sqrt{d}}{2} \end{vmatrix}^2 = d. \tag{6}$$

In both cases one can therefore put the integer ω into the form:

$$\omega = \frac{x + y\sqrt{\Delta}}{2}, \tag{7}$$

with the condition that

$$4N(\omega) = x^2 - \Delta y^2 \equiv 0 \pmod 4 \tag{8}$$

shall hold.

The two forms of the minimal basis which we have obtained, namely $(1, \sqrt{d})$ and $\left(1, \frac{-1+\sqrt{d}}{2}\right)$, are therefore:

$$\begin{aligned} (1,0) &= \left(1, \tfrac{1}{2}\sqrt{\Delta}\right), && \Delta \equiv 0, \\ &= \left(1, \frac{-1+\sqrt{\Delta}}{2}\right), && \Delta \equiv 1. \end{aligned} \pmod 4. \tag{9}$$

Here the sign of $\sqrt{\Delta}$ may be fixed in any manner whatever, but it must then be kept fixed throughout the same discussion. We will assume once and for all:

3. If Δ is positive, $\sqrt{\Delta}$ shall likewise be positive, and if Δ is negative, $-i\sqrt{\Delta}$ shall be positive, or $\sqrt{\Delta}$ positive imaginary.

From (1), (2) there follows further:

4. The fundamental number of a quadratic field is always $\equiv 0$ or $\equiv 1$ with respect to the modulus 4, and has, apart from 4, no quadratic divisor. It is therefore a discriminant, and indeed a fundamental discriminant.

§ 91. Ideals and Forms in Quadratic Fields.

If $\mathfrak{a}$ is an ideal of the quadratic field Ω, α_1, α_2 a positive basis, and

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 \tag{1}$$

a basis form of $\mathfrak{a}$, then

$$N(\lambda) = (\alpha_1 t_1 + \alpha_2 t_2)(\alpha_1' t_1 + \alpha_2' t_2) = N(\mathfrak{a})\, T, \tag{2}$$

and $N(\mathfrak{a})$ is the greatest common divisor of the three integral rational numbers

$$\alpha_1\alpha_1', \quad \alpha_1\alpha_2' + \alpha_2\alpha_1', \quad \alpha_2\alpha_2'.$$

If we therefore set

$$\begin{aligned} \alpha_1\alpha_1' &= a\, N(\mathfrak{a}), \\ \alpha_1\alpha_2' + \alpha_2\alpha_1' &= b\, N(\mathfrak{a}), \\ \alpha_2\alpha_2' &= c\, N(\mathfrak{a}), \end{aligned} \tag{3}$$

then

$$T = at_1^2 + bt_1t_2 + ct_2^2 \tag{4}$$

is the form belonging to the basis α_1, α_2. We denote it, when the notation for the variables is of no consequence, by

$$\varphi = (a, b, c). \tag{5}$$

It is a primitive quadratic form whose discriminant

$$\Delta = b^2 - 4ac \tag{6}$$

is equal to the fundamental number of the field Ω (§ 87). In order to form the basis α_1, α_2 according to Bd. II, § 163, one seeks first the least natural number a_{11} divisible by $\mathfrak{a}$, and then the least natural number a_{12} for which a rational number a_{12} satisfying the condition

$$a_{12} + a_{22}\theta \equiv 0 \pmod{\mathfrak{a}} \tag{7}$$

can be determined. Since the congruence (7) can be satisfied for $a_{22} = a_{11}$, namely by $a_{12} = 0$, then it follows that a_{11} is divisible by a_{22}, and we set $a_{11} = a_{22}a$. Then we have the basis of $\mathfrak{a}$:

$$\begin{aligned} \alpha_1 &= a_{22}a, \\ \alpha_2 &= a_{12} + a_{22}\theta. \end{aligned} \tag{8}$$

Then, by Bd. II, § 164, (12):

$$N(\mathfrak{a}) = a_{22}^2\, a, \tag{9}$$

and if for $(1, \theta)$ we take the basis § 90, (9), then

$$\theta + \theta' = 0 \text{ or } = -1,$$

$$\theta\theta' = -\tfrac{1}{4}\Delta \text{ or } = \frac{1-\Delta}{4}. \tag{10}$$

According as $\Delta \equiv 0$, or $\equiv 1 \pmod 4$, and

$$(\theta - \theta')^2 = \Delta.$$

In these two cases there then results, when one substitutes (8) in (3) and takes account of (9):

$$a_{22}b = 2a_{12} \text{ or } = 2a_{12} - a_{22},$$

and consequently in both cases:

$$\alpha_1 = a_{22}a, \quad \alpha_2 = a_{22}\,\frac{b+\sqrt{\Delta}}{2}, \tag{11}$$

and the form

$$T = (a,\, b,\, c)$$

here has a positive first coefficient.

If we set

$$\omega = -\frac{\alpha_2}{\alpha_1} = -\frac{b+\sqrt{\Delta}}{2a}, \tag{12}$$

then ω is a root of the quadratic equation:

$$a\omega^2 + b\omega + c = 0 \tag{13}$$

and

$$\lambda = a_{22}a(x - \omega y) \tag{14}$$

is a functional corresponding to the ideal $\mathfrak{a}$, and at the same time a basis form of $\mathfrak{a}$. a is the least positive integer for which $a\omega$ becomes an integer.

The necessary and sufficient condition for two integers α_1, α_2 of the field Ω to form the basis of an ideal consists in this, that first of all α_1, α_2 is a basis of the field Ω, and that, if ω_1, ω_2 is a minimal basis of this field,

$$\alpha_1\omega_1, \quad \alpha_2\omega_1, \quad \alpha_1\omega_2, \quad \alpha_2\omega_2$$

by the linear form

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2$$

are representable with integral t_1, t_2 (Bd. II, § 163, 164). For then, if α is representable by λ, every product $\omega\alpha$ is also representable by λ. In our case this condition reduces to the requirement that $\alpha_1\theta, \alpha_2\theta$ must be representable by λ. But if (a, b, c) is any quadratic form of the discriminant Δ with positive first coefficient a, then we may also take $\frac{1}{2}(b + \sqrt{\Delta})$ for θ, and since

$$\left(\frac{b+\sqrt{\Delta}}{2}\right)^2 = -ac + b\,\frac{b+\sqrt{\Delta}}{2}$$

holds, (11) is always the basis of an ideal in Ω. Here the positive number a_{22} may be assumed arbitrarily. If we set $a_{22} = 1$, then

$$\lambda = a(x - \omega y) \tag{15}$$

is the basis form of an ideal $\mathfrak{a}$ completely determined by (a, b, c), whose norm is equal to a, and a basis of this ideal is

$$(a,\ a\omega). \tag{16}$$

§ 92. Prime Ideals in the Quadratic Field.

A prime p is in the quadratic field Ω either still indecomposable itself, and is then a prime ideal of the second degree, or p splits into two prime ideals of the first degree $\mathfrak{p}, \mathfrak{p}'$. We have accordingly two cases to distinguish:

$$\text{a)} \quad p = \mathfrak{p} \text{ in } \Omega \text{ indecomposable:} \quad N(\mathfrak{p}) = p^2,$$

$$\text{b)} \quad p = \mathfrak{p}\mathfrak{p}' \qquad\qquad N(\mathfrak{p}) = N(\mathfrak{p}') = p,$$

and in the latter case the two prime ideals $\mathfrak{p}, \mathfrak{p}'$ may be either different from one another or else equal, so that we can distinguish yet a third case:

$$\text{c)} \quad p = \mathfrak{p}^2, \quad N(\mathfrak{p}) = p.$$

It is now a question of ascertaining the conditions under which the one or the other of these cases occurs.

If $N(\mathfrak{p}) = p$, then p is the number of incongruent numbers (mod $\mathfrak{p}$) in Ω, and consequently in this case every number in Ω is congruent to one of the rational numbers r:

$$0,\ 1,\ 2,\ \ldots,\ p-1 \tag{1}$$

If on the other hand $N(\mathfrak{p}) = p^2$, then there are also numbers which, with respect to $\mathfrak{p}$, are not congruent to any rational number.

If, however, $(1, \theta)$ is a basis of Ω and if, in order to comprise both cases of § 90, (9),

$$\theta = \frac{-\Delta + \sqrt{\Delta}}{2} \tag{2}$$

is assumed, then in case a)

$$\theta \text{ not congruent with a number } r,$$

in case b) or c)

$$\theta \equiv r \pmod{\mathfrak{p}},$$

so that, if one sets $-x = 2r + \Delta$, it follows:

1. The necessary and sufficient condition for the occurrence of one of the cases b), c) is this, that there is a rational integer which is at once even or odd with Δ, and for which

$$\frac{x + \sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}}. \tag{3}$$

If $\mathfrak{p}'$ is the ideal conjugate to $\mathfrak{p}$, then

$$\frac{-x + \sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}'}, \tag{4}$$

and from this it follows

$$x^2 \equiv \Delta \pmod{4p}. \tag{5}$$

If conversely the congruence (5) is satisfied by $x = r$, then

$$\frac{r + \sqrt{\Delta}}{2} \cdot \frac{-r + \sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}},$$

and consequently either $x = r$ or $x = -r$ satisfies condition (3).

The two ideals $\mathfrak{p}, \mathfrak{p}'$ will be identical with one another when the two numbers (3) and (4), and consequently their sum $\sqrt{\Delta}$ as well, are divisible by $\mathfrak{p}$, and hence Δ by p. Then $x \equiv 0$ or $x \equiv p \pmod{2p}$ is the only root of (5).

We thus have the result:

2. The prime factor $\mathfrak{p}$ of the natural prime p is of the second degree when the congruence (5) has no solution, of the first degree when (5) has a solution, and p is the square of $\mathfrak{p}$ when (5) has only one solution, that is, when p divides Δ.

The latter is in agreement with the general theorem Bd. II, § 174.

With the use of the symbol (Δ, p), which we introduced in § 85, we can also give these theorems the form:

3. If $(\Delta, p) = -1$, then p is indecomposable in Ω.
„ $(\Delta, p) = +1$, then p splits into two different conjugate prime ideals.
„ $(\Delta, p) = 0$, then p is the square of a prime ideal.

§ 93. Representation of Numbers as Ideal Norms.

A prime p is the norm of an ideal only when $(\Delta, p) = +1$ or $= 0$, but not when $(\Delta, p) = -1$. In the last case it is only p^2 that is a norm, namely that of p. In general a positive rational integer m can be represented as a norm in more than one way. We will denote the number of these representations for the moment by $\psi(m)$ and determine it more closely.

We first take m to be a prime power p^k. If then $(\Delta, p) = +1$ and $p = \mathfrak{p}\mathfrak{p}'$, one can represent p^k in the following way:

$$p^k = N(\mathfrak{p}^k),\ N(\mathfrak{p}^{k-1}\mathfrak{p}'),\ N(\mathfrak{p}^{k-2}\mathfrak{p}'^2),\ \ldots,\ N(\mathfrak{p}'^k), \tag{1}$$

and therefore

$$(\Delta, p) = +1: \quad \psi(p^k) = k + 1. \tag{2}$$

But if $(\Delta, p) = -1$, then, when k is even, $p^k = N(p^{1/2\,k})$, and when k is odd, p is not representable as a norm. Thus

$$(\Delta, p) = -1: \quad \psi(p^k) = 0 \text{ or } = 1, \tag{3}$$

according as k is odd or even.

If finally $(\Delta, p) = 0$, so that p is contained in Δ, then $p = \mathfrak{p}^2$ and $p^k = N(\mathfrak{p}^k)$; hence:

$$(\Delta, p) = 0: \quad \psi(p^k) = 1. \tag{4}$$

These three cases can be combined into a single formula: for the number p^k has the following $k+1$ divisors:

$$1,\ p,\ p^2,\ \ldots,\ p^k. \tag{5}$$

And therefore the formulae (2), (3), (4) give, when $m = p^k$, and n runs through the divisors of m,

$$\psi(m) = \overset{n}{\Sigma}(\Delta, n). \tag{6}$$

This expression, however, holds generally. For if m_1 and m_2 are relatively prime and if

$$m_1 = N(\mathfrak{m}_1), \quad m_2 = N(\mathfrak{m}_2),$$

then

$$m = m_1 m_2 = N(\mathfrak{m}_1 \mathfrak{m}_2).$$

From a representation of m_1, m_2 as norms one can therefore derive a representation of m as a norm.

If conversely

$$m_1 m_2 = N(\mathfrak{m}) = \mathfrak{m}\,\mathfrak{m}',$$

where $\mathfrak{m}$, $\mathfrak{m}'$ are conjugate ideals, then, if m_1 and m are divisible by a prime ideal $\mathfrak{p}$, m_1 must also be divisible by $\mathfrak{p}\,\mathfrak{p}'$, and hence $m_1 = N(\mathfrak{m}_1)$ as well, and likewise $m_2 = N(\mathfrak{m}_2)$.

Accordingly, if m_1 and m_2 are relatively prime:

$$\psi(m_1)\,\psi(m_2) = \psi(m_1 m_2). \tag{7}$$

On the other hand, if n_1 and n_2 run through the divisors of m_1 and m_2,

$$\Sigma(\Delta,\, n_1)\,\Sigma(\Delta,\, n_2) = \Sigma(\Delta,\, n_1 n_2),$$

and with this formula (6) is established in general. We state this as a theorem:

4. A natural number m can be represented in the quadratic field Ω with the fundamental number Δ in

$$\Sigma(\Delta,\, n)$$

different ways as the norm of an ideal, when n runs through all the divisors of m.

§ 94. The quadratic law of reciprocity.

From the decomposition of the primes in the quadratic field Dedekind derives a new proof of the law of reciprocity of quadratic residues[23].

By Bd. II, § 167, 3. the necessary and sufficient condition that the prime ideal $\mathfrak{p}$ dividing the prime p (in any field Ω) be of the first degree is this, that for every integer ω of the field Ω

$$\omega^p \equiv \omega \pmod{\mathfrak{p}}, \tag{1}$$

[23] Dirichlet-Dedekind, Zahlentheorie, 4. Aufl. (S. 636, Anm.).

and in our case of the quadratic field this reduces to the requirement that

$$\theta^p \equiv \theta \pmod{\mathfrak{p}} \tag{2}$$

must hold.

If we take $\Delta = -4$, then $\theta = i = \sqrt{-1}$, and the congruence (2) becomes

$$i^{p-1} \equiv 1 \pmod{\mathfrak{p}}.$$

Now since $(1+i)(1-i) = 2$, so that for an odd p neither 2 nor $(1 \pm i)$ can be divisible by $\mathfrak{p}$, we must have $i^{p-1} = 1$, and hence $p-1$ must be divisible by 4.

Accordingly there follows from § 92, 2.:

1) The congruence

$$x^2 \equiv -4 \pmod{4p},$$

or, what is the same thing,

$$x^2 \equiv -1 \pmod{p}$$

has solutions then and only then, when $p \equiv 1 \pmod 4$.

If we take, secondly, $\Delta = 8$, then $\theta = \sqrt{2}$, or, if $r = \sqrt{i}$ is an 8th root of unity,

$$\theta = r + r^{-1},$$

and by the binomial theorem the condition (2) can be represented thus:

$$r^p + r^{-p} \equiv r + r^{-1} \pmod{\mathfrak{p}}.$$

This requires:

$$(r^p - r)(r^p - r^{-1})\, r^{-p} \equiv 0 \pmod{\mathfrak{p}},$$

hence:

$$r^p \equiv r \text{ or } r^p \equiv r^{-1} \pmod{\mathfrak{p}},$$

$$r^{p\pm 1} \equiv 1, \quad i^{\frac{p\pm 1}{2}} - 1 \equiv 0 \pmod{\mathfrak{p}},$$

and from this it follows as above, for an odd p:

$$p \equiv \pm 1 \pmod 8.$$

2) The congruence

$$x^2 \equiv 2 \pmod{p}$$

is therefore solvable then and only then, when $p \equiv 1$ or $p \equiv -1 \pmod 8$.

Finally we set, if q is an odd prime,

$$\Delta = \pm q \tag{3}$$

and determine the sign so that $\pm q \equiv 1 \pmod 4$. In this case

$$\theta = \frac{-1 + \sqrt{\pm q}}{2},$$

and we can represent this expression by qth roots of unity. Let r denote an imaginary qth root of unity, and let a run through the quadratic residues, b through the non-residues of q. We set, as in Bd. I, § 179

$$A = \sum^{a} r^a, \quad B = \sum^{b} r^b \tag{4}$$

and have, by the formulas (3) given there, which are derived without the use of the law of reciprocity:

$$\theta = A, \quad \theta' = B. \tag{5}$$

The sign of $\sqrt{+q}$ depends on the choice of r, but does not come into consideration here. Now, by the polynomial theorem:

$$A^p \equiv \Sigma r^{pa}, \quad B^p \equiv \Sigma r^{pb},$$

hence

$$A^p \equiv A, \quad B^p \equiv B, \quad \text{if } \left(\frac{p}{q}\right) = +1,$$

and

$$A^p \equiv B, \quad B^p \equiv A, \quad \text{if } \left(\frac{p}{q}\right) = -1.$$

Since A is not congruent to B when p is different from q, it follows from this and from 2. of the preceding section:

3) The congruence

$$x^2 \equiv \pm q \pmod{p}$$

has solutions then and only then, when p is a quadratic residue of q.

And this is the law of reciprocity with its supplementary theorems.

§ 95. Equivalent Forms and Ideals in the Quadratic Field.

If the two ideals [§ 91, (16)]

$$\mathfrak{a} = (a, a\,\omega), \quad \mathfrak{a}' = (a', a'\,\omega') \tag{1}$$

are equivalent, then there is an (integral or non-integral) number η in Ω for which $\eta\,\mathfrak{a}' = \mathfrak{a}$, and this number η is determined by the ideals $\mathfrak{a}$, $\mathfrak{a}'$ themselves up to a factor which is a numerical unit is determined. There must therefore be four rational integers α, β, γ, δ which satisfy the condition

$$\begin{aligned} \eta\,\mathfrak{a}' &= \mathfrak{a}\,(\alpha - \gamma\,\omega), \\ \eta\,\mathfrak{a}'\,\omega' &= \mathfrak{a}\,(-\beta + \delta\,\omega) \end{aligned} \tag{2}$$

If we denote for the moment by η_1, ω_1, ω_1' the numbers conjugate to η, ω, ω', then accordingly

$$\mathfrak{a}'^2\,\eta\,\eta_1 \begin{vmatrix} 1, & \omega' \\ 1, & \omega_1' \end{vmatrix} = \mathfrak{a}^2 \begin{vmatrix} \alpha, & -\gamma \\ -\beta, & \delta \end{vmatrix} \begin{vmatrix} 1, & \omega \\ 1, & \omega_1 \end{vmatrix},$$

hence:

$$\mathfrak{a}'^2\,\eta\,\eta_1\,(\omega' - \omega_1') = \mathfrak{a}^2\,(\alpha\delta - \beta\gamma)\,(\omega - \omega_1),$$

and since, by § 91, (12):

$$\mathfrak{a}\,(\omega - \omega_1) = \mathfrak{a}'\,(\omega' - \omega_1') = -\sqrt{\Delta}$$

holds, there follows from this, if

$$\varepsilon = \alpha\delta - \beta\gamma$$

is put,

$$\mathfrak{a}\,\varepsilon = \mathfrak{a}'\,\eta\,\eta_1; \tag{3}$$

if therefore, as we have already assumed, $\mathfrak{a}$ and $\mathfrak{a}'$ are positive and (according to the sharpened concept of equivalence) $\eta\eta_1$ is positive, then ε is positive as well.

But if we interchange $\mathfrak{a}$ with $\mathfrak{a}'$, then η passes into $1 : \eta$, and ε may pass into ε'. Accordingly there follows also

$$\mathfrak{a}'\,\varepsilon' = \frac{\mathfrak{a}}{\eta\,\eta_1}, \tag{4}$$

and consequently $\varepsilon\varepsilon' = 1$. But since ε and ε' are positive integral rational numbers, it follows from this that $\varepsilon = \varepsilon' = 1$. Hence by (2)

$$\omega' = \frac{-\beta - \delta\,\omega}{\alpha - \gamma\,\omega}, \qquad \omega = \frac{\alpha\,\omega' + \beta}{\gamma\,\omega' + \delta}, \tag{5}$$

$$\alpha\delta - \beta\gamma = 1.$$

Now ω and ω' are the roots of the two forms

$$(a,\, b,\, c), \quad (a',\, b',\, c') \qquad [\S\,91,\ (13)],$$

and the equivalence of these forms is expressed by (5). We therefore have, in agreement with the general theory (§ 88), the theorem:

5. The quadratic forms belonging to equivalent ideals are equivalent.

From this one recognises the significance of the sharpened concept of equivalence: for if one wished to admit negative values of $N(\eta)$ as well, then $\varepsilon = -1$ would also be possible, and the corresponding forms could be improperly equivalent.

Now let $\varphi = (a, b, c)$, $\varphi' = (a', b', c')$ be, conversely, two equivalent forms with positive first coefficients a, a', and

$$\omega = -\frac{b + \sqrt{\Delta}}{2a}, \quad \omega' = -\frac{b' + \sqrt{\Delta}}{2a'}\,. \tag{6}$$

Then there is a linear substitution $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ with determinant $+1$ by which (a, b, c) passes into (a', b', c').

We then have

$$\begin{aligned} a' &= a\alpha^2 + b\alpha\gamma + c\gamma^2, \\ b' &= 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + c\gamma\delta, \\ c' &= a\beta^2 + b\beta\delta + c\delta^2, \end{aligned} \tag{7}$$

$$\omega = \frac{\alpha\omega' + \beta}{\gamma\omega' + \delta}, \quad \omega' = \frac{\delta\omega - \beta}{-\gamma\omega + \alpha}\,, \tag{8}$$

and a short calculation shows that ω and ω' are corresponding roots, i. e. roots with the same sign of $\sqrt{\Delta}$.

Accordingly, for the ideals $\mathfrak{a}, \mathfrak{a}'$ belonging to φ and φ' we obtain two basis forms:

$$\begin{aligned} \lambda &= a(x - \omega y), \\ \lambda' &= a'(x' - \omega' y'), \end{aligned}$$

and by (8) we have

$$\lambda = \frac{a[(\delta x - \beta y) + (\gamma x - \alpha y)\omega']}{\gamma\omega' + \delta}\,.$$

If we therefore set

$$x' = \delta x - \beta y, \quad y' = -\gamma x + \alpha y,$$

it follows that

$$\lambda = \frac{a}{a'(\gamma\omega' + \delta)}\,\lambda',$$

and since $a : a'(\gamma\omega' + \delta)$ is a number in Ω, it follows that $\mathfrak{a}$ and $\mathfrak{a}'$ are equivalent.

We thus have once more, in agreement with the general theorem § 88, 4. and at the same time in closer determination:

6. If (a, b, c) (a', b', c') are two equivalent forms with positive first coefficients, and if ω, ω' are corresponding roots of these forms, then the ideals $(a, a\omega)$ and $(a', a'\omega')$ are equivalent.

For a negative discriminant the first coefficients of equivalent forms always have the same sign, and one considers only those forms in which this sign is positive.

With a positive discriminant there exist in every class of forms forms with positive first coefficients, and therefore classes of forms and classes of ideals correspond to one another in a one-to-one manner.

A basis of the principal ideal η is $(\eta, \eta\theta)$ and

$$\left.\begin{array}{lll} \theta = \dfrac{\sqrt{\Delta}}{2}, & \theta^2 = \dfrac{\Delta}{4}, & \Delta \equiv 0 \\ \theta = \dfrac{-1+\sqrt{\Delta}}{2}, & \theta^2 = \dfrac{\Delta-1}{4} - \theta, & \Delta \equiv 1 \end{array}\right\} \pmod 4.$$

If x, y are rational integers and

$$\eta = x + y\theta,$$

then this basis is, in the two cases:

$$\begin{aligned} (\alpha_1, \alpha_2) &= \left(x + y\theta,\ \frac{\Delta y}{4} + \theta x\right), \\ &= \left(x + y\theta,\ y\,\frac{\Delta-1}{4} + (x-y)\theta\right) \end{aligned}$$

and the substitution determinant A [§ 87, (1)] is

$$x^2 - \frac{\Delta}{4}y^2, \quad x^2 - xy + \frac{1-\Delta}{4}y^2 = N(\eta).$$

The basis (α_1, α_2) is therefore positive if the norm of η is positive (otherwise α_1 would have to be interchanged with α_2). From the basis form

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 = \eta(t_1 + \theta t_2)$$

one obtains

$$N(\lambda) = N(\eta)\, T,$$

and thereupon T becomes the principal form

$$(9) \qquad (1,\ 0,\ \tfrac{\Delta}{4}), \quad \left(1,\ -1,\ \frac{+1-\Delta}{4}\right).$$

If $N(\eta)$ were negative, one would have obtained for T the forms

$$\left(-1,\ 0,\ \frac{-\Delta}{4}\right), \quad \left(-1,\ +1\ \frac{-1-\Delta}{4}\right)$$

and these are equivalent to (9) only if the Pell equation $t^2 - \Delta u^2 = -4$ is solvable.

Twelfth Chapter.
Orders in the quadratic field[1]).

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§ 96. Discriminants of the orders.

1. Definition. If Q is a natural number, then the totality of the integers of the quadratic field Ω that are congruent to a rational number with respect to the modulus Q is called an order, and Q is called the conductor of this order. The order with the conductor Q is denoted by $[Q]$. From this definition it follows:

1. All rational integral numbers belong to every order;

2. Sum, difference and product of two numbers of the order belong to the same order;

i. e. within the order one can carry out addition, subtraction and multiplication unconditionally, but not division in general.

Let the totality of all integers of the field Ω likewise form an order (of conductor 1), the maximal order, which would accordingly be denoted by [1].

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[1]) Dedekind introduced the expression order in the general theory of algebraic numbers, because in the special case of the quadratic field they correspond to the Gauss orders of quadratic forms. Hilbert uses for it the expression „Ring“ or „Zahlring“. Cf. Dirichlet-Dedekind, Vorlesungen über Zahlentheorie (§ 172 of the third, § 170 of the fourth edition). Dedekind: Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers, Festschrift zur Säkularfeier von Gauss' Geburtstag, Braunschweig 1877. Hilbert: Die Theorie der algebraischen Zahlen, Bericht der Deutschen Mathematischen Vereinigung von 1894/95, Kap. IX.

If Δ is the fundamental number of the field Ω, we put, according as Δ is even or odd:

$$\begin{aligned} &\theta_0 = \tfrac{1}{2}\sqrt{\Delta}, \qquad \text{or} \\ &\theta_0 = \frac{-1+\sqrt{\Delta}}{2}. \end{aligned} \tag{1}$$

Then by § 90 every integer of the field Ω is representable in the form

$$\omega = x + y\theta_0 \tag{2}$$

where x and y are integers, and this number is then and only then congruent to a rational number a with respect to the modulus Q, when y is divisible by Q, because then

$$\frac{x - a + y\theta_0}{Q}$$

must be an integer. Accordingly all numbers of the order $[Q]$ are contained in the form:

$$\omega = x + yQ\theta_0,$$

or, in the two cases distinguished in (1), if

$$D = Q^2\Delta \tag{3}$$

is put:

$$\begin{aligned} &x - y\frac{\sqrt{D}}{2}, && \Delta \text{ even}, \\ &x - \frac{Q}{2}y + y\frac{\sqrt{D}}{2}, && \Delta \text{ odd}, D \text{ even}, \\ &x - \frac{Q-1}{2}y + y\frac{-1+\sqrt{D}}{2}, && D \text{ odd}. \end{aligned} \tag{4}$$

If we therefore put

$$\theta = \frac{\sqrt{D}}{2}, \qquad D \equiv 0 \pmod 4$$
$$\theta = \frac{-1+\sqrt{D}}{2}, \quad D \equiv 1 \pmod 4 \tag{5}$$

and in the last two formulae (4) replace the arbitrary integers

$$x - \frac{Q}{2}y, \quad x - \frac{Q-1}{2}y$$

by x, we obtain the result:

2. Theorem. Every number of the order $[Q]$, and only these, are contained in the form

$$\omega = x + y\theta, \tag{6}$$

in which x, y are arbitrary rational integers.

We call the system

$$(1, \theta) \tag{7}$$

a basis of the order $[Q]$.

By substituting (5) one can also represent the numbers (6) in the form:

$$\omega = \frac{x + y\sqrt{D}}{2}, \tag{8}$$

where the x, y are integers which satisfy the condition

$$x^2 - y^2 D \equiv 0 \pmod 4. \tag{9}$$

The number D is a discriminant whose associated fundamental discriminant is Δ. We call it the discriminant of the order $[Q]$.

§ 97. Orders and Ideals.

Let $\mathfrak{m}$ be an ideal of the quadratic field Ω. If all integers divisible by $\mathfrak{a}$ are divisible by a rational number m, then the ideal $\mathfrak{a} = m\mathfrak{m}$ is divisible by m; we have

$$N(\mathfrak{a}) = m^2 N(\mathfrak{m}),$$

and the rational number $mN(\mathfrak{m})$, which is less than $N(\mathfrak{a})$, is divisible by $\mathfrak{a}$.

If m is the greatest natural number by which $\mathfrak{a}$ is divisible, we shall call m the divisor of $\mathfrak{a}$. If the divisor $= 1$, the ideal is called primary, otherwise derived. One obtains all ideals derived from a primary ideal $\mathfrak{m}$ by multiplying $\mathfrak{m}$ by arbitrary natural numbers, and all ideals derived from a primary ideal are equivalent to one another.

3. Theorem. If $\mathfrak{a}$ is a primary ideal, then the norm of $\mathfrak{a}$ is at the same time the least natural number divisible by $\mathfrak{a}$.

For let us denote by a the least natural number divisible by $\mathfrak{a}$; the norm of $\mathfrak{a}$ is then in any case divisible by a. We put

$$N(\mathfrak{a}) = ma,$$

and if $\mathfrak{a}'$ is the ideal conjugate to $\mathfrak{a}$, then, since by Bd. II, § 164 the norm of an ideal is equal to the product of the conjugate ideals, it follows that

$$N(\mathfrak{a}) = \mathfrak{a}\mathfrak{a}' = ma. \tag{1}$$

Since a is to be divisible by $\mathfrak{a}$, we set

$$a = \mathfrak{a}\mathfrak{m}' = \mathfrak{a}'\mathfrak{m}$$

and obtain from (1):

$$\mathfrak{a} = m\mathfrak{m}. \tag{2}$$

Thus m divides the divisor of the ideal $\mathfrak{a}$, and with this Theorem 3. is proved, since, if $\mathfrak{a}$ is primary, its divisor must be $= 1$.

4. Theorem. If $\mathfrak{m}$ is a primary ideal of the field Ω, then every integer in Ω is congruent to a rational number with respect to the modulus $\mathfrak{m}$.

For by Bd. II, § 165, $N(\mathfrak{m})$ is in general the number of the numbers in Ω that are incongruent with respect to $\mathfrak{m}$. If now $N(\mathfrak{m}) = a$ is at the same time the least natural number divisible by $\mathfrak{m}$, then the a numbers

$$0, 1, 2, \ldots, a-1$$

are all incongruent, and among them all classes of numbers modulo $\mathfrak{m}$ are represented.

Let now $[Q]$ be an order with the conductor Q and the discriminant D, and $(1, \theta)$ the basis of this order. Further let $\mathfrak{m}$ be a primary ideal coprime to Q. By Theorem 4. there is then a rational integer t which satisfies the condition

$$\theta + t \equiv 0 \pmod{\mathfrak{m}}$$

We set

$$\begin{array}{ll} t = \dfrac{b}{2}, & \text{if } D \equiv 0 \\ t = \dfrac{1+b}{2}, & \text{„ } \quad D \equiv 1 \end{array} \pmod 4 \tag{3}$$

and obtain from § 96, (5):

$$\theta + t = \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}, \tag{4}$$

wherein b is a rational integer which by (3) satisfies the condition

$$b \equiv D \pmod 2$$

Moreover let

$$N(\mathfrak{m}) = a, \tag{5}$$

so that a is the least natural number divisible by $\mathfrak{m}$.

By the congruence (4) the number b is determined only with respect to the modulus $2a$. For if b, b' are two numbers which satisfy the condition (4), then $\frac{1}{2}(b' - b)$ is divisible by a.

Since further

$$N\left(\frac{b + \sqrt{D}}{2}\right) = \frac{b^2 - D}{4}$$

is a rational integer divisible by $\mathfrak{m}$, then it must be divisible by a; if we set it $= ac$, then c is a rational integer and

$$D = b^2 - 4ac, \tag{6}$$

$$b^2 \equiv D \pmod{4a}, \tag{7}$$

and if the congruence (7) is satisfied, then every number b' congruent to b with respect to the modulus $2a$ satisfies the same congruence.

From this we conclude:

5. Theorem. Every primary ideal $\mathfrak{m}$ coprime to Q whose norm $= a$ yields us, through (4), one and only one root b of the congruence (7), provided that not individual numbers, but classes of numbers with respect to the modulus $2a$, are regarded as roots.

The two numbers

$$a, \quad \frac{b+\sqrt{D}}{2} \tag{8}$$

cannot, if a is relatively prime to Q, both be divisible by one and the same rational prime; for otherwise

$$\frac{b+\sqrt{D}}{2} - \frac{b-\sqrt{D}}{2} = \sqrt{D}$$

would have to be divisible by p; hence D would be divisible by p^2, and p would have to be a divisor of Q, contrary to the assumption.

For an odd p this is evident, but for $p = 2$ it would follow:

$$b \equiv 0 \pmod 2, \quad b^2 - D \equiv 0 \pmod{16},$$

and hence

$$D \equiv 0, 4 \pmod{16}.$$

Thus $D/4$ would still be a discriminant and 2 would have to divide Q. With this it is proved:

6. Theorem. If a is a natural number coprime to Q and b is a root of (7), then one obtains a definite primary ideal $\mathfrak{m}$ as the greatest common divisor of the two numbers (8).

Now we prove:

7. Theorem. Every number μ of the order $[Q]$ that is divisible by the primary ideal $\mathfrak{m}$ and coprime to Q is representable in the form

$$\mu = ax + \frac{b+\sqrt{D}}{2}y \tag{9}$$

once and only once, if x, y are rational integers. If x, y are regarded as variconsidered, then μ is called a basis form of the ideal $\mathfrak{m}$ in the order $[Q]$.

For the proof we observe that, by (4), every number of the order $[Q]$ is contained in the form

$$\omega = z + \frac{b+\sqrt{D}}{2}y, \tag{10}$$

where y and z are rational integers. Now if $\mathfrak{m}$ is primary, then

$$N(\mathfrak{m}) = a \tag{11}$$

is the least natural number divisible by $\mathfrak{m}$, and hence if ω is to be divisible by $\mathfrak{m}$, then z must be divisible by a. If therefore we set $z = ax$, we obtain the form (9). Conversely it is evident that every number of this form is divisible by $\mathfrak{m}$.

That the representation of a number μ in this form is possible in only one way follows from the fact that $\sqrt{D}$ is irrational.

The form μ is therefore called a basis form of the ideal $\mathfrak{m}$ in the order Q. From it one obtains a basis form of the same ideal in the field Ω:

$$\lambda = ax + \frac{b'+\sqrt{\Delta}}{2}y,$$

if b' is determined from the congruence

$$b'Q \equiv b \pmod{2a}$$

which is always solvable, since Q is relatively prime to a, and for even Q also b is even.

From (9) there results

$$N(\mu) = a(ax^2 + bxy + cy^2). \tag{12}$$

The number μ which one obtains when definite numbers are put in (9) for x, y we can decompose into ideal factors:

$$\mu = \mathfrak{a}\mathfrak{m},$$

if

$$N(\mathfrak{a}) = ax^2 + bxy + cy^2 \tag{13}$$

holds, and the following theorem is valid:

8. Theorem. If μ is relatively prime to Q, and x, y have no common divisors, then the ideal $\mathfrak{a}$ is primary.

We prove it thus: Let p be a rational prime dividing $\mathfrak{a}$; then μ is divisible by p, and from this it follows first of all that y must be divisible by p; for a number ω of the form (10) can be divisible by a prime p that does not divide Q only if both y and $z = ax$ are divisible by p. Accordingly, since by assumption x and y

are relatively prime, x cannot be divisible by p, and $\mathfrak{a}$ must be divisible by p.

If we put $\mathfrak{a} = p^k\mathfrak{a}_1$, where $\mathfrak{a}_1$ is no longer divisible by p, and denote by $\mathfrak{p}$ a prime ideal dividing p, then by (11) $\mathfrak{p}$ must divide $\mathfrak{m}$ or $\mathfrak{m}'$, and since $\mathfrak{m}$ was assumed to be primary, $\mathfrak{p}$ cannot be $= p$. Therefore $N(\mathfrak{p}) = p = \mathfrak{p}\mathfrak{p}'$, and if $\mathfrak{m}$ is divisible by $\mathfrak{p}$, it cannot at the same time be divisible by $\mathfrak{p}'$. Consequently $\mathfrak{m}$ is divisible by $\mathfrak{p}^k$. Hence, by the definition of $\mathfrak{m}$ (Theorem 6), $\dfrac{b+\sqrt{D}}{2}$ is also divisible by $\mathfrak{p}^k$, and consequently $\dfrac{b+\sqrt{D}}{2}y$ at least by $\mathfrak{p}^{k+1}$, while ax and consequently μ exactly by $\mathfrak{p}^k$, whereas on the other hand $\mathfrak{a}\mathfrak{m} = \mu$ is likewise divisible at least by $\mathfrak{p}^{k+1}$, in which lies a contradiction. The like holds also when $p = \mathfrak{p}^2$, that is, in Δ (but not in Q). Consequently no rational prime can be contained in $\mathfrak{a}$, and $\mathfrak{a}$ is primary.

From what has been proved so far we can formulate the following theorem:

9. Theorem. If

$$\mu = ax + \frac{b+\sqrt{D}}{2}y = \mathfrak{m}\mathfrak{a},$$

$$\mu' = ax' + \frac{b+\sqrt{D}}{2}y' = \mathfrak{m}\mathfrak{a}'$$

are two numbers coprime to Q, if neither x and y nor x' and y' have a common divisor, if

$$N(\mathfrak{a}) = N(\mathfrak{a}') = A,$$

and if for one and the same number B

$$\frac{B+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{a}}$$
$$\equiv 0 \pmod{\mathfrak{a}'},$$

---

then the two ideals $\mathfrak{a}$ and $\mathfrak{a}'$ are identical, and the numbers μ, μ' differ only by a unit factor.

For $\mathfrak{a}$ as well as $\mathfrak{a}'$ are primary by 8., and by 6. are defined as the greatest common divisor of

$$A \text{ and } \frac{B+\sqrt{D}}{2}.$$

Thirteenth Chapter.
Equivalence with respect to Number Groups.

---

§ 98. Number groups in the orders.

In order to investigate more closely the relations of the quadratic irrational numbers to the orders of the quadratic forms, we derive from the numbers of the field Ω number groups according to the following points of view.

1. We take an arbitrary natural number Q and exclude from the integers in Ω in the first place all those that are not coprime to Q. From the remaining ones we also take arbitrary quotients.

The integral and fractional numbers η so obtained we call foreign to Q. Let their totality be denoted by O. They form a group in so far as multiplication and division of two of these numbers always yield numbers of the same system.

2. We now proceed in the same way with the numbers of the order $[Q]$, that is, we exclude all numbers of this order that are not relatively prime to Q, and we also form the quotients of any two of the numbers that remain. We denote these numbers in η' integral and fractional numbers of the order $[Q]$. The numbers O' are all contained in O and likewise form a group with respect to multiplication and division. We call it the group of the order $[Q]$, or briefly an order group.

Every number η in O can, by multiplication by an integral rational number n coprime to Q, be transformed into an integer ω.

If η' is a number in O', one can make the denominator rational and finds that $n\eta'$ is an integer of the order $[Q]$, and consequently is congruent to a rational number with respect to the modulus Q.

Since n is coprime to Q, one can set this number $= nr$, where r is a rational number coprime to Q. Accordingly every number η' has the property

$$\eta' \equiv r \pmod{Q}, \tag{1}$$

where r is a rational integer and the congruence is to be understood in the sense $n\eta' \equiv nr$.

The congruences (1) can be multiplied and divided if by r^{-1} one understands the integer satisfying the condition $r \cdot r^{-1} \equiv 1 \pmod{Q}$.

Conversely, an integral or fractional number η' which satisfies a condition (1) belongs to the group O'. For if $n\eta' = \omega \equiv nr$ is a number in $[Q]$, then $\eta' = \omega : n$ is a quotient of two such numbers.

3. In O' there is now again contained a group O_0, which consists of all numbers η_0 that satisfy the condition

$$\eta_0 \equiv 1 \pmod{Q} \tag{2}$$

and the numbers of O_0 likewise yield again, through multiplication and division, numbers of O_0.

We thus have three kinds of number groups, each of which is contained in the preceding one:

1. O contains all numbers in Ω prime to Q.

2. O' contains the integral and fractional numbers of the order $[Q]$ prime to Q.

3. O_0 contains the number classes in O' that are congruent to unity with respect to the modulus Q.

The groups O, O', O_0 are infinite Abelian groups. But if one takes a complete system of residues of rational numbers coprime to Q, $r_1, r_2, \ldots, r_\mu$, then every number in O' can be represented as the product of one of these r_i with a number in O_0, and one can therefore set, in the mode of expression used in Bd. II, § 2:

$$O' = r_1 O_0 + r_2 O_0 + \cdots + r_\mu O_0, \tag{3}$$

and μ is the index of the divisor O_0 of O:

$$\mu = (O', O_0), \tag{4}$$

hence a finite number, namely:

$$\mu = \varphi(Q) = Q \, \Pi \left(1 - \frac{1}{q}\right), \tag{5}$$

where $\varphi(Q)$ is the symbol customary in number theory for the number of number classes with respect to the modulus Q with numbers coprime to Q,

and q in the product Π runs through the primes, distinct from one another, that divide Q.

If we denote by

$$\varrho_1, \varrho_2, \ldots, \varrho_\nu$$

a complete system of representatives of the numbers coprime to Q in Ω, taken to the modulus Q, then in the same way

$$O = \varrho_1 O_0 + \varrho_2 O_0 + \cdots + \varrho_\nu O_0 \tag{6}$$

and

$$\nu = \psi(Q) = (O, O_0) \tag{7}$$

is the index of the divisor O_0 with respect to O.

The number $\psi(Q)$ we have determined in general in § 168, (3) of Band II, and also for the case in which an ideal takes the place of Q. Since here $N(Q) = Q^2$, we have:

$$\psi(Q) = Q^2 \, \Pi \left(1 - \frac{1}{N(\mathfrak{q})}\right), \tag{8}$$

when $\mathfrak{q}$ runs through the ideal prime divisors of Q that are distinct from one another.

Now (§ 92)

1. $N(\mathfrak{q}) = q$, when $(\Delta, q) = 0$,
2. $N(\mathfrak{q}) = q$, when $(\Delta, q) = +1$,
3. $N(\mathfrak{q}) = q^2$, when $(\Delta, q) = -1$.

We shall denote these three kinds of primes by q_1, q_2, q_3, and observe that every prime q_2 is the norm of two different prime ideals $\mathfrak{q}$. Accordingly there follows from (8)

$$\psi(Q) = Q^2 \, \Pi \left(1 - \frac{1}{q_1}\right) \Pi \left(1 - \frac{1}{q_2}\right)^2 \Pi \left(1 - \frac{1}{q_3^2}\right). \tag{9}$$

Now the index of O' with respect to O is, by Bd. II, § 2 (4):

$$j = \frac{\nu}{\mu} = (O, O') = \frac{(O, O_0)}{(O', O_0)} = \frac{\psi(Q)}{\varphi(Q)}$$

and

$$\varphi(Q) = Q\,\Pi\left(1-\frac{1}{q_1}\right)\Pi\left(1-\frac{1}{q_2}\right)\Pi\left(1-\frac{1}{q_3}\right);$$

consequently there results:

$$(O,O') = Q\,\Pi\left(1-\frac{1}{q_2}\right)\Pi\left(1+\frac{1}{q_3}\right),$$

for which one may then also put (according to 1., 2., 3.):

$$j = (O,O') = Q\,\Pi\left(1-\frac{(\Delta,q)}{q}\right). \tag{10}$$

In this the product Π refers to all primes q that divide Q.

We can therefore select a system of representatives

$$\sigma_1,\ \sigma_2,\ \ldots,\ \sigma_j$$

in O in such a way that

$$O = \sigma_1 O' + \sigma_2 O' + \cdots \sigma_j O' \tag{11}$$

holds.

§ 99. Equivalence in the Orders.

10. Definition. We will now call two ideals $\mathfrak{a}$, $\mathfrak{a}'$ equivalent with respect to the order $[Q]$ if

$$\mathfrak{a}' = \eta'\mathfrak{a} \tag{1}$$

holds, and η' denotes a number in O'. $\mathfrak{a}$ and $\mathfrak{a}'$ are here always assumed to be coprime to Q.

If one takes a number $\mu = \mathfrak{a}\mathfrak{m}$ in $[Q]$ divisible by $\mathfrak{a}$ in such a way that $\mathfrak{m}$ becomes a primary ideal, and sets $\mu' = \eta'\mu$, then we have:

$$\begin{aligned} \mu &= \mathfrak{m}\mathfrak{a}, \\ \mu' &= \mathfrak{m}\mathfrak{a}', \end{aligned} \tag{2}$$

and we can also explain the equivalence of $\mathfrak{a}$ and $\mathfrak{a}'$ with respect to $[Q]$ by the fact that $\mathfrak{a}$ and $\mathfrak{a}'$ are transformed into numbers of the order $[Q]$ by multiplication with one and the same ideal $\mathfrak{m}$.

By the ideal $\mathfrak{m}$ there is determined, according to § 97, a family of parallel quadratic forms (a,b,c)[24], in which a and b are determined from

$$a = N(\mathfrak{m}), \quad \frac{b+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}} \tag{3}$$

And so this family of forms does not change, by (2), when a is replaced by an equivalent ideal a'.

But if in (2) one takes another ideal $\mathfrak{m}_1$ in place of the ideal $\mathfrak{m}$ and sets

$$\mu_1 = \mathfrak{m}_1\mathfrak{a}, \tag{4}$$

[24] By a family of parallel forms one understands the system

$$(a,\ b+2la,\ c+lb+l^2a),$$

when l runs through all rational integers.

then

$$\frac{\mu}{\mu_1} = \frac{\mathfrak{m}}{\mathfrak{m}_1} = \eta$$

is a number in $[O']$, and hence $\mathfrak{m}_1$ is equivalent to $\mathfrak{m}$.

If therefore one puts

$$a_1 = N(\mathfrak{m}_1) = \mathfrak{m}_1\mathfrak{m}_1', \quad \frac{b_1 + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}_1}, \tag{5}$$

then one obtains another quadratic form (a_1, b_1, c_1), of which we can show that it is equivalent to (a, b, c).

We put, for brevity:

$$\omega = \frac{b + \sqrt{D}}{2a}, \quad \omega_1 = \frac{b_1 + \sqrt{D}}{2a_1}$$

and remark that

$$\eta a_1 = \eta\mathfrak{m}_1\mathfrak{m}_1' = \mathfrak{m}\mathfrak{m}_1',$$

$$\eta\omega_1 a_1 = \frac{\mathfrak{m}}{\mathfrak{m}_1}\,\frac{b_1 + \sqrt{D}}{2}$$

are numbers in $[Q]$ divisible by $\mathfrak{m}$ [by (5)]. Accordingly, by § 97, 7., the rational integers α, β, γ, δ can be so determined that:

$$\begin{aligned} \eta a_1 &= a(\alpha - \gamma\omega), \\ \eta a_1\omega_1 &= a(-\beta + \delta\omega), \end{aligned} \tag{6}$$

and one proves exactly as in § 95 that

$$\alpha\delta - \beta\gamma = 1$$

must hold, and hence that the two forms

$$(a, b, c), \quad (a_1, b_1, c_1)$$

are equivalent. We have therefore:

11. Theorem. Every ideal class with respect to $[Q]$ that is represented by an ideal $\mathfrak{a}$ corresponds to a class of forms A represented by the form (a, b, c), and conversely.

In order to obtain the class of forms A, one takes a primary ideal $\mathfrak{m}$ of the class A^{-1} reciprocal to A and puts $a = N(\mathfrak{m})$, $\frac{b+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}$. And if conversely the form (a, b, c) is given, then the greatest common divisor $\mathfrak{m}$ of

$$a, \quad \frac{b + \sqrt{D}}{2}$$

is an ideal of the class A^{-1}.

§ 100. Ideal classes with respect to the orders.

The concept of the equivalence of ideals, and the theorems on class numbers, can be presented most clearly if

one represents the ideals, according to Bd. II, § 169, by the functionals discussed in Bd. II, § 153 ff., because in doing so one can make use of the ordinary rules of multiplication and division[25].

We restrict ourselves here once and for all, in the field Ω, to numbers and ideals that are coprime to an arbitrarily assumed rational integer Q (the conductor of an order).

[25] Cf. also the author's memoir „Über Zahlengruppen in algebraischen Körpern", first communication, Mathematische Annalen 48, 438.

Under this supposition the integral and fractional numbers of the field Ω form the group previously denoted by O. We extend this group by the addition of the functionals and denote the group so extended by $\overline{O}$. To every element of $\overline{O}$ there then corresponds an integral or fractional ideal, and to those elements whose quotient is a functional unit there corresponds the same ideal.

If η is a number in O and ε is any functional unit, then a principal ideal corresponds to the product $\varepsilon\eta$. Accordingly we denote

by $\overline{E}$ the group of functional units,
„ E „ „ „ numerical units,

and by

$\overline{E}O$ the principal class of the functionals.

If we take a system of representatives

$$\varphi_1,\ \varphi_2,\ \ldots,\ \varphi_h \tag{1}$$

of non-equivalent integral functionals in Ω, then every integral or fractional functional Φ will be representable in the form:

$$\Phi = \varphi_i\overline{\omega},$$

where $\overline{\omega}$ is a functional from $\overline{E}O$, and the classes of ideals in Ω are the subsidiary groups of $\overline{E}O$ in $\overline{O}$. Accordingly the class number h is the index of the divisor $\overline{E}O$ of $\overline{O}$:

$$h = (\overline{O},\ EO). \tag{2}$$

Here, in the sharper division into classes, one has to restrict oneself in the case of O to the numbers with positive norm.

If we now consider the division of the ideals into classes according to the order $[Q]$, then we have to take the functional group $\overline{E}O'$ as the principal class, and the class number is

$$h' = (\overline{O},\ \overline{E}O').$$

In order to derive from this the relation between h and h', we make use of the general group theorems of Bd. II, § 2, which we formulate and supplement as follows:

12. Theorem. If A is a group and B a divisor of A of finite index (A, B), and further C a divisor of B of finite index (B, C), then, by Bd. II, § 2, (4):

$$(A, C) = (A, B)(B, C). \tag{4}$$

If A is again a group with the divisor B of index μ, we set

$$A = a_1B + a_2B + \cdots + a_\mu B, \tag{5}$$

where $a_1, a_2, \ldots, a_\mu$ is a complete system of representatives of A with respect to B.

Now let C be a group of elements which are composable with the elements of A (e. g. in our case simply by multiplication); then from (5) there follows:

$$AC = a_1BC + a_2BC + \cdots + a_uBC. \tag{6}$$

Now one and the same element can occur in the two subsidiary groups a_1BC, a_2BC only if $a_1a_2^{-1}$ is contained in BC, but not in B.

From this we formulate the theorem:

13. Theorem. If A is a group, B a divisor of A of finite index (A, B), and further C a group composable with A of such a kind that A and BC contain no common elements apart from those of B, then

$$(AC,\ BC) = (A, B). \tag{7}$$

The assumption of this theorem can also be expressed thus:

B is the intersection of A and BC,

and it is satisfied, e. g., when C is contained in B, that is, when $B = BC$, and also when C has no element in common with A.

14. Theorem. If A is a group composed of the two groups A' and B

$$A = A'B \tag{8}$$

and B' the intersection of A' and B, then

$$(A, B) = (A', B'), \tag{9}$$

provided that these indices are finite.

For by (8), in every subsidiary group aB to B in A an element a' from A' is contained, and one can therefore represent these subsidiary

groups also by $a'B$, where a' is contained in A'. If then a'_1B, a'_2B are different subsidiary groups to B in A, then a'_1B', a'_2B' are different subsidiary groups to B' in A', and conversely; for $a'_1a'^{-1}_2$ is contained in B' then and only then, when it is at the same time contained in B, and from this follows formula (9).

We now apply these theorems to the expressions (2) and (3) for h and h'. By (4) we have

$$(\overline{O}, \overline{E}O') = (\overline{O}, \overline{E}O)(\overline{E}O, \overline{E}O'), \tag{10}$$

hence

$$h' = h(\overline{E}O, \overline{E}O'). \tag{11}$$

We now denote, as already above, by E the group of the numerical units in O. Then $\overline{E}E = \overline{E}$, since the numerical units are contained among the functional ones, and every number that is contained in $\overline{E}O'$ is contained in EO'. Consequently EO' is the intersection of O and $\overline{E}O'$. If we therefore apply (7), putting $O, EO', \overline{E}$ in place of A, B, C, then B is the intersection of A and BC, and we obtain:

$$(\overline{E}O, \overline{E}O') = (\overline{E}O, \overline{E}EO') = (O, EO'). \tag{12}$$

Further, by (4):

$$(O, EO')(EO', O') = (O, O'), \tag{13}$$

and if we denote by E' the group of the numerical unit in O'

$$(EO', O') = (EO', E'O') = (E, E')$$

(by 2), for E' is the intersection of $E'O'$ and E.

Thus by (13) we have

$$(O, EO')(E, E') = (O, O')$$

and by (12)

$$(\overline{E}O, \overline{E}O')(EE') = (O, O'),$$

hence finally by (11):

$$(E, E')h' = (O, O')h. \tag{14}$$

15. Theorem. Formula (14) holds unchanged if O' denotes not precisely the order group, but any number group of finite index (O, O') contained in O,

provided we call two ideals $\mathfrak{a}, \mathfrak{a}'$ equivalent according to O' if their quotient $\mathfrak{a}'/\mathfrak{a} = \eta'$ is a number in O'.

If E' is the group of the units contained in O', and if (E, E') has a finite value, then the class number h' is finite and determined by the formula (12).

If, for example, O is the group of all numbers in Ω (apart from 0), and O' the group of the numbers with positive norm, then, provided there are any numbers with negative norm at all, $(O, O') = 2$. If there are then units with negative norm, $(E, E') = 2$ as well. But if all units have positive norm, then $(E, E') = 1$, and we obtain $h' = h$ in the first case, $2h' = h$ in the second case.

The equivalence with respect to O is then the general equivalence, that with respect to O' the sharpened one.

Let us return to the orders $[Q]$ and understand by O the numbers coprime to Q (if necessary only those with positive norm); then we have determined (O, O') in the preceding section.

If the discriminant D and its stem Δ are negative, then in general, both in E and in E', there are only the two units ± 1. In the two exceptional cases $\Delta = -3, -4$, E contains six and four units, but E', if $Q > 1$, only two, and thus $(E, E') = 1$ in general, $(E, E') = 3$ in the case $\Delta = -3$, $(E, E') = 2$ in the case $\Delta = -4$. Hence in these cases we have, by § 98, (10):

$$\lambda h' = Q\Pi\left(1 - \frac{(\Delta, q)}{q}\right) h, \tag{15}$$

in which:

$$\begin{aligned} &\lambda = 2, \quad \text{for } \Delta = -4 \\ &\lambda = 3, \quad \text{„ } \Delta = -3 \\ &\lambda = 1, \quad \text{„ } \Delta < -4 \end{aligned} \tag{16}$$

This is the relation which was first derived by Gauss and later confirmed by Dirichlet in another way.

If, for example, $\Delta \equiv 1 \pmod 4$, $Q = 2$, then in Gaussian notation h is the class number for the forms of the second kind, h' that for the forms of the first kind. We have

$$\begin{aligned} Q\left(1 - \frac{(\Delta, q)}{2}\right) &= 1, \quad \Delta \equiv 1 \pmod 8, \\ &= 3, \quad \Delta \equiv 5 \pmod 8. \end{aligned}$$

Thus

$$\begin{aligned} h' &= h, \quad \Delta \equiv 1 \\ &= 3h, \quad \Delta \equiv 5 \end{aligned} \pmod 8,$$

and only in the exceptional case $\Delta = -3$ does $h' = h$ occur.

If D is positive, then (E, E') still has to be determined.

Let

$$\varepsilon = \frac{t + u\sqrt{D}}{2}$$

be the fundamental unit in O, and λ the smallest positive exponent for which

$$\varepsilon^\lambda = \frac{t + u\sqrt{D}}{2}$$

is contained in O'. Then E consists of the powers $\pm\varepsilon^\nu$, and E' of the powers $\pm\varepsilon^{\lambda\nu}$, and we have

$$(E, E') = \lambda,$$

and formula (15) gives the class number h'. In general no further determination can be given for λ.

If again $\Delta \equiv 1 \pmod 4$ and $Q = 2$, then $\lambda = 1$ or $= 3$; namely $= 1$ if u is even in (17), and $= 3$ if u is odd. The latter can occur only when $\Delta \equiv 5 \pmod 8$, and consequently for $\Delta \equiv 1 \pmod 8$ we always have $\lambda = 1$, while for $\Delta \equiv 1 \pmod 8$ λ can be either $= 1$ or $= 3$.

Fourteenth Chapter.
Composition of Forms and Ideals.

§ 101. Composition in the Orders.

Let Δ be the field discriminant and $D = Q^2\Delta$ the discriminant of the order $[Q]$.

If $T = (a, b, c)$ is a primitive quadratic form of the discriminant D with a positive and coprime to Q, then from it we can determine a basis form λ in O of an ideal $\mathfrak{a}$ coprime to Q, in the following way:

We set

$$\lambda = at_1 + \frac{b + \sqrt{D}}{2} t_2, \tag{1}$$

$$N(\lambda) = a(at_1^2 + bt_1t_2 + ct_2^2). \tag{2}$$

If we denote the ideal determined by the functional λ, that is, the greatest common divisor of

$$a, \frac{b + \sqrt{D}}{2} \tag{3}$$

by $\mathfrak{a}$, then from (2) it follows that

$$N(\mathfrak{a}) = a. \tag{4}$$

Here a is the least natural number divisible by $\mathfrak{a}$. For if this least number is a_1, then a_1 must be a divisor of a, say $a = a_1a_2$. Here a_1, a_2, b can have no common divisor; for, in the first place, they cannot all three be even, since otherwise we should have $a = 4a'$, $b = 2b'$, $c = c'$,

$$D = 4(b'^2 - 4a'c)$$

and 4 would have to divide Q, contrary to the assumption that a is coprime to Q; and likewise an odd common prime divisor of a_1, b, a_2 would divide Q.

If we therefore set

$$\lambda_1 = a_1t_1 + \frac{b + \sqrt{D}}{2} t_2,$$

then

$$N(\lambda_1) = a_1(a_1t_1^2 + b\,t_1t_2 + a_2c\,t_2^2),$$

and $(a_1,\ b,\ a_2c)$ is likewise primitive. Since λ_1 is now divisible by $\mathfrak{a}$, a_1 must be divisible by $N(\mathfrak{a})$, i. e. by a, and must therefore be $= a$. Hence $\mathfrak{a}$ is primary.

If in (1) we fix the $\sqrt{D}$ once and for all, e. g. as positively real or positively imaginary, then the linear form λ is uniquely determined by the quadratic form T. The form of the conjugate ideal would be obtained not by changing the sign of $\sqrt{D}$, but that of b.

Let $\mathfrak{a}$ and $\mathfrak{b}$ be any two ideals coprime to the conductor Q of the order $[Q]$, which we shall now denote by O, and let

$$\begin{aligned} \mu &= \alpha_1u_1 + \alpha_2u_2, \\ \nu &= \beta_1v_1 + \beta_2v_2 \end{aligned} \tag{5}$$

be basis forms of $\mathfrak{a}$ and $\mathfrak{b}$ in O. From $\mathfrak{a}$ and $\mathfrak{b}$ we form the product

$$\mathfrak{a}\,\mathfrak{b} = \mathfrak{c}$$

with the basis form in O:

$$\lambda = \gamma_1 t_1 + \gamma_2 t_2. \tag{6}$$

The quadratic forms belonging to μ, ν, λ shall be denoted by U, V, T.

1. We call λ composed or compounded out of μ and ν, and T out of U and V.

We set:

$$\begin{aligned} T &= (a,\ b,\ c),\\ U &= (a_1,\ b_1,\ c_1),\\ V &= (a_2,\ b_2,\ c_2), \end{aligned} \tag{7}$$

and our problem is solved if T can be derived from U and V.

If

$$\begin{array}{lll} \mathfrak{a} & \text{equivalent to} & \mathfrak{a}',\\ \mathfrak{b} & " \quad " & \mathfrak{b}', \end{array}$$

then also

$$\mathfrak{a}\,\mathfrak{b} \quad " \quad " \quad \mathfrak{a}'\mathfrak{b}',$$

hence

$$\mathfrak{c} \quad " \quad " \quad \mathfrak{c}',$$

and if therefore U, V are replaced by equivalent forms U', V', then an equivalent form T' also takes the place of T.

If we denote by A, B, C the classes in O to which λ, μ, ν belong, and likewise denote the classes of the quadratic forms U, V, T, then C is also said to be compounded from A and B, and one sets symbolically

$$C = AB. \tag{8}$$

This composition of the classes is uniquely determined once any three representatives U, V, T of them are given, and it is therefore enough if we choose U, V in their classes in any suitable way.

In every ideal class there are ideals that are relatively prime to any given ideal. We therefore first take $\mathfrak{a}$ in A primary, but otherwise arbitrary, then $\mathfrak{b}$ likewise primary and relatively prime, not only to $\mathfrak{a}$, but to a_1. Then a_1 and a_2 are relatively prime (for the greatest common divisor d of a_1 and a_2 would be relatively prime to $\mathfrak{b}$, and consequently $a_2 : d$ would be divisible by $\mathfrak{b}$). But from this it follows that $d = 1$, because a_2 was to be the least natural number divisible by $\mathfrak{b}$.

Once a_1, a_2 have been determined in this way, one can determine b in accordance with the two congruences

$$\begin{aligned} b &\equiv b_1 \pmod{2a_1},\\ &\equiv b_2 \pmod{2a_2} \end{aligned}$$

and then

$$b^2 - D = 4\,a_1 a_2 c$$

is divisible by $4a_1a_2$. We therefore set

$$\begin{aligned} U &= (a_1,\ b,\ a_2 c),\\ V &= (a_2,\ b,\ a_1 c). \end{aligned} \tag{9}$$

Now a_1a_2 is the least rational integer divisible by $\mathfrak{a}\mathfrak{b}$, and $\mathfrak{a}\mathfrak{b}$ is the greatest common divisor of

$$a_1 a_2, \quad \frac{b - \sqrt{D}}{2};$$

consequently

$$T = (a_1 a_2,\ b,\ c) \tag{10}$$

is the form compounded from U and V, and the linear forms λ, μ, ν corresponding to the quadratic forms T, U, V are:

$$\lambda = a_1 a_2\, t_1 + \frac{b - \sqrt{D}}{2}\, t_2,$$
$$\mu = a_1\, u_1 + \frac{b - \sqrt{D}}{2}\, u_2,$$
$$\nu = a_2\, v_1 + \frac{b - \sqrt{D}}{2}\, v_2.$$

From this it follows, with regard to

$$\left(\frac{b - \sqrt{D}}{2}\right)^2 = a_1 a_2 c + b\,\frac{b - \sqrt{D}}{2}\ :$$

$$\mu\nu = a_1 a_2 (u_1 v_1 - c\, u_2 v_2) + \frac{b - \sqrt{D}}{2}\,(a_1 u_1 v_2 + a_2 u_2 v_1 + b\, u_2 v_2). \tag{11}$$

If, then, one makes the bilinear substitution:

$$\begin{aligned} t_1 &= u_1 v_1 - c\, u_2 v_2, \\ t_2 &= a_1 u_1 v_2 + a_2 u_2 v_1 + b\, u_2 v_2, \end{aligned} \tag{12}$$

there results

$$\lambda = \mu\nu, \tag{13}$$

and consequently

$$T = UV. \tag{14}$$

In the requirement that a_1 and a_2 are to be relatively prime there lies at times a certain inconvenience, e. g. when it is a question of the composition of a form with itself.

Let us then drop the hypothesis that $\mathfrak{a}$ is relatively prime to $\mathfrak{b}$, and a_1 to a_2, but hold fast to the assumption that

$$b^2 - D = 4\, a_1 a_2 c \tag{15}$$

is divisible by $4a_1a_2$; then relation (11) still holds, and it follows that $\mathfrak{ab}$ is the greatest common divisor of

$$a_1 a_2, \quad a_1\,\frac{b - \sqrt{D}}{2}, \quad a_2\,\frac{b - \sqrt{D}}{2}, \quad b\,\frac{b - \sqrt{D}}{2}$$

(Bd. II, § 160, 2).

If now a_1, a_2, b have no common divisor, then the equation $k_1 a_1 + k_2 a_2 + kb = 1$ can be satisfied by rational integers k, and accordingly $\mathfrak{ab}$ is likewise the greatest common divisor of

$$a_1 a_2, \quad \frac{b - \sqrt{D}}{2}. \tag{16}$$

2. It is to be proved that $a_1 a_2$ is the smallest natural number divisible by $\mathfrak{ab}$.

For if we set this smallest number $= a$, then $a_1 a_2$ is divisible by a, say

$$a_1 a_2 = a\, a'.$$

If we form the two forms

$$\lambda = a_1 a_2\, t_1 + \frac{b - \sqrt{D}}{2}\, t_2,$$
$$\lambda' = a\, t_1 + \frac{b - \sqrt{D}}{2}\, t_2,$$

then it follows that:

$$N(\lambda) = a_1 a_2 (a_1 a_2\, t_1^2 + b\, t_1 t_2 + c\, t_2^2),$$
$$N(\lambda') = a(a\, t_1^2 + b\, t_1 t_2 + a' c\, t_2^2).$$

The two forms $(a_1 a_2,\ b,\ c)$, $(a,\ b,\ a'c)$ are primitive (because a common prime divisor of $a_1 a_2$, b, c would, on account of (15), have to divide Q), and for the absolute norm it follows that:

$$N_a(\lambda) = a_1 a_2 = N(\mathfrak{a}\mathfrak{b}),$$
$$N_a(\lambda') = a;$$

but since λ' is divisible by $\mathfrak{a}\mathfrak{b}$, a must be divisible by $a_1 a_2$, and consequently $a' = 1$.

From this it follows:

3. If a_1, a_2, b are without common divisor and

$$D = b^2 - 4\, a_1 a_2 c,$$

then the form

$$T = (a_1 a_2,\ b,\ c)$$

is compounded from the two forms:

$$U = (a_1,\ b,\ c\, a_2), \quad V = (a_2,\ b,\ c\, a_1)$$

Applying this to two opposite forms

$$U = (a,\ b,\ c), \quad V = (c,\ b,\ a)$$

there results

$$T = (ac,\ b,\ 1),$$

and this is equivalent to the principal form $\left(1,\ 0,\ \frac{-D}{4}\right)$ or $\left(1,\ 1,\ \frac{1-D}{4}\right)$.

4. Two opposite classes, compounded, give the principal class.

In order to compose a class A repeatedly with itself, one chooses $\mathfrak{a}$ in A so that it is relatively prime to D, and denotes by $(a,\ b,\ c)$ the corresponding form. Then a is relatively prime to b. For if d were a divisor of a and b, then d would also be contained in D, hence relatively prime to $\mathfrak{a}$. Then $a : d$ too would be divisible by a, which, if $d > 1$, contradicts the definition of a.

Now one can, however, choose b so that

$$b^2 \equiv D \pmod{4a^n} \tag{17}$$

holds, for an arbitrarily large n. For one can alter b by an arbitrary multiple of $2a$. If we therefore assume (17) to be satisfied for an exponent n, and accordingly put

$$b^2 = D + 4a^n c, \tag{18}$$

then it follows that:

$$(b + 2ha^n)^2 - D \equiv 4a^n(c + hb) \pmod{4a^{n+1}},$$

and hence, if one determines h from the congruence $c + hb \equiv 0 \pmod{a}$, there results a b satisfying the congruence (17) for $n + 1$.

Since now b is relatively prime to a, it follows by 2. that, if $n \geqq m$, then a^m is the least natural number divisible by $\mathfrak{a}^m$, and from this there results the following rule of composition:

5. If the form $(a, b, a^n c) = U$ satisfies the condition (18), and if $m \leqq n$, then

$$U^m = (a^m, b, a^{n-m}c).$$

With this the classes of quadratic forms of the discriminant Δ are united into an Abelian group which is isomorphic with the group of ideal classes, and at the same time a way has been indicated by which one can actually carry out the composition in these groups by means of suitably chosen representatives[1]).

§ 102. Composition of the Orders.

An order $O = [Q]$ in the field Ω is completely determined by the conductor Q. Let O_1, O_2 be two orders with the conductors Q_1, Q_2, and let the greatest common divisor of Q_1 and Q_2 be Q. If then M is the least common multiplum of Q_1 and Q_2, then

$$Q_1 Q_2 = QM. \tag{1}$$

6. The order O belonging to the conductor Q is called composed or compounded from O_1 and O_2.

We set symbolically

$$O = O_1 O_2. \tag{2}$$

The discriminants of these three orders are

$$D = Q^2\Delta, \quad D_1 = Q_1^2\Delta, \quad D_2 = Q_2^2\Delta. \tag{3}$$

If Q_1 and Q_2 are relatively prime, then the order composed from the two is the maximal order O_0.

To carry out the composition of the orders one can proceed just as in the preceding section. We set

$$Q_1 = Qm_1, \quad Q_2 = Qm_2, \tag{4}$$

and then m_1, m_2 are relatively prime to one another.

Let $\mathfrak{a}$ and $\mathfrak{b}$ then denote two ideals which are relatively prime to $Q_1 Q_2$.

If then

$\mathfrak{a}$ equivalent $\mathfrak{a}'$ with respect to O_1,
$\mathfrak{b}$ „ $\mathfrak{b}'$ „ O_2,

then

$$\frac{\mathfrak{a}}{\mathfrak{a}'} = \eta_1 \text{ a number in } O_1,$$
$$\frac{\mathfrak{b}}{\mathfrak{b}'} = \eta_2 \text{ „ „ „ } O_2,$$

i. e. η_1 is congruent to a rational number with respect to the modulus Q_1, and η_2 with respect to the modulus Q_2. Hence both, and consequently their product, are congruent to a rational number with respect to the modulus Q and belong to the order Q. Accordingly we have also

$$\mathfrak{c} = \mathfrak{a}\mathfrak{b} \text{ equivalent to } \mathfrak{c}' = \mathfrak{a}'\mathfrak{b}' \text{ with respect to } O.$$

[1]) The unit in the group of classes is formed by the principal class. Concerning the composition of quadratic forms the following are to be mentioned: Gauss, „Disquisitiones arithmeticae“, Art. 235 ff. Dirichlet, De formarum binariarum secundi gradus compositione (1851), Werke Bd. II, S. 105. Dirichlet-Dedekind, Vorlesungen über Zahlentheorie, 4. Aufl., § 145 f. Dedekind, Crelles Journal, Bd. 129. H. Weber, Göttinger Nachrichten, 9. Februar 1907.

Let a_1 be the least natural number divisible by $\mathfrak{a}$, and let $\mathfrak{b}$ be relatively prime to a_1 (and consequently also to $\mathfrak{a}$); and if a_2 is the least natural number divisible by b, then a_1 and a_2 are also relatively prime. One can then always, whatever b_1, b_2 may be, satisfy the congruences

$$b_1 \equiv m_1 b \pmod{2a_1},$$
$$b_2 \equiv m_2 b \pmod{2a_2}$$

and accordingly assume the basis forms of $\mathfrak{a}$ and $\mathfrak{b}$ in the shape:

$$\mu = a_1 u_1 + m_1 \frac{b - \sqrt{D}}{2} u_2,$$
$$\nu = a_2 v_1 + m_2 \frac{b - \sqrt{D}}{2} v_2,$$

and

$$b^2 - D = 4a_1 a_2 c$$

is divisible by $4a_1a_2$. Then a_1a_2 is the least natural number divisible by $\mathfrak{ab}$, and

$$\lambda = a_1 a_2 t_1 + \frac{b - \sqrt{D}}{2} t_2$$

is a basis form of $\mathfrak{c}$.

Accordingly, the composition of the corresponding quadratic forms of the discriminants D_1, D_2:

$$U = (a_1, m_1 b, m_1^2 a_2 c),$$
$$V = (a_2, m_2 b, m_2^2 a_1 c), \tag{5}$$

gives the form T of the discriminant D:

$$T = (a_1 a_2, b, c). \tag{6}$$

Fifteenth Chapter.
Genera of the Quadratic Forms.

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§ 103. Representation of Numbers by Quadratic Forms.

A natural number m is called properly representable by the primitive form (a, b, c) with the discriminant D if there are two relatively prime numbers x, y which satisfy the equation

$$ax^2 + bxy + cy^2 = m \tag{1}$$

and one can then find a form (m, n, l) equivalent to (a, b, c) whose discriminant

$$D = n^2 - 4ml \tag{2}$$

is. If m is representable by the form (a, b, c), then it is also representable by every equivalent form, and hence by every form of the class A to which (a, b, c) belongs. We then call m representable by the class A. If A, A' are two classes and m, m' two mutually coprime numbers which are properly representable by this class, then the product mm' is properly representable by the composed class AA'. This follows from the compositions of the forms (12), (14), § 101.

From (2) there resulted:

$$n^2 \equiv D \pmod{4m}. \tag{3}$$

If (m, n, l) is imprimitive, then D cannot be a fundamental discriminant. As hitherto, we denote the fundamental discriminant associated with D by Δ and put

$$D = Q^2\Delta,$$

and the greatest common divisor of m, n, l divides Q^2. In what follows we therefore assume that m is relatively prime to Q, and are then certain that the form (m, n, l) determined by (2) is primitive.

Conversely, if the congruence (3) is satisfied, one may set $D = n^2 - 4ml$ and obtains a form of discriminant D, (m, n, l), by which m is properly representable.

1. The necessary and sufficient condition for a number m coprime to Q to be properly representable by a form of discriminant D therefore consists in this, that the congruence

$$x^2 \equiv D \pmod{4m} \tag{4}$$

be solvable.

From a form (a, b, c) one obtains a family of equivalent forms:

$$(a,\ b + 2\lambda a,\ C), \tag{5}$$

in which λ is an arbitrary integer and C is determined by the equation

$$D = (b + 2\lambda a)^2 - 4aC$$

and turns out to be equal to $a\lambda^2 + b\lambda + c$. The system (5) is called a family of parallel forms.

If the congruence (4) is satisfied, then also

$$(x + 2\lambda m)^2 \equiv D \pmod{4m},$$

and we shall regard the whole family of roots of (4) which are congruent with respect to the modulus $2m$ as one root. In this sense let the number of roots of (4), which is accordingly certainly finite, be denoted by $\psi(D, m)$.

2. Every root of (4) gives rise to a family of parallel forms by which the number m is properly representable.

The number $\psi(D, m)$ is easily determined. If first m', m'' are relatively prime, then

$$\psi(D, m'),\ \psi(D, m'') = \psi(D, m'm''). \tag{6}$$

For if

$$x'^2 \equiv D \pmod{4m'}, \quad x''^2 \equiv D \pmod{4m''}, \tag{7}$$

and

$$(8)\qquad \begin{aligned} x &\equiv x' \pmod{2m'},\\ &\equiv x'' \pmod{2m''}, \end{aligned}$$

then $(x^2 - D) : 4$ is divisible by m' and by m'', hence also by $m'm''$, and the congruence

$$(9)\qquad x^2 \equiv D \pmod{4m'm''}$$

is satisfied. Conversely, every root of the congruence (9) gives one root of each of the two congruences (7). Thus one obtains all

roots of (9) and only these, if in (8) one combines the roots x', x'' of (7) in pairs; and from this (6) follows.

By (6) it only remains to determine the function $\psi(D, m)$ for the case that $m = q^k$ is a power of a prime q that does not divide Q. We need the symbol (D, n) in the sense explained in § 85, and we distinguish three cases:

1) If q divides Δ, then $(D, q) = 0$, and the congruence (3) has a solution only when $k = 1$; namely, according as D is even or odd, $x \equiv 0$ or $\equiv q \pmod{2q}$, while there is no solution when $k > 1$, and this holds also for $q = 2$. For in that case Q would have to be odd and Δ divisible by 4, consequently x even. But the congruence

$$\left(\frac{x}{2}\right)^2 \equiv Q^2 \frac{\Delta}{4} \pmod{2^k}$$

is solvable only for $k = 1$, since $\Delta : 4$ is not a discriminant.

2) If $(D, q) = -1$, then the congruence (3) is not solvable, because in that case D is a quadratic non-residue of q or (for $q = 2$) of 8.

3) If $(D, q) = +1$, then D is a quadratic residue of q (or of 8), and the congruence,

$$x^2 \equiv D \mod 4q^k$$

has, as is known from number theory, t w o r o o t s.

We collect these results in a survey as follows:

$$\begin{array}{llll} \text{If } 1) & (D,q) = 0, & \text{then} & \psi(D,q) = 1, \quad \psi(D,q^k) = 0 \quad (k > 1),\\ \text{„ } 2) & (D,q) = -1, & \text{„ „} & \psi(D,q^k) = 0 \\ \text{„ } 3) & (D,q) = +1, & \text{„ „} & \psi(D,q^k) = 2 \end{array} \Bigg\} \quad (k \geqq 1).$$

to which there is still to be added:

$$\psi(D, 1) = 1.$$

By (5), therefore, $\psi(D, m)$ is always equal to zero when a prime divides m for which $(D, q) = -1$, or when a prime factor of D divides m more than once.

In the other cases $\psi(D, q)$ is a power of 2 whose exponent is equal to the number of primes that divide m, but not D.

Between the symbols $\psi(D, m)$ and (D, m) there subsists the following general relation, always under the assumption that m is relatively prime to Q:

Let ε run through all the divisors of m (1 and m included) and e^2 through all the quadratic divisors of m; then

$$(10)\qquad \sum^{e^2} \psi\left(D, \frac{m}{e^2}\right) = \sum^{\varepsilon} (D, \varepsilon).$$

If this relation is correct for two numbers m', m'' that have no common divisor, then it follows for $m = m'm''$. For if e', ε' and e'', ε'' have the same meaning for m' and m'' as e, ε have for m, then $\varepsilon = \varepsilon'\varepsilon''$ are the divisors and $e^2 = e'^2 e''^2$ the quadratic divisors of $m = m'm''$. But by (6) and § 85, (8) we have:

$$\sum^{e'^2}\sum^{e''^2} \psi\left(D, \frac{m'}{e'^2}\right)\psi\left(D, \frac{m''}{e''^2}\right) = \sum^{e^2} \psi\left(D, \frac{m}{e^2}\right),$$

$$\sum^{\varepsilon'}\sum^{\varepsilon''} (D, \varepsilon')(D, \varepsilon'') = \Sigma(D, \varepsilon).$$

Accordingly we need only prove the relation (10) under the assumption that $m = q^k$ is a prime power.

But under this assumption we have:

$$\frac{n}{e^2} = q^{k-2s}, \quad 0 \leqq s \leqq \frac{k}{2},$$
$$\varepsilon = q^s, \qquad s = 0, 1, 2, \ldots, k.$$

If now

1. $(D, q) = 0$, then $\sum^{s} \psi(D, q^{k-2s}) = 1$,
2. $(D, q) = -1$, „ „ $\sum^{s} \psi(D, q^{k-2s}) = 1$ (k even), $= 0$ (k odd),
3. $(D, q) = 1$, „ „ $\sum^{s} \psi(D, q^{k-2s}) = k + 1$,

and in the three cases just as large a value is obtained for the sum

$$\sum_{0,k}^{s} (D, q^s);$$

for in case 1. only the term of this sum corresponding to $s = 0$ has the value 1, the others vanish; in case 2. the terms corresponding to the even s have the value $+1$, the others the value -1, and in case 3. all $k + 1$-terms have the value $+1$.

The relation (10) is thus proved in general.

§ 104. Characters and Genera of Quadratic Forms.

By a fundamental-discriminant divisor δ of a discriminant D we shall understand the following (§ 84):

1. δ is a fundamental discriminant dividing D.

2. The quotient $D : \delta = D_1$ is itself again a discriminant.

An odd fundamental-discriminant divisor can contain no prime factors other than such as divide D, and none of them more than once; on the other hand, any product of distinct odd primes dividing D, furnished with a sign such that $\delta \equiv 1 \pmod 4$, is always a fundamental-discriminant divisor.

Besides these, there occur among the fundamental-discriminant divisors:

$$\begin{aligned} &-4, \text{ if } D \equiv 0, -4 \pmod{16}, \\ &+8, \text{ „} D \equiv 0, 8 \pmod{32}, \\ &-8, \text{ „} D \equiv 0, -8 \pmod{32}, \end{aligned} \tag{1}$$

If, then, we denote the odd fundamental-discriminant divisors by δ_1, all the fundamental-discriminant divisors δ come out as follows:

$$
\begin{aligned}
&1)\ \delta = \delta_1, && D \equiv \ \ 1 \pmod 4,\\
& && \equiv \ \ 4 \pmod{16},\\
&2)\ \delta = \delta_1,\ -4\delta_1, && D \equiv -4 \pmod{16},\\
& && \equiv 16 \pmod{32},\\
&3)\ \delta = \delta_1,\ +8\delta_1, && D \equiv \ \ 8 \pmod{32},\\
&4)\ \delta = \delta_1,\ -8\delta_1, && D \equiv -8 \pmod{32},\\
&5)\ \delta = \delta_1,\ -4\delta_1, && D \equiv \ \ 0 \pmod{32}.\\
&\qquad 8\delta_1,\ -8\delta_1,
\end{aligned}
$$

The number of fundamental-discriminant divisors (1 being counted as a fundamental-discriminant divisor) is accordingly always a power of 2.

3. If we set it equal to 2^λ, then in the cases 1), 2), 3), 4) λ is equal to the number of distinct primes dividing D, and in the case 5) greater by one.

By applying a symbolic multiplication, the fundamental-discriminant divisors of D can be made into an Abelian group.

For if δ_1, δ_2 are two fundamental-discriminant divisors of D, then the product $\delta_1\delta_2$ is likewise a discriminant, but not always a fundamental discriminant. We denote by δ the fundamental discriminant associated with of $\delta_1\delta_2$ and set symbolically

$$\delta = \delta_1\delta_2. \tag{2}$$

If δ_1, δ_2 have no common divisor, then this symbolic multiplication is an actual one. But if they do have a common divisor, then a square factor has still to be cast out of the true product $\delta_1\delta_2$. If we set $\delta_1 = \delta_2$ in (2), then $\delta = 1$, that is, every element of the group of the δ is reciprocal to itself, and together with (2) there holds:

$$\delta_1 = \delta\delta_2. \tag{3}$$

The elements of this group can be represented in the following way by a basis (Bd. II, § 11):

$$\delta = \delta_1^{\varepsilon_1}\delta_2^{\varepsilon_2}\dots\delta_\lambda^{\varepsilon_\lambda}, \tag{4}$$

in which $\varepsilon_1, \varepsilon_2, \dots \varepsilon_\lambda$ have the values 0 or 1, and λ has the meaning given above. $\delta_1, \delta_2, \dots \delta_\lambda$ are, in case 1), the odd prime discriminants $\pm p$ that divide D; in case 2) there is added to these -4, in case 3) 8, in case 4) -8, in case 5) 8 and -8 (or -4 and 8 or -4 and -8).

With the help of the fundamental divisors the characters of the primitive quadratic forms are now defined.

In a class of forms A we choose a representative (a, b, c) in which a is positive and relatively prime to D. If then a number m coprime to D is representable by A, we have

$$m = ax^2 + bxy + cy^2,$$

$$4am = (2ax + by)^2 - Dy^2. \tag{5}$$

Now let δ be a fundamental divisor of D and m relatively prime to δ; then, first for an odd δ, by § 85, (14), (16):

$$(\delta, m) = (\delta, a). \tag{6}$$

But this relation holds also for $\delta = -4\delta_1, +8\delta_1, -8\delta_1$. For in these cases a and m are odd, and by (1) and (5):

$$\begin{aligned} &\delta = -4\delta_1, \quad D \equiv 0, -4 \pmod{16}, \quad am \equiv 1 \pmod 4, \\ &\delta = 8\delta_1, \quad\ D \equiv 0, 8 \pmod{32}, \quad\ am \equiv \pm 1 \pmod 8, \\ &\delta = -8\delta_1, \quad D \equiv 0, -8 \pmod{32}, \quad am \equiv 1, 3 \pmod 8, \end{aligned}$$

from which, by § 85, equation (6) follows for these cases also, and it is thereby proved in general.

4. The value of the symbol (δ, m) therefore depends not on the number m, but only on the class of forms A by which m is representable. It is called the character of this class belonging to the fundamental divisor δ, and is denoted by $\chi(\delta, A)$.

First of all, for every class A whatever and for arbitrary fundamental divisors $\delta, \delta_1, \delta_2$, we have:

$$\chi(1, A) = 1, \tag{7}$$

$$\chi(\delta_1, A)\chi(\delta_2, A) = \chi(\delta_1\delta_2, A), \tag{8}$$

$$\chi(\delta, A_0) = 1, \tag{9}$$

in which $\delta_1\delta_2$ is the symbolic product defined above, and A_0 denotes the principal class by which the number 1 is representable.

If for a class A the characters $\chi(\delta_1, A)$, $\chi(\delta_2, A)$ ... $\chi(\delta_\lambda, A)$ are given, then by (4) and (8) all the $\chi(\delta, A)$ are determined. Each of these characters, however, can be $= +1$ or $= -1$, and so there result 2^λ possible determinations of the characters. But these are not independent of one another.

For from (5) it follows that

$$Dy^2 \equiv (2ax + by)^2 \pmod{4m},$$

and from this it follows, if one assumes m relatively prime to a, and consequently to y:

$$(Dy^2, m) = (\Delta, m) = 1.$$

From this it results that for every class A

$$\chi(\Delta, A) = +1. \tag{10}$$

Now to every fundamental divisor δ there can be determined a definite complementary fundamental divisor δ', different from δ, in such a way that, in the sense of the symbolic multiplication,

$$\delta\delta' = \Delta \tag{11}$$

holds. The complement of δ' is again δ, and every prime divisor which divides δ and δ' at the same time must divide Q.

From this there follows, by (10), for every class A:

$$\chi(\delta, A) = \chi(\delta', A), \tag{12}$$

and accordingly the number of possible determinations of the $\chi(\delta, A)$ is now only $2^{\lambda-1}$.

One unites into a genus (Genus) all classes of forms A in which all the characters $\chi(\delta, A)$ agree, and arrives at the result:

5. The number of existing genera is at most equal to $2^{\lambda-1}$.

Whether the number of existing genera is really so great is a deeper question, to which we shall return later.

If the class A is compounded from the class A', A'', then symbolically

$$A = A'A''.$$

Let m', m'' be representable by the class A', A'', and let us take m', m'' relatively prime; then $m = m'm''$ is representable by A (§ 103), and from this there results for the characters the formula:

$$\chi(\delta, A'A'') = \chi(\delta, A')\chi(\delta, A''). \tag{13}$$

By 1., the symbol $\chi(\delta, A)$ is determined by any number m coprime to δ and properly representable by A, and we therefore define the characters of these numbers m by

$$\chi(\delta, m) = \chi(\delta, A). \tag{14}$$

It is, however, expedient to free oneself also from the assumption that m be relatively prime to δ, and this can be done in the following way. The assumption that m be relatively prime to Q is, however, to be retained.

If m is properly representable by the class A, then we can find in A a representative as follows:

$$\varphi = (m, B, C) \tag{15}$$

We decompose m into two factors

$$m = nn', \tag{16}$$

so that n and n' are coprime and n is relatively prime to δ, n' relatively prime to δ'.

This is always possible, and mostly in several ways, since by assumption m is relatively prime to Q, and therefore m, δ, δ' have no common divisor. One may, for example, take into n the primes dividing δ' in powers as high as those in which they are contained in m, and besides these the prime powers that are coprime to δ.

Then the form φ is compounded from the two forms

$$(n, B, Cn') \qquad (n', B, Cn)$$

whose characters, by (12) and 4., are determined by

$$(\delta, n), \quad (\delta'\ n')$$

and hence, by (13):

$$\chi(\delta, A) = (\delta, n)(\delta', n'). \tag{17}$$

In this, however, the requirement that m be properly representable by A can also be given up; for a common divisor of x, y (relatively prime to Q) gives a

quadratic divisor of m, which one again decomposes into two coprime quadratic factors, of which the one is taken to n, the other to n', and thereby (δ, n) and (δ', n') are not altered.

Accordingly one can determine a character $\chi(\delta, n)$ of an arbitrary number coprime to Q in the following way.

One sets:

$$n = \varepsilon n' \tag{18}$$

and lets ε denote the product of the powers dividing n of those primes which at the same time divide δ; then ε is relatively prime to δ' and n' relatively prime to δ, and one sets:

$$\chi(\delta, n) = (\delta', \varepsilon)(\delta, n'), \tag{19}$$

by which $\chi(\delta, n)$ is uniquely determined and always different from zero. If then (a, b, c) is a representative of the class A, then

$$\chi(\delta, ax^2 + bxy + cy^2) = \chi(\delta, A), \tag{20}$$

in which x, y are arbitrary integers with or without a common divisor, for which $ax^2 + bxy + cy^2$ becomes coprime to Q.

Concerning the symbol $\chi(\delta, n)$ so defined there holds a series of theorems which we have still to derive.

6. If n_1 and n_2 are any two numbers coprime to Q, then

$$\chi(\delta, n_1)\chi(\delta, n_2) = \chi(\delta, n_1 n_2). \tag{21}$$

For if we set, according to (18),

$$n_1 = \varepsilon_1 n_1', \quad n_2 = \varepsilon_2 n_2',$$

then

$$n_1 n_2 = \varepsilon_1 \varepsilon_2 n_1' n_2',$$

and this passes into (18) if one sets $n = n_1 n_2$, $\varepsilon = \varepsilon_1 \varepsilon_2$, $n' = n_1' n_2'$. Thus (21) follows from § 103 (8).

7. If one lets e run through the sequence of divisors of n, and sets $n = ee'$, then

$$\chi(\delta, n)\overset{e}{\Sigma}(D, e) = \overset{e}{\Sigma}(\delta, e)(\delta', e'). \tag{22}$$

If formula (22) holds for $n = n_1$ and $n = n_2$, then it holds, under the assumption that n_1, n_2 are relatively prime, also for $n = n_1 n_2$ [by (21)]; it therefore needs to be proved only for a prime power $n = p^k$, when p does not divide Q

divides. In this case, however, $(D, p^s) = (\Delta, p^s)$ [§ 85 (9)], and so what has to be proved is:

$$\chi(\delta, p^k)\sum_{0,k}^{s}(\Delta, p^s) = \sum_{0,k}^{s}(\delta, p^s)(\delta', p^{k-s}).$$

If now p does not divide δ, then

$$\chi(\delta, p^k) = (\delta, p^k),$$

and if on the right one multiplies each term under the summation sign by $(\delta, p^{k-s})^2 = 1$, one obtains, in agreement with the left-hand side,

$$(\delta, p^k)\sum^{s}(\Delta, p^{k-s}).$$

But if p divides δ, and consequently does not divide δ', then $\chi(\delta, p^k) = (\delta', p^k)$; and if one multiplies in the same way by $(\delta', p^s)^2$, the same result follows, whereby (22) is proved.

8. If $\chi(\delta, n)$ is different from $\chi(\delta', n)$, then

$$\Sigma(D, \bar{e}) = 0, \quad \Sigma(\delta, e)(\delta', e') = 0. \tag{23}$$

For since the two sums do not change when δ is interchanged with δ', this follows from (22).

9. If δ_1, δ_2 are two stem divisors of D and $\delta_1\delta_2$ their symbolic product, then

$$\chi(\delta_1, n)\,\chi(\delta_2, n) = \chi(\delta_1\delta_2, n). \tag{24}$$

Because of (21) this formula needs to be proved only for the case that n is a prime.

But symbolically we have

$$(\delta_1\delta_2)' = \delta_1'\delta_2 = \delta_1\delta_2',$$

and from this (24) follows easily from the definition (19).

§ 105. Application of the Legendre Symbol.

If one wishes to use the more familiar notation of Legendre and Jacobi, the matter takes the following shape. If l denotes an odd prime and a a number not divisible by l, then according to Legendre:

$$\left(\frac{a}{l}\right) = +1, \tag{1}$$

when a is a quadratic residue of l,

$$\left(\frac{a}{l}\right) = -1,$$

when a is a quadratic non-residue of l, and by the extended definition of this symbol introduced by Jacobi,

$$\left(\frac{a}{l\,l'\,l''\ldots}\right) = \left(\frac{a}{l}\right)\left(\frac{a}{l'}\right)\left(\frac{a}{l''}\right)\cdots \tag{2}$$

If furthermore a is odd, then

$$\left(\frac{-1}{a}\right) = (-1)^{\frac{a-1}{2}}, \quad \left(\frac{-2}{a}\right) = (-1)^{\frac{a^2-1}{8}}, \tag{3}$$

and from this, if $a, b, \ldots$ are relatively prime to m:

$$\left(\frac{a}{m}\right)\left(\frac{b}{m}\right)\cdots = \left(\frac{ab\ldots}{m}\right); \tag{4}$$

if D is a discriminant and a a number coprime to $2D$ which is properly representable by a form φ of the discriminant D, and if further l is an odd prime dividing D, then the value of the symbol

$\left(\frac{a}{l}\right)$ depends not on the particular number a, but only on the class of forms A to which φ belongs. The same holds for

$$(5)\qquad \left.\begin{array}{ll}\left(\dfrac{-1}{a}\right) & D\equiv 0,-4\\ \left(\dfrac{2}{a}\right) & D\equiv 0,8\\ \left(\dfrac{-2}{a}\right) & D\equiv 0,-8\end{array}\right\}\pmod{32}.$$

From this the characters of the classes of forms can be composed.

In a later application we shall have to take special account of the case in which D is divisible by 4. We therefore set:

$$(6)\qquad D=-4m^{26},$$

and for this case we obtain, if $l,l',\ldots$ are the different odd primes dividing m, the following characters:

$$(7)\qquad \begin{array}{ll}\left(\dfrac{a}{l}\right),\ \left(\dfrac{a}{l'}\right),\ldots & m\equiv 3 \pmod 4,\\ \left(\dfrac{-1}{a}\right),\ \left(\dfrac{a}{l}\right),\ \left(\dfrac{a}{l'}\right),\ldots & m\equiv 1 \pmod 4,\\ & m\equiv 4 \pmod 8,\\ \left(\dfrac{2}{a}\right),\ \left(\dfrac{a}{l}\right),\ \left(\dfrac{a}{l'}\right),\ldots & m\equiv 6 \pmod 8,\\ \left(\dfrac{-2}{a}\right),\ \left(\dfrac{a}{l}\right),\ \left(\dfrac{a}{l'}\right),\ldots & m\equiv 2 \pmod 8,\\ \left(\dfrac{-1}{a}\right),\ \left(\dfrac{2}{a}\right),\ \left(\dfrac{a}{l}\right),\ \left(\dfrac{a}{l'}\right),\ldots & m\equiv 0 \pmod 8.\end{array}$$

The number of these fundamental characters is, as one sees, equal to the number λ determined more closely in § 104, 3., and the number of sign combinations is equal to 2^λ.

The dependence among these characters may be represented as follows.

If

$$m=n^2m',\quad D=Q^2\Delta=-4n^2m'$$

and n^2 is the greatest square number dividing m, then

$$(8)\qquad \begin{array}{ll}\Delta\equiv -m', & \text{if } m'\equiv 3 \pmod 4,\\ \Delta\equiv -4m' & \text{„ } \quad m'\equiv 1,2 \pmod 4;\end{array}$$

in the first case m' is odd and not equal to 1, hence equal to one of the primes l or equal to a product of several of them, and from

$$\left(\frac{\Delta}{a}\right)=\left(\frac{-m'}{a}\right)=\left(\frac{a}{m'}\right)=1,$$

in which the law of reciprocity of quadratic residues has been applied, a relation between the characters results.

In the second case, for odd m',

$$\left(\frac{-m'}{a}\right)=\left(\frac{-1}{a}\right)\left(\frac{a}{m'}\right)=1,$$

[26] In the notation of Gauss: forms of the first kind of the determinant $-m$.

and this is again a product of several of the characters (7), and if finally $m' = 2m''$ is even, then

$$\left(\frac{-m'}{a}\right) = \left(\frac{\pm 2}{a}\right)\left(\frac{a}{m''}\right),$$

where the sign $\pm$ is so determined that $\pm m'' \equiv 1 \pmod 4$. Thus again the number of genera is at most $= 2^{\lambda-1}$.

§ 106. The Genera of the Ideal Classes.

In § 100 we saw how, from a group of numbers in the field Ω, we can derive divisions of the ideals into classes, and how the class number is to be determined for these divisions.

Here we wish to consider the division into genera from this point of view as well.

In order to be able to extend our considerations at once to the orders, we first exclude from the region of the natural numbers all those numbers which have a common divisor with some arbitrarily assumed number S.

Thereupon we take a positive rational modulus m, and include in S, among others, all the primes that are contained in m.

Thereby all numbers are excluded that have a common divisor with m.

From the remaining numbers and their quotients one can form a group of rational fractions, which we shall denote by Z.

If we replace (following Gauss) a number a^{-1} by the integer which, multiplied by a, yields a number congruent to unity, then we can carry over the theory of congruence with respect to the modulus m without further ado to the fractional numbers Z.

If then a/b and a_1/b_1 are fractions in Z, then $a/b \equiv a_1/b_1$, whenever $ab_1 \equiv a_1 b$ in the ordinary sense.

If, then, we unite all mutually congruent numbers in Z into one class, Z splits into $\varphi(m)$ classes of numbers, each of which can be represented by an integral rational number.

If M is the group of the numbers a from Z which satisfy the condition

$$a \equiv 1 \pmod m$$

then the classes of numbers (modulo m) are the subsidiary groups to M, and we have

$$(Z, M) = \varphi(m). \tag{1}$$

We now consider, as in § 98, the groups O of the integral and fractional numbers ω of the quadratic field Ω, or of an order $[Q]$ of this field, of all of which we

exclude numbers which are not relatively prime to S in the numerator or in the denominator. (In the case of the orders, the prime factors of the conductor Q must be contained in S.)

We now form from M a divisor A, into which we take all numbers a that satisfy a congruence

$$N(\omega) \equiv a \pmod m \tag{2}$$

that is, for which there exists a number ω in O whose norm is congruent to a.

1. These numbers are called norm residues for the modulus m.

From the definition of the norm residues it follows that the norm residues form a group with respect to multiplication and division.

It follows further that every square and every quadratic residue (modulo m) is at the same time a norm residue, and hence that every norm non-residue is at the same time a quadratic non-residue. From the group nature of A it follows:

The product of a norm residue and a norm residue is a norm residue.

The product of a norm residue and a norm non-residue is a norm non-residue.

The group of norm residues always has a finite index (Z, A), for it unites within itself whole classes of numbers with respect to the modulus m, and the number of these is finite.

§ 107. Composition of the norm residue groups.

We wish to consider two moduli m_1, m_2 that are relatively prime to one another, and we denote the groups of norm residues of m_1 and m_2 by A_1 and A_2. If we set for the moment

$$(Z, A_1) = \mu$$

and decompose

$$Z = a_1 A_1 + a_2 A_1 + \cdots + a_\mu A_1, \tag{1}$$

then we may replace the $a_1, a_2, \ldots, a_\mu$ by arbitrary numbers congruent to them with respect to the modulus m_1. Since m_1 and m_2 are assumed to be relatively prime, rational x_i can be determined, whatever the rational numbers c_i may be, from the congruences

$$a_i + x_i m_1 \equiv c_i \pmod{m_2}$$

and one can therefore choose the a_i so that they are contained in A_2. One can therefore represent every number z in Z as a product

a number in A_1 with a number in A_2, which we express by

$$Z = A_1 A_2 \tag{2}$$

in symbolic form.

The intersection A of the groups A_1 and A_2 contains all and only those numbers which are at the same time norm residues of m_1 and of m_2, and we prove that the group of the norm residues m

$$m = m_1 m_2 \tag{3}$$

is A.

For if a is a number which is contained at the same time in A_1 and in A_2, then there are two numbers ω_1, ω_2 in O for which the conditions

$$\begin{aligned} N(\omega_1) &\equiv a \pmod{m_1}, \\ N(\omega_2) &\equiv a \pmod{m_2} \end{aligned} \tag{4}$$

are satisfied. Two rational numbers x_1, x_2 can now be determined in such a way that

$$\begin{aligned} x_1 &\equiv 1 \pmod{m_1}, \quad x_2 \equiv 0 \pmod{m_1}, \\ &\equiv 0 \pmod{m_2}, \qquad \equiv 1 \pmod{m_2}, \end{aligned}$$

and if we then accordingly put

$$\omega = x_1 \omega_1 + x_2 \omega_2$$

we have

$$(5)\qquad \begin{aligned} N(\omega) \equiv N(\omega_1) &\equiv a \pmod{m_1},\\ \equiv N(\omega_2) &\equiv a \pmod{m_2}, \end{aligned}$$

and since m_1 and m_2 are relatively prime, we also have

$$(6)\qquad N(\omega) \equiv a \pmod{m},$$

so that a is a norm residue of m. Conversely, a norm residue of m is at the same time a norm residue of every divisor of m, and hence also of m_1 and m_2.

From this there now follows, by means of Theorems 13 and 14, § 100:

$$(Z, A_1) = (A_1A_2, A_1) = (A_1A_2, A_1A) = (A_2, A),$$
$$(Z, A) = (Z, A_2)(A_2, A) = (Z, A_1)(Z, A_2),$$

and we have the relation:

$$(7)\qquad (Z, A) = (Z, A_1)(Z, A_2).$$

With this the question of the norm residues is reduced to the case in which the modulus is a prime power.

§ 108. Norm residues of prime powers.

Let O be an order group in the field Ω (§ 98) and P, the groups of the rational numbers which are relatively prime to the discri-

minant D of this order and at the same time norm residues of a prime p with respect to this order, i. e. of the numbers a which satisfy the condition

$$(1)\qquad N(\omega) \equiv a \pmod{p}$$

for a number ω of the order O. First of all the following theorem holds:

1. If p does not divide D, then every number a in Z is a norm residue.

This needs to be proved only for the case that a is a quadratic non-residue, since every quadratic residue is at the same time a norm residue.

So let $\mathfrak{p}$ be a prime ideal of the field Ω dividing p.

If then D is a quadratic residue of p, then $\mathfrak{p}$ is of the first degree (§ 92), and every number is congruent to a rational number with respect to the modulus $\mathfrak{p}$.

Since p does not divide D, the ideal $\mathfrak{p}'$ conjugate to $\mathfrak{p}$ is different from $\mathfrak{p}$, and we can find a number ω in O satisfying the congruences:

$$\begin{aligned} \omega &\equiv a \pmod{\mathfrak{p}},\\ &\equiv 1 \pmod{\mathfrak{p}'}\ ^{1)} \end{aligned}$$

(Bd. II, § 166). If then ω' is conjugate to ω, then

$$\omega \equiv a, \quad \omega' \equiv 1 \pmod{\mathfrak{p}},$$

and consequently

$$N(\omega) \equiv a \pmod{\mathfrak{p}}.$$

If however D is a quadratic non-residue of p, then $\mathfrak{p} = p$, and $D^{\frac{p-1}{2}} \equiv -1 \pmod{p}$, hence $\sqrt{D}^p \equiv -\sqrt{D}$ and for every number in O

$$\begin{aligned} \omega^p &\equiv \omega' \pmod{p}, \\ N(\omega) &\equiv \omega^{p+1} \pmod{p}. \end{aligned} \tag{2}$$

If now γ is a primitive root of p in the field Ω, then $c \equiv \gamma^{p+1} \pmod{p}$ is a primitive root of p in the field of rational numbers (Bd. II, § 167).

If we therefore put

$$a \equiv \gamma^{(p+1)x}, \quad \omega \equiv \gamma^x \pmod{p}, \tag{3}$$

then by (2) $N(\omega) \equiv a \pmod{p}$, and our theorem is thus proved.

---

[1] If one puts $\omega = a + \xi\pi Q$, where π is a number divisible by $\mathfrak{p}$ but not by $\mathfrak{p}'$, then the number ξ can be determined from the congruence $a + \xi\pi Q \equiv 1 \pmod{\mathfrak{p}'}$, and ω belongs to the order O on account of the factor Q.

2. If p is an odd prime dividing D, then a is a norm residue then and only then, when a is a quadratic residue of p.

This follows immediately from the condition for the norm residues:

$$x^2 - Dy^2 \equiv 4a \pmod{p} \tag{4}$$

,

which in this case reduces to $x^2 \equiv 4a \pmod{p}$.

The prime 2 does not come into consideration, because $x^2 \equiv a \pmod{2}$ is always solvable; the moduli 4 and 8, however, do. If D is odd, then $D \equiv 1 \pmod{4}$, and the condition for a norm residue a is:

$$\frac{x^2 - Dy^2}{4} \equiv a \pmod{4} \text{ or } \pmod{8},$$

and this congruence is solvable for every odd a (by even x, y), hence:

3. If 4 does not divide D, then every odd number a is a norm residue of 4 and of 8.

If D is divisible by 4, then the condition for a norm residue a is the possibility of the congruence:

$$x^2 - \frac{D}{4}y^2 \equiv a \pmod{4} \text{ or } \pmod{8}. \tag{5}$$

Since a is odd, x and $Dy^2/4$ cannot both be odd.

If one goes through the individual cases accordingly, there results

$$\frac{D}{4} \equiv 0 \pmod 8, \quad a \equiv 1 \pmod 4, \quad a \equiv 1 \pmod 8,$$
$$\frac{D}{4} \equiv 1 \pmod 8, \quad \text{no condition for } a,$$
$$\frac{D}{4} \equiv 2 \pmod 8, \quad a \equiv +1,\ -1 \pmod 8,$$
$$\frac{D}{4} \equiv 3 \pmod 8, \quad a \equiv 1 \pmod 4, \quad a \equiv +1,\ 5 \pmod 8,$$
$$\frac{D}{4} \equiv 4 \pmod 8, \quad a \equiv 1 \pmod 4, \quad a \equiv +1,\ 5 \pmod 8,$$
$$\frac{D}{4} \equiv 5 \pmod 8, \quad \text{no condition for } a,$$
$$\frac{D}{4} \equiv 6 \pmod 8, \quad a \equiv +1,\ 3 \pmod 8,$$
$$\frac{D}{4} \equiv 7 \pmod 8, \quad a \equiv 1 \pmod 4, \quad a \equiv +1,\ 5 \pmod 8.$$

Accordingly we have:

4. If $\frac{1}{4}D \equiv 1 \pmod 8$, then every odd number is a norm residue of 4 and of 8.

If $\frac{1}{4}D \equiv 0 \pmod 8$, then only the numbers of the form $4n+1$ are norm residues of 4, and the numbers of the form $8n+1$ norm residues of 8.

If $\frac{1}{4}D \equiv 2 \pmod 4$, then all odd numbers are norm residues of 4.

If $\frac{1}{4}D \equiv 2$, or $\equiv 6 \pmod 8$, then in the first case the numbers $8n+1$, $8n-1$, in the second case the numbers $8n+1$, $8n+3$ are norm residues of 8.

If $\frac{1}{4}D \equiv 3, 4, 7 \pmod 8$, then all and only the numbers of the form $4n+1$ are norm residues of 4 and of 8.

5. If p is an odd prime, and an a not divisible by p is a norm residue of p^k (for any positive exponent k), then a is also a norm residue of p^{k+1}.

For the proof let

$$N(\omega) = a + p^k b,$$
$$\omega_1 = \omega\,(1 + x\,p^k),$$

where x is rational. Consequently

$$\begin{aligned} N(\omega_1) &= (a + p^k b)(1 + xp^k)^2, \\ &\equiv a + p^k(b + ax) \pmod{p^{k+1}}, \end{aligned}$$

and if x is determined from the congruence $b + ax \equiv 0 \pmod p$, then it follows that

$$N(\omega_1) \equiv a \pmod{p^{k+1}}.$$

6. If D is odd, the same still holds for $p = 2$ as well.

For in this case every odd a is a norm residue of 8. We therefore put

$$N(\omega) = a + 2^k b, \qquad k \geqq 3,$$
$$\omega_1 = \omega\left(1 + 2^k x\,\frac{1+\sqrt{D}}{2}\right),$$
$$N(\omega_1) \equiv (a + 2^k b)(1 + 2^k x) \pmod{2^{k+1}},$$

and if one assumes $ax + b \equiv 0 \pmod 2$, then it follows that

$$N(\omega_1) \equiv a \pmod{2^{k+1}}.$$

7. If D is even and a a norm residue of $2^k \geqq 8$, then a is also a norm residue of 2^{k+1}.

For let us put

$$N(\omega) = a + 2^k b,$$
$$\omega_1 = \omega(1 + 2^{k-1}x),$$

where x is rational. Then $2k - 2 > k$, and consequently

$$N(\omega_1) \equiv (a + 2^k b)(1 + 2^k x) \pmod{2^{k+1}};$$
$$\equiv a + 2^k(b + ax);$$

if therefore

$$ax + b \equiv 0 \pmod 2,$$

then

$$N(\omega_1) \equiv a \pmod{2^{k+1}}.$$

From all this there now results the following theorem:

8. If a number r in Z is a quadratic residue of every odd prime dividing D, and if moreover, for even D,

$$\text{a)} \begin{cases} r \equiv 1 \pmod 8 & \text{if } \dfrac{D}{4} \equiv 0 \pmod 8, \\ r \equiv +1, -1 \pmod 8 & \text{„ } \dfrac{D}{4} \equiv 2 \pmod 8, \\ r \equiv 1, 3 \pmod 8 & \text{„ } \dfrac{D}{4} \equiv 6 \pmod 8, \end{cases}$$

$$\text{b) } r \equiv +1, -5 \pmod 4 \quad \text{„ } \frac{D}{4} \equiv 3, 4, 7 \pmod 8,$$

then r is a norm residue of any arbitrary modulus m coprime to r.

These numbers r form a group R, which we call the group of absolute norm residues.

If b is a fixed quadratic non-residue of the odd prime p, and z a number in Z, then either z or $b^{-1}z$ is contained in the group of norm residues. So if P is the group of norm residues of p^k, then

$$(Z, P) = 1, \quad \text{when } p \text{ does not divide } D,$$
$$(Z, P) = 2, \quad \text{when } p \text{ divides } D,$$

further, if L is the group of norm residues of 2^k:

$$(Z, L) = 1, \quad D \equiv 1 \pmod 4,$$
$$D \equiv 4, 20 \pmod{32},$$
$$(Z, L) = 2, \quad D \equiv 8, 12, 16, 24, 28 \pmod{32},$$
$$(Z, L) = 4, \quad D \equiv 0 \pmod{32}\,^{1}).$$

---

[1]) There is no need to fear that the symbol (Z, P) for the index of a divisor of a group, explained in § 99 and in Bd. II, § 21, will be confused with the extended Legendre-Jacobi symbol similarly denoted in § 85.

9. If, then, we denote by ν the number of the different odd primes dividing D, then by § 98, 12. we have

$$(Z, R) = 2^{\nu}, \quad D \equiv 1 \pmod 4, \quad D \equiv 4 \pmod{16},$$
$$(Z, R) = 2^{\nu+1}, \quad D \equiv 8, 12, 16, 24, 28 \pmod{32},$$
$$(Z, R) = 2^{\nu+2}, \quad D \equiv 0 \pmod{32},$$

or in general

$$(Z, R) = 2^{\lambda},$$

where λ has the same meaning as in § 104, 3.

This number agrees, as one sees, with the number of the fundamental divisors of D, and from 8. it follows that an absolute norm residue r retains this property when r is increased by a multiple of D.

From this the theorem:

10. The absolute norm residues are contained in a finite number of arithmetic progressions which advance by multiples of D.

The odd primes dividing D, and in cases a) and b) of No. 8 the number 8 or 4, are called the characteristic primes and prime powers of the order.

§ 109. The Genera of Ideals.

According to the foregoing we can decompose the totality of the rational numbers z, excluding those which are not relatively prime to D in the numerator or in the denominator, into μ classes (subsidiary groups), where

$$(Z, R) = \mu \tag{1}$$

is the value determined above. Accordingly we have:

$$Z = R + R_1 + R_2 \ldots + R_{\mu-1}. \tag{2}$$

If a number z belongs to one of these subsidiary groups, then all numbers congruent to z with respect to the modulus D belong to the same subsidiary group.

11. The principal class R of the norm residues is a group with respect to multiplication and division.

If r belongs to the principal class R, then two numbers z and rz belong to one and the same class.

The totality of the integral and fractional ideals $\mathfrak{a}$ of the field Ω that are coprime to D also form a group $\overline{O}$.

12. We divide the ideals $\mathfrak{a}$ into genera $G, G_1, G_2, \ldots$ by uniting all ideals $\mathfrak{a}_\nu$ whose norms $N(\mathfrak{a}_\nu)$ are contained in one class R_ν into a genus G_ν.

13. The ideals whose norms are at the same time norm residues r belong to the principal genus G, and the principal genus is itself a group.

If we wish to take the orders into account at the same time, we must also admit equivalence with respect to the order $[Q]$ (§ 100), which of course (for $Q = 1$) includes general equivalence. Then the following holds:

14. The ideals of the principal class belong to the principal genus, and equivalent ideals have the same genus.

For if $\mathfrak{a}$ is equivalent to $\mathfrak{a}_1$, then there is a number η in $[Q]$ for which $\mathfrak{a} = \eta\mathfrak{a}_1$; consequently $N(\mathfrak{a}) = N(\eta)N(\mathfrak{a}_1)$, and $N(\eta)$ is a number in R. Consequently $\mathfrak{a}$ and $\mathfrak{a}_1$ belong, by 13., to the same genus. The genera therefore comprise not merely the individual ideals, but the ideal classes.

In order that a number a from Z can be an ideal norm, a certain condition must be fulfilled. For if

$$a = N(\mathfrak{a}), \tag{3}$$

and $\mathfrak{a}$ is in the first place a primary integral ideal, then by § 91

$$\Delta = b^2 - 4ac, \tag{4}$$

where b, c are integral rational numbers, and consequently

$$(\Delta, a) = +1. \tag{5}$$

If $\mathfrak{a}$ is not primary, but equal to $m\mathfrak{a}_0$ with primary $\mathfrak{a}_0$, and $a_0 = N(\mathfrak{a}_0)$, then $a = m^2 a_0$, and consequently, by § 85, (15), condition (5) must be satisfied here also; and the same follows for fractional numbers as well, if numerator and denominator are relatively prime to D, provided one regards a in (5) as an integral representative of its class with respect to the modulus D.

Since there are now numbers a for which $(\Delta, a) = -1$, not every number in Z can be the norm of an ideal, and the number of the genera is therefore smaller than μ. It must be a divisor of μ, i.e. a power of 2, and consequently we have the theorem:

15. The number g of the genera is at most equal to

$$\frac{1}{2}(Z, R),$$

that is, to half of the number determined in 9.

That this number is really the exact number of the genera we shall prove later. We further remark on this the following:

The fact that for every Δ there are numbers a for which $(\Delta, a) = -1$ follows from the law of reciprocity for quadratic residues. If one can establish the existence of such numbers a without presupposing this law, then conversely a proof of the law of reciprocity can be derived from it. In this way Gauss derived his second proof of this law. (Cf. Dirichlet-Dedekind, § 152 ff.)

If r is a norm residue and δ any stem divisor of D, then

$$N(\omega) = \frac{x^2 - y^2 D}{4} \equiv r \pmod{\delta} \tag{6}$$

and consequently, if r is relatively prime to δ,

$$(\delta, r) = +1. \tag{7}$$

For odd δ this is evident; for the even ones it follows easily in the various cases of § 108. We therefore set quite generally, when z is relatively prime to D:

$$\chi(z) = (\delta, z). \tag{8}$$

Then $(\delta, r) = 1$ for every number r of the group of absolute norm residues, and in general

$$\chi(z)\chi(z_1) = \chi(zz_1). \tag{9}$$

For all ideals $\mathfrak{a}$ of the principal genus which are coprime to D, we have

$$\chi[N(\mathfrak{a})] = +1,$$

and from this it follows that $\chi[N(\mathfrak{a})]$ has one and the same value $\chi(G)$ for all ideals $\mathfrak{a}$ of one genus.

These functions are therefore the characters of the group of the genera.

Sixteenth Chapter.

Class Number in Quadratic Fields.

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§ 110. Fundamental Units in the Orders.

The theory of units, which we have set forth in general in Bd. II, § 191, takes on a simpler shape for the quadratic fields, but must on the other hand be extended and elaborated for the orders.

The integers of the order O with the discriminant D are, by § 96, (6), contained in the form $x_1 + x_2\theta$, where

$$\theta = \tfrac{1}{2}\sqrt{D}, \qquad \text{if } D \equiv 0 \pmod 4,$$
$$\theta = \frac{1+\sqrt{D}}{2}, \quad \text{„} \quad D \equiv 1 \pmod 4,$$

and x_1, x_2 are integral rational numbers.

Such a number ε is a unit if

$$N(\varepsilon) = (x_1 + x_2\theta)(x_1 + x_2\theta') = \pm 1 \tag{1}$$

holds. From this there results the condition

$$x_1^2 - x_2^2 D = \pm 4. \tag{2}$$

If D is negative, then only the positive sign can stand here, and in general there are only the two solutions $x = \pm 2$, $y = 0$ solutions:

$$\begin{array}{lll} D = -4, & x_1 = \pm 2, & x_2 = 0, \\ & x_1 = 0, & x_2 = \pm 1. \\ D = -3, & x_1 = \pm 2, & x_2 = 0, \\ & x_1 = \pm 1, & x_2 = \pm 1. \end{array}$$

From this it follows:

1. For a negative D there are in general only the two units

$$\varepsilon = +1, \qquad \varepsilon = -1,$$

and for two special cases

$$\begin{array}{ll} D = -4: & \varepsilon = +1, -1, +i, -i, \\ D = -3: & \varepsilon = +1, -1, \\ & -\dfrac{1+\sqrt{-3}}{2}, -\dfrac{1-\sqrt{-3}}{2}, +\dfrac{1+\sqrt{-3}}{2}, +\dfrac{1-\sqrt{-3}}{2} \end{array}$$

The numbers of the order O different from zero are then associated in twos, or, in the two exceptional cases, in fours or in sixes.

2. If D is positive, then there is a solution of the Pell equation:

$$T^2 - U^2 D = \pm 4,$$

in which T and U have positive values that are as small as possible. If then

$$\varepsilon = \frac{T + U\sqrt{D}}{2},$$

all units are contained in the form $\pm\varepsilon^{\pm\nu}$, where ν runs through the sequence of numbers $0, 1, 2, \ldots$.

If equation (2) is solvable for the negative sign, then $N(\varepsilon) = -1$, and $\pm\varepsilon^{\pm\nu}$ has a positive norm for even ν and a negative norm for odd ν. But if (2) is solvable only for the upper sign, then all units have positive norm. Among the numbers y of the order O which are not units, however, there are always such with negative norm.

If $N(\varepsilon) = -1$, then for every number y in O with negative norm there is an associated one with positive norm. But if $N(\varepsilon) = +1$, then the numbers y split into two classes, of which the one has positive and the other negative norm, and no number of the one class is associated with one of the other class.

Accordingly we consider only the numbers y of the order O with positive norm, among which y and $\pm\varepsilon^{\pm\nu}y$ are associated only when

$$\varepsilon = \frac{T + U\sqrt{D}}{2}$$

the fundamental unit with positive norm, and hence T, U the least positive solution of

$$T^2 - U^2 D = +4 \tag{3}$$

.

For a positive discriminant there are infinite families of associated numbers, and the point is to single out a definite one from each of these families by a suitable determination. For this the general considerations of Bd. II, § 195 furnish us the means, which we now apply to our special case.

Let, according to § 98,

$$\lambda = a\,t_1 + \frac{b + \sqrt{D}}{2}\,t_2$$

be a basis form in the order $[Q]$ of an ideal $\mathfrak{a}$, which we shall assume to be primary, and let

$$\begin{aligned} y &= a\,x_1 + \frac{b + \sqrt{D}}{2}\,x_2 \\ y' &= a\,x_1 + \frac{b - \sqrt{D}}{2}\,x_2 \end{aligned} \tag{4}$$

with integral rational x_1, x_2 be an integer y in $[Q]$ divisible by $\mathfrak{a}$ and of positive norm, together with its conjugate y'; and since we have assumed $\mathfrak{a}$ to be primary, we have

$$a = N(\mathfrak{a}). \tag{5}$$

Let further

$$\begin{aligned} \varepsilon &= \frac{T + U\sqrt{D}}{2}, \\ \varepsilon' &= \frac{T - U\sqrt{D}}{2} \end{aligned} \tag{6}$$

be the fundamental unit in O together with its conjugate, both of which are positive and have positive norm.

We determine the numbers ξ_1 and ξ_2 from the linear equations:

$$\begin{aligned} \xi_1 \log\varepsilon + \xi_2 &= \log|y|, \\ \xi_1 \log\varepsilon' + \xi_2 &= \log|y'|, \\ \xi_1 \log\frac{\varepsilon}{\varepsilon'} &= \log\left|\frac{y}{y'}\right|, \\ \xi_2 &= \tfrac{1}{2}\log|yy'|, \end{aligned} \tag{7}$$

where $|y|$ denotes the absolute value of y.

If y is replaced by an associated number $\pm\varepsilon^k y$, then ξ_1 passes into $\xi_1 + k$, where k is a positive or negative integer, and one can therefore always choose y among the associated numbers in one way, so that

$$0 \leqq \xi_1 < 1 \tag{8}$$

holds. The number y so determined is then the reduced number.

If we observe further that the equations (7) do not change when y is replaced by $-y$, it follows that:

3. In a family of associated numbers there are always two and only two reduced numbers, which differ only in sign.

From (7) there results, if ξ_1 satisfies the condition (8):

$$0 \leqq \log\left|\frac{y}{y'}\right| < \log\frac{\varepsilon}{\varepsilon'}, \tag{9}$$

and from this

$$1 \leqq \left|\frac{y}{y'}\right| < \frac{\varepsilon}{\varepsilon'}, \tag{10}$$

and conversely there follow from (10) again the equations (7) with the condition (8).

Since we have moreover assumed that y is to have a positive norm, y, y' have the same sign, and this sign is positive if x_2 is positive. For then $y - y' = x_2\sqrt{D}$ is positive, and since $|y| > |y'|$, both y and y' must be positive.

In the special case $x_2 = 0$, y and y' are positive if we assume x_1 to be positive. The ambiguity still remaining in 3. is thereby removed.

Since therefore y and y' are positive, it follows from (10):

$$y'\varepsilon - y\varepsilon' > 0. \tag{11}$$

From which, by (4) and (6):

$$2aUx_1 - (T - bU)x_2 > 0, \qquad x_2 \geqq 0. \tag{12}$$

And these conditions, which we call the isolation conditions, already imply that y', and consequently $N(y)$, is positive. For by (3) we have

$$T > U\sqrt{D},$$

and accordingly there follows from (12):

$$U[2ax_1 + (b - \sqrt{D})x_2] > 0.$$

Thus it is proved:

4. If $\mathfrak{a}$ is an ideal coprime to Q, then every pair of numbers x_1, x_2 satisfying the conditions (12) furnishes us a positive number y divisible by $\mathfrak{a}$ with positive norm of the order $[Q]$

$$y = a\,x_1 + \frac{b + \sqrt{D}}{2}x_2, \tag{13}$$

and among these numbers no two are associated.

§ 111. The Dirichlet Limit Formula.

If t denotes a positive constant, then by the condition:

$$N(y) = a\,(a\,x_1^2 + b\,x_1x_2 + c\,x_2^2) \leqq t, \tag{1}$$

and, when the discriminant is positive, by the isolation conditions, there is bounded, in a plane in which x_1, x_2 are rectangular coordinates, a region F_t which for negative discriminant is bounded by an ellipse, and for positive discriminant by an arc of a hyperbola and by straight lines. The points whose coordinates x_1, x_2 are integral rational numbers are called lattice points.

To a whole system of associated numbers whose norm is positive and less than t there then corresponds, for positive discriminant, one lattice point in F_t, and for negative discriminant two, and in the two exceptional cases four and six lattice points.

If Z_t denotes the number of lattice points in the region F_t, and V the area of the region F_t, then [Bd. II, § 194, (6)]:

$$V = Z_t\,t^{-1} + R_t\,t^{-\frac{1}{2}}, \tag{2}$$

in which R_t does not become infinite as t increases infinitely.

For negative discriminant one obtains V from the well-known area of the ellipse:

$$V = \frac{2\pi}{a\sqrt{-D}}, \tag{3}$$

and for positive discriminant one obtains it most simply by introducing, according to § 110, (7), in place of x_1, x_2 the variables of integration ξ_1, ξ_2, whose limits are $0, 1$ and $-\infty, 0$:

$$V = \frac{\log\varepsilon}{a\sqrt{D}}. \tag{4}$$

Accordingly, from (2) there results:

$$\begin{aligned} \mathrm{Lim}_{t=\infty}\frac{Z_t}{t} &= \frac{\log\varepsilon}{a\sqrt{D}}, \qquad D > 0, \\ &= \frac{2\pi}{a\sqrt{-D}}, \quad D < 0. \end{aligned} \tag{5}$$

But according to Bd. II, § 196, 4. we have

$$\mathrm{Lim}_{s=1}\sum\frac{s-1}{(Ny)^s} = \mathrm{Lim}_{t=\infty}\frac{Z_t}{t},$$

and thus from (5) there results:

$$\begin{aligned} \mathrm{Lim}_{s=1}\sum\frac{s-1}{(Ny)^s} &= \frac{1}{a\sqrt{D}}\log\varepsilon, \quad D > 0, \\ &= \frac{2\pi}{a\sqrt{-D}}, \qquad D < 0. \end{aligned} \tag{6}$$

Here y runs through the sequence of the integers of the order $[Q]$ that are divisible by a primary ideal $\mathfrak{a}$ coprime to Q, where, however, of a family of associated numbers only a single representative, which may now be chosen arbitrarily, is to be retained (for negative D two or four or six).

Here, associated numbers (with respect to the order $[Q]$) means numbers whose quotient is a unit of the order Q.

We now wish to exclude from the numbers y all those that are not relatively prime to Q. Then the sum (6) is modified in the following way: If r is a prime factor of Q, then all numbers of the order $[Q]$ that are not coprime to r are divisible by r, since they are indeed congruent to a rational number with respect to the modulus r.

If we therefore set

$$y = r\, y_1, \tag{7}$$

then y_1 is divisible by $\mathfrak{a}$, but it need not belong to the order $[Q]$; it need only belong to the order $\left[\frac{Q}{r}\right] = [Q_1]$. If, for positive discriminant, ε_1 is the fundamental unit of the order $[Q_1]$, then

$$\varepsilon = \varepsilon_1^{\lambda} \tag{8}$$

is a power of ε_1, and we have $\sqrt{D} = r\sqrt{D_1}$. The numbers

$$y = r\, y_1\, \varepsilon_1^t, \quad t = 0, 1, 2, \ldots, \lambda - 1 \tag{9}$$

all have the same norm; they are associated according to $[Q_1]$, but not according to $[Q]$. Accordingly we obtain from (6):

$$\mathrm{Lim}_{s=1} \sum \frac{s-1}{(N y_1)^s} = \frac{r}{a\sqrt{D}} \frac{\log \varepsilon}{\lambda};$$

in order to obtain from this the sum relating to the numbers (9), one has, since $N(\varepsilon_1) = 1$, $N(r) = r^2$, to multiply by λ and to divide by r^2. One obtains in this way:

$$\mathrm{Lim} \sum \frac{s-1}{(Ny)^s} = \frac{1}{r\, a\sqrt{D}} \log \varepsilon.$$

If one subtracts this from the sum (6) and proceeds in the same way with all prime factors r of Q, there results:

$$\mathrm{Lim} \sum \frac{s-1}{(Ny)^s} = \prod \Big(1 - \frac{1}{r}\Big) \frac{1}{a\sqrt{D}} \log \varepsilon, \qquad D > 0, \tag{10}$$

and still more simply for a negative discriminant:

$$= \prod \Big(1 - \frac{1}{r}\Big) \frac{\pi}{a\sqrt{-D}}, \qquad D < 0,$$

in which the sum extends only over those numbers y that are coprime to Q, and the product Π relates to all prime factors of r. Only in the two exceptional cases $D = -4$, $D = -3$ is the right-hand side of the last formula still to be divided by 2 or by 3.

If one decomposes y into ideal factors

$$y = \mathfrak{a}\, \mathfrak{m}, \tag{11}$$

then $\mathfrak{m}$ runs through all ideals of a definite ideal class according to the order $[Q]$ that are coprime to Q. This class is, if $\mathfrak{a}$ belongs to the class A,

$$M = A^{-1},$$

and among the numbers

$$N(y) = a\,N(\mathfrak{m})$$

in the case of the positive discriminant each $N(\mathfrak{m})$ occurs only once, in the case of a negative discriminant twice or four or six times.

Accordingly there results from (10):

$$\text{(12)}\qquad \begin{aligned} \operatorname{Lim}_{s=1} \sum \frac{s-1}{(N\mathfrak{m})^s} &= \prod\Big(1-\frac{1}{r}\Big)\frac{1}{\sqrt{D}}\log\varepsilon, \quad D>0 \\ &= \prod\Big(1-\frac{1}{r}\Big)\frac{\pi}{\sqrt{-D}}, \qquad D<0 \end{aligned}$$

and in the two exceptional cases $D=-4$, $D=-3$ the last expression is to be divided by 2 or by 3.

Here $\mathfrak{m}$ runs through the totality of the ideals of a class (with respect to $[Q]$), and one sees that this limiting value does not depend on this particular class.

It is sometimes expedient to exclude from the sum on the left-hand side of formula (12) not merely those ideals of $\mathfrak{m}$ which have a divisor in common with Q, but also those which are not coprime to Δ. Formula (12) holds unchanged under this assumption if by the r of the right-hand side one understands not only the prime divisors of Q, but all the prime divisors of D.

For if $\mathfrak{r}$ is a prime ideal which divides D but not Q, then $N(\mathfrak{r}) = r_1$ is a natural prime which divides D but not Q, and $N(\mathfrak{rm}) = r_1\,N(\mathfrak{m})$. If, then, one further removes from the sum on the left-hand side of y the terms corresponding to the $\mathfrak{rm}$, then on the right the factor

$\Big(1-\frac{1}{r_1}\Big)$ is added.

5. The limiting value in formula (12) depends only on the discriminant D, not on the particular class M, and is always different from zero.

§ 112. Class Number.

We now assume a function $F(z)$, which moreover need only be defined for integer values of z, and of which we suppose that the infinite series

$$\text{(1)}\qquad \overset{\mathfrak{a}}{\Sigma} F(N\mathfrak{a}),$$

in which $\mathfrak{a}$ runs through the totality of the ideals of the field Ω (or only a part of them), is unconditionally convergent. When it is a question of orders $[Q]$, we wish to exclude from the $\mathfrak{a}$ all those which are not relatively prime to Q.

Any natural number m coprime to Q then occurs among the $N(\mathfrak{a})$, by § 93, 4.

$$\overset{n}{\Sigma}(\Delta, n)$$

times, if n runs through the divisors of m, and we obtain the formula:

$$\text{(2)}\qquad \sum F(N\mathfrak{a}) = \overset{m}{\Sigma}\overset{n}{\Sigma}(\Delta, n)\,F(m),$$

in which m runs through all the positive integers relatively prime to Q, and n, for each m, runs through the divisors of m.

If we arrange the right side of (2) according to n, then for a given n all the multiples of n occur among the m, and we can also set:

$$\Sigma F(N\mathfrak{a}) = \overset{m}{\Sigma}\overset{n}{\Sigma}(\Delta, n)\, F(mn), \tag{3}$$

in which now m and n, independently of one another, assume all natural numerical values that have no divisor in common with Q.

If in particular F is such that

$$F(mn) = F(m)\, F(n) \tag{4}$$

holds, then we can also represent (3) in this way:

$$\overset{\mathfrak{a}}{\Sigma} F(N\mathfrak{a}) = \overset{m}{\Sigma} F(m) \overset{n}{\Sigma}(\Delta, n)\, F(n). \tag{5}$$

In the sums over m and over n those numbers are excluded that have a divisor in common with Q.

Likewise, on the left side those ideals are excluded that are not relatively prime to Q.

We can apply formula (5) first of all in order to determine the class number from the formulae of the preceding section. In this way one also obtains the ratio of the class number in the field to the class number with respect to the order $[Q]$. But since we have already determined this ratio in another way (§ 100), we shall here restrict ourselves to the class number of the field, i.e. we shall set $Q = 1$.

If in (5) we set $F(z) = z^{-s}$, then the presupposed unconditional convergence takes place as long as $s > 1$. But if we multiply by $s - 1$ and let s pass over into 1, we can make use of § 111, (12).

We apply this formula in the first version, in which $\mathfrak{m}$ runs through the ideals of the class M coprime to Q, and r denotes the prime factors of Q. These therefore drop out entirely for $Q = 1$, and $\mathfrak{m}$ in the summation formula:

$$\begin{aligned} \sum \frac{s-1}{(N\mathfrak{m})^s} &= \frac{1}{\sqrt{\Delta}} \log \varepsilon, \quad \Delta > 0 \\ &= \frac{\pi}{\sqrt{-\Delta}}, \qquad \Delta < 0 \end{aligned} \tag{6}$$

is no longer subject to any restriction with regard to divisibility .

Every part of the sum on the left side of formula (5) which refers to an ideal class then gives the same contribution to the sum, and we obtain:

$$\begin{aligned} \mathrm{Lim}_{s=1}(s-1) \sum \frac{1}{m^s} \sum \frac{(\Delta, n)}{n^s} &= h\, \frac{1}{\sqrt{\Delta}} \log \varepsilon, \quad \Delta > 0 \\ &= h\, \frac{\pi}{\sqrt{-\Delta}}, \qquad \Delta < 0. \end{aligned} \tag{7}$$

Here h is the class number of the field Ω, and in order not to have to cite the two exceptional cases $\Delta = -4$, $\Delta = -3$ again and again, we shall stipulate that in these two cases, in which the class number is 1, the value $\frac{1}{2}$ or $\frac{1}{3}$ is to be understood by h.

As often as n runs through a complete system of residues to the modulus Δ , we have $\Sigma(\Delta, n) = 0$ [§ 85, (22)], and consequently the sum $\Sigma(\Delta, n)$, when the n are arranged in order of magnitude, always remains finite, however far n may increase. Accordingly the theorem Bd. II, § 196, 1. is applicable, according to which

$$\mathrm{Lim}_{s=1} \sum \frac{(\Delta, n)}{n^s} = \sum \frac{(\Delta, n)}{n}$$

holds. On the other hand

$$\mathrm{Lim}(s-1)\Sigma\frac{1}{m^s} = 1$$

(Bd. II, § 196, p. 723), and from (7) there results:

$$(8) \qquad \begin{aligned} h \log \varepsilon &= \sqrt{\Delta} \sum_{1,\infty}^{n} \frac{(\Delta, n)}{n}, \qquad \Delta > 0 \\ h\pi &= \sqrt{-\Delta} \sum_{1,\infty}^{n} \frac{(\Delta, n)}{n}, \quad \Delta < 0. \end{aligned}$$

The sum

$$\sigma = \Sigma \frac{(\Delta, n)}{n}$$

can be represented in finite form. For, by § 86, (2):

$$(9) \qquad \sqrt{\Delta}\,(\Delta, n) = \overset{k}{\Sigma}(\Delta, k)\, e^{-\frac{2nk\pi i}{\Delta}},$$

in which k runs through a complete system of residues to the modulus Δ and, for a negative discriminant, $\sqrt{\Delta} = i\sqrt{-\Delta}$ is to be set. Accordingly it follows:

$$\sigma = \frac{1}{\sqrt{\Delta}} \overset{k}{\sum}(\Delta, k) \sum^{n} \frac{1}{n}\, e^{-\frac{2nk\pi i}{\Delta}},$$

and since σ is real:

$$\begin{aligned} \Delta > 0. \qquad \sigma &= \frac{1}{\sqrt{\Delta}} \overset{k}{\sum}(\Delta, k) \sum^{n} \frac{1}{n} \cos \frac{2nk\pi}{\Delta}, \\ \Delta < 0. \qquad \sigma &= \frac{1}{\sqrt{-\Delta}} \overset{k}{\sum}(\Delta, k) \sum^{n} \frac{1}{n} \sin \frac{mk\pi}{-\Delta}. \end{aligned}$$

The two infinite series in n have, however, determinate values, which Abel derived in his memoir on the binomial series from the power development of $\log(1 - z)$, and which follow also from the theory of Fourier series. If k is then taken positive and between 0 and $\pm\Delta$, there results:

$$\begin{aligned} \sum \frac{1}{n} \cos \frac{2nk\pi}{\Delta} &= -\log\left(2 \sin \frac{k\pi}{\Delta}\right), \\ \sum \frac{1}{n} \sin \frac{2nk\pi}{-\Delta} &= \frac{\pi}{2}\left(1 - \frac{2k}{-\Delta}\right). \end{aligned}$$

From (8) there follows therefore [with regard to $\Sigma(\Delta, k) = 0$]:

$$
(10)\qquad
\begin{aligned}
h\log\varepsilon &= -\overset{k}{\Sigma}(\Delta,k)\log\sin\frac{k\pi}{\Delta}, && \Delta>0,\\
h &= \frac{1}{\Delta}\overset{k}{\Sigma}(\Delta,k)\,k, && \Delta<0.
\end{aligned}
$$

For the case of the negative discriminant one can recast the expression for h in such a way that it takes the form of an integer.

If in the first place Δ is odd ($\equiv 1 \bmod 4$), one can decompose the sequence of numbers k thus:

$$
\begin{matrix} v, & -\Delta - v, \\ 2v, & -\Delta - 2v, \end{matrix}
\qquad 0 < v < -\frac{\Delta}{2},
$$

and one thus obtains, since here $(\Delta,-v) = -(\Delta,v)$ (§ 85), two forms of the expression h:

$$
\begin{aligned}
h &= \frac{2}{\Delta}\overset{v}{\Sigma}(\Delta,v)\,v + \overset{v}{\Sigma}(\Delta,v),\\
h &= \frac{4}{\Delta}\Sigma(\Delta,2v)\,v + \overset{v}{\Sigma}(\Delta,v),
\end{aligned}
$$

from which, if one multiplies the first by 2 and the second by $(\Delta,2)$ and subtracts:

$$
(11)\qquad [2-(\Delta,2)]\,h = \overset{v}{\Sigma}(\Delta,v).
$$

If, then, $(\Delta,2) = -1$, the right-hand side must be divisible by 3.

For the case of an even Δ one decomposes k into

$$
v, \quad -\frac{\Delta}{2}+v, \qquad 0 < v < \frac{\Delta}{2}
$$

and applies the formula:

$$
\left(\Delta,\ -\frac{\Delta}{2}+v\right) = -(\Delta,v),
$$

which follows easily from the theorems of § 85.

Then one finds first:

$$
h = \frac{1}{2}\overset{v}{\Sigma}(\Delta,v),
$$

and if one decomposes the v once more into

$$
(12)\qquad
\begin{aligned}
&\mu, \quad -\frac{\Delta}{2}-\mu: \qquad 0<\mu\leqq\frac{-\Delta}{4},\\
&h = \Sigma(\Delta,\mu).
\end{aligned}
$$

The formula for positive discriminants can be transformed by means of the theory of the division of the circle, which we do not wish to go into here.

§ 113. The Number of Genera.

Formula (5) of the preceding section permits the determination of the exact number of genera after Dirichlet. By δ we understand any stem divisor of D, and we set:

$$(1)\qquad \begin{aligned} F(z) &= \frac{(\delta, z)}{z^s}, \quad \text{where } z \text{ is relatively prime to } D, \\ F(z) &= 0, \qquad \text{if } z \text{ and } D \text{ have a common divisor.} \end{aligned}$$

Then $(\delta, N\mathfrak{a})$ is one of the characters of the genus determined by $\mathfrak{a}$. If we therefore denote this character by (δ, A), it follows from § 112, (5):

$$(2)\qquad \overset{A}{\Sigma}(\delta, A)\overset{\mathfrak{a}}{\Sigma}\frac{1}{(N\mathfrak{a})^s} = \Sigma\frac{(\delta, m)}{m^s}\Sigma\frac{(\delta', n)}{n^s}.$$

In this, $\delta' = \delta\Delta$, and since n is relatively prime to D, it can also be replaced by the stem of $\delta\Delta$ (according to the symbolic multiplication in § 104). If $\delta = 1$, then $\delta' = \Delta$, and if $\delta = \Delta$, then $\delta' = 1$.

Apart from the two cases $\delta = 1$, $\delta = \Delta$, the sums on the right-hand side of (2) therefore retain a finite value for $s = 1$ (which, according to the preceding section, are expressible by class numbers of the discriminants δ, δ').

On the left-hand side of (2), in the sum over $\mathfrak{a}$ the symbol $\mathfrak{a}$ runs through the totality of the ideals of the individual classes,

and the sum $\overset{\mathfrak{a}}{\Sigma}\frac{s-1}{(N\mathfrak{a})^s}$, extended over a class A, has for $s = 1$ a definite limiting value different from zero, which by § 111, 5. is the same for all classes. From this it follows by 2.:

$$(3)\qquad \overset{A}{\Sigma}(\delta, A) = 0,$$

except when $\delta = 1$ or $= \Delta$,

$$(4)\qquad \overset{A}{\Sigma}(\delta, A) = h,$$

when $\delta = 1$ or $= \Delta$ [because $(\Delta, A) = 1$, § 104, (10)].

Further, for every class A_0 of the principal genus $(\delta, A_0) = 1$, while for every class A which does not belong to the principal genus there is at least one δ_1 such that $(\delta_1, A) = -1$. From this it follows, if 2^ν is the number of fundamental divisors,

$$(5)\qquad \begin{aligned} \overset{\delta}{\Sigma}(\delta, A_0) &= 2^\nu, \\ (\delta_1, A)\overset{\delta}{\Sigma}(\delta, A) &= \overset{\delta}{\Sigma}(\delta\delta_1, A), \end{aligned}$$

and since $\delta\delta_1$ runs through the totality of the fundamental divisors together with δ (if one reduces $\delta\delta_1$ according to the symbolic multiplication in § 104):

$$(\delta_1, A)\overset{\delta}{\Sigma}(\delta, A) = \overset{\delta}{\Sigma}(\delta, A),$$

and hence, when $(\delta_1, A) = -1$:

$$(6)\qquad \overset{\delta}{\Sigma}(\delta, A) = 0.$$

Accordingly we can determine the double sum

$$\overset{\delta}{\Sigma}\overset{A}{\Sigma}(\delta, A)$$

in two ways. By (3), (4) there results:

$$\sum_{\delta}\sum_{A}(\delta, A) = 2h, \tag{7}$$

and by (5), (6):

$$\sum_{\delta}\sum_{A}(\delta, A) = 2^{\nu} g, \tag{8}$$

where g denotes the number of classes of the principal genus (and consequently of every genus). The comparison gives:

$$h = 2^{\nu-1} g, \tag{9}$$

and therefore the theorem:

6. The number of genera is exactly equal to $2^{\nu-1}$, i. e. equal to $\frac{1}{2}(Z, R)$, § 108, 9.

THIRD BOOK.

COMPLEX MULTIPLICATION.

Seventeenth Chapter.
Elliptic Functions and Quadratic Forms.

§ 114. Singular periods of the doubly periodic functions.

If we denote by $\varphi(u)$ a doubly periodic function with the periods ω_1, ω_2, then, if n denotes an integer, $\varphi(nu)$ has the same periods, and upon this rests the multiplication of the elliptic functions which was considered in the sixth chapter. The question now arises whether a multiplier μ cannot be determined in yet another way, so that $\varphi(\mu u)$ has the periods ω_1, ω_2; that is, whether a multiplication exists also with a non-integer multiplier. This will be the case then and only then, when for a system of integers a, b, c, ∂ the equations hold:

$$\mu\,\omega_1 = a\,\omega_1 + b\,\omega_2,$$
$$\mu\,\omega_2 = c\,\omega_1 + \partial\,\omega_2.$$

If, in order to explore more closely the requirement contained in them, we put:

$$\omega = \frac{\omega_2}{\omega_1}$$

and assume that ω has a positive imaginary part, then it follows:

$$\mu = a + b\omega,$$

$$\omega = \frac{c + \partial\omega}{a + b\omega}. \tag{1}$$

So long as ω is regarded as variable, this equation is only possible if $b = c = 0$, $a = \partial$, and μ is therefore an integer.

If, on the other hand, equation (1) is not an identity, then ω is the root of a quadratic equation with integral coefficients:

$$b\,\omega^2 + (a - \partial)\,\omega - c = 0, \tag{2}$$

hence, if we put

$$a\,\partial - b\,c = n$$

$$m = -4\,b\,c - (a - \partial)^2 = 4\,n - (a + \partial)^2$$

then we have

$$m \equiv 0 \quad \text{or} \quad \equiv -1 \pmod 4,$$

and we obtain:

$$\omega = \frac{-a + \partial + \sqrt{-m}}{2\,b},$$

$$\mu = \frac{a + \partial + \sqrt{-m}}{2},$$

$$n = \frac{a + \partial + \sqrt{-m}}{2} \cdot \frac{a + \partial - \sqrt{-m}}{2},$$

$$\mu^2 - (a + \partial)\,\mu + n = 0.$$

In order that ω may have a positive imaginary part, m, and all the more therefore n, must be positive, and μ is a complex, integral, algebraic number. This explains the designation complex multiplication.

If now, conversely, ω satisfies a quadratic equation

$$A\,\omega^2 + B\,\omega + C = 0 \tag{3}$$

in which A, B, C are integers, and the discriminant

$$D = B^2 - 4\,A\,C \tag{4}$$

is negative, then ω is not real, and in one of the two roots of (3) the imaginary part is positive. This one shall be taken for ω. Then ω is called a root of the quadratic form (A, B, C).

Without prejudice to generality, A, B, C may be assumed to be without a common divisor and A, C positive. Infinitely many relations of the form (1) [or (2)] can then be set up for ω. One has only to put, when x is an integer different from zero:

$$\begin{aligned} b &= A\,x, \\ c &= -\,C\,x, \\ a - \partial &= B\,x. \end{aligned} \tag{5}$$

If one further sets

$$a + \partial = y, \tag{6}$$

then it follows:

$$\begin{aligned} a &= \frac{y + B\,x}{2}, \\ b &= A\,x, \\ c &= -C\,x, \\ \partial &= \frac{y - B\,x}{2}, \end{aligned} \tag{7}$$

from which, if

$$a\,\partial - b\,c = n$$

is set, there results:

$$4\,n = y^2 - D\,x^2. \tag{8}$$

The number n is thus decomposed into the two complex integer factors

$$n = \frac{y + x\sqrt{D}}{2} \cdot \frac{y - x\sqrt{D}}{2}$$

. For ω and μ there is found further:

$$\omega = \frac{-B + \sqrt{D}}{2\,A} = \frac{2\,C}{-B - \sqrt{D}},$$

$$\mu = a + b\,\omega = \frac{y + \sqrt{D}\,x}{2}.$$

The numbers x, y introduced here are bound to the single condition that $y + B\,x$, and consequently $y - B\,x$ as well, be even numbers, so that a, ∂ become integers according to (7). If therefore B is even, then y must be even, while x is arbitrary; if B is odd, then x and y are to be assumed both even or both odd. By (4) this distinction can also be expressed thus:

1) If $D \equiv 0 \pmod 4$, then $y \equiv 0 \pmod 2$,
2) If $D \equiv 1 \pmod 4$, then $y \equiv x \pmod 2$.

If, for given A, B, C, the numbers a, b, c, ∂ are to be without a common divisor, then still other conditions come in:

From (5), (6) it follows that every common divisor of a, b, c, ∂ is also a divisor of x and y; conversely every common divisor of x and y is also a divisor of b, c, $2\,a$, $2\,\partial$. If therefore a, b, c, ∂ are to be without a common divisor, then x and y

no greater common divisor than 2. If x and y have the greatest common divisor 2, then b and c are even, and a, b, c, ∂ are then and only then relatively prime, when a and ∂, and hence also $n = a\,\partial - b\,c$, are odd.

If we disregard the uninteresting case $n = 1$, equation (8) has no solution satisfying these conditions in which $x = 0$. But if we change the signs of x, y at the same time, then by (10) a, b, c, ∂ pass over into $-a$, $-b$, $-c$, $-\partial$, and equation (1) is not altered. The transformation $\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$ therefore remains unchanged.

For brevity we now call a solution of equation (8) a proper solution, and $n = \frac{1}{4}(y^2 - D\,x^2)$ a proper representation of n, if x and y satisfy the following conditions:

x is positive.

The greatest common divisor of x and y is 1 or 2.

If it is $= 2$, then n is odd.

If now ω satisfies a given equation (3), then by every system $\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$ a proper solution of (8) is uniquely determined. For by (5) x is the greatest common divisor of b, c, $a - \partial$, and y is determined by (6); and conversely, from a proper solution of (8) the system a, b, c, ∂ is completely determined by (7). But the question is still to be decided whether two different proper solutions x, y and x', y' can lead to equivalent systems $\begin{pmatrix} a, & b \\ c, & \partial \end{pmatrix}$, $\begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix}$. By equivalent systems are here to be understood, according to § 28, § 53, two such for which

$$\begin{pmatrix} a', & b' \\ c', & \partial' \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} a. & b \\ c, & \partial \end{pmatrix} \tag{9}$$

holds, when $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ is a linear transformation, and hence

$$\alpha\,\delta - \beta\,\gamma = 1. \tag{10}$$

If we compose both sides of equation (9) on the right with $\begin{pmatrix} \partial, & -b \\ -c, & a \end{pmatrix}$, there results:

$$\begin{pmatrix} a'\,\partial - b'\,c, & -a'\,b + b'\,a \\ c'\,\partial - \partial'\,c, & -c'\,b + \partial'\,a \end{pmatrix} = \begin{pmatrix} n\,\alpha, & n\,\beta \\ n\,\gamma, & n\,\delta \end{pmatrix},$$

or, if for a, b, c, ∂ we put the expressions (7) and for a', b', c', ∂' the corresponding ones

$$a' = \frac{y' + B\,x'}{2}, \quad b' = A\,x',$$

$$c' = -\,C\,x', \qquad \partial' = \frac{y' - B\,x'}{2}$$

we obtain:

$$\begin{pmatrix} \dfrac{y\,y' - D\,x\,x' + B(x'y - y'x)}{4}, & A\,\dfrac{x'y - y'x}{2} \\ -\,C\,\dfrac{x'y - y'x}{2}, & \dfrac{y\,y' - D\,x\,x' - B(x'y - y'x)}{4} \end{pmatrix} = \begin{pmatrix} n\,\alpha, & n\,\beta \\ n\,\gamma, & n\,\delta \end{pmatrix}.$$

From this it follows:

$$\begin{aligned} 4\,n\,\alpha &= y\,y' - D\,x\,x' + B(x'y - y'x), \\ 2\,n\,\beta &= A(x'y - y'x), \\ 2\,n\,\gamma &= -\,C(x'y - y'x), \\ 4\,n\,\delta &= y\,y' - D\,x\,x' - B(x'y - y'x), \end{aligned}$$

$$\begin{aligned} 2\,n(\alpha - \delta) &= B(x'y - y'x), \\ 2\,n(\alpha + \delta) &= y\,y' - D\,x\,x'. \end{aligned}$$

Now since A, B, C have no common divisor, it follows that $x'y - y'x$ must be divisible by $2n$, and by the last equation $y\,y' - D\,x\,x'$ is divisible by $2n$ as well. We therefore set

$$\begin{aligned} x'y - y'x &= 2\,n\,\xi, \\ y\,y' - D\,x\,x' &= 2\,n\,\eta \end{aligned} \tag{11}$$

and obtain:

$$\alpha = \frac{\eta + B\,\xi}{2}, \qquad \beta = A\,\xi,$$

$$\gamma = -\,C\,\xi, \qquad \delta = \frac{\eta - B\,\xi}{2},$$

from which, by (10), it follows:

$$\eta^2 - D\,\xi^2 = 4. \tag{12}$$

If we solve the equations (11) for x' and y', it follows:

$$\begin{aligned} 2\,x' &= \xi\,y + \eta\,x, \\ 2\,y' &= \eta\,y + D\,\xi\,x. \end{aligned} \tag{13}$$

If $-D > 4$, then (12) has only the two solutions

$$\xi = 0, \qquad \eta = \pm 2, \tag{14}$$

and from this it follows, since x and x' are to be positive:

$$x' = x, \quad y' = y, \quad (\eta = +2).$$

But if $-D = 4$, then (12) has in addition the solution

$$\xi = \pm 1, \quad \eta = 0,$$

and from (13) it follows:

$$y' = \mp 2\, x,$$
$$x' = \pm \frac{y}{2},$$

and the sign of η is to be determined so that x' becomes positive. If finally $-D = 3$, then besides (14) we have the solutions

$$\xi = \pm 1, \quad \eta = \pm 1,$$

in which at first both signs are arbitrary. But then from (13) it follows:

$$\pm 2\, x' = x \pm y,$$
$$\pm 2\, y' = y \mp 3\, x,$$

and one can choose the signs in two ways so that x' becomes positive. We thus have the theorem:

I. For a given equation (3), every proper solution of the equation (8) leads to a transformation $\left(\begin{smallmatrix} a, & b \\ c, & \partial \end{smallmatrix}\right)$ of the nth order. Different solutions of (8) lead in general to non-equivalent systems $\left(\begin{smallmatrix} a, & b \\ c, & \partial \end{smallmatrix}\right)$. In the two exceptional cases $D = -4$ and $D = -3$, in each case two or three different solutions of (8) lead to equivalent systems.

If we denote by k a number which is in general equal to the number of proper solutions of (8), but for $D = -4$ and $D = -3$ is half or a third of the number of these solutions, then we can state Theorem I. as follows:

II. From an equation (3) we can derive k non-equivalent systems $\left(\begin{smallmatrix} a, & b \\ c, & \partial \end{smallmatrix}\right)$ which satisfy the equation (1).

§ 115. The singular values of the invariant $j(\omega)$.

The question that first claims our interest is that of the values of the modular functions of ω for the

particular values of the argument ω that we considered in the preceding section. On these depend the moduli of the elliptic functions which admit a complex multiplication. After Kronecker they are called singular moduli. Accordingly we shall speak also of the singular values of the modular functions, and in particular of the singular invariants $j(\omega)$, and we understand by these the values that these functions take when ω is the root (with positive imaginary part) of a quadratic form with negative discriminant.

If ω satisfies equation (3), § 114, and a, b, c, ∂, n are determined as in the preceding section, then there follows first of all:

$$j(\omega) = j\left(\frac{c + \partial\, \omega}{a + b\, \omega}\right), \tag{1}$$

and if therefore

$$F_n(u, v) = 0 \tag{2}$$

is the invariant equation belonging to the degree of transformation n (§ 69), then (2) is satisfied if a, b, c, ∂ are without a common divisor and

$$u = j(\omega), \quad v = j\left(\frac{c + \partial\,\omega}{a + b\,\omega}\right) = u \tag{3}$$

is set. Thereupon we arrive at the first principal theorem of this theory:

III. The singular value

$$u = j(\omega)$$

is a root of the algebraic equation:

$$F_n(u, u) = 0. \tag{4}$$

If conversely u is a root of equation (4), then, when ω is determined from the equation $j(\omega) = u$, one of the values of v that satisfy equation (2) is likewise $= u$, and there subsists therefore an equation of the form (1); by § 53 this has the consequence that ω is equivalent to $\dfrac{c + \partial\,\omega}{a + b\,\omega}$, and from this there results an equation of the form § 114, (3). Thus we have the complement to III.:

IV. Every root of (4) is a singular invariant.

If τ is a variable with positive imaginary part, and v likewise a variable, then by § 69 $F_n[v, j(\tau)]$ splits into the linear factors

$$v - j\left(\frac{c + \partial\,\tau}{a + b\,\tau}\right),$$

in which $\left(\begin{smallmatrix} a, & b \\ c, & \partial \end{smallmatrix}\right)$ runs through a complete system of representatives of non-equivalent transformations of degree n, and consequently $F_n(u, u)$ can be decomposed into the factors

$$j(\tau) - j\left(\frac{c + \partial\,\tau}{a + b\,\tau}\right) \tag{5}$$

when $u = j(\tau)$ is set.

But from Theorem II., § 114, it follows that, if we set $\tau = \omega$, k of the factors (5) vanish. We can, however, conclude still further that the quotient

$$\frac{j(\tau) - j\left(\dfrac{c + \partial\,\tau}{a + b\,\tau}\right)}{j(\tau) - j(\omega)} \tag{6}$$

remains finite and different from zero for $j(\tau) = j(\omega)$. For if numerator and denominator are differentiated with respect to τ, and $\tau = \omega$ is then set, the limiting value results:

$$1 - \frac{n}{(a + b\,\omega)^2} = \frac{2\sqrt{D}\,x}{y + \sqrt{D}\,x},$$

which is different from zero.

The invariant $j(\omega)$ remains unchanged when ω is replaced by an equivalent number, but it has a different value for every other number ω. It therefore belongs not to the individual form (A, B, C), but to the whole class of equivalent forms, and is for that reason called the class invariant.

Further, the number k depends not on the coefficients A, B, C, but only on D and n, and is therefore the same for all primitive forms of discriminant D. We now introduce the following

function of the variable u: let $\omega_1, \omega_2, \ldots, \omega_h$ denote a complete system of non-equivalent numbers of discriminant D, that is, the system of the roots of a system of non-equivalent forms (A, B, C) of discriminant D. We set

$$H_m(u) = [u - j(\omega_1)]\,[u - j(\omega_2)]\,\ldots\,[u - j(\omega_h)], \tag{7}$$

in which, as we shall now often do,

$$m = -D \tag{8}$$

has been set, so that m denotes a positive integer.

This function, which is very important for what follows, is called the class function. The roots of the equation

$$H_m(u) = 0 \tag{9}$$

are the class invariants of the discriminant $D = -m$, and this equation is therefore called the class equation.

The degree h of the class equation is equal to the number of primitive classes of the quadratic forms of the discriminant D. If we now consider the quotients (6), then from all this there results:

V. The function $F_n(u, u)$ is divisible by $[H_m(u)]^k$, and the quotient is relatively prime to $H_m(u)$.

Accordingly the function $F_n(u, u)$ can be decomposed into factors, and it follows, if C is a constant equal to the coefficient of the highest power u^N of u:

$$F_n(u, u) = C\,H_m^k(u)\,H_{m'}^{k'}(u)\,H_{m''}^{k''}(u)\ \ldots, \tag{10}$$

if the product extends over all the positive integers $m, m', m'', \ldots$ for which the equation § 114, (8)

$$4n = y^2 + m\,x^2 \tag{11}$$

admits proper solutions, and $k, k', k'', \ldots$ denotes in each case the number of these solutions (with the modification indicated there in the two exceptional cases $D = -4$, $D = -3$).

The degree N of the equation $F_n(u, u)$ is to be determined first.

If we choose the representatives in (5) as in § 69, then

$$F_n[j(\tau), j(\tau)] = \Pi\left[j(\tau) - j\left(\frac{c + \partial\,\tau}{a}\right)\right]. \tag{12}$$

Here a, ∂ denoted all the pairs of positive integers satisfying the condition $a\,\partial = n$, and c runs through a system of residues with respect to the modulus a, with the exclusion of those values which are not relatively prime to the greatest common divisor e of a and ∂, so that the number of values of c belonging to a decomposition of n into the two factors a and ∂ is equal to $a\,\varphi(e)$:e. If now N is the degree of $F_n(u, u)$ and C the coefficient of the highest power of u, then the development begins of $F_n[j(\tau), j(\tau)]$ in powers of $q = e^{\pi i \tau}$ with Cq^{-2N} [§ 69, (4)].

If then we develop the factors of the right-hand side of (12) in the same way, we can determine both C and N.

For a factor of the right-hand side of (1)

$$j(\tau) - j\left(\frac{c+\partial\tau}{a}\right)$$

we have the following beginnings of the development

$$\begin{array}{lll} 1. & -q^{-\frac{2\partial}{a}} e^{-\frac{2\pi i c}{a}}, & \partial > a, \\ 2. & q^{-2}, & \partial < a, \\ 3. & q^{-2}\left(1 - e^{-\frac{2\pi i c}{a}}\right), & \partial = a. \end{array}$$

Let us first assume that n is not a square; then case 3. does not occur, and we have

$$N = \sum_{\partial>a}^{\partial} \frac{\partial}{e}\varphi(e) + \sum_{\partial<a}^{a} \frac{a}{e}\varphi(e),$$

or, what is the same,

$$N = 2\sum_{\partial>a}^{\partial} \frac{\partial}{e}\varphi(e). \tag{13}$$

By 1., C is in any case an nth root of unity, and since it is at the same time a rational number, it must be $= \pm 1$.

If then n is a square, $\varphi(\sqrt{n})$ factors of the form 3. occur, and there results:

$$N = 2\sum_{\partial>a}^{\partial} \frac{\partial}{e}\varphi(e) + \varphi(\sqrt{n}). \tag{14}$$

Here too C is indeed different from zero, but not equal to ± 1. In this case we do not need to determine the value of C more closely. (It is always, as follows from the theory of the division of the circle, a divisor of $\sqrt{n}$).

If in particular n is a prime, then

$$N = 2n. \tag{15}$$

One can always dispose of x and y in such a way that, the conditions of § 114 being observed, $n = \frac{1}{4}(y^2 + mx^2)$ does not become a square. To this end let x be taken divisible by 4, and y by 2 but not by 4, and moreover let y have no odd common divisor with mx^2. If then $\frac{1}{4}(y^2 + mx^2)$ is a square M^2, then

it has an odd square prime divisor p^2 which does not divide y^2, and we have

$$(y + 4\lambda p)^2 + mx^2 = 4M^2 + 8\lambda yp + 16\lambda^2 p^2,$$

and this number, if λ is not divisible by p, is indeed divisible by p, but not by p^2, and so cannot be a square. Moreover, λ can still be so disposed of that $y + 4\lambda p$ has no odd common divisor with x, so that the conditions of § 114 are satisfied. Accordingly, for every m, $j(\omega)$ satisfies an equation $F_n(u, u)$ in which $C = \pm 1$, and from this it follows, by Bd. II, § 149:

VI. The class invariants are integral algebraic numbers.

§ 116. Class Number Relations.

If we equate the degree, as it results from what precedes for both parts of equation (10), § 115, we obtain the formula

$$N = hk + h'k' + h''k'' + \cdots, \tag{1}$$

$$\sum hk = 2\sum^{\partial} \frac{\partial}{e}\varphi(e) + \varphi(\sqrt{n}), \tag{2}$$

in which $\varphi(\sqrt{n}) = 0$ is to be set when n is not a square, and N is determined by (13), (14) or (15) of the preceding section. This is the Kronecker class number relation, to which we wish to give a somewhat more convenient form[27].

Beside n we consider all the values n' that arise from n by removing a square factor, form equation (2) for them, and add together all the results so obtained. Here only the value $n' = 1$ is to be excluded, in case n is a square, because $F_1(u, u) = 0$.

Let δ^2, then, be any square factor of n different from n itself, and

$$n = n'\delta^2.$$

If we decompose n' into two factors $n' = a'\partial'$ and denote by e' the greatest common divisor of ∂' and a', then we have

$$\Sigma N' = 2\sum^{\delta} \sum_{\partial' > a'}^{\partial'} \frac{\partial'}{e'}\varphi(e') + \sum^{n'} \varphi(\sqrt{n'}), \tag{3}$$

if δ^2 runs through all the square factors of n (with the possible exception of n itself).

If we now set

$$\partial'\delta = \partial, \quad a'\delta = a, \quad e'\delta = e,$$

then $a\partial = n$, and e is the greatest common divisor of a and ∂; at the same time $\partial' > a'$ has as its consequence that $\partial > a$. Conversely, from every decomposition $a\partial$ of n and from every divisor e' of e one obtains a decomposition $a'\partial'$ of n', where

$$\delta = \frac{e}{e'}, \quad \partial' = \frac{\partial e'}{e}, \quad a' = \frac{ae'}{e}$$

holds. Accordingly

$$\Sigma N' = 2\sum_{\partial > a}^{\partial} \frac{\partial}{e} \sum \varphi(e') + \sum^{n'} \varphi(\sqrt{n'}). \tag{4}$$

In this we now make use of the number-theoretic relation: $\sum \varphi(\partial) = n$, in which the sum refers to all divisors of n. If we first assume that n is not a square, then none of the n' are squares either, and $\varphi(\sqrt{n'}) = 0$. Hence, since $\sum \varphi(e') = e$, we obtain

$$\Sigma N' = 2\sum_{\partial > \sqrt{n}} \partial. \tag{5}$$

If on the other hand n is a square, then e' still runs through all the divisors of e, but $\sqrt{n'}$ runs through all the divisors of $\sqrt{n}$, with the exception of 1; from this there follows for this case:

$$\Sigma N' = 2\sum_{\partial > \sqrt{n}} \partial + \sqrt{n} - 1. \tag{6}$$

In both cases ∂ runs through all the divisors of n that are greater than $\sqrt{n}$.

We have now still to form the sum of the expressions Σkh for the various values of n' or δ. But if we set $n = \delta^2 n'$, then from every proper solution of

$$4n' = y'^2 + mx'^2 \tag{7}$$

[27] The class number relations, of which the one derived here is only the simplest case, were discovered by Kronecker (Crelles Journal, Bd. 57) and considerably generalised by Gierster (Mathematische Annalen, Bd. 21, 22) and Hurwitz (ibid. Bd. 25).

we obtain, by multiplication by δ^2, a solution (improper, if $\delta > 1$) of

$$4n = y^2 + mx^2, \tag{8}$$

namely

$$y = \delta y', \quad x = \delta x'.$$

If, conversely, any solution x, y of (8) is given, then $4n$ is divisible by the square of every common divisor of x and y. We denote by δ the greatest common divisor of x, y whose square is at the same time a divisor of n, and which is therefore either equal to the greatest common divisor of x, y or equal to half of it, and we set

$$y = \delta y', \quad n = \delta x',$$

and thus obtain a proper solution of (7).

If, then, we put for $-m$ all discriminants for which equation (8) has solutions at all, and understand by k the number of these solutions (with positive x), and further by $h(m)$ the class number of primitive forms of discriminant $-m$, then the formula results

$$\Sigma k\, h(m) = 2 \sum \partial, \tag{9}$$

if n is not a square,

$$= 2 \sum \partial + \sqrt{n} - 1,$$

if n is a square.

Here it must still be recalled that in the two exceptional cases $m = 3$, $m = 4$ only the third part or the half of the number of proper solutions of (8) was understood by k in § 115.

We will now do justice to this circumstance by understanding by k the total number of solutions of (8), but setting

$$h(3) = \tfrac{1}{3}, \quad h(4) = \tfrac{1}{2}$$

[instead of $h(3) = 1$, $h(4) = 1$].

It is further to be observed that in forming the sum (9) the solutions belonging to the value $n' = 1$ have not been counted. This comes into consideration only in the case where n is a square. Equation (7):

$$4 = y'^2 + mx'^2$$

then has only the following proper solutions:

$$\begin{array}{lllll} m = 3: & x' = 1, & y' = \pm 1, & k = 2, & h = \tfrac{1}{3}, \\ m = 4: & x' = 1, & y' = 0, & k = 1, & h = \tfrac{1}{2}, \end{array}$$

and hence, if one wished to remove this exception, there would still have to be added to (9):

$$\Sigma kh, \text{ referred to } n' = 1, = \tfrac{1}{2} + \tfrac{2}{3} = \tfrac{7}{6}.$$

Accordingly the formula now reads

$$\Sigma k\, h(m) = 2\Sigma \partial, \tag{10}$$

n not a square,

$$= 2\Sigma\partial + \sqrt{n} + \tfrac{1}{6},$$

n a square.

We decompose the sum on the left side of (10) into its individual parts by taking each solution of equation (8) separately. We then obtain a sum of expressions $h(m)$, in which, however, the same m occurs as often as (8) has solutions. One of the solutions, however, gives

$$m = \frac{4n - y^2}{x^2},$$

and this occurs once if $y = 0$, and twice (with $+y$ and $-y$) if y is different from zero. But the number of primitive classes of the discriminant $-m$ is equal to the number of non-primitive classes of the discriminant $-x^2m$ of divisor x; and so, if we now understand by $h(m)$ (in distinction from the previous formula) the total number of classes of the discriminant $-m$ (primitive and imprimitive), then we obtain from (10):

$$
\begin{aligned}
&h(4n) + 2h(4n-1) + 2h(4n-4) + 2h(4n-9) + \cdots \\
&= 2\sum_{\partial > \sqrt{n}} \partial, \quad (n \text{ not a square}), \\
&= 2\sum_{\partial > \sqrt{n}} \partial + \sqrt{n} + \tfrac{1}{6}, \quad (n \text{ a square}).
\end{aligned} \tag{11}
$$

In this, the sum of the $h(4n - y^2)$ on the left is to be continued as long as $4n - y^2$ remains positive, and on the right ∂ runs through all divisors of n that are greater than $\sqrt{n}$.

It is, however, to be observed here that the forms $(x, 0, x)$ are to be reckoned only with $\frac{1}{2}$ each, and (x, x, x) with $\frac{1}{3}$, because they arise from the representations of $4n$ by the discriminant $m = -4, -3$.

§ 117. Arithmetic Nature of the Class Function $H_m(u)$.

We now return once more to equation (10), § 115, in order to derive from it the nature of the coefficients of the class equation $H_m(u) = 0$. First, however, we consider the two special cases: $m = 4$ and $m = 3$.

If we set, as in § 54, (4)

$$\gamma_2(\omega) = \sqrt[3]{j(\omega)}, \quad \gamma_3(\omega) = \sqrt[2]{j(\omega) - 1728},$$

then it follows from § 54, (14):

$$\gamma_3(i) = -\gamma_3(i), \quad \text{hence} \quad \gamma_3(i) = 0.$$

Furthermore, by the same formula, if we set

$$\varrho = \frac{-1 + i\sqrt{3}}{2}, \qquad \varrho + 1 = \frac{-1}{\varrho}$$

then:

$$\gamma_2(\varrho + 1) = e^{-\frac{2\pi i}{3}}\gamma_2(\varrho) = \gamma_2\left(-\frac{1}{\varrho}\right) = \gamma_2(\varrho),$$

consequently $\gamma_2(\varrho) = 0$, and we obtain

$$
\begin{aligned}
&H_3(u) = u, && j(\varrho) = 0, \\
&H_4(u) = u - 1728, && j(i) = 1728.
\end{aligned} \tag{1}
$$

The two functions $H_3(u)$ and $H_4(u)$ therefore have integral rational numerical coefficients. This property we can now establish in general for all H_m by complete induction.

First we observe that, once it has been proved that $H_m(u)$ has rational coefficients, it follows at once that the coefficients are integral rational numbers (the first = 1). For these coefficients are integral algebraic numbers (by § 115, VI.), and consequently, if they are rational, integral rational numbers. Now we consider the largest among the values of m that occur in equation (10), § 115; these are

$$
\begin{aligned}
&m = 4n, && x = 1, \quad y = 0, && k = 1, \\
&m = 4n - 1, && x = 1, \quad y = \pm 1, && k = 2.
\end{aligned}
$$

Accordingly $F_n(u,u)$ contains the two factors $H_{4n}(u)$, $H_{4n-1}(u)^2$, and otherwise nothing but factors $H_m(u)$ in which $m < 4n-1$. If we assume that it has already been proved of the latter that they have rational coefficients, it follows that

$$H_{4n}(u)\, H_{4n-1}(u)^2 = \Phi(u)$$

has rational coefficients, and hence, in order to find $H_{4n-1}(u)$, one has to seek the greatest common divisor of $\Phi(u)$ and $\Phi'(u)$, which is done by rational calculation. But thereby $H_{4n}(u)$ too is found by rational means, and since $H_3(u)$ and $H_4(u)$ have rational coefficients, it follows in general:

VII. The class functions $H_m(u)$ have integral rational numerical coefficients.

§ 118. Composition of the Quadratic Forms.

Since in the algebraic theory of the class equation we must rely upon the theory of the composition of the quadratic forms, we preface the following remarks about it:

Two primitive forms $\varphi_1 = (a_1, b_1, c_1)$, $\varphi_2 = (a_2, b_2, c_2)$ of discriminant $-m$ are called concordant (einig, concordantes) if $a_1, a_2, \frac{1}{2}(b_1 + b_2)$ are without a common divisor; φ_1 and φ_2 are therefore certainly concordant if a_1, a_2 are already without a common divisor, and since, when only the classes k_1, k_2 of φ_1, φ_2 are given, one can always choose φ_1, φ_2 in their classes in this way, it follows that in any two classes (which may also be identical) one can always find two concordant forms. If, then, this assumption holds, it follows from

$$\tfrac{1}{4}(b_1^2 - b_2^2) = \tfrac{1}{2}(b_1 - b_2) \cdot \tfrac{1}{2}(b_1 + b_2) = a_1 c_1 - a_2 c_2,$$

that $\frac{1}{2}(b_1 - b_2)$ is divisible by the greatest common divisor δ of a_1, a_2, and from this, that the two congruences

$$b' \equiv b_1 \pmod{2a_1}, \quad b' \equiv b_2 \pmod{2a_2} \tag{1}$$

can always be satisfied.

For from the first of them it follows that $b' = b_1 + 2\lambda a_1$, and one has therefore to determine λ from the congruence

$$\tfrac{1}{2}(b_1 - b_2) + a_1 \lambda \equiv 0 \pmod{a_2}$$

which is always solvable. If μ is the least common multiple of a_1 and a_2, so that $a_1 a_2 = \mu\delta$, then by (1) b' may still be replaced by $b = b' + 2\mu h$, where h is an arbitrary integer. Now since $\frac{1}{4}(b_1^2 + m)$ is divisible by a_1, and $\frac{1}{2}(b_2^2 + m)$ by a_2, $\frac{1}{4}(b'^2 + m)$ is divisible by μ, and if we determine h from the congruence

$$\frac{b'^2 + m}{4\mu} + b'h \equiv 0 \pmod{\delta}$$

which is always solvable, since b' is by assumption relatively prime to δ, then $b^2 + m = 4a_1 a_2 c$ becomes divisible by $4a_1 a_2$. The two classes k_1, k_2 are then represented by

$$\varphi_1 = (a_1, b, a_2 c), \quad \varphi_2 = (a_2, b, a_1 c),$$

and the form

$$\varphi = \varphi_1 \varphi_2 = (a_1 a_2, b, c)$$

is compounded from $\varphi_1 \varphi_2$. The form φ belongs to a class k which is said to be compounded from k_1, k_2.

Let us assume $\varphi = (a, b, c)$ in such a way that a is relatively prime to m; then b is also relatively prime to a (because of $m = 4ac - b^2$). If $a = a_1 a_2$, then (a_1, b, ca_2), (a_2, b, ca_1) are likewise concordant primitive forms, and one obtains the composition

$$(a, b, c) = (a_1, b, ca_2),\ (a_2, b, ca),$$

and by repeated application of this theorem one arrives at the result:

I. The form $\varphi = (a, b, c)$ can, if a is relatively prime to m, be composed of forms whose first coefficients are primes, namely the prime factors of a.

One of these components, say

$$(p, b, acp^{-1}),$$

can be replaced by an equivalent form $(p, b', c'p^g)$ whose third coefficient $c'p^g$ is divisible by an arbitrarily high power of p.

To prove this we need the symbol $\sim$ for equivalence, and we have

$$(p, b, cp^{g-1}) \sim (p, b + 2\lambda p^g, c'p^g),$$

in which c', on account of the equality of the discriminants, is to be determined from

$$pc' = c + \lambda b + \lambda^2 p^g$$

Now since b is not divisible by p, λ can be determined from the congruence $c + \lambda b \equiv 0 \pmod{p}$, and c' turns out to be an integer. With this, having regard to (I.), the following is proved:

II. In every class k of primitive forms of discriminant D one can find a representative φ which can be composed of forms

$$P = (p, b, p^g c),$$

in which p is a prime not dividing D, and g is an arbitrary exponent.

The form P can easily be composed with itself, for, in the sense of the compositions, when $g > \nu$:

$$(p^\nu, b, p^{g-\nu}c)(p, b, p^g c) = (p^{\nu+1}, b, p^{g-\nu-1}c),$$

and consequently, by the inference from $\nu - 1$ to ν:

$$P^\nu = (p^\nu, b, p^{g-\nu}c). \tag{2}$$

In the chain of compositions

$$P,\ P^2,\ P^3,\ \ldots \tag{3}$$

there may occur a form that belongs to the principal class, that is, one equivalent to $(1,\ 0,\ \frac{1}{4}m)$ or $[1,\ 1,\ \frac{1}{4}(m+1)]$. If this happens for P^ε, then p^ε is representable by the principal form, and there is a proper representation

$$4\,p^\varepsilon = y^2 - D\,x^2. \tag{4}$$

If conversely such a representation is possible, then there is a form $(p^\varepsilon,\ b,\ c)$ in the principal class.

If ε is the least positive exponent for which equation (4) is properly solvable, then ε is called the index of p, and p belongs to the exponent ε.

If

$$P = (p,\ b,\ p^{\varepsilon-1}c), \tag{5}$$

then

$$b^2 \equiv D \pmod{4\,p^\varepsilon},$$

and when this condition is satisfied, then in the form $(p,\ b,\ c')$ of discriminant D the third coefficient is divisible by $p^{\varepsilon-1}$. The form (5) is preserved if we replace b by any number b' congruent to it with respect to the modulus $2\,p^\varepsilon$. If therefore a is not divisible by p, and

$$\varphi = (a,\ B,\ C)$$

is any form of discriminant D concordant with P, then the two congruences

$$x \equiv b \pmod{2\,p^{\varepsilon}}, \quad x \equiv B \pmod{2\,a}$$

can be satisfied at the same time, and if one then puts this x in place of b and B, it follows that:

III. If l is the class of P and k an arbitrary class of discriminant D, then in l and k one can choose the representatives:

$$(p,\, b,\, a\,c\,p^{\varepsilon-1}), \quad (a,\, b,\, c\,p^{\varepsilon}), \tag{6}$$

and in the compounded class lk one obtains a representative

$$(a\,p,\, b,\, p^{\varepsilon-1}c). \tag{7}$$

If therefore

$$\omega = \frac{-b+\sqrt{-m}}{2\,a}$$

is a root of the class k, then $\omega\colon p$ is a root of the class lk.

§ 119. The Discriminant of the Invariant Equation.

We now raise the question when an invariant equation

$$F_p(v,\, u) = 0, \tag{1}$$

in which p is a prime, has two or more equal roots v for $u = j(\omega)$, or when among the $p+1$ quantities

$$j(p\,\omega), \quad j\left(\frac{\omega+c}{p}\right), \quad c = 0,\, 1,\, \ldots,\, p-1 \tag{2}$$

two or more occur that are equal to one another. This takes place then and only then, when among the $p+1$ quantities

$$p\,\omega, \quad \frac{\omega+c}{p} \tag{3}$$

two equivalent ones occur. Accordingly ω must in any case satisfy a quadratic equation with integral coefficients and negative discriminant $D = -m$, which we assume in the form:

$$A\,\omega^2 + B\,\omega + C = 0, \quad 4\,A\,C - B^2 = m > 0, \tag{4}$$

in which A, B, C are without a common divisor.

If we replace ω by an equivalent number, and hence (4) by an equivalent equation, the quantities (2) are only interchanged among themselves. The question as to the number of equal roots of (1) is therefore not affected by this, and we may accordingly assume that A is not divisible by p. If now, in the first place, $p\,\omega$ is equivalent to $\dfrac{\omega+c}{p}$, then:

$$\frac{\omega+c}{p} = \frac{\gamma+\delta\,p\,\omega}{\alpha+\beta\,p\,\omega}, \quad \alpha\,\delta - \beta\,\gamma = 1. \tag{5}$$

If we write this equation thus:

$$\beta\,p\,\omega^2 + (\alpha+\beta\,c\,p-\delta\,p^2)\,\omega + c\,\alpha - \gamma\,p = 0, \tag{6}$$

then it follows by comparison with (4) (since A was to be indivisible by p) that α is divisible by p, and must therefore be $= p\,\alpha'$, and that an integer x exists which satisfies the conditions:

$$\beta = A\,x, \quad \alpha' + \beta\,c - \delta\,p = B\,x, \quad c\,\alpha' - \gamma = C\,x. \tag{7}$$

If we further put

$$\alpha' - \beta\, c + \delta\, p = y, \tag{8}$$

then it follows:

$$\begin{aligned} 2\,\alpha' &= B\,x + y, \\ 2\,(\beta\, c - \delta\, p) &= B\,x - y. \end{aligned} \tag{9}$$

and from

$$p\,\alpha'\,\delta - \beta\,\gamma = 1 \tag{10}$$

there then follows

$$y^2 + m\,x^2 = 4. \tag{11}$$

The value $x = 0$ is to be excluded, because otherwise $\beta = 0$ would have to hold, which contradicts equation (10), and we may, without loss of generality, assume x to be positive. Since moreover m must be either $\equiv 0$ or $\equiv 3$ with respect to the modulus 4, there remain for the satisfaction of (11) only the following two possibilities:

$$1. \quad m = 3, \quad x = 1, \quad y = \pm 1.$$

Since for the discriminant -3 there is only one class of forms, we may assume $A = 1$, $B = 1$, $C = 1$, that is, we may set for ω the imaginary third root of unity $e^{\frac{2\pi i}{3}}$, and we then obtain among the quantities (3) three equivalent ones by taking $y = +1$ and $= -1$:

$$p\,\omega, \quad \frac{\omega}{p}, \quad \frac{\omega + 1}{p}, \tag{12}$$

as is also recognised from the equations

$$\frac{\omega}{p} = \frac{-1}{p + p\,\omega}, \quad \frac{\omega + 1}{p} = \frac{-1}{p\,\omega}$$

from which the equivalence of the three quantities (12) is evident.

$$2. \quad m = 4, \quad x = 1, \quad y = 0.$$

Because here too there is only one class of forms, we may assume $B = 0$, $A = C = 1$, that is, $\omega = i$, and we find the two equivalent values:

$$p\,\omega, \quad \frac{\omega}{p},$$

as also follows from the equation

$$p\,\omega = \frac{-p}{\omega}$$

Further, let two of the values (3)

$$\frac{\omega + c}{p}, \quad \frac{\omega + c'}{p}$$

be equivalent, so that an equation holds:

$$\frac{\omega + c}{p} = \frac{\gamma\, p + \delta\,(\omega + c')}{\alpha\, p + \beta\,(\omega + c')}, \quad \alpha\,\delta - \beta\,\gamma = 1, \tag{13}$$

or

$$\beta\,\omega^2 + (\beta\,c' + \beta\,c + \alpha\,p - \delta\,p)\,\omega + \beta\,c\,c' + \alpha\,c\,p - \delta\,c'\,p - \gamma\,p^2 = 0, \tag{14}$$

whence by comparison with (4):

$$\begin{aligned} \beta &= A\,x, \\ \beta\,c' + \beta\,c + \alpha\,p - \delta\,p &= B\,x, \\ \beta\,c\,c' + \alpha\,c\,p - \delta\,c'\,p - \gamma\,p^2 &= C\,x. \end{aligned} \tag{15}$$

and if one again sets

$$\beta\,c' - \beta\,c + \alpha\,p + \delta\,p = y \tag{16}$$

then:

$$\begin{aligned} 2\,(\beta\,c' + \alpha\,p) &= B\,x + y, \\ 2\,(\beta\,c - \delta\,p) &= B\,x - y. \end{aligned} \tag{17}$$

From this

$$y^2 + m\,x^2 = 4\,p^2. \tag{18}$$

If x is divisible by p, then y must also be divisible by p, and by cancellation of the factor p^2 we come back to equation (11), which, as we have seen, is solvable only in the cases $m = 3$ and $m = 4$. We therefore assume further that x is indivisible by p.

If m is divisible by p, then y is also divisible by p, and β is, by (15), not divisible by p. But then it follows from (16): $c \equiv c' \pmod{p}$, and both therefore correspond to the same root of (1).

So let p now divide neither x nor m. Then it does not divide y either, and by (17) c and c' are determined from the congruences

$$\beta\,c' \equiv \frac{B\,x + y}{2}, \quad \beta\,c \equiv \frac{B\,x - y}{2} \pmod{p} \tag{19}$$

and once c, c' are determined, α and δ result from (17) as integers. With this the first two equations (15) are satisfied, and γ is obtained from the last equation (15) likewise as an integer.

Indeed we have

$$\gamma\,p^2 = \beta\,c\,c' + (\alpha\,c - \delta\,c')\,p - C\,x \equiv 0 \pmod{p^2}. \tag{20}$$

For from (17) and (15) we have

$$\begin{aligned} \beta^2\,c\,c' + \beta\,(\alpha\,c - \delta\,c')\,p - \alpha\,\delta\,p^2 - A\,C\,x^2 &= \beta\,[\beta\,c\,c' + (\alpha\,c - \delta\,c')\,p - C\,x] - \alpha\,\delta\,p^2 \\ &= -\frac{y^2 + m\,x^2}{4} = -p^2; \end{aligned}$$

consequently the second term of (20) is divisible by p^2, and if one substitutes γp^2 for it, it follows that

$$\alpha\delta - \beta\gamma = 1.$$

We thus arrive at the result:

IV. The invariant equation $F_p(v, u)$ has multiple roots always then and only then, when $u = j(\omega)$, where ω is the root of a quadratic equation whose discriminant is not divisible by p, but which admits a solution of equation (18).

For a given p there is only a finite number of values m which meet this requirement, since $m < 4p^2$ must hold. The values $m = 3$, $m = 4$ are contained among them as special cases.

Since by (18) $-m$ is a quadratic residue of p, p is representable by a form of discriminant $-m$; but p^2 is representable by the principal form of a discriminant. The class of forms by which p is representable is therefore itself either the principal class, or, compounded with itself, it gives the principal class; it is therefore two-sided. We can therefore also state Theorem IV. in the following way:

V. The roots of the discriminant of the invariant equation $F_p(v, u)$ are the singular invariants $j(\omega)$, where ω is the root of a quadratic equation of negative discriminant D for which the prime p has the index 1 or 2.

Eighteenth Chapter.
Galois Group of the Class Equation.

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§ 120. Relations between the class invariants of the same discriminant.

For what follows it is convenient to have a notation by which the dependence of the class invariant on the class of forms is expressed more simply. We therefore set, when k denotes an arbitrary primitive class of discriminant $-m$, which according to § 118, III. is represented by (a, b, cp^{ε}):

$$(k) = j\left(\frac{-b+\sqrt{-m}}{2a}\right) = j(\omega).$$

When one or more of the (k) occur under a function symbol, we shall also omit the brackets, writing for example $f(k, k', \ldots)$ for $f[(k), (k'), \ldots]$.

If therefore

$$(k) = j(\omega), \tag{1}$$

and l is a class represented by $(p, b, acp^{\varepsilon-1})$, then by § 118, III.:

$$(lk) = j\left(\frac{\omega}{p}\right), \tag{2}$$

and so the two quantities

$$u = (k), \quad v = (lk)$$

satisfy the invariant equation

$$F_p(v, u) = 0. \tag{3}$$

The quantity $v = (lk)$ is, however, in addition a root of the class equation

$$H_m(v) = 0, \tag{4}$$

and it is first to be established how many roots equation (3) has in common with (4).

But all the roots of (3) are, by § 69:

$$j(p\omega), \quad j\left(\frac{\omega+\lambda}{p}\right), \tag{5}$$

where λ runs through a system of residues with respect to the modulus p. The first of these does not belong to the roots of (4), for $\omega' = p\omega$ satisfies the equation:

$$a\omega'^2 + pb\omega' + p^{\varepsilon+2}c = 0,$$

which is primitive and of discriminant $-mp^2$. If, however, we set $p\omega' = \omega + \lambda$, we obtain for ω' the equation:

$$ap^2\omega'^2 + p(b - 2a\lambda)\omega' + (a\lambda^2 - b\lambda + cp^{\varepsilon}) = 0, \tag{6}$$

and these equations likewise have the discriminant $-mp^2$. But there are two among them which are imprimitive with the divisor p, namely those which correspond to the values

$$\lambda \equiv 0, \quad a\lambda - b \equiv 0 \pmod{p} \tag{7}$$

The first is the root of the form $(ap, b, cp^{\varepsilon-1})$, the second that of the form

$$(ap,\ b - 2a\lambda,\ C),$$

and since $b - 2a\lambda \equiv b \pmod{2a}$ and [by (7)] $\equiv -b \pmod{2p}$, this form is compounded from (a, b, cp^{ε}) and $(p, -b, acp^{\varepsilon-1})$, which belongs to the class l^{-1} reciprocal to l. From this it follows:

I. The equations (3) and (4) have only two roots in common with one another, namely the class invariants (lk) and $(l^{-1}k)$, and if the class l is two-sided, they have only one root in common.

If, then, one sets $u = (k)$ in (3), the greatest common divisor of (3) and (4) is of the second degree, and if l is two-sided, of the first degree. In the former case (lk) and $(l^{-1}k)$ are the roots of a quadratic equation whose coefficients depend rationally on u, and whose form is for the rest dependent only on the class l, not on k. To what has thus been proved we can give the following expression:

II. We have

$$(lk) + (l^{-1}k) = f_l(k), \tag{8}$$

or, if l is two-sided,

$$(lk) = f_l(k), \tag{9}$$

in which $f_l(k)$ denotes a rational function of (k), whose rational numerical coefficients depend only on the class l, not on the class k.

We now consider the sequence of class invariants

$$\ldots, (l^{-2}k), (l^{-1}k), (k), (lk), (l^2k), (l^3k), \ldots, \tag{10}$$

which can be continued arbitrarily forwards and backwards. In this sequence, if l belongs to the exponent ε, then $(l^{\nu}k)$ is identical with $(l^{\nu+\varepsilon}k)$, while all the terms lying between them differ from it and from one another. If l is two-sided, the sequence (10) contains only two different terms, each of which is rationally expressible by the other. Otherwise we conclude from (8) that every term of the sequence (10) is rationally expressible by the two preceding ones (or also by the two following ones) in the form

$$(l^{\nu+1}k) = -(l^{\nu-1}k) + f_l(l^{\nu}k). \tag{11}$$

By a repeated application of this formula one arrives at the theorem that every term of the sequence (10) can be rationally expressed by any two consecutive terms of the same sequence.

But if l is two-sided, then all the terms of the sequence (10) are rationally expressible by one of them.

§ 121. Separation of the opposite classes.

It is now a question of separating the two roots of the quadratic equation from one another in the case of an l that is not two-sided as well, which is possible by the adjunction of $\sqrt{D}$.

If the class l is not two-sided, the two class invariants (lk), $(l^{-1}k)$ are different from one another, and therefore (§ 118, III.)

$$v = (lk) = j\left(\frac{\omega}{p}\right)$$

is a simple root of the transformation equation

$$F_p(u, v) = 0$$

(§ 119, V.). If we therefore put

$$M = \left(\frac{\eta\left(\frac{\omega}{p}\right)}{\eta(\omega)}\right)^{24}, \tag{1}$$

then it follows by § 72 that M is rationally expressible by

$$j(\omega) \quad \text{and} \quad j\left(\frac{\omega}{p}\right),$$

is therefore expressible rationally through (k) and (lk), and from § 72 it follows that M is an integral algebraic number.

We apply this theorem to each pair of consecutive terms of the sequence (7) and obtain, if we set

$$\omega_1 = \frac{\omega}{p}, \quad \omega_2 = \frac{\omega_1}{p}, \quad \ldots, \quad \omega_\varepsilon = \frac{\omega_{\varepsilon-1}}{p}$$

and if φ denotes a rational function completely determined by the class l:

$$\begin{aligned} M &= \left(\frac{\eta(\omega_1)}{\eta(\omega)}\right)^{24} && = \varphi(k,\, lk), \\ M_1 &= \left(\frac{\eta(\omega_2)}{\eta(\omega_1)}\right)^{24} && = \varphi(lk,\, l^2k), \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots && \\ M_{\varepsilon-1} &= \left(\frac{\eta(\omega_\varepsilon)}{\eta(\omega_{\varepsilon-1})}\right)^{24} && = \varphi(l^{\varepsilon-1}k,\, k). \end{aligned} \tag{2}$$

By multiplication of all these equations there follows:

$$\left(\frac{\eta(\omega_\varepsilon)}{\eta(\omega)}\right)^{24} = \varphi(k,\, lk)\,\varphi(lk,\, l^2k)\,\ldots\,\varphi(l^{\varepsilon-1}k,\, k). \tag{3}$$

But by § 120 the right-hand side of this equation can be represented as a rational function $\Phi(k, lk)$ of (k) and (lk) (with rational numerical coefficients). On the other hand, since l belongs to the exponent ε, l^ε is equivalent to the principal form, and so $l^\varepsilon k$ to k and ω_ε to ω; there is therefore an equation:

$$\omega_\varepsilon = \frac{\gamma + \delta\,\omega}{\alpha + \beta\,\omega}, \quad \alpha\delta - \beta\gamma = 1, \tag{4}$$

and $\eta(\omega_\varepsilon) : \eta(\omega)$ is to be determined according to § 38. If, as we have assumed, ω satisfies the equation

$$a\,\omega^2 + b\,\omega + c\,p^\varepsilon = 0, \tag{5}$$

then one obtains from (4), since $\omega_\varepsilon = \omega/p^\varepsilon$:

$$\beta\,\omega^2 + (\alpha - p^\varepsilon\delta)\,\omega + \gamma\,p^\varepsilon = 0,$$

which by comparison with (5) gives

$$\beta = a\,x, \quad \alpha - p^\varepsilon\delta = b\,x, \quad \gamma = -c\,x,$$

and if one sets $\alpha + p^\varepsilon\delta = y$:

$$\alpha = \frac{y + b\,x}{2}, \quad \beta = a\,x, \tag{6}$$

$$p^\varepsilon\gamma = -c\,x, \quad p^\varepsilon\delta = \frac{y - b\,x}{2},$$

hence

$$4p^\varepsilon = y^2 + mx^2.$$

From this x and y are determined up to sign, and x may be assumed positive, and the sign of y follows from the congruence

$$y \equiv bx \pmod{p^\varepsilon}. \tag{7}$$

If, on the other hand, one considers the composition $l^{-1}k$, then one has to compose the form (a, b, cp^ε) with $(p, -b, acp^{\varepsilon-1})$; one determines b' from the two congruences:

$$b' \equiv b \pmod{2a}, \qquad b' \equiv -b \pmod{2p^\varepsilon},$$

and everything is to be carried out in just the same way, except that in place of (7) the congruence

$$y \equiv -bx \pmod{p^\varepsilon} \tag{8}$$

enters, that is, y receives the opposite sign.

But according to (5) and (6)

$$\alpha + \beta\omega = \frac{y + (2a\omega + b)x}{2} = \frac{y + \sqrt{-m}\,x}{2}, \tag{9}$$

in which, if a and x and consequently β are assumed positive, $\sqrt{-m}$ is to be taken positive imaginary.

But by § 38, (4)

$$\left(\frac{\eta(\omega_\varepsilon)}{\eta(\omega)}\right)^{24} = \left(\frac{y + \sqrt{-m}\,x}{2}\right)^{12},$$

and consequently one obtains by (3)

$$\Phi(k, lk) = \left(\frac{y + \sqrt{-m}\,x}{2}\right)^{12}. \tag{10}$$

In the same way there follows, on account of (16):

$$\Phi(k, l^{-1}k) = \left(\frac{y - \sqrt{-m}\,x}{2}\right)^{12}, \tag{11}$$

and equation (10) is therefore not satisfied when (lk) is replaced by $(l^{-1}k)$, except when

$$\left(\frac{y + \sqrt{-m}\,x}{2}\right)^{12} = \left(\frac{y - \sqrt{-m}\,x}{2}\right)^{12} \tag{12}$$

holds.

But this is not possible if, as we have assumed, p does not divide m. For if one decomposes p in the field $\Omega = \Re(\sqrt{D})$ into the two conjugate prime ideals $\mathfrak{p}, \mathfrak{p}'$, distinct from one another, then the one divides $\frac{1}{2}(y + \sqrt{-m}\, x)$, the other

$\frac{1}{2}(y - \sqrt{-m}\, x)$, and these two quantities must therefore be relatively prime. Consequently equation (12) cannot hold.

The quadratic equation whose two roots are (lk), $(l^{-1}k)$ therefore has only one root in common with (18), and from this there results the important theorem:

III. After the adjunction of $\sqrt{D}$, (lk) can be expressed rationally by (k) in the form:

$$(lk) = f_l(k, \sqrt{D}), \tag{13}$$

where f_l is a rational function whose form is determined by the class l alone.

The change of the sign of $\sqrt{D}$ has the effect that (lk) passes into $(l^{-1}k)$, thus:

$$f_l(k, \sqrt{D}) = f_{l^{-1}}(k, -\sqrt{D}).$$

If l is two-sided, then by I. the same formula holds, only that $\sqrt{D}$ then does not occur in f_l.

If l' is a second class of the same nature as l, then one can apply formula (13) to $l'k$ and obtains:

$$(l\,l'k) = f_l(l'k, \sqrt{D}),$$

which, with the use of

$$(l'k) = f_{l'}(k, \sqrt{D})$$

passes into an equation of the form:

$$(l\,l'k) = f_{ll'}(k, \sqrt{D})$$

Since on the left side l can be interchanged with l', it follows that

$$f_{ll'}(k, \sqrt{D}) = f_{l'l}(k, \sqrt{D}),$$

so that $f_{ll'}$ depends only on the composed class ll'. In the same way one can derive

$$(l\,l'\,l''k) = f_{ll'l''}(k, \sqrt{-m}),$$

and since by § 118, II. every arbitrary class s of the discriminant D can be composed out of such classes l, formula (13), together with the consequences drawn from it, is no longer bound to the special assumption made about l, that the first coefficient be a prime.

We have then, if s, k denote two arbitrary classes of the discriminant D, the formulae:

$$(sk) = f_s(k, \sqrt{D}), \tag{14}$$

$$(s^{-1}k) = f_s(k, -\sqrt{D}). \tag{15}$$

If k is the principal class, then

$$2\omega = \sqrt{D} \text{ or } 2\omega = -1 + \sqrt{D},$$

and since $\sqrt{D}$ is purely imaginary, $k = j(\omega)$ is real by § 69 (4). Accordingly there follows from (14) and (15), since we now have $(sk) = (s)$:

The invariants of opposite classes are conjugate imaginary, the invariants of two-sided classes are real.

If we return to an arbitrary class k and put sk in place of k in formula (15), it follows that:

$$(k) = f_s(sk, -\sqrt{D}), \tag{16}$$

or, putting $sk = k'$ and writing f for f_s, by (21)

$$\begin{aligned} (k') &= f(k, \sqrt{D}), \\ (k) &= f(k', -\sqrt{D}), \end{aligned} \tag{17}$$

where now k, k' may be any two classes of the determinant D.

We are hereby in a position to determine the Galois group of the class equation, or at least, to begin with, a group of permutations in which the group of the class equation is contained as a divisor.

Let us assume first that $\sqrt{D}$ has been adjoined to the domain of rationality of the rational numbers, and let $(k), (k'), (k'') \ldots$ denote all the class invariants of the discriminant D, while

$$R(k, k', k'', \ldots)$$

is any rational function of these quantities. By our theorem this function can be expressed rationally by one of the quantities (k), thus say:

$$R(k, k', k'', \ldots) = R'(k).$$

If, further, (s) denotes an arbitrary class invariant of the discriminant D, and if the function R has the property of remaining unchanged under all h permutations

$$\begin{pmatrix} k, & k', & k'', & \ldots \\ sk, & sk', & sk'', & \ldots \end{pmatrix},$$

whose totality we will denote by $\mathfrak{S}$, then $R'(k) = R'(sk)$, and is therefore rationally expressible.

The Galois group of the class equation after the adjunction of $\sqrt{D}$ is therefore contained in the system $\mathfrak{S}$,

and since $\mathfrak{S}$ is an Abelian group, the class equation is an Abelian one.

Let us assume that the function R has real (rational) coefficients; its rational expression will nevertheless have the form $a + b\sqrt{D}$, in which a, b are rational numbers. The imaginary part will drop out then and only then, when R also remains unchanged under the interchange of all the classes $k, k', k'', \ldots$ with their opposites. From this it follows that, without adjunction of $\sqrt{D}$, the Galois group of the class equation is contained in the system of $2h$ permutations which one obtains by converting every class in $\mathfrak{S}$ into its opposite. Only in the particular case in which all the classes are two-sided (which occurs only for a finite number of determinants) is this latter system identical with $\mathfrak{S}$, and the class equation is then an Abelian one without adjunction of $\sqrt{D}$.

The algebraic field which is formed from the rational functions of a class invariant is therefore, apart from the exceptional case last mentioned, not a normal field, but is different from its conjugate fields. One obtains a normal field, on the other hand, if one adjoins the square root $\sqrt{D}$ to the field of the class invariants. For every rational function of all the roots of the equation can be expressed rationally by one of them and $\sqrt{D}$.

§ 122. Irreducibility.

We now consider the algebraic number field

$$\mathfrak{K} = \mathfrak{R}(k, \sqrt{D}), \tag{1}$$

which is composed of a class invariant (k) of the discriminant D and of $\sqrt{D}$; by the preceding section this field contains all the class invariants $(k), (k'), (k''), \ldots$ belonging to the same discriminant. This field we call the class field of the discriminant D. The quadratic field $\mathfrak{R}(\sqrt{D})$ we denote, as hitherto, by Ω; if $D = Q^2\Delta$ and Δ is the stem of D, it is identical with $\mathfrak{R}(\sqrt{\Delta})$. Let h be the class number of the discriminant D.

1. Let (k) be any one of the h class variants, and let

$$\Phi(k) = (k)^\nu + \alpha_1(k)^{\nu-1} + \alpha_2(k)^{\nu-2} \cdots + \alpha_\nu = 0 \tag{2}$$

be the equation of lowest degree in Ω whose root is (k). Since (k) is an integer, the α_i must also be integers

of the field Ω (Bd. II, § 149). If t denotes a variable, then $\Phi(t)$ is a divisor of the class function $H_m(t)$, and consequently the discriminant of $\Phi(t)$ is also a divisor of the discriminant of $H_m(t)$.

If ξ is an arbitrary integer of the field $\mathfrak{K}$, then the relative trace with respect to Ω (Bd. II, § 175):

$$\mathfrak{S}\left(\frac{\xi}{t-(k)}\,\Phi(t)\right) = \varphi(t)$$

is an integral function of t, of degree at most $\nu - 1$, whose coefficients are integers of Ω. If we therefore let t pass over into (k), it follows that:

$$\xi = \frac{\varphi(k)}{\Phi'(k)}, \tag{3}$$

and from this:

$$b\xi = \beta_0 + \beta_1(k) + \beta_2(k)^2 + \cdots + \beta_{\nu-1}(k)^{\nu-1}, \tag{4}$$

in which $\beta_0, \beta_1, \ldots, \beta_{\nu-1}$ are integers in Ω, and b denotes an integral rational number, for which the discriminant of equation (2):

$$b = N[\Phi'(k)]$$

may be taken. The field discriminant then divides b, and none of the primes not dividing b is divisible in Ω or in $\mathfrak{K}$ by the square of a prime ideal (Bd. II, § 174). The number b does not depend on the particular number ξ.

An integer ξ can be represented in the form (4) in only one way, since otherwise (k) would satisfy an equation of lower than the νth degree, which contradicts the assumption; and from this it follows that ξ is divisible by an integer α of the field Ω only if all the coefficients $\beta_0, \beta_1, \ldots, \beta_{\nu-1}$ are divisible by α. For if one represents ξ/α in the form (4), the coefficients in this representation must likewise be integers.

If we replace (k) in (3) by another root of equation (2), then ξ passes over into a conjugate value, which is likewise an integer.

2. By Fermat's theorem we have, for every arbitrary prime p, when a denotes an integral rational number:

$$a^p \equiv a \pmod{p}. \tag{5}$$

Furthermore, if p does not divide $2D$, then by § 85

$$D^{\frac{p-1}{2}} \equiv (D, p) \pmod{p},$$

and hence:

$$\sqrt{D}^p \equiv (D, p)\sqrt{D} \pmod{p}. \tag{6}$$

If therefore α denotes, as above, an integer of the field Ω:

$$\alpha = \frac{x + y\sqrt{\Delta}}{2},$$

then, if one multiplies by $2Q$, writes x for Qx, and denotes by α' the number conjugate to α:

$$2Q\alpha = x + y\sqrt{D}, \quad 2Q\alpha' = x - y\sqrt{D},$$

and from this it follows by Fermat's theorem and by (6):

$$\begin{array}{ll} \alpha^p \equiv \alpha, & (D,p) = +1, \\ \alpha^p \equiv \alpha', & (D,p) = -1, \end{array} \pmod{p}. \tag{7}$$

From this it now follows, if $f(k, \sqrt{-m})$ is any number of the form (4), by application of the polynomial theorem:

$$\begin{array}{ll} f(k,\sqrt{D})^p \equiv f(k^p, \sqrt{D}), & (D,p) = +1, \\ f(k,\sqrt{D})^p \equiv f(k^p, -\sqrt{D}), & (D,p) = -1, \end{array} \pmod{p}, \tag{8}$$

in which k^p stands for $(k)^p$, and so has the meaning of an actual power.

3. If $\mathfrak{P}$ is a prime ideal in $\mathfrak{K}$ that divides p, then the norm of $\mathfrak{P}$ is a power of p:

$$N(\mathfrak{P}) = p^f. \tag{9}$$

The exponent f is the degree of the prime ideal $\mathfrak{P}$. For every arbitrary integer ξ of the field $\mathfrak{K}$ we have

$$\xi^{N(\mathfrak{P})} \equiv \xi \pmod{\mathfrak{P}}, \tag{10}$$

and if conversely p^f is the lowest power of p for which the congruence

$$\xi^{p^f} \equiv \xi \pmod{\mathfrak{P}} \tag{11}$$

is satisfied by every number ξ, then f is the degree of the ideal $\mathfrak{P}$ (Bd. II, § 167).

The degrees of the prime ideals contained in the prime p are now to be determined.

Since p does not divide the fundamental number of the field $\mathfrak{K}$, p is not divisible by the square of a prime ideal.

4. If $(k), (k'), \ldots, (k^{(h-1)})$ denote all the roots of the equation

$$H_m(u) = 0,$$

then for a variable t

$$H_m(t) = [t - (k)][t - (k')] \ldots [t - (k^{(h-1)})],$$

and this function has integral rational coefficients. Hence by (5) we have:

$$[H_m(t)]^p \equiv H_m(t^p) \pmod{p},$$

and so, if we set $t = (k)$:

$$[(k)^p - (k)][(k)^p - (k')] \ldots [(k)^p - (k^{(h-1)})] \equiv 0 \pmod{p}.$$

If, then, $\mathfrak{P}$ denotes a prime ideal in $\mathfrak{K}$ that divides p, at least one of the factors of the left side must be divisible by $\mathfrak{P}$, and from this there follows a congruence

$$(k)^p \equiv (k') \pmod{\mathfrak{P}}, \tag{12}$$

in which (k') is some one of the class invariants, and it may also be identical with (k).

5. If first

$$(D,p) = -1,$$

then $\mathfrak{P}$ can in no case be of the first degree, since already

$$\begin{aligned} \sqrt{D}^p &\equiv -\sqrt{D} \\ \sqrt{D}^{p^2} &\equiv \sqrt{D} \end{aligned} \pmod{\mathfrak{P}} \tag{13}$$

holds. But if in (12) (according to § 121)

$$(k') = f(k, \sqrt{D}), \quad k = f(k', -\sqrt{D}),$$

then it follows from (13) that

$$(k')^p \equiv f(k^p, -\sqrt{D}) \equiv f(k', -\sqrt{D}) \equiv k,$$

$$(k)^{p^2} \equiv k \pmod{\mathfrak{P}},$$

and consequently, for all integers ξ of the field $\mathfrak{K}$,

$$\xi^{p^2} \equiv \xi \pmod{\mathfrak{P}}, \tag{14}$$

and $\mathfrak{P}$ is of the second degree.

6. If secondly

$$(D,p) = +1,$$

then there are, as was proved in § 118, two opposite classes l, l^{-1} (which, if l is two-sided, are identical with one another), which can be represented by forms with the first coefficient p; and if (k) is any class invariant, then by § 120 the invariant equation for the degree of transformation p is:

$$F_p(u,v) = 0 \tag{15}$$

is satisfied for

$$u = (k), \quad v = (lk) \quad \text{and} \quad v = (l^{-1}k).$$

But from what was proved at the end of § 69 [formula (32)], the congruence follows from this:

$$[(k)^p - (lk)][(lk)^p - (k)] \equiv 0 \pmod{p}. \tag{16}$$

If now $\mathfrak{P}$ is a prime ideal that divides p, then one of the two factors on the left side of (17) must be divisible by $\mathfrak{P}$. Which of the two assumptions thus possible we pursue further is immaterial, since the one passes over into the other when k is interchanged with $l^{-1}k$ and l with l^{-1}. So let

$$(k)^p \equiv (lk) \pmod{\mathfrak{P}}. \tag{17}$$

If the congruence (17) is satisfied for any one of the conjugate values (k), then it holds for every other one as well. For by § 121:

$$(k') = f(k, \sqrt{D}), \quad (lk') = f(lk, \sqrt{D}),$$

hence

$$(k')^p \equiv f(k^p, \sqrt{D}) \equiv f(lk, \sqrt{D}) \equiv (lk') \pmod{\mathfrak{P}}.$$

It therefore follows from (14), if k is replaced by $l^{-1}k$:

$$(l^{-1}k)^p \equiv (k) \pmod{\mathfrak{P}}, \tag{18}$$

that is, according as the one or the other of the two classes l belonging to p is put for l, the one or the other factor of (16) is divisible by $\mathfrak{P}$. By repeated application of (17) there results, for any arbitrary positive exponent π,

$$(k)^{p^\pi} \equiv (l^\pi k) \pmod{\mathfrak{P}} \tag{19}$$

(inference from π to $\pi+1$). If now l belongs to the exponent ε, or ε is the index of p, then by (19):

$$(k)^{p^\varepsilon} \equiv (k) \pmod{\mathfrak{P}}, \tag{20}$$

and p^ε is the lowest power of p that satisfies this condition. For if a still lower power of p satisfies the congruence (19), it follows from (20) that two different class invariants (k), (k') are congruent with respect to the modulus $\mathfrak{P}$. Their difference $(k)-(k')$ would then be divisible by $\mathfrak{P}$, and consequently the discriminant of H_m divisible by p, contrary to the hypothesis.

From this it now follows again, by (7), that the congruence

$$\xi^{p^\varepsilon} \equiv \xi \pmod{\mathfrak{P}} \tag{21}$$

is satisfied for every integer ξ in $\mathfrak{K}$, and hence the theorem:

A prime p splits in the field $\mathfrak{K}$ into prime ideals all different from one another, whose degree is equal to the index of p.

We wish to prove further that among the different prime ideals $\mathfrak{P}$ dividing p there is always one for which, of the two congruences following from (16), the congruence (17) holds.

To this end we remark: if we transform all the numbers of the field $\mathfrak{K}$ into the conjugate imaginary ones, interchanging $\sqrt{D}$ with $-\sqrt{D}$ and every class invariant (k) with the opposite one (k^{-1}), then all the numbers of an ideal $\mathfrak{A}$ pass over into the numbers of a conjugate imaginary ideal $\mathfrak{A}'$, which may also be identical with $\mathfrak{A}$, and the norm of the ideal $\mathfrak{A}$ is equal to the norm of the ideal $\mathfrak{A}'$. If $\mathfrak{A}$ is a prime ideal, then $\mathfrak{A}'$ is a prime ideal too.

If now the congruence (17) is not satisfied, then by (16)

$$(lk)^p \equiv (k) \pmod{\mathfrak{P}}$$

must hold, and this congruence too remains in force when k is replaced by another class k'. If, then, we put $l^{-1}k^{-1}$ in place of k, it follows that:

$$(k^{-1})^p \equiv (l^{-1}k^{-1}) \pmod{\mathfrak{P}}.$$

Since therefore $(k^{-1})^p-(l^{-1}k^{-1})$ is a number in $\mathfrak{P}$, $(k)^p-(lk)$ is contained in the ideal $\mathfrak{P}'$ conjugate to $\mathfrak{P}$, and we have

$$(k)^p \equiv (lk) \pmod{\mathfrak{P}'},$$

which arises from (17) by the interchange of $\mathfrak{P}$ with $\mathfrak{P}'$.

7. From these theorems the irreducibility of the class equation, even after adjunction of $\sqrt{D}$, is a simple consequence.

If k, k' are any two classes of the discriminant D, then by § 118 we can compose k' in the following manner:

$$k' = kll'l'' \ldots,$$

so that the classes $l, l', l'', \ldots$ represent the primes $p, p', p'', \ldots$. By what has just been proved, we can therefore choose the prime divisors $\mathfrak{P}, \mathfrak{P}', \mathfrak{P}''$ of these primes in such a way that

$$\begin{aligned} (k)^p &\equiv (kl) \pmod{\mathfrak{P}}, \\ (kl)^{p'} &\equiv (kll') \pmod{\mathfrak{P}'}, \\ (kll')^{p''} &\equiv (kll'l'') \pmod{\mathfrak{P}''} \end{aligned} \tag{22}$$

. . . . . .

Let us now assume that the class equation splits, so that for a variable t

$$H_m(t) = H_1(t)H_2(t),$$

then, since k, k' were arbitrary classes, we may choose (k) among the roots of $H_1(t)$ and (k') among those of $H_2(t)$. In the chain (22) the initial term (k) therefore belongs to the roots (k_1) of $H_1(t)$, and the final term (k') to the roots (k_2) of $H_2(t)$. Since the passage from the roots of the one factor to those of the other must occur at least once in the chain (22), there is a pair of roots $(k_1), (k_2)$ which, for some prime p and a prime ideal $\mathfrak{P}$ dividing it, satisfies the congruence

$$(k_1)^p \equiv (k_2) \pmod{\mathfrak{P}}$$

Now since $H_1(k_1) = 0$, it follows by the inference often applied that

$$H_1(k_1)^p \equiv H_1(k_1^p) \equiv H_1(k_2) \equiv 0 \pmod{\mathfrak{P}}.$$

Since now

$$H_1(k_2) = \prod^{k_1}[(k_2) - (k_1)]$$

one of the differences $(k_2) - (k_1)$ must be divisible by $\mathfrak{P}$, which is not possible, since p does not divide the discriminant of H.

With this the irreducibility of the class equation is proved; the degree of the field $\mathfrak{K}$ is established as equal to twice the class number h, and the group $\mathfrak{S}$ found in the preceding section is recognised as the true group of the class equation.

8. Accordingly we can decompose all primes p, apart from a finite number of exceptions (those which divide $2D$ or the discriminant of H) into their prime factors in the field $\mathfrak{K}$.

If $(D, p) = +1$, then p splits and field Ω into two conjugate prime ideals of the first degree. If p is of index ε, then every prime ideal $\mathfrak{p}$ dividing p is of degree ε. If therefore

$$p = \mathfrak{P}_1\mathfrak{P}_2 \ldots \mathfrak{P}_r,$$

then by 5. $N(p) = p^{2h}$, and consequently

$$2h = r\varepsilon.$$

The number of prime factors into which p splits in the field $\mathfrak{K}$ is therefore equal to $2h/\varepsilon$.

If further $(D, p) = -1$, then the prime factors of p are of the second degree, and their number is therefore $= h$.

9. We draw one further interesting conclusion from these results. Let $(D, p) = +1$ and let ε be the index of p. Then $4p^\varepsilon$ is representable by the form $x^2 - Dy^2$, and

$$p^\varepsilon = \mu\mu', \tag{23}$$

if for brevity

$$\mu = \frac{x + y\sqrt{D}}{2}, \quad \mu' = \frac{x - y\sqrt{D}}{2} \tag{24}$$

is set. We now know that p splits into nothing but distinct prime ideals of degree ε. Since p is odd, and x, y have at most the common divisor 2, μ, μ' have no common prime factor in $\mathfrak{K}$. To every prime ideal $\mathfrak{P}$ that divides μ there corresponds a prime ideal $\mathfrak{P}'$ distinct from it (the conjugate one), which arises from $\mathfrak{P}$ by putting, for all numbers of $\mathfrak{P}$, the conjugate imaginary numbers. Thus μ is divisible not only by $\mathfrak{P}$, but by $\mathfrak{P}^\varepsilon$.

According to § 121 we have

$$\left(\frac{\eta(\omega_1)}{\eta(\omega)}\right)^2\left(\frac{\eta(\omega_2)}{\eta(\omega_1)}\right)^2\cdots\left(\frac{\eta(\omega_\varepsilon)}{\eta(\omega_{\varepsilon-1})}\right)^2=\varrho\mu, \tag{25}$$

and by § 72, 4 the factors P^2 on the left side of this expression satisfy an equation $P^2\varphi(P^2,\gamma_2,\gamma_3)=p$, in which $\varphi(P^2,\gamma_2,\gamma_3)$ is an integral algebraic number.

From this it follows, however, that P^2 cannot be divisible by a higher than the first power of $\mathfrak{P}$, while μ is nevertheless divisible by $\mathfrak{P}^\varepsilon$. We therefore conclude from (25) that each of the numbers P^2 must be divisible by the first power of $\mathfrak{P}$, and hence that all these numbers are associated. If, then, ϱ denotes any algebraic unit, we have

$$\frac{x+y\sqrt{-m}}{2}=\varrho P^{2\varepsilon},$$

so μ is really represented as the εth power of a number existing in the field $\mathfrak{K}$, and P^2 itself is a representation of a prime ideal in the field Ω by a principal ideal in the field $\mathfrak{K}$.

§ 123. Relations between class invariants in the different orders.

Between the class invariants of different discriminants $D=Q^2\Delta$ with the same stem Δ there subsist algebraic relations, which we have now to seek out.

Let p be an arbitrary prime ($p=2$ also not excluded), and

$$D'=p^2D. \tag{1}$$

Let $-D=m$, $-D'=m'$, hence $m'=p^2m$, and let

$$H_m(u)=0, \tag{2}$$

$$H_{m'}(v)=0 \tag{3}$$

be the class equations belonging to the discriminants $-m,-m'$.

$$v=j(\omega')$$

shall be an arbitrary root of the second, and

$$A'\omega'^2+B'\omega'+C'=0,\quad B'^2-4A'C'=D' \tag{4}$$

the primitive quadratic equation which ω' satisfies. A' may be assumed, as is permissible, to be indivisible by p.

We now consider the invariant equation

$$F_p(u,v)=0, \tag{5}$$

whose roots are:

$$u=j(p\omega'),\quad j\left(\frac{\omega'+c}{p}\right),\quad c=0,1,2,\ldots,p-1. \tag{6}$$

The argument of these functions:

$$\omega=p\omega'\quad\text{or}\quad=\frac{\omega'+c}{p}$$

satisfies a quadratic equation to be derived from (4), namely:

$$\begin{gathered}\omega = p\omega', \quad A'\omega^2 + B'p\omega + C'p^2 = 0,\\ \omega = \frac{\omega' + c}{p},\\ A'p^2\omega^2 + (B' - 2A'c)p\omega + A'c^2 - B'c + C' = 0.\end{gathered} \tag{7}$$

The discriminants of these equations are $p^2D' = p^4D$, and the first of them is always primitive, the discriminant

of the second equation reduces to D only when it is imprimitive and

$$\begin{gathered}B' - 2A'c \text{ by } p,\\ A'^2c - B'c + C' \text{ by } p^2\end{gathered} \tag{8}$$

is divisible, and then the corresponding number (6) satisfies at the same time equation (5) and equation (2). But this is the case, as we shall show at once, only for one value c, and consequently the value u in question can be expressed rationally in terms of v.

For if in the first place p is odd, the first congruence (8):

$$2A'c - B' \equiv 0 \pmod{p}$$

has only one root, and for this

$$(2A'c - B')^2 - p^2D = 4A'(A'c^2 - B'c + C') \equiv 0 \pmod{p^2}.$$

If $p = 2$, then D', and consequently B', is even, and hence $B' - 2A'c$ is always divisible by 2. We then have

$$\left(A'c - \frac{B'}{2}\right)^2 - D = A'(A'c^2 - B'c + C'),$$

and since $D \equiv 0$ or $\equiv 1 \pmod 4$, the left side here is divisible by 4 if it is assumed that $A'c - \frac{B'}{2} \equiv D \pmod 2$.

If, conversely, one starts from an equation

$$A\omega^2 + B\omega + C = 0, \quad B^2 - 4AC = D \tag{9}$$

of discriminant D, takes A relatively prime to p, and sets $\omega' = p\omega$, then ω' satisfies the primitive equation

$$A\omega'^2 + Bp\omega' + Cp^2 = 0 \tag{10}$$

of discriminant $D' = p^2D$; and if we therefore set $v = j(\omega')$ then $u = j(\omega)$ is rationally expressible in terms of v. From this, then, the theorem follows:

1. Every class invariant for the discriminant D is rationally expressible in terms of a class invariant for the discriminant p^2D.

In order, however, to answer the question how many values of v lead to the same value of $u = j(\omega)$, we observe that the various values of v which do this must be roots of equation (5) and (3), and hence must have one of the forms

$$j(p\omega), \quad j\left(\frac{\omega + c}{p}\right) \tag{11}$$

are contained. Now from (9) we have:

$$\begin{gathered}\omega' = p\omega, \quad A\omega'^2 + Bp\omega' + Cp^2 = 0,\\ \omega' = \frac{\omega + c}{p},\\ Ap^2\omega'^2 + (B - 2Ac)p\omega' + Ac^2 - Bc + C = 0,\end{gathered} \tag{12}$$

whose discriminant is D'. But among the quantities (11) only those will satisfy equation (3) for which the corresponding equation (12) is primitive.

The first of the equations (12) is, by our assumption, always primitive; of the remaining ones only those are non-primitive for which

$$Ac^2 - Bc + C \equiv 0 \pmod{p}, \tag{13}$$

and of the quantities (11) just as many drop out, as not being solutions of (3), as the congruence (13) has solutions. If for the moment we denote the number of these solutions by t, then the number of the values of v that belong to the same value of u is $(p + 1 - t)$.

If we first take $p = 2$, then, since A is odd, we have the following cases:

$$\begin{aligned} &B \equiv 0, && c \equiv C \pmod 2, && t = 1,\\ &B \equiv 1, \quad C \equiv 0, && c \equiv 0, 1 \pmod 2, && t = 2,\\ &B \equiv 1, \quad C \equiv 1 && \pmod 2, && t = 0, \end{aligned}$$

(because $c^2 - c$ is always even). This may be summarized as follows:

$$\begin{aligned} &D \equiv 0 \pmod 4, && t = 1,\\ &D \equiv 1 \pmod 8, && t = 2,\\ &D \equiv 5 \pmod 8, && t = 0. \end{aligned}$$

Using the symbol (D, p) (§ 85) we may therefore put:

$$t = (D, p) + 1. \tag{14}$$

If p is odd, the congruence (13) is equivalent to

$$(2Ac - B)^2 \equiv D \pmod{p},$$

and this has one solution $(2Ac - B \equiv 0)$ when p divides D, that is, when $(D, p) = 0$; two solutions when D is a quadratic residue of p, that is, when $(D, p) = +1$; and no solution when D is a quadratic non-residue, that is, when $(D, p) = -1$. So formula (14) holds here also.

Let us therefore provisionally assume, what we shall at once examine more closely, that among the $p + 1$ quantities

$$p\,\omega, \quad \frac{\omega + c}{p} \tag{15}$$

no two equivalent ones, then to each value of u there belong $[p - (D, p)]$ values v, and if we denote the degrees of (2) and (3) by h and h', then the following relation results (in agreement with the formulae of § 100):

$$h' = [p - (D, p)]\, h. \tag{16}$$

According to § 122, h and h' are the class numbers of the discriminants D, D'.

It thus remains to ask whether, among the quantities (15), after the exclusion of those which fall away by reason of (13), equivalent ones can still occur. If in the first place $p\omega$ is equivalent to $\frac{\omega+c}{p}$, then:

$$\frac{\omega + c}{p} = \frac{\gamma + \delta p\omega}{\alpha + \beta p\omega}, \quad \alpha\delta - \beta\gamma = 1, \tag{17}$$

or:

$$\beta p\,\omega^2 + (\alpha + \beta cp - \delta p^2)\omega + c\alpha - \gamma p = 0. \tag{18}$$

This equation must arise from (9) by multiplication with an integer g, and it therefore follows that:

$$\beta p = gA, \quad \alpha + \beta cp - \delta p^2 = gB, \quad c\alpha - gp = gC.$$

Since A was assumed to be not divisible by p, it follows from this that g, and consequently α, is divisible by p. If we set $g = px$, $\alpha = p\alpha'$, then it follows that:

$$\beta = Ax, \quad \alpha' + \beta c - \delta p = Bx, \quad c\alpha' - \gamma = Cx. \tag{19}$$

If we further set

$$\alpha' - \beta c + \delta p = y, \tag{20}$$

then we obtain:

$$\begin{aligned} 2\alpha' &= Bx + y, \\ 2(\beta c - \delta p) &= Bx - y, \end{aligned} \tag{21}$$

and from

$$p\alpha'\delta - \beta\gamma = 1 \tag{22}$$

there then follows

$$y^2 - Dx^2 = 4. \tag{23}$$

The value $x = 0$ is to be excluded, since otherwise $\beta = 0$ would have to hold, which contradicts equation (22), and we may, without loss of generality, assume x to be positive. Since moreover D is, with respect to the modulus 4, either $\equiv 0$ or $\equiv 1$, then

there remain only the following two possibilities for the fulfilment of (23):

1. $D = -3,\ x = 1,\ y = \pm 1.$

Since for the discriminant -3 there is only one class of forms, we may assume $A = 1$, $B = 1$, $C = 1$, i.e. we may set for ω the imaginary third root of unity $e^{\frac{2\pi i}{3}}$, and we obtain among the quantities (15) three equivalent ones by assuming $y = +1$, and $= -1$:

$$p\,\omega, \quad \frac{\omega}{p}, \quad \frac{\omega + 1}{p}, \tag{24}$$

as is also recognised from the equations:

$$\frac{\omega}{p} = \frac{-1}{p + p\omega}, \quad \frac{\omega + 1}{p} = \frac{-1}{p\omega}$$

from which the equivalence of the three quantities (24) is evident.

2. $D = -4,\ x = 1,\ y = 0.$

We may likewise assume $B = 0$, $A = C = 1$, i.e. $\omega = i$, and we find the two equivalent values:

$$p\,\omega, \quad \frac{\omega}{p}, \tag{25}$$

as also follows from the equation:

$$p\,\omega = \frac{-p}{\omega}$$

Let, further, two of the values (3)

$$\frac{\omega + c}{p}, \quad \frac{\omega + c'}{p}$$

be equivalent, so that an equation holds:

$$\frac{\omega + c}{p} = \frac{\gamma p + \delta(\omega + c')}{\alpha p + \beta(\omega + c')}, \quad \alpha\delta - \beta\gamma = 1, \tag{26}$$

or

$$\begin{aligned} \beta\,\omega^2 + (\beta c' + \beta c + \alpha p - \delta p)\omega \ + \\ \beta c c' + \alpha c p - \delta c' p - \gamma p^2 = 0, \end{aligned} \tag{27}$$

from which, by comparison with (9),

$$\begin{aligned} \beta &= Ax, \\ \beta c' + \beta c + \alpha p - \delta p &= Bx, \\ \beta c c' + \alpha c p - \delta c' p - \gamma p^2 &= Cx, \end{aligned} \tag{28}$$

follows, and if one again sets

$$\beta c' - \beta c + \alpha p + \delta p = y \tag{29}$$

then:

$$\begin{aligned} 2(\beta c' + \alpha p) &= Bx + y, \\ 2(\beta c - \delta p) &= Bx - y. \end{aligned} \tag{30}$$

From this

$$y^2 - Dx^2 = 4p^2. \tag{31}$$

From (28) there result the congruences

$$\begin{aligned} &\beta c c' \equiv Cx, \quad \beta(c + c') \equiv Bx, \quad \beta \equiv Ax \\ &Dx^2 \equiv \beta^2 (c - c')^2 \pmod{p}. \end{aligned} \tag{32}$$

If c' is to be different from c modulo p, then according to this neither x nor D can be divisible by p, and it follows from (32)

$$\begin{aligned} A(c+c') &\equiv B, \\ Acc' &\equiv C, \end{aligned} \tag{33}$$

and this leads, by elimination of c', to the congruence (13) for c, and so to the value of c that is to be excluded.

If we summarise this, we see that among the quantities (15) equivalent ones occur only in the two exceptional cases $D = -3$, $D = -4$, and indeed in the case $D = -3$ three of them at a time, in the case $D = -4$ two of them at a time, are equivalent.

In these cases, therefore, the right-hand side of formula (16) must still be divided by 3 or by 2, and the formula thus remains correct even then if, as already earlier, for $D = -3, -4$ one understands by h not 1, but $\frac{1}{3}$ and $\frac{1}{2}$.

§ 124. Class Fields and Order Fields.

If $D = Q^2\Delta$ is a discriminant and Δ its stem, then by the results of the preceding section the class invariant u of the discriminant Δ can be expressed rationally by the class invariant v of the discriminant D, and to every u there belong r values of v, if r is determined by formula (16), § 123. These values of v are the roots of an equation of degree r, whose coefficients depend rationally on u. The quantities v therefore belong to a field $\mathfrak{K}(D)$ over $\mathfrak{K}(\Delta)$, which is none other than the class field of the discriminant D.

With reference to the quadratic fields $\Omega = \mathfrak{R}(\sqrt{\Delta})$ both fields are Abelian. We wish, if a more exact investi-

gation is necessary, to call the field $\mathfrak{K}(\Delta)$ the class field of Ω, $\mathfrak{K}(D)$ the class field for the discriminant D, or the order field for the conductor Q.

According to the results of § 122, perhaps with a finite number of exceptions,

> a prime ideal $\mathfrak{p}$ of the field Ω is decomposable in the field $\mathfrak{K}(\Delta)$ into prime ideals of the first degree then and only then, when $\mathfrak{p}$ is of the first degree in Ω and belongs to the principal class, and is therefore equal to a complex prime
>
> $$\pi = \frac{x + y\sqrt{\Delta}}{2}$$
>
> .

If $\mathfrak{p}$ is to be decomposed in $\mathfrak{K}(D)$ into prime ideals $\mathfrak{P}$ of the first degree, there is the further requirement that π belong to the order determined by Q, i. e. that it be congruent to a rational number with respect to the modulus Q (§ 96).

Nineteenth Chapter.

Calculation of the Class Invariants.

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§ 125. The Class Invariant γ_2.

We have already remarked above that, besides the function $j(\omega)$, other modular functions are also suited to the formation of class invariants, and often give occasion to simpler results. Among these there first presents itself the function $\gamma_2(\omega)$, which is defined by $\sqrt[3]{j(\omega)}$.

We have seen in § 71 that between the functions

$$u = \gamma_2(\omega), \quad v = \gamma_2\left(\frac{c+\partial\omega}{a}\right), \tag{1}$$

if $a\partial = n$ is not divisible by 3, and c is assumed to be divisible by 3, a transformation equation

$$\Phi_n(u, v) = 0 \tag{2}$$

subsists. But the function Φ_n depends only on the two arguments vu^{-n}, u^3, and hence in $\Phi_n(u, u) = 0$ there occurs, if $n \equiv 1 \pmod 3$, only u^3, that is, $j(\omega)$; in this case therefore this equation can yield no other result than the invariant equation. It is otherwise, however, in the case

$$n \equiv -1 \pmod 3, \tag{3}$$

which we shall now assume.

If one of the values is to become $v = u$, a relation must subsist

$$\frac{\gamma+\delta\omega}{\alpha+\beta\omega} = \frac{c+\partial\omega}{a}, \quad \alpha\delta - \beta\gamma = 1, \tag{4}$$

and at the same time [§ 54, (15)]

$$-\alpha\beta + \alpha\gamma + \beta\delta - \alpha^2\beta\delta \equiv 0 \pmod 3 \tag{5}$$

must hold. Thus in any case ω is a root of a quadratic equation:

$$A\omega^2 + B\omega + C = 0, \quad D = B^2 - 4AC. \tag{6}$$

If, conversely, such an equation holds, then by comparison with (4) there follows, in the manner applied earlier, when x, y are integers:

$$\begin{aligned} \partial\beta &= Ax, \\ c\alpha - a\gamma &= Cx, \\ c\beta + \partial\alpha - a\delta &= Bx, \\ c\beta - \partial\alpha - a\delta &= y, \end{aligned} \tag{7}$$

from which:

$$\begin{aligned} 2\partial\alpha &= Bx - y, \\ 2(c\beta - a\delta) &= Bx + y, \\ 4n &= y^2 - Dx^2. \end{aligned} \tag{8}$$

If we assume A relatively prime to $3n$, then $\partial = 1$, $a = n$ must be taken; for by (7), under this assumption x must be divisible by ∂, and consequently also $c\alpha - a\gamma$ and $c\beta - a\delta$, from which it follows that a, c too would have to be divisible by ∂, whereas a, c, ∂ have no common divisor. Thus

$$(9) \qquad \begin{aligned} 2\alpha &= Bx - y, & \beta &= Ax, \\ 2n\gamma &= c(Bx - y) - 2Cx, & 2n\delta &= 2cAx - Bx - y. \end{aligned}$$

Besides the restrictions stated in § 114, 1), 2), the numbers x, y are here subject, on account of (3), to the further condition that $y^2 - Dx^2 \equiv -1 \pmod 3$. This excludes the possibility that D is divisible by 3. x must be different from zero, and we may assume it positive without restricting the generality. (A simultaneous change of sign of x and y produces only a change of sign of $\alpha, \beta, \gamma, \delta$, which is of no consequence.)

If $D \equiv -1 \pmod 3$, then x and y must be indivisible by 3; if $D \equiv 1 \pmod 3$, then y is divisible by 3 and x is not divisible. For the rest, the numbers x, y may be assumed arbitrarily. Then c is determined from the two (mutually compatible) congruences:

$$\begin{aligned} cAx &\equiv \frac{Bx + y}{2} \\ c\,\frac{Bx - y}{2} &\equiv Cx \end{aligned} \pmod n$$

and, since n is indivisible by 3, it can be subjected to the condition

$$c \equiv 0 \pmod 3$$

as well.

Now by (7) β is not divisible by 3, and therefore the congruence (5) reduces, on multiplication by β, to:

$$\alpha + \delta \equiv 0 \pmod 3$$

or, by (7) and (3), to

$$(10) \qquad B \equiv 0 \pmod 3.$$

If in (6) we replace ω by $\omega \pm 1$, then B passes over into $B \pm 2A$, and of the three numbers $B, B + 2A, B - 2A$ one and only one is divisible by 3. We therefore assume that B itself is divisible by 3; it then follows that of the three values

$$(11) \qquad \gamma_2(\omega), \quad \gamma_2(\omega + 1) = e^{-\frac{2\pi i}{3}}\gamma_2(\omega), \quad \gamma_2(\omega - 1) = e^{\frac{2\pi i}{3}}\gamma_2(\omega)$$

only the first satisfies the equation

$$(12) \qquad \Phi_n(u, u) = 0$$

Equation (12) and

$$(13) \qquad u^3 - j(\omega) = 0$$

therefore have only one common root, and this one is rationally expressible through $j(\omega)$. With this it is proved:

1. $\gamma_2(\omega)$ is a class invariant when ω is the root of a quadratic form whose discriminant and first coefficient are not divisible by 3, while the middle coefficient is divisible by 3.

If we apply this result to the case $n = 2$, we obtain first of all from equation (5), § 71:

$$\Phi_2(u, u) = u^4 - 2u^3 - 495u^2 + 2^4 \cdot 3^3 \cdot 5^3 = 0. \tag{14}$$

The equation

$$y^2 - Dx^2 = 8$$

is, since $D \equiv 0$ or $\equiv 1 \pmod 4$ must hold, solvable only for three negative values of D, namely

$$\begin{aligned} &D = -8, \quad x = 1, \quad y = 0, \\ &D = -7, \quad x = 1, \quad y = \pm 1, \\ &D = -4, \quad x = 1, \quad y = \pm 2. \end{aligned} \tag{15}$$

Corresponding to these three values of D one may choose the following for equation (6):

$$\begin{aligned} &D = -8, \qquad \omega^2 + 2 = 0, \\ &D = -7, \quad \omega^2 + 3\omega + 4 = 0, \\ &D = -4, \qquad \omega^2 + 1 = 0 \end{aligned}$$

or

$$\begin{aligned} &D = -8, \quad \omega = \sqrt{-2}, \\ &D = -7, \quad \omega = \frac{-3 + \sqrt{-7}}{2}, \\ &D = -4, \quad \omega = i. \end{aligned}$$

The value $\gamma_2(i)$ is, however, known (by § 117), namely $= 12$, and therefore the left side of (14) must be divisible by $u - 12$. Since this factor is known, the others are easily found:

$$u^4 - 2u^3 - 495u^2 + 2^4 \cdot 3^3 \cdot 5^3 = (u - 12)(u - 20)(u + 15)^2.$$

Since $\gamma_2(\omega)$, just like $j(\omega)$, has a positive value for a purely imaginary ω, the factor $u - 20$ must correspond to the value $D = -8$, and we obtain:

$$\gamma_2(i) = 12, \tag{16}$$

$$\gamma_2(\sqrt{-2}) = 20, \tag{17}$$

$$\gamma_2\left(\frac{-3 + \sqrt{-7}}{2}\right) = -15. \tag{18}$$

Besides -7 there exist five further odd negative discriminants not divisible by 3:

(19) $$-11, \quad -19, \quad -43, \quad -67, \quad -163^{1)},$$

for which there is only one class of forms. For these, then, by our theorem:

$$-\gamma_2\left(\frac{-3+\sqrt{-m}}{2}\right)$$

is an integral rational number Z.

In order to find these rational numbers, we shall seek limits between which they must lie, and which differ from one another by less than a unit.

By § 54, (5), (8) we have:

$$\gamma_2(\omega) = f(\omega)^{16} - \frac{16}{f(\omega)^8} = f_1(\omega)^{16} + \frac{16}{f_1(\omega)^8} = f_2(\omega)^{16} + \frac{16}{f_2(\omega)^8},$$

further, by § 34, (19):

$$f_2\left(\frac{-3+\omega}{2}\right) f(\omega) = e^{-\frac{\pi i}{8}}\sqrt{2},$$

---

1) That no further discriminants of this kind exist can so far be inferred only from the fact that, as far as the computation of the class numbers has been carried, no further numbers of this kind have been found [cf. the table of class numbers of Gauss (Werke, Bd. II, S. 450)].

hence

$$-\gamma_2\left(\frac{-3+\omega}{2}\right) = f(\omega)^8 - \frac{256}{f(\omega)^{16}}.$$

If one therefore sets:

(20) $$q = e^{-\pi\sqrt{m}},$$

one obtains [§ 24, (11)]:

$$Z = -\gamma_2\left(\frac{-3+i\sqrt{m}}{2}\right)$$

$$= q^{-\frac{1}{3}} \prod_{1,\infty}^{\nu} (1+q^{2\nu-1})^8 - \frac{256\, q^{\frac{2}{3}}}{\prod_{1,\infty}^{\nu} (1+q^{2\nu-1})^{16}}.$$

We now make use of the determination of limits valid for every properly fractional positive x:

$$1 + x < e^x < \frac{1}{1-x}, \quad e^{-x} > 1 - x,$$

and obtain:

$$1 < \Pi(1+q^{2\nu-1}) < e^{\frac{q}{1-q^2}},$$

from which

$$q^{-\frac{1}{3}} - 256\, q^{\frac{2}{3}} < Z < q^{-\frac{1}{3}} e^{\frac{8q}{1-q^2}} - 256\, q^{\frac{2}{3}} e^{-\frac{16q}{1-q^2}},$$

and by enlarging the upper limit still further, one may also put in its place:

$$Z < q^{-\frac{1}{3}} - 256\, q^{\frac{2}{3}} + \frac{8\, q^{\frac{2}{3}}}{1-8\,q-q^2} + \frac{2^{12}\, q^{\frac{5}{3}}}{1-q^2}.$$

For $m = 11$, q is approximately $= 2^{-15}$, from which one sees that the difference of the two limits:

$$\frac{8\,q^{\frac{2}{3}}}{1-8\,q-q^2}+\frac{2^{12}\,q^{\frac{5}{3}}}{1-q^2}$$

is already very small for $m = 11$, and therefore still more so for the larger values of m. Z is therefore the integer lying next above

$$q^{-\frac{1}{3}}-256\,q^{\frac{2}{3}}$$

and this value, for the larger m already $q^{-\frac{1}{3}}$, come extraordinarily close to an integer. From this one computes these numbers[28]:

$$-\gamma_2\left(\frac{-3+\sqrt{-11}}{2}\right)=32,$$

$$-\gamma_2\left(\frac{-3+\sqrt{-19}}{2}\right)=96=32\cdot 3,$$

$$-\gamma_2\left(\frac{-3+\sqrt{-43}}{2}\right)=960=64\cdot 15,$$

$$-\gamma_2\left(\frac{-3+\sqrt{-67}}{2}\right)=5280=32\cdot 3\cdot 5\cdot 11,$$

$$-\gamma_2\left(\frac{-3+\sqrt{-163}}{2}\right)=640320=64\cdot 3\cdot 5\cdot 23\cdot 29.$$

In these numbers the decomposability into comparatively small primes is noteworthy.

For the discriminant -27 also there exists only one class. But since this discriminant is divisible by 3, it is not $\gamma_3(\omega)$, but only $j(\omega)$, that is a rational number. One finds by a similar calculation:

$$-j\left(\frac{-1+\sqrt{-27}}{3}\right)=3\cdot 2^{15}\cdot 5^3.$$

§ 126. **The Class Invariants $f(\omega)^{24}$.**

The roots of the cubic equation

$$(u-16)^3-u\,j(\omega)=0 \tag{1}$$

are, as we have seen earlier in general (§ 54),

$$u=f(\omega)^{24},\quad -f_1(\omega)^{24},\quad -f_2(\omega)^{24},$$

or

$$f(\omega)^{24},\quad f(\omega+1)^{24},\quad f\left(1-\frac{1}{\omega}\right)^{24}. \tag{2}$$

Now let ω again be the root of the quadratic equation:

$$A\omega^2+B\omega+C=0 \tag{3}$$

[28] One does well to fix in mind, once and for all, for these and for many of the calculations described later, the Briggian logarithm

$$\log(\pi\log e)=0{,}134\,934\,184\,0$$

for which, moreover, 7 decimal places are usually sufficient.

with the negative discriminant:

$$D = B^2 - 4AC, \tag{4}$$

in which A, B, C have no common divisor, and let $j(\omega)$ be the class invariant. If we set

$$\omega' = \omega + 1, \quad \omega'' = 1 - \frac{1}{\omega},$$

then

$$\begin{aligned} &A\omega'^2 + (B - 2A)\omega' + (A - B + C) = 0, \\ &A + B + C - (B + 2C)\omega'' + C\omega''^2 = 0. \end{aligned} \tag{5}$$

The three arguments of (2) are therefore the roots of equivalent quadratic forms.

We distinguish the three cases:

1. $D \equiv 0 \pmod 4$. Here B is even, and A and C cannot both be even; of the three forms (3), (5) one therefore has two odd outer coefficients, while the other two have one even and one odd outer coefficient.

2. If $D \equiv 1 \pmod 8$, then B is odd, $B^2 \equiv 1 \pmod 8$, $AC \equiv 0 \pmod 2$, hence at least one of the two coefficients is even, and among the three forms (3), (5) there is one, but also only one, that has two even outer coefficients.

3. If $D \equiv 5 \pmod 8$, then $4AC \equiv 4 \pmod 8$ and the two outer coefficients A and C are odd.

If it now proves possible to set up an equation with integral coefficients which only one of the three roots (2) satisfies, it follows from this that this one root is rationally expressible by $j(\omega)$ and is therefore a class invariant.

Between the functions

$$u = f(\omega)^{24}, \quad v = f\left(\frac{c + \partial\omega}{a}\right)^{24} \quad (c \equiv 0) \pmod 2 \tag{6}$$

there exists, for an odd degree of transformation n (§§ 73, 74), a transformation equation:

$$\Phi_n(u, v) = 0,$$

and the equation:

$$\Phi_n(u, u) = 0 \tag{7}$$

is therefore satisfied, by § 53, only when for one of the quantities (6):

$$\frac{c + \partial\omega}{a} = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \tag{8}$$

in which

$$\begin{aligned} &\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} \pmod 2 \\ \text{or} \quad &\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \equiv \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} \pmod 2 \end{aligned} \tag{9}$$

is a linear transformation belonging to the first or to the second class (§ 36).

The comparison of (8) with (3) leads, however, as above, to the conditions:

$$
\begin{aligned}
&\beta\partial = Ax,\\
&\alpha c - \gamma a = Cx,\\
&\beta c - \delta a + \alpha\partial = Bx,\\
&\beta c - \delta a - \alpha\partial = y,\\
&\beta c - \delta a = \frac{Bx+y}{2}, \quad \alpha\partial = \frac{Bx-y}{2},\\
&4n = y^2 - Dx^2.
\end{aligned} \tag{10}
$$

If we assume A and n to have no common divisor, then $\partial = 1$, $a = n$ must hold; for a prime dividing x and ∂ would have to divide y as well, and consequently $\alpha c - \gamma a$ and $\beta c - \delta a$, and, since $\alpha\delta - \beta\gamma = 1$, also a and c, whereas a, c, ∂ are to have no common divisor. From (10), α, β, γ, δ, c are determined in the same way as in the preceding section. There result first of all the two congruences

$$cAx \equiv \frac{Bx+y}{2}, \quad c\,\frac{Bx-y}{2} \equiv Cx \pmod{n},$$

of which the one follows from the other if x is coprime to n, and the condition

$$c \equiv 0 \pmod{2}$$

is compatible with them. Further it follows that:

$$
\begin{aligned}
&\alpha = \frac{Bx-y}{2}, \quad n\gamma = -Cx + c\,\frac{Bx-y}{2},\\
&\beta = Ax, \qquad\qquad n\delta = -\frac{Bx+y}{2} + Acx.
\end{aligned} \tag{11}
$$

If now D, and hence B, is odd, then, since by (9) α and δ are either both even or both odd, x and y must be even; accordingly β and γ are even, and α and δ must be odd. Consequently, of the two numbers $\frac{1}{2}x$, $\frac{1}{2}y$ one must be even and one odd. But this is independent of how A and C behave with respect to the modulus 2.

If, therefore, one of the three quantities (2) satisfies the equation (7), then so do the other two, and we cannot separate these three from one another.

We must therefore assume B to be even, and for simplification we set:

$$D = -4m, \tag{12}$$

in which m is a positive integer.

We distinguish:

1. If $D \equiv 4 \pmod{8}$, we set

$$
\begin{aligned}
n = m, \quad x = 1, \quad y = 0,\\
\alpha = \frac{B}{2}, \quad \beta = A,\\
\gamma \equiv -C, \quad \delta \equiv \frac{B}{2} \pmod{2}.
\end{aligned}
$$

The congruence (9) therefore requires, since not all three coefficients A, B, C can be even, that the two outer coefficients be odd and the middle one divisible by 4; and this occurs only for one of the three forms (3), (5). Consequently one of the three functions (2) satisfies equation (7), and is therefore a class invariant.

2. $D \equiv 0 \pmod 8$,

$$n = m+1, \quad x = 1, \quad y = \pm 2,$$

$$\alpha = \frac{B}{2} - 1, \quad \beta = A,$$

$$\gamma \equiv -C, \quad \delta \equiv \frac{B}{2} + 1 \pmod 2.$$

Here too, then, the two outer coefficients must be odd and $B \equiv 2 \pmod 4$, and everything then stands as above. With this it is proved:

1. If ω is the root of a primitive quadratic equation with negative discriminant D divisible by 4 and with odd outer coefficients, then $f(\omega)^{24}$ is a class invariant for the discriminant D.

The assumption that A is relatively prime to n, i. e. to $m+1$ or to m, may afterwards be given up as unessential.

For if we apply a linear transformation (S) to ω, then, for odd A, C, equation (3) passes into an equivalent equation in which the outer coefficients are both odd then and only then, when (S) belongs to the first or the second class, that is, when $f(\omega)^{24}$ remains unchanged under (S). One can then further determine (S) so that the first coefficient in the transformed equation (3) becomes relatively prime to any number one pleases, and hence also to n.

This procedure is not applicable to the case $m = 1$, because $\Phi_n(u, u)$ vanishes identically for $n = 1$, $\omega = i$ (but also only for this value); for $m = 1$, however, we have already found earlier (§ 83):

$$f(i)^{24} = 64,$$

so that in this case too the theorem stated holds.

If m is odd, then for odd A, C the middle coefficient B is divisible by 4; if, on the other hand, m is even, then under the same assumption B is not divisible by 4. But by the substitution $\omega + 1$ for ω we achieve, in the case of an even m as well, that B becomes divisible by 4; but then C also becomes even. By the interchange $(\omega, \omega+1)$, however, $f(\omega)^{24}$ passes into $-f_1(\omega)^{24}$, so that we can also give our theorem the following expression:

2. If ω is the root of the quadratic equation:

$$A\omega^2 + B\omega + C = 0,$$

in which A is odd and B is even, then, according as $D \equiv 4$ or $D \equiv 0 \pmod 8$, $f(\omega)^{24}$ or $f_1(\omega)^{24}$ is a class invariant.

Since A and C cannot both be even, only three cases are possible, of which, by the interchanges

$$(\omega, \omega+1), \quad \left(\omega, -\frac{1}{\omega}\right)$$

the second is reduced to the first and the third to the second. In these cases, then, we have:

(13)
$$\begin{array}{lll} A \equiv 1, & C \equiv 1 \pmod 2, & \text{class invariant } f(\omega)^{24}, \\ A \equiv 1, & C \equiv 0 & \text{„ „ } f_1(\omega)^{24}, \\ A \equiv 0, & C \equiv 1 & \text{„ „ } f_2(\omega)^{24}. \end{array}$$

In particular, therefore, if we take as our basis the principal form of the discriminant $-4m$, $(1, 0, m)$, then $f(\sqrt{-m})^{24}$ for odd, and $f_1(\sqrt{-m})^{24}$ for even m, is a class invariant. These two numbers have a real positive value, and in the calculations that follow further below they shall be taken into consideration by preference.

From equation (1) we conclude (Bd. II, § 154, 11.), since $j(\omega)$ is an integral algebraic number, that $f(\omega)^{24}$ must also be an integral algebraic number, and the form of equation (1) shows that the norm of this number is a power of 2.

§ 127. **The Powers of $f(\omega)$ as Class Invariants.**

By the definition of $\gamma_2(\omega)$ we have:

$$f(\omega)^8 = \frac{f(\omega)^{24} - 16}{\gamma_2(\omega)}. \tag{1}$$

If ω is the root of an equation:

$$A\omega^2 + 2B\omega + C = 0, \quad AC - B^2 = m, \tag{2}$$

whose discriminant $-4m$ is not divisible by 3, then we may assume, passing if necessary to an equivalent equation, that A, C are odd, that A is indivisible by 3 and that B is divisible by 3. But then, by the two preceding sections, $f(\omega)^{24}$ and $\gamma_2(\omega)$ are class invariants, and from (1) we obtain the theorem:

3. If ω is the root of a quadratic form of negative discriminant not divisible by 3, whose two outer coefficients are odd and whose middle coefficient is divisible by 6, then

$$f(\omega)^8$$

is a class invariant.

In order to investigate the question whether still lower powers of $f(\omega)$ can also be class invariants, we make use of the formula:

$$f(\omega + 2r)^6 = i^{-r} f(\omega)^6, \tag{3}$$

in which r is an integer; the values

$$\omega_r = \omega + 2r$$

are the roots of equivalent quadratic equations

$$A\omega_r^2 + 2B_r\omega_r + C_r = 0,$$

in which

$$B_r = B - 2Ar, \tag{4}$$

and here r can be determined according to the modulus 4 in such a way that, for odd m:

$$B_r \equiv 0, 2, 4, 6 \pmod 8 \tag{5}$$

and for even m:

$$B_r \equiv 1, 3, 5, 7 \pmod 8. \tag{6}$$

If we now succeed in finding an equation with rational coefficients that, among the four values (3), is satisfied either only

by the one corresponding to the value $r = 0$, or by the two corresponding to the values $r = 0, 2$, then it follows that $f(\omega)^6$ or $f(\omega)^{12}$ are class invariants.

If moreover m is indivisible by 3, then one may assume the B_r all divisible by 3, and it then follows, by combination with Theorem 1., that also

$$f(\omega)^8 f(\omega)^{-6} = f(\omega)^2 \quad \text{or} \quad f(\omega)^{12} f(\omega)^{-8} = f(\omega)^4$$

are contained among the class invariants.

Now that our next goal has thus been indicated, we make use of the result of § 73, that between

$$v = f\left(\frac{c + \partial\omega}{a}\right)^3, \quad u = f(\omega)^3$$

a modular equation:

$$\Phi_n(u, v) = 0 \tag{7}$$

holds, when

$$a\partial = n, \quad c \equiv 0 \pmod{16}$$

The equation derived from this

$$\Phi_n(u, u) = 0 \tag{8}$$

is satisfied then and only then, when

$$\frac{c + \partial\omega}{a} = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \tag{9}$$

where $\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$ is a linear transformation of the first or second class which [according to § 40, (12)] satisfies the condition:

$$\left(\frac{2}{\alpha - \beta}\right) e^{-\frac{\pi i}{8}(\alpha-\beta)(\alpha+\beta+\gamma-\delta)} = 1,$$

or the condition equivalent to it:

$$\alpha\gamma + \beta\delta + 2\alpha^2 - 2\alpha\beta - 2\alpha\delta \equiv 0 \pmod{16}. \tag{10}$$

But from (9), if ω satisfies equation (2), there follows, as we have seen in the preceding section,

$$\begin{array}{ll} n = y^2 + mx^2, & \\ \alpha = Bx - y, & n\gamma = -Cx + c(Bx - y), \\ \beta = Ax, & n\delta = -Bx - y + Acx. \end{array} \tag{11}$$

If we first assume m to be odd, then we may set $n = m$, hence $x = 1$, $y = 0$, and since c is divisible by 16, it follows:

$$\alpha = B, \quad \beta = A, \quad \gamma \equiv -mC, \quad \delta \equiv -mB \pmod{8},$$

hence (10) gives:

$$B\left(m\,\frac{A + C}{2} - (m + 1)\,B + A\right) \equiv 0 \pmod{8}. \tag{12}$$

If now $m \equiv 3 \pmod 4$, then $A + C \equiv 0 \pmod 4$ and the expression standing in the parenthesis in (12) is odd; hence (12) is satisfied only under the assumption that

$$B \equiv 0 \pmod{8}.$$

In order to avoid repetitions, we shall assume once and for all, in the theorems to be stated in what follows, that ω is the root of such an equation (3) of discriminant $-4m$ in which A is odd and B divisible by 8. With this the theorem is proved:

4. If $m \equiv 3 \pmod 4$, then $f(\omega)^6$ is a class invariant.

If further $m \equiv 5 \pmod 8$, the congruence (12) reduces to:

$$B\left(\frac{5C - A}{2} + 2B\right) \equiv 0 \pmod{8},$$

and this condition is fulfilled if B is divisible by 4, but it is not fulfilled if B is divisible only by 2 and not by 4. From this it follows:

5. If $m \equiv 5 \pmod 8$, then $f(\omega)^{12}$ is a class invariant.

If $m \equiv 1 \pmod 8$, nothing can be concluded from the congruence (12). This agrees with the circumstance that in this case $\Phi_m(u, u)$ depends only on u^8. In order to reach similar results here as well, we apply the transformation of the second order. We take in (7):

$$n \equiv 1 \pmod 8 \tag{13}$$

and set:

$$u\,v = \pm\sqrt{2}^3,$$

i. e.:

$$f\left(\frac{c+\partial\omega}{a}\right)^3 f(\omega)^3 = \pm\sqrt{2}^3,$$

or [§ 34, (18)]:

$$f\left(\frac{c+\partial\omega}{a}\right)^3 = \pm f\left(\frac{\omega-1}{\omega+1}\right)^3. \tag{14}$$

The equation (7) thereby passes into an equation:

$$\Phi_n\left(u, \frac{\pm\sqrt{2}^3}{u}\right) = 0, \tag{15}$$

or, more precisely, according to the choice of sign, into two equations which, apart from the product

$$\sqrt{2}\, f(\omega)^2$$

contain only rational numerical coefficients. One sees the latter from the fact that in equation (7) there occur only such products $u^h v^k$ in which $h + k$ is even (cf. the beginning of § 74).

Relation (14) now requires that ω satisfy an equation

$$\frac{c+\partial\omega}{a} = \frac{\gamma(\omega+1)+\delta(\omega-1)}{\alpha(\omega+1)+\beta(\omega-1)} \tag{16}$$

in which $\begin{pmatrix}\alpha, & \beta\\ \gamma, & \delta\end{pmatrix}$ is a linear transformation belonging to the first or the second class. But in order that one of the two equations (14) be actually satisfied, there is further required, by § 40, (12):

$$(\alpha-\beta)(\alpha+\beta+\gamma-\delta) \equiv 0 \pmod 8. \tag{17}$$

Now if ω is a root of the quadratic equation (2), it follows from (16), as above:

$$\begin{aligned}
\partial(\alpha+\beta) &= Ax,\\
\partial(\alpha-\beta) &= Bx+y,\\
c(\alpha-\beta)-a(\gamma-\delta) &= Cx,\\
c(\alpha+\beta)-a(\gamma+\delta) &= Bx-y,\\
2n &= y^2+mx^2.
\end{aligned} \tag{18}$$

If now, as was assumed,

$$m \equiv 1 \pmod 8,$$

we set:

$$n = \frac{m+1}{2} \quad \text{or} \quad \frac{m+9}{2},$$

according as $m \equiv 1$ or $\equiv 9 \pmod{16}$, so that $n \equiv 1 \pmod 8$ as well; then one has to set $x = 1$, $y = \pm 1$ or $= \pm 3$, and if we assume A relatively prime to n, then $\partial = 1$, $a = n$, and from (18) it follows:

$$\begin{aligned}
\alpha+\beta &= A, & n(\gamma-\delta) &= -C+c(B+y),\\
\alpha-\beta &= B+y, & n(\gamma+\delta) &= -B+y+cA.
\end{aligned}$$

Accordingly, since $B + y$ is odd and $c \equiv 0 \pmod{16}$, condition (17) gives:

$$A \equiv C \pmod{8},$$

which is only possible if $B \equiv 0 \pmod{4}$. But in order to decide which of the two equations (15) is fulfilled, it depends, according to § 40, (12), on whether

$$(\alpha - \beta)^2 - 1 + (\alpha - \beta)(\alpha + \beta + \gamma - \delta)$$

or, what is the same thing,

$$(B + y)^2 - 1 + (B + y)(A - nC)$$

is divisible by 16 or only by 8. But if in this expression B is increased by an odd multiple of 4, then it is altered by an odd multiple of 8, from which it is to be concluded that, according as $B \equiv 0$ or $\equiv 4 \pmod{8}$, the one or the other of the two equations (15) is satisfied.

With this it is proved:

6. If $m \equiv 1 \pmod{8}$, then $\sqrt{2}\, f(\omega)^6$ is a class invariant.

In the cases where $m \equiv 3 \pmod{4}$, we can go one step further. In § 73 we became acquainted, besides the Schlaefli modular equations, also with the equations

$$\Phi_n(u_1, v_1) = 0 \tag{19}$$

which subsist between

$$u_1 = f_1(\omega)^3, \quad v_1 = \left(\frac{2}{a}\right) f_1\left(\frac{c + \partial\omega}{a}\right)^3 \tag{20}$$

and from § 73, (10), (15) it follows that, if $n \equiv 7 \pmod{8}$ is assumed, $\Phi_n(u_1, v_1)$ can be rationally represented by

$$u_1 v_1, \quad u_1^8 + v_1^8$$

If, then, in (19) we put

$$v_1 = f_1\left(\frac{c + \partial\omega}{a}\right)^3 = \pm f_2(\omega)^3, \quad u_1 = f_1(\omega)^3 \tag{21}$$

there results

$$u_1 v_1 = \frac{\pm\sqrt{2}^3}{f(\omega)^3},$$

and $u_1^8 + v_1^8$ can be expressed rationally by $f(\omega)^{24}$ [§ 34, (11)], that is,

$$\Phi_n(u_1, v_1) = 0$$

passes over into one or, according to the choice of the sign, two rational equations for

$$\xi = \sqrt{2}\, f(\omega)^3$$

and these we wish to denote by

$$F(\pm\xi) = 0 \tag{22}$$

But equation (22) again has as its consequence an equation of the form (21), so that the signs agree in both, and equation (21) requires, by § 40, (7):

$$\begin{aligned} &\frac{c + \partial\omega}{a} = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \quad \delta \equiv 0 \pmod{2}, \\ &\gamma^2 - 1 + \gamma(2\alpha - \delta) \equiv 0 \text{ or } \equiv 8 \pmod{16}, \end{aligned} \tag{23}$$

and in fact, according as the one or the other congruence holds, the one or the other sign applies in (21) and (22).

If we take $m = n$, and therefore $m \equiv -1 \pmod 8$, then from (2) and (23) there results, exactly as above:

$$\beta = A, \quad n\gamma = -C + cB,$$
$$\alpha = B, \quad n\delta = -B + cA,$$

and hence, from the second congruence (23):

$$C^2 + CB - 1 \equiv 0 \text{ or } \equiv 8 \pmod{16},$$

from which it follows first of all that B must in any case be divisible by 8; but if we increase B by an odd multiple of 8, then one of these congruences changes into the other, and consequently one sign in (21) changes into the other. For a definite ω, therefore, only one of the two equations (22) holds, and it follows:

7. If $m \equiv 7 \pmod 8$, then $\sqrt{2}\, f(\omega)^3$ is a class invariant.

If we may trust an induction confirmed by all known examples, then the following theorem also holds:

8. If $m \equiv 3 \pmod 8$, then $f(\omega)^3$ is a class invariant.

The general proof of this is, however, still lacking.

If $m \equiv 2 \pmod 4$, we apply the same procedure as above in the case $m \equiv 1 \pmod 8$.

We put:

$$2n = m + y^2, \tag{24}$$

and take $y = 0$ or $= \pm 2$, so that $n \equiv 1 \pmod 4$ results.

If we then set in equation (19)

$$v_1 = f_2\left(\frac{\omega}{2}\right)^3 = \frac{\sqrt{2}^3}{f_1(\omega)^3}, \quad u_1 = f_1(\omega)^3 \tag{25}$$

then there results an equation which contains only $\sqrt{2}\, f_1(\omega)^6$ and rational coefficients. (19), however, is satisfied only [§ 40, (7)] if

$$\frac{c + \partial\omega}{a} = \frac{2\gamma + \delta\omega}{2\alpha + \beta\omega}, \quad \delta \equiv 0 \pmod 2 \tag{26}$$

and in addition

$$\gamma^2 - 1 + \gamma(2\alpha - \delta) \equiv 0 \pmod{16}; \tag{27}$$

from (26) and (24) it follows:

$$\begin{matrix} 2\alpha = B + y, & 2n\gamma \equiv -C \\ \beta = A, & 2n\delta \equiv -(B - y) \end{matrix} \pmod{16},$$

and from this one concludes as above:

9. If $m \equiv 2 \pmod 4$, then $\sqrt{2}\, f_1(\omega)^6$ is a class invariant.

Again all the examples indicate that, if $m \equiv 4 \pmod 8$, $\sqrt{2}\, f_1(\omega)^{12}$ is a class invariant. But here too the proof is still lacking. If m is divisible by 8, no reduction occurs.

In cases 4. to 8. there follows, by 3., a further reduction to the third root, if m is not divisible by 3. Accordingly we obtain the following cases:

$$m \equiv \pm 1 \pmod 3,$$

			class invariant
1)	$m \equiv 3$	(mod 4)	$f(\omega)^2$,
2)	$m \equiv 5$	(mod 8)	$f(\omega)^4$,
3)	$m \equiv 1$	"	$\sqrt{2}\, f(\omega)^2$,
4)	$m \equiv 7$	"	$\sqrt{2}\, f(\omega)$,
5)	$m \equiv 3$	"	$f(\omega)$,
6)	$m \equiv 2$	"	$\sqrt{2}\, f_1(\omega)^2$,
7)	$m \equiv 4$	"	$\sqrt{2}\, f_1(\omega)^4$,

of which, to be sure, the cases 5) and 7) are inferred only by induction.

The class invariants $f(\omega)$ are particularly well suited to numerical calculation, because among them all they yield the simplest results. We therefore give first a larger series of examples, which can easily be derived on the basis of our developments so far and which at the same time teach the methods that can also be used for further calculations of this kind.

In these calculations one is frequently faced with the task of decomposing reducible integral rational functions into factors. Such a decomposition, once found, is of course immediately

verifiable; but finding the factors is also easy, at least as far as the calculations have so far been carried, since in later cases one can often use results found earlier and moreover knows the general form of the factors. Examples will make this clear.

§ 128. The first cases of the calculation of $f(\sqrt{-m})$.

If in the cubic equation:

$$u^3 - \gamma_2(\omega)\, u - 16 = 0, \tag{1}$$

whose roots according to § 54 are

$$f(\omega)^8, \quad -f_1(\omega)^8, \quad -f_2(\omega)^8$$

we put $\omega = i$, hence [§ 125, (16)], $\gamma_2 = 12$, then it follows:

$$u^3 - 12u - 16 = 0,$$

an equation whose only positive root is $u = 4$, so that in agreement with § 126 we obtain:

$$f(i) = \sqrt[4]{2}; \tag{2}$$

if we put $\omega = \sqrt{-2}$, hence $\gamma_2 = 20$ [§ 125, (17)], then from (1) we obtain the equation:

$$u^3 - 20u - 16 = 0$$

with the only rational root -4. But since, by the theorems of the two preceding sections, $f_1(\sqrt{-2})^8$ must be rational, we have:

$$f_1(\sqrt{-2}) = \sqrt[4]{2}. \tag{3}$$

We further put $\omega = e^{\frac{2\pi i}{3}}$, hence $\gamma_2(\omega) = 0$ (§ 117), and obtain from (1):

$$u^3 = 16.$$

This equation is satisfied by [§ 34, (19)]:

$$u = -f_2\left(\frac{-1+\sqrt{-3}}{2}\right)^8 = \frac{16\, e^{\frac{2\pi i}{3}}}{f(\sqrt{-3})^8},$$

from which the real positive value:

$$f(\sqrt{-3}) = \sqrt[3]{2}. \tag{4}$$

In order to be able to apply the remaining results of § 125, we put

$$m = 7, \quad 11, \quad 19, \quad 43, \quad 67, \quad 163,$$

and use the formula that follows from § 34, (19):

$$f(\omega)\, f_2\left(\frac{-3+\omega}{2}\right) = e^{-\frac{\pi i}{8}}\sqrt{2}.$$

Accordingly, if we put

$$f(\sqrt{-m}) = x$$

then by (1) x is the real positive root of the equation

$$x^{24} + \gamma_2\left(\frac{-3+\sqrt{-m}}{2}\right)x^{16} - 2^8 = 0. \tag{5}$$

For $m = 7$ we obtain by § 125, (18)

$$f(\sqrt{-7}) = \sqrt{2}, \tag{6}$$

while in the other cases, after the values for γ_2 have been substituted, equation (5) splits first into two factors of degree 12, these again into two factors of degree 6, and finally these into two of degree 3. The equations so obtained are cubic equations for x^8, x^4, x^2, x, of which in each case we retain only the one that has a real positive root, and which is in the end satisfied by $f(\sqrt{-m})$ itself.

To find the first decomposition we put equation (5) into the form:

$$(x^{12} - ax^4)^2 - (bx^8 + c)^2 = 0,$$

and have to determine the integers a, b, c from the equations:

$$2a + b^2 = -\gamma_2, \quad 2bc = a^2, \quad c^2 = 2^8$$

so that $c = \pm 16$; the equation of degree 12 with positive root therefore reads:

$$x^{12} - bx^8 - ax^4 - 16 = 0.$$

The numbers a, b are easily determined from the equations above, and in this way one finally finds the cubic equations sought. For $m = 11$, for example, one obtains successively

$$\begin{aligned} x^{12} - 8x^8 + 16x^4 - 16 &= 0, \\ x^6 - 4x^2 - 4 &= 0, \\ x^3 - 2x^2 + 2x - 2 &= 0. \end{aligned}$$

In the five cases of § 125 one finds the following equations:

$$x = f(\sqrt{-11}), \quad x^3 - 2x^2 + 2x - 2 = 0, \tag{7}$$

$$x = f(\sqrt{-19}), \quad x^3 - 2x - 2 = 0, \tag{8}$$

$$x = f(\sqrt{-43}), \quad x^3 - 2x^2 - 2 = 0, \tag{9}$$

$$x = f(\sqrt{-67}), \quad x^3 - 2x^2 - 2x - 2 = 0, \tag{10}$$

$$x = f(\sqrt{-163}), \quad x^3 - 6x^2 + 4x - 2 = 0. \tag{11}$$

§ 129. Application of the Transformation of the Second Order to the Calculation of Class Invariants.

The transformation of the second order can be applied, repeating it several times if necessary, to all those discriminants $-4m$ for which $y^2 + mx^2$ is a power of 2, thus for example to $m = 7$, 15, 23, 31.

These calculations are, however, laborious, and we shall find simpler ways of reaching the goal in these cases. The transformation of the second order renders better service in finding, from a known class invariant, a new one belonging to a discriminant which is four times the former.

The following formulae lead to this: According to § 34:

$$f_1(\omega)^8 + f_2(\omega)^8 = f(\omega)^8,$$

$$f_1(\omega)^8 f_2(\omega)^8 = \frac{16}{f(\omega)^8},$$

from which

$$f_1(\omega)^8 - f_2(\omega)^8 = \frac{\sqrt{f(\omega)^{24} - 64}}{f(\omega)^4}.$$

The sign of the root changes only for $f(\omega)^{24} = 64$, that is, for $\omega = i$, and so long as $-i\omega$ is real and greater than 1 it is to be taken positive, since the left-hand side becomes positively infinite for a vanishing q. From this it follows:

$$2f_2(\omega)^8 = \frac{f(\omega)^{12} - \sqrt{f(\omega)^{24} - 64}}{f(\omega)^4},$$

or

$$f_2(\omega)^8 f(\omega)^4 \left[f(\omega)^{12} + \sqrt{f(\omega)^{24} - 64} \right] = 32,$$

and in the same way:

$$f_2(\omega)^8 f_1(\omega)^4 \left[f_1(\omega)^{12} + \sqrt{f_1(\omega)^{24} + 64} \right] = 32.$$

Further, according to § 34, (16):

$$f_1(2\omega)\, f_2(\omega) = \sqrt{2},$$

from which

$$\begin{aligned} 2f_1(2\omega)^8 &= f(\omega)^4 \left[f(\omega)^{12} + \sqrt{f(\omega)^{24} - 64} \right], \\ &= f_1(\omega)^4 \left[f_1(\omega)^{12} + \sqrt{f_1(\omega)^{24} + 64} \right]. \end{aligned} \tag{1}$$

These formulae can serve, when $f(\omega)$ or $f_1(\omega)$ is known, to calculate $f_1(2\omega)$, and so $f_1(\sqrt{-4m})$ from $f(\sqrt{-m})$

or $f_1(\sqrt{-m})$. In this way one finds very easily, if one takes $f(i)$, $f_1(\sqrt{-2})$, $f(\sqrt{-3})$, $f(\sqrt{-7})$ from the formulae of the preceding section:

$$f_1(\sqrt{-4})^8 = 8, \tag{2}$$

$$f_1(\sqrt{-16})^8 = 8\sqrt{2}(1 + \sqrt{2})^2, \tag{3}$$

$$f_1(\sqrt{-8})^8 = 8(1 + \sqrt{2}), \tag{4}$$

$$f_1(\sqrt{-32})^8 = 32(1 + \sqrt{2})^2 + 8\sqrt{2}\sqrt{8(1 + \sqrt{2})^4 + (1 + \sqrt{2})}. \tag{5}$$

If we set:

$$f_1(\sqrt{-32})^8 = 8x,$$

then x is a root of the biquadratic class equation:

$$(x^2 - 24x - 2)^2 - 8(8x + 1)^2 = 0, \tag{6}$$

which by the adjunction of $\sqrt{2}$ splits into two quadratic equations

$$x^2 - 8(1 \pm \sqrt{2})^2 x - 2(1 \pm \sqrt{2}) = 0$$

Furthermore we find in this way:

$$f_1(\sqrt{-12})^4 = 2\sqrt[6]{2}(1 + \sqrt{3}), \tag{7}$$

$$f_1(\sqrt{-28})^4 = 2\sqrt{2}(3 + \sqrt{7}). \tag{8}$$

We shall apply this method to other cases later on.

§ 130.

Calculation of Class Invariants from the Schlaefli Modular Equations.

I. If in one of the equations between $u = f(\omega)$ and $v = f(n\omega)$, or between $u = f_1(\omega)$, $v = f_1(n\omega)$ (§ 73), we substitute for u one of the known values of $f(\sqrt{-m})$ or $f_1(\sqrt{-m})$, then we obtain an equation which $f(\sqrt{-n^2m})$ or $f_1(\sqrt{-n^2m})$ satisfies.

Under certain circumstances this equation still contains extraneous factors, which have to be sought out and removed.

If n is a prime, then by § 123 we obtain complete information about these superfluous factors. If n divides m, then in the equation in question a linear factor belonging to the determinant $-m$ is to be separated off; if $-m$ is a quadratic non-residue of n, then superfluous factors are not present at all; if $-m$ is a quadratic residue of n, then a factor belonging to the discri-

minant $-4m$ is to be separated off, and if $m = 1$, then the equation remaining after the separation of the extraneous factors is a square.

As examples of these cases we take:

1. $m = 3,\ n = 3,$ separation of a linear factor.
2. $m = 2,\ n = 5,$ no extraneous factor.
3. $m = 2,\ n = 3,$ separation of a quadratic factor.
4. $m = 1,\ n = 3,$ square.
5. $m = 1,\ n = 5,$ square after separation of a quadratic factor.
6. $m = 1,\ n = 7,$ square.

In case 1. one has to set, in the formula of § 73 relating to the degree of transformation $n = 3$:

$$u^3 = 2, \quad v^3 = f(\sqrt{-27})^3 = 2x,$$

hence

$$A = x^2 + \frac{1}{x^2}, \quad B = 4x - \frac{2}{x},$$

and consequently:

$$x^4 - 4x^3 + 2x + 1 = (x - 1)(x^3 - 3x^2 - 3x - 1) = 0,$$

so that for $m = 27$ one obtains:

$$f(\sqrt{-27})^3 = 2x, \quad x^3 - 3x^2 - 3x - 1 = 0. \tag{1}$$

In case 2. we set, in system II. of § 73 ($n = 5$):

$$u_1 = f_1(\sqrt{-2}) = \sqrt[4]{2}, \quad v_1 = f_1(\sqrt{-50}) = \sqrt[4]{2}\,x,$$

$$A_1 = \frac{1}{x^3} - x^3, \quad B_1 = 2\Big(x^2 + \frac{1}{x^2}\Big),$$

so that (without an extraneous divisor) one obtains the equation of degree 6:

$$\frac{1}{x^3} - x^3 + 2\Big(x^2 + \frac{1}{x^2}\Big) = 0, \quad f_1(\sqrt{-50}) = \sqrt[4]{2}\,x \tag{2}$$

which, for

$$y = \frac{1}{x} - x$$

reduces to the third degree:

$$y^3 + 2y^2 + 3y + 4 = 0. \tag{3}$$

In case 3. we set, in System II. of § 73 ($n = 3$):

$$u_1 = \sqrt[4]{2}, \quad v_1^3 = f_1(\sqrt{-18})^3 = \sqrt[4]{2}\,x,$$

$$A_1 = \frac{2}{x^2} - \frac{x^2}{2}, \quad B_1 = 2x + \frac{4}{x},$$

$$x^4 - 4x^3 - 8x - 4 = (x^2 + 2)(x^2 - 4x - 2),$$

hence

$$x^2 - 4x - 2 = 0, \quad f_1(\sqrt{-18})^3 = \sqrt[4]{2}\,x. \tag{4}$$

The solution of (4) gives:

$$x = 2 + \sqrt{6}. \tag{5}$$

In case 4. we have:

$$u = f(i) = \sqrt[4]{2}, \quad v^3 = f(\sqrt{-9})^3 = \sqrt[4]{2}\,x,$$

$$A = \frac{x^2}{2} + \frac{2}{x^2}, \quad B = 2x - \frac{4}{x},$$

$$x^4 - 4x^3 + 8x + 4 = (x^2 - 2x - 2)^2 = 0,$$

hence

$$x^2 - 2x - 2 = 0, \quad f(\sqrt{-9})^3 = \sqrt[4]{2}\,x, \tag{6}$$

$$x = 1 + \sqrt{3}. \tag{7}$$

In case 5.:

$$u = f(i) = \sqrt[4]{2}, \quad v = f(\sqrt{-25}) = \sqrt[4]{2}\,x,$$

$$A = x^3 + \frac{1}{x^3}, \quad B = 2\Big(x^2 - \frac{1}{x^2}\Big),$$

$$x^3 + \frac{1}{x^3} - 2\Big(x^2 - \frac{1}{x^2}\Big) = \Big(x + \frac{1}{x}\Big)\Big(x - \frac{1}{x} - 1\Big)^2 = 0.$$

$$x - \frac{1}{x} - 1 = 0, \quad f(\sqrt{-25}) = \sqrt[4]{2}\,x, \tag{8}$$

$$x = \frac{1 + \sqrt{5}}{2}. \tag{9}$$

Finally, for $n = 7$ we set:

$$u = \sqrt[4]{2}, \quad v = f(\sqrt{-49}) = \frac{x}{\sqrt[4]{2}}$$

and find

$$\begin{aligned} 0 &= (x^8 - 4x^7 + 28x^4 - 32x + 16) \\ &= (x^4 - 2x^3 - 2x^2 - 4x + 4)^2, \end{aligned}$$

from which, by solving a quadratic equation, one readily obtains:

$$x + \frac{2}{x} = 1 + \sqrt{7}, \quad x = \sqrt[4]{2}\, f(\sqrt{-49}). \tag{10}$$

In the same way the class invariants listed in the table at the end have been computed for

$$m = 75, \quad 36, \quad 100, \quad 63, \quad 175$$

and these computations can also be continued still further.

II. The Schlaefli modular equations can be turned to account in various other ways as well for the computation of class invariants.

If we set

$$\omega = -\frac{m}{\omega}, \quad \omega = \sqrt{-m},$$

then

$$f(\omega) = f\Big(\frac{\omega}{m}\Big) = f(\sqrt{-m}); \tag{11}$$

if therefore in the modular equation for the transformation degree m

$$u = v$$

is set, there results an equation among whose roots $u = f(\sqrt{-m})$ occurs. In the system I. of § 73 we must then always set $A = 2$, and the formulae result:

$$m = 3, \quad B = u^6 - \frac{8}{u^6} = 2,$$

$$m = 5, \quad B = u^4 - \frac{4}{u^4} = 2,$$

$$m = 7, \quad B = u^6 + \frac{8}{u^6} = 9,$$

$$m = 11, \quad B = u^2 - \frac{2}{u^2}, \quad B^5 - B^3 - 2B = 2,$$

$$m = 13, \quad B = u^{12} - \frac{64}{u^{12}} = 9 \cdot 2^5,$$

$$m = 17, \quad B = u^8 + \frac{16}{u^8}, \quad B^2 - 68B - 544 = 0,$$

$$m = 19, \quad B = u^6 - \frac{8}{u^6}, \quad B^3 - 38B^2 + 252B - 648 = 0.$$

The cases $m = 3, 7, 11, 19$ yield no new results, but can be used for the verification of those found earlier; from $m = 5, 13$ we obtain by the solution of a quadratic equation:

$$f(\sqrt{-5})^4 = 1 + \sqrt{5}, \tag{12}$$

$$f(\sqrt{-13})^4 = 3 + \sqrt{13}. \tag{13}$$

For $m = 17$ the class number is 4, and hence the equation given above, which is of degree 4 with respect to u^8, is the class equation. If one solves it with respect to B, one obtains:

$$u^8 + \frac{16}{u^8} = 34 + 10\sqrt{17},$$
$$u^4 + \frac{4}{u^4} = 5 + \sqrt{17},$$
$$\left(u^2 + \frac{2}{u^2}\right)^2 = \frac{(1+\sqrt{17})^2}{2}.$$

If therefore one puts

$$\sqrt{2}\,x = f(\sqrt{-17})^2$$

one obtains for x the quadratic equation:

$$x + \frac{1}{x} = \frac{1+\sqrt{17}}{2}. \tag{14}$$

III. For ω we put the root of the quadratic equation:

$$2\omega^2 + 2r\omega + n = 0,$$

in which n denotes an odd integer and r an integer whose square is less than $2n$; thus:

$$2\omega + 2r = -\frac{n}{\omega}, \tag{15}$$

$$\omega = \frac{-r + \sqrt{-m}}{2}, \quad m = 2n - r^2. \tag{16}$$

Then, by § 34:

$$f_2(\omega) f_1(2\omega + 2r) = e^{-\frac{r\pi i}{12}}\sqrt{2}, \tag{17}$$

and by (15), (16):

$$f_2(\omega) f_2\left(\frac{\omega}{n}\right) = e^{-\frac{r\pi i}{12}}\sqrt{2}, \tag{18}$$

$$f_2(\omega) = \frac{e^{-\frac{r\pi i}{12}}\sqrt{2}}{f_1(r + \sqrt{-m})}; \tag{19}$$

we therefore put, according as r (and hence m) is even or odd,

$$x = f_1(\sqrt{-m}), \quad x = f(\sqrt{-m}), \tag{20}$$

then

$$f_2(\omega) = \frac{e^{-\frac{r\pi i}{24}}\sqrt{2}}{x}, \quad f_2\left(\frac{\omega}{n}\right) = e^{-\frac{r\pi i}{24}} x, \tag{21}$$

and these values are to be substituted for u_1, v_1 in the system II., § 73, in order to obtain an equation for x. By setting the various admissible values for r, one obtains from (16):

$$\begin{aligned}
&n = 3, && m = 6,\ 5,\ 2,\\
&n = 5, && m = 10,\ 9,\ 6,\ 1,\\
&n = 7, && m = 14,\ 13,\ 10,\ 5,\\
&n = 11, && m = 22,\ 21,\ 18,\ 13,\ 6,\\
&n = 13, && m = 26,\ 25,\ 22,\ 17,\ 10,\ 1,\\
&n = 17, && m = 34,\ 33,\ 30,\ 25,\ 18,\ 9,\\
&n = 19, && m = 38,\ 37,\ 34,\ 29,\ 22,\ 13,\ 2.
\end{aligned}$$

As a simple example we choose $n = 7$ and obtain from § 73, II.:

$$\frac{x^8}{4} + \frac{4}{x^8} = 7 + 4\sqrt{2}\cos\frac{r\pi}{4}.$$

From this there results for $r = 0$:

$$\frac{x^2}{\sqrt{2}} + \frac{\sqrt{2}}{x^2} = 1 + \sqrt{2}, \quad x = f_1(\sqrt{-14}), \tag{22}$$

for $r = 1$ the already known

$$f(\sqrt{-13})^4 = 3 + \sqrt{13}, \tag{23}$$

for $r = 2$:

$$f_1(\sqrt{-10})^2 = \frac{1+\sqrt{5}}{\sqrt{2}}. \tag{24}$$

As a second example we take further $n = 13$ and obtain:

$$A_1 = \frac{\sqrt{2}}{x^2} - \frac{x^2}{\sqrt{2}}, \quad B_1 = 16\cos\frac{r\pi}{2}; \tag{25}$$

for $r = 0, 4$, hence $m = 26, 10$, there results from this $B_1 = 16$, and consequently, by § 73:

$$A_1^7 - 6A_1^5 + A_1^3 + 20A_1 + 16 = 0,$$

while for $r = 2$, hence $m = 22$, in this equation A_1 is to be changed into $-A_1$. But one easily finds the decomposition:

$$A_1^7 - 6A_1^5 + A_1^3 + 20A_1 + 16 = (A_1+1)^2(A_1-2)^2(A_1^3 + 2A_1^2 + A_1 + 4). \tag{26}$$

If $-i\omega > \sqrt{2}$, then $f_1^2(\omega) > \sqrt{2}$ as well; for as $-i\omega$ increases from $\sqrt{2}$ to ∞, $f_1(\omega)^2$ likewise increases monotonically[29], from $\sqrt{2}$ to ∞, and consequently A_1 is negative for $r = 0, 2, 4$. The first factor $A_1 + 1$ vanishes, as appears from the already known result (24), for $r = 4$; hence the third factor $A_1^3 + 2A_1^2 + A_1 + 4$ vanishes for $r = 0$, while $A_1 + 2$ vanishes for $r = 2$. We thus obtain from (25):

$$f_1(\sqrt{-10})^2 = \sqrt{2}\,y, \quad y - \frac{1}{y} = 1, \tag{27}$$

$$f_1(\sqrt{-22})^2 = \sqrt{2}\,y, \quad y - \frac{1}{y} = 2, \tag{28}$$

$$f_1(\sqrt{-26})^2 = \sqrt{2}\,y, \quad y^6 - 2y^5 - 2y^4 + 2y^2 - 2y - 1 = 0. \tag{29}$$

[29]For from § 54, (11) there follows, by the substitution $(\omega, 1 - \frac{1}{\omega})$:

$$d\,f_1(\omega)^8 = -\frac{\pi i}{3}\vartheta_{10}^4\left[f(\omega)^8 + f_1(\omega)^8\right]d\omega.$$

IV. As an example of another kind of use of the Schlaefli modular equations, one which leads to the otherwise hardly accessible class invariant $f(\sqrt{-41})$, let the following serve. We set:

$$2n\omega^2 + 2r\omega + n = 0, \tag{30}$$

$$\omega = \frac{-r + \sqrt{-m}}{2n}, \quad m = 2n^2 - r^2. \tag{31}$$

Here r may denote any number which makes m positive, but which has no divisor in common with n [since otherwise (30) is imprimitive]. Then, by § 34:

$$f_2\left(\frac{\omega}{n}\right) = f_1[2(n\omega + r)] = \frac{e^{-\frac{r\pi i}{12}}\sqrt{2}}{f_2(n\omega)}, \tag{32}$$

$$f_1[2(n\omega + r)] = f_1(r + \sqrt{-m}) = e^{-\frac{r\pi i}{24}}x,$$

hence:

$$f_2\left(\frac{\omega}{n}\right) = e^{-\frac{r\pi i}{24}}x,$$

$$f_2(n\omega) = \frac{e^{-\frac{r\pi i}{24}}\sqrt{2}}{x},$$

if, according as r is even or odd,

$$x = f_1(\sqrt{-m}) \quad \text{or} \quad = f(\sqrt{-m}) \tag{33}$$

is set. If, then, in the modular equation belonging to n, II., § 73, we take:

$$u_1 = f_2(\omega),$$

$$v_1 = \left(\frac{2}{n}\right) f_2(n\omega) = \left(\frac{2}{n}\right) \frac{e^{-\frac{r\pi i}{24}}\sqrt{2}}{x}$$

or

$$v_1 = f_2\left(\frac{\omega}{n}\right) = e^{-\frac{r\pi i}{24}}x,$$

there result two equations, from which by elimination of u_1 one derives an equation for x. In order to carry out this calculation for $n = 5$, we set:

$$\begin{aligned} x f_2(\omega) &= \sqrt{2}\, e^{-\frac{5r\pi i}{24}}\xi, \\ \frac{x}{f_2(\omega)} &= e^{-\frac{5r\pi i}{24}}\eta, \end{aligned} \tag{34}$$

from which, corresponding to the two assumptions for v_1, one obtains for an odd r the equations:

$$\begin{aligned} \xi^3 + \frac{1}{\xi^3} + 2\left(\eta^2 - \frac{1}{\eta^2}\right) &= 0, \\ \eta^3 + \frac{1}{\eta^3} + 2\left(\xi^2 - \frac{1}{\xi^2}\right) &= 0. \end{aligned} \tag{35}$$

These equations hold for $r = 1, 3, 7$, which belong to the values $m = 49, 41, 1$. If one subtracts the two equations (35), one can cast off the factor $\xi - \eta$, which can vanish only for $m = 1$:

$$[(\xi + \eta)^2 - \xi\eta]\left(1 - \frac{1}{\xi^3\eta^3}\right) - 2(\xi + \eta)\left(1 + \frac{1}{\xi^2\eta^2}\right) = 0,$$

and if one adds the two equations (35):

$$[(\xi+\eta)^3 - 3\,\xi\eta(\xi+\eta)]\left(1+\frac{1}{\xi^3\eta^3}\right)$$
$$+\ 2[(\xi+\eta)^2 - 2\,\xi\eta]\left(1-\frac{1}{\xi^2\eta^2}\right) = 0.$$

But by (34):

$$\sqrt{2}\,\xi\eta = x^2,$$

and one thus obtains an equation for x^2 when one eliminates $(\xi+\eta)$. By the ordinary elimination procedure this equation is found, when one sets

$$z = \frac{x^2}{\sqrt{2}} + \frac{\sqrt{2}}{x^2}$$

in the form

$$z^6 - 9\,z^5 + 20\,z^4 + 6\,z^3 - 19\,z^2 - 17\,z - 6 = 0 \tag{36}$$

But this still contains the factor relating to the determinant -49. For this factor one has [by formula (10)]:

$$z = 2 + \sqrt{7}.$$

The left side of (36) must therefore be divisible by

$$z^2 - 4\,z - 3$$

and the carrying out of the division yields the factor valid for the determinant -41:

$$z^4 - 5\,z^3 + 3\,z^2 + 3\,z + 2 = 0, \tag{37}$$
$$z = \frac{f(\sqrt{-41})}{\sqrt{2}} + \frac{\sqrt{2}}{f(\sqrt{-41})}.$$

§ 131. Calculation of class invariants from the irrational forms of the modular equations.

In an extraordinarily simple way the irrational forms of the modular equations lead in many cases to the setting up of class equations.

1. In § 75 we have seen that, if $m \equiv -1 \pmod 8$, then between the two functions:

$$2A = f(\omega)f(m\omega) + (-1)^{\frac{m+1}{8}}[f_1(\omega)f_1(m\omega) + f_2(\omega)f_2(m\omega)],$$
$$B = \frac{2}{f_1(\omega)f_1(m\omega)} + \frac{2}{f_2(\omega)f_2(m\omega)} + \frac{(-1)^{\frac{m+1}{8}}2}{f(\omega)f(m\omega)}$$

an algebraic equation exists, and these equations are set up there for $m = 7,\ 23,\ 31,\ 47,\ 71$.

If we put in them:

$$\omega = \frac{-1}{\sqrt{-m}}, \quad m\omega = \sqrt{-m}, \quad f(\sqrt{-m}) = \sqrt{2}\,x,$$

then by § 34 we obtain:

$$A = x^2 + \frac{(-1)^{\frac{m+1}{8}}}{x}, \quad B = 4\,x + \frac{(-1)^{\frac{m+1}{8}}}{x}.$$

From this there results, for example, for $m = 23$, where $A = 1$, the equation:

$$f(\sqrt{-23}) = \sqrt{2}\,x, \quad x^3 - x - 1 = 0 \tag{1}$$

and for $m = 31$:

$$x^9 - 4\,x^6 + 3\,x^3 - 1 = 0.$$

This is in the first place a cubic equation for x^3; but from it the cubic equation for x itself is easily split off:

$$f(\sqrt{-31}) = \sqrt{2}\,x, \quad x^3 - x^2 - 1 = 0. \tag{2}$$

In like manner the equations result:

$$f(\sqrt{-47}) = \sqrt{2}\,x, \quad x^5 - x^3 - 2\,x^2 - 2\,x - 1 = 0, \tag{3}$$

$$f(\sqrt{-71}) = \sqrt{2}\,x, \quad x^7 - 2\,x^6 - x^5 + x^4 + x^3 + x^2 - x - 1 = 0. \tag{4}$$

In the last case, $m = 71$, the factor $(x + 1)^2$, which is foreign to the problem, has been separated off from the equation of degree 9 that is immediately obtained.

2. In order to arrive at a further mode of calculation, we apply the transformation of the second order. We have first of all in general [for a variable ω, § 34, (17)]:

$$f(2\omega)^8 + f_2(2\omega)^8 = 2\,\frac{f(\omega)^8 + f_1(\omega)^8}{f_2(\omega)^4},$$
$$f(2\omega)^8 - f_2(2\omega)^8 = \frac{16}{f_2(\omega)^8},$$

from which:

$$f(2\omega)^8 = \frac{f(\omega)^8 + f_1(\omega)^8}{f_2(\omega)^4} + \frac{8}{f_2(\omega)^8},$$
$$f_2(2\omega)^8 = \frac{f(\omega)^8 + f_1(\omega)^8}{f_2(\omega)^4} - \frac{8}{f_2(\omega)^8};$$

from this

$$f(\omega)^8 f(2\omega)^8 + f_1(\omega)^8 f_2(2\omega)^8 = 8 + \frac{[f(\omega)^8 + f_1(\omega)^8]^2}{f_2(\omega)^4},$$

which by § 34 can easily be brought into the form:

$$[f(\omega)^4 f(2\omega)^4 + f_1(\omega)^4 f_2(2\omega)^4]^2 = 16 + f_2(\omega)^{12} + \frac{64}{f_2(\omega)^{12}},$$

and from this the root can be extracted:

$$f(\omega)^4 f(2\omega)^4 + f_1(\omega)^4 f_2(2\omega)^4 = f_2(\omega)^6 + \frac{8}{f_2(\omega)^6}. \tag{5}$$

Now let ω satisfy the quadratic equation:

$$2\,\omega^2 + 2\,r\,\omega + n = 0 \tag{6}$$

with odd n, hence:

$$\omega = \frac{-r + \sqrt{-m}}{2}, \quad m = 2\,n - r^2. \tag{7}$$

If then

$$\sqrt{2}\,x = f(\sqrt{-m})^2, \quad \text{or } = f_1(\sqrt{-m})^2 \tag{8}$$

is put, according as r is odd or even, then there results

$$f_2(\omega)^2 = e^{-\frac{r\pi i}{12}} \frac{\sqrt{2}}{x}. \tag{9}$$

Furthermore, from

$$2\,\omega = -\frac{n}{\omega} - 2\,r$$

there follows, by the formulae of § 34:

$$f(2\omega) = f\left(\frac{\omega}{n}\right) e^{\frac{r\pi i}{12}}$$

$$\begin{aligned} f_1(2\omega) &= f_2\left(\frac{\omega}{n}\right) e^{\frac{r\pi i}{12}} \\ f_2(2\omega) &= f_1\left(\frac{\omega}{n}\right) e^{-\frac{r\pi i}{6}}, \end{aligned} \tag{10}$$

so that equation (5) gives:

$$f(\omega)^4 f\left(\frac{\omega}{n}\right)^4 + (-1)^r f_1(\omega)^4 f_1\left(\frac{\omega}{n}\right)^4 = e^{-\frac{r\pi i}{12}} \sqrt{8} \left(x^3 + \frac{e^{-\frac{r\pi i}{2}}}{x^3}\right). \tag{11}$$

From (10) there follows still further:

$$f_2(\omega) f_2\left(\frac{\omega}{n}\right) = \sqrt{2}\, e^{-\frac{r\pi i}{12}}, \tag{12}$$

$$f(\omega) f\left(\frac{\omega}{n}\right) f_1(\omega) f_1\left(\frac{\omega}{n}\right) = \sqrt{2}\, e^{\frac{r\pi i}{12}}. \tag{13}$$

If we assume $n = 23$ and $r = 0$, then equation § 75, (8) gives, with the use of (12):

$$f(\omega) f\left(\frac{\omega}{n}\right) - f_1(\omega) f_1\left(\frac{\omega}{n}\right) = 2 + \sqrt{2}, \tag{14}$$

from which, by squaring twice and using (11) and (13), it follows that:

$$x^3 + \frac{1}{x^3} = 36 + 26\sqrt{2};$$

from this one derives the simpler equation:

$$x + \frac{1}{x} = 3 + \sqrt{2}, \quad \sqrt{2}\, x = f_1^2(\sqrt{-46}). \tag{15}$$

3. Finally we make one further application of the transformation equations of § 76 for a composite degree of transformation, to the determinant -39.

If we set

$$u = f(\sqrt{-39}), \quad v = f\left(\sqrt{\frac{-13}{3}}\right), \quad \frac{u^2 + v^2}{uv} = z,$$

then in equation (22), § 76, which relates to $n = 39$, we have to set:

$$A = \frac{u^2}{v^2} - 2\,\frac{v}{u}, \quad B = \frac{v^2}{u^2} - 2\,\frac{u}{v},$$

and the equation mentioned gives:

$$z^3 + z^2 - 5\,z - 6 = 0,$$

from which, after casting off the factor $z+2$, the following arises:

$$z^2 - z - 3 = 0, \quad z = \frac{1+\sqrt{13}}{2}. \tag{16}$$

Between u and v, however, there subsists on the other hand a transformation equation of the third order (§ 73):

$$\frac{u^6v^6 - 8}{u^3v^3} = \frac{u^{12}+v^{12}}{u^6v^6} = z^6 - 6\,z^4 + 9\,z^2 - 2,$$

which by means of (16) can be brought to the form $z^4 - 2 = \dfrac{27+7\sqrt{13}}{2}$.

From this one obtains

$$\begin{aligned} u^3v^3 &= 4\,(3+\sqrt{13}),\\ u^6+v^6 &= 4\,(17+5\sqrt{13}),\\ u^3-v^3 &= \sqrt{2}\,(3+\sqrt{13}), \end{aligned}$$

so that, if $u^3 = \sqrt{8}\,x$ is set, the quadratic equation for x follows:

$$x^2 - \frac{3+\sqrt{13}}{2}\,(x+1) = 0, \quad f(\sqrt{-39})^3 = \sqrt{8}\,x. \tag{17}$$

We now wish to apply our results to two problems which belong to the general theory of transformation of the seventh chapter.

§ 132. The Schläfli Modular Equation for the 23rd Degree of Transformation.

Every class equation occurs as a divisor in a great number of equations of transformation, and can therefore, if it is known in another way, be used for the calculation of equations of transformation. We take as an example the 23rd degree of transformation.

According to § 73 there exists, if

$$u = f(\omega), \quad v = f(23\,\omega), \quad f\left(\frac{c+\omega}{23}\right), \quad c \equiv 0 \pmod{48} \tag{1}$$

has been set, an equation between

$$\begin{aligned} A &= \left(\frac{u}{v}\right)^{12} + \left(\frac{v}{u}\right)^{12}\\ B &= uv + \frac{2}{uv}, \end{aligned} \tag{2}$$

and it has already been shown in § 74, (14) that this equation has the form:

$$A = B^{11} + m_1B^{10} + \cdots + m_{10}B + m_{11}, \tag{3}$$

in which the coefficients $m_1, m_2 \ldots, m_{11}$ are rational numbers. Instead of now calculating these rational numerical coefficients, as was done there, from the developments of u and v in powers of q, we seek to determine the roots of equation (3) for $u = v$ from complex multiplication.

By multiplication with $u^{12}v^{12}$ equation (3) passes over into a form which does not contain u, v in the denominator, and which we denote for the moment by

$$F(u,v)=0 \tag{4}$$

so that for an indeterminate x and ω:

$$F'[f(\omega),x]=[x-f(23\,\omega)]\prod\left[x-f\left(\frac{c+\omega}{23}\right)\right]. \tag{5}$$

We now investigate for which values of ω the function:

$$\begin{aligned}F(u,u)&=F[f(\omega),f(\omega)]\\&=[f(\omega)-f(23\,\omega)]\prod\left[f(\omega)-f\left(\frac{c+\omega}{23}\right)\right]\end{aligned} \tag{6}$$

vanishes. First of all, the factor

$$f(\omega)-f\left(\frac{\omega}{23}\right)$$

evidently vanishes for

$$u=f(\sqrt{-23})=f\left(\frac{\sqrt{-23}}{23}\right).$$

If we further understand by r one of the numbers ±2 or ±4, and correspondingly by

$$m=23-r^2$$

either 19 or 7, then of the factors of (6) t w o vanish in each case for $\omega=\sqrt{-m}$, namely:

$$f(\omega)-f\left(\frac{24\,r+\omega}{23}\right),$$

for we have

$$\begin{aligned}f\left(\frac{24\,r+\sqrt{-m}}{23}\right)&=e^{\frac{r\pi i}{24}}f\left(\frac{r+\sqrt{-m}}{23}\right)\\&=e^{\frac{r\pi i}{24}}f\left(\frac{1}{r-\sqrt{-m}}\right)=f(\sqrt{-m}).\end{aligned}$$

We add further that

$$\frac{f(\omega)-f\left(\frac{24\,r+\omega}{23}\right)}{f(\omega)-f(\sqrt{-m})}$$

tends to a finite limit for $\omega=\sqrt{-m}$, as one recognises by differentiation with respect to ω (§ 54), and from this it follows that $F(u,u)$ is divisible by $[u-f(\sqrt{-m})]^2$.

If we set $u=v$, then

$$A=2,\quad B=u^2+\frac{2}{u^2},$$

and if we set $u=f(\sqrt{-m})$, $m=7,19,23$, then from § 128, (6), (8), § 131, (1) we obtain by elimination of u the equations for B:

$$\begin{aligned} m &= 7, && B - 3 = 0, \\ m &= 19, && B^2 - 6B^2 + 10B - 6 = 0, \\ m &= 23, && B^3 - 5B^2 + 4B - 1 = 0. \end{aligned}$$

The two last cubic equations are irreducible, since they have no rational root, and from this the required modular equation (3) follows without calculation:

$$A - 2 = (B-3)^2 (B^3 - 6B^2 + 10B - 6)^2 (B^3 - 5B^2 + 4B - 1). \tag{7}$$

In the memoir: „Ein Beitrag zur Transformationstheorie der elliptischen Funktionen mit einer Anwendung auf Zahlentheorie“, Math. Annalen 43, 185, I have derived by the same route the transformation equation for the 47th degree of transformation as well.

§ 133. The resolvents of degrees 7 and 11 for the 7th and 11th degree of transformation.

We have seen in § 82 that for the 7th and 11th degree of transformation there exist resolvents of degrees 7 and 11. In the formation of these equations too the theory of complex multiplication can render useful service.

When $n = 7$ or $= 11$, we consider the transformation equation

$$F_n(u, v) = 0,$$

whose roots, if we set $u = f(\omega)$, are

$$v_\infty = f(n\,\omega), \quad v_c = f\left(\frac{\omega + c}{n}\right) \tag{1}$$

where c may be taken as index of v with respect to the modulus n, while under the symbol f it must be assumed divisible by 48.

These equations are, according to § 73:

$$n = 7, \quad v^8 - u^7 v^7 + 7u^4 v^4 - 8\,uv + u^8 = 0, \tag{2}$$

$$n = 11, \quad v^{12} - u^{11} v^{11} + 11\,u^9 v^9 - 44\,u^7 v^7 + 88\,u^5 v^5 - 88\,u^3 v^3 + 32\,uv + u^{12} = 0. \tag{3}$$

In § 73 we investigated the influence of the three permutations

$$\begin{aligned} (c') &= (\omega, \omega + 2), \\ (c'') &= \left(\omega, \frac{-1}{\omega}\right), \\ (c''') &= \left(\omega, \frac{\omega - 1}{\omega + 1}\right), \end{aligned} \tag{4}$$

by which u passes over into

$$e^{-\frac{\pi i}{12}} u, \quad u, \quad \frac{\sqrt{2}}{u},$$

upon the roots of the equations (2) and (3). This influence results from the compositions:

$$\begin{pmatrix} n, & 0 \\ c, & 1 \end{pmatrix} \begin{pmatrix} 1, & 0 \\ 2, & 1 \end{pmatrix} = \begin{pmatrix} n, & 0 \\ c', & 1 \end{pmatrix}, \tag{5}$$

$$c' \equiv c + 2 \pmod{n}; \tag{6}$$

$$\begin{pmatrix} n, & 0 \\ c, & 1 \end{pmatrix} \begin{pmatrix} 0, & 1 \\ -1, & 0 \end{pmatrix} = \begin{pmatrix} -c'', & n \\ -\dfrac{c\,c''+1}{n}, & c \end{pmatrix} \begin{pmatrix} n, & 0 \\ c'', & 1 \end{pmatrix}, \tag{7}$$

$$c\,c'' \equiv -1 \pmod{n}; \tag{8}$$

$$\begin{pmatrix} n, & 0 \\ c, & 1 \end{pmatrix} \begin{pmatrix} 1, & 1 \\ -1, & 1 \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} 1, & 1 \\ -1, & 1 \end{pmatrix} \begin{pmatrix} n, & 0 \\ c''', & 1 \end{pmatrix}, \tag{9}$$

$$c''' \equiv \frac{c-1}{c+1} \pmod{n}, \tag{10}$$

$$\alpha - \beta = 1 - c''', \quad \gamma + \delta = c + 1,$$
$$\alpha + \beta = n, \quad n\,(\gamma - \delta) = (c-1) - c'''\,(c+1),$$

and hence, by § 34 and § 40, (12), the root v_c passes, under the substitutions (c'), (c''), (c''') and with regard to

$$2 \equiv 2\,n^2 \pmod{48},$$

over into:

$$e^{-\frac{\pi i n}{12}} v_{c'}, \quad v_{c''}, \quad \left(\frac{2}{n}\right) \frac{\sqrt{2}}{v_{c'''}}, \tag{11}$$

and this holds also for $c = \infty$. The permutations of the indices which result in this way are:

$$\begin{aligned} n = 7, \qquad c &= \infty, 0, 1, 2, 3, 4, 5, 6, \\ c' &= \infty, 2, 3, 4, 5, 6, 0, 1, \\ c'' &= 0, \infty, 6, 3, 2, 5, 4, 1, \\ c''' &= 1, 6, 0, 5, 4, 2, 3, \infty. \end{aligned} \tag{12}$$

$$\begin{aligned} n = 11, \qquad c &= \infty,\ 0,\ 1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ \ 8,\ 9,\ 10, \\ c' &= \infty,\ 2,\ \ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ 10,\ 0,\ \ 1, \\ c'' &= 0,\ \infty,\ 10,\ 5,\ 7,\ 8,\ 2,\ 9,\ 3,\ \ 4,\ 6,\ \ 1, \\ c''' &= 1,\ 10,\ \ 0,\ 4,\ 6,\ 5,\ 8,\ 7,\ 9,\ \ \ 2,\ 3,\ \infty. \end{aligned} \tag{13}$$

As roots of the resolvents of degree 7 and of degree 11 we may, following § 82, take in each case one of the two following functions:

$$
\begin{aligned}
&n = 7,\\
&w_\nu = \frac{(v_\infty - v_\nu)(v_{\nu+1} - v_{\nu+3})(v_{\nu+2} - v_{\nu+6})(v_{\nu+4} - v_{\nu+5})}{i\sqrt{7}\,u^4},\\
&w'_\nu = \frac{(v_\infty - v_\nu)(v_{\nu+1} - v_{\nu+5})(v_{\nu+2} - v_{\nu+3})(v_{\nu+4} - v_{\nu+6})}{-i\sqrt{7}\,u^4},
\end{aligned}
\tag{14}
$$

$$
\begin{aligned}
&n = 11, \quad w_\nu =\\
&\frac{(v_\infty - v_\nu)(v_{\nu+1} - v_{\nu+2})(v_{\nu+3} - v_{\nu+6})(v_{\nu+4} - v_{\nu+8})(v_{\nu+5} - v_{\nu+10})(v_{\nu+9} - v_{1+7})}{i\sqrt{11}\,u^6},\\
&w'_\nu =\\
&\frac{(v_\infty - v_\nu)(v_{\nu+1} - v_{\nu+6})(v_{\nu+3} - v_{\nu+7})(v_{\nu+4} - v_{\nu+2})(v_{\nu+5} - v_{\nu+8})(v_{\nu+9} - v_{\nu+10})}{-i\sqrt{11}\,u^6},
\end{aligned}
\tag{15}
$$

in which $\sqrt{7}$ and $\sqrt{11}$ are to be taken positive.

There now emerges, first of all, the following remarkable difference between the two cases.

By the two permutations (c'), (c'') the w_ν are, in the case $n = 7$, only permuted among themselves, namely in the following manner:

$$
\begin{array}{rl}
& w_0,\ w_1,\ w_2,\ w_3,\ w_4,\ w_5,\ w_6,\\
(c') & w_2,\ w_3,\ w_4,\ w_5,\ w_6,\ w_0,\ w_1,\\
(c'') & w_0,\ w_2,\ w_1,\ w_6,\ w_4,\ w_5,\ w_3,
\end{array}
$$

while [according to (13)] in the case $n = 11$ the quantities w at the same time change their sign under the permutation (c'), so that the following permutations occur:

$$
\begin{array}{rl}
& w_0,\ w_1,\ w_2,\ w_3,\ w_4,\ w_5,\ w_6,\ w_7,\ w_8,\ w_9,\ w_{10},\\
(c') & -w_2,\ -w_3,\ -w_4,\ -w_5,\ -w_6,\ -w_7,\ -w_8,\ -w_9,\ -w_{10},\ -w_0,\ -w_1,\\
(c'') & w_0,\ w_9,\ w_6,\ w_3,\ w_5,\ w_4,\ w_2,\ w_8,\ w_7,\ w_1,\ w_{10},
\end{array}
$$

and it is the same with w'.

From this it now follows (§ 54) that the w_ν are roots of an equation of degree 7 or of degree 11:

$$
w^n + A_1 w^{n-1} + A_2 w^{n-2} + \cdots + A_n = 0 \tag{16}
$$

are, but that therein, in the case $n = 7$, all the A_i depend rationally on $f(\omega)^{24}$, whereas in the case $n = 11$ the A_i with even index likewise depend rationally on $f(\omega)^{24}$, while those with odd index are the product of $f(\omega)^{12}$ with a rational function of $f(\omega)^{24}$. Apart from rational numbers, the numerical coefficients contain only the irrationality $i\sqrt{7}$ or $i\sqrt{11}$ respectively, and accordingly we will denote these equations by

$$
\begin{aligned}
&\Phi[w, f(\omega)^{24}, \quad i\sqrt{7}] = 0,\\
&\Phi[w, f(\omega)^{12}, \quad i\sqrt{11}] = 0.
\end{aligned}
\tag{17}
$$

If we change the signs of i and w at the same time, then, as we have seen in § 82, there results an equation whose roots are the quantities w'.

In the application of the substitution (c''') also a difference between the two cases shows itself.

For it follows from (13), taking into account that by (2) the product of all the v_ν has the value u^8, that in the case $n = 7$ the w_ν are merely permuted among one another by the substitution (c'''), and indeed in the following manner:

$$\begin{pmatrix} w_0, & w_1, & w_2, & w_3, & w_4, & w_5, & w_6 \\ w_5, & w_3, & w_6, & w_2, & w_4, & w_0, & w_1 \end{pmatrix},$$

and from this it follows that in this case the coefficients A_ν all remain unchanged under the interchange:

$$\left(f(\omega)^{24},\ \frac{2^{12}}{f(\omega)^{24}} \right).$$

Moreover, in the denominator of these coefficients only powers of $f(\omega)^{24}$ occur (§ 73), and the powers of $f(\omega)^{24}$ rise to the same height as those of $f(\omega)^{-24}$.

In the case $n = 11$ the quantities w_ν pass over into the quantities w'_ν under the permutation (c'''), and indeed in the following order:

$$\begin{pmatrix} w_0, & w_1, & w_2, & w_3, & w_4, & w_5, & w_6, & w_7, & w_8, & w_9, & w_{10} \\ w'_8, & w'_2, & w'_7, & w'_0, & w'_9, & w'_5, & w'_6, & w'_4, & w'_3, & w'_{10}, & w'_1 \end{pmatrix}.$$

By (17) we can express this property by the equation identical with respect to w:

$$-\Phi[-w, f(\omega)^{12}, -i\sqrt{11}] = \Phi\left(w, \frac{2^6}{f(\omega)^{12}}, i\sqrt{11} \right),$$

and from this there result the following properties of the coefficients A_ν:

A_ν remains unchanged under the simultaneous interchange:

$$\left(f(\omega)^{12}, \frac{64}{f(\omega)^{12}} \right), \quad \left(i\sqrt{11}, -i\sqrt{11} \right),$$

and for even ν it contains only the even powers of $f(\omega)^{12}$, for odd ν only the odd ones.

Here too only powers of $f(\omega)$ occur in the denominator, and the powers of $f(\omega)$ rise to the same height as those of $f(\omega)^{-1}$.

In order to come to a clear view of the degrees of the functions A_ν, we consider the beginnings of the developments in ascending powers of $q = e^{\pi i \omega}$.

The development of $f(\omega)$ begins with $q^{-\frac{1}{24}}$, that of v_∞ with $q^{-\frac{n}{24}}$, that of v_c with $q^{-\frac{1}{24n}} e^{-\frac{2\pi i c}{48} \frac{1}{n}}$; but in the last exponential quantity, if we reduce c to its least residue (mod n), $\frac{1}{48}$ is also to be taken according to the modulus n, and so for $n = 7, 11$

$$\frac{1}{48} \equiv -1 \pmod 7, \qquad \frac{1}{48} \equiv 3 \pmod{11}$$

is to be set. If accordingly we now set

$$\varrho = e^{\frac{2\pi i}{7}} \quad \text{or} \quad = e^{-\frac{6\pi i}{11}}$$

then we obtain from (14), (15) as the beginnings of the development for $n = 7$:

$$i\sqrt{7}\, w_\nu = q^{-\frac{1}{7}} \varrho^{3\nu} (\varrho - \varrho^3)(\varrho^2 - \varrho^6)(\varrho^4 - \varrho^5) + \cdots,$$

for $n = 11$:

$$i\sqrt{11}\, w_\nu = q^{-\frac{5}{22}} \varrho^{5\nu} (\varrho - \varrho^2)(\varrho^3 - \varrho^6)(\varrho^4 - \varrho^8)(\varrho^5 - \varrho^{10})(\varrho^9 - \varrho^7) + \cdots$$

But for $n = 7$:

$$(\varrho - \varrho^3)(\varrho^2 - \varrho^6)(\varrho^4 - \varrho^5) = -\varrho - \varrho^2 - \varrho^4 + \varrho^3 + \varrho^5 + \varrho^6,$$

and for $n = 11$:

$$(\varrho - \varrho^2)(\varrho^3 - \varrho^6)(\varrho^4 - \varrho^8)(\varrho^5 - \varrho^{10})(\varrho^9 - \varrho^7) =$$
$$-\varrho - \varrho^3 - \varrho^4 - \varrho^5 - \varrho^9 + .\varrho^2 + \varrho^6 + \varrho^8 + \varrho^7 + \varrho^{10}.$$

Since 1, 2, 4 are the quadratic residues of 7, and 1, 3, 4, 5, 9 the quadratic residues of 11, these two sums can be determined according to Bd. I, § 179, and in the two cases they give:

$$i\sqrt{7} \quad \text{and} \quad -i\sqrt{11},$$

so that

$$\begin{aligned} w_\nu &= q^{-\frac{1}{7}}\varrho^{3\nu} + \cdots, \qquad n = 7, \\ w_\nu &= -q^{-\frac{5}{22}}\varrho^{5\nu} + \cdots, \quad n = 11. \end{aligned} \tag{18}$$

From this an upper limit can be derived, up to which the powers of $f(\omega)$ in A_ν can at most rise, provided one observes further that the A_ν are representable as integral rational (and symmetric) functions of the w_ν.

For $n = 7$ it follows in this way, since all exponents of $f(\omega)$ must be divisible by 24, that A_1, A_2, A_3, A_4, A_5, A_6 are constants, while A_7 contains only the first power of $f(\omega)^{24}$, and that with the coefficient -1 [by (18)].

For $n = 11$, if $f(\omega)^{12\lambda}$ is the highest power of $f(\omega)$ occurring in A_ν, then

$$\lambda \leqq \frac{5\nu}{11}$$

must hold, and moreover λ must be even or odd at the same time as ν. Hence A_2, A_4 are constant, A_6, A_8 contain the highest power $f(\omega)^{24}$, and A_{10} contains $f(\omega)^{48}$ as well. Since the A with odd index contain only odd powers of $f(\omega)^{12}$, we have $A_1 = 0$, while A_3, A_5 contain $f(\omega)^{12}$, A_7, A_9 contain up to $f(\omega)^{36}$, and in A_{11} the function $f(\omega)$ rises to the 60th power.

For the case $n = 7$ we will now determine the constants completely by means of complex multiplication, and for this the consideration of the values of the w_ν for $\omega = i$ suffices.

For $\omega = i$ we get

$$u = f(i) = \sqrt[4]{2},$$

and if we introduce this value into equation (2) and put

$$\sqrt[4]{2}\, v = x$$

we obtain:

$$x^8 - 4x^7 + 28x^4 - 32x + 16 = 0. \tag{19}$$

The left-hand side of this equation is the square of the expression:

$$x^4 - 2x^3 - 2x^2 - 4x + 4, \tag{20}$$

so that the values of v for $\omega = i$ accordingly become equal to one another in pairs. From § 119, (15), (17) it follows, if we set $\omega = i$, and therefore $A = 1$, $C = 1$, $B = 0$, $m = 4$, $p = x = 7$, $y = 0$, that for $cc' \equiv -1 \pmod 7$ we get $v_c = v_{c'}$, so that:

$$v_\infty = v_0, \quad v_1 = v_6, \quad v_2 = v_3, \quad v_4 = v_5. \tag{21}$$

The expression (20) can be decomposed into the two quadratic factors:

$$x^2 - (1 + \sqrt{7})\, x + 2, \qquad x^2 - (1 - \sqrt{7})\, x + 2$$

so that the eight values of x become equal in pairs, each to one of the four quantities

$$\frac{1+\sqrt{7}}{2} \pm \frac{\sqrt[4]{7}}{\sqrt{2}}, \qquad \frac{1-\sqrt{7}}{2} \pm i\,\frac{\sqrt[4]{7}}{\sqrt{2}} \tag{22}$$

and it still remains to assign these four values to the individual v_ν. For this we remark that v_∞, v_0 are real for $\omega = i$, and that v_1, v_6 must likewise be real, because they are at the same time equal to one another and conjugate imaginary.

v_ν arises from v_0 by multiplying q by a certain root of unity; but since the development of v_0 in powers of q contains only positive coefficients [cf. § 24, (11)], it follows that v_0 must be greater than v_1, and consequently

$$\begin{aligned} \sqrt[4]{2}\,v_\infty = \sqrt[4]{2}\,v_0 &= \frac{1+\sqrt{7}}{2} + \frac{\sqrt[4]{7}}{\sqrt{2}}, \\ \sqrt[4]{2}\,v_1 = \sqrt[4]{2}\,v_6 &= \frac{1+\sqrt{7}}{2} - \frac{\sqrt[4]{7}}{\sqrt{2}}. \end{aligned} \tag{23}$$

For the two other pairs of roots there results:

$$\begin{aligned} \sqrt[4]{2}\,v_2 = \sqrt{2}\,v_3 &= \frac{1-\sqrt{7}}{2} + i\,\frac{\sqrt[4]{7}}{\sqrt{2}}, \\ \sqrt[4]{2}\,v_4 = \sqrt{2}\,v_5 &= \frac{1-\sqrt{7}}{2} - i\,\frac{\sqrt[4]{7}}{\sqrt{2}}. \end{aligned} \tag{24}$$

That the sign of i is correctly chosen in these expressions is inferred in the following way:

From (23) it is to be seen that for $\omega = i$, and therefore for $u = \sqrt{2}$, the root w_0 vanishes, and that consequently A_7 must vanish for this value. By the properties established above, A_7 is hereby completely determined, namely:

$$A_7 = -\left(u^{12} - \frac{64}{u^{12}}\right)^2, \tag{25}$$

from which it follows that for no other value of u^{24} than 64 do two of the values v_ν become equal to one another. This equality therefore occurs only for $\omega = i$ and for certain values of ω equivalent to i, and hence not for any other purely imaginary value of ω.

The beginning of the development in ascending powers of q now gives:

$$\begin{aligned} v_2 - v_5 &= f\left(\frac{\omega - 2\,.\,48}{7}\right) - f\left(\frac{\omega + 2\,.\,48}{7}\right) \\ &= 2\,i\,q^{-\frac{1}{7\,.\,24}} \sin\frac{4\pi}{7} + \cdots, \end{aligned} \tag{26}$$

so that for a sufficiently large real value of $-i\omega$ the left side of (26) is positive imaginary. If now $-i\omega$ goes along a real path from ∞ to 1, then, as we have seen above, $v_2 - v_5$ does not pass through zero. Hence for $\omega = i$ also the difference $v_2 - v_5$ must be positive imaginary, as was assumed in (24).

If, then, one substitutes the values (23), (24) into (14), there results for $\omega = i$:

$$w_0 = 0, \quad w_5 = 0$$

$$\begin{aligned} w_1 = w_2 = w_3 = w_6 &= \frac{7 + i\sqrt{7}}{2}, \\ w_4 &= -\frac{i\sqrt{7}\,(1+i\sqrt{7})^2}{4}. \end{aligned} \tag{27}$$

Accordingly, for the case $n = 7$ the resolvent can be formed completely. It reads:

$$w^2\left(w-\frac{7+i\sqrt{7}}{2}\right)^4\left(w+\frac{i\sqrt{7}\,(1+i\sqrt{7})^2}{4}\right) = \left(u^{12}-\frac{64}{u^{12}}\right)^2, \tag{28}$$

and it takes a simpler shape when

$$w+\frac{i\sqrt{7}\,(1+i\sqrt{7})^2}{4} = z^2$$

is set:

$$z\left(z^2-\frac{i\sqrt{7}\,(1+i\sqrt{7})^2}{4}\right)\left(z^2+i\sqrt{7}\right)^2 = u^{12}-\frac{64}{u^{12}}. \tag{29}$$

For the case $n = 11$ the corresponding calculation has not yet been carried out. If one uses the mode of consideration of § 119 to seek out the values of ω for which one of the quantities w_ν vanishes, one obtains comparatively simply:

$$\omega = i, \quad \omega = \sqrt{-7}, \quad \omega = \frac{1+\sqrt{-7}}{1-\sqrt{-7}},$$

$$u = \sqrt[4]{2}, \quad u = \sqrt{2}, \quad u = 1,$$

and from this

$$A_{11} = \left(u^{12}-\frac{2^6}{u^{12}}\right)^3\left(u^{12}-\frac{2^{12}}{u^{12}}\right)\left(u^{12}-\frac{1}{u^{12}}\right).$$

Twentieth Chapter.

The Multiplier Equation in Complex Multiplication.

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§ 134. The Class Invariant $\gamma_3(\omega)$.

We have now to test the function

$$\gamma_3(\omega) = \sqrt{j(\omega)-12^3}$$

for its property as a class invariant.

We have first, for a variable ω, the multiplier equation of the 1st level (§ 72, 3), according to which, if

$$n \equiv 3, \quad c \equiv 0 \pmod 4, \tag{1}$$

the ν quantity,

$$M = (-1)^{\frac{a-1}{2}}\partial^3\left(\frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}\right)^6\gamma_3(\omega), \tag{2}$$

are merely permuted among one another by the linear transformations. If we therefore set

$$u = j(\omega), \quad v = j\left(\frac{c+\partial\omega}{a}\right), \tag{3}$$

then between u, v there subsists the invariant equation

$$F(v, u) = 0, \tag{4}$$

and we have

$$F(x, u) = \Pi(x - v),$$

when the product sign extends over the ν systems of values of a, b, c. The sum

$$F(x, u) \sum \frac{M}{x - v} = \psi(x, u) \tag{5}$$

is, for an indeterminate x, unchanged by linear transformation, and therefore a rational function of u. But since the function has, for every finite value of $j(\omega)$, a finite

value, then it is an integral function of $j(\omega)$. If in (5) we let x pass over into v, there results

$$M = \frac{\psi(v, u)}{F'(v)}, \tag{6}$$

in which $\psi(v, u)$ is an integral rational function of u and v, with rational numerical coefficients.

If we now let ω in (6) pass over into the root of a quadratic equation whose discriminant

$$D = B^2 - 4AC$$

is negative:

$$A\omega^2 + B\omega + C = 0, \tag{7}$$

then one or more of the values v can become equal to u. If this is the case for only one of the ν values v, then $F'(v)$ will not vanish for $v = u$, and by (6) we can express M rationally through this singular value of u.

But the equation $u = v$ carries with it the condition:

$$\frac{c + \partial\omega}{a} = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}.$$

From this it follows:

$$\begin{gathered} \partial\beta = Ax, \qquad \partial\alpha = \frac{Bx - y}{2}, \\ c\beta - a\delta = \frac{Bx + y}{2}, \quad c\alpha - a\gamma = Cx, \end{gathered} \tag{8}$$

$$4n = y^2 - Dx^2, \tag{9}$$

in which x can be assumed positive.

We assume A odd and relatively prime to D. If $D \equiv 1 \pmod 4$, we put $n = -D$ and obtain from (9):

$$y^2 + n(x^2 - 4) = 0.$$

According to this, the value $x = 1$ is admissible only when $-D$ is three times a square number. If we set this case aside for the moment, there remains only

$$x = 2, \quad y = 0,$$

and from (8) there results:

$$\begin{gathered} \partial = 1, \qquad \beta = 2A, \quad \alpha = B, \\ a = n, \qquad \gamma \equiv 2C, \quad \delta \equiv B \pmod 4, \\ \alpha + \beta\omega = \sqrt{D}, \end{gathered}$$

and by the last two equations (8) c is determined to the modulus n.

By § 38 (15) we therefore have

$$M = (-1)^{A+C+\frac{B+1}{2}} \sqrt{D}^3 \, \gamma_3(\omega), \tag{10}$$

and here there occurs only one of the values $v = u$; accordingly M, and consequently also $\sqrt{D}\,\gamma_3(\omega)$, is rationally expressible through $j(\omega)$.

In order to dispose of the exceptional case as well, we set

$$n = -D = 3\,m^2.$$

Then (9) becomes:

$$y^2 + 3\,m^2(x^2 - 4) = 0,$$

and this has the three solutions:

$$x = 2, \quad y = 0,$$
$$x = 1, \quad y = \mp 3\,m.$$

For $x = 2$ the equations (8) again give the formula (10):

$$M_1 = (-1)^{A+C+\frac{B+1}{2}} \sqrt{D}^3 \, \gamma_3(\omega),$$

and the two other cases give

$$\partial = 1, \qquad a = n,$$
$$\alpha = \frac{B \pm 3\,m}{2}, \qquad \beta = A,$$
$$\gamma \equiv C, \qquad \alpha + \delta \equiv B \pmod 4,$$
$$\alpha + \beta\omega = \frac{\sqrt{D} \pm 3\,m}{2}.$$

From this one obtains by § 38 (15) for M_2, M_3 the values

$$M_2 = (-1)^{\frac{B-1}{2}} \left(\frac{3\,m + \sqrt{D}}{2}\right)^3 \gamma_3(\omega),$$

$$M_3 = (-1)^{\frac{B-1}{2}} \left(\frac{-3\,m + \sqrt{D}}{2}\right)^3 \gamma_3(\omega).$$

Now it is true that the quantities M_1, M_2, M_3 are not rationally expressible through $j(\omega)$ individually, but, as we shall show at once, their symmetric functions are, for example their sum, and from this it follows for this case also that $\sqrt{D}\,\gamma_3(\omega)$ is a class invariant.

From the assumption we have made, that A is relatively prime to $2\,D$, we can free ourselves afterwards, since $\gamma_3(\omega)$ changes at most its sign under all linear transformers. We then have the theorem:

1. If $D \equiv 1 \pmod 4$, then $\sqrt{D}\,\gamma_3(\omega)$ is a class invariant of the discriminant D.

To establish the theorem completely, we must still prove that the symmetric functions of M_1, M_2, M_3 are, in the exceptional case last considered, rational functions of $j(\omega)$. This requires us to investigate what becomes of (6) when, for a singular value u of v, the denominator $F'(v)$ vanishes, and, since M remains finite, the numerator as well. We again let ω be variable to begin with, and denote the roots of (4) by

$$v_1, v_2, v_2, \ldots, v_\nu. \tag{11}$$

Now let, for the particular value of ω under consideration,

$$v_1,\ v_2,\ \ldots,\ v_\lambda \tag{12}$$

become equal to one another, while the remaining $v_{\lambda+1}, \ldots, v_\nu$ stay different from them. We take any symmetric function of the quantity (12), for example, for an indeterminate t:

$$\sigma = (t - v_1),\ (t - v_2),\ \ldots,\ (t - v_\lambda); \tag{13}$$

if we carry out in σ all permutations of the quantities (11), we determine r values

$$\sigma_1,\ \sigma_2,\ \ldots,\ \sigma_r, \tag{14}$$

when

$$r = \frac{\nu!}{\lambda!\,(\nu - \lambda)!}$$

denotes the number of combinations of ν quantities taken λ at a time (without repetition). We can dispose of t in such a way that, even for the singular value ω, only one of the quantities (14) becomes equal to the first σ. The σ are the roots of an equation

$$\Phi(x, u) = 0$$

of degree r, and for the singular value $x = \sigma$ the quantity $\Phi'(\sigma)$ remains different from zero.

Let us now consider a symmetric function S of the quantity corresponding to (12)

$$M_1,\ M_2,\ \ldots,\ M_\lambda,$$

then, through the ν combinations of the numbers a, ∂, c that lead to the values (11), it likewise takes r values $S_1, S_2, \ldots, S_r$, and the sum

$$\Phi(x, u) \sum \frac{S}{x - \sigma} = \Theta(x, u)$$

is a rational integral function of x and u. From this it follows, on letting x pass into σ:

$$S = \frac{\Theta(\sigma, u)}{\Phi'(u)}.$$

For the singular value all the quantities (12) now become equal to one another and equal to u; σ is therefore a rational function of u, and with this the desired proof is completed.

As noteworthy examples we choose the cases of § 125:

$$\gamma_3\left(\frac{-1+\sqrt{-3}}{2}\right) = 24\sqrt{-3},$$
$$\gamma_3\left(\frac{-1+\sqrt{-7}}{2}\right) = 27\sqrt{-7},$$
$$\gamma_3\left(\frac{-1+\sqrt{-11}}{2}\right) = 7.8\sqrt{-11},$$
$$\gamma_3\left(\frac{-1+\sqrt{-19}}{2}\right) = 8.27\sqrt{-19},$$
$$\gamma_3\left(\frac{-1+\sqrt{-27}}{2}\right) = 8.11.23\sqrt{-3},$$
$$\gamma_3\left(\frac{-1+\sqrt{-43}}{2}\right) = 8.7.81\sqrt{-43},$$
$$\gamma_3\left(\frac{-1+\sqrt{-67}}{2}\right) = 7.8.9.31\sqrt{-67},$$
$$\gamma_3\left(\frac{-1+\sqrt{-163}}{2}\right) = 8.27.7.11.19.127\sqrt{-163}.$$

Here too the decomposability of the rational factors into comparatively small primes is striking. We give here a few further examples in which it is not $\sqrt{D}$ but $\sqrt{-D}$ that occurs.

In § 128 we found:

$$f_1(\sqrt{-2})^4 = 2, \quad f(\sqrt{-3})^3 = 2, \quad f(\sqrt{-7})^2 = 2.$$

To these we apply the formulae (§ 129):

$$f_1^8 - f_2^8 = \frac{\sqrt{f^{24}-64}}{f^4},$$

$$f^8 + f_2^8 = \frac{\sqrt{f_1^{24}+64}}{f_1^4},$$

and find

$$f(\sqrt{-2})^8 + f_2(\sqrt{-2})^8 = 4\sqrt{2},$$
$$f_1(\sqrt{-3})^8 - f_2(\sqrt{-3})^8 = \frac{4\sqrt{3}}{\sqrt[3]{2}},$$
$$f_1(\sqrt{-7})^8 - f_2(\sqrt{-7})^8 = 6\sqrt{7}.$$

From this one obtains $\varkappa^2$ and $\varkappa'^2$ from the formulae [§ 54, (3)]

$$\varkappa^2 = \frac{f_2(\omega)^8}{f(\omega)^8}, \quad \varkappa'^2 = \frac{f_1(\omega)^8}{f(\omega)^8},$$

hence

$$\omega = \sqrt{-2}: \quad \varkappa^2 = (\sqrt{2}-1)^2,$$
$$\omega = \sqrt{-3}: \quad \varkappa^2 = \frac{2-\sqrt{3}}{4} = \frac{(1-\sqrt{3})^2}{8},$$
$$\omega = \sqrt{-7}: \quad \varkappa^2 = \frac{8-3\sqrt{7}}{16} = \frac{(3-\sqrt{7})^2}{32}.$$

§ 135. The class invariant $\varkappa^2$ and $\varkappa$.

It has been shown in § 126 that, if the discriminant D is divisible by 4, and the quadratic equation whose root is ω has odd outer coefficients, then $f(\omega)^{24}$ is a class invariant for the discriminant D.

But by § 54 and § 34 we have

$$\varkappa^2\varkappa'^2 = \frac{16}{f(\omega)^{24}}, \quad \varkappa^2 = \frac{f_2(\omega)^8}{f(\omega)^8} = \frac{16}{f(\omega)^8\, f_1(2\omega)^8}. \tag{1}$$

If now ω satisfies the equation

$$A\,\omega^2 + B\,\omega + C = 0 \tag{2}$$

with the discriminant

$$D = B^2 - 4\,A\,C,$$

then $\omega' = 2\,\omega$ satisfies the equation

$$A\,\omega'^2 + 2\,B\,\omega' + 4\,C = 0$$

with the discriminant $4\,D$, and by § 126, (13) therefore $f_1(2\omega)^{24}$ is a class invariant for this discriminant. Thus $\varkappa^2\varkappa'^2$ and $\varkappa^6$ are rationally expressible by class invariants of the discriminant D and $4\,D$.

The identical equation

$$\frac{\varkappa^2\varkappa'^2 + \varkappa^4\varkappa'^4 + 2\,\varkappa^6}{1 + \varkappa^6} = \varkappa^2 \tag{3}$$

shows that the same holds also for the modulus $\varkappa^2$.

Since $\varkappa^2$ passes under every linear substitution into a rational (linear fractional) function of $\varkappa^2$, the assumption that the coefficients A, C are to be odd can now also be dropped, and we have the theorem:

2. If the discriminant $D \equiv 0 \pmod 4$, then $\varkappa^2(\omega)$ is a class invariant of the discriminant $4\,D$.

It has further been shown in § 134 that for $D \equiv 1 \pmod 4$ $\sqrt{D}\,\gamma_3(\omega)$ is a class invariant. But we have (§ 54)

$$\gamma_3(\omega) = \frac{8(2 + \varkappa^2\varkappa'^2)(\varkappa'^2 - \varkappa^2)}{\varkappa^2\varkappa'^2}. \tag{4}$$

If ω satisfies equation (2), and one sets:

$$\omega = \frac{\omega' - 1}{\omega' + 1}, \tag{5}$$

then for ω' the equation results:

$$(A + B + C)\omega'^2 - 2(A - C)\omega' + (A - B + C) = 0. \tag{6}$$

If $D \equiv 5 \pmod 8$, then A, B, C are odd, and if $D \equiv 1 \pmod 8$, we may assume A and C to be even, and consequently in both cases (6) is a primitive equation for ω' of discriminant $4D$. Therefore, by § 126, 1., $f(\omega')^{24}$ is a class invariant for this discriminant, and since now by § 34 (18)

$$f(\omega)f(\omega') = \sqrt{2}$$

holds, the same is true of $f(\omega)^{24}$. From this it follows that $\varkappa^2\varkappa'^2$ is a class invariant for $4D$, and since by (4) $\varkappa^2$ can be expressed rationally in terms of γ_3 and $\varkappa^2\varkappa'^2$, it follows that:

3. If the discriminant $D \equiv 1 \pmod 4$, then $\varkappa^2$ is rationally expressible by a class invariant of the discriminant $4D$ and by $\sqrt{D}$.

If $j(\omega)$ is a class invariant of the discriminant D, then $j(2\omega)$, $j(\frac{1}{2}\omega)$ are class invariants for $4D$, and we have:

$$\begin{aligned} j(2\omega) &= \frac{(f_1(2\omega)^{24}+16)^3}{f_1(2\omega)^{24}} = \frac{[256+f_2(\omega)^{24}]^3}{f_2(\omega)^{48}} \\ &= \frac{16(16\varkappa'^2+\varkappa^4)^3}{\varkappa^8\varkappa'^2}, \\ j\left(\frac{\omega}{2}\right) &= \frac{\left[f_2\left(\frac{\omega}{2}\right)^{24}+16\right]^3}{f_2\left(\frac{\omega}{2}\right)^{24}} = \frac{[256+f_1(\omega)^{24}]^3}{f_1(\omega)^{48}} \\ &= \frac{16(16\varkappa^2+\varkappa'^4)^3}{\varkappa^2\varkappa'^8}. \end{aligned} \tag{7}$$

From this it follows that conversely the class invariants of the discriminant $4D$ are also rationally expressible in terms of $\varkappa^2$.

In order to be able to regard $\varkappa(\omega)$ itself as a class invariant, one applies the Gauss transformation (§ 9, § 32). According to it:

$$\varkappa^2\left(\frac{\omega}{2}\right) = \frac{4\varkappa(\omega)}{[1+\varkappa(\omega)]^2},$$

from which:

$$\varkappa(\omega) = \frac{\varkappa^2\left(\frac{\omega}{2}\right)[1+\varkappa^2(\omega)]}{4-2\varkappa^2\left(\frac{\omega}{2}\right)}.$$

Thus $\varkappa(\omega)$ is expressed rationally by $\varkappa^2(\omega)$ and $\varkappa^2\left(\frac{\omega}{2}\right)$, and $\omega' = \frac{\omega}{2}$ satisfies the equation

$$4A\omega'^2 + 2B\omega' + C = 0. \tag{8}$$

If C is an even number, which we may assume when $D \equiv 0 \pmod 4$ or $\equiv 1 \pmod 8$, then the factor 2 can be cancelled in (8), and the discriminant of (8) is equal to D. But if C is odd, which is necessary when $D \equiv 5 \pmod 8$, then the discriminant of (8) is equal to $4D$. From this it follows by 2. and 3.:

4. a) If $D \equiv 0 \pmod 4$, then $\varkappa(\omega)$ is rationally expressible by the class invariants of the discriminant $4D$.

b) If $D \equiv 1 \pmod 8$, then $\varkappa(\omega)$ is expressible by $\sqrt{D}$ and by the class invariants of the discriminant $4D$.

c) If $D \equiv 5 \pmod 8$, then $\varkappa(\omega)$ is rationally expressible by $\sqrt{D}$ and by the class invariants of the discriminant $16D$.

§ 136. Quadratic Degrees of Transformation.

When the degree of transformation n is an odd square number, then by § 72, 5. one has for a variable ω:

$$M = (-1)^{\frac{a-1}{2}} \partial^3 \left(\frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)} \right)^6, \tag{1}$$

when $a\partial = n$ and c is divisible by 8, an integral algebraic function of the field $\mathfrak{R}(v,u)$, where

$$v = j\left(\frac{c+\partial\omega}{2}\right), \qquad u = j(\omega). \tag{2}$$

Now let a root of the primitive quadratic equation

$$A\omega^2 + B\omega + C = 0, \tag{3}$$

with negative discriminant

$$D = B^2 - 4AC. \tag{4}$$

be substituted in it for ω, and we assume once and for all that A is positive and relatively prime to $2Dn$, which is no restriction. If now $v = u$ is to hold, the necessary and sufficient condition for it is:

$$\frac{c+\partial\omega}{a} = \frac{\gamma+\delta\omega}{\alpha+\beta\omega}, \tag{5}$$

$$\alpha\delta - \beta\gamma = 1. \tag{6}$$

If one compares (5) with (3), it follows (§ 114) that there must be two positive integers x, y, the first of which may be assumed positive, and which satisfy the conditions:

$$\begin{aligned} \partial\beta &= Ax, & c\beta + \partial\alpha - a\delta &= Bx, \\ c\alpha - a\gamma &= Cx, & -c\beta + \partial\alpha + a\delta &= y, \\ 2\partial\alpha &= Bx + y, \\ 2(c\beta - a\delta) &= Bx - y, \end{aligned} \tag{7}$$

and from this, on account of (6):

$$4n = y^2 - Dx^2. \tag{8}$$

A prime divisor of ∂ would, since A is assumed relatively prime to n, have to divide x and y. It would then also divide

$$c\alpha - a\gamma \quad \text{and} \quad c\beta - a\delta,$$

and consequently divide a, ∂, c. But since these three numbers have no common divisor, $\partial = 1$ must hold, and from (7) there results:

$$\begin{aligned} \alpha &= \frac{Bx+y}{2}, & \beta &= Ax, \\ c\alpha - n\gamma &= Cx, & c\beta - n\delta &= \frac{Bx-y}{2}; \end{aligned} \tag{9}$$

from this it follows that x and y have no odd common divisor, and that, if x and y are even, $\frac{1}{2}(Bx+y)$ must be odd. Since we may moreover assume x positive,

only proper solutions of (8) come into consideration (§ 114). But by (9) every proper solution leads to a system of values α, β, and, through the congruences

$$c\alpha \equiv Cx, \qquad c\beta \equiv \frac{Bx - y}{2} \pmod{n} \tag{10}$$

to a determinate value of $c \pmod{n}$, which may in addition be assumed divisible by 8. If conversely x, y satisfy these conditions, then α and β have no common divisor, as appears from

$$\alpha \frac{Bx - y}{2} - \beta Cx = -n$$

For according to this, an odd divisor of α and β would have to divide n, and would therefore, since A was assumed relatively prime to n, divide x and consequently y.

It follows further, if we leave aside the two exceptional cases $D = -3$, $D = -4$, which do not concern us here at all, because for these $j(\omega)$ is rational, that different solutions of (8) also lead to different values c. For if we assume that one and the same value of c could lead to two different systems $(\alpha, \beta, \gamma, \delta)$, then from (5) a relation of the form

$$\omega = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}$$

would follow, and this substitution, if it is not the identical one, is possible only for the two exceptional cases mentioned.

Accordingly the number of the v-values in (2) which by (3) become equal to u is as great as the number of the proper solutions of the equation (8).

If the equation (8) has only one such solution, the corresponding value of M is rationally expressible by u. But if it has several solutions, the corresponding values of M are the roots of a rational equation:

$$\Phi(M, u) = 0, \tag{11}$$

whose degree with respect to M is just as great as the number of these solutions (§ 134).

Furthermore u satisfies the class equation

$$H_{-D}(u) = 0.$$

We can represent $\Phi(M, u)$ as an integral function of u, and can then replace this function by its greatest common divisor with $H_{-D}(u)$.

We may assume α and β to be positive. For β this follows from the assumptions already made. For α we can always achieve it by increasing B by a multiple of $2A$, whereby we arrive at an equivalent (parallel) form. In this way $Bx + y$ can be made positive.

If we therefore determine the quotient

$$\frac{\eta\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right)}{\eta(\omega)} = E\begin{pmatrix}\alpha, & \beta \\ \gamma, & \delta\end{pmatrix}$$

by § 38, (15), then, since β is even or odd according as x is even or odd, it follows that:

$$\begin{aligned} x \equiv 0 \pmod 2: \quad & M = \left(\frac{\beta}{\alpha}\right) i^{3\frac{\alpha-1}{2}} e^{\frac{\pi i}{4}\alpha(\gamma-\beta)} \sqrt{\alpha + \beta\omega}^{\,3}, \\ x \equiv 1 \pmod 2: \quad & M = \left(\frac{\alpha}{\beta}\right) i^{3\frac{1-\beta}{2}} e^{\frac{\pi i}{4}\beta(\alpha+\delta)} \sqrt{-i(\alpha + \beta\omega)}^{\,3}. \end{aligned} \tag{12}$$

The square roots

$$\sqrt{\alpha+\beta\omega}=\sqrt{\frac{y+x\sqrt{D}}{2}},$$
$$\sqrt{-i(\alpha+\beta\omega)}=\sqrt{-i\frac{y+x\sqrt{D}}{2}} \tag{13}$$

have positive real part. These values of M therefore satisfy equation (11). If equal ones occur among them, the degree of equation (11) can be lowered by rational means through the separation of multiple roots.

Since equation (11) is to be applied to the decomposition of the class equation according to the signs of M, it is of importance to decide whether among the values of M in (12) belonging to the same ω there occur any that differ only in sign.

We investigate when two different ones among the values (12), say M, M', have the same 8th power. If

$$M^8=M'^8, \tag{14}$$

then, if ϱ is a twelfth root of unity and $x,y;\ x',y'$ are two different solutions of equation (8), we must have:

$$y'+x'\sqrt{D}=\varrho(y+x\sqrt{D}). \tag{15}$$

From this it follows that ϱ must satisfy a quadratic equation, and hence that

$$\varrho=\pm i \quad \text{or} \quad \varrho=\frac{\pm1\pm\sqrt{-3}}{2} \tag{16}$$

must hold, because these are the only non-real twelfth roots of unity which satisfy a quadratic equation, and $\varrho=\pm1$ would lead to $x=x'$, $y=y'$. Accordingly $\sqrt{D}$ must belong to a field which is determined by one of these irrationalities ϱ, that is, we must have

$$D=-4m^2 \quad \text{or} \quad D=-3m^2 \tag{17}$$

In the first case it follows from (15)

$$y'=\pm2mx,$$
$$2mx'=\mp y, \tag{18}$$

and hence, by (8):

$$n=m^2(x^2+x'^2),$$

and since n was assumed to be odd, m must also be odd; by (18) y,y' are even, and because of (8) so also are x,x', and from (12) it follows, since γ and β are even, that:

$$M^4=(\alpha+\beta\omega)^6=\left(\frac{y+ixm}{2}\right)^6,$$

and from (18):

$$M'^4=(\alpha'+\beta'\omega)^6=\left(\frac{y'+ix'm}{2}\right)^6=-\left(\frac{y+ixm}{2}\right)^6,$$

$$M'^4=-M^4. \tag{19}$$

Thus M and M' have not equal but opposite 4th powers.

In the second of the exceptional cases (17) there follows in the same way from (15):

$$\begin{aligned} \pm 2y' &= y - 3mx, \\ \pm 2mx' &= y + mx. \end{aligned} \tag{20}$$

Thereupon it follows, if one takes the upper signs,

$$m(x'y - y'x) = 2n, \tag{21}$$

and from this one concludes that m must be odd; for if m were even, then y and y' would have to be even, and by (21) $2n$ would be divisible by 4, and hence n by 2. This case is therefore not to be considered further, if we assume an even D.

With this the following theorem is proved:

1. Let $D \equiv 0 \pmod 4$ be a negative discriminant, and let x, y be two numbers without a common

odd divisor, and in such a way that, if x is even, $y \equiv 2 \pmod 4$, and further in such a way that

$$n = \tfrac{1}{4}(y^2 - Dx^2)$$

is an odd square number. Let M further be determined by formula (12). Then the function $H_{-D}(u)$ has a divisor

$$\Phi(M, u), \tag{22}$$

which with none of the functions

$$\Phi(-M, u), \quad \Phi(iM, u), \quad \Phi(-i, Mu) \tag{23}$$

has a root in common.

§ 137. Reduction of odd discriminants to even ones.

According to a theorem stated by Kronecker, the class equation can, upon the adjunction of square roots, be decomposed into as many factors as there are genera of the classes of forms for the class equation in question, and indeed in such a way that each of these factors is satisfied only by those class invariants for which the corresponding quadratic forms (A, B, C) belong to one and the same genus. We obtain this decomposition from Theorem 1, proved in the preceding section.

That this theorem, in the form in which we have stated it, refers only to even discriminants D is no restriction for the proof of Kronecker's theorem.

For if $D \equiv 1 \pmod 8$, then by § 123 the class field $\mathfrak{K}(D)$ is identical with the class field $\mathfrak{K}(4D)$. But if $D \equiv 5 \pmod 8$, then the degree of $\mathfrak{K}(4D)$ is three times as great as that of $\mathfrak{K}(D)$; the latter, however, is contained in the former, and the number of genera for $\mathfrak{K}(D)$ and $\mathfrak{K}(4D)$ is the same (§ 104).

If $u = j(\omega)$ is a class invariant for $D \equiv 1 \pmod 8$, then

$$v = j(\omega') = j(2\omega), \quad j\left(\frac{\omega}{2}\right), \quad j\left(\frac{\omega+1}{2}\right) \tag{1}$$

are class invariants of $4D$, and u is a rational function of v:

$$u = \varphi(v), \tag{2}$$

which remains unchanged, whichever of the three quantities (2) one may put for v. All characters of the forms whose roots are ω and ω' have the same value.

If in (2) one replaces v by another class invariant of the same discriminant $4D$ and of the same genus, then u too passes over into another class invariant of the discriminant D, but likewise remains in the same genus.

Accordingly, if in (2) one lets v run through the class invariants of one genus of the discriminant $4D$, then u runs through the invariants of the corresponding genus of the discriminant D [taking each of these values three times, and for $D \equiv 1 \pmod 8$ only once], and the symmetric functions of these u are contained in the same domain of rationality as the quantities v. If therefore the class function $H_{-4D}(v)$ is decomposable into factors according to the genera, the same holds of $H_{-D}(u)$[1].

§ 138. Splitting of the Class Equation according to the Genera.

In order to apply Theorem 1., § 136, one sets

$$D = B^2 - 4AC = -4m, \tag{1}$$

and decomposes m into two factors

$$m = m'm'', \tag{2}$$

where it is assumed that m'' is odd and without square divisor.

Now in Theorem 1., § 136, we make the assumption:

$$\begin{aligned} x &= 4, \quad y = 2(4m' - m''), \\ n &= (4m' + m'')^2. \end{aligned} \tag{3}$$

$$\begin{aligned} \alpha &= 2B + 4m' - m'', \\ \alpha m'' &= 4AC - (B - m'')^2, \\ \beta &= 4A. \end{aligned} \tag{4}$$

$$\begin{aligned} \alpha + \beta\omega &= 2i\sqrt{m'm''} + 4m' - m'', \\ &= (2\sqrt{m'} + i\sqrt{m''})^2, \end{aligned} \tag{5}$$

in which the square roots are to be taken positive. Then, since B is even, we have

$$\alpha m'' \equiv -1 \pmod 4, \quad \left(\frac{\beta}{\alpha}\right) = \left(\frac{A}{\alpha}\right),$$

and consequently, by the law of reciprocity and by (4):

$$\left(\frac{\alpha m''}{A}\right) = \left(\frac{-1}{A}\right) = \left(\frac{-1}{A}\right)\left(\frac{A}{\alpha m''}\right),$$

hence:

$$\left(\frac{A}{\alpha m''}\right) = 1.$$

$$\left(\frac{\beta}{\alpha}\right) = \left(\frac{A}{m''}\right), \tag{6}$$

$$\beta \equiv 4, \quad \gamma \equiv 4\,C \pmod 8, \quad \alpha \equiv 1 \pmod 2 \quad [\S\ 136, (7)], \tag{7}$$

[1]The carrying out of the corresponding considerations for an odd D would indeed also be possible, but would make numerous distinctions and prolixities necessary. Here is a point where the Gaussian notation and distinction of forms of the first and second kind, of which I still made use in the first edition, would bring with it a certain simplification of the expression. This is connected with the exceptional position which the number 2 occupies in the whole theory of elliptic functions, and which in the Weierstrassian theory recedes somewhat, but yet does not entirely disappear. On the other hand, this singling out of the number 2 is again the source of great simplifications, especially in the numerical results.

hence:

$$e^{\frac{\pi i\alpha}{4}(\gamma-\beta)} = (-1)^{C-1},$$

further:

$$B + 2m' \equiv 2C \pmod 4,$$

hence:

$$\frac{\alpha-1}{2} \equiv -\frac{m''+1}{2} + 2C \pmod 4,$$

$$i^{3\frac{\alpha-1}{2}} = (-1)^C i^{-3\frac{m''+1}{2}},$$

and accordingly, finally, by § 136, (12), (13):

$$M = -\left(\frac{A}{m''}\right) i^{-3\frac{m''+1}{2}} \left(2\sqrt{m'} + i\sqrt{m''}\right)^3,$$

for which one may also put:

$$M = -\left(\frac{A}{m''}\right)\left(2\, i^{-\frac{m''+1}{2}}\sqrt{m'} + i^{-\frac{m''-1}{2}}\sqrt{m''}\right)^3. \tag{8}$$

By Theorem 1. (§ 136), $H_{-D}(u)$ is then divisible by a function

$$\Phi(M, u),$$

which is relatively prime to $\Phi(-M, u)$. If we put

$$\mu = -\left(2\, i^{\frac{m''+1}{2}}\sqrt{m'} + i^{\frac{m''-1}{2}}\sqrt{m''}\right)^3, \tag{9}$$

then for every class invariant of D one of the two equations

$$\Phi(\mu, u) = 0, \quad \Phi(-\mu, u) = 0 \tag{10}$$

satisfied, and indeed the first or the second, according as

$$\left(\frac{A}{m''}\right) = +1 \quad \text{or} \quad \left(\frac{A}{m''}\right) = -1 \tag{11}$$

holds.

2. From this it follows that the sign $\left(\frac{A}{m''}\right) = \pm 1$ does not depend on the particular form (A, B, C), but only on the class represented by this form, and is therefore a character of the class of forms. Now the invariant of a two-sided class is real, and the invariants of opposite classes are conjugate imaginary. But for two opposite forms (A, B, C), $(A, -B, C)$ the sign (11) is the same, and from this it follows that $\Phi(\mu, u)$ has either real or conjugate imaginary roots, and consequently real coefficients, and therefore does not change when μ is replaced by the conjugate imaginary value μ'. Now the two cases $m'' \equiv 1$, $m'' \equiv 3 \pmod 4$ must be distinguished, because it depends upon this which term in (9) is imaginary. In the one case i occurs only in the combination $i\sqrt{m'}$, in the other in $i\sqrt{m''}$, and accordingly we have:

$$\begin{aligned}\Phi(\mu, u) = \tfrac{1}{2}\left[\Phi(\mu, u) + \Phi(\mu', u)\right] &= \Psi(\sqrt{m''}, u) : m'' \equiv 1 \\ &= \Psi(\sqrt{m'}, u) : m'' \equiv 3\end{aligned} \pmod 4. \tag{12}$$

The function Ψ, and consequently Φ also, has real coefficients and contains only one of the two square roots $\sqrt{m''}, \sqrt{m'}$. By our assumption $\sqrt{m''}$ is always irrational, while $\sqrt{m'}$ is rational only when m' is a square. Since $H(u)$ has rational coefficients, it must be divisible, when $m'' \equiv 1$,

by $\Psi(\sqrt{m''}, n)$ and $\Psi(-\sqrt{m''}, n)$, and when $m'' \equiv 3$ and m' is not a square, by $\Psi(\sqrt{m'}, n)$ and $\Psi(-\sqrt{m'}, n)$. Now M changes its sign through the simultaneous changes of sign:

$$\begin{matrix}(i,-i),(\sqrt{m''},-\sqrt{m''}) : m'' \equiv 1 \\ (i,-i),(\sqrt{m'},-\sqrt{m'}) : m'' \equiv 3\end{matrix} \pmod 4, \tag{13}$$

and consequently, unless $m'' \equiv 3$ and at the same time m' is a square, we have in both cases:

$$H(u) = \Phi(M,u)\,\Phi(-M,u), \tag{14}$$

and this decomposition is impossible only when $m'' \equiv 1 \pmod 4$ and at the same time m' is a square or $m' = 1$.

3. Each of the two signs (11) occurs in equally many classes of the discriminant $D = -4m'm''$ and the class function is, by adjunction of

$$\begin{matrix}\sqrt{m''}, & \text{if } m'' \equiv 1 \\ \sqrt{m''}, & ,, \quad m'' \equiv 3\end{matrix} \pmod 4$$

decomposable into two factors of degree $\frac{1}{2}h$, except when $m'' \equiv 3$ and m' is a square.

In this exceptional case, if we set $Q^2 = 4m'$, then $-m'' = \Delta$ is the stem of

$$D = Q^2\Delta,$$

and for every number A representable by a form of the discriminant D and coprime to D we have:

$$(D, A) = \left(\frac{A}{m''}\right) = +1$$

and $H(u)$ is identical with $\Phi(M,u)$.

If one decomposes m in a second manner into two factors

$$m = m_1' m_1'', \tag{15}$$

where m_1'' satisfies the same conditions as m'', then one obtains in the same way a decomposition

$$H(u) = \Phi(M_1,u)\,\Phi(-M_1,u), \tag{16}$$

and by seeking out the greatest common divisor of $\Phi(M,u)$ and $\Phi(M_1,u)$, one obtains a decomposition of $H(u)$ into four factors:

$$H(u) = \Phi_1(u)\,\Phi_2(u)\,\Phi_3(u)\,\Phi_4(u),$$

provided that a square root of a prime occurs in $\Phi(M,u)$ which is not contained in $\Phi(M_1,u)$. One continues in this way and gradually decomposes $H(u)$ into factors whose number is a power of 2.

We wish to consider the application to the individual cases somewhat more closely.

1) If $m \equiv 3 \pmod 4$, then all characters are contained in the form

$$\left(\frac{A}{m''}\right)$$

and formula (8) suffices to separate all genera from one another. If here $m'' \equiv 3$, then $m' \equiv 1 \pmod 4$, and from (12) it follows that in the partial equations only the square roots of such numbers occur as are of the form $4n+1$

are. If, then, we denote by $p, p', p'', \ldots$ the prime factors of m of the form $4n+1$, and by $q, q', q'', \ldots$ the prime factors of m of the form $4n+3$, then the following square roots occur in the partial equations:

$$\sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \quad \sqrt{qq'}, \sqrt{qq''}, \ldots;$$

so that if only one q is contained in m, then $\sqrt{q}$ does not occur in the partial equations. Since in this case at least one q must divide m, the number of square roots required for the decomposition is equal to the number of primes dividing m, diminished by 1, and hence equal to the number of independent characters corresponding to this case.

2) If

$$m \equiv 1 \pmod 4, \quad m \equiv 6, 2 \pmod 8, \quad m \equiv 4 \pmod{16},$$

then, by § 105, the characters

$$\left(\frac{-1}{A}\right), \left(\frac{2}{A}\right), \left(\frac{-2}{A}\right), \left(\frac{-1}{A}\right)$$

can respectively be expressed by the characters of the form

$$\left(\frac{A}{m''}\right)$$

and so in these cases too formula (8) suffices to separate all the genera from one another.

Formulae (9), (12) teach that in these cases the following adjunctions are necessary for the complete separation of the genera:

$$\begin{array}{ll} m \equiv 1 \pmod 4, & \sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \sqrt{q}, \sqrt{q'}, \sqrt{q''}, \ldots \\ m \equiv 6 \pmod 8, & \sqrt{2}, \sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \sqrt{qq'}, \sqrt{qq''}, \ldots \\ m \equiv 2 \pmod 8, & \sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \sqrt{2q}, \sqrt{2q'}, \sqrt{2q''}, \ldots \\ m \equiv 4 \pmod{16}, & \sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \sqrt{q}, \sqrt{q'}, \sqrt{q''}, \ldots \end{array} \tag{17}$$

For in the case $m \equiv 1 \pmod 4$ the number of the q is in any event even. If, then, we put $m = qr^2m''$, $m' = qr^2$ and understand by r^2 the greatest square number dividing r^2m'', then $m'' \equiv 3 \pmod 4$, and formula (12) gives the adjunction of $\sqrt{q}$; thus in this case, for the perfect separation of the genera,

$$\sqrt{p}, \sqrt{p'}, \sqrt{p''}, \ldots, \quad \sqrt{q}, \sqrt{q'}, \sqrt{q''}, \ldots$$

are to be adjoined, and the number of square roots is equal to the number of primes dividing m, again in agree-

ment with the number of independent characters. In a similar way the remaining cases of (16) result.

In the cases that still remain, namely $m \equiv 12 \pmod{16}$ and $m \equiv 0 \pmod 8$, the foregoing decomposition is indeed still applicable, but in the partial equations so obtained two or four genera are still united in each, corresponding to the characters

$$\left(\frac{-1}{A}\right), \left(\frac{2}{A}\right).$$

We therefore derive a second transformation formula like (8), by satisfying the equation

$$4n = y^2 - Dx^2$$

(8), § 136, in the following way:

$$m = m'm'' \equiv 0 \pmod 4,$$

$$x = 2, \quad y = 2(m' - m''), \quad n = (m' + m'')^2,$$

where again m'' is assumed odd and divisible by no square, but may also be $= 1$; and from § 136, (9) one obtains

$$\alpha = B + m' - m'', \quad \beta = 2A, \quad \gamma \equiv -2C \pmod 8,$$

$$\alpha m'' = AC - \left(\tfrac{1}{2}B - m''\right)^2,$$

$$\sqrt{\alpha+\beta\omega} = \sqrt{m'} + i\sqrt{m''}.$$

If for simplicity we take $B \equiv 0 \pmod 8$, which if necessary is attained by passing to a parallel form, then $m' \equiv C \pmod 8$, $\alpha \equiv C - m'' \pmod 8$, and consequently

$$i^{3\frac{\alpha-1}{2}} e^{\frac{\pi i}{4}\alpha(\gamma-\beta)} = (-1)^{\frac{1}{4}C}(-1)^{\frac{A+1}{2}} i^{3\frac{m''-1}{2}},$$

$$\left(\frac{\beta}{\alpha}\right) = \left(\frac{2}{m''}\right)\left(\frac{A}{m''}\right)\left(\frac{2}{\alpha m''}\right)\left(\frac{A}{\alpha m''}\right).$$

$$\left(\frac{A}{\alpha m''}\right) = 1, \quad \left(\frac{2}{\alpha m''}\right) = (-1)^{\frac{1}{4}C},$$

and from this there follows, by § 136, (12), (13):

$$M = -\left(\frac{2}{m''}\right)\left(\frac{A}{m''}\right)\left(\frac{-1}{A}\right)\left(i^{\frac{m''-1}{2}}\sqrt{m'} + i^{\frac{m''+1}{2}}\sqrt{m''}\right)^3. \tag{18}$$

This formula completes the preceding one for the case $m \equiv 12 \pmod{16}$, and shows that in this case also the complete splitting of the class equation takes place through adjunction of

$$\sqrt{p},\ \sqrt{p'},\ \sqrt{p''},\ \ldots,\ \sqrt{q},\ \sqrt{q'},\ \sqrt{q''},\ \ldots.$$

If now m is divisible by a still higher power of 2 than the second, then, when the exponent of 2 is odd, one can by § 105, (7) reduce all the characters to such as are of the form

$$\left(\frac{A}{m''}\right),\ \left(\frac{-1}{A}\right)\left(\frac{A}{m''}\right)$$

Put

$$m = 2r^2 m'',$$

where again by r^2 is understood the greatest square number dividing $\frac{1}{2}m$, and apply formula (8) or (18) according as $m'' \equiv 1$ or $\equiv 3 \pmod 4$. One then obtains the complete splitting of the class equation upon adjunction of the square roots of all the primes dividing m, with the exception of $\sqrt{2}$.

But if the exponent of the highest power of 2 dividing m is an even number, then (18) also does not yet suffice for the complete decomposition of the class equation. Here again one can indeed split the function H by the formulae (8), (18) through the adjunction of all the $\sqrt{p}$ and $\sqrt{q}$, but $\sqrt{2}$ is not thereby brought in, and so two genera always still remain contained in each such partial equation. In order to separate these also, we derive yet a third formula, on the assumption that m is divisible by 8.

We set

$$m = m'm'',$$

$$x = 1, \quad y = \tfrac{1}{2}m' - 2m'', \quad n = \left(\tfrac{1}{4}m' + m''\right)^2.$$

$$\sqrt{-i(\alpha+\beta\omega)} = e^{-\frac{\pi i}{4}}\left(\tfrac{1}{2}\sqrt{m'} + i\sqrt{m''}\right),$$

where again m'' is odd and divisible by no square. Here it follows from § 136:

$$M = (-1)^{\frac{m}{8}} e^{\frac{\pi i}{4}Am''}\left(\frac{A}{m''}\right)\left(i^{-\frac{m''+1}{2}}\tfrac{1}{2}\sqrt{m'} + i^{-\frac{m''-1}{2}}\sqrt{m''}\right)^3. \tag{19}$$

After the decomposition according to the characters $\left(\frac{A}{m''}\right)$ has been settled by the formulae (8) and (18), it is now only a question of the decomposition according to the characters

$$\left(\frac{-1}{A}\right),\ \left(\frac{2}{A}\right),$$

that is, according to the behaviour of A with respect to the modulus 8. It is therefore sufficient to set $m'' = 1$ in formula (19), whereby one obtains the following four values of M:

$$
\begin{aligned}
M &= (-1)^{\frac{m}{8}} \frac{1+i}{\sqrt{2}} \left(1 - i\tfrac{1}{2}\sqrt{m}\right)^3, && A \equiv 1 \pmod 8,\\
M &= (-1)^{\frac{m}{8}} \frac{-1+i}{\sqrt{2}} \left(1 - i\tfrac{1}{2}\sqrt{m}\right)^3 = M', && A \equiv 3 \pmod 8,\\
M &= (-1)^{\frac{m}{8}} \frac{-1-i}{\sqrt{2}} \left(1 - i\tfrac{1}{2}\sqrt{m}\right)^3 = M'', && A \equiv 5 \pmod 8,\\
M &= (-1)^{\frac{m}{8}} \frac{1-i}{\sqrt{2}} \left(1 - i\tfrac{1}{2}\sqrt{m}\right)^3 = M''', && A \equiv 7 \pmod 8.
\end{aligned}
\tag{20}
$$

The four values of M arise from the first through the interchanges:

$$
\begin{array}{llll}
M, & \sqrt{2}, & i, & \sqrt{m},\\
M', & -\sqrt{2}, & -i, & -\sqrt{m},\\
M'', & -\sqrt{2}, & i, & \sqrt{m},\\
M''', & \sqrt{2}, & -i, & -\sqrt{m}.
\end{array}
\tag{21}
$$

If now $\Phi(M, u) = 0$ is, in the same sense as above, the equation holding between u and M, then the four functions

$$\Phi(M, u), \quad \Phi(-M, u), \quad \Phi(iM, u), \quad \Phi(-iM, u)$$

are without a common divisor, and if $\Phi(M, u)$ is contained in $H(u)$, then the three other functions $\Phi(-M, u)$, $\Phi(iM, u)$, $\Phi(-iM, u)$ are contained in $H(u)$ as well, since the signs of the irrationalities (20) can be changed arbitrarily. It is therefore

$$H(u) = \Phi(M, u)\,\Phi(-M, u)\,\Phi(iM, u)\,\Phi(-iM, u),$$

and thus each of the four cases

$$A \equiv 1, \quad A \equiv 3, \quad A \equiv 5, \quad A \equiv 7 \pmod 8$$

occurs in equally many classes of forms of the discriminant $-4m$.

An exception occurs only when m is a square or the double of a square, because in the first case $\sqrt{m}$ is rational and therefore only the interchange (M, M'') is admitted, while in the second case $\sqrt{2}$ and $\sqrt{m}$ change their sign simultaneously, so that only the interchange (M, M') is admissible.

But these interchanges already suffice to split the partial equation further by adjunction of $\sqrt{2}$.

In the first case, according to § 105, only the two congruences

$$A \equiv 1, \quad A \equiv 5 \pmod 8,$$

in the second case only the two congruences

$$A \equiv 1, \quad A \equiv 3 \pmod 8$$

occur, and again in equally many classes of forms. The complete decomposition of the class equation according to the genera always requires, when m is divisible by 8, the adjunction of the roots of all the primes dividing m, including 2.

One notices that these considerations contain a new proof of the theorems on the existence of the genera proved by Gauss and Dirichlet, though only for negative discriminants.

§ 139. Examples.

For the actual calculation of the splitting of the class equation the formulae of the preceding section are applicable only to a limited extent, because of the too high degree of the transformation equations. We shall afterwards bring into application, in one case, a method that is at least closely related to them. If the class equation is known, the decomposition can usually be found more easily in a direct way, by making an ansatz of known form with indeterminate coefficients; thus from the formulae § 130, (2), (29), (37) one finds the following partial equations, in which the positive sign of the square root corresponds to the principal genus:

$$m = 50, \quad f_1\Big(\sqrt{-50}\Big) = \sqrt[4]{2}\, x,$$

$$x^3 - x^2 = \frac{1+\sqrt{5}}{2}(x+1),$$

$$m = 26, \quad f_1\Big(\sqrt{-26}\Big)^2 = \sqrt{2}\, x,$$

$$x^3 - x^2 = \frac{3+\sqrt{13}}{2}(x+1),$$

$$m = 41, \quad z = \frac{f(\sqrt{-41})^2}{\sqrt{2}} + \frac{\sqrt{2}}{f(\sqrt{-41})^2},$$

$$z^2 - \frac{5+\sqrt{41}}{2}z + \frac{7+\sqrt{41}}{2} = 0.$$

But in order to give an application of our general method, we assume $m \equiv 3 \pmod 8$ and set $D = -m$. We satisfy the equation:

$$4n = y^2 + mx^2,$$

by setting

$$m = m'm'',$$

$$x = 1, \quad y = \frac{m' - m''}{2}, \quad 4n = \left(\frac{m' + m''}{2}\right)^2,$$

whereby y, and consequently n, comes out odd. Further, by § 136, (13), we have

$$\alpha + \beta\omega = \left(\frac{\sqrt{m'} + i\sqrt{m''}}{2}\right)^2.$$

If, under the assumption that n is not divisible by 3, we set

$$M = \left(\frac{c}{e}\right) i^{\frac{a-1}{2}} \sqrt{\partial}\, \frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}, \tag{1}$$

then M can be determined by the application of § 38, (15), and one finds, if m'' denotes that one of the two factors m', m'' which is congruent to 1 modulo 4:

$$M = -\left(\frac{A}{m''}\right)(-1)^{\frac{m'-m''-2}{8}} e^{-\frac{\pi i}{3}A(m'-m'')}\, \frac{\sqrt{m''} - i\sqrt{m'}}{2}. \tag{2}$$

To make an application of this, we set $n = 25$, hence

$$20 = m' + m'',$$

and consequently

$$\begin{aligned} m'' &= 1, \quad m' = 19, \quad m = 19, \\ m'' &= 17, \quad m' = 3, \quad m = 51, \\ m'' &= 13, \quad m' = 7, \quad m = 91, \\ m'' &= 9, \quad m' = 11, \quad m = 99. \end{aligned}$$

If we take the form

$$\left(1,\ \frac{m'-m''}{2},\ n\right)$$

as the basis, then we must set $A=1$, and (2) gives:

$$\begin{aligned}
&m=19, \quad M=-\frac{1-i\sqrt{19}}{2},\\
&m=51, \quad M=e^{-\frac{\pi i}{3}}\frac{\sqrt{17}-i\sqrt{3}}{2},\\
&m=91, \quad M=\frac{\sqrt{13}-i\sqrt{7}}{2},\\
&m=99, \quad M=e^{\frac{\pi i}{3}}\frac{3-i\sqrt{11}}{2}.
\end{aligned} \tag{3}$$

Now by § 72, (27), (28) M satisfies a transformation equation which, when

$$\begin{aligned}
\chi &= M^5+5M^4+15M^3+25M^2+25M\\
&= M^3\left[\left(M+\frac{5}{M}\right)^2+5\left(M+\frac{5}{M}\right)+5\right]
\end{aligned}$$

is set, takes the shape:

$$j(\omega)=\gamma_2(\omega)^3=\frac{(\chi^2+10\chi+5)^3}{\chi}. \tag{4}$$

For $m=19$ one obtains from this the value already found above (§ 125)

$$\gamma_2\left(\frac{-3+\sqrt{-19}}{2}\right)=-96,$$

and for the three remaining values of m one obtains

$$\begin{aligned}
&j\left(\frac{-7+\sqrt{-51}}{2}\right)=-2^{14}\cdot 27\left(6263+1519\sqrt{17}\right),\\
&\gamma_2\left(\frac{-3+\sqrt{-91}}{2}\right)=-48\left(227+63\sqrt{13}\right),\\
&j\left(\frac{-1+\sqrt{-99}}{2}\right)=-2^{12}\left(4\,591\,804\,316+799\,330\,532\sqrt{33}\right).
\end{aligned} \tag{5}$$

From this one can arrive at cubic equations for

$$f\left(\sqrt{-51}\right)^{24}, \quad f\left(\sqrt{-91}\right)^{8}, \quad f\left(\sqrt{-99}\right)^{24}$$

by making use of the formulae (§ 34, § 54):

$$f_2\left(\frac{-r+\sqrt{-m}}{2}\right)=\frac{\sqrt{2}\,e^{-\frac{r\pi i}{24}}}{f(\sqrt{-m})},$$

$$\gamma_2\left(\frac{-r+\sqrt{-m}}{2}\right)e^{-\frac{r\pi i}{3}}=\frac{f(\sqrt{-m})^{24}-16^2}{f(\sqrt{-m})^{16}},$$

in which $r=7,3,1$; $m=51,91,99$ are to be set. If, then, one sets for $r=3$:

$$f(\sqrt{-m})=x,$$

one obtains:

$$x^{24} + \gamma_2(\omega)x^{16} - 16^2 = 0, \tag{6}$$

and for $r = 7, 1$:

$$f(\sqrt{-m})^3 = 2x,$$

for these two cases:

$$x^{24} - \left[3 - 2^{-8}j(\omega)\right] x^{16} + 3x^8 - 1 = 0, \tag{7}$$

where for $\gamma_2(\omega)$ and $j(\omega)$ the values (5) are to be substituted. These equations can be decomposed still further, and for x itself one obtains much simpler equations:

$$\begin{array}{ll} f(\sqrt{-51})^3 = 2x, & x^3 - 4x^2 - x - 1 = \sqrt{17}\,x^2, \\ f(\sqrt{-91}) = x, & x^3 - 2x^2 - x - 2 = \sqrt{13}\,x, \\ f(\sqrt{-99})^3 = 2x, & x^3 - 13x^2 - 4x - 1 = \sqrt{33}\,(2x^2 + x)\,. \end{array} \tag{8}$$

The representation of the class invariants $j(\omega)$ by a single square root is always possible whenever two genera are present and one class in each genus. These values of m are found in Gauss's table of class numbers (Werke, Bd. II, S. 450 ff.) under the designation II, 3, and of these the ones to be selected are those which are $\equiv 3 \pmod 8$. Their number is in all probability finite, as the table just mentioned shows; they are the numbers:

$$m = 35, \quad 51, \quad 75, \quad 91, \quad 99, \quad 115, \quad 123,$$

$$147, \quad 187, \quad 235, \quad 267, \quad 403, \quad 427.$$

If the classes of forms split into an arbitrary number of genera, but each genus contains only one class, then by the theorem proved in § 138 all class invariants can be expressed by square roots. Euler and Gauss concluded by induction that there is only a finite number of such discriminants, namely 65[30]. In the memoir „Zur komplexen Multiplikation elliptischer Funktionen“ (Mathematische Annalen, Bd. 33) I have computed all these numbers. They are also contained in the table of class invariants which is appended to this book.

Twenty-first Chapter.
The Norms of the Class Invariants $f(\omega)$.

§ 140. Convergence of an Infinite Series.

We consider an arbitrary quadratic form

$$\psi(x, y) = Ax + 2Bxy + Cy^2 \tag{1}$$

with negative discriminant

$$-m = B^2 - AC, \tag{2}$$

[30] Euler, Nouv. Mém. de Berlin 1776, S. 338. Gauss, Disq. art. 303. Euler arrived at these numbers by a different route, namely from the problem of determining large primes (Cf. the Straßburg dissertation of Peter Meyer: Beweis eines von Euler entdeckten Satzes, betreffend die Bestimmung von Primzahlen, Straßburg 1906). These numbers are:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 18, 21, 22, 24, 25, 28, 30, 33, 37, 40, 42, 45, 48, 57, 58, 60, 70, 72, 78, 85, 88, 93, 102, 105, 112, 120, 130, 133, 165, 168, 177, 190, 210, 232, 240, 253, 273, 280, 312, 330, 345, 357, 385, 408, 462, 520, 760, 840, 1320, 1365, 1848.

in which, for the present, the coefficients A, B, C are not necessarily to be regarded as integers; they are only required to be real. If we take x, y as rectangular coordinates in a plane, then to the integral values of x, y there correspond the lattice points. The equation

$$\psi(x, y) = t$$

represents, for a constant t, an ellipse, and the number $Z(t)$ of those values of ψ, for integral x, y, which are smaller than t is equal to the number of lattice points lying in the interior of this ellipse. According to Bd. II, § 194, (5), the limiting value of the ratio $Z(t) : t$ for an infinitely large t approaches the area of the ellipse $\psi = 1$, namely the value $\pi : \sqrt{m}$. Thus we have

$$\text{Lim}\,\frac{Z(t)}{t} = \frac{\pi}{\sqrt{m}}. \tag{3}$$

If, then, in the infinite series

$$S = \sum^{x,y} \frac{1}{\psi(x, y)^s} \tag{4}$$

we let the numbers x, y run through all integral values, with the exception of the combination $x = 0$, $y = 0$, then, according to Bd. II, § 196, 4, S is convergent for $s > 1$, and we have

$$\lim_{s=1}(s-1)S = \frac{\pi}{\sqrt{m}}. \tag{5}$$

In what follows it shall be shown, by direct transformation of the sum S, that

$$S - \frac{\pi}{(s-1)\sqrt{m}}$$

has a finite limiting value for $s = 1$, and this limiting value shall be determined.

§ 141. The Kronecker limit formula.

We arrange the terms of the series

$$S = \Sigma^{x,y} \frac{1}{\psi(x, y)^s}$$

first in such a way that we separate off the terms corresponding to the vanishing y, and take together, two at a time, those of the remaining ones which correspond to opposite values of x and y.

Thus we obtain:

$$S = M_s + N_s, \tag{1}$$

if for brevity

$$M_s = 2\Sigma^{y}_{1,\infty}\Sigma^{x}_{-\infty,\infty} \frac{1}{(Ax^2 + 2Bxy + Cy^2)^s}, \tag{2}$$

$$N_s = 2\Sigma^{x}_{1,\infty} \frac{1}{(Ax^2)^s} \tag{3}$$

is set. The value N, which N_s takes for $s = 1$, can be determined directly, since the series remains unconditionally convergent for $s = 1$, and it gives:

$$N = \frac{2}{A}\Sigma^{x}_{1,\infty}\frac{1}{x^2} = \frac{\pi^2}{3A}. \tag{4}$$

In order to ascertain the behaviour of M_s for $s = 1$, we decompose the function $\psi(x, y) = Ax^2 + 2Bxy + Cy^2$ into two conjugate imaginary linear factors:

$$Ax^2 + 2Bxy + Cy^2 = A(x + \omega_1 y)(x - \omega_2 y),$$

if

$$\omega_1 = \frac{B + i\sqrt{m}}{A}, \quad \omega_2 = \frac{-B + i\sqrt{m}}{A} \tag{5}$$

are so defined that $\sqrt{m}$ is positive.

By this means

$$M = \frac{2}{A^s} \Sigma_{1,\infty}^{y} \Sigma_{-\infty,\infty}^{x} \frac{1}{(x + \omega_1 y)^s (x - \omega_2 y)^s}. \tag{6}$$

Now, by a well-known theorem from the theory of the Γ-functions, if k denotes an arbitrary quantity with positive imaginary part:

$$\frac{1}{(-ik)^s} = \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty e^{2\pi i k \xi} \, \xi^{s-1} \, d\xi,$$

in which the power $(-ik)^s$ is uniquely defined by the convention that, when $-ik = \varrho e^{i\Theta}$ is set, ϱ is taken positive and the angle Θ between $-\frac{\pi}{2}$ and $+\frac{\pi}{2}$, one is to set $(-ik)^s = \varrho^s e^{i\Theta s}$.

If we here replace ik by the conjugate imaginary quantity $-ik'$, at the same time choosing a new letter of integration η, and multiply the two equations, there follows:

$$\frac{1}{(kk')^s} = \frac{(2\pi)^{2s}}{\Gamma(s)\Gamma(s)} \int_0^\infty \int_0^\infty e^{2\pi i (k\xi - k'\eta)} (\xi\eta)^{s-1} \, d\xi \, d\eta. \tag{7}$$

In this we substitute for k, k' the two conjugate factors $x + \omega_1 y$, $x - \omega_2 y$ of ψ, and introduce the integral expression (7) into (6).

In this way we obtain:

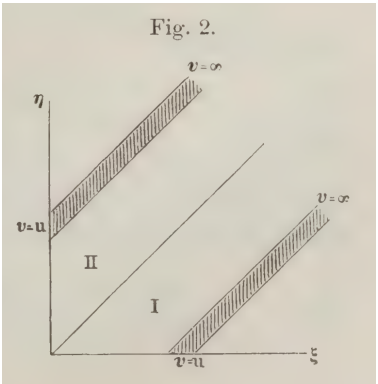
$$M_s = \frac{(2\pi)^{2s}}{A_s \Gamma(s)\Gamma(s)} \Sigma^{x,y} 2 \int_0^\infty \int_0^\infty e^{2\pi i [x(\xi - \eta) + y(\omega_1 \xi + \omega_2 \eta)]} (\xi\eta)^{s-1} \, d\xi \, d\eta. \tag{8}$$

We now regard the product of the two integrals on the right-hand side of this expression, which for brevity we denote by

$$W = 2 \int_0^\infty \int_0^\infty e^{2\pi i [x(\xi - \eta) + y(\omega_1 \xi + \omega_2 \eta)]} (\xi\eta)^{s-1} \, d\xi \, d\eta \tag{9}$$

as a double integral which, if ξ, η are interpreted as rectangular coordinates in the plane, extends over the positive quadrant of the coordinate system.

Fig. 2.



In order to transform the double integral, we divide the domain of integration by a straight line bisecting the angle into the two parts I, II (see Fig. 2), and substitute in the first part

$$\xi - \eta = u, \qquad \xi + \eta = v,$$

in the second part:

$$\xi - \eta = -u, \qquad \xi + \eta = v.$$

If one then integrates, as the figure indicates, first with respect to v at constant u, one obtains:

$$W = \int_0^\infty e^{2\pi i x u}\, du \int_u^\infty e^{\pi i y[u(\omega_1-\omega_2)+v(\omega_1+\omega_2)]} \left(\frac{v^2-u^2}{4}\right)^{s-1} dv,$$

$$+ \int_0^\infty e^{-2\pi i x u}\, du \int_u^\infty e^{\pi i y[-u(\omega_1-\omega_2)+v(\omega_1+\omega_2)]} \left(\frac{v^2-u^2}{4}\right)^{s-1} dv,$$

or, if one sets, as an abbreviation,

$$\varphi_\pm(u) = \int_u^\infty e^{\pi i y[v(\omega_1+\omega_2)\pm u(\omega_1-\omega_2)]} \left(\frac{v^2-u^2}{4}\right)^{s-1} dv \tag{10}$$

then

$$W = \int_0^\infty e^{2\pi i x u}\varphi_+(u)\, du + \int_0^\infty e^{-2\pi i x u}\varphi_-(u)\, du. \tag{11}$$

We decompose these integrals taken with respect to u into parts of such a kind that their limits are the sequence of the integers; we therefore set, since x is an integer:

$$\begin{aligned} \int_0^\infty e^{\pm 2\pi i x u}\varphi_\pm(u)\, du &= \Sigma_{0,\infty}^{\nu} \int_\nu^{\nu+1} e^{\pm 2\pi i x u}\varphi_\pm(u)\, du, \\ &= \Sigma_{0,\infty}^{\nu} \int_0^1 e^{\pm 2\pi i x u}\varphi_\pm(u+\nu)\, du, \\ &= \int_0^1 e^{\pm 2\pi i x u} f(u)\, du, \end{aligned} \tag{12}$$

if again

$$f(u) = \Sigma_{0,\infty}^{\nu}\varphi_\pm(u+\nu) \tag{13}$$

is set.

If we now first carry out the summation with respect to x, we can make use of the fundamental formula from the theory of Fourier series:

$$\begin{aligned} \Sigma_{-\infty,\infty}^{x} \int_0^1 e^{\pm 2\pi i x u} f(u)\, du &= \tfrac{1}{2}[f(0)+f(1)] \\ &= \tfrac{1}{2}\varphi_\pm(0) + \Sigma_{1,\infty}^{\nu}\varphi_\pm(\nu), \end{aligned} \tag{14}$$

and thus obtain, since by (10)

$$\varphi_+(0) = \varphi_-(0) = \varphi(0) = \int_0^\infty e^{\pi i y v(\omega_1+\omega_2)} \left(\frac{v}{2}\right)^{2s-2} dv \tag{15}$$

holds, from (11), (12) and (14):

$$\Sigma_{-\infty,\infty}^{x} W = \varphi(0) + \Sigma_{1,\infty}^{\nu}[\varphi_+(\nu) + \varphi_-(\nu)]. \tag{16}$$

If we introduce this into (8), then M_s splits into two parts:

$$M_s = P_s + Q_s, \tag{17}$$

if

$$P_s = \frac{(2\pi)^{2s}}{A^s \Gamma(s)\Gamma(s)} \Sigma^y_{1,\infty} \varphi(0), \tag{18}$$

$$Q_s = \frac{(2\pi)^{2s}}{A^s \Gamma(s)\Gamma(s)} \Sigma^y_{1,\infty} \Sigma^\nu_{1,\infty} [\varphi_+(\nu) + \varphi_-(\nu)] \tag{19}$$

are set.

By (15), if for v we introduce a new variable of integration t by the equation

$$\pi i v(\omega_1 + \omega_2) = -\frac{2\pi v\sqrt{m}}{A} = -t$$

and carry out the summation over y, we have:

$$\Sigma^y_{1,\infty} \varphi(0) = \frac{2A^{2s-1}}{(4\pi\sqrt{m})^{2s-1}} \int_0^\infty \frac{t^{2s-2}\, dt}{e^t - 1},$$

and therefore:

$$P_s = \frac{(2\pi)^{2s} A^{s-1}}{(4\pi\sqrt{m})^{2s-1}\Gamma(s)^2}\, 2 \int_0^\infty \frac{t^{2s-2}\, dt}{e^t - 1}. \tag{20}$$

From this the limiting value P can easily be determined. For the integral

$$\int_0^\infty t^{2s-2} e^{-t} \left(\frac{1}{1 - e^{-t}} - \frac{1}{t} \right) dt, \tag{21}$$

is continuous up to $s = 1$, since the expression standing within the parenthesis retains a finite value for $t = 0$ and $t = \infty$, and it therefore has the limiting value:

$$\int_0^\infty e^{-t} \left(\frac{1}{1 - e^{-t}} - \frac{1}{t} \right) dt = -\Gamma'(1) = 0{,}5772\ldots$$

If, then, we decompose the integral (21) into its two parts and set

$$\int_0^\infty t^{2s-3} e^{-t}\, dt = \Gamma(2s-2) = \frac{\Gamma(2s-1)}{2(s-1)},$$

there follows:

$$\operatorname*{Lim}_{s=1} \left(\int_0^\infty \frac{t^{2s-2}\, dt}{e^t - 1} - \frac{\Gamma(2s-1)}{2(s-1)} \right) = -\Gamma'(1),$$

and if therefore the development in powers of $s - 1$:

$$\Gamma(2s-1) = 1 + 2\Gamma'(1)(s-1) + \cdots$$

is substituted,

$$\operatorname*{Lim}_{s=1} \left\{ 2 \int_0^\infty \frac{t^{2s-2}\, dt}{e^t - 1} - \frac{1}{s-1} \right\} = 0. \tag{22}$$

One obtains further, by development in powers of $s - 1$:

$$\frac{(2\pi)^{2s} A^{s-1}}{(4\pi\sqrt{m})^{2s-1}\Gamma(s)^2} = \frac{\pi}{\sqrt{m}} \left[1 + (s-1) \left(\log \frac{A}{4m} - 2\Gamma'(1) \right) + \cdots \right] \tag{23}$$

so that, by (20):

$$P_s = \frac{(2\pi)^{2s} A^{s-1}}{(4\pi\sqrt{m})^{2s-1}\Gamma(s)^2}\left(2\int_0^\infty \frac{t^{2s-2}\,dt}{e_t - 1} - \frac{1}{s-1}\right) + \frac{\pi}{\sqrt{m}(s-1)} + \frac{\pi}{\sqrt{m}}\left(\log\frac{A}{4m} - 2\Gamma'(1)\right) + \cdots, \tag{24}$$

in which the terms that still follow vanish with $s-1$. From this it follows:

$$\operatorname{Lim}\left(P_s' - \frac{\pi}{\sqrt{m}(s-1)}\right) = \frac{\pi}{\sqrt{m}}\left(\log\frac{A}{4m} - 2\Gamma'(1)\right). \tag{25}$$

There still remains the part Q_s to be investigated, which for $s = 1$ takes a finite limiting value Q; this we can determine at once by substituting the value $s = 1$.

For in (10), as long as u and y are greater than zero, after substituting the value $s = 1$, the integration

carried out, and with regard to the value $2i\sqrt{m}$ of $A(\omega_1 + \omega_2)$ it gives:

$$\varphi_\pm(\nu) = \frac{A}{2\pi\sqrt{m}\,y}\, e^{2\pi i y\nu\omega},$$

in which $\omega = \omega_1$ or $= \omega_2$ is to be set, according as the upper or the lower sign is taken in $\varphi_\pm(\nu)$. By carrying out the summation with respect to y there follows from this:

$$\sum_{1,\infty}^{\nu}\sum_{1,\infty}^{y}\varphi_\pm(\nu) = -\frac{A}{2\pi\sqrt{m}}\sum_{1,\infty}^{\nu}\log\left(1 - e^{2\pi i\nu\omega}\right),$$

and the sum with respect to ν can be reduced to the function $\eta(\omega)$, since by § 24, (8):

$$\eta(\omega) = e^{\frac{\pi i\omega}{12}}\prod_{1,\infty}^{\nu}\left(1 - e^{2\pi i\nu\omega}\right)$$

holds. Accordingly there results, always for $s = 1$,

$$\sum_{1,\infty}^{\nu}\sum_{1,\infty}^{y}\varphi_\pm(\nu) = -\frac{A}{2\pi\sqrt{m}}\left(\log\eta(\omega) - \frac{\pi i\omega}{12}\right).$$

If we introduce this result into (19), after having set $s = 1$ there, we obtain the limiting value of Q_s:

$$Q = -\frac{2\pi}{\sqrt{m}}\log\eta(\omega_1)\eta(\omega_2) - \frac{\pi^2}{3A}. \tag{26}$$

Accordingly we obtain from (4), (25), (26) the following limit formula, the derivation of which was the aim of this discussion:

$$\operatorname*{Lim}_{s=1}\sum^{x,y}\frac{1}{(Ax^2 + 2Bxy + Cy^2)^s} - \frac{\pi}{(s-1)\sqrt{m}} = -\frac{2\pi\Gamma'(1)}{\sqrt{m}} + \frac{\pi}{\sqrt{m}}\log\frac{A}{4m} - \frac{2\pi}{\sqrt{m}}\log\eta(\omega_1)\eta(\omega_2). \tag{27}$$

From (27) there follows, in agreement with § 140, (5):

$$\operatorname{Lim}(s-1)\sum^{x,y}\frac{1}{\psi^s} = \frac{\pi}{\sqrt{m}},$$

hence depending only on m, not on the particular form ψ. If we now understand $A, 2B, C$ to be integers and let ψ run through a complete system of representatives of the classes of forms belonging to the discriminant $-4m$, then it follows

$$\operatorname*{Lim}_{s=1}(s-1)\sum^{\psi}\sum^{x,y}\frac{1}{\psi^s}=\frac{\pi h}{\sqrt{m}}, \tag{28}$$

where h is the class number belonging to the discriminant $-4m$.

From (27) one derives a simpler formula for the function

$$f(\omega)=e^{-\frac{\pi i}{24}}\,\frac{\eta\left(\frac{\omega+1}{2}\right)}{\eta(\omega)}=e^{\frac{\pi i}{24}}\,\frac{\eta\left(\frac{\omega-1}{2}\right)}{\eta(\omega)}.$$

For if one replaces

$$\begin{aligned} &A \text{ by } 2A,\\ &B \text{ „ } A+B,\\ &C \text{ „ } \tfrac{1}{2}(A+2B+C), \end{aligned}$$

then $m=AC-B^2$ remains unchanged, and ω_1, ω_2 pass over into

$$\omega_1'=\frac{\omega_1+1}{2},\quad \omega_2'=\frac{\omega_2-1}{2}.$$

If one therefore sets

$$\begin{aligned} &\psi=(A,\,2B,\,C),\\ &\psi'=[2A,\,2(A+B),\,\tfrac{1}{2}(A+2B+C)], \end{aligned} \tag{29}$$

then there results from (27)

$$\operatorname*{Lim}_{s=1}\left(\sum^{x,y}\frac{1}{\psi^s}-\sum^{x,y}\frac{1}{\psi'^s}\right)=\frac{2\pi}{\sqrt{m}}\log\frac{f(\omega_1)f(\omega_2)}{\sqrt{2}}. \tag{30}$$

We now take not only $A, 2B, C$, but also A, B, C, to be integers without common divisor, and A odd. Then ψ is a primitive form of discriminant $-4m$, and we consider first the two cases

$$m\equiv 1,\quad m\equiv 2 \pmod 4.$$

If we assume, which is no further restriction, that $B\equiv m+1 \pmod 2$, that is, B even or odd according as m is odd or even, then there results from

$$\begin{aligned} &m=AC-B^2:\\ &A-C\equiv 0 \pmod 4 \quad m \text{ odd},\\ &A+C\equiv 0 \text{ „} \qquad\quad m \text{ even}, \end{aligned}$$

and the form ψ' is primitive of discriminant $-4m$. By the composition of quadratic forms we have:

$$\psi'=\psi_0\psi,$$

if

$$\begin{aligned} \psi_0&=\left(2,\,2,\,\tfrac{m+1}{2}\right) \quad (m \text{ odd}),\\ &=(2,\,0,\,\tfrac{m}{2}) \qquad (m \text{ even}), \end{aligned}$$

and consequently ψ' runs through a system of representatives of the discriminant $-4m$ at the same time as ψ. But ω_1 runs through the same sequence of values as ω_2, though in a different order, and accordingly summation of formula (30) gives:

$$\Pi\,\frac{f(\omega)}{\sqrt[4]{2}}=1, \tag{31}$$

if the product Π refers to the roots with positive imaginary part of a complete system of forms with the discriminant $-4m$.

In what follows we shall, for the sake of brevity, call this sequence of values of the ω a complete system of roots of the discriminant $-4m$.

§ 142. The Norms of the Class Invariants $f(\omega)$.

We let ω run through a complete system of roots of the discriminant $-4m$, and we assume that in the form $(A, 2B, C)$, whose root is ω, A, C are odd and A is relatively prime to m, which involves no further restriction; then by § 126 the 24th powers of $f(\omega)$ are class invariants and their norm is a power of 2.

In a two-sided class there is always a representative of one of the two forms:

$$(A,\ 0,\ C), \quad (A,\ 2B,\ A),$$

in which A may be assumed to be odd. In the first case ω is purely imaginary and consequently [§ 24, (11)]:

$$f(\omega), \quad f_1(\omega), \quad f_2(\omega)$$

are real and positive; in the second case ω and $1 : \omega$ are conjugate imaginary, and consequently:

$$f(\omega) = f\left(-\frac{1}{\omega}\right)$$

is real and

$$f_1(\omega) = f_2\left(-\frac{1}{\omega}\right), \quad f_2(\omega) = f_1\left(-\frac{1}{\omega}\right)$$

are conjugate imaginary, and by the formula $f(\omega)f_1(\omega)f_2(\omega) = \sqrt{2}$ here too $f(\omega)$ is positive.

We may therefore always assume the representative $(A, 2B, C)$ to be so chosen that A and C are odd and that $f(\omega)$ becomes real and positive for a two-sided class; if

we further denote two opposite classes by $(A, \pm 2B, C)$, then the corresponding values of $f(\omega)$ are conjugate imaginary, their product therefore positive, and it follows accordingly from these determinations that the product

$$\Pi f(\omega)$$

has a positive real value which is a power of 2. Denoting the class number by h, we set this power $= 2^{h\tau}$, so that

$$\Pi \frac{f(\omega)}{2^{\tau}} = 1. \tag{1}$$

It will be our task to determine this exponent τ. But we first premise the following remark, which lends this task an increased interest.

In consequence of the equation [§ 54, (8)]:

$$f(\omega)^{24} - \gamma_2(\omega) f(\omega)^8 - 16 = 0 \tag{2}$$

it follows that, if ω is the root of a form of the discriminant $-4m$ belonging to the class k, $f(\omega)$ is an integral algebraic number; and since $f(\omega)^8$ in (2) may also be replaced by $-f_1(\omega)^8$, $-f_2(\omega)^8$, the quantities $f_1(\omega)$, $f_2(\omega)$ are likewise integral algebraic numbers. Consequently so is

$$\frac{\sqrt{2}}{f(\omega)} = f_1(\omega) f_2(\omega).$$

If p is an odd prime of which $-m$ is a quadratic residue, and if p is representable by the forms of the class l (of the discriminant $-4m$), then, upon a suitable determination of c according to § 118,

$$\frac{c + \omega}{p}$$

is the root of a form belonging to the compounded class lk, and (unless it happens that $p = 3$) c may be assumed divisible by 48. If we therefore set:

$$u = f(\omega), \quad v = f\left(\frac{c+\omega}{p}\right),$$

then uv as well as $2 : uv$ is an integral algebraic number.

If therefore we set, following § 74,

$$B = (uv)^s + \left(\frac{2}{p}\right)\frac{2^s}{(uv)^s},$$
$$A = \left(\frac{u}{v}\right)^r + \left(\frac{v}{u}\right)^r$$

(where now A, B are of course not to be confused with the coefficients of the quadratic form), then B is an integral algebraic number, and by § 74, (14) A is the root of an algebraic equation whose coefficients are integral algebraic numbers. Hence A too is an integral algebraic number.

But since the two quotients $u^r : v^r$ and $v^r : u^r$ are the roots of the quadratic equation

$$x^2 - Ax + 1 = 0$$

it follows that

$$\frac{u}{v} \quad \text{and} \quad \frac{v}{u}$$

are integers, and, since they are reciprocal to one another, units.

Thus u and v are associated numbers.

Since one can now, by § 118, pass by repeated compositions with classes l (by which primes are representable) from every class k to every other class k' of the same discriminant, and since two numbers associated with a third are associated with one another, we have the theorem:

If one puts for ω the h roots of the forms of a system of representatives of the discriminant $-4m$, then the h numbers $f(\omega)$ are associated with one another; and from this, by (1), at once the remarkable theorem, which may easily be confirmed in all the examples:

$f(\omega) : 2^\tau$ is an integer, and indeed a unit.

1. The determination of the exponents τ is possible by elementary means, if $m \equiv 1$ or $\equiv 2 \pmod 4$ or $m \equiv 3 \pmod 8$.

If in the equation with odd outer coefficients

$$A\omega^2 + 2B\omega + C = 0 \tag{3}$$

with the discriminant $4(B^2 - AC) = -4m$ we make the substitution

$$\omega = \frac{\omega' - 1}{\omega' + 1}, \quad \omega' = -\frac{\omega + 1}{\omega - 1}, \tag{4}$$

we obtain the equation

$$\frac{A + 2B + C}{2}\,\omega'^2 - (A - C)\,\omega' + \frac{A - 2B + C}{2} = 0, \tag{5}$$

which likewise has the discriminant $-4m$, and in which, if $m \equiv 1$ or $\equiv 2 \pmod 4$, the two outer coefficients are odd; for we have:

$$\text{for } m \equiv 1 \pmod 4, \quad B \equiv 0 \pmod 2, \quad A + C \equiv 2 \pmod 4,$$
$$\text{for } m \equiv 2 \pmod 4, \quad B \equiv 1 \pmod 2, \quad A + C \equiv 0 \pmod 4.$$

From this it follows that

$$f(\omega) \quad \text{and} \quad f\left(\frac{\omega-1}{\omega+1}\right),$$

apart from 24th roots of unity, run through the same sequence of values. For if one replaces ω' by an equivalent number, then, if the outer coefficients are to remain odd, the substitution must belong to the first or the second class (§ 36), and from (4) it then follows that ω too passes into an equivalent number. But if two forms (5) are equivalent, then the corresponding forms (3) are equivalent as well, and conversely.

Accordingly

$$\Pi f(\omega) f\left(\frac{\omega-1}{\omega+1}\right) = 2^{2h\tau},$$

while on the other hand [§ 34, (18)]:

$$f(\omega) f\left(\frac{\omega-1}{\omega+1}\right) = \sqrt{2},$$

from which there results:

$$\tau = \frac{1}{4}, \quad m \equiv 1,\ 2 \pmod 4, \tag{6}$$

in agreement with the result of the preceding section [§ 141, (31)].

2. If next $m \equiv 3 \pmod 8$, then to a root ω' of a form of discriminant $-m$ there correspond in each case three roots of forms of discriminant $-4m$:

$$2\omega', \quad \frac{\omega'}{2}, \quad \frac{\omega'+1}{2},$$

and the 24th powers of the quantities:

$$f_1(2\omega') = \frac{\sqrt{2}}{f_2(\omega')},$$
$$f_2\left(\frac{\omega'}{2}\right) = \frac{\sqrt{2}}{f_1(\omega')},$$
$$f_2\left(\frac{\omega'+1}{2}\right) = \frac{\sqrt{2}}{f(\omega')},$$

whose product is $= 2$, are class invariants of the discriminant $-4D$. Accordingly:

$$\Pi f(\omega) = 2^{\frac{1}{3}h},$$

hence

$$\tau = \frac{1}{3}, \quad m \equiv 3 \pmod 8. \tag{7}$$

3. For the case $m \equiv 7 \pmod 8$ we cannot determine the value of τ in this simple way. For this the limit formula derived in the preceding section is required. We first treat the two cases $m \equiv 3 \pmod 4$ uniformly and set:

$$\varphi = ax^2 + bxy + cy^2 = (a, b, c),$$

$$b^2 - 4ac = -m, \tag{8}$$

$$S' = \sum^{x,y} \frac{1}{\varphi(x,y)^s}, \tag{9}$$

in which x, y run through all integer values, with the exception of the combination $0, 0$.

We decompose the sum S into four partial sums

$$S'_{00}, \quad S'_{01}, \quad S'_{10}, \quad S'_{11},$$

so that x, y run through only even values in S'_{00}, and only odd values in S'_{11}. In S'_{10}, x runs through the odd numbers and y the even ones, and conversely in S'_{01}. Then

$$S' + 2S'_{00} = (S'_{01} + S'_{00}) + (S'_{10} + S'_{00}) + (S'_{11} + S'_{00}). \tag{10}$$

If we replace

$$\begin{array}{llllll} \text{in} & S'_{00} & x, y & \text{by} & 2x,\ 2y, \\ \text{„} & S'_{01} + S'_{00} & x, y & \text{„} & 2x,\ y, \\ \text{„} & S'_{10} + S'_{00} & x, y & \text{„} & x,\ 2y, \\ \text{„} & S'_{00} + S'_{11} & x, y & \text{„} & x - y,\ x + y, \end{array}$$

then the new x, y are subject to no further restriction than that they do not both vanish at the same time. If we therefore set:

$$\begin{aligned} \varphi &= (a, b, c), \\ \varphi_1 &= (4a, 2b, c), \\ \varphi_2 &= (a, 2b, 4c), \\ \varphi_3 &= (a + b + c,\ 2(c - a),\ a - b + c), \end{aligned} \tag{11}$$

then

$$\begin{aligned} 4^s S'_{00} &= S', \\ S'_{01} + S'_{00} &= \overset{x,y}{\Sigma} \varphi_1^{-s}, \\ S'_{10} + S'_{00} &= \overset{x,y}{\Sigma} \varphi_2^{-s}, \\ S'_{00} + S'_{11} &= \overset{x,y}{\Sigma} \varphi_3^{-s}. \end{aligned} \tag{12}$$

We assume a to be odd and let φ run through a complete system of representatives of the discriminant $-m$. If then $m \equiv 3 \pmod 8$, then c is odd, and φ_1, φ_2, φ_3 together run through a system of representatives of the discriminant $-4m$. For among the $3h$ roots of these forms

$$\frac{\omega}{2}, \quad 2\,\omega, \quad 1 - \frac{2}{\omega + 1}$$

there occur, by § 123, no equivalent ones.

If on the other hand $m \equiv 7 \pmod 8$, then c is even, $\frac{1}{2}\varphi_1$, $\frac{1}{2}\varphi_3$ each run through a system of representatives of the discriminant $-m$, while φ_2 runs through a system of representatives of the discriminant $-4m$.

If then ψ runs through a system of representatives of the discriminant $-4m$, and we set

$$S = \overset{x,y}{\Sigma} \frac{1}{\psi^s}, \tag{13}$$

then there results from (10):

$$\begin{aligned} \Big(1 + \frac{2}{4^s}\Big) \overset{\varphi}{\Sigma} S' &= \overset{\psi}{\Sigma} S, \quad m \equiv 3 \\ \Big(1 + \frac{2}{4^s} - \frac{2}{2^s}\Big) \overset{\varphi}{\Sigma} S' &= \overset{\psi}{\Sigma} S, \quad m \equiv 7 \end{aligned} \pmod 8. \tag{14}$$

If in formulae (29), (30) of the preceding section we set

$$\psi' = 2\,\varphi, \quad \psi = (A, 2B, C),$$

thus:

$$\begin{aligned} A &= a, \\ B &= b - a, \\ C &= 4c - 2b + a, \end{aligned}$$

then there results:

$$S - \frac{1}{2^s} S' = \frac{2\pi}{\sqrt{m}} \log \frac{f(\omega_1) f(\omega_2)}{\sqrt{2}},$$

and when ψ runs through a system of representatives of the discriminant $-4m$, then φ runs three times, or only once, through a system of representatives of the discriminant $-m$, according as $m \equiv 3$ or $\equiv 7 \pmod 8$. We therefore obtain, if ε in these two cases is 3 or 1:

$$\operatorname*{Lim}_{s=1} \Big(\overset{\psi}{\Sigma} S - \frac{\varepsilon}{2^s} \overset{\varphi}{\Sigma} S' \Big) = \frac{4\pi}{\sqrt{m}} \log \Pi \frac{f(\omega)}{\sqrt[4]{2}}, \tag{15}$$

in which ω runs through the roots of the forms ψ. And from this there results, by (14):

$$\begin{aligned} \operatorname*{Lim}_{s=1} (4^s + 2 - 3.\,2^s) \overset{\psi}{\Sigma} S &= \frac{24\pi}{\sqrt{m}} \log \Pi \frac{f(\omega)}{\sqrt[4]{2}}, \quad m \equiv 3 \pmod 8, \\ \operatorname*{Lim}_{s=1} (4^s + 2 - 3.\,2^s) \overset{\psi}{\Sigma} S &= \frac{8\pi}{\sqrt{m}} \log \Pi \frac{f(\omega)}{\sqrt[4]{2}}, \quad m \equiv 7 \pmod 8. \end{aligned} \tag{16}$$

But we have [§ 141, (28)]:

$$\operatorname{Lim} (s-1) \overset{\psi}{\Sigma} S = \frac{\pi h}{\sqrt{m}}, \quad \operatorname{Lim} \frac{4^s + 2 - 3.\,2^s}{s-1} = 2 \log 2,$$

hence:

$$\begin{aligned} \log \Pi \frac{f(\omega)}{\sqrt[3]{2}} &= 0, \quad m \equiv 3 \pmod 8, \\ \log \Pi \frac{f(\omega)}{\sqrt{2}} &= 0, \quad m \equiv 7 \pmod 8. \end{aligned} \tag{17}$$

The first of these formulae gives the result already derived in another way; the second gives the new one:

$$\tau = \tfrac{1}{2}, \quad m \equiv 7 \pmod 8. \tag{18}$$

4. There still remains the determination of τ in the case where m is divisible by a higher power of 2. In order to carry out this determination as well, let

$$m = 4m', \tag{19}$$

$$\omega' = \frac{-B + i\sqrt{m}}{A}, \quad m' = AC - B^2, \tag{20}$$

so that ω' is the root of the quadratic form $(A, 2B, C)$ of discriminant $-m$, in which A and C may be assumed to be o d d. Then

$$\omega_1 = 2\,\omega', \quad \omega_2 = \frac{\omega'}{2} \tag{21}$$

are roots of the forms

$$(A, 4B, 4C), \quad (4A, 4B, C), \tag{22}$$

and by § 126, (13) the 24th powers of

$$f_1(\omega_1) = \frac{\sqrt{2}}{f_2(\omega')}, \quad f_2(\omega_2) = \frac{\sqrt{2}}{f_1(\omega')}$$

are class invariants of the discriminant $-4m$.

If ω' runs through a complete system of roots of the discriminant $-m$, then ω_1 and ω_2 together run through a complete system of roots of the discriminant $-4m$ (§ 123), and we obtain,

when ω runs through a complete system of roots of forms with odd outer coefficients, with the use of the formula:

$$f_1(\omega')f_2(\omega')f(\omega') = \sqrt{2},$$

if h' is the class number belonging to the discriminant $-m'$:

$$\Pi f(\omega) = \Pi f_1(\omega_1)f_2(\omega_2) = \frac{2^h}{\Pi f_1(\omega')f_2(\omega')} = \sqrt{2}^{h'}\Pi f(\omega'), \tag{23}$$

in which h, h' denote the class numbers for the discriminants $-4m$, $-m$. We then have (§ 123):

$$h = 2h'.$$

If therefore, as above, τ, τ' are so determined that

$$\frac{f(\omega)}{2^\tau}, \quad \frac{f(\omega')}{2^{\tau'}}$$

become units, then

$$\Pi f(\omega) = 2^{h\tau}, \quad \Pi f(\omega') = 2^{h'\tau'},$$

and from this, by (23):

$$\tau = \frac{\tau'}{2} + \frac{1}{4}, \tag{24}$$

a formula which holds also for $m' = 1$, where $h' = h$ and the two values $2\,\omega'$ and $\omega' : 2$ are equivalent.

By repeated application of this formula there results, when

$$m = 4^\lambda m'$$

holds, for an arbitrary positive λ:

$$\tau = \frac{\tau'}{2^\lambda} + \frac{1}{2} - \frac{1}{2^{\lambda+1}}.$$

If we summarise what has here been proved, we can say that the following quantities are algebraic units:

$$\frac{f(\omega)}{\sqrt[4]{2}}, \quad m \equiv 1, 2 \pmod 4,$$

$$\frac{f(\omega)}{\sqrt[3]{2}}, \quad m \equiv 3 \pmod 8,$$

$$\frac{f(\omega)}{\sqrt{2}}, \quad m \equiv 7 \pmod 8,$$

$$\frac{f(\omega)}{2^{\frac{1}{2}-\frac{1}{2^{\lambda+2}}}}, \quad m = 4^\lambda m', \quad m' \equiv 1, 2 \pmod 4,$$

$$\frac{f(\omega)}{2^{\frac{1}{2}-\frac{1}{3.\,2^{\lambda+1}}}}, \quad m = 4^\lambda m', \quad m' \equiv 4 \pmod 8,$$

$$\frac{f(\omega)}{\sqrt{2}}, \quad m = 4^\lambda m', \quad m' \equiv 7 \pmod 8.$$

§ 143. Partial norms of $f(\omega)$.

We make one further application of the limit formula (30) of § 141 to the determination of the product

$$\Pi f(\omega), \tag{1}$$

in which the product sign Π extends not over a complete system of roots, but only over the roots ω of the classes of forms of one genus. We restrict ourselves here, however, to the case in which

$$\Delta = -4m \tag{2}$$

is a fundamental discriminant, so that m has no square divisor and is either $\equiv 1$ or $\equiv 2 \pmod 4$, or, what is the same thing,

$$\Delta \equiv -4 \text{ or } \equiv 8 \pmod{16}. \tag{3}$$

Let δ be a fundamental divisor of Δ different from Δ and 1, and let $\chi(\delta, k)$ be the character of the class k corresponding to this fundamental divisor.

If δ' is the fundamental divisor to δ that is complementary to δ, then in this case

$$\begin{aligned} \delta\,\delta' &= \Delta, \\ \chi(\delta, k) &= \chi(\delta', k), \end{aligned}$$

and we thus obtain all the genera if we put for δ the odd fundamental divisors, which is what we shall do here. Now let, as in § 141, (29), (30):

$$\begin{aligned} \psi_0 &= \Big(2, 2, \frac{m+1}{2}\Big),\ \Big(2, 0, \frac{m}{2}\Big), \\ \psi &= (A, 2B, C), \\ \psi' &= \psi_0\,\psi. \end{aligned} \tag{4}$$

If then k, k_0, k' are the classes to which ψ, ψ_0, ψ' belong, then

$$\chi(\delta, k') = \chi(\delta, k_0)\,\chi(\delta, k),$$

hence

$$\begin{aligned} \chi(\delta, k) &= (\delta, A), \\ \chi(\delta, k') &= (\delta, 2)(\delta, A). \end{aligned} \tag{5}$$

If ω_1 is a root of the class k, then ω_2 is a root of the opposite class k^{-1}; but in these two classes the characters $\chi(\delta, k)$ and $\chi(\delta, k^{-1})$ are the same. If we therefore multiply the formula § 141, (30) by $\chi(\delta, k)$ and form the sum over all classes k, it follows that:

$$\overset{k}{\Sigma}\chi(\delta,k)\Big(\overset{x,y}{\Sigma}\frac{1}{\psi^s} - \overset{x,y}{\Sigma}\frac{1}{\psi'^s}\Big) = \frac{4\pi}{\sqrt{m}}\overset{k}{\Sigma}\chi(\delta,k)\log\frac{f(\omega)}{\sqrt[4]{2}}, \tag{6}$$

and on account of (5) one may write for the left-hand side:

$$[1-(\delta,2)]\overset{k}{\Sigma}\frac{\chi(\delta,k)}{1}\overset{x,y}{\Sigma}\frac{1}{\psi^s}.$$

Accordingly we have:

$$\begin{aligned} (\delta,2) = +1\colon &\quad \frac{2\pi}{\sqrt{m}}\overset{k}{\Sigma}\chi(\delta,k)\log\frac{f(\omega)}{\sqrt[4]{2}} = 0, \\ (\delta,2) = -1\colon &\quad \frac{2\pi}{\sqrt{m}}\overset{k}{\Sigma}\chi(\delta,k)\log\frac{f(\omega)}{\sqrt[4]{2}} = \overset{k}{\Sigma}\chi(\delta,k)\overset{x,y}{\Sigma}\frac{1}{\psi^s}, \end{aligned} \tag{7}$$

where on the right the limiting value for $s = 1$ is to be taken.

In § 113 we proved the formula:

$$\overset{k}{\Sigma}\chi(\delta,k)\overset{x,y}{\Sigma}\frac{1}{\psi^s} = \Sigma\frac{(\delta,n)}{n^s}\Sigma\frac{(\delta',n)}{n^s}. \tag{8}$$

If we denote by $K(\delta)$ the class number for the discriminant δ, then, as was proved in § 112, (8):

$$\begin{aligned}\sum^{n}\frac{(\delta,n)}{n} &= \frac{\pi}{\sqrt{-\delta}}K(\delta) \qquad \delta<0,\\ \sum^{n}\frac{(\delta,n)}{n} &= \frac{\log\varepsilon}{\sqrt{\delta}}K(\delta) \qquad \delta>0,\end{aligned} \tag{9}$$

in which the square roots are to be taken positive, and

$$\varepsilon = \frac{T+U\sqrt{\delta}}{2}$$

is the fundamental positive unit of the quadratic field with the fundamental number δ. For the two exceptional cases $\delta = -3$, $\delta = -4$ these formulae hold if by K one understands not the

class number itself, but the third part or the half of it.

Now δ' is always even and of opposite sign to δ, and $4m = -\delta\delta'$; accordingly there result from (7) the formulae:

$$\begin{aligned}2\overset{k}{\Sigma}\chi(\delta,k)\log\frac{f(\omega)}{\sqrt[4]{2}} &= 0, && \delta\equiv 1 \pmod 8,\\ &= K(\delta)K(\delta')\log\varepsilon, && \delta\equiv 5 \pmod 8,\end{aligned} \tag{10}$$

in which

$$\begin{aligned}\varepsilon &= \frac{T+U\sqrt{\delta}}{2}, && \delta>0\equiv 1 \pmod 8,\\ &= \frac{T+U\sqrt{\delta'}}{2}, && \delta<0\equiv 5 \pmod 8\end{aligned} \tag{11}$$

is to be set. The first of the formulae (10) still holds for $\delta = 1$ as well.

Since now the character $\chi(\delta,k)$ has the same value for all classes of a genus and for every δ, the products (1) are determined by (10) and (11). For by § 113

$$\overset{\delta}{\Sigma}\chi(\delta,k) = 0,$$

except when k is the principal class, and for this class the sum is equal to the number g of the genera. One obtains, for example, for the roots ω of the principal genus

$$\left(\Pi\frac{f(\omega)}{\sqrt[4]{2}}\right)^{2g} = \Pi\,\varepsilon^{K(\delta)K(\delta')},$$

in which g denotes the number of the genera, and the product on the left is to be extended over all roots ω of the principal genus, the product on the right over all fundamental divisors δ of Δ which are $\equiv 5 \pmod 8$.

In order to form the product of the class invariants for a genus other than the principal genus, one has to multiply formula (8), before the summation, by $\chi(\delta,k^{-1})$, if k denotes a class of the genus in question.

The application of formula (10) requires the knowledge of the class numbers of positive and negative discriminants and of the numbers T, U. The class numbers have been computed to a wide

extent by Gauss and published from his posthumous papers in the second volume of the Werke, pp. 450 to 476. For the solutions T, U of the Pell equation, Legendre's „Theorie des nombres", or also the „Canon Pellianus" of Degen, contains a table.

There exist 63 negative discriminants with the property that each genus contains only a single class; that there are no more, indeed even that their number is only a finite one, can of course so far only be inferred by induction, not proved. The majority of these discriminants, which have already been collected in § 139, are $\equiv 1, 2 \pmod 4$ and without square divisor, or $\equiv 8 \pmod{16}$.

For the former the class invariants can be completely computed by formula (10), and a similar formula, into whose formation we do not wish to enter here, also leads to the goal for the discriminants divisible by 8.

For the isolated discriminants of this kind which do not fit in here, the class invariants $f(\omega)$ can be computed by one of the other methods.[31]

In order to carry out the calculation on a simple example that is easy to survey, we choose $m = 105 = 3.5.7$. If we omit the values of δ which are $\equiv 1 \pmod 8$, since these give no contribution in the sum of the formulae (10), then we have to put:

$$\begin{aligned} \delta &= -3, \quad 5, \quad 21, \quad -35, \\ \delta' &= 140, \quad -84, \quad -20, \quad 12 \end{aligned}$$

and obtain, since $g = 8$:

$$\begin{aligned} 16 \log \frac{f(\sqrt{-105})}{\sqrt[4]{2}} &= K(-3)\,K(140) \log \frac{T + U\sqrt{140}}{2}, \\ &\quad + K(5)\,K(-84) \log \frac{T + U\sqrt{84}}{2}, \\ &\quad + K(21)\,K(-20) \log \frac{T + U\sqrt{20}}{2}, \\ &\quad + K(-35)\,K(12) \log \frac{T + U\sqrt{12}}{2}. \end{aligned}$$

But we have

$$\begin{aligned} K(-3) = \tfrac{1}{3}, \quad K(-84) = 4, \quad K(-20) = 2, \quad K(-35) = 2, \\ K(+140) = 4, \quad K(5) = 1, \quad K(21) = 2, \quad K(12) = 2^{1)}, \end{aligned}$$

as one finds from the tables of Gauss, or here also easily by direct enumeration.

Further, according to the tables of Legendre or of Degen:

$$\begin{aligned} \log \left(\frac{T + \sqrt{140}\,U}{2} \right) &= \log(6 + \sqrt{35}) = \log \frac{(\sqrt{5} + \sqrt{7})^2}{2}, \\ \log \left(\frac{T + \sqrt{5}\,U}{2} \right) &= \log \left(\frac{1 + \sqrt{5}}{2} \right)^2 = \tfrac{2}{3} \log(2 + \sqrt{5}), \\ \log \frac{T + \sqrt{21}\,U}{2} &= \log \frac{5 + \sqrt{21}}{2} = \tfrac{1}{3} \log(55 + 12\sqrt{21}) \\ &= \log \left(\frac{\sqrt{3} + \sqrt{7}}{2} \right)^2, \\ \log \left(\frac{T + \sqrt{12}\,U}{2} \right) &= \log(2 + \sqrt{3}), \end{aligned}$$

[31] Cf. the author's memoir: „Zur komplexen Multiplikation elliptischer Funktionen". Mathematische Annalen, Bd. 23. I here call attention to an error in the Gauss table which I found on the occasion of these computations: Gauss' Werke, Bd. II, S. 475, the positive determinant 136 must have the designation IV, 2, not IV, 1.

[1)] According to the notation of Gauss the class numbers of the second kind are to be taken.

and from this one obtains:

$$\left(\frac{f(\sqrt{-105})}{\sqrt[4]{2}}\right)^{12} = (2+\sqrt{5})^2(55+12\sqrt{21})(6+\sqrt{35})(2+\sqrt{3})^3,$$

or, what means the same thing:

$$\sqrt{2}^{13} f(\sqrt{-105})^6 = (1+\sqrt{5})^3(\sqrt{3}+\sqrt{7})^3(\sqrt{5}+\sqrt{7})(1+\sqrt{3})^3.$$

§ 144. Calculation of some further class invariants.

Next after the discriminants for which only one class of forms occurs in each genus, which we considered in the preceding section, the simplest results are given by those which contain two classes of forms in each genus, and among these again by those for which only two genera occur. The class invariants for these discriminants $-4m$ are the roots of quadratic equations whose coefficients contain only one square root. We again assume fundamental discriminants, and obtain from the table of Gauss the following values of m:

$$\begin{aligned} &m = 14,\ 34,\ 46,\ 82,\ 142 \equiv \pm 2 \pmod{16},\\ &m = 17,\ 49,\ 73,\ 97,\ 193 \equiv 1 \pmod{8}. \end{aligned}$$

In these cases the classes of forms of the principal genus can be represented, for an even m, by

$$(1,0,m), \quad \left(2,0,\frac{m}{2}\right), \tag{1}$$

for an odd m by

$$(1,0,m), \quad \left(2,1,\frac{m+1}{2}\right), \tag{2}$$

of which the latter is equivalent to

$$\left(\frac{m+1}{2},\ \frac{m-1}{2},\ \frac{m+1}{2}\right). \tag{3}$$

For the forms (1) the class invariants are, by § 127, 6.,

$$\frac{f_1(\sqrt{-m})^2}{\sqrt{2}} \quad \text{and} \quad \frac{1}{\sqrt{2}} f_2\left(\frac{\sqrt{-m}}{2}\right)^2 = \frac{\sqrt{2}}{f_1(\sqrt{-m})^2} \qquad (\S\ 34)$$

and for the forms (2), (3) (§ 127, 3.)

$$\frac{f(\sqrt{-m})^2}{\sqrt{2}} \quad \text{and} \quad \frac{1}{\sqrt{2}} f\left(\frac{1+\sqrt{-m}}{1-\sqrt{-m}}\right)^2 = \frac{\sqrt{2}}{f(\sqrt{-m})^2},$$

and by § 142 these are integral algebraic numbers.

If then we set, corresponding to the two cases:

$$\sqrt{2}\,x = f_1(\sqrt{-m})^2, \quad f(\sqrt{-m})^2, \tag{4}$$

then

$$x + \frac{1}{x}$$

is an integral algebraic number which contains only one square root, and from § 138 we obtain information as to which square root occurs in it.

It is $\sqrt{2}$ when $m \equiv 6 \pmod 8$, hence for $m = 14, 46, 142$; it is $\sqrt{\frac{m}{2}}$ when $m \equiv 2 \pmod 8$, hence for $m = 34, 82$; and further $\sqrt{m}$ in the case of an odd m (with the exception of $m = 49$, where $\sqrt{7}$ takes its place).

If then we set

$$x + \frac{1}{x} = a + b\sqrt{p}, \tag{5}$$

then a, b are rational numbers which have at most the denominator 2. But a and b must also be positive. For if in (5) we change the sign of $\sqrt{p}$, there arises another quadratic

equation whose roots are the class invariants belonging to the second genus, and which are therefore conjugate imaginary, while the roots of (5) are real. From this there follows the determination of quantity:

$$(a + b\sqrt{p})^2 > 4 > (a - b\sqrt{p})^2;$$

and so a and b must have the same sign. But since x is positive by (4), a and b must both be positive.

In order actually to find a and b, one needs only to compute the expression on the left-hand side of (5), by (4), to a few decimals, and for this it is amply sufficient to put

$$f(\sqrt{-m}) = f_1(\sqrt{-m}) = e^{\frac{\pi\sqrt{m}}{24}}$$

and to compare the decimals so found with the decimals of $\sqrt{p}$, in order to obtain b at once and thereupon a. The calculation is exceedingly simple and gives the following results:

$$\begin{aligned}
m = 14, &\quad x + \frac{1}{x} = 1 + \sqrt{2},\\
m = 17, &\quad x + \frac{1}{x} = \frac{1 + \sqrt{17}}{2},\\
m = 34, &\quad x + \frac{1}{x} = \frac{3 + \sqrt{17}}{2},\\
m = 46, &\quad x + \frac{1}{x} = 3 + \sqrt{2},\\
m = 49, &\quad x + \frac{1}{x} = 2 + \sqrt{7},\\
m = 73, &\quad x + \frac{1}{x} = \frac{5 + \sqrt{73}}{2},\\
m = 82, &\quad x + \frac{1}{x} = \frac{15 + \sqrt{41}}{2},\\
m = 97, &\quad x + \frac{1}{x} = \frac{9 + \sqrt{97}}{2},\\
m = 142, &\quad x + \frac{1}{x} = 9 + 5\sqrt{2},\\
m = 193, &\quad x + \frac{1}{x} = 13 + \sqrt{193}.
\end{aligned}$$

Twenty-Second Chapter.

Cayley's Development of the Modular Functions.

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§ 145. Limiting values for $s = 1$.

In this chapter a function-theoretic application of the limit formula is to be given. Here, therefore, ω does not denote a quadratic irrational number, but a variable with positive imaginary part. The modular functions belong to the functions with natural boundaries, that is: as one approaches the boundary of convergence, the singular points lie everywhere dense upon that boundary, here therefore upon the axis of the real ω, so that these functions cannot be continued analytically beyond this boundary. Cayley gave, in his last investigations, developments of the modular functions which are remarkable because they make evident the behaviour of the functions on approach to the boundary. The key to these developments is the limit formula § 141, (27)[1].

If one puts in this formula:

$$\begin{aligned}(1)\qquad &\operatorname*{Lim}_{s=1}\sum^{x,y}\frac{1}{(Ax^2+2Bxy+Cy^2)^s}-\frac{\pi}{(s-1)\sqrt{m}}\\ &=-\frac{2\pi\Gamma'(1)}{\sqrt{m}}+\frac{\pi}{\sqrt{m}}\log\frac{A}{4m}-\frac{2\pi}{\sqrt{m}}\log\eta(\omega_1),\ \eta(\omega_2),\\ &\omega_1=\frac{B+i\sqrt{m}}{A},\quad \omega_2=\frac{-B+i\sqrt{m}}{A},\end{aligned}$$

$$(2)\qquad \omega_1=\alpha+\beta i,\quad \omega_2=-\alpha+\beta i,\quad \beta>0$$

there results

$$A=1,\quad B=\alpha,\quad C=\alpha^2+\beta^2,\quad m=\beta^2,$$

and formula 1 gives:

$$\begin{aligned}(3)\qquad \sum^{x,y}\frac{1}{[(x-\alpha y)^2+\beta^2y^2]^s}&=\sum^{x,y}\frac{1}{(x+\omega_1y)^s(x-\omega_2y)^s}\\ &=\frac{\pi}{(s-1)\beta}-\frac{2\pi\Gamma'(1)}{\beta}-\frac{\pi}{\beta}\log 4\beta^2-\frac{2\pi}{\beta}\log\eta(\omega_1)\eta(\omega_2).\end{aligned}$$

The symbol Lim has here been omitted for the sake of brevity.

We put

$$(4)\qquad S=\sum^{x,y}\frac{1}{[(x-\alpha y)^2+\beta^2y^2]^s},$$

in which x, y run through all integer values with the exception of the combination $0, 0$. S is an unconditionally convergent series so long as $s>1$.

Now we divide the terms of S into three kinds, according as the x, y are even or odd, and put

$$\begin{aligned}(5)\qquad S_0&=\sum^{x,y}\frac{1}{[(x-\alpha y)^2+\beta^2y^2]^s},\quad x\equiv y\pmod 2,\\ S_1&=\sum^{x,y}\frac{1}{[(x-\alpha y)^2+\beta^2y^2]^s},\quad x\equiv 0\pmod 2,\\ S_2&=\sum^{x,y}\frac{1}{[(x-\alpha y)^2+\beta^2y^2]^s},\quad y\equiv 0\pmod 2.\end{aligned}$$

[1] Cayley's first paper is found in the Comptes Rendus of the Paris Academy of 1893, Bd. 161. Further matter in a correspondence with the author of this book, which is published in the 47th volume of the Mathematische Annalen.

In S_0, then, the numbers x, y are either both even or both odd; in S_1, x runs through the even numbers and y through all integers; in S_2, y is even and x arbitrary. If one considers the sum $S_0 + S_1 + S_2$, all the terms of S occur in it, and indeed three times over those terms in which x and y are both even. If one takes the factor 4^{-s} out of the sum of these terms, then S itself remains, and there results accordingly:

$$S_0 + S_1 + S_2 = \left(1 + \frac{2}{4^s}\right) S. \tag{6}$$

If one passes to the limit $s = 1$, one has to put

$$\frac{1}{4^s} = \frac{1}{4} - \frac{s-1}{2} \log 2 \ldots, \tag{7}$$

and obtains from (3) the formula valid for $s = 1$:

$$S_0 + S_1 + S_2 = \frac{3}{2} S - \frac{\pi}{\beta} \log 2. \tag{8}$$

If we replace α and β in S by 2α and 2β, this has the same effect as if we replaced y by $2y$; that is, S passes into S_2, and accordingly we obtain from (3):

$$\begin{aligned} 2\, S_2 = {} & \frac{\pi}{(s-1)\beta} - \frac{2\pi\, \Gamma'(1)}{\beta} - \frac{\pi}{\beta} \log 4\beta^2 \\ & - \frac{2\pi}{\beta} \log 2\, \eta(2\omega_1)\, \eta(2\omega_2). \end{aligned} \tag{9}$$

If one replaces α, β by $\frac{1}{2}\alpha$, $\frac{1}{2}\beta$ and multiplies by 4^{-s}, the effect is the same as if one had replaced x by $2x$, and one obtains S_1. Thus from (3), having regard to (7):

$$\begin{aligned} 2\, S_1 = {} & \frac{\pi}{(s-1)\beta} - \frac{2\pi\, \Gamma'(1)}{\beta} - \frac{\pi}{\beta} \log 4\beta^2 \\ & - \frac{2\pi}{\beta} \log \eta\left(\frac{\omega_1}{2}\right) \eta\left(\frac{\omega_2}{2}\right). \end{aligned} \tag{10}$$

If finally one replaces α, β by $\frac{1}{2}(\alpha + 1)$, $\frac{1}{2}\beta$ and then $2x - y$ by x, there results in the same way:

$$\begin{aligned} 2\, S_0 = {} & \frac{\pi}{(s-1)\beta} - \frac{2\pi\, \Gamma'(1)}{\beta} - \frac{\pi}{\beta} \log 4\beta^2 \\ & - \frac{2\pi}{\beta} \log \eta\left(\frac{1+\omega_1}{2}\right) \eta\left(\frac{-1+\omega_2}{2}\right), \end{aligned} \tag{11}$$

and accordingly, having regard to the formulae [§ 34, (9)]:

$$f(\omega) = e^{-\frac{\pi i}{24}} \frac{\eta\left(\frac{\omega+1}{2}\right)}{\eta(\omega)} = e^{\frac{\pi i}{24}} \frac{\eta\left(\frac{\omega-1}{2}\right)}{\eta(\omega)},$$

$$f_1(\omega) = \frac{\eta\left(\frac{\omega}{2}\right)}{\eta(\omega)}, \qquad f_2(\omega) = \sqrt{2}\, \frac{\eta(2\omega)}{\eta(\omega)},$$

by subtracting the formulae (9), (10), (11) from (3):

$$S - 2\, S_0 = \frac{2\pi}{\beta} \log f(\omega_1)\ f(\omega_2),$$

$$S - 2\, S_1 = \frac{2\pi}{\beta} \log f_1(\omega_1)\ f_1(\omega_2),$$

$$S - 2\, S_2 = \frac{2\pi}{\beta} \log f_2(\omega_1)\ f_2(\omega_2),$$

from which, by addition according to § 34, (11), relation (8) follows again.

In the difference $2S_0 - S$ there now occur exactly the same terms as in S; only those in which $x + y$ is odd receive the negative sign, and we can therefore also write:

$$(12)\qquad \begin{aligned} \frac{-2\pi}{\beta}\log f(\omega_1)\,f(\omega_2) &= \operatorname*{Lim}_{s=1}\sum \frac{(-1)^{x+y}}{[(x-\alpha y)^2+\beta^2y^2]^s},\\ \frac{-2\pi}{\beta}\log f_1(\omega_1)\,f_1(\omega_2) &= \operatorname*{Lim}_{s=1}\sum \frac{(-1)^{x}}{[(x-\alpha y)^2+\beta^2y^2]^s},\\ \frac{-2\pi}{\beta}\log f_2(\omega_1)\,f_2(\omega_2) &= \operatorname*{Lim}_{s=1}\sum \frac{(-1)^{y}}{[(x-\alpha y)^2+\beta^2y^2]^s}, \end{aligned}$$

in which x, y now assume all integral pairs of values with the exception of $0, 0$. These three formulae can be derived from one of them by § 34, (13), (14), for example by replacing, in the second,

$$(13)\qquad \begin{array}{lllll} \omega_1,\ \omega_2 & \text{by} & \omega_1+1, & \omega_2-1,\\ & \text{and by} & \dfrac{-1}{\omega_1}, & \dfrac{-1}{\omega_2}, \end{array}$$

that is

$$\begin{array}{lllll} \alpha,\ \beta & \text{by} & \alpha+1, & \beta,\\ & \text{and by} & \dfrac{-\alpha}{\alpha^2+\beta^2}, & \dfrac{\beta}{\alpha^2+\beta^2}. \end{array}$$

§ 146. A Theorem on the Convergence of Series.

If one wished to interchange the symbols Lim and Σ with one another in the expressions (12), § 145, that is, to put $s = 1$ under the summation sign without more ado, one would not obtain unconditionally convergent series, and it must therefore be investigated in what sense these formulae then still remain valid.

In order to carry out this investigation, we will first derive a general theorem from the theory of series, which is a sharpening of the theorem Bd. II, § 196, 3. Let

$$(1)\qquad \mu_1 \leqq \mu_2 \leqq \mu_3 \leqq \mu_4 \ldots$$

be a sequence of infinitely many positive numbers, and let $Z(t)$ denote the number of these numbers that are not greater than t. This $Z(t)$ is to have a finite value for every t, from which it then follows that the μ_n increase to infinity with n. But we will further assume that

$$(2)\qquad \frac{Z(t)}{t} = a + \frac{\theta}{\sqrt{t}}$$

be, where a is a constant (independent of t), and θ a function of t which remains enclosed within finite limits.

If we assume first that the μ_n are all different from one another, then $Z(\mu_n) = n$, and from (2) it follows that:

$$(3)\qquad \frac{n}{\mu_n} = a + \frac{\theta}{\sqrt{\mu_n}};$$

from this it follows that $n : \mu_n$ remains finite, and consequently, with an altered meaning of θ:

$$\frac{n}{\mu_n} = a + \frac{\theta}{\sqrt{n}},$$

or, by the binomial theorem, for any positive s:

$$(4)\qquad \frac{n^s}{\mu_n^s} = a^s\left(1 + \frac{\theta}{\sqrt{n}}\right),$$

in which all the quantities denoted by θ have finite values.

But these formulae still hold even when equal ones occur among the μ_n. For let, say (as in Bd. II, § 196, 3.)

$$\mu_{m+1}, \mu_{m+2}, \ldots, \mu_{m+l} = \mu_n$$

be equal to one another and

$$m < n \leqq m + l,$$

then

$$Z(\mu_n - 0) = m, \qquad Z(\mu_n) = m + l,$$

$$\frac{Z(\mu_n - 0)}{\mu_n} < \frac{n}{\mu_n} \leqq \frac{Z(\mu_n)}{\mu_n},$$

$$a + \frac{\theta'}{\sqrt{\mu_n}} < \frac{n}{\mu_n} < a + \frac{\theta}{\sqrt{\mu_n}},$$

from which (3) follows and (4) can be derived as above.

The theorem we wish to prove runs as follows:

Let $\varepsilon_1, \varepsilon_2, \varepsilon_3, \ldots$ be an infinite sequence of positive or negative, but finite quantities and

$$\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n = \gamma_n \sqrt{n}, \tag{5}$$

so constituted that γ_n remains, in absolute value, below a finite constant.

Then

$$\sigma = \frac{\varepsilon_1}{\mu_1^s} + \frac{\varepsilon_2}{\mu_2^s} + \frac{\varepsilon_3}{\mu_3^s} + \cdots \tag{6}$$

is convergent and a continuous function of s, so long as

$$s > \tfrac{1}{2}$$

holds.

Because of (4) this theorem need only be proved for $\mu_n = n$, since the series $\Sigma \varepsilon_n \theta / n^{s+\frac{1}{2}}$ converges unconditionally and therefore has this property by known elementary theorems.

Let us therefore set

$$\sigma = \frac{\varepsilon_1}{1^s} + \frac{\varepsilon_2}{2^s} + \frac{\varepsilon_3}{2^s} + \cdots \tag{7}$$

and, by (5), $\varepsilon_n = \gamma_n \sqrt{n} - \gamma_{n-1} \sqrt{n-1}$; then

$$\sigma = \frac{\varepsilon_1}{1^s} + \gamma_2 \sqrt{2} \left(\frac{1}{2} - \frac{1}{3^s} \right) + \gamma_3 \sqrt{3} \left(\frac{1}{3^s} - \frac{1}{4^s} \right) + \cdots \tag{8}$$

Since now, for an infinitely large n,

$$\sqrt{n} \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right) = \frac{s}{n^{s+\frac{1}{2}}}$$

holds, the series (8) is unconditionally convergent as long as $s > \frac{1}{2}$, and from this our assertion follows for this series also by the same elementary theorems.

§ 147. Development of f, f_1, f_2.

The series (12), § 145 are now in this case. Let us consider, for example, the second of them, from which, as we have seen, the others can be derived, and let us set for $\mu_1, \mu_2, \mu_3, \ldots$ the values, arranged in order of magnitude, of the function

$$(x-\alpha y)^2+\beta^2 y^2 = n. \tag{1}$$

We take xy as rectangular coordinates in the plane and overlay the plane with two lattices, setting in the one lattice the even integers for the x, in the other the odd ones.

The number of lattice points that lie in the interior of the ellipse (1) or on its periphery, and for which therefore $\mu_n \leqq n$, we denote for the two kinds by $Z^0(n), Z^1(n)$.

Let us set

$$x = \xi\sqrt{n}, \qquad y = \eta\sqrt{n},$$

then (1) goes over into

$$(\xi-\alpha\eta)^2+\beta^2\eta^2 = 1, \tag{2}$$

and the area of this ellipse is π/β, and by Bd. II, § 194, 1.:

$$\begin{aligned} \frac{Z^0(t)}{t} &= \frac{\pi}{2\,\beta}+\frac{\gamma^0}{\sqrt{t}}, \\ \frac{Z^1(t)}{t} &= \frac{\pi}{2\,\beta}+\frac{\gamma^1}{\sqrt{t}}, \end{aligned} \tag{3}$$

in which γ^0 and γ^1 remain finite for $t=\infty$. The factor $\frac{1}{2}$ in $\pi/2\,\beta$ comes from the fact that here the meshes of the lattice are rectangles of area 2. This, however, is in agreement with the formula § 146, (2).

From (3) there follows

$$Z^0(t) - Z^1(t) = \gamma\sqrt{t}, \tag{4}$$

in which γ likewise remains finite.

Now we can write the sum occurring in formula (12), § 145

$$\sigma = \sum \frac{(-1)^x}{[(x-\alpha y)^2+\beta^2 y^2]^s}$$

in the following way:

$$\sigma = \frac{\varepsilon_1}{\mu_1^s}+\frac{\varepsilon_2}{\mu_2^s}+\frac{\varepsilon_3}{\mu_3^s}+\cdots,$$

in which the $\varepsilon_1, \varepsilon_2, \varepsilon_3, \ldots$ have only the values ± 1 and

$$\begin{aligned} \varepsilon_1+\varepsilon_2+\varepsilon_3+\cdots+\varepsilon_n &= Z^0(n)-Z^1(n), \\ &= \gamma\sqrt{n}. \end{aligned} \tag{5}$$

With this the assumptions of our theorem § 146 are satisfied.

Accordingly σ is a continuous function of s for $s > \frac{1}{2}$, and hence in particular also for $s=1$, and from § 145, (12) we obtain:

$$\begin{aligned} -\frac{2\pi}{\beta}\log f(\omega_1)\,f(\omega_2) &= \sum \frac{(-1)^{x+y}}{(x-\alpha y)^2+\beta^2 y^2}, \\ -\frac{2\pi}{\beta}\log f_1(\omega_1)\,f_1(\omega_2) &= \sum \frac{(-1)^{x}}{(x-\alpha y)^2+\beta^2 y^2}, \\ -\frac{2\pi}{\beta}\log f_2(\omega_1)\,f_2(\omega_2) &= \sum \frac{(-1)^{y}}{(x-\alpha y)^2+\beta^2 y^2}. \end{aligned} \tag{6}$$

These formulae can also be represented in yet another way. We introduce the notation:

$$
\begin{aligned}
\sum \frac{1}{(x-\alpha y)^2+\beta^2 y^2} &= S_{00} : x \text{ even},\ y \text{ even},\\
&= S_{01} : x \text{ even},\ y \text{ odd},\\
&= S_{10} : x \text{ odd},\ y \text{ even},\\
&= S_{11} : x \text{ odd},\ y \text{ odd}.
\end{aligned} \tag{7}
$$

Then

$$
\begin{aligned}
\frac{2\pi}{\beta}\log f(\omega_1)\cdot f(\omega_2) &= -S_{00}+S_{10}+S_{01}-S_{11},\\
\frac{2\pi}{\beta}\log f_1(\omega_1)\, f_1(\omega_2) &= -S_{00}+S_{10}-S_{01}+S_{11},\\
\frac{2\pi}{\beta}\log f_2(\omega_1)\, f_2(\omega_2) &= -S_{00}-S_{10}+S_{01}+S_{11},
\end{aligned} \tag{8}
$$

from which by addition:

$$
\frac{2\pi}{\beta}\log 2 = -3\,S_{00}+S_{10}+S_{01}+S_{11},
$$

and if one eliminates S_{00} by means of this:

$$
\begin{aligned}
\frac{2\pi}{\beta}\log f(\omega_1)\, f(\omega_2) &= \frac{2\pi}{3\,\beta}\log 2+\tfrac{2}{3}S_{10}+\tfrac{2}{3}S_{01}-\tfrac{4}{3}S_{11},\\
\frac{2\pi}{\beta}\log f_1(\omega_1)\, f_1(\omega_2) &= \frac{2\pi}{3\,\beta}\log 2+\tfrac{2}{3}S_{10}-\tfrac{4}{3}S_{01}+\tfrac{2}{3}S_{11},\\
\frac{2\pi}{\beta}\log f_2(\omega_1)\, f_2(\omega_2) &= \frac{2\pi}{3\,\beta}\log 2-\tfrac{4}{3}S_{10}+\tfrac{2}{3}S_{01}+\tfrac{2}{3}S_{11}.
\end{aligned} \tag{9}
$$

These sums, however, are to be understood in such a way that in all of them at once

$$
(x-\alpha y)^2+\beta^2 y^2 < n
$$

shall hold, and that n then grows to infinity. Each individual sum S thereby becomes infinite, but their combinations, as they occur in these formulae, receive finite limiting values.

We wish further to denote by S'_{01}, S'_{10}, S'_{11} the sums which have the same expression as S_{01}, S_{10}, S_{11} according to (6), only with the difference that x and y are to have no common divisor.

If we set

$$
\begin{gathered}
S_{10}+S_{01}-2\,S_{11}=T_n,\\
(x-\alpha y)^2+\beta^2 y^2 < n,\\
\operatorname*{Lim}_{n=\infty} T_n = T
\end{gathered}
$$

and denote by $T_n^{(p)}$ the sum which arises from T_n when all terms are removed in which x and y are both divisible by the odd prime p, then there results:

$$
T_n^{(p)} = T_n - \frac{1}{p^2}\,T_{np^{-2}}
$$

and by passage to the limit $n=\infty$:

$$
T^{(p)} = T\left(1-\frac{1}{p^2}\right).
$$

If one proceeds in this way with all primes and sets

$$S'_{10} + S'_{01} - 2\,S'_{11} = T',$$

then it follows that

$$T' = T\,\Pi\left(1 - \frac{1}{p^2}\right),$$

in which the product Π extends over all odd primes p. Now, by development in ascending powers of p^{-2}, one obtains:

$$\begin{aligned}\frac{1}{\Pi(1-p^{-2})} &= 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots, \\ &= \frac{\pi^2}{8},\end{aligned}$$

and consequently

$$T = \frac{\pi^2}{8}\,T',$$

and by proceeding in the same way with the two other sums, there follows from (9):

$$\begin{aligned}\log f(\omega_1)\,f(\omega_2) &= \tfrac{1}{3}\log 2 + \frac{\beta\pi}{24}\,(S'_{10} + S'_{01} - 2\,S'_{11}), \\ \log f_1(\omega_1)\,f_1(\omega_2) &= \tfrac{1}{3}\log 2 + \frac{\beta\pi}{24}\,(S'_{10} - 2\,S'_{01} + S'_{11}), \\ \log f_2(\omega_1)\,f_2(\omega_2) &= \tfrac{1}{3}\log 2 + \frac{\beta\pi}{24}\,(-2\,S'_{10} + S'_{01} + S'_{11}).\end{aligned} \tag{10}$$

These formulae simplify essentially if one sets $\alpha = 0$, and hence

$$\omega_1 = \omega_2 = \omega = i\,\beta$$

that is, if one assumes a purely imaginary ω. Then

$$S'_{10} = \sum \frac{1}{x^2 - \omega^2 y^2} \qquad \begin{array}{l} x \text{ odd, } y \text{ even,} \\ xy \text{ relatively prime.}\end{array}$$

If one separates off the term $x = \pm 1$, $y = 0$ and takes the remaining ones together four at a time, then there results

$$S'_{10} = 2 + 4\sum \frac{1}{x^2 - \omega^2 y^2},$$

in which x, y run through positive values only.

One proceeds in the same way with the other sums. If one then sets

$$\begin{aligned}\sum \frac{\omega}{y^2\omega^2 - x^2} &= S_1 \qquad x, y \text{ odd,} \\ &= S_2 \qquad x \text{ odd, } y \text{ even,} \\ &= S_3 \qquad x \text{ even, } y \text{ odd,}\end{aligned} \tag{11}$$

x, y positive and relatively prime, then it follows:

$$\begin{aligned}\beta\,S'_{10} &= -2\,i\,\omega + 4\,i\,S_2, \\ \beta\,S'_{01} &= +\frac{2\,i}{\omega} + 4\,i\,S_3, \\ \beta\,S'_{11} &= 4\,i\,S_1,\end{aligned}$$

and consequently there results from (10), for a purely imaginary ω:

$$\begin{aligned}
\log f(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\left(-\omega + \frac{1}{\omega} - 4\,S_1 + 2\,S_2 + 2\,S_3\right),\\
\log f_1(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\left(-\omega - \frac{2}{\omega} + 2\,S_1 + 2\,S_2 - 4\,S_3\right),\\
\log f_2(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\left(2\,\omega + \frac{1}{\omega} + 2\,S_1 - 4\,S_2 + 2\,S_3\right).
\end{aligned} \tag{12}$$

§ 148. Elementary Derivation of the Developments.

One can also arrive at the foregoing developments by the following route. By a known formula of analysis:

$$\sum_{1,\infty}^{x} \frac{\omega}{\omega^2 y^2 - x^2} = -\frac{1}{2\,\omega y^2} + \frac{\pi i}{2\,y}\,\frac{q^{2y}+1}{q^{2y}-1}, \tag{1}$$

if $q = e^{\pi i \omega}$[1]. From this, by development in powers of q:

$$\sum_{1,\infty}^{x} \frac{\omega}{\omega^2 y^2 - x^2} = -\frac{1}{2\,\omega y^2} - \frac{\pi i}{y}\left(\frac{1}{2} + \sum_{1,\infty}^{n} q^{2ny}\right). \tag{2}$$

If one here replaces y by $\frac{1}{2}y$, it follows:

$$\sum_{1,\infty}^{x} \frac{2\,\omega}{\omega^2 y^2 - 4x^2} = -\frac{1}{\omega y^2} - \frac{\pi i}{y}\left(\frac{1}{2} + \sum_{1,\infty} q^{ny}\right), \tag{3}$$

and if one subtracts (2) from (3):

$$\sum \frac{(-1)^x\,\omega}{\omega^2 y^2 - x^2} = -\frac{1}{2\,\omega y^2} - \frac{\pi i}{y}\sum_{1,\infty}^{x} q^{(2n-1)y}. \tag{4}$$

If we sum these formulae once again with respect to y from 1 to ∞, and make use of the formulae which are partly known and partly follow from § 24, (11):

$$\sum^{y}\frac{1}{y^2} = \frac{\pi^2}{6}, \quad \sum^{y}\frac{(-1)^y}{y^2} = -\frac{\pi^2}{12}, \quad \sum\frac{(-1)^y}{y} = -\log 2,\ .$$

$$\sum^{n}\sum^{y}\frac{q^{(2n-1)y}}{y} = -\log \Pi(1-q^{2n-1}) = -\log f_1(\omega) - \frac{\pi i\omega}{24},$$

$$\sum^{n}\sum^{y}\frac{(-1)^y q^{2ny}}{y} = -\log \Pi(1+q^{2n}) = -\log f_2(\omega) + \frac{\pi i\omega}{12} - \tfrac{1}{2}\log 2,$$

$$\sum^{n}\sum^{y}\frac{(-1)^y q^{(2n-1)y}}{y} = -\log \Pi(1+q^{2n-1}) = -\log f(\omega) - \frac{\pi i\omega}{24},$$

[1] From the development:

$$\operatorname{cotg} z = \frac{1}{z} + \frac{2\,z}{z^2-\pi^2} + \frac{2\,z}{z^2-4\pi^2} + \frac{2\,z}{z^2-9\pi^2} + \cdots$$

then there results:

$$
\begin{aligned}
\sum_{1,\infty}^{y}\sum_{1,\infty}^{x}\frac{(-1)^{x+y}\omega}{\omega^2y^2-x^2} &= \frac{\pi^2}{24\omega}-\frac{\pi^2\omega}{24}+\pi i\log f(\omega),\\
\sum^{y}\sum^{x}\frac{(-1)^{x}\omega}{\omega^2y^2-x^2} &= -\frac{\pi^2}{12\omega}-\frac{\pi^2\omega}{24}+\pi i\log f_1(\omega),\\
\sum^{y}\sum^{x}\frac{(-1)^{y}\omega}{\omega^2y^2-x^2} &= \frac{\pi^2}{24\omega}+\frac{\pi^2\omega}{12}+\pi i\log f_2(\omega)\,.
\end{aligned}
\tag{5}
$$

The double sums on the left side of these formulae are to be understood in such a way that the summation with respect to x is to be carried out first, and then the summation with respect to y. But we may also sum in such a way that we take

$$
0<x<m,\quad 0<y<n,\quad n=\infty\,,\quad \frac{m}{n}=\infty \tag{6}
$$

and then let m and n grow to infinity so that $m:n$ becomes infinite. That both give the same can be shown in various ways, for example by comparing the sums with integrals.

If, then, we set, as in § 147, (10), binding x, y to the condition (6):

$$
\begin{aligned}
\sum^{x,y}\frac{\omega}{\omega^2y^2-x^2} &= S_0 \qquad x,\ y \text{ even},\\
&= S_1 \qquad x,\ y \text{ odd},\\
&= S_2 \quad x \text{ odd},\ y \text{ even},\\
&= S_3 \quad x \text{ even},\ y \text{ odd},
\end{aligned}
$$

it follows from (5):

$$
\begin{aligned}
S_0+S_1-S_2-S_3 &= \frac{\pi^2}{24\omega}-\frac{\pi^2\omega}{24}+\pi i\log f(\omega),\\
S_0-S_1-S_2+S_3 &= -\frac{\pi^2}{12\omega}-\frac{\pi^2\omega}{24}+\pi i\log f_1(\omega),\\
S_0-S_1+S_2-S_3 &= \frac{\pi^2}{24\omega}+\frac{\pi^2\omega}{12}+\pi i\log f_2(\omega).
\end{aligned}
$$

From this by addition:

$$
3S_0-S_1-S_2-S_3=\pi i\log\sqrt{2},
$$

and if S_0 is eliminated:

$$
\begin{aligned}
\tfrac{4}{3}S_1-\tfrac{2}{3}S_2-\tfrac{2}{3}S_3 &= \frac{\pi^2}{24\omega}-\frac{\pi^2\omega}{24}-\frac{\pi i}{3}\log\sqrt{2}+\pi i\log f(\omega),\\
-\tfrac{2}{3}S_1-\tfrac{2}{3}S_2+\tfrac{4}{3}S_3 &= -\frac{\pi^2}{12\omega}-\frac{\pi^2\omega}{24}-\frac{\pi i}{3}\log\sqrt{2}+\pi i\log f_1(\omega),\\
-\tfrac{2}{3}S_1+\tfrac{4}{3}S_2-\tfrac{2}{3}S_3 &= \frac{\pi^2}{24\omega}+\frac{\pi^2\omega}{12}-\frac{\pi i}{3}\log\sqrt{2}+\pi i\log f_2(\omega).
\end{aligned}
$$

One can now pass over, just as before, from the sums S to the S', in which x, y are relatively prime, and finds:

$$
\begin{aligned}
\log f(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\Big(-\omega + \frac{1}{\omega} - 4S_1' + 2S_2' + 2S_3'\Big),\\
\log f_1(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\Big(-\omega - \frac{2}{\omega} + 2S_1' + 2S_2' - 4S_3'\Big),\\
\log f_2(\omega) &= \tfrac{1}{3}\log\sqrt{2} + \frac{\pi i}{24}\Big(-\omega - \frac{1}{\omega} + 2S_1' - 4S_2' + 2S_3'\Big).
\end{aligned}
\tag{7}
$$

These formulae agree in form with § 147, (11), which is remarkable in so far as the manner of the passage to the limit is a different one in the two cases.

§ 149. Developments for the function $\log\eta(\omega)$.

Let us consider the sum

$$
U = \sum^{x,y} \frac{1}{(x+\omega y)^{2s}}, \tag{1}
$$

extended over all x, y with the exception of the combination 0, 0.

By separating off the terms with $y = 0$, one may write for it:

$$
U = 2\sum_{1,\infty}^{x} \frac{1}{x^{2s}} + 2\sum_{1,\infty}^{y}\sum_{-\infty,\infty}^{x} \frac{1}{(x+\omega y)^{2s}}; \tag{2}
$$

next we have

$$
\begin{aligned}
\frac{\Gamma(2s)}{[-2\pi i(x+\omega y)]^{2s}} &= \int_0^\infty e^{2\pi i(x+y\omega)\xi}\xi^{2s-1}\,d\xi,\\
&= \int_0^1 e^{2\pi i x\xi}\sum_{0,\infty}^{n} e^{2\pi i y\omega(\xi+n)}(\xi+n)^{2s-1}\,d\xi.
\end{aligned}
$$

If one now sums over x and applies the Fourier series:

$$
\sum^{x}\int_0^1 e^{2\pi i x\xi} f(\xi)\,d\xi = \tfrac{1}{2}[f(0)+f(1)],
$$

then it follows:

$$
\frac{\Gamma(2s)}{(-2\pi i)^{2s}}\sum_{-\infty,\infty}^{x}\frac{1}{(x+\omega y)^{2s}} = \sum_{1,\infty}^{n} e^{2\pi i y\omega n}\,n^{2s-1},
$$

and by summation over y:

$$
\begin{aligned}
\frac{\Gamma(2s)}{(-2\pi i)^{2s}}\sum_{1,\infty}^{y}\sum_{-\infty,\infty}^{x}\frac{1}{(x+\omega y)^{2s}} &= \sum_{1,\infty}^{n}\frac{e^{2\pi i\omega n}}{1-e^{2\pi i\omega n}}\,n^{2s-1}\\
&= -\frac{1}{2\pi\imath}\sum_{1,\infty}^{n}\frac{d\log(1-e^{2\pi i\omega n})}{d\omega}\,n^{2s-2}.
\end{aligned}
$$

Accordingly we obtain, by passing to the limit and having regard to the definition of the function $\eta(\omega)$ [§ 24, (8)]:

$$
\operatorname*{Lim}_{s=1}\sum_{1,\infty}^{y}\sum^{x}\frac{1}{(x+\omega y)^{2s}} = -2\pi i\,\frac{d\log\eta(\omega)}{d\omega} - \frac{\pi^2}{6}.
$$

Furthermore we have

$$\operatorname{Lim} \sum \frac{1}{x^{2s}} = \frac{\pi^2}{6},$$

and accordingly from (1) and (2) it follows:

$$\operatorname*{Lim}_{s=1} \sum^{x,y} \frac{1}{(x+\omega y)^{2s}} = -4\pi i \, \frac{d \log \eta(\omega)}{d\omega} \tag{3}$$

[to be summed as indicated for (1)]. If one wished here to pass to the limit under the summation sign, one would obtain:

$$-4\pi i \, \frac{d \log \eta(\omega)}{d\omega} = \sum^{x,y} \frac{1}{(x+\omega y)^2}, \tag{4}$$

but then one would have a sum that is only conditionally convergent, and a precise manner of passing to the limit would have to be laid down. One would have to assume in the xy-plane a closed curve within which the points x, y lie, and then let this curve move out to infinity. The value of the sum will depend on the nature of this curve. Perhaps the ellipse $|x + \omega y| = n$ presents itself for this. In itself formula (4) has no meaning.

FOURTH BOOK.

CLASS FIELDS.

Twenty-Third Chapter.

The Division Field.

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§ 150. **The Homogeneous Weierstrass Functions.**

In § 114 we set out from the question under what suppositions a doubly periodic function $\varphi(u)$ with the periods ω_1, ω_2 admits a multiplication $\varphi(\mu u)$, that is, under what circumstances $\mu\omega_1, \mu\omega_2$ can be expressed linearly and with integral coefficients by ω_1, ω_2; and we had arrived at the result that, for variable ω_1, ω_2, this is possible only when μ is an integral rational number, and that only when

$$\omega = \frac{\omega_2}{\omega_1} \tag{1}$$

is an imaginary quadratic irrational number can μ also be a complex number of the same field as ω. What now matters is to explore somewhat more closely the formulae of complex multiplication that follow from this assumption.

We will first investigate, as the formally simpler case, the complex multiplication of the Weierstrass elliptic function $\wp(u)$, but we must then also take into consideration the complex multiplication of the Jacobi elliptic functions.

In § 37, (8), § 46, (13) we derived the formulae for the homogeneity of the functions $\sigma(u)$, $\wp(u)$ and of the invariants $g_2, g_3, 16\,G = g_2^3 - 27\,g_3^2$, which we here collect once more:

$$\begin{aligned}
\sigma(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \lambda\ \sigma(u, \omega_1, \omega_2),\\
\wp(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \lambda^{-2}\,\wp(u, \omega_1, \omega_2),\\
g_2(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-4}\,g_2(\omega_1, \omega_2),\\
g_3(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-6}\,g_3(\omega_1, \omega_2),\\
G(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-12}\,G(\omega_1, \omega_2),
\end{aligned} \tag{2}$$

and the functions § 46, (12), (15), (16):

$$\begin{aligned}
\frac{4\,.\,27\,g_2^3}{G} &= j(\omega),\\
\frac{4\,.\,27\,.\,27\,g_3^2}{G} &= j(\omega) - 27\,.\,64,\\
\omega_1^{12}\,G &= 2^8\,\pi^{12}\,\eta(\omega)^{24}
\end{aligned} \tag{3}$$

depend on the ratio (1).

In § 54, (4), (5) the single-valued functions of ω:

$$\begin{aligned}
\gamma_2(\omega) &= \sqrt[3]{j(\omega)} = \frac{f(\omega)^{24} - 16}{f(\omega)^8},\\
\gamma_3(\omega) &= \sqrt{j(\omega) - 27\,.\,64} = \frac{[f(\omega)^{24} + 8]\,[f_1(\omega)^8 - f_2(\omega)^8]}{f(\omega)^8}
\end{aligned} \tag{4}$$

are further defined.

From the last equation (3) one recognises [for instance from the development of $\eta(\omega)$, § 24, (8)] that G cannot vanish so long as ω has a positive imaginary part, and consequently g_2, g_3 also cannot both vanish at once.

From the series developments of § 56 there results:

$$\sigma(u) = \sum^{m,n} a_{m,n} \left(\tfrac{1}{2} g_2\right)^m (2g_3)^n \frac{u^{2\nu+1}}{(2\nu+1)!}, \tag{5}$$
$$\nu = 2m + 3n,$$

in which the rational coefficients $a_{m,n}$ can have only powers of 3 in the denominator (according to Schwarz-Weierstrass they are integers, but that is of no consequence here).

Likewise

$$\wp(u) = \sum b_{m,n} \left(\tfrac{1}{2} g_2\right)^m (2g_3)^n \frac{u^{2\nu-2}}{(2\nu+1)!} \tag{6}$$

with rational coefficients $b_{m,n}$. In order to obtain functions of two variables, we make the following substitution:

If neither of the two invariants g_2, g_3 vanishes, we put:

$$u = \sqrt{\frac{g_2 g_3}{G}}\, w, \tag{7}$$

hence

$$g_2^m g_3^n u^{2\nu+1} = \left(\frac{g_2^3}{G}\right)^{m+n} \left(\frac{g_2^2}{G}\right)^{m+2n} \sqrt{\frac{g_2 g_3}{G}}\, w^{2\nu+1},$$

and thereby there results from (5):

$$\sqrt{\frac{G}{g_2 g_3}}\, \sigma(u) = \sum A_{m,n}\, j^{m+n} (j - 27\,.\,64)^{m+2n} \frac{w^{2\nu+1}}{(2\nu+1)!} \tag{8}$$

in which the $A_{m,n}$ are rational numbers that can contain in the denominator only powers of 2 and of 3.

In the same way there results from the series development (6) for the $\wp$-function:

$$\frac{g_2 g_3}{G}\, \wp(u) = \sum^{m,n} B_{m,n}\, j^{m+n} (j - 27\,.\,64)^{m+2n} \frac{w^{2\nu-2}}{(2\nu+1)!}, \tag{9}$$

in which the $B_{m,n}$ are rational numerical coefficients. That these too have only powers of 2 and 3 in the denominator is made probable by the first cases, but does not come into consideration for us now.

If g_2 or g_3 is equal to zero, the substitution (7) is not usable. If first

$$g_3 = 0,$$

then in the series for $\sigma(u)$ and $\wp(u)$ those terms drop out in which n is positive. We then have

$$\sigma(u) = \sum a_{m,0}\, g_2^m \frac{u^{4m+1}}{(4m+1)!},$$
$$\wp(u) = \sum b_{m,0}\, g_2^m \frac{u^{4m-2}}{(2m+1)!}.$$

We then set

$$u = g_2^{-\frac{1}{4}}\, w \tag{10}$$

and obtain

$$\begin{aligned} g_2^{\frac{1}{4}}\, \sigma(u) &= \sum^{m} A_m \frac{w^{4m+1}}{(4m+1)!}, \\ g_2^{-\frac{1}{2}}\, \wp(u) &= \sum^{m} B_m \frac{w^{4m-2}}{(4m+1)!}, \end{aligned} \tag{11}$$

where A_m, B_m are again rational numbers.

If finally

$$g_2 = 0,$$

then we set

$$u = g_3^{-\frac{1}{6}}\, w, \tag{12}$$

and obtain

$$\begin{aligned} g_3^{\frac{1}{6}}\, \sigma(u) &= \sum^{n} A_n \frac{w^{6n+1}}{(6n+1)!}, \\ g_3^{-\frac{1}{3}}\, \wp(u) &= \sum^{n} B_n \frac{w^{6n-2}}{(6n+1)!}. \end{aligned} \tag{13}$$

In the developments (8), (9), (11), (13) we have

$$A_{00} = 1, \quad B_{00} = 1, \quad A_0 = 1, \quad B_0 = 1. \tag{14}$$

§ 151. **The Complex Multiplication of the Function** $\wp(u)$.

We now assume that the periods ω_1, ω_2 of the function $\wp(u)$ satisfy a quadratic equation

$$A\omega_2^2 + B\omega_2\omega_1 + C\omega_1^2 = 0, \tag{1}$$

in which A, B, C are integral rational numbers without a common divisor. Let A be positive, and let the discriminant of the quadratic form (A, B, C)

$$B^2 - 4AC = \Delta \tag{2}$$

be a negative fundamental discriminant, i. e. let Δ contain no square factor after whose separation a discriminant remains (§ 84).

If we set

$$\omega = \frac{\omega_2}{\omega_1}, \tag{3}$$

then from (1) there follows:

$$2A\omega = -B + \sqrt{\Delta}, \tag{4}$$

and if we assume $\sqrt{\Delta}$ to be positive imaginary, then ω receives a positive imaginary part and can serve as the modulus of a ϑ-function. Δ is the fundamental number of an imaginary quadratic field, which we denote by Ω[1)].

The class field $\mathfrak{R}(\Delta)$ is determined by the singular invariant $j(\omega)$, and its relative degree with respect to Ω is equal to the class number of the discriminant Δ. The integers of the quadratic field $\Omega = \mathfrak{R}(\sqrt{\Delta})$ are of the form:

$$\mu = \frac{x + y\sqrt{\Delta}}{2}, \tag{5}$$

in which x and y are integral rational numbers that satisfy the condition

$$x \equiv By \pmod 2.$$

For the number μ defined by (5) we determine four integral rational numbers a, b, c, ∂:

$$\begin{aligned} a &= \frac{x + By}{2}, \quad & b &= Ay, \\ c &= -Cy, & \partial &= \frac{x - By}{2}, \end{aligned} \tag{6}$$

from which:

$$a\partial - bc = \frac{x^2 - \Delta y^2}{4} = \mu\mu' = m, \tag{7}$$

if

$$\mu' = \frac{x - y\sqrt{\Delta}}{2} \tag{8}$$

is the number of the field Ω conjugate to μ.

If we take y different from zero, then from (1), by multiplication by y, there follows:

$$b\omega_2^2 + (a - \partial)\omega_2\omega_1 - c\omega_1^2 = 0,$$

and for this one can also write:

$$(a\omega_1 + b\omega_2)\omega_2 = (c\omega_1 + \partial\omega_2)\omega_1$$

or

$$\omega = \frac{c + \partial\omega}{a + b\omega}. \tag{9}$$

By (4) and (5) there follows from (6):

$$\begin{aligned} a + b\omega &= \frac{x + (2A\omega + B)y}{2} = \mu, \\ c + \partial\omega &= \mu\omega, \end{aligned} \tag{10}$$

[1)] The assumption that Δ is a fundamental discriminant is made here for the sake of simplification. In my memoir "Über Zahlgruppen in algebraischen Körpern" (Mathematische Annalen, Bd. L) this assumption is not made. One then obtains more general fields, which belong to the orders, as in § 124.

or in homogeneous form:

$$\begin{aligned} \mu\omega_1 &= a\omega_1 + b\omega_2, \\ \mu\omega_2 &= c\omega_1 + \partial\omega_2, \end{aligned} \tag{11}$$

and by solving these linear equations:

$$\begin{aligned} \mu'\omega_1 &= \quad \partial\omega_1 - b\omega_2, \\ \mu'\omega_2 &= -c\omega_1 + a\omega_2. \end{aligned} \tag{12}$$

From this it follows that $\wp(\mu u)$ is a doubly-periodic function with the periods ω_1, ω_2, and since it is moreover an even function of u, it can be represented rationally by $\wp(u)$ (§ 21). If, then, we denote by R and P integral rational functions of $\wp(u)$ without common divisor, then

$$\wp(\mu u) = \frac{R}{P}. \tag{13}$$

Since the quotient $\wp(\mu u) : \wp(u)$ remains finite for $u = 0$, i. e. for $\wp(u) = \infty$, the degree of R is higher by one unit than the degree of the denominator.

§ 152. The Poles of the Function $\wp(\mu u)$.

In order to establish the degree of the functions R and P, we must count the zeros of P, that is, the points of infinity of $\wp(\mu u)$ which are not at the same time points of infinity of $\wp(u)$. But $\wp(\mu u) : \wp(u)$ becomes infinite then and only then, when

$$u = \frac{h_1\omega_1 + h_2\omega_2}{\mu} \tag{1}$$

holds, where h_1, h_2 are integers and where $h_1\omega_1 + h_2\omega_2$, but not $(h_1\omega_1 + h_2\omega_2) : \mu$, is a period. If we therefore set

$$\wp(\mu u) = \frac{R_\mu[\wp(u)]}{P_\mu[\wp(u)]}, \tag{2}$$

and denote by g the roots of $P_\mu(x)$, then

$$g = \wp\left(\frac{h_1\omega_1 + h_2\omega_2}{\mu}\right). \tag{3}$$

The number g does not change when h_1 and h_2 are altered by multiples of $m = \mu\mu'$, because by this the argument of the $\wp$-functions is, according to (11), § 151, altered by a period.

Since the functions $\wp(u)$ and $\wp(\mu u)$ are not altered when we subject the period ω_1, ω_2 to a linear transformation with determinant 1, we do not restrict the generality by assuming that A is relatively prime to m; and on account of the periodicity of $\wp(u)$ we may therefore also set:

$$g = \wp\left(\frac{h_1\omega_1 + h_2 A\omega_2}{\mu}\right).$$

On account of the homogeneity of $\wp(u)$ we may now set $\omega_1 = 1$, $\omega_2 = \omega$, and obtain, according to § 151, (4):

$$A\omega = \frac{-B + \sqrt{\Delta}}{2},$$

and accordingly we have

$$h_1\omega_1 + h_2 A\omega_2 = h_1 + h_2\left(\frac{-B + \sqrt{\Delta}}{2}\right). \tag{4}$$

If we therefore set

$$h_1 + h_2\frac{-B+\sqrt{\Delta}}{2} = \nu$$

then from (3) there results:

$$g = \wp\left(\frac{\nu}{\mu}\right), \tag{5}$$

where ν, like μ, is an integer of the field Ω that is not divisible by μ.

If this condition is fulfilled, then g is a root of $P_\mu(x)$. But since $\wp(\mu u)$ becomes infinite in the second order in the u-plane for $u \equiv 0$ (modd. ω_1, ω_2), $P_\mu(x)$ is divisible by $(x-g)^2$, unless

$$\wp(u) - g$$

itself vanishes in the second order for $u = \nu : \mu$; on account of § 46, (18) this occurs when

$$\wp'\left(\frac{\nu}{\mu}\right) = 0$$

holds, that is, when $\nu : \mu$ is a half period, and thus

$$g = e_1, e_2, e_3 \tag{6}$$

results.

1. By (4), if $\omega_1 = 1$, $\omega_2 = \omega$ is set, every integer of the field Ω is a period of $\wp(u)$. The converse does not hold; but since $A\omega$ is an integer in Ω, every period must be converted into an integer by multiplication by A.

From this it follows that two values

$$g = \wp\left(\frac{\nu}{\mu}\right), \qquad g' = \wp\left(\frac{\nu'}{\mu}\right)$$

are equal to one another then and only then, when

$$\nu \equiv \pm\nu' \pmod{\mu}$$

holds. For if $g = g'$, then either

$$\frac{\nu+\nu'}{\mu} \quad \text{or} \quad \frac{\nu-\nu'}{\mu}$$

must be converted into an integer by multiplication by A. But A is relatively prime to μ, and consequently one of these two numbers must itself be an integer. The number of incongruent values of ν, however, is equal to $N(\mu) = m$ (Bd. II, § 165).

If we therefore let ν run through a complete system of residues with respect to the modulus μ, excluding zero, we obtain every value of g twice, except when

$$2\nu \equiv 0 \pmod{\mu}, \tag{7}$$

and in this case $\wp\left(\frac{\nu}{\mu}\right)$ is one of the values e. From this it follows:

$$P_\mu(x) = \Pi\left[x - \wp\left(\frac{\nu}{\mu}\right)\right], \tag{8}$$

and $P_\mu(x)$ is of degree $m-1$.

In the product $P_\mu(x)$ every factor $x-g$ occurs twice, except when g is equal to one of the e, that is, when the congruence (7) can be satisfied for a ν different from zero.

1) If μ is relatively prime to 2, and hence if m is odd, then the congruence (7) has only the root $\nu=0$. In that case $P_\mu(x)$ is a square, and if $S(x)$ is a function of degree $\frac{1}{2}(m-1)$,

$$P_\mu(x)=S^2,\quad m\equiv 1 \pmod 2. \tag{9}$$

If μ has a divisor in common with 2, three cases are to be distinguished.

2) If μ is divisible by 2, then ν, if it is to satisfy the congruence (7), must be a multiple of $\frac{1}{2}\mu$; and since there are in Ω three numbers different from zero that are incongruent with respect to the modulus 2, (7) has three roots. Accordingly the three values e_1, e_2, e_3 occur among the g, and we have:

$$P_\mu(x)=\wp'(u)^2S^2,\quad m\equiv 0 \pmod 2, \tag{10}$$

where S is an integral function of degree $\frac{1}{2}m-2$.

3) If 2 splits in the field Ω into two (equal or different) prime ideals, that is, if $\Delta\equiv 0 \pmod 4$ or $\equiv 1 \pmod 8$, then μ can be divisible by one of these prime factors without being divisible by 2. In that case the congruence (7) has only one root different from zero, and among the numbers g only one of the three numbers (6), say e, occurs. In this case

$$P_u(x)=[\wp(u)-e]\,S^2, \tag{11}$$

where S is a function of degree $\frac{1}{2}m-1$.

With regard to the two special cases $g_3=0$ and $g_2=0$ the following is still to be remarked.

4) If $g_3=0$, then $\Delta=-4$, $j(\omega)=0$, and from the differential equation § 46, (18) there follows:

$$\wp'(u)^2=4\wp(u)^3-g_2\wp(u),$$

by which, having regard to the initial term of the development, the $\wp$-function is uniquely determined.

If one then replaces u by iu, it follows that:

$$-[\wp'(iu)]^2=4\wp^3(iu)-g_2\wp(iu),$$

and from this:

$$\wp(iu)=-\wp(u). \tag{12}$$

The quantities g are therefore opposite to one another in pairs, one of the three quantities e_1, e_2, e_3 is equal to zero, and the other two are likewise opposite and equal. From this it follows that in formulae (9) to (11) the function S can in this case contain only the even powers of $\wp(u)$.

5) If $g_2=0$, then $\Delta=-3$, $j(\omega)=64\,.\,27$, and, when an imaginary third root of unity is denoted by ϱ, we have:

$$\wp(u)=\varrho\,\wp(\varrho^2u)=\varrho^2\,\wp(\varrho u), \tag{13}$$

and the roots g of S arrange themselves in threes in the manner:

$$g, \quad \varrho g, \quad \varrho^2 g. \tag{14}$$

Among these three values two are equal to one another only if all three vanish. From (13), however, it follows that this occurs when u becomes congruent to one of the values $\pm 1 : \sqrt{-3}$. For when $\Delta = -3$, $1, \varrho, \varrho^2$ are periods of $\wp(u)$, and since now

$$\frac{\varrho - \varrho^2}{\sqrt{-3}} = 1$$

holds, it follows by (13) that:

$$\wp\left(\frac{\varrho}{\sqrt{-3}}\right) = \wp\left(\frac{\varrho^2}{\sqrt{-3}}\right) = 0,$$

and the value zero therefore occurs among the g then and only then, when m is divisible by 3. From this it follows:

If $\Delta = -3$ and $m \equiv 0 \pmod 3$, then $S : \wp(u)$ is a rational function of $\wp(u)^3$.

If $\Delta = -3$ and $m \equiv 1 \pmod 3$, then S itself is a rational function of $\wp(u)^3$.

[Since m must have the form $\frac{1}{4}(x^2 + 3y^2)$, m cannot here be $\equiv -1 \pmod 3$.]

§ 153. **The Function** $\tau(u)$.

Following § 150 (6), (8), (9), (10), (12), (13) we now introduce a function $\tau = \tau(u)$ which depends on only two arguments w, ω, by setting:

$$\begin{aligned} &\text{a)} \quad \tau(u) = \frac{g_2 g_3}{G}\wp(u), \quad \text{in general,} \\ &\text{b)} \qquad = \frac{\wp(u)^2}{g_2}, \quad \text{if } g_3 = 0, \\ &\text{c)} \qquad = \frac{\wp(u)^3}{g_3}, \quad \text{if } g_2 = 0. \end{aligned} \tag{1}$$

And if then μ is, as above, an integer of the field Ω, we can set in all three cases:

$$\tau(\mu u) = \frac{\Psi(\tau)}{\Phi(\tau)}, \tag{2}$$

in which Φ, Ψ are integral functions of τ which, apart from constant factors, agree in case (1) a) with the R, P of the preceding section, and in the cases (1) b) and (1) c) arise from these by raising to the square or to the cube [§ 152, 4), 5)].

We arrange the fraction (2) so that the highest power of τ in the denominator Φ has the coefficient 1. The remaining coefficients of Φ are then rationally composed of quantities of the form

$$\tau\left(\frac{\nu}{\mu}\right) = \tau\left(\frac{\nu\mu'}{m}\right),$$

and since $\nu\mu'$ is a period of τ, these quantities belong to the roots of the division equation of the periods, and are therefore, since the modulus of the corresponding elliptic functions is an algebraic number, themselves algebraic numbers (§ 58, § 61).

From this it follows:

2. The coefficients in $\Phi(\tau)$ are algebraic numbers.

We extend the fraction $\Psi : \Phi$ by multiplication of numerator and denominator by the product of the conjugate values $\Phi', \Phi'', \Phi''', \ldots$ and obtain:

$$\tau(\mu u) = \frac{Z}{N}, \tag{3}$$

in which now Z and N are integral functions of τ which may contain a common divisor, while N has rational numerical coefficients.

By § 150, (8), (10), (13) the development of $\tau(u)$ in ascending powers of u begins with a negative power of u. In the equation

$$\tau(\mu u)\, N = Z \tag{4}$$

we arrange the function Z in descending powers of τ, develop both sides in ascending powers of u, and compare the coefficients of equally high powers of u. We then obtain, for the determination of the coefficients of Z, a sequence of linear equations, each following one of which contains only one new one of these coefficients, and from this these coefficients can be calculated successively and rationally. If one now attends to the form of the developments of § 150, it follows that the coefficients of Z are rational functions of $j(\omega)$ and $\sqrt{\Delta}$.

If one frees N and Z from common factors by the Euclidean algorithm, one comes back to the functions Ψ, Φ, and it follows, if by the class field we understand the totality of the rational functions of $\sqrt{\Delta}$ and $j(\omega)$:

3. The coefficients of the functions $\Phi(\tau)$, $\Psi(\tau)$ in (2) belong to the class field.

§ 154. **The Division Field.**

If we free the function $\Phi(x)$ (by rational computation) from all factors occurring in it more than once, there remains an integral function $T_\mu(x)$ of x whose coefficients belong to the class field, and the roots of $T_\mu(x)$ are all those different from one another among the quantities

$$\tau\left(\frac{\nu}{\mu}\right). \tag{1}$$

Two of these quantities

$$\tau\left(\frac{\nu}{\mu}\right), \quad \tau\left(\frac{\nu'}{\mu}\right)$$

are equal to one another,

$$\begin{aligned} \text{in case (1) a), when} \quad & \frac{\nu}{\mu} \equiv \pm\frac{\nu'}{\mu}, \\ \text{(1) b), when} \quad & \frac{\nu}{\mu} \equiv \pm\frac{\nu'}{\mu} \equiv \pm i\frac{\nu'}{\mu}, \\ \text{(1) c), when} \quad & \frac{\nu}{\mu} \equiv \pm\frac{\nu'}{\mu}, \quad \pm\varrho\frac{\nu'}{\mu}, \quad \pm\varrho^2\frac{\nu'}{\mu}, \end{aligned}$$

in which the congruence is referred to the periods, and thus (§ 152, 1.)

$$\begin{array}{rll} \text{in case a)}, & \text{when } \nu \equiv \pm\nu', & \\ \text{b)}, & \text{when } \nu \equiv \pm\nu',\ \pm i\nu' & \pmod{\mu}, \\ \text{c)}, & \text{when } \nu \equiv \pm\nu',\ \pm\varrho\nu',\ \pm\varrho^2\nu'. & \end{array} \tag{2}$$

Now let

$$\mu = \mathfrak{a}\,\mathfrak{m}, \quad \mu_1 = \mathfrak{a}_1\,\mathfrak{m} \tag{3}$$

be two integers with the greatest common ideal divisor $\mathfrak{m}$ in the field Ω. Then T_μ and T_{μ_1} have a common linear divisor if

$$\tau\left(\frac{\nu}{\mu}\right) = \tau\left(\frac{\nu_1}{\mu_1}\right), \tag{4}$$

and by (2) this takes place then and only then, when

$$\mu\nu_1 \equiv \varepsilon\mu_1\nu \pmod{\mu\mu_1}, \tag{5}$$

in which ε is a unit of the field Ω, and thus $\varepsilon = \pm 1$ in general, $\varepsilon = \pm 1, \pm i$ when $\Delta = -4$, and $\varepsilon = \pm 1, \pm\varrho, \pm\varrho^2$ when $\Delta = -3$. But from (5) it follows that ν must be divisible by $\mathfrak{a}$, and ν_1 by $\mathfrak{a}_1$.

If we think of ν as given and ν_1 as sought, then the congruence (5) is possible only if $\mu_1\nu$ is divisible by μ; and since $\mathfrak{a}$ and $\mathfrak{a}_1$ are relatively prime, ν must be divisible by $\mathfrak{a}$. If α is an integer in Ω divisible by $\mathfrak{a}$, and so constituted that $\alpha : \mathfrak{a}$ is relatively prime to $\mathfrak{m}$, then we may accordingly put

$$\nu \equiv \alpha\xi \pmod{\mu} \tag{6}$$

(Bd. II, § 166, 7.), and we obtain all the values $\tau\left(\frac{\nu}{\mu}\right)$ that are different from one another, and each of them two, four, or six times, if we let ξ run through a complete system of residues with respect to the modulus $\mathfrak{m}$ (with exclusion of zero). One then obtains all the common roots of T_μ and T_{μ_1} in the form

$$\tau_\xi = \tau\left(\frac{\alpha\xi}{\mu}\right). \tag{7}$$

The greatest common divisor $D_\mathfrak{m}$ of T_μ and T_{μ_1} therefore has all the quantities (7) as roots. From $D_\mathfrak{m}$ a divisor $T_\mathfrak{m}$ can still be separated off by rational means, one that has as roots only those among the quantities (7) in which ξ is relatively prime to $\mathfrak{m}$, if one frees $D_\mathfrak{m}$ of all the factors that it has in common with a $D_{\mathfrak{m}'}$, where $\mathfrak{m}'$ is a divisor of $\mathfrak{m}$. Since in this

consideration in (3) $\mathfrak{m}$ can denote any arbitrary ideal in Ω, we arrive at the following principal theorem:

> 4. If $\mathfrak{m}$ is an arbitrary ideal of the field Ω, then there exists a function $T_\mathfrak{m}$ in Ω whose roots are those of the numbers τ_ξ that are distinct from one another, when ξ runs through a system of incongruent numbers coprime to $\mathfrak{m}$.

To determine the degree of the function $T_\mathfrak{m}$, we observe that two quantities $\tau_\xi, \tau_{\xi'}$ are equal to one another only when

$$\xi \equiv \varepsilon\xi' \pmod{\mathfrak{m}} \tag{8}$$

holds. Thus just as many of the τ_ξ become equal to one another as there are units incongruent modulo $\mathfrak{m}$; in general these are two, for $\Delta = -4$ they are four, and for $\Delta = -3$ they are six. These numbers are, however, reduced still further when different ones among the units ε are congruent with respect to the modulus $\mathfrak{m}$, that is, when $1 - \varepsilon$ is divisible by $\mathfrak{m}$. But this can occur only when $\mathfrak{m}$ is a divisor of 2 or of 3. Hence:

5. The degree of the function $T_{\mathfrak{m}}$ is equal to the number $\psi(\mathfrak{m})$ (Bd. II, § 168) of the numbers in Ω that are incongruent with respect to the modulus $\mathfrak{m}$ and coprime to $\mathfrak{m}$, divided by the number of units in Ω that are incongruent with respect to $\mathfrak{m}$.

The roots of the function $T_{\mathfrak{m}}$ determine an algebraic field $\mathfrak{T}_{\mathfrak{m}}$ over the class field, whose relative degree is at most equal to the degree of $T_{\mathfrak{m}}$, and both degrees are equal when $T_{\mathfrak{m}}$ is irreducible.

6. This field $\mathfrak{T}_{\mathfrak{m}}$ we shall call the division field for the modulus $\mathfrak{m}$.

These division fields play a similar role for the field Ω to that played by the cyclotomic fields for the field of rational numbers, except that here the class field is still interposed between them, and to this there is no analogue in the field of rational numbers, just as there is none in any one-class field.

By the multiplication theorem (§ 151),

$$\tau(\xi u) = \theta_{\xi}[\tau(u)] \tag{9}$$

can be expressed rationally in terms of $\tau(u)$. If we set $u = \xi' \frac{\alpha}{\mu}$, then it follows:

$$\tau_{\xi\xi'} = \tau\left(\frac{\xi\xi'\alpha}{\mu}\right) = \theta_{\xi}(\tau_{\xi'}), \tag{10}$$

and consequently, by interchange of ξ with ξ':

$$\theta_{\xi}(\tau_{\xi'}) = \theta_{\xi'}(\tau_{\xi}), \tag{11}$$

and hence, according to Bd. I, § 169:

7. The division field is relatively Abelian with respect to the class field.

The numbers ξ, taken with respect to the modulus $\mathfrak{m}$, form a group under multiplication, and the relative group of the division field is isomorphic with this group (or at least with a divisor of this group. The irreducibility of $T_{\mathfrak{m}}$ will be proved further on, whereby the group of $\mathfrak{T}_{\mathfrak{m}}$ itself is then established).

§ 155. **Multiplication of the elliptic functions for an odd multiplier.**

For the demonstration of the existence of the division field, the use of the Weierstrass function and of the τ-function derived from it is very expedient, and it would be most convenient if the further investigation of this field could also be based upon it. But for that the arithmetic nature of the multiplication formulae is not yet sufficiently known, and it is therefore for the present still necessary to consider the complex multiplication and division of the Jacobi elliptic functions as well.

Here we wish to give the multiplication formulae which we derived in § 57 yet a somewhat different shape.

We consider throughout, to begin with, the multiplier m as an odd number. We set, if

$$\varkappa = \frac{\vartheta_{10}}{\vartheta_{00}} \tag{1}$$

denotes the modulus of the elliptic functions, following Kronecker[1]:

$$\lambda = 4\left(\varkappa + \frac{1}{\varkappa}\right) = 4\,\frac{f(\omega)^4}{f_2(\omega)^4} + 4\,\frac{f_2(\omega)^4}{f(\omega)^4} \quad [\S\ 54, (3)], \tag{2}$$

and develop this expression in ascending powers of

$$q = e^{\pi i \omega}.$$

The form of this development can be derived from the product formulae [§ 24, (11)]:

$$\lambda = q^{-\frac{1}{2}}\,\Pi\left(\frac{1+q^{2\nu-1}}{1+q^{2\nu}}\right)^4 + 16\,q^{\frac{1}{2}}\ \Pi\left(\frac{1+q^{2\nu}}{1+q^{2\nu-1}}\right)^4$$

and one finds:

$$\lambda = q^{-\frac{1}{2}}(1 + \lambda_1 q + \lambda_2 q^2 + \lambda_3 q^3 + \cdots), \tag{3}$$

in which the constants $\lambda_1, \lambda_2, \lambda_3, \ldots$ are integral rational numbers.

If we further set

$$x = \sqrt{\varkappa}\ \operatorname{sn}\left(\frac{w}{\sqrt{\varkappa}}, \varkappa\right), \tag{4}$$

then, by § 45, (9), (4) remains unchanged when $\varkappa$ is interchanged with $1 : \varkappa$. Now x satisfies the differential equation

$$\left(\frac{dx}{dw}\right)^2 = 1 - \tfrac{1}{4}\lambda x^2 + x^4. \tag{5}$$

If, then, one develops x by Taylor's theorem into a series in ascending powers of w:

$$x = w + X_1 w^3 + X_2 w^5 \ldots, \tag{6}$$

then the coefficients $X_1, X_2, \ldots$ are integral rational functions of λ with rational numerical coefficients, as follows from (5) by the method of indeterminate coefficients.

If one uses the notation of § 42, then

$$x = \sqrt{\varkappa}\ \operatorname{sn} v = \frac{\vartheta_{11}(u)}{\vartheta_{01}(u)}, \tag{7}$$

and, by § 57, IV., with a somewhat modified meaning of the A, B, C, D:

$$\begin{aligned} \sqrt{\varkappa}\ \operatorname{sn} mv &= \frac{xA(x)}{D(x)}, \\ \operatorname{cn} mv &= \sqrt{1 - \frac{x^2}{\varkappa}}\ \frac{B(x)}{D(x)}, \\ \operatorname{dn} mv &= \sqrt{1 - \varkappa x^2}\ \frac{C(x)}{D(x)}. \end{aligned} \tag{8}$$

Here $A(x), B(x), C(x), D(x)$ are integral rational functions of x of degree $\frac{1}{2}(m^2-1)$, and it is to be seen from the recursion formulae § 57, (13), (14) that the functions A, D and the product BC are integral rational functions of λ, whose

numerical coefficients are rational. That they are integers cannot be seen, on account of the denominator 2 which appears at first.

[1] Zur Theorie der elliptischen Funktionen. Sitzungsbericht der Berliner Akademie vom 29. Juli 1886.

In order to investigate this more closely, we consider the roots of the function $A(x)$: they are, if we set:

$$\Omega_{h,h'} = \frac{2hK + 2h'iK'}{m},$$

$$x_{h,h'} = \sqrt{\varkappa}\,\mathrm{sn}\,\Omega_{h,h'} = \frac{\vartheta_{11}\left(\frac{h+h'\omega}{m}, \omega\right)}{\vartheta_{01}\left(\frac{h+h'\omega}{m}, \omega\right)}, \tag{9}$$

in which h, h' may be any integers whatever; only they must not both be equal to zero. One obtains all the roots of $A(x)$ by letting h, h' each run through a complete system of residues with respect to the modulus m, apart from the combination 0,0. It is then

$$x_{h,h'} = \frac{-i \sum\limits_{-\infty,\infty}^{\nu} (-1)^\nu e^{\frac{\pi i h}{m}(2\nu+1)} q^{\left(\frac{2h+1}{2} + \frac{h'}{m}\right)^2}}{\sum\limits_{-\infty,\infty}^{\nu} (-1)^\nu e^{\frac{2\pi i \nu h}{m}} q^{\left(\nu + \frac{h'}{m}\right)^2}}. \tag{10}$$

In order to develop this expression in ascending powers of q, one must first seek out the lowest power of q in the denominator, that is, the term in which $\nu + \frac{h'}{m}$ becomes as small as possible. This occurs when ν is the integer lying nearest to $-h'/m$. Since m is odd, there is only one such number ν, and formula (10) takes the shape:

$$x_{h,h'} = \frac{Q_1}{1 + Q_2} = Q_1 - Q_1 Q_2 + Q_1 Q_2^2 - \ldots, \tag{11}$$

in which Q_1 and Q_2 are series ordered according to ascending powers of q, whose coefficients are integral algebraic numbers (cyclotomic numbers).

Accordingly $x_{h,h'}$ can be developed in ascending powers of q, and the coefficients of this development are integral algebraic numbers.

If S is a symmetric function of the $x_{h,h'}$ with rational integral coefficients, then it too can be developed in the same way in powers of q. On the other hand, by the fundamental theorem on symmetric functions S is rationally expressible through

the coefficients of $A(x)$, and is therefore an integral rational function of λ with rational numerical coefficients of the form:

$$S = S_0\lambda^s + S_1\lambda^{s-1} + S_2\lambda^{s-2} + \cdots + S_s, \tag{12}$$

in which the $S_0, S_1, S_2, \ldots$, are rational numbers. That they are integers is just what is to be proved. But this follows very simply if in (12) one substitutes the development (3) for λ and then compares with one another the coefficients of equally high powers of q on both sides. For one obtains, for the determination of the $S_0, S_1, S_2, \ldots$, a sequence of linear equations, each following one of which contains only one new S_i, and that with the coefficient 1, while on the other side of these equations there stand integral algebraic numbers. Accordingly the $S_0, S_1, S_2, \ldots$, are integral algebraic numbers, and since they are rational, they are also integral rational numbers.

Since we know moreover, by § 57, II and (7), $A(0)$ and $A(\infty)$, there results for $A(x)$ the expression:

$$A(x) = \pm x^{m^2-1} + a_1 x^{m^2-3} + a_2 x^{m^2-5} + \cdots + m, \tag{13}$$

in which the $a_1, a_2, \ldots$, are integral functions of λ with integral coefficients.

Further, by § 57, (7):

$$D(x) = \pm x^{m^2-1} A\left(\frac{1}{x}\right),$$

and consequently:

$$\pm D(x) = \pm 1 + a_1 x^2 + a_2 x^4 + \cdots + m x^{m^2-2}. \tag{14}$$

[The sign $\pm$ in (13) and (14) is determined according to § 57 by$(-1)^{\frac{m-1}{2}}$].

For the functions B and C the relation holds:

$$B(x) = x^{m^2-1} C\left(\frac{1}{x}\right), \tag{15}$$

and their roots are therefore reciprocal to one another. The first and the last coefficient are equal to unity in B and in C.

The roots of B are:

$$y_{h,h'} = \sqrt{\varkappa}\,\mathrm{sn}(\Omega_{h,h'} + K) = \sqrt{\varkappa}\,\frac{\mathrm{cn}\,\Omega_{h,h'}}{\mathrm{dn}\,\Omega_{h,h'}}$$

$$= \frac{\vartheta_{10}\left(\frac{h+h'\omega}{m}, \omega\right)}{\vartheta_{00}\left(\frac{h+h'\omega}{m}, \omega\right)} \tag{16}$$

and can be developed in the following way:

$$\frac{\sum\limits_{-\infty,\infty}^{\nu} q^{\left(\nu+\frac{1}{2}+\frac{h'}{m}\right)^2} e^{\frac{(2\nu+1)\pi i}{m}h}}{\sum\limits_{-\infty,\infty}^{\nu} q^{\left(\nu+\frac{h'}{m}\right)^2} e^{\frac{2\nu\pi i}{m}h}}. \tag{17}$$

In the denominator of this expression there occurs only one term with the lowest power of q; one obtains it by putting for ν the integer that lies nearest to the fraction $-h'/m$.

In the numerator too there occurs in general only one lowest term, whose coefficient is a unit. In the particular case where h'/m is an integer, however, two equal lowest terms occur in the numerator, namely (for $h' = 0$) $\nu = 0, -1$. Together these give

$$q^{\frac{1}{4}}\, 2\cos\frac{h\pi}{m}$$

and $2\cos\dfrac{h\pi}{m}$ is an algebraic unit. [To be inferred from Bd. I, § 144, (19).]

Accordingly both the expression (17) and its reciprocal value can be developed in ascending powers of q in such a way that the coefficient of the first term has the value 1 and all the remaining coefficients are integral algebraic numbers. From this one concludes, just as in regard to $A(x)$, that $B(x)\,C(x)$ can be represented in the following way:

$$B(x)\,C(x) = x^{2m^2-2} + b_1 x^{2m^2-4} + \cdots + 1, \tag{18}$$

in which the $b_1, b_2 \ldots$ are integral rational functions of λ with integral rational numerical coefficients. Moreover the coefficients b that are equally distant from the beginning and from the end are equal to one another.

Let us consider further the equation arising from (8):

$$xA(x) - \sqrt{\varkappa}\,\mathrm{sn}\,mv\,D(x) = 0, \tag{19}$$

which in regard to x is of degree m^2. Its roots are the quantities:

$$\sqrt{\varkappa}\,\mathrm{sn}\,v, \quad \sqrt{\varkappa}\,\mathrm{sn}(v + \Omega_{h,h'}).$$

The product of these roots is, apart from the sign, equal to the independent term in this equation, and hence equal to

$$\pm\sqrt{\varkappa}\,\mathrm{sn}\,mv,$$

and from this there results:

$$\pm\frac{\sqrt{\varkappa}\,\mathrm{sn}\,mv}{\sqrt{\varkappa}\,\mathrm{sn}\,v}=\prod^{h,h'}\sqrt{\varkappa}\,\mathrm{sn}(v+\Omega_{h,h'}). \tag{20}$$

§ 156. Transition to the singular moduli.

We now assume that ω satisfies a quadratic equation:

$$a\omega^2+b\omega+c=0 \tag{1}$$

whose fundamental discriminant

$$\Delta=b^2-4ac \tag{2}$$

is negative.

If then $j(\omega)$ is the invariant, the quantity λ [§ 155, (2)] satisfies the equation of degree 6:

$$\lambda^6-9\cdot16\lambda^4-[j(\omega)-27\cdot2^8]\lambda^2+64[j(\omega)-27\cdot64]=0, \tag{3}$$

whose coefficients are integral algebraic numbers. Consequently λ is itself an integral algebraic number (Bd. II, § 154, 11.)[32] and in § 135 we have seen that λ is a class invariant of the discriminant 4Δ or 16Δ.

We therefore adjoin the number λ to the class field $\mathfrak{K}(\Delta)$ and obtain a class field:

$$\begin{aligned}&\mathfrak{L}=\mathfrak{K}(4\Delta),\ \text{if}\ \Delta\equiv0\ (\mathrm{mod}\ 4),\ \text{or}\ \equiv1\ (\mathrm{mod}\ 8).\\&\mathfrak{L}=\mathfrak{K}(16\Delta),\qquad\Delta\equiv5\ (\mathrm{mod}\ 8).\end{aligned} \tag{4}$$

If h is the class number of Δ, and therefore the relative degree of $\mathfrak{K}(\Delta)$ with respect to the quadratic field $\Omega=\mathfrak{R}(\sqrt{\Delta})$, and h' the relative degree of $\mathfrak{L}$ with respect to Ω, then (§ 100, § 123):

$$\begin{aligned}&h'=h,\quad\text{if}\ \Delta\equiv1\ (\mathrm{mod}\ 8),\\&h'=2h,\quad\text{„}\quad\Delta\equiv0\ (\mathrm{mod}\ 4),\\&h'=6h,\quad\text{„}\quad\Delta\equiv5\ (\mathrm{mod}\ 8).\end{aligned} \tag{5}$$

In the first of these three cases $\mathfrak{L}$ is therefore identical with $\mathfrak{K}(\Delta)$.

Through the substitution of the singular modulus ω the coefficients $a_1,a_2\ldots,b_1,b_2,\ldots$, in § 155, (13), (18) pass into integral algebraic numbers, and from this it follows, Bd. II, § 154, 11., that the roots of $A(x)$, namely,

$$\sqrt{\varkappa}\,\mathrm{sn}\,\Omega_{h,h'},\quad\text{are integers,} \tag{6}$$

and that the roots of $B(x)$

$$\sqrt{\varkappa}\,\frac{\mathrm{cn}\,\Omega_{h,h'}}{\mathrm{dn}\,\Omega_{h,h'}},\quad\text{are units.} \tag{7}$$

Now for variable $\varkappa$ and v we have [§ 45, (5)]:

$$\mathrm{dn}\,(v,\varkappa)=\frac{\mathrm{dn}\,(iv,\varkappa')}{\mathrm{cn}\,(iv,\varkappa')}. \tag{8}$$

[32] It is this circumstance that particularly recommends the introduction of λ in place of $\varkappa$, because $\varkappa$ is in general not an integer, but only $4\varkappa$ is.

If we here set:

$$v = \Omega_{h,h'} = \frac{2hK + 2h'iK'}{m},$$

$$iv = \frac{-2h'K' + 2hiK}{m} = \Omega'_{-h',h},$$

then the Ω' have the same meaning for $\varkappa'$ as the Ω have for $\varkappa$, i. e. they correspond to the interchange of $(\omega, -1/\omega)$, and it follows from (7) and (8) that also

$$\frac{\operatorname{dn} \Omega_{h,h'}}{\sqrt{\varkappa'}} \text{ is a unit.}$$

Thus also

$$\frac{1}{\sqrt{\varkappa'}} \operatorname{dn} \Omega_{h,h'} = \frac{\vartheta_{00}\left(\frac{h+h'\omega}{m}\right)}{\vartheta_{01}\left(\frac{h+h'\omega}{m}\right)},$$

(9)

$$\sqrt{\frac{\varkappa}{\varkappa'}} \operatorname{cn} \Omega_{h,h'} = \frac{\vartheta_{10}\left(\frac{h+h'\omega}{m}\right)}{\vartheta_{01}\left(\frac{h+h'\omega}{m}\right)} \text{ are algebraic units.}$$

From formula § 155, (20) there follows for $v = 0$:

$$\pm m = \overset{h,h'}{\Pi}\ \sqrt{\varkappa} \operatorname{sn} \Omega_{h,h'} = \Pi\, \varkappa_{h,h'}. \tag{10}$$

1. The integers $\varkappa_{\mu,\mu'}$ are therefore divisors of the number m.

We now set, when m and n are two odd numbers:

$$\Omega_{h,h'} = \frac{2hK + 2h'iK'}{m},$$

$$H_{l,v'} = \frac{2lK + 2l'iK'}{n},$$

and substitute in formula § 155, (20):

$$v = H_{l,v'}.$$

This gives:

$$\frac{\sqrt{\varkappa} \operatorname{sn} m\, H_{l,v'}}{\sqrt{\varkappa} \operatorname{sn} H_{l,v'}} = \overset{h,h'}{\Pi}\ \sqrt{\varkappa} \operatorname{sn} (H_{l,v'} + \Omega_{h,h'}), \tag{11}$$

in which

$$H_{l,v'} + \Omega_{h,h'} = \frac{2(lm + hn)K + 2(lm + h'n)iK'}{mn},$$

and consequently the factors of the product (11):

$$\sqrt{\varkappa} \operatorname{sn} (H_{l,v'} + \Omega_{h,h'})$$

are integers [by (6)]. If in (11) one again interchanges m and n, it follows that:

$$\frac{\sqrt{\varkappa} \operatorname{sn} n\, \Omega_{h,h'}}{\sqrt{\varkappa} \operatorname{sn} \Omega_{h,h'}} \text{ is an integer.} \tag{12}$$

This holds for arbitrary odd n, and hence also, when n is relatively prime to m, for $n^{-1} \pmod m$; and so it is likewise true that

$$\frac{\sqrt{\varkappa} \operatorname{sn} n^{-1}\, \Omega_{h,h'}}{\sqrt{\varkappa} \operatorname{sn} \Omega_{h,h'}} \text{ is an integer.}$$

If here one replaces h, h' by nh, nh', it follows that also

$$\frac{\sqrt{\varkappa}\,\mathrm{sn}\,\Omega_{h,h'}}{\sqrt{\varkappa}\,\mathrm{sn}\,n\,\Omega_{h,h'}} \text{ is an integer,}$$

and therefore the quotient (12) is a unit, or, expressed otherwise:

2. If n runs through a sequence of odd numbers coprime to m, then the integers:

$$\sqrt{\varkappa}\,\mathrm{sn}\,n\,\Omega_{h,h'} \tag{13}$$

are associated with one another.

§ 157. Complex Multipliers.

Let ω satisfy, as in the preceding section, the quadratic equation

$$a\omega^2 + b\omega + c = 0 \tag{1}$$

with negative fundamental discriminant

$$\Delta = b^2 - 4ac,$$

so let:

$$\omega = \frac{-b + \sqrt{\Delta}}{2a}. \tag{2}$$

As in the preceding section, we base our considerations upon the function

$$\sqrt{\varkappa}\,\mathrm{sn}\,v = \frac{\vartheta_{11}(u)}{\vartheta_{01}(u)} = \boldsymbol{s}(u) \tag{3}$$

which we denote, as a function of $u = v/2K$, by $\boldsymbol{s}(u)$. This function then has the double periodicity expressed by

$$\begin{aligned} \boldsymbol{s}(u+1) &= -\boldsymbol{s}(u), \\ \boldsymbol{s}(u+\omega) &= \boldsymbol{s}(u) \end{aligned} \tag{4}$$

All periods of $\boldsymbol{s}(u)$ are contained in the form $2m + n\omega$, in which m, n are integral rational numbers, and they therefore pass over into integers on multiplication by a.

Two numbers u, u' that differ only by a period are called congruent with respect to the modulus $2, \omega$:

$$u' \equiv u \pmod{2, \omega}.$$

Since the function $\mathrm{sn}\,v$ takes a value only twice in the period parallelogram, we have

$$\boldsymbol{s}(u) = \boldsymbol{s}(u'),$$

only when either

$$\begin{aligned} &u \equiv u' \\ &\text{or} \qquad\qquad \pmod{2, \omega} \\ &u \equiv 1 - u' \end{aligned} \tag{5}$$

holds, and $\boldsymbol{s}(u)$ vanishes only when

$$u \equiv 0, 1 \pmod{2, \omega}$$

holds.

Since every fractional number in Ω can be transformed into an integer by multiplication with an integral rational number, the following theorem results from § 156, (6):

1. If η is any fractional number of the field Ω, then $\boldsymbol{s}(\eta)$ is an algebraic number, and if in particular η can be represented in such a way that its denominator is relatively prime to 2, then $\boldsymbol{s}(\eta)$ is an integral algebraic number.

Now let

$$\mu = y + x\sqrt{\Delta} \tag{6}$$

be an integer in Ω, in which x, y are integral rational numbers[33], and

$$\mu' = y - x\sqrt{\Delta},$$

the number conjugate to μ. Further,

$$m = \mu\mu' = y^2 - \Delta x^2 \tag{7}$$

is the norm of μ, which we assume to be odd.

It is then, by (2),

$$\begin{aligned} \mu &= y + bx + 2a\omega x, \\ \mu\omega &= -2cx + (y - bx)\omega, \\ m &= (y + bx)(y - bx) + 4acx^2, \end{aligned} \tag{8}$$

from which it follows that

$$y + bx \text{ and } y - bx$$

are odd.

From this there results, by (4):

$$\begin{aligned} \boldsymbol{s}[\mu(u+1)] &= -\boldsymbol{s}(\mu u), \\ \boldsymbol{s}[\mu(u+\omega)] &= \boldsymbol{s}(\mu u). \end{aligned} \tag{9}$$

The function $\boldsymbol{s}(\mu u)$ therefore has the same periods as $\boldsymbol{s}(u)$, and since $\boldsymbol{s}(\mu u)$ is an odd function of u, it can be expressed rationally by $\boldsymbol{s}(u)$. We denote by $A(\boldsymbol{s})$, $D(\boldsymbol{s})$ integral rational functions of $\boldsymbol{s} = \boldsymbol{s}(u)$, which moreover contain only the even powers of $\boldsymbol{s}$, and set:

$$\boldsymbol{s}(\mu u) = \mu\, \boldsymbol{s}(u)\, \frac{A(\boldsymbol{s})}{D(\boldsymbol{s})}. \tag{10}$$

The values of $\boldsymbol{s}(u)$ for which $\boldsymbol{s}(\mu u)$ becomes zero or infinite are, by 3., algebraic numbers, and since the coefficients of A and D can be composed out of these, these too are algebraic numbers. In order to determine their nature more closely, we think of the fraction (10) as first extended in such a way that D receives rational coefficients. This is brought about by multiplying numerator and denominator by the product of all the factors conjugate to the denominator. We can therefore put $A(\boldsymbol{s})$ and $D(\boldsymbol{s})$ into the form:

$$\begin{aligned} A(\boldsymbol{s}) &= A_1 + A_3\boldsymbol{s}^2 + A_5\boldsymbol{s}^4 + \cdots, \\ D(\boldsymbol{s}) &= D_1 + D_3\boldsymbol{s}^2 + D_5\boldsymbol{s}^4 + \cdots, \end{aligned} \tag{11}$$

in which the $D_1, D_3, \ldots$, are by our assumption rational numbers, while $A_1, A_3, \ldots$, are to be determined. If we set, as in § 155, (4), (6):

$$w = \sqrt{\varkappa}\, v = 2\sqrt{\varkappa}\, Ku,$$

[33] In the case of an even Δ, μ can accordingly be any integer from Ω. In the case of an odd Δ, μ is a number of the order [2].

then $\boldsymbol{s}(u)$ goes over into $\sqrt{\varkappa}\ \mathrm{sn}\left(\dfrac{w}{\sqrt{\varkappa}}\right)$, and equation (10) gives:

$$\frac{D\left[\sqrt{\varkappa}\ \mathrm{sn}\left(\dfrac{\omega}{\sqrt{\varkappa}}\right)\right]\ \sqrt{\varkappa}\ \mathrm{sn}\left(\dfrac{\mu w}{\sqrt{\varkappa}}\right)}{\mu\ \mathrm{sn}\left(\dfrac{w}{\sqrt{\varkappa}}\right)} = A_1 + A_3\boldsymbol{s}^2 + A_5\boldsymbol{s}^4 + \cdots \tag{12}$$

If one develops the left side in powers of w, the coefficients are rationally expressible through λ and $\sqrt{\Delta}$, and hence belong to the field

$$\mathfrak{L} = \mathfrak{R}(\lambda, \sqrt{\Delta}).$$

If one develops the right side in the same way and orders it by the powers of w, then A_ν appears first in the coefficient of $w^{\nu-1}$, and indeed carrying the factor 1. Accordingly A_ν can be determined once the earlier coefficients have already been determined, and it follows that all these coefficients belong to the field $\mathfrak{L}$.

Since afterwards the numerator and denominator of the fraction A/D can again be freed from common factors by rational calculation, the following theorem results:

2. If in the expression (8) the numerator and denominator $A(\boldsymbol{s})$, $D(\boldsymbol{s})$ are freed from common factors, then these functions belong to the field $\mathfrak{L}$.

In order to represent the function $A(x)$, we take the coefficient of the highest power of x equal to 1 and investigate for which values of u the function $\boldsymbol{s}(\mu u)$ vanishes. We denote a value u for which $\boldsymbol{s}(\mu u)$ vanishes by $2\varrho/\mu$, and then obtain as the condition for the vanishing, by (5), one of the two congruences

$$2\varrho \equiv 0, 1 \pmod{2, \omega}.$$

Thus ϱ is a number of the field Ω which may indeed be fractional, but can have no other denominators than a divisor of $2a$. On the other hand one may alter ϱ by a multiple of μ without the value $\boldsymbol{s}(2\varrho/\mu)$ being changed, and since only the latter is of consequence, and since the congruence

$$\varrho \equiv \frac{\xi}{2a} \pmod{\mu},$$

where ξ is an integer in Ω, can be satisfied by an integer ϱ, we may assume ϱ to be an integer if we presuppose that the first coefficient a in (1) is odd and relatively prime to μ. We hold fast to this assumption from now on.

If, then, we let ϱ run through a complete system of residues with respect to the modulus μ, with exclusion of zero, we obtain all the values of $\boldsymbol{s}$ for which $A(\boldsymbol{s})$ vanishes, in the form:

$$\boldsymbol{s}\left(\frac{2\varrho}{\mu}\right). \tag{13}$$

It remains to be shown that no two of the numbers (13) are equal to one another. If we had

$$\boldsymbol{s}\left(\frac{2\varrho}{\mu}\right) = \boldsymbol{s}\left(\frac{2\varrho'}{\mu}\right),$$

without $\varrho \equiv \varrho' \pmod{\mu}$ holding, then by (5)

$$\frac{2(\varrho + \varrho')}{\mu} \equiv 1 \pmod{2, \omega}$$

would have to hold. Hence $a(1+\omega)$ would have to be divisible by 2, and hence also, if ω is conjugate to ω', $a(1+\omega')$ divisible by 2, and consequently

$$a(\omega - \omega') = \sqrt{\Delta} \equiv 0 \pmod{2}.$$

Thus Δ would be divisible by 4, and $\frac{1}{4}\Delta = \frac{1}{4}a^2(\omega-\omega')^2$ would likewise still be a discriminant, which contradicts the assumption that Δ is a fundamental discriminant[34]. Accordingly the quantities (13), whose number amounts to $m-1$ (Bd. II, § 165), are all different from one another, and there results

$$A(x) = \prod^{\varrho} \left[x - \boldsymbol{s}\left(\frac{2\varrho}{\mu}\right)\right]. \tag{14}$$

By § 44, (20) the function $\boldsymbol{s}(u)$ satisfies the equation:

$$\boldsymbol{s}\left(u+\frac{\omega}{2}\right) = \frac{1}{\boldsymbol{s}(u)},$$

and, because of (8) and (4):

$$\boldsymbol{s}\left[\mu\left(u+\frac{\omega}{2}\right)\right] = \frac{(-1)^{cx}}{\boldsymbol{s}(\mu u)},$$

and accordingly there results from (10), if u is replaced by $u+\frac{\omega}{2}$:

$$\frac{(-1)^{cx}}{\boldsymbol{s}(\mu u)} = \frac{1}{\boldsymbol{s}(u)}\,\frac{A\left(\frac{1}{\boldsymbol{s}}\right)}{D\left(\frac{1}{\boldsymbol{s}}\right)} = \frac{(-1)^{cx}D(\boldsymbol{s})}{\boldsymbol{s}(u)A(\boldsymbol{s})}, \tag{15}$$

$$A(\boldsymbol{s})A\left(\frac{1}{\boldsymbol{s}}\right) = (-1)^{cx}D(\boldsymbol{s})D\left(\frac{1}{\boldsymbol{s}}\right). \tag{16}$$

Since A and D are without common divisors, we have, for a variable x and a constant h:

$$h\,x^{m-1}A\left(\frac{1}{x}\right) = D(x),$$

and, on replacing x by $1/x$:

$$h\,A(x) = x^{m-1}D\left(\frac{1}{x}\right),$$

and by (16):

$$h^2 = (-1)^{cx}.$$

Thus

$$D(x) = \varepsilon\,x^{m-1}A\left(\frac{1}{x}\right), \tag{17}$$

where $\varepsilon = \pm 1$ or $= \pm i$. By (16) this depends on whether cx is even or odd[35].

If we therefore use the notation (10) again, but now under the hypothesis that A and D are without common divisors, we obtain:

$$\varepsilon\,\boldsymbol{s}(\mu u) = \frac{A_1\boldsymbol{s} + A_3\boldsymbol{s}^3 + \cdots A_{m-2}\boldsymbol{s}^{m-2} + \boldsymbol{s}^m}{1 + A_{m-2}\boldsymbol{s}^2 + \cdots A_3\boldsymbol{s}^{m-3} + A_1\boldsymbol{s}^{m-1}}, \tag{18}$$

and the $A_1, A_3, \ldots, A_{m-1}$ are integers of the field $\mathfrak{L}$, for they are the symmetric fundamental functions of the integers (13).

[34] b would have to be even and $a-b+c$ divisible by 4, and $\frac{1}{4}\Delta = \left(\frac{b}{2}-a\right)^2 - a(a-b+c)$ would be a discriminant. For an arbitrary discriminant D the same end could be attained by adding to the numerator of the expressions (13) a power of 2 as a factor.

[35] From § 138 it follows that i is contained in the field $\mathfrak{L}$; (17) is therefore not in contradiction with Theorem 2.

In particular, if one sets $u = 0$, it follows that:

$$A_1 = \varepsilon\mu, \tag{19}$$

and from this:

$$\varepsilon\mu = \prod^{\varrho} \boldsymbol{s}\left(\frac{2\varrho}{\mu}\right). \tag{20}$$

The $\boldsymbol{s}\left(\frac{2\varrho}{\mu}\right)$ are therefore divisors of μ.

From (10) it follows that:

$$\boldsymbol{s}A(\boldsymbol{s}) - \boldsymbol{s}(\mu u)D(\boldsymbol{s}) = 0, \tag{21}$$

and this, when $\boldsymbol{s}(\mu u)$ is regarded as given, is an equation for $\boldsymbol{s}(u)$ of degree m, whose roots are:

$$\boldsymbol{s}(u),\ \boldsymbol{s}\left(u + \frac{2\varrho}{\mu}\right).$$

The product of these roots is therefore, since $\boldsymbol{s}^m$ has the coefficient 1, equal to the negative independent term, i. e. it is:

$$\varepsilon\,\frac{\boldsymbol{s}(\mu u)}{\boldsymbol{s}(u)} = \prod^{\varrho} \boldsymbol{s}\left(u + \frac{2\varrho}{\mu}\right), \tag{22}$$

and from this, for $u = 0$, as above:

$$\varepsilon\mu = \prod^{\varrho} \boldsymbol{s}\left(\frac{2\varrho}{\mu}\right). \tag{23}$$

Now let η be a fractional number in Ω with the odd ideal denominator $\mathfrak{a}$; then it follows from (22) that $\boldsymbol{s}(\mu\eta)/\boldsymbol{s}(\eta)$ is an integer, if μ is an arbitrary number (6) in Ω. If we take μ relatively prime to $\mathfrak{a}$ and set $\mu'\mu \equiv 1 \pmod{\mathfrak{a}}$, then we may replace η by $\mu'\eta$ and find that $\boldsymbol{s}(\eta)/\boldsymbol{s}(\mu'\eta)$ is likewise an integer, in which μ' is just as arbitrary as μ. Consequently $\boldsymbol{s}(\eta)/\boldsymbol{s}(\mu\eta)$ is also an integer and hence a unit.

We have therefore the theorem:

3. If η is a number in Ω with odd ideal denominator, and if μ runs through a sequence of numbers relatively prime to $\mathfrak{a}$, then the numbers

$$\boldsymbol{s}(\mu\eta)$$

are associated with one another.

Yet a further consequence follows from this if one sets $u = 2\varrho_1/\mu_1$ in (22):

$$\frac{\boldsymbol{s}\left(\frac{2\mu\varrho_1}{\mu_1}\right)}{\boldsymbol{s}\left(\frac{2\varrho_1}{\mu_1}\right)} = \prod^{\varrho} \boldsymbol{s}\left(\frac{2\varrho_1}{\mu_1} + \frac{2\varrho}{\mu}\right). \tag{24}$$

If now μ and ϱ_1 are relatively prime to μ_1, then by 3. the left side is a unit, and consequently all the factors of the right side

$$\boldsymbol{s}\left(\frac{2\varrho_1}{\mu_1} + \frac{2\varrho}{\mu_2}\right), \tag{25}$$

which are indeed integers, must be units.

But every $\boldsymbol{s}(\eta)$ can be brought into this form if

$$\eta = \frac{\nu}{\mu}$$

is a fractional number in Ω whose denominator is odd and divisible by two distinct prime ideals. For in this case one can decompose the denominator μ of η into two ideals $\mathfrak{a}_1\mathfrak{a}_2$

which are relatively prime to one another, and one can then determine two odd integers α_1, α_2 in Ω that are relatively prime to one another, of which the one is divisible by $\mathfrak{a}_1$, the other by $\mathfrak{a}_2$. Then $\eta\alpha_1\alpha_2$ is an integer in Ω, and by Bd. II, § 166, (4) one can determine two integers ϱ_1, ϱ_2 in such a way that

$$\frac{\nu\,\alpha_1\alpha_2}{\mu} = \varrho_1\alpha_2 + \varrho_2\alpha_1$$

and therefore

$$\boldsymbol{s}\left(\frac{2\nu}{\mu}\right) = \boldsymbol{s}\left(\frac{2\varrho_1}{\alpha_1} + \frac{2\varrho_2}{\alpha_2}\right)$$

holds. But this is of the form (25). From it we obtain the theorem:

4. If a fractional number η in Ω with odd denominator contains two or more distinct prime factors in the denominator, then $\boldsymbol{s}(\eta)$ is a unit.

§ 158. Decomposition of the Function $\boldsymbol{A}(x)$.

Now let

$$\begin{aligned} \mu &= y + x\sqrt{\Delta}, \\ \mu_1 &= y_1 + x_1\sqrt{\Delta} \end{aligned} \tag{1}$$

be two odd integers in Ω of the form § 157, (6). Let $\mathfrak{m}$ be the greatest common ideal divisor of μ and μ_1. If one therefore sets

$$\begin{aligned} \mu &= \mathfrak{m}\mathfrak{a}, \\ \mu_1 &= \mathfrak{m}\mathfrak{a}_1 \end{aligned} \tag{2}$$

then $\mathfrak{a}$ and $\mathfrak{a}_1$ are two equivalent ideals without a common divisor.

The functions $\boldsymbol{A}(x)$ that correspond to the two numbers μ, μ_1 we denote by

$$\boldsymbol{A}_\mu(x),\ \boldsymbol{A}_{\mu_1}(x),$$

and we ask what common roots these functions have, that is, when the equation:

$$\boldsymbol{s}\left(\frac{2\varrho}{\mu}\right) = \boldsymbol{s}\left(\frac{2\varrho_1}{\mu_1}\right) \tag{3}$$

can be satisfied, when ϱ is taken with respect to the modulus μ, and ϱ_1 with respect to the modulus μ_1.

Since the congruence

$$\frac{2\varrho}{\mu} + \frac{2\varrho_1}{\mu_1} \equiv 1 \pmod{2, \omega}$$

is not possible, as one shows exactly as above (p. 589), the following is necessary and sufficient for equation (3):

$$\frac{2(\mu_1\varrho - \mu\varrho_1)}{\mu\mu_1} \equiv 0 \pmod{2, \omega}. \tag{4}$$

From this it follows:

$$\mu\varrho_1 \equiv \mu_1\varrho \pmod{\mu\mu_1}, \tag{5}$$

and this is possible only if

$$\varrho \equiv 0 \pmod{\mathfrak{a}}, \quad \varrho_1 \equiv 0 \pmod{\mathfrak{a}_1}. \tag{6}$$

If conversely the conditions (5) and (6) are satisfied, then:

$$\frac{2\varrho}{\mu} - \frac{2\varrho_1}{\mu_1}$$

is an integer and equation (3) is satisfied.

Now take an integer in Ω divisible by $\mathfrak{a}$:

$$\alpha = \mathfrak{a}\mathfrak{c}, \tag{7}$$

in which $\mathfrak{c}$ is relatively prime to μ and μ_1. If then ϱ is divisible by $\mathfrak{a}$, one can determine an integer ξ with respect to the modulus $\mathfrak{m}$ from the congruence

$$\varrho \equiv \alpha\xi \pmod{\mu}$$

(Bd. II, § 166, 7.); then

$$\alpha_1 = \frac{\alpha\mu_1}{\mu} = \mathfrak{a}_1\mathfrak{c} \tag{8}$$

is an integer divisible by $\mathfrak{a}_1\mathfrak{c}$ and by no other ideal, and from (5) there results

$$\varrho_1 \equiv \alpha_1\xi \pmod{\mu_1}.$$

5. We thus obtain the common roots of $\boldsymbol{A}_\mu(x)$, $\boldsymbol{A}_{\mu_1}(x)$ in the form:

$$\boldsymbol{s}\left(\frac{2\alpha\xi}{\mu}\right) = \boldsymbol{s}\left(\frac{2\alpha_1\xi}{\mu_1}\right), \tag{9}$$

where ξ runs through a complete system of residues with respect to the modulus $\mathfrak{m}$.

Here $\mathfrak{m}$ can denote any arbitrary odd ideal in Ω.

For to every such ideal one can choose two numbers μ, μ_1 of the form (2), and then choose ω among the equivalent numbers in such a way that $\mathfrak{a}$ becomes odd and relatively prime to $\mu\mu_1$.

If one seeks the greatest common divisor of $\boldsymbol{A}_\mu$ and $\boldsymbol{A}_{\mu_1}$, one obtains an integral function $\boldsymbol{A}_\mathfrak{m}$ whose coefficients are integers in $\mathfrak{L}$ and whose roots are the quantities (9); and if one finally frees $\boldsymbol{A}_\mathfrak{m}$ of the factors that it has in common with any $\boldsymbol{A}_{\mathfrak{m}'}$ in which $\mathfrak{m}'$ is a proper divisor of $\mathfrak{m}$, one obtains an integral function:

$$T_\mathfrak{m}(x) = x^\nu - \tau_{\nu-1}x^{\nu-1} + \tau_{\nu-2}x^{\nu-2} - \cdots \pm \tau_0, \tag{10}$$

whose roots are only those among the quantities (8) in which ξ is relatively prime to $\mathfrak{m}$.

The degree ν of this function is to be determined according to Bd. II, § 168. We shall call it the ideal division function. Its coefficients are integers of the field $\mathfrak{L}$, and in particular

$$\tau_0 = \prod_{\xi} \boldsymbol{s}\left(\frac{2\alpha\xi}{\mu}\right), \tag{11}$$

in which ξ runs through a complete system of numbers incongruent to one another and coprime to $\mathfrak{m}$.

The roots of $T_\mathfrak{m}(x)$ are equal and opposite, and we obtain an equation of degree $1/2\,\nu$ for the quantity

$$\boldsymbol{s}\left(\frac{2\alpha\xi}{\mu}\right)^2.$$

By adjunction of these quantities to the field $\mathfrak{L}$ there arises a field $\mathfrak{T}_\mathfrak{m}$, which we likewise call a division field. It results from the division field of § 154 by adjunction of x. The group of this field is contained in the group of the numbers $\xi \pmod{\mathfrak{m}}$, as follows easily from the multiplication formulae.

We note further that the general division field $\mathfrak{T}_\mathfrak{m}$ can be reduced, by application of the addition theorem, to the case where $\mathfrak{m}$ is a power of a prime ideal in Ω.

§ 159. Prime Ideals.

6. If η is a fractional number of the field Ω which in reduced form has the odd ideal denominator $\mathfrak{a}$, and if $\mathfrak{m}$ is an ideal relatively prime to $\mathfrak{a}$, then the integer $\boldsymbol{s}(\eta)$ is relatively prime to $\mathfrak{m}$.

For let there be taken in Ω an integer μ divisible by $\mathfrak{a}$ and coprime to $2\mathfrak{m}$. Then $\mu\eta$ is an integer. In the product § 157, (23)

$$\mu = \overset{\varrho}{\Pi}\, \boldsymbol{s}\left(\frac{2\varrho}{\mu}\right) \tag{1}$$

there occurs a number 2ϱ which is congruent to $\mu\eta$ with respect to the modulus μ. Consequently $\boldsymbol{s}(\eta)$ is a divisor of μ and hence relatively prime to $\mathfrak{m}$, as was to be proved.

Let us now assume that in the formula (11), § 158

$$\tau_0 = \overset{\xi}{\Pi}\, \boldsymbol{s}\left(\frac{2\alpha\xi}{\mu}\right) \tag{2}$$

$\mathfrak{m}$ is a prime ideal of the field Ω, and let us accordingly set $\mathfrak{m} = \mathfrak{p}$, so that ξ runs through a complete system of residues with respect to the modulus $\mathfrak{p}$, with the exclusion of zero.

If we then take μ so that it is divisible only by the first power of $\mathfrak{p}$ and is not divisible by some arbitrarily chosen other prime ideal $\mathfrak{p}'$, then by Theorem 6. τ_0 cannot be divisible by $\mathfrak{p}'$. Now τ_0 depends only on the ideal $\mathfrak{p}$ and is independent of how the number μ is otherwise taken. Consequently by Theorem 6. τ_0 is divisible by no prime ideal different from $\mathfrak{p}$. But neither can $\mathfrak{p}$ divide τ_0 in a higher than the first power, for the factors of the product (1) all occur among the factors of the product (2), and consequently μ is divisible by τ_0. Accordingly we may set outright

$$\tau_0 = \mathfrak{p} \tag{3}$$

and since τ_0 is a number in the field $\mathfrak{L}$, it follows:

7. Every odd prime ideal of the field Ω is a field $\mathfrak{L}$, a principal ideal.

The assumption that μ should be relatively prime to $\mathfrak{a}$ likewise involves no restriction of this theorem. For, with $\mathfrak{p}$ given, we can select from among the equivalent forms (a, b, c), without changing λ, one for which this requirement is fulfilled.

This does not, however, yet prove that $\mathfrak{p}$ is also a principal ideal in the class field $\mathfrak{K}(\Delta)$ itself. At least, this follows only in the case $\Delta \equiv 1 \pmod 8$, where $\mathfrak{K}$ is identical with $\mathfrak{L}$[36].

By Theorem 3., § 157 the factors of the product (2) are all associated with one another. The number of these factors is $N(\mathfrak{p}) - 1$, and accordingly we may also set:

$$\mathfrak{p} = \left[\boldsymbol{s}\left(\frac{2\alpha}{\mu}\right)\right]^{N(\mathfrak{p})-1}. \tag{4}$$

§ 160. Prime ideals of the first degree in $\mathfrak{T}_\mathfrak{m}$.

[36] Cf. on this § 122, where the case of the prime 2 is also settled.

What now matters above all is to seek out the primes p that are decomposable in the field $\mathfrak{L}$ into prime ideals of the first degree. By § 122 and § 156 these are the primes p that are representable by the principal form of the discriminant $D = 4\Delta$ or 16Δ. These are of the form:

$$\begin{aligned} &1) \quad p = y^2 - \Delta x^2, \qquad \Delta \equiv 0 \pmod 4 \text{ or } \equiv 1 \pmod 8, \\ &2) \quad p = y^2 - 4\Delta x^2, \qquad \Delta \equiv 5 \pmod 8. \end{aligned}$$

If $\Delta \equiv 1 \pmod 8$, then the third coefficient of the form (a, b, c) must be even. If $\Delta \equiv 0 \pmod 4$, then we may assume c even, by passing if necessary to the parallel form $(a, b + 2a, a + b + c)$. If therefore p splits into the two prime factors π, π', then

$$(1) \qquad \begin{aligned} &1) \quad \pi = y + x\sqrt{\Delta}, \\ &2) \quad \pi = y + 2x\sqrt{\Delta}, \end{aligned}$$

and in order to apply the multiplication formula § 157, (10), (17) to $\mu = \pi$, we have in the latter case to replace x by $2x$. We may therefore in all cases assume cx even, and hence $\varepsilon = \pm 1$.

The conditions for the modulus 2 that lie in the representations (1), 1), 2) can be summarised in the single one:

$$\pi \equiv 1 \pmod 2.$$

If we set, when $N(\pi) = p$, for the moment by way of abbreviation:

$$(2) \qquad \begin{aligned} \varphi(x) &= A_1 x + A_3 x^3 + \cdots + A_{p-2} x^{p-2} + x^p, \\ \psi(x) &= 1 + A_{p-2} x^2 + \cdots + A_3 x^{p-3} + A_1 x^{p-1}, \end{aligned}$$

then by § 157, (18), (19)

$$(3) \qquad A_1 = \pm\pi,$$

and

$$(4) \qquad \pm \boldsymbol{s}(\pi u) = \frac{\varphi(\boldsymbol{s})}{\psi(\boldsymbol{s})},$$

and from this there follows by differentiation with respect to u:

$$(5) \qquad \pm \frac{\pi \boldsymbol{s}'(\pi u)}{\boldsymbol{s}'(u)} = \frac{\varphi'(\boldsymbol{s})\psi(\boldsymbol{s}) - \varphi(\boldsymbol{s})\psi'(\boldsymbol{s})}{\psi(\boldsymbol{s})^2}.$$

But by § 155, (5)

$$\boldsymbol{s}'(u)^2 = 1 - \tfrac{1}{4}\lambda \boldsymbol{s}^2 + \boldsymbol{s}^4,$$

and from (5) there results, by squaring and making use of (4):

$$\pi^2 \frac{4\psi(\boldsymbol{s})^4 - \lambda\varphi(\boldsymbol{s})^2\psi(\boldsymbol{s})^2 + 4\psi(\boldsymbol{s})^4}{4 - \lambda\boldsymbol{s}^2 + 4\boldsymbol{s}^4} = [\varphi'(\boldsymbol{s})\psi(\boldsymbol{s}) - \varphi'(\boldsymbol{s})\psi(\boldsymbol{s})]^2.$$

Since on the right there stands an integral function of $\boldsymbol{s}$, the numerator on the left-hand side must be divisible by the denominator (which also follows from consideration of the zero points), and it follows:

$$\pi^2 \Phi = [\varphi'(\boldsymbol{s})\psi(\boldsymbol{s}) - \varphi(\boldsymbol{s})\psi'(\boldsymbol{s})]^2,$$

where Φ is an integral function of $\boldsymbol{s}$ whose coefficients are integers in $\mathfrak{L}$. Hence Φ must be the square of an integral function whose coefficients are likewise integral algebraic numbers (Bd. I, § 2; Bd. II, § 159), and from this there results, for a variable x:

$$(6) \qquad \varphi'(x)\psi(x) - \psi'(x)\varphi(x) \equiv 0 \pmod \pi.$$

From this it can be proved that the coefficients $A_1, A_3, A_5, \ldots, A_{p-2}$ are all divisible by π.

Of A_1 we know this already. Let us then assume that it has been proved for

$$A_1, A_3, \ldots, A_{2\nu-1} \quad (2\nu+1 < p),$$

then it follows:

$$\begin{aligned} \varphi(x) &\equiv A_{2\nu+1}x^{2\nu+1} + \cdots \\ \varphi'(x) &\equiv (2\nu+1)A_{2\nu+1}x^{2\nu} + \cdots \qquad (\text{mod } \pi) \\ \psi(x) &\equiv 1 + A_{p-2}x^2 + \cdots \\ \psi'(x) &\equiv 2A_{p-2}x + \cdots \end{aligned}$$

If we now seek in (6) the coefficient of $x^{2\nu}$, it follows:

$$(2\nu+1)A_{2\nu+1} \equiv 0 \ (\text{mod } \pi).$$

Since $2\nu+1 < p$ and p is a prime, $2\nu+1$ is relatively prime to π, and it follows:

$$A_{2\nu+1} \equiv 0 \ (\text{mod } \pi). \tag{7}$$

If in (4) we put for u one of the values $2\alpha\xi/\mu$, § 158, (9), then $\boldsymbol{s}(u)$ becomes a root of the ideal division equation § 158, (10), and it follows:

$$\pm\boldsymbol{s}(\pi u) \equiv [\boldsymbol{s}(u)]^p \ (\text{mod } \pi),$$

and if we raise to a square:

$$\boldsymbol{s}(\pi u)^2 \equiv [\boldsymbol{s}(u)]^{2p} \ (\text{mod } \pi). \tag{8}$$

In particular, therefore, we have

$$\boldsymbol{s}(u)^2 \equiv \boldsymbol{s}(u)^{2p} \ (\text{mod } \pi), \tag{9}$$

if

$$\pi \equiv \pm 1 \ (\text{mod } \mathfrak{m}) \tag{10}$$

is. The number $\boldsymbol{s}(u)^2$ generates the field $\mathfrak{T}_\mathfrak{m}$. Moreover, by § 122, for every number ϱ in $\mathfrak{L}$:

$$\varrho \equiv \varrho^p \quad (\text{mod } \mathfrak{P}), \tag{11}$$

if $\mathfrak{P}$ is a prime ideal in $\mathfrak{L}$ that divides π, and consequently, for every integer τ in $\mathfrak{T}_\mathfrak{m}$, if $\mathfrak{P}'$ is a prime ideal in $\mathfrak{T}_\mathfrak{m}$:

$$\tau^p \equiv \tau \quad (\text{mod } \mathfrak{P}'), \tag{12}$$

if the conditions (10) and (1) are satisfied.

From this it follows:

8. In order that a prime π existing in Ω should split in the field $\mathfrak{T}_\mathfrak{m}$ into prime ideals of the first degree, it is necessary and sufficient that

$$\begin{aligned} \pi &\equiv 1 \quad (\text{mod } 2), \\ \pi &\equiv \pm 1 \quad (\text{mod } \mathfrak{m}). \end{aligned}$$

A certain finite number of primes (those which divide the discriminants of the equations that occur) is here left out of account.

§ 161. Number Groups and Ideal Groups.

As in § 98 and § 106 I denote by O the totality of the integral and fractional numbers of the quadratic field Ω, after the exclusion of all numbers that are not coprime to an arbitrarily assumed S. S may be a rational number, a number in Ω, or an ideal as well. I will call it the excludent. In the three memoirs „über Zahlgruppen in algebraischen Körpern“ [1] I have gathered together, under the name number groups, certain systems of numbers out of O which were defined by the requirement that the product and the quotient of any two numbers of such a group were contained in the same group. Such a number group can never contain the number zero, but certainly contains the number 1 [2].

In the same sense in which we speak of number groups we may also speak of ideal groups, which likewise may have their excludent. Here fractional ideals naturally occur as well, and in dealing with them one makes use, not without advantage, of the mode of representation by means of the functionals (Bd. II, § 169).

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[1] Mathematische Annalen Bd. 48, 49, 50.

[2] Fueter calls these groups „Zahlstrahlen“. Cf. Fueter, Die Theorie der Zahlstrahlen I, II; Crelle, Bd. 130, 132.

The totality of the integral and fractional numbers of the field Ω, to which we assign an arbitrary excludent S, forms the number group O, and the totality of the ideals $\overline{O}$ in the same sense an ideal group.

Likewise the totality of the numerical units forms a number group E, and the totality of the functional units an ideal group $\overline{E}$.

The multiplication of two groups A and B into the product AB is effected by multiplying every element of A by every element of B. If A is a number group, then $\overline{E}A$ is an ideal group. But this contains only principal ideals and can therefore be called a principal ideal group.

If the whole group O splits into a finite number of cosets with respect to A:

$$O = A\alpha_1 + A\alpha_2 + \cdots + A\alpha_j, \tag{1}$$

then the number j is called the index of A and is denoted by

$$j = (O, A) \tag{2}$$

(as in § 100).

We state the first theorem:

1. If A is a number group with finite index j, and η any number in O, then η^j is contained in A.

For of the unlimited sequence of powers $1, \eta, \eta^2, \eta^3, \ldots$, two different terms must be contained in the same coset $A\alpha_i$, and their quotient, which is of course also a power of η, is contained in A. The lowest exponent r for which η^r is contained in A proves, in the well-known way, to be a divisor of j.

2. If A is a number group with finite index, then the ideal group $\overline{O}$ can likewise be decomposed with respect to $\overline{E}A$ into a finite number of cosets:

$$\begin{aligned} \overline{O} &= \overline{A}_1 + \overline{A}_2 + \cdots + \overline{A}_h, \\ &= A\mathfrak{a}_1 + A\mathfrak{a}_2 + \cdots + A\mathfrak{a}_h, \end{aligned} \tag{3}$$

and indeed we can therein choose $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_h$ as integral ideals in $\overline{O}$, since by 1. we can convert every fractional ideal in $\overline{O}$ into an integral ideal by multiplication with a number in A. We need only choose for η a number that is divisible by the denominator of the ideal, and multiply by a sufficiently high power of η.

The number

$$h = (\overline{O}, \overline{E}A) \tag{4}$$

is called the class number of the field Ω with respect to A, and the system $A\mathfrak{a}_1, A\mathfrak{a}_2, \ldots, A\mathfrak{a}_h$ the ideal classes with respect to A. We can prove Theorem 2 by the general theorems on groups in § 100, 12. According to this:

$$\begin{aligned}(\overline{O},\, \overline{E}A) &= (\overline{O},\, \overline{E}O)\,(\overline{E}O,\, \overline{E}A),\\ (\overline{E}O,\, \overline{E}A) &= (\overline{E}O,\, \overline{E}EA) = (O,\, EA),\\ (O,\, EA)\,(EA,\, A) &= (O,\, A),\\ (EA,\, A) &= (E,\, E'),\end{aligned}$$

if E' denotes the group of numerical units contained in A. Consequently

$$(\overline{O},\, \overline{E}A) = (\overline{O},\, \overline{E}O)\,\frac{(O,\, A)}{(E,\, E')}, \tag{5}$$

and $(\overline{O},\, \overline{E}O)$ is the number of absolute ideal classes in Ω, hence a finite number. In the quadratic field with negative discriminant the number $(E,\, E')$ is equal to 1 if -1 occurs in A, equal to 2 if -1 does not occur in A, and it can rise to 4 or to 6 only in the two exceptional cases $\Delta = -4$, $\Delta = -3$.

If $\mathfrak{a}_1, \mathfrak{a}_1'$ belong to a class $\overline{A_1}$, and $\mathfrak{a}_2, \mathfrak{a}_2'$ to a class $\overline{A_2}$, then the products $\mathfrak{a}_1\mathfrak{a}_2$ and $\mathfrak{a}_1'\mathfrak{a}_2'$ belong to the same class, which we call the class $\overline{A_1}\,\overline{A_2}$. The classes can therefore be composed and form a finite Abelian group, in which the principal class $\overline{E}A$ is the unit.

A number group A shall be called a congruence group if it consists not merely of individual numbers, but of whole number classes with respect to a modulus M; that is, if α belongs to A, then all numbers congruent to α with respect to the modulus M shall also belong to A; M shall be called the modulus of the congruence group. A congruence group always has a finite index, since the number classes with respect to the modulus M are already finite in number. Instead of M one may also choose any multiple of M as the modulus. For this reason we also gain no more general concept if we take an ideal for M. We may even assume M to be a natural number.

If an ideal $\mathfrak{m}$ is relatively prime to M, then in every number class represented by α one can determine a number α_0 from the congruence

$$\alpha_0 = \alpha + M\vartheta \equiv 0 \pmod{\mathfrak{m}} \tag{6}$$

where ϑ is an integer in Ω.

From this it follows that every prime ideal $\mathfrak{e}$ dividing the excludent must divide M, since otherwise α_0 would be divisible by $\mathfrak{e}$ for $\mathfrak{m} = \mathfrak{e}$.

If α and M have a common divisor, then all numbers of the form $\alpha + M\xi$, in which ξ is an arbitrary number, have the same common divisor with M. But in A there are also numbers that are relatively prime to M, e. g. the number 1. The following theorem then holds:

3. If $\mathfrak{m}$ is relatively prime to M, then in A one can determine an integer $\alpha_0 = \mathfrak{m}\mathfrak{n}$ divisible by $\mathfrak{m}$ in such a way that $\mathfrak{n}$ becomes relatively prime to M and to an arbitrarily given ideal $\mathfrak{c}$.

For if by (6) one has determined a number divisible by $\mathfrak{m}$,

$$\alpha_0 = \mathfrak{m}\mathfrak{n}$$

which is coprime to M, then one obtains other such numbers:

$$\alpha_0' = \alpha_0 + M\xi = \mathfrak{m}\mathfrak{n}',$$

if

$$\xi = \mathfrak{m}\mathfrak{r}$$

is an arbitrary integer divisible by $\mathfrak{m}$. Now if $\mathfrak{q}$ is a prime ideal that divides $\mathfrak{c}$ but not α_0, let $\mathfrak{r}$ be assumed divisible by $\mathfrak{q}$; then $\mathfrak{n}'$ is not divisible by $\mathfrak{q}$. But if $\mathfrak{q}$ divides $\mathfrak{c}$ and α_0, let $\mathfrak{r}$ be assumed not divisible by $\mathfrak{q}$; then $\mathfrak{n}'$ is likewise not divisible by $\mathfrak{q}$.

From this it follows:

4. If $\mathfrak{m}$ is an arbitrary ideal coprime to M, then in every ideal class $\overline{A_\nu}$ there are ideals that are relatively prime to $\mathfrak{m}$.

To prove this, take an ideal $\mathfrak{b}$ in the class $(\overline{A_\nu})^{-1}$ and determine by 3. a number

$$\alpha = \mathfrak{a}\mathfrak{b},$$

so that $\mathfrak{a}$ is relatively prime to $\mathfrak{m}$; then $\mathfrak{a}$ belongs to the class $\overline{A_\nu}$ and satisfies the requirement of Theorem 4.

§ 162. The ideals of the principal class divisible by an ideal.

By Ω there shall now be understood the quadratic field with negative fundamental number Δ.

In this field let A be a congruence group with the modulus M and the excludent S. The group A contains, as we have seen, infinitely many integers. We ask for the number $T(t)$ of the integers in A that are divisible by some given integral ideal $\mathfrak{m}$ coprime to M, and whose norm is not greater than the positive number t.

Let (α_1, α_2) be a basis of $\mathfrak{m}$; then by § 91, (11):

$$\alpha_1 = a_{22}\, a, \quad \alpha_2 = a_{22}\, \frac{b + \sqrt{\Delta}}{2}. \tag{1}$$

Here let a_{22} denote the greatest integral rational number dividing $\mathfrak{m}$, and $a_{22}\, a$ the least integral rational number divisible by $\mathfrak{m}$; b is a root of the congruence:

$$b^2 \equiv \Delta \pmod{4a},$$

and we have:

$$\Delta = b^2 - 4ac. \tag{2}$$

$$N(\mathfrak{m}) = a_{22}^2\, a. \tag{3}$$

Since $\mathfrak{m}$ has been assumed relatively prime to M, we can, by § 161, (6), determine in every number class with respect to M occurring in A a number α_0 divisible by $\mathfrak{m}$. This number has the form

$$\alpha_0 = x_0\alpha_1 + y_0\alpha_2,$$

and if we set

$$\begin{aligned} \alpha &= \alpha_0 + M(x\alpha_1 + y\alpha_2) \\ &= (x_0 + Mx)\alpha_1 + (y_0 + My)\alpha_2 \end{aligned} \tag{4}$$

and let x, y run through all positive and negative integral rational numbers, then α runs through the totality of the integers of the group A that are divisible by $\mathfrak{m}$ and congruent to α_0 with respect to M.

In order to obtain all the integers in A divisible by $\mathfrak{m}$, we have to seek a complete system of residues with respect to M in Ω and to select from among them the numbers α_0 contained in A that are divisible by $\mathfrak{m}$.

If the norm of this number α is to be not greater than t, then, if we denote the quadratic form

$$ax^2 + bxy + cy^2$$

by $\varphi(x, y)$, the condition for this is:

$$\varphi(x_0 + Mx,\ y_0 + My) \leqq \frac{t}{a_{22}^2\, a}, \qquad [\S\,91, (2), (4)].$$

Let us take in a plane a rectangular coordinate system X, Y, and let us call lattice points those points with the coordinates

$$X = (x_0 + Mx)\, a_{22} \sqrt{\frac{a}{t}},$$

$$Y = (y_0 + My)\, a_{22} \sqrt{\frac{a}{t}},$$

where x_0, y_0 are kept fixed and x, y run through all integral rational numbers; then by this lattice the plane is divided into squares whose area is

$$\delta^2 = M^2\, \frac{a_{22}^2\, a}{t}$$

and if T_0 is the number of lattice points that lie within the ellipse

$$\varphi(X, Y) = 1$$

or on its boundary, then $T_0\, \delta^2$ is, for an infinite t, equal to the area of this ellipse, and hence equal to $\frac{\pi}{\sqrt{-\Delta}}$.

If one takes account further of the elements of area δ^2 that are cut through by the periphery of the ellipse, from which one obtains a measure for the error of this determination of area, then

$$T_0 = \frac{\pi t}{M^2\, a_{22}^2\, a\, \sqrt{-\Delta}} + R_0 \sqrt{t}, \tag{5}$$

where R_0 is a quantity that remains finite as t increases infinitely [Bd. II, § 194, (6)].

We obtain such an expression for every α_0, and if we add all these expressions together, we obtain, having regard to (3), the theorem:

If A is a congruence group with the modulus M and $\mathfrak{m}$ an ideal in Ω coprime to M, and further T the number of integral principal ideals contained in $\overline{E}A$ whose norm is not greater than t, then

$$T = \frac{gt}{N(\mathfrak{m})} + R\sqrt{t}, \tag{6}$$

where g is a number independent of $\mathfrak{m}$ and t and different from zero, and R is a function of t that is finite as A increases infinitely.

The constant g is more closely determined by (5):

$$g = \frac{\gamma\, \pi}{M^2 \sqrt{-\Delta}}, \tag{7}$$

where γ is the number of classes of numbers contained in A with respect to the modulus M.

If by T we wish to understand the number of integral ideals of the principal class $\overline{E}A$ whose norm is not greater than t, then (6) still holds in that case too; only, if -1 occurs in A, the number g has to be halved, and in the two exceptional cases $\Delta = -4$, $\Delta = -3$ it may possibly have to be divided further by 4 or by 6.

§ 163. The Dirichlet Sums.

We now understand by $\overline{O}$ the ideal group of all the ideals in Ω (with regard to the excludent S); let A be a congruence group and

$$\overline{A} = \overline{E}A, \tag{1}$$

the group of the corresponding principal ideals. We decompose $\overline{O}$ according to § 161, (3) into the classes with respect to A:

$$\overline{O} = \overline{A}_1 + \overline{A}_2 + \cdots + \overline{A}_h, \tag{2}$$

and consider the sums:

$$S_i = \sum^{\mathfrak{a}_i} \frac{1}{(N\mathfrak{a}_i)^s}, \tag{3}$$

in which s denotes a positive variable > 1, and $\mathfrak{a}_i$ runs through all the integral ideals of a class $\overline{A}_i$.

We transform the sums S_i in the following way:

Since $N(\mathfrak{a}_i)$ must be one of the numbers 1, 2, 3, ... we denote by a_ν the number of ideals of the class $\overline{A}_i$ whose norm $= \nu$, and obtain

$$S_i = a_1 + \frac{a_2}{2^s} + \frac{a_3}{3^s} + \frac{a_4}{4^s} + \cdots \tag{4}$$

where, if a number ν does not occur among the $N(\mathfrak{a}_i)$, one is to set $a_\nu = 0$.

If we denote by $Z(\nu)$ the number of ideals of the class $\overline{A}_i$ whose norm is not greater than ν, then for $Z(\nu)$ we obtain from formula (6) of the preceding section a limiting value:

By § 161, 4. we take in the class $(A_i)^{-1}$ an ideal $\mathfrak{m}$ coprime to M, and in

$$\mathfrak{m}\,\mathfrak{a}_i = \overline{\alpha} \tag{5}$$

we let $\mathfrak{a}_i$ run through the ideals of the class $\overline{A}_i$, and consequently $\overline{\alpha}$ through the ideals of the principal class divisible by $\mathfrak{m}$; if then

$$N(\mathfrak{a}_i) \leqq \nu,$$

then

$$N(\mathfrak{m})\,N(\mathfrak{a}_i) = N(\alpha) \leqq \nu\,N(\mathfrak{m}).$$

If, as before, T denotes the number of ideals of the principal class divisible by $\mathfrak{m}$ whose norm is not greater than t, then, if we set $t = \nu N(\mathfrak{m})$,

$$Z(\nu) = T,$$

and from formula § 162, (6) there results

$$Z(\nu) = g\nu + R_\nu\sqrt{\nu}, \tag{6}$$

in which g denotes a number that is the same for all classes $\overline{A}_i$, and R_ν a function of ν that does not become infinite as ν increases to infinity.

By the meaning of a_ν we have

$$a_\nu = Z(\nu) - Z(\nu-1), \quad Z(0) = 0 \tag{7}$$

and consequently

$$S_i = \sum_{0,\infty}^{\nu} \frac{Z(\nu) - Z(\nu-1)}{\nu^s}. \tag{8}$$

If we substitute the expression (6) for $Z(\nu)$, there follows:

$$S_i = g \sum \frac{1}{\nu^s} + \sum R_\nu \sqrt{\nu} \left(\frac{1}{\nu^s} - \frac{1}{(\nu+1)^s} \right). \tag{9}$$

The second term of this sum:

$$\sum R_\nu \sqrt{\nu} \left(\frac{1}{\nu^s} - \frac{1}{(\nu+1)^s} \right)$$

is unconditionally convergent so long as $s > \frac{1}{2}$, and is therefore a continuous function of s which, for $s = 1$, becomes

$$\sum_{1,\infty} \frac{R_\nu}{\sqrt{\nu}\,(\nu+1)}$$

Indeed,

$$\mathrm{Lim}_{\nu=\infty}\, \nu^{s+\frac{1}{2}} R_\nu \sqrt{\nu} \left(\frac{1}{\nu^s} - \frac{1}{(\nu+1)^s} \right) = s \,\mathrm{Lim}\, R_\nu$$

finite. The first sum in (9) is

$$\sum \frac{1}{\nu^s} = \frac{1}{s-1} + C_s, \tag{10}$$

where C_s remains finite for $s = 1$, passing namely into the value of Euler's constant 0,5772... (Bd. II, S. 723).

If this is taken into account, there follows from (9) the result:

5. If S_i has the meaning (3), then

$$S_i = \frac{g}{s-1} + C_i, \tag{11}$$

where g is a constant different from zero and independent of the particular class $\overline{A}_i$, and C_i is a function of s depending on $\overline{A}_i$ which remains finite for $s = 1$.

The group of degree h of the ideal class $\overline{A}_i$ is an Abelian one, and there are therefore h characters $\chi_\kappa(\overline{A}_i)$, which are all hth roots of unity, among them the principal character χ_1, which is $= 1$ for every class. Likewise, for the principal class $\overline{A}_1$ each of the characters has the value $+1$. For these characters the theorems hold:

$$
\begin{aligned}
&\sum^{i}\chi_\kappa(\overline{A}_i)=0, \quad \sum^{\kappa}\chi_\kappa(\overline{A}_i)=0,\\
&\kappa=2,3,\ldots,h, \quad i=2,3,\ldots,h,\\
&\sum^{i}\chi_1(\overline{A}_i)=h, \quad \sum^{\kappa}\chi_\kappa(\overline{A}_1)=h
\end{aligned}
\tag{12}
$$

(Bd. II, § 13). If the integral ideal $\mathfrak{a}$ in $\overline{O}$ belongs to the class $\overline{A}$, then we set for every χ:

$$\chi(\mathfrak{a})=\chi(\overline{A}), \tag{13}$$

and have thus determined the characters according to the group A for every ideal in $\overline{O}$. For any two ideals $\mathfrak{a}$ and $\mathfrak{b}$ from $\overline{O}$ the relation holds:

$$\chi(\mathfrak{a})\,\chi(\mathfrak{b})=\chi(\mathfrak{ab}). \tag{14}$$

Now from S_i we form the h sum:

$$Q_\kappa=\sum^{i}\chi_\kappa(\overline{A}_i)\,S_i, \tag{15}$$

for which, by (3) and (13), we may also write:

$$Q=\sum^{\mathfrak{a}}\frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s}. \tag{16}$$

In this, χ may be any of the characters χ_κ, and the sum extends over all ideals $\mathfrak{a}$ of $\overline{O}$. By (11), if we set

$$G_\kappa=\sum^{i}\chi_\kappa(\overline{A}_i)\,C_i$$

then

$$
\begin{aligned}
Q_1&=\frac{hG}{s-1}+G_1,\\
Q_\kappa&=G_\kappa, \quad \kappa>1.
\end{aligned}
\tag{17}
$$

where the $G_1, G_2, \ldots, G_h$ are functions of s which retain a finite value for $s=1$.

The sums Q can be transformed in the same way that we have already come to know in § 197 of the second volume.

Accordingly, if $\mathfrak{p}$ denotes any prime ideal in $\overline{O}$:

$$
\begin{aligned}
\frac{1}{1-\chi(\mathfrak{p})N(\mathfrak{p})^{-s}}&=1+\chi(\mathfrak{p})N(\mathfrak{p})^{-s}+\chi(\mathfrak{p})^2N(\mathfrak{p})^{-2s}+\cdots\\
&=1+\frac{\chi(\mathfrak{p})}{N(\mathfrak{p})^s}+\frac{\chi(\mathfrak{p}^2)}{N(\mathfrak{p}^2)^s}+\frac{\chi(\mathfrak{p}^3)}{N(\mathfrak{p}^3)^s}+\cdots,
\end{aligned}
$$

and since all ideals $\mathfrak{a}$ can be represented as products of such ideals $\mathfrak{p}$, it follows that:

(18) $$Q = \Pi \frac{1}{1 - \chi(\mathfrak{p}) N(\mathfrak{p})^{-s}},$$

where the infinite product Π extends over all prime ideals $\mathfrak{p}$ in $\overline{O}$.

To every ideal $\mathfrak{a}$ in $\overline{O}$ there belongs a certain exponent φ, that is, there is a certain lowest positive exponent φ for which $\mathfrak{a}^{\varphi}$ is contained in the principal class $\overline{A}_1$. The number φ is always a divisor of the class number h. We set:

(19) $$h = \varepsilon\varphi.$$

If in this sense $\mathfrak{a}$ belongs to the exponent φ, then all the characters $\chi_k(\mathfrak{a})$ are φth roots of unity, and every φth root of unity occurs ε times among them (Bd. II, § 198, 1.).

If, then, $\mathfrak{p}$ belongs to the exponent φ, we have

$$\begin{aligned}&[1 - \chi_1(\mathfrak{p}) N(\mathfrak{p})^{-s}][1 - \chi_2(\mathfrak{p}) N(\mathfrak{p})^{-s}] \cdots [1 - \chi_h(\mathfrak{p}) N(\mathfrak{p})^{-s}] \\ &\quad = [1 - N(\mathfrak{p})^{-\varphi s}]^{\varepsilon},\end{aligned}$$

and from (18) it follows that:

(20) $$Q_1 Q_2 \cdots Q_h = \Pi \frac{1}{[1 - N(\mathfrak{p})^{-\varphi s}]^{\varepsilon}}.$$

The question now concerns the limiting value which these expressions approach when $s = 1$.

As regards the left-hand side, it follows from (17) that

(21) $$\begin{cases}\operatorname{Lim}(s-1)\, Q_1 = hg \text{ finite and different from zero,} \\ \operatorname{Lim} Q_2\, Q_3 \ldots Q_h \text{ finite (perhaps zero)}\end{cases}$$

is. To assess the right-hand side we observe that the prime ideals $\mathfrak{p}$ are of the first or of the second degree, and that accordingly $N(\mathfrak{p})$ is equal to a natural prime or to the square of a natural prime. Further, we rely on the well-known theorem of analysis according to which an infinite product

$$(1 - q_1)\,(1 - q_2)\,(1 - q_3) \ldots$$

converges unconditionally and has a limiting value different from zero, if

$$q_1 + q_2 + q_3 + \cdots$$

is an unconditionally convergent series, and none of the $q_i = 1$.

We accordingly divide the prime ideals $\mathfrak{p}$ into two kinds, $\mathfrak{p}_1, \mathfrak{p}_2$, in such a way that the $\mathfrak{p}_1$ are to be of the first degree and to belong to the exponent $\varphi = 1$ (hence $\varepsilon = h$). The $\mathfrak{p}_2$ are to be either of the second degree, or to belong to a higher exponent φ. Thus the $\mathfrak{p}_1$ are the prime ideals of the first degree belonging to the principal class $\overline{A}_1$.

Then

$$N(\mathfrak{p}_1)^{\varphi} = N(\mathfrak{p}_1) = p_1$$

is a natural prime, and

$$N(\mathfrak{p}_2)^{\varphi} = p_2^k$$

is at least the 2nd (at most the $2h$th) power of a prime. By the analytic theorem just mentioned, the product

$$\Pi\,[1 - N(\mathfrak{p}_2)^{-s\varphi}]^{\varepsilon}$$

is then finite and different from zero for $s = 1$, and from (20), (21) there follows the theorem:

6. The limiting value

$$\lim_{s=1} \Pi \frac{s-1}{\Pi\left[1 - N(\mathfrak{p}_1)^{-s}\right]^h}$$

is finite when $\mathfrak{p}_1$ runs through the prime ideals of the first degree of the principal class $\overline{A}_1$.

A particular interest attaches to the question whether this limiting value can vanish.

§ 164. The Class Field.

To the number group A in the quadratic field Ω there shall exist an algebraic field over Ω, to be denoted by $\mathfrak{K}$ or $\mathfrak{K}(A)$, of which we shall assume only:

7. Definition of the class field.

The prime ideals $\mathfrak{p}_1$ of the first degree of the principal class $\overline{A}_1$, and only these, shall split again in the field $\mathfrak{K}(A)$ into prime ideals of the first degree.

Thus it is only the prime ideals in Ω, which we designated earlier by $\mathfrak{p}_1$, that split in $\mathfrak{K}$ into prime ideals of the first degree.

Any finite number of primes may thereby be excepted. They are then included in the excludent.

If n is the relative degree of the field $\mathfrak{K}$ over Ω, and p a natural prime which the ideals $\mathfrak{p}_1$ and $\mathfrak{P}$ in Ω and $\mathfrak{K}$ divide, and if one denotes the absolute norms in the fields Ω and $\mathfrak{K}$ by $N_\Omega, N_\mathfrak{K}$, then

$$N_\mathfrak{K}(\mathfrak{P}) = N_\Omega(\mathfrak{p}_1) = p, \quad N_\mathfrak{K}(\mathfrak{p}_1) = p^n,$$

and consequently $\mathfrak{p}_1$ must split in the field $\mathfrak{K}$ into n prime ideals $\mathfrak{P}$:

$$\mathfrak{p}_1 = \mathfrak{P}_1\mathfrak{P}_2 \ldots \mathfrak{P}_n. \tag{1}$$

The prime ideals which divide the fundamental numbers of the fields Ω or $\mathfrak{K}$ shall always be excluded. Then the $\mathfrak{P}_1, \mathfrak{P}_2, \ldots, \mathfrak{P}_n$ are distinct from one another.

Whether such a field $\mathfrak{K}$ exists remains for the present undecided. We will investigate what can be concluded from the definition.

Since n ideals $\mathfrak{P}$ divide every ideal $\mathfrak{p}_1$, we have

$$\prod \left[1 - N_\mathfrak{K}(\mathfrak{p}_1)^{-s}\right]^n = \prod \left[1 - N_\mathfrak{K}(\mathfrak{P})^{-s}\right], \tag{2}$$

in which the first product extends over all ideals $\mathfrak{p}_1$ of the field Ω, the second over all ideals $\mathfrak{P}$ of the first degree of the field $\mathfrak{K}$; and by Theorem I, § 197 of Band II, which holds for any field whatever,

$$(s-1)\prod \frac{1}{\left[1 - N_\mathfrak{K}(\mathfrak{P})^{-s}\right]} \tag{3}$$

has for $s = 1$ a finite limiting value different from zero, which is there reduced to the class number in the field $\mathfrak{K}$. If we put, for brevity,

$$P = \prod \frac{1}{\left[1 - N(\mathfrak{p}_1)^{-s}\right]}, \tag{4}$$

then according to (2) and (3), for $s = 1$

$$(s-1)\,P^n \text{ finite and different from zero,} \tag{5}$$

and according to § 163, 6.:

$$(s-1)\,P^h \text{ finite,} \tag{6}$$

and if one eliminates P from these, it follows that:

$$(s-1)^{n-h} \text{ finite.}$$

Therefore

$$n \overline{\overline{>}} h. \tag{7}$$

From this follows the theorem:

8. The relative degree n of the class field cannot be smaller than the class number h of the group. If $n = h$, then $(s-1)P^h$ is not zero for $s = 1$, and the sums $Q_2, Q_3, \ldots, Q_h$ are likewise different from zero.

If an equation of degree h can be formed in Ω from which a field can be derived that exhibits the characteristic properties of the class field, and if the degree of this field is not greater than h, then it follows that it cannot be smaller either, and so must be $= h$. With this the irreducibility of the equation in question is established.

On this rests the demonstration of the irreducibility of the cyclotomic equation which we gave in § 198 of the II. Band. There the field of rational numbers took the place of Ω. In the next section we shall make further applications of this.

Whether there are groups for which the degree of the class field is greater than the class number remains undecided. I consider it improbable.

We prove further that there can be at most one class field. Every such field is generated by the adjunction of an algebraic number ξ to Ω.

Let ξ be an integer generating the class field $\mathfrak{K} = \Omega(\xi)$, one which therefore satisfies an equation irreducible in Ω

$$f(\xi) = \xi^n + \alpha_1\xi^{n-1} + \alpha_2\xi^{n-2} + \cdots + \alpha_n = 0 \tag{8}$$

in which the $\alpha_1, \alpha_2, \ldots, \alpha_n$ are integers in Ω.

Every integer η of the field $\mathfrak{K}$ can (Bd. I, § 151) be represented in the form:

$$\eta = \frac{\varphi(\xi)}{f'(\xi)},$$

where $\varphi(\xi)$ has integral coefficients in Ω. One can put this in the form:

$$\gamma\eta = \gamma_0 + \gamma_1\xi + \gamma_2\xi^2 + \cdots + \gamma_{n-1}\xi^{n-1}, \tag{9}$$

where $\gamma, \gamma_0, \gamma_1, \ldots, \gamma_{n-1}$ are integers in Ω, and such that the first, γ, is a divisor of the discriminant of f. The prime factors of γ we exclude by means of the excludent.

Let

$\mathfrak{P}$ be a prime ideal of the first degree in $\mathfrak{K}$,

$\mathfrak{p}_1$ the prime ideal of the first degree in Ω that is divisible by $\mathfrak{P}$,

p the norm of $\mathfrak{p}_1$ and of $\mathfrak{P}$;

then for every integer α in Ω:

$$\alpha^p \equiv \alpha \pmod{\mathfrak{p}_1}$$

and

$$\xi^p \equiv \xi \pmod{\mathfrak{P}}, \tag{10}$$

hence, by (9), for every integer η in $\mathfrak{K}$:

$$\eta^p \equiv \eta \pmod{\mathfrak{P}}.$$

9. The necessary and sufficient condition for $\mathfrak{K}(\xi)$ to be a class field therefore consists in this, that for every prime ideal $\mathfrak{P}$ dividing a $\mathfrak{p}_1$, and only for this one, the congruence (10) is satisfied.

Let now

$$\mathfrak{K} = \Omega(\xi), \quad \mathfrak{K}' = \Omega(\xi')$$

be two class fields of the same group A. First of all, it readily follows that they must be of the same degree. For the condition (5) must be satisfied for both. If, then, n and n' are the degrees, $(s-1)^{n'-n}$ must be finite and different from zero for $s = 1$, which is only possible if $n = n'$.

By composition of $\mathfrak{K}$ and $\mathfrak{K}'$ we derive a third field:

$$\mathfrak{K}'' = \Omega(\xi, \xi'),$$

of which we shall show that it too is a class field.

We can generate the field $\mathfrak{K}''$ by means of a number:

$$\xi'' = \alpha\xi + \alpha'\xi', \tag{11}$$

in which α, α' are integers in Ω.

Let $\mathfrak{P}''$ be a prime ideal in $\mathfrak{K}''$ dividing some $\mathfrak{p}_1$, and let $\mathfrak{P}, \mathfrak{P}'$ be the prime ideals in $\mathfrak{K}, \mathfrak{K}'$ divisible by $\mathfrak{P}''$, and $N(\mathfrak{p}_1) = p$. Then by 9.:

$$\xi^p \equiv \xi \pmod{\mathfrak{P}}, \quad \xi'^p \equiv \xi' \pmod{\mathfrak{P}'}, \tag{12}$$

and consequently, by (11):

$$\xi''^p \equiv \alpha^p\xi^p + \alpha'^p\xi'^p \equiv \xi'' \pmod{\mathfrak{P}''}. \tag{13}$$

If conversely the congruence (13) is satisfied, then the congruences (12) follow from it again, because ξ and ξ' can be expressed rationally [in the form (9)] by means of ξ''.

Thus $\mathfrak{K}''$ is likewise a class field, and its degree is just as high as that of $\mathfrak{K}$ and $\mathfrak{K}'$. Hence ξ and ξ' are primitive numbers of $\mathfrak{K}''$, and consequently all these fields are identical (Bd. I, § 151, 3.). If $\mathfrak{K}$ is a class field, then the fields conjugate to $\mathfrak{K}$ are also class fields and are therefore identical with $\mathfrak{K}$.

With this the theorem is proved:

10. For a given number group there is only one class field, and this is a normal field.

We now wish to become more closely acquainted with the most important properties and applications of the class field in the cases arising from complex multiplication.

Let us assume that the number group A contains another group A' as a divisor of finite index (A, A'), and that these two groups possess class fields:

$$\begin{aligned} \mathfrak{K} &= \mathfrak{K}(A) = \Omega(\xi), \\ \mathfrak{K}' &= \mathfrak{K}(A') = \Omega(\xi'), \end{aligned} \tag{14}$$

then it can be proved that the field $\mathfrak{K}$ is contained in $\mathfrak{K}'$. For if $\mathfrak{p}_1, \mathfrak{p}_1'$ are the prime ideals of the first degree in Ω of the principal classes $\overline{A}_1, \overline{A}_1'$, then the $\mathfrak{p}_1'$ are contained among the $\mathfrak{p}_1$. If therefore p is the prime which $\mathfrak{p}_1$ and $\mathfrak{p}_1'$ divide, then by 9.:

$$\begin{aligned} &\xi^p \equiv \xi \pmod{\mathfrak{P}} \text{ for all } \mathfrak{p}_1 \text{ and consequently for all } \mathfrak{p}_1', \\ &\xi'^p \equiv \xi' \pmod{\mathfrak{P}'} \text{ for all } \mathfrak{p}_1', \end{aligned} \tag{15}$$

if $\mathfrak{P}$ and $\mathfrak{P}'$ are prime ideals of the first degree in $\mathfrak{K}$ and $\mathfrak{K}'$.

We now form the field:

(16) $$\mathfrak{K}'' = \Omega(\xi, \xi') = \Omega(\xi''),$$

by setting

(17) $$\xi'' = \alpha\xi + \alpha'\xi'$$

and we denote by $\mathfrak{P}''$ a prime ideal of the first degree of this field that divides some $\mathfrak{p}_1'$.

Let $\mathfrak{P}, \mathfrak{P}'$ be, as above, the prime ideals in $\mathfrak{K}, \mathfrak{K}'$ divisible by $\mathfrak{P}''$. Then by (15) and (17)

$$\xi''^p \equiv \xi'' \pmod{\mathfrak{P}''},$$

and consequently $\mathfrak{K}''$ is a class field of A'.

Accordingly $\mathfrak{K}''$ is identical with $\mathfrak{K}'$; ξ' is a primitive number of $\mathfrak{K}''$, and consequently every number in $\mathfrak{K}''$, and hence ξ as well, can be expressed rationally by ξ'.

From this it follows:

11. If a number group A contains another number group A' as a divisor of finite index, and if the class fields $\mathfrak{K}(A)$, $\mathfrak{K}(A')$ exist, then $\mathfrak{K}(A)$ is contained in $\mathfrak{K}(A')$.

§ 165. Prime Ideals in the Classes.

We now return to the functions Q which we represented in § 163, (16) by infinite series and in § 163, (18) by infinite products:

(1) $$Q_\varkappa = \Pi \frac{1}{1 - \chi_\varkappa(\mathfrak{p})\, N(\mathfrak{p})^{-s}}$$

in which $\chi_\varkappa$ denotes one of the characters of the group and $\mathfrak{p}$ runs through all prime ideals in $\overline{O}$. We develop the logarithm of this product in the following way:

(2) $$\log Q_\varkappa = \sum \frac{\chi_\varkappa(\mathfrak{p})}{N(\mathfrak{p})^s} + \frac{1}{2}\sum \frac{\chi_\varkappa(\mathfrak{p}^2)}{N(\mathfrak{p})^{2s}} + \frac{1}{3}\sum \frac{\chi_\varkappa(\mathfrak{p}^3)}{N(\mathfrak{p})^{3s}} + \cdots$$

We do not need to know the multiple of $2\pi i$ that is not more closely defined in the $\log Q$, since it could change with s only discontinuously; and since continuous functions of s stand on the right and on the left, as long as $s > 1$, this multiple does not change with s at all.

Let $\overline{A}_i$ be one of the classes of the ideal group $\overline{A}$ and $\overline{A}_i'$ the opposite class. Then by (12), (13), (14), § 163:

$$\sum^{k} \chi_\varkappa(\overline{A}_i')\, \chi_\varkappa(\mathfrak{a}) = h \text{ or } = 0,$$

according as the ideal $\mathfrak{a}$ is contained in the class A_i or in some other class. If we therefore multiply formula (2) by $\chi_\varkappa(\overline{A}_i')$ and sum with respect to $\varkappa$, it follows that:

(3) $$\frac{1}{h}\sum^{\varkappa} \chi_\varkappa(A_i') \log Q_k = \sum^{(1)} \frac{1}{N(\mathfrak{p})^s} + \frac{1}{2}\sum^{(2)} \frac{1}{N(\mathfrak{p})^{2s}} + \frac{1}{3}\sum^{(3)} \frac{1}{N(\mathfrak{p})^{3s}} + \cdots,$$

in which the sums $\sum^{(1)}, \sum^{(2)}, \sum^{(3)}, \ldots$ extend over all prime ideals $\mathfrak{p}$ whose first, second, third, ... power is contained in the class $\overline{A}_i$.

We pass in (3) to the limit $s = 1$; the sums $\sum^{(2)}, \sum^{(3)}, \ldots$ retain a finite value in doing so, since their unconditional convergence does not cease for $s = 1$. The same holds of that part of the sum $\sum^{(1)}$ which relates to prime ideals of higher than the first degree, and so it remains questionable only whether the sum

$$\sum \frac{1}{N(\mathfrak{p})^s},$$

extended over all prime ideals of the first degree of the class $\overline{A_i}$, remains finite or becomes infinite. The former would be the case if there were in $\overline{A_i}$ no prime ideals of the first degree at all, or only a finite number of them.

But if we assume the existence of a class field of degree h, then from § 164, 8. and § 163, (21) it follows that

$$\log Q_2,\ \log Q_3,\ \ldots,\ \log Q_h$$

retain finite limiting values, while $\log Q_1$ becomes infinite. The left-hand side of (3) therefore becomes infinite, and the right-hand side cannot remain finite.

Thus it is proved:

12. If a class field exists whose degree is not higher than the class number, then every class contains infinitely many prime ideals of the first degree.

§ 166. **Prime Ideals in the Ideal Classes.**

If we take as group first of all the totality of the numbers O of the field Ω without zero, then the classes are the ideal classes of the field Ω, which, as we have seen, can be uniquely assigned to the classes of quadratic forms of the discriminant Δ

. If we take as our group the order $O' = [Q]$ with the conductor Q, which is defined by the congruence condition that every number in O' shall be congruent to a rational number modulo Q, then we have a congruence group whose excludent and modulus is $= Q$. The classes correspond to the classes of forms of the discriminant $D = Q^2\Delta$. The second case includes the first within itself (for $Q = 1$). The class field is here the class field $\mathfrak{K}(D)$, which, to distinguish it from more general ones, we shall call the order field.

For if (k) is a class invariant, then (§ 122)

$$(k)^p \equiv (k) \pmod{\mathfrak{P}}$$

is satisfied only when p is representable by the principal class of the discriminant D. Then p splits in Ω into two conjugate prime ideals $\mathfrak{p}$, $\mathfrak{p}'$, which are representable by numbers in O or in O', and hence are principal ideals.

The field $\mathfrak{K}(\Delta)$ is the class field of Ω in the narrower sense, the one that corresponds to the ideal classes of this field, and it may also be designated the principal class field $\mathfrak{K}(\Omega)$.

In every ideal class, both with respect to O and with respect to O', there are therefore infinitely many prime ideals of the first degree. In other words: if $\mathfrak{a}$ is an arbitrary ideal (coprime to Q), then there are infinitely many prime ideals $\mathfrak{p}$ which are equivalent to $\mathfrak{a}$ with respect to O'. Then, by § 95, the forms corresponding to the ideals $\mathfrak{a}$ and $\mathfrak{p}$ are also equivalent, and p is representable by the form belonging to $\mathfrak{a}$. With this the theorem is proved:

1. By every primitive form of the discriminant D infinitely many primes are representable[37].

[37]This theorem, for negative as well as for positive discriminants, was first proved by Dirichlet, but the proof was published only in one special case (Bericht der Berliner Akademie 1840, Werke Bd. I, S. 497). The detailed proof

§ 167. **Primes in Linear Forms.**

We now define a number group A in Ω as follows:

Let $\mathfrak{m}$ be an integral ideal in Ω, and let A consist of all integral and fractional numbers α in Ω that are coprime to $\mathfrak{m}$ and satisfy the condition:

$$\alpha \equiv 1 \pmod{\mathfrak{m}}. \tag{1}$$

If we take the prime divisors of $\mathfrak{m}$ into the excludent, and denote by O the group of the numbers in Ω, then O splits with respect to the modulus $\mathfrak{m}$ into $\psi(\mathfrak{m})$ classes of numbers, in which $\psi(\mathfrak{m})$ has the meaning it has in Bd. II, § 168, namely, when $\mathfrak{p}$ runs through the prime divisors of $\mathfrak{m}$:

$$\psi(\mathfrak{m}) = N(\mathfrak{m})\ \Pi\Big(1 - \frac{1}{N(\mathfrak{p})}\Big). \tag{2}$$

To the group A there belong not merely individual numbers, but classes of numbers with respect to the modulus $\mathfrak{m}$. Accordingly:

$$(O, A) = \psi(\mathfrak{m}). \tag{3}$$

In order to determine the class number for our group, we apply the theorems of § 161.

According to these, if $\overline{E}$ denotes the group of the functional units and E the group of the numerical units:

$$(\overline{O}, \overline{E}A) = (\overline{O}, \overline{E}O)\,(\overline{E}O, \overline{E}A),$$

$$(\overline{E}O, \overline{E}A) = (\overline{E}O, \overline{E}EA) = (O, EA).$$

Further

$$(O, EA)\,(EA, A) = (O, A) = \psi(\mathfrak{m}).$$

$(EA, A) = e$ is the number of the units incongruent with respect to $\mathfrak{m}$, and

$$(\overline{O}, \overline{E}O) = H$$

is the class number of the field Ω,

$$(\overline{O}, \overline{E}A) = h$$

is the class number with respect to A. From this there results:

$$h = \frac{H\,\psi(\mathfrak{m})}{e}. \tag{4}$$

The number e is in general equal to the number of the units in Ω, hence $= 2$, and in the two exceptional cases $\Delta = -4$, $\Delta = -3$ we have $e = 4$ and $e = 6$. The number e is smaller when among the units there occur congruent ones (modulo $\mathfrak{m}$), which is only possible when $\mathfrak{m}$ is a divisor of 2 or of 3.

If E denotes the group of units, or also a group contained in it, then we obtain the same value of the class number and the same class field if we let EA take the place of A.

This holds not only for this particular group, but generally. We could therefore, e. g., also have defined the numbers α of the group A by

$$\alpha \equiv \pm 1 \pmod{\mathfrak{m}}$$

---

was given by H. Weber (Mathematische Annalen XX, 1882). A generalisation likewise originating with Dirichlet, to which we shall come back in the next section, was proved by A. Meyer (Crelles Journ., Bd. 103). The proof given here in the text rests on a different foundation, but for the present it of course applies (so long as the class field for positive discriminants is not known) only to negative discriminants.

since $E = +1, -1$ is a group also in the exceptional cases $\Delta = -4$, $\Delta = -3$.

Now it can be shown that the division field $\mathfrak{T}_\mathfrak{m}$ is the class field for A. If we first exclude all prime factors which divide the discriminant of the division field, and denote by τ any number of the division field, then by § 160

$$\tau^p \equiv \tau \pmod{\mathfrak{p}} \tag{5}$$

is satisfied only when $\mathfrak{p}$ is a prime ideal of the first degree in Ω which belongs to the principal class. But that is the criterion for the class field. By § 154, however, the degree of the division field is at most equal to the expression h given in (4); consequently it is exactly equal to h, and the division equation is irreducible. Moreover, it is thereby proved that in each of the ideal classes $\overline{A}_\nu$ with respect to A there are infinitely many prime ideals of the first degree.

One can give this theorem the following expression:

If β is an arbitrary number in O, and $\mathfrak{a}$ an arbitrary ideal in $\overline{O}$, then by

$$\overline{A}_\nu = \mathfrak{a}\beta A \tag{6}$$

a class with respect to A is determined (§ 161), and this class contains infinitely many prime ideals of the first degree. By (6) such a prime ideal can be represented in the form:

$$\mathfrak{p} = \mathfrak{a}\beta\alpha, \tag{7}$$

where $\alpha \equiv 1 \pmod{\mathfrak{m}}$. If we therefore set

$$\alpha\beta = \pi,$$

then

$$\pi \equiv \beta \pmod{\mathfrak{m}}.$$

From this it follows:

2. If β is a number in O, $\mathfrak{a}$ an ideal in $\overline{O}$, then there are infinitely many, in general fractional, numbers π in O which satisfy the condition

$$\pi \equiv \beta \pmod{\mathfrak{m}} \tag{8}$$

holds, and for which

$$\mathfrak{p} = \mathfrak{a}\pi \tag{9}$$

becomes a prime ideal.

If one takes $\mathfrak{a} = 1$, then π becomes an integer, and hence a prime existing in Ω. For this particular case the theorem reads:

3. In the field Ω there are infinitely many existing primes π whose norm is a natural prime, and which are congruent, with respect to an arbitrary modulus $\mathfrak{m}$, to an arbitrary number β coprime to $\mathfrak{m}$.

If we take for the modulus $\mathfrak{m}$ an integer μ in Ω, we obtain the following theorem:

4. If μ is an integer in Ω, then in the linear form

$$\mu\xi + \beta,$$

in which ξ runs through the integers of Ω and β is relatively prime to μ, infinitely many primes are contained.

Here one has a beautiful generalisation of the theorem on primes in arithmetic progressions (Bd. II, § 198)[38].

[38] For the case $\Delta = -4$ this proof has been given by H. Weber, Crelles Journ., Bd. 129 (Dirichlet-Band), for $\Delta = -3$ in a Strasbourg dissertation by H. Bresslau (Straßburg 1907).

§ 168. Reduction of the class equation in the cyclotomic fields.

We define a further group A by the following determination:

Let Q, m be two integral rational numbers whose prime factors we take up into the excludent S. Let O be, as in § 98, the group of the numbers in Ω, and O' the group of the numbers of the order $[Q]$, and let A consist of all integral and fractional numbers α in O' for which the congruence condition

$$N(\alpha) \equiv 1 \pmod{m} \tag{1}$$

holds. First of all we have the class number

$$h = (\overline{O}, \overline{E}A) = (\overline{O}, \overline{E}O')\,(\overline{E}O', \overline{E}A) \tag{2}$$

to determine. The first factor is the class number of the quadratic forms of the order O' [§ 100, (3)], which we denote by H'. We have therefore

$$h = H'(\bar{E}O', \bar{E}A) = H'(O', A) \tag{3}$$

(by § 100, 13.).

It remains to determine (O', A).

If $\varrho_1, \varrho_2, \ldots, \varrho_\mu$ denotes a complete system of representatives of O' with respect to A, then the numbers

$$N(\varrho_1),\ N(\varrho_2)\ldots,\ N(\varrho_\mu) \tag{4}$$

must all be different with respect to the modulus m, and every norm of a number in O is congruent with respect to m to one of these numbers. If therefore Z is the group of rational numbers, M the group of those numbers z from Z which satisfy the condition

$$z \equiv 1 \pmod{m}$$

and R_m the group of norm residues with respect to m, then (4) is a complete system of representatives of R_m with respect to M, and consequently

$$\mu = (O', A) = (R_m, M).$$

For this one may also put (§ 100, 12.):

$$(O', A) = \frac{(Z, M)}{(Z, R_m)}. \tag{5}$$

Now we have

$$(Z, M) = \varphi(m) = m\Pi\Big(1 - \frac{1}{r}\Big), \tag{6}$$

where r runs through the prime factors of m, and consequently

$$h = \frac{H'\varphi(m)}{(Z, R_m)}. \tag{7}$$

We obtain the class field $\mathfrak{K}(A)$ of our group by adjoining to the class field $\mathfrak{K}(D)$ a primitive mth root of unity:

$$\mathfrak{K}(A) = \mathfrak{K}(D, \varrho). \tag{8}$$

For the condition that a prime ideal $\mathfrak{P}$ of the field $\mathfrak{K}(D, \varrho)$ dividing a prime p be of the first degree is, if (k) denotes a class invariant of the discriminant D:

$$(k)^p \equiv (k), \quad \varrho^p \equiv \varrho \pmod{\mathfrak{P}}, \tag{9}$$

provided that all disturbing primes, e.g. those in the discriminants of the equations defining the numbers (k) and (ϱ), have been taken up into the excludent S.

The first of the conditions (9) requires that p be properly representable by the principal form of the discriminant D (§ 122), the second that

$$p \equiv 1 \pmod{m} \tag{10}$$

hold.

In order that a prime ideal $\mathfrak{p}$ of the first degree in Ω, which does not divide S, be contained in $\bar{E}A$, it is necessary and sufficient that

1. $\mathfrak{p}$ be a principal ideal, i. e. an existing number π, and that this number π be contained in A, i. e. that

2. $N(\pi) = p \equiv 1 \pmod{m}$

hold. These conditions agree exactly with the conditions (9), and consequently $\mathfrak{K}(D, \varrho)$ is the class field of the group A.

The degree of the field $\mathfrak{K}(A)$ is therefore not merely at most, but exactly equal to h.

The number (Z, R_m) is $= 1$ if none of the characteristic primes of § 108 divides m, and then the degree of the field $\mathfrak{K}(A)$ is equal to $H'\varphi(m)$. If, conversely, we include in m all the characteristic primes and prime powers, then $R_m = R$ becomes the group of absolute norm residues, and we have

$$(Z, R) = 2g,$$

if g is the number of genera of the discriminant D (§ 113, 7.). If, then, H'_g denotes the number of classes of one genus, then $H' = gH'_g$, and formula (7) gives:

$$h = \tfrac{1}{2} H'_g \varphi(m). \tag{11}$$

By § 138 the class function $H_{-D}(u)$ splits, through the adjunction of square roots, into as many factors as there are genera of classes of forms, and the degree of each of these factors is H'_g.

Now if the class equation were still further decomposable, through the adjunction of any roots of unity whatever, than according to the genera, then one could assume m in such a way that these roots of unity too, and also $\sqrt{\Delta}$, were contained in the field $\mathfrak{R}(\varrho)$.

If then h'' is the lowest degree to which the class equation can be reduced by the adjunction of roots of unity, then the absolute degree of the field $\mathfrak{K}(D, \varrho)$ is at most equal to $h''\varphi(m)$

and the relative degree with respect to Ω at most equal to $\frac{1}{2}h''\varphi(m)$, and from this it follows by (11) that h'' cannot be smaller than H'_g. Thus it is proved:

5. The class equation $H_{-D}(u)$ is, in the field that contains all roots of unity, not further decomposable than into the factors corresponding to the genera.

§ 169. Relation of the division fields to the class field.

In the last two sections we have considered two kinds of groups, of which the first led to the division fields of the elliptic functions, the second to the order fields in combination with roots of unity. Between these there exists a relation:

The first group A consisted of the numbers α in O which satisfied the condition

$$\alpha \equiv \pm 1 \pmod{\mathfrak{m}} \tag{1}$$

where $\mathfrak{m}$ is any ideal in Ω; the second group A' consisted of the numbers α' of the order $[Q]$ which satisfied the condition

$$N(\alpha') \equiv 1 \pmod{m} \tag{2}$$

where m and Q denote natural numbers.

If we assume that $\mathfrak{m}$ is a prime ideal or a power of a prime ideal, then we can choose Q and m in such a way that the group A' is contained in the group A.

For if r is a rational number, then, since α' belongs to the order $[Q]$:

$$\alpha' \equiv r \pmod{Q}, \quad N(\alpha') \equiv r^2 \pmod{Q}, \tag{3}$$

and if Q is divisible by m:

$$\alpha' \equiv r \pmod{m}, \tag{4}$$

hence by (2) and (3)

$$r^2 \equiv 1 \pmod{m}$$

$$(r-1)(r+1) \equiv 0 \pmod{m}, \tag{5}$$

we therefore assume Q to be divisible by m, and m by $\mathfrak{m}$; then by (5), if $\mathfrak{m}$, as we have assumed, is a prime ideal or a power of such a one, either

$$r \equiv 1 \text{ or } r \equiv -1 \pmod{\mathfrak{m}}, \tag{6}$$

and from (4) it follows that:

$$\alpha' \equiv \pm 1 \pmod{\mathfrak{m}}; \tag{7}$$

consequently A' is contained in A, and by § 164, 11. $\mathfrak{K}(A)$ is therefore contained in $\mathfrak{K}(A')$. Since now, by § 158, every division field can be composed of such fields whose divisor is a prime ideal or a power of a prime ideal, it follows that:

6. The division field of the elliptic functions can be reduced to order fields and cyclotomic fields[39].

FIFTH BOOK.

ALGEBRAIC FUNCTIONS.

Twenty-fourth Chapter.

Algebraic Functions of One Variable.

§ 170. Introductory.

The theory of algebraic functions of one variable is the starting point of the general investigations of Abel on the new transcendents, which have since received the name „Abelian functions“ and which are the highest generalisation of the elliptic functions. The principal problems of this theory have been brought to a certain conclusion by the papers of Riemann, Weierstrass, Clebsch, Brill and Noether. In particular, in the conception of many-sheeted (Riemann) surfaces Riemann has created a means for the investigation of these functions that is extraordinarily effective through its intuitive clarity.

[39] A part of this theorem is proved by another route in the Straßburg dissertation of Daniel Bauer, “Über den Teilungskörper der elliptischen Funktionen mit singulärem Modul”. Straßburg 1903.

All these investigations, however, which rest on the one hand upon the theory of functions, on the other hand, as with Clebsch, Brill, Noether, upon the methods of rational algebra (theory of forms and invariants, or of algebraic curves), must always make certain restrictions; they must exclude certain „exceptional cases“ and content themselves with the treatment of the so-called general cases. Not subject to this restriction is the theory of algebraic functions worked out on the analogy of number theory by Dedekind and myself, of which a survey is to be given here[1].

[1] On the literature concerning algebraic and Abelian functions let the following be mentioned here: Abel, Mém. sur une propriété générale d'une classe très-étendue de fonctions transcendentes (presented to the Paris Academy in 1826, Werke, new edition by Sylow and Lie, Bd. I, S. 145). Riemann, Theorie der Abelschen Funktionen, Crelles Journ., Bd. 54, 1857, Werke 2. Aufl., Nr. 88. Weierstrass, Vorlesungen 1875/76, Werke, Bd. IV. Clebsch, Über die

§ 171. Definition of algebraic functions.

A variable θ is called an algebraic function of an independent variable z when the two variables are connected with one another by an algebraic equation

$$F(\theta, z) = 0 \tag{1}$$

In this, F denotes an expression of the form

$$F(\theta, z) = a_0\theta^n + a_1\theta^{n-1} + \cdots + a_{n-1}\theta + a_n, \tag{2}$$

whose coefficients $a_0, a_1, \ldots, a_n$ are integral rational functions of z without a common divisor. About the constant coefficients in these expressions we make no further assumption than that they are real or complex numbers[1].

We assume the function $F(\theta, z)$ to be irreducible in the sense that it shall not split into factors which are themselves rational functions of θ and z.

Every integral function $G(\theta, z)$ can, according to Bd. I, § 20, be decomposed in only one way into irreducible factors (apart from constant factors). We say that the function $G(\theta, z)$

Anwendung der Abelschen Funktionen in der Geometrie, Crelles Journ., Bd. 63, 1864. Clebsch and Gordan, Theorie der Abelschen Funktionen, Leipzig 1866. Brill and Noether, Über die algebraischen Funktionen und ihre Anwendung in der Geometrie, Mathematische Annalen VII, 1874. Brill and Noether, Die Entwickelung der Theorie der algebraischen Funktionen, Bericht der Deutschen Mathematiker-Vereinigung 1894. Appell et Goursat, Théorie des fonctions algébriques, Paris 1895. On the number-theoretic methods: Kronecker, Über die Diskriminante algebraischer Funktionen, Crelles Journ., Bd. 91, 1881, and: Festschrift zu Kummers Doktor-Jubiläum, Crelles Journ., Bd. 92, 1881. Dedekind and Weber, Theorie der algebraischen Funktionen einer Veränderlichen, Crelles Journ., Bd. 92, 1879. Hensel and Landsberg, Theorie der algebraischen Funktionen einer Variablen, Leipzig 1902. The form of the theory presented in what follows rests upon the concept of functionals introduced in Bd. II, § 153, which in this form had already been applied, in a Strassburg dissertation prompted by Wellstein and written by Rehfeld (1906), to the treatment of a special problem of the theory of functions.

[1] No gap would remain anywhere if we wished to restrict ourselves here to algebraic numbers. If one regards θ, z as Cartesian coordinates in the plane, then equation (1) represents an algebraic curve, and one can apply the theory of these curves, as Clebsch and Gordan and Brill and Noether have done. To be sure, only the real values of these variables have a really intuitive meaning here.

shall vanish then and only then, when the function $F(\theta, z)$ occurs among its irreducible factors. This is the definition of vanishing, according to which, if one wished to be entirely correct, every

equation $G(\theta, z) = 0$ would have to be understood as a congruence with respect to the modulus F. In order not to be prolix, however, we shall not employ this mode of expression here.

If we divide equation (1) by a_0, it takes the form

$$f(\theta, z) = \theta^n + b_1\theta^{n-1} + b_2\theta^{n-2} + \cdots + b_{n-1}\theta + b_n, \tag{3}$$

in which the $b_1, b_2, \ldots, b_n$ are integral or fractional rational functions of z.

The system of all integral and fractional rational functions $\Phi(\theta, z)$ of θ and z in which the denominator is not divisible by F, and so does not vanish, has the property of reproducing itself by addition, subtraction, multiplication and division (except by 0), and therefore forms a field of algebraic functions (Bd. I, § 146), which I shall denote by Ω.

The degree n of the irreducible equation (2) or (3), that is, of the equation of lowest degree which θ satisfies, is called the degree of the algebraic field with respect to θ, or also the degree of θ.

If $\varphi(\theta)$ is an integral function of θ whose coefficients are integral or fractional rational functions of z, then by division one can form an equation:

$$\varphi(\theta) = f(\theta)q(\theta) + r(\theta),$$

in which $q(\theta)$ and $r(\theta)$ are functions of the same kind as φ, but of which $r(\theta)$ is at most of degree $n - 1$. By reason of (3) we then have

$$\varphi(\theta) = r(\theta). \tag{4}$$

If $\varphi(\theta)$ is not divisible by $f(\theta)$, then the two functions have no rational divisor in common, and one can determine two functions of the field $f_1(\theta), \varphi_1(\theta)$ in such a way that

$$f(\theta)f_1(\theta) + \varphi(\theta)\varphi_1(\theta) = 1$$

holds (Bd. I, § 6). Since now $f(\theta) = 0$, there follows from this:

$$\varphi_1(\theta) = \frac{1}{\varphi(\theta)}. \tag{5}$$

One can therefore replace every fractional function of θ by an integral function of θ, and so obtains the theorem:

1. Every function ζ of the field Ω can be put in one single way into the form:

$$\zeta = x_0 + x_1\theta + x_2\theta^2 + \cdots + x_{n-1}\theta^{n-1}, \tag{6}$$

whose coefficients $x_0, x_1, x_2, \ldots, x_{n-1}$ are rational functions of z.

That this is possible at all is inferred from (4) and (5); that it is possible in only one way follows from the irreducibility of $f(\theta)$; for if one had two different representations of one and the same quantity ζ in the form (6), their difference would yield an equation for θ of a degree lower than n.

If one chooses n functions of the field Ω arbitrarily:

$$\begin{aligned}
\eta_1 &= x_0^{(1)} + x_1^{(1)}\theta + \cdots + x_{n-1}^{(1)}\theta^{n-1}, \\
\eta_2 &= x_0^{(2)} + x_2^{(2)}\theta + \cdots + x_{n-1}^{(2)}\theta^{n-1}, \\
&\cdots\cdots\cdots\cdots \\
\eta_n &= x_0^{(n)} + x_1^{(n)}\theta + \cdots + x_{n-1}^{(n)}\theta^{n-1},
\end{aligned} \tag{7}$$

with the single restriction that the determinant

$$\Sigma \pm x_0^{(1)} x_1^{(2)} \cdots x_{n-1}^{(n)}$$

is not identically zero, then by elimination of the powers of θ from (6) and (7) every function ζ in Ω can be put into the form:

$$\zeta = y_1\eta_1 + y_2\eta_2 + \cdots + y_n\eta_n, \tag{8}$$

whose coefficients y_i are rational functions of z.

Such a system of functions $\eta_1, \eta_2, \ldots, \eta_n$ is called a basis of the field Ω.

In order that a system of functions $\eta_1, \eta_2, \ldots, \eta_n$ form a basis of Ω, it is necessary and sufficient that no equation of the form

$$y_1\eta_1 + y_2\eta_2 + \cdots + y_n\eta_n = 0$$

subsist in which the coefficients $y_1, y_2, \ldots, y_n$ do not all vanish identically.

Thus the functions

$$1, \theta, \theta^2, \ldots, \theta^{n-1}$$

form a basis of Ω.

The transition from one basis $\eta_1, \eta_2, \ldots, \eta_n$ to another $\eta_1', \eta_2', \ldots, \eta_n'$ takes place through a linear substitution:

$$(\eta_1', \eta_2', \ldots, \eta_n') = L(\eta_1, \eta_2, \ldots, \eta_n) \tag{9}$$

or, written more briefly,

$$(\eta_i') = L(\eta_i), \tag{10}$$

in which L has the meaning

$$L = \begin{pmatrix} l_{1,1}, & l_{1,2}, & \ldots, & l_{1,n} \\ l_{2,1}, & l_{2,2}, & \ldots, & l_{2,n} \\ \cdot & \cdot & \cdot & \cdot \\ l_{n,1}, & l_{n,2}, & \ldots, & l_{n,n} \end{pmatrix} \quad \text{(Bd. II, § 41).} \tag{11}$$

The elements $l_{i,k}$ of this substitution are rational functions of z.

The system (η') is again a basis then and only then, when the determinant $|L|$ of this substitution does not vanish.

§ 172.
Norms and Traces.

If, for the representation of the functions in Ω, one chooses an arbitrary basis $\eta_1, \eta_2, \ldots, \eta_n$, then, ζ being an arbitrary function of the field, one can set, since the products $\zeta\eta_i$ are contained in the field:

$$\begin{aligned} \zeta\eta_1 &= y_{1,1}\eta_1 + y_{1,2}\eta_2 + \cdots + y_{1,n}\eta_n, \\ \zeta\eta_2 &= y_{2,1}\eta_1 + y_{2,2}\eta_2 + \cdots + y_{2,n}\eta_n, \\ &\cdots\cdots\cdots\cdots \\ \zeta\eta_n &= y_{n,1}\eta_1 + y_{n,2}\eta_2 + \cdots + y_{n,n}\eta_n, \end{aligned} \tag{1}$$

and from this one obtains, by elimination of the η:

$$\begin{vmatrix} y_{1,1} - \zeta, & y_{1,2}, & \ldots, & y_{1,n} \\ y_{2,1}, & y_{2,2} - \zeta, & \ldots, & y_{2,n} \\ \cdot & \cdot & \cdot & \cdot \\ y_{n,1}, & y_{n,2}, & \ldots, & y_{n,n} - \zeta \end{vmatrix} = 0, \tag{2}$$

an equation which, ordered according to powers of ζ, takes the shape:

$$\varphi(\zeta) = \zeta^n + b_1\zeta^{n-1} + \cdots + b_{n-1}\zeta + b_n = 0, \tag{3}$$

in which $b_1, b_2, \ldots, b_n$ are rational functions of z. These coefficients are independent of the choice of the basis η_i. For the equations (1), by which the transition from the basis η_i

to the basis $\xi\eta_i$ is effected, we can represent, following the notation (9), (10) of the preceding section, in this way:

$$(\xi\eta_i) = Y(\eta_i), \tag{4}$$

and accordingly there results, by transition to the basis η_i':

$$(\xi\eta_i') = LYL^{-1}(\eta_i').$$

The functions $b_1, b_2, \ldots, b_n$ are therefore completely determined by the function ξ.

The function

$$(-1)^n b_n = \begin{vmatrix} y_{1,1}, & y_{1,2}, & \cdots, & y_{1,n} \\ y_{2,1}, & y_{2,2}, & \cdots, & y_{2,n} \\ \cdot & \cdot & \cdot & \cdot \\ y_{n,1}, & y_{n,2}, & \cdots, & y_{n,n} \end{vmatrix} \tag{5}$$

is called the norm of the function ξ and is denoted by $N(\xi)$. Concerning it the following theorems hold:

1. If ξ is not identically zero, then $N(\xi)$ is different from zero.

For if the determinant of the system (1) vanishes, then rational functions $y_1, y_2, \ldots, y_n$, not all of which vanish, can be determined so that

$$\xi(y_1\eta_1 + y_2\eta_2 + \cdots + y_n\eta_n) = 0$$

holds, and this requires, since $\eta_1, \eta_2, \ldots, \eta_n$ is a basis, that $\xi = 0$.

2. If ξ is a rational function of z, then its norm is the nth power of this function.

For if ξ is rational, the equations (1) reduce to the identities $\xi\eta_i = \xi\eta_i$. Accordingly all terms in the determinant (5) vanish with the exception of the diagonal terms, and these all become equal to ξ. We express this theorem by the formula:

$$N(a) = a^n.$$

3. If ξ, ξ' are two functions in Ω, then the theorem holds

$$N(\xi\xi') = N(\xi)N(\xi'). \tag{6}$$

For if, according to (4):

$$(\xi\eta_i) = Y(\eta_i), \quad (\xi'\eta_i) = Y'(\eta_i),$$

then

$$(\xi\xi'\eta_i) = YY'(\eta_i),$$

and from this formula (6) follows, because the determinant of a composed linear substitution, here YY', is equal to the product of the determinant of the components.

4. From 2. and 3. it follows, when ζ is different from zero:

$$N(\zeta)\, N\left(\frac{1}{\zeta}\right) = 1,$$

and from this, for any two ζ, ζ':

$$N\left(\frac{\zeta'}{\zeta}\right) = \frac{N(\zeta')}{N(\zeta)}. \tag{7}$$

5. If t is an indeterminate or variable quantity, and $\varphi(t)$ the function which by (3) becomes zero for $t = \zeta$, then

$$\varphi(t) = N(t - \zeta). \tag{8}$$

This follows from (1) and (2). For if in (1) one replaces ζ by $\zeta - t$, this amounts to the same thing as replacing $y_{i,i}$ by $y_{i,i} - t$ and leaving the remaining $y_{i,k}$ unchanged.

If we decompose the function $\varphi(t)$, which is a rational function of t and z, into a product of irreducible factors $\varphi_1(t), \varphi_2(t), \ldots$ then one of these factors must vanish for $t = \zeta$. Let this be $\varphi_1(t)$, and let

$$\varphi_1(t) = t^e + b'_1 t^{e-1} + \cdots + b'_{e-1} t + b'_e \tag{9}$$

be of degree e. Then $\varphi_1(\zeta) = 0$ is the equation of lowest degree which ζ satisfies, and from ζ one derives a field Ω_1 which is contained in Ω and has the degree e. Every number η of the field Ω_1 can be represented, and that in only one way, in the form:

$$\eta = x_0 + x_1\zeta + \cdots + x_{e-1}\zeta^{e-1}. \tag{10}$$

Let further

$$\theta^f + \eta_1\theta^{f-1} + \cdots + \eta_{f-1}\theta^{f-1} + \eta_f = 0 \tag{11}$$

be the equation of lowest degree with coefficients in Ω_1 which θ satisfies, so that f is the relative degree of Ω with respect to Ω_1.

By means of the ef functions

$$\zeta^h\theta^k \qquad \begin{matrix} h = 0, 1, \ldots, e-1 \\ k = 0, 1, \ldots, f-1 \end{matrix} \tag{12}$$

every function of the field can be expressed linearly, and between them there subsists no linear equation with coefficients depending rationally on z. For otherwise θ or ζ would arise from an equation of lower degree than f or e. Ac-

cordingly the functions (12) form a basis of the field Ω, and it follows that

$$ef = n. \tag{13}$$

Thus e and f are divisors of n.

If

$$\xi_1, \xi_2, \ldots, \xi_e \tag{14}$$

denotes a basis of the field Ω_1, then the n quantities:

$$\xi_1\theta^k, \xi_2\theta^k, \ldots, \xi_e\theta^k, \qquad k = 0, 1, 2, \ldots, f-1 \tag{15}$$

form a basis of Ω. If one forms the substitution of the e quantities:

$$(\xi\xi_i) = Y_1(\xi_i), \tag{16}$$

then

$$(-1)^e b'_e = |Y_1| = N_1(\xi) \tag{17}$$

is the norm of ξ in the field Ω_1, and if one forms the substitution for $(\xi\xi_i\theta_k)$ in the field Ω with the basis (15), there results

$$(\xi\xi_i\theta^k) = Y_1(\xi_i)\, Y_1(\xi_i\theta) \ldots Y_1(\xi_i\theta^{e-1}),$$

from which

$$N(\xi) = |Y_1|^f. \tag{18}$$

From the substitution determinant $|Y_1|$ one derives the function $\varphi_1(t)$ by formula (2), and so it follows from (18):

$$N(t-\xi) = [\varphi_1(t)]^f, \tag{19}$$

and from this:

6. The function $\varphi(t) = N(t-\xi)$ is either irreducible or a power of an irreducible function.

If the exponent f of this power is greater than 1, then the field Ω is imprimitive, Ω_1 is a subfield, and Ω is a field of degree f over Ω_1 (Bd. I, § 151).

If $f = 1$, $e = n$, then ξ is called a primitive function of the field Ω, and Ω_1 is identical with Ω.

We return to equation (2) or (3), which, as we have seen, is independent of the choice of the basis, and define the coefficient

$$-b_1 = y_{1,1} + y_{2,2} + \cdots + y_{n,n} = S(\xi) \tag{20}$$

as the trace of the function ξ. For the trace there result from the definition the following fundamental theorems, in which ξ, ξ' denote any two functions in Ω, and x a rational function:

$$\begin{aligned} S(0) &= 0, \\ S(1) &= n, \\ S(x\xi) &= x\, S(\zeta), \\ S(\zeta+\zeta') &= S(\zeta) + S(\zeta'). \end{aligned} \tag{21}$$

All traces are rational functions of z.

§ 173. Discriminants.

If $(\eta_1, \eta_2, \ldots, \eta_n) = (\eta_i)$ is a system of n functions in the field Ω, then the n^2 traces $S(\eta_i\eta_k)$ are rational functions of z. We define as the discriminant of the system (η_i) the determinant

$$\Delta(\eta_1, \eta_2, \ldots, \eta_n) = \begin{vmatrix} S(\eta_1\eta_1), & S(\eta_1\eta_2), & \ldots, & S(\eta_1\eta_n) \\ S(\eta_2\eta_1), & S(\eta_2\eta_2), & \ldots, & S(\eta_2\eta_n) \\ \cdots & \cdots & \cdots & \cdots \\ S(\eta_n\eta_1), & S(\eta_n\eta_2), & \ldots, & S(\eta_n\eta_n) \end{vmatrix}, \tag{1}$$

for which we also write, more briefly, $\Delta(\eta_i)$.

We prove the theorem:

7. The discriminant $\Delta(\eta_i)$ is then and only then different from zero, when (η_i) is a basis of the field Ω.

In order to prove it, let us first assume that $\Delta(\eta_i) = 0$. Then, by an elementary theorem on determinants, the rational functions $y_1, y_2, \ldots, y_n$ can be so determined, without their being all zero, that

(2)
$$y_1 S(\eta_1\eta_k) + y_2 S(\eta_2\eta_k) + \cdots + y_n S(\eta_n\eta_k) = S[\eta_k(y_1\eta_1 + y_2\eta_2 + \cdots + y_n\eta_n)] = 0 \qquad (k = 1, 2, \ldots, n)$$

holds. If $x_1, x_2, \ldots, x_n$ denotes an arbitrary system of rational functions, and one sets

$$\begin{aligned} y_1\eta_1 + y_2\eta_2 + \cdots + y_n\eta_n &= \eta, \\ x_1\eta_1 + x_2\eta_2 + \cdots + x_n\eta_n &= \xi, \end{aligned} \tag{3}$$

then from (2), by multiplication by x_k and summation, there follows:

$$S(\xi\eta) = 0. \tag{4}$$

Now if (η_i) is a basis, then ξ can be any function whatever in Ω, and hence also $1/\eta$, and then formula (4) gives the contradictory result $S(1) = 0$; the discriminant of a basis (η_i) can therefore not vanish.

In order to prove the converse as well, we assume that (η_i) is a basis and that

$$(\eta_i') = X(\eta_i) \tag{5}$$

is a linear substitution. The system (η_i') is then and only then likewise a basis, when the determinant $|X|$ of this substitution is different from zero (§ 171). If one sets, in accordance with (5),

$$\eta_h' = x_{h,1}\eta_1 + x_{h,2}\eta_2 + \cdots + x_{h,n}\eta_n,$$

then

$$S(\eta_h'\eta_k') = \sum^{i\,i'} x_{h,i}x_{k,i'}S(\eta_i\eta_{i'}),$$

and by forming the determinant from this:

$$\Delta(\eta_1', \eta_2', \ldots, \eta_n') = |X|^2\Delta(\eta_1, \eta_2, \ldots, \eta_n). \tag{6}$$

If, then, $\Delta(\eta_i')$ is different from zero, $|X|$ cannot vanish, and (η_i') is a basis. With this, 7. is proved.

From (6) there follows further, by the definition of the norm in § 172, if one sets $\eta_i' = \zeta\eta_i$, where ζ is an arbitrary function in Ω:

$$\Delta(\zeta\eta_i) = N(\zeta)^2\Delta(\eta_i). \tag{7}$$

§ 174. The Power Sums.

The traces of the powers of θ

$$s_k = S(\theta^k) \tag{1}$$

are nothing other than the power sums, which can be calculated by the Newton formulae. But since we may not speak here of the „roots“ of the equation $f(\theta) = 0$, which we do not yet possess, these formulae must be derived directly from the definition of the trace.

If $f(\theta) = 0$ is the equation defining the field Ω, we set

$$f(t) = t^n + a_1t^{n-1} + \cdots + a_{n-1}t + a_n \tag{2}$$

and form the quotient:

$$\frac{f(t)}{t-\theta} = \eta_0 + \eta_1 t + \eta_2 t^2 + \cdots + \eta_{n-1}t^{n-1}, \tag{3}$$

in which

$$\begin{aligned} \eta_0 &= a_{n-1} + a_{n-2}\theta + \cdots + a_1\theta^{n-2} + \theta^{n-1}, \\ \eta_1 &= a_{n-2} + a_{n-3}\theta + \cdots + \theta^{n-2}, \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ \eta_{n-2} &= a_1 + \theta, \\ \eta_{n-1} &= 1. \end{aligned} \tag{4}$$

Between these functions there hold the relations:

$$
\begin{aligned}
\theta\,\eta_0 &= -a_n\eta_{n-1},\\
\theta\,\eta_1 &= \eta_0 - a_{n-1}\eta_{n-1},\\
\theta\,\eta_2 &= \eta_1 - a_{n-2}\eta_{n-1},\\
&\cdots\cdots\cdots\cdots\cdots\\
\theta\,\eta_{n-1} &= \eta_{n-2} - a_1\eta_{n-1},
\end{aligned}
\tag{5}
$$

and the $\eta_0, \eta_1, \ldots, \eta_{n-1}$ form a basis of Ω, since by (4) the powers of θ can be expressed linearly by the η_i.

If, then, ζ is an arbitrary function in Ω, one can set:

$$\zeta = y_0\eta_0 + y_1\eta_1 + \cdots + y_{n-1}\eta_{n-1}. \tag{6}$$

From $y_0, y_1, \ldots, y_{n-1}$ we derive a continuation of the sequence of these rational functions $y_n, y_{n+1}, y_{n+2}, \ldots$ by setting:

$$a_n y_r + a_{n-1}y_{r+1} + \cdots + a_1 y_{r+n-1} + y_{r+n} = 0 \qquad (r = 0, 1, 2, \ldots). \tag{7}$$

Then from (5) and (6) there results:

$$
\begin{aligned}
\zeta\theta &= \eta_0 y_1 + \eta_1 y_2 + \cdots + \eta_{n-2}y_{n-1},\\
&\quad - \eta_{n-1}(a_n y_0 + a_{n-1}y_1 + \cdots + a_1 y_{n-1}),\\
&= y_1\eta_0 + y_2\eta_1 + \cdots + y_n\eta_{n-1},
\end{aligned}
$$

and in general, for a positive r:

$$\zeta\theta^r = y_r\eta_0 + y_{r+1}\eta_1 + \cdots + y_{r+n-1}\eta_{n-1}, \tag{8}$$

as one easily confirms from (5) and (7) by the inference from r to $r+1$. If one orders this sum according to powers of θ, there results an expression of the form

$$\zeta\theta^r = x_{0,r} + x_{1,r}\theta + x_{2,r}\theta^2 + \cdots + x_{n-1,r}\theta^{n-1}, \tag{9}$$

wherein

$$
\begin{aligned}
x_{0,r} &= y_r a_{n-1} + y_{r+1}a_{n-2} + \cdots + y_{r+n-2}a_1 + y_{r+n-1},\\
x_{1,r} &= y_r a_{n-2} + y_{r+1}a_{n-3} + \cdots + y_{r+n-1},\\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\
x_{n-2,r} &= y_r a_1 + y_{r+1},\\
x_{n-1,r} &= y_r.
\end{aligned}
\tag{10}
$$

But by the definition of the trace [§ 172, (20)] we have

$$
\begin{aligned}
S(\zeta) &= x_{0,0} + x_{1,1} + \cdots + x_{n-1,n-1},\\
&= y_0 a_{n-1} + 2y_1 a_{n-2} + 3y_2 a_{n-3} + \cdots + n y_{n-1}.
\end{aligned}
\tag{11}
$$

If one wishes to apply this expression to $\zeta = \eta_r$, one has to set $y_r = 1$ and the remaining $y = 0$, and obtains

$$S(\eta_r) = (r+1)a_{n-r-1}, \quad S(\eta_{n-1-r}) = (n-r)a_r,$$

and consequently, so long as $r \leqq n$, by (4) with the notation (1):

$$(n-r)a_r = a_r s_0 + a_{r-1}s_1 + \cdots + a_1 s_{r-1} + s_r, \tag{12}$$

and if one takes the trace of $\theta^r f(\theta) = 0$:

$$0 = a_n s_r + a_{n-1}s_{r+1} + \cdots + a_1 s_{r+n-1} + s_{r+n}. \tag{13}$$

These are the Newtonian formulae (Bd. I, § 46).

If we denote by

$$f'(t) = nt^{n-1} + (n-1)a_1 t^{n-2} + \cdots + 2a_{n-2}t + a_{n-1}$$

the derived function of $f(t)$, multiply equation (12) by θ^{n-r-1} and sum from $r = 0$ to $r = n-1$, then it follows that:

$$f'(\theta) = \eta_0 s_0 + \eta_1 s_1 + \cdots + \eta_{n-1}s_{n-1}, \tag{14}$$

and if, for any positive k, one multiplies equations (12) by $\theta^{n-r+k-1}$ and takes in as many of equations (13) besides as are needed until $n - r + k - 1$ begins to become negative, there results:

$$\theta^k f'(\theta) = \eta_0 s_k + \eta_1 s_{k+1} + \cdots + \eta_{n-1}s_{k+n-1}. \tag{15}$$

To form the norm of $f'(\theta)$, one would have first to express, in these equations, the basis $\eta_0, \eta_1, \ldots, \eta_{n-1}$ by means of the substitution (2) in terms of $(1, \theta, \theta^2, \ldots, \theta^{n-1})$, then to form equation (15) for $k = 0, 1, \ldots, n-1$ and to take the determinant of this system. Instead of this one may take the determinant with respect to the η, and then multiply by the substitution determinant:

$$\begin{vmatrix} a_{n-1}, & a_{n-2}, & \ldots, & a_1, & 1 \\ a_{n-2}, & a_{n-3}, & \ldots, & 1, & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1, & 0, & \ldots, & 0, & 0 \end{vmatrix} = (-1)^{\frac{1}{2}n(n+1)}$$

In this way one obtains

$$N[f'(\theta)] = (-1)^{\frac{1}{2}n(n+1)} \begin{vmatrix} s_0, & s_1, & \ldots, & s_{n-1} \\ s_1, & s_2, & \ldots, & s_n \\ \cdots & \cdots & \cdots & \cdots \\ s_{n-1}, & s_n, & \ldots, & s_{2n-2} \end{vmatrix}. \tag{16}$$

But this determinant is, by the definition of the discriminant (§ 173):

$$\Delta(1, \theta, \theta^2, \ldots, \theta^{n-1}),$$

and so we have:

$$Nf'(\theta) = (-1)^{\frac{1}{2}n(n+1)}\Delta(1, \theta, \theta^2, \ldots, \theta^{n-1}). \tag{17}$$

In the discussion of this chapter we have, in accordance with the principal aim of the investigation, regarded the coefficients $a_1, a_2, \ldots, a_n$ as rational functions of z. Everything, however, remains entirely unchanged if we fix upon any domain of rationality whatever for these quantities.

§ 175. Integral Functions of z.

Every function ω of the field Ω satisfies, as we have seen, an equation of lowest degree:

$$\omega^e + b_1\omega^{e-1} + b_2\omega^{e-2} + \cdots + b_e = 0, \tag{1}$$

whose coefficients $b_1, b_2, \ldots, b_e$ are rational functions of z.

1. If these coefficients are integral rational functions of z, then ω is called an integral algebraic function, or briefly an integral function, of z.

For the integral algebraic functions the same theorems hold as for the integers (Bd. II, § 149).

If we set

$$\varphi(t) = t^e + b_1t^{e-1} + b_2t^{e-2} + \cdots + b_e, \tag{2}$$

then, as we have seen in § 172,

$$N(t - \omega) = \varphi(t)^f \quad (ef = n) \tag{3}$$

is an integral power of $\varphi(t)$. If we therefore arrange $N(t - \omega)$ according to powers of t, the coefficients all become integral rational functions again, and hence in particular:

2. The norm and the trace of an integral function ω are integral rational functions of z.

3. A rational function of z is an integral algebraic function only when it is an integral rational function.

For if $\omega = -b$ is rational, then $e = 1$ and $\omega + b = 0$ is the equation of lowest degree for ω, and hence ω is integral only when b is integral and rational.

4. Every function η in Ω can be transformed into an integral algebraic function by multiplication by a rational function of z different from zero.

For every function η satisfies an equation of lowest degree:

$$b_0\eta^e + b_1\eta^{e-1} + \cdots + b_{e-1}\eta + b_e = 0, \tag{4}$$

and if one multiplies by b_0^{e-1} and sets $b_0\eta = \omega$, one obtains an equation of the form (1).

5. A function ω of the field Ω is an integral function if it satisfies any equation

$$\psi(\omega) = \omega^m + c_1\omega^{m-1} + \cdots + c_{m-1}\omega + c_m = 0 \tag{5}$$

in which $c_1, c_2, \ldots, c_m$ are integral rational functions, even if this is not the equation of lowest degree.

For if $\varphi(\omega) = 0$ is the equation of lowest degree for ω, then $\psi(t)$ must, for a variable t, be divisible by $\varphi(t)$. That is simply the content of equation (5). We have therefore

$$\psi(t) = \chi(t)\varphi(t),$$

in which $\psi(t)$ is likewise an integral function of t. If now, as was assumed, $\psi(t)$ has integral coefficients, then by a theorem of Gauss the coefficients of $\varphi(t)$ and of $\chi(t)$ must also be integral functions. Cf. Bd. I, § 2, § 20, § 148 [1]).

From this there follows the principal theorem below:

6. Sum, difference, product of two integral functions in Ω are again integral functions.

For if ω', ω'' are two integral functions in Ω which satisfy the equations:

$$\begin{aligned} \omega'^{n'} + b'_1\omega'^{n'-1} + \cdots + b'_{n'-1}\omega' + b'_{n'} &= 0, \\ \omega''^{n''} + b''_1\omega''^{n''-1} + \cdots + b''_{n''-1}\omega'' + b''_{n''} &= 0, \end{aligned} \tag{6}$$

then let

$$\omega_1, \omega_2, \ldots, \omega_m$$

denote the products

$$\omega'^{\nu'}\omega''^{\nu''}; \quad \nu' = 0, 1, \ldots, n'-1, \quad \nu'' = 0, 1, \ldots, n''-1,$$

and let ω denote one of the three functions $\omega' \pm \omega''$, $\omega'\omega''$; then with the help of equations (6) one may set:

$$\begin{aligned} \omega\omega_1 &= x_{1,1}\omega_1 + \cdots + x_{1,m}\omega_m, \\ &\cdots\cdots\cdots\cdots\cdots \\ \omega\omega_m &= x_{m,1}\omega_1 + \cdots + x_{m,m}\omega_m, \end{aligned} \tag{7}$$

[1]) In this proof the decomposability of integral rational functions into linear factors need not be assumed, but only the decomposability into irreducible factors. One may therefore still stipulate an arbitrary domain of rationality for the numerical coefficients. In what follows, however, we shall nevertheless have to assume the decomposability of an integral rational function of z into linear factors, i. e. the fundamental theorem of algebra.

where $m = n'n''$, and the $x_{\nu,\nu'}$ are integral rational functions of z. From (7) there results, by elimination of the ω_i:

$$\begin{vmatrix} x_{1,1} - \omega, & x_{1,2}, & \ldots, & x_{1,m} \\ x_{2,1}, & x_{2,2} - \omega, & \ldots, & x_{2,m} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ x_{m,1}, & x_{m,2}, & \ldots, & x_{m,m} - \omega \end{vmatrix} = 0,$$

and this is an equation for ω of the form (5).

By repeated application of this theorem it follows that every integral rational function of integral functions is again an integral function.

7. An integral function ω is called divisible by an integral function ω' if there exists a third integral function ω'' such that

(8) $$\omega = \omega'\omega''$$

holds.

From this definition it follows at once:

If ω is divisible by ω' and ω' is divisible by ω'', then ω is also divisible by ω''.

If ω' and ω'' are divisible by ω, then $\omega' \pm \omega''$ is also divisible by ω.

If $\omega_1, \omega_2, \omega_3, \ldots$ are divisible by ω and $\omega_1', \omega_2', \omega_3', \ldots$ are arbitrary integral functions, then

$$\omega_1\omega_1' + \omega_2\omega_2' + \omega_3\omega_3' + \cdots$$

is also divisible by ω.

§ 176. Minimal basis and field discriminant.

Since every function of the field Ω can be converted by multiplication with an integral rational function into an integral algebraic function of z, there also exist field bases consisting of integral functions (e.g., in the notation of § 171, (2), the powers of $a_0\theta$).

If now

(1) $$\omega_1,\ \omega_2,\ \ldots,\ \omega_n$$

is such a basis consisting of integral functions, then every in the form

(2) $$\omega = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n$$

contained function, when the $x_1, x_2, \ldots, x_n$ are integral rational functions, an integral function in Ω. But it is not thereby said that conversely, in the form (2) with integral coefficients x_i, all integral functions in Ω are contained.

If, then, there are integral functions in the form (2) in which the x_i are not all integral rational functions, we can find an integral function:

$$\frac{x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n}{z - c},$$

in which the x_i are integral and not all divisible by $z - c$. If we reduce the x_i to their constant residues (with respect to $z - c$), there results an integral function ω in the form:

(3) $$\omega = \frac{c_1\omega_1 + c_2\omega_2 + \cdots + c_n\omega_n}{z - c},$$

in which the constants $c_1, c_2, \ldots, c_n$ certainly do not all vanish. If, say, c_1 is different from zero, we can express ω_1 by $\omega, \omega_2, \ldots, \omega_n$ and obtain a new integral basis of Ω:

(4) $$\omega, \omega_2, \ldots, \omega_n.$$

But for the discriminant there results, by § 173, (6), the relation:

(5) $$\Delta(\omega, \omega_2, \ldots, \omega_n) = \frac{c_1^2}{(z-c)^2}\Delta(\omega_1, \omega_2, \ldots, \omega_n).$$

Both discriminants are integral rational functions of z, but the discriminant of the basis (4) is of lower degree than the discriminant of the basis (1).

If we continue in this way to lower the degree of the discriminant, we must finally arrive at a basis $\omega_1, \omega_2, \ldots, \omega_n$ consisting of integral functions and having the property that in the form (2), with integral rational x_i, all and only the integral functions in Ω are expressed.

8. Definition: An integral basis $\omega_1, \omega_2, \ldots, \omega_n$ of the field Ω is called a minimal basis if in the form

(6) $$\omega = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n$$

all integral functions of the field Ω can be represented with integral rational $x_1, x_2, \ldots, x_n$.

Such a minimal basis therefore always exists.

9. A system of integral functions derived from a minimal basis $\omega_1, \omega_2, \ldots, \omega_n$

(7) $$\omega'_\nu = x_{\nu,1}\omega_1 + x_{\nu,2}\omega_2 + \cdots + x_{\nu,n}\omega_n,$$
$$\nu = 1, 2, \ldots, n$$

is then and only then a minimal basis, when the determinant

(8) $$X = \Sigma \pm x_{1,1}x_{2,2}\ldots x_{n,n}$$

is a constant different from zero.

For if this determinant, which is a rational function of z, has a linear factor $z - c$, then the constants $c_1, c_2, \ldots, c_n$ can be so determined that the n integral rational functions

$$c_1x_{1,\nu} + c_2x_{2,\nu} + \cdots + c_nx_{n,\nu}$$

become divisible by $z - c$ without all the c_i vanishing. But then

$$\frac{c_1\omega'_1 + c_2\omega'_2 + \cdots + c_n\omega'_n}{z-c}$$

is an integral function, and $\omega'_1, \omega'_2, \ldots, \omega'_n$ is not a minimal basis.

For the discriminant one obtains from (7), according to § 173, (6)

(9) $$\Delta(\omega'_1, \omega'_2, \ldots, \omega'_n) = X^2\Delta(\omega_1, \omega_2, \ldots, \omega_n).$$

From this it follows:

10. The discriminant of a minimal basis is, apart from a constant factor, independent of the particular choice of the basis.

Among all discriminants of bases formed from integral functions, the discriminant of a minimal basis has the lowest degree (hence the name minimal basis). It is an integral rational function of z uniquely determined by the field itself, apart from a constant factor. It is therefore also called the discriminant of the field Ω and is denoted by $\Delta(\Omega)$.

Twenty-fifth Chapter.
Functionals.

§ 177. Rational Functionals.

We now carry over the concept of the functional, which served us in the 17th chapter of the second volume for the foundation of the theory of algebraic numbers, to the algebraic functions.

We therefore adjoin arbitrary variables to the field Ω and calculate with them according to the rules of literal calculation. There thus arises an extended field $\overline{\Omega}$, whose elements are called functionals. The field $\overline{\Omega}$ itself is called the functional field. The variables here are not to be taken, in the sense of analysis, as symbols for variable numbers, but as mere symbols of calculation, and are to be well distinguished from the variables z, θ of the field Ω. We wish to call these auxiliary variables the functional variables.

From the field Z of the rational functions of z there arises, by adjunction of the variables, the field $\overline{Z}$ of the rational functionals.

An integral rational function of the variables:

$$\Phi(u, v, w, \ldots)$$

with integral rational functions of z as coefficients is called an integral function in Z. The greatest common divisor of the coefficients of Φ is called the divisor of the function, and the function is called primitive if the coefficients have no common divisor. The primitive rational functions and their quotients are also called units in the field $\overline{Z}$.

The quotients of integral functions in Z are the functionals in $\overline{Z}$. Every functional $\overline{A}$ in $\overline{Z}$ can be put into the form:

$$\overline{A} = \frac{\Phi_1}{\Phi_2} = a\frac{E_1}{E_2} = aE, \tag{1}$$

where Φ_1, Φ_2, E_1, E_2 are integral functions in Z, among them the units E_1, E_2, and E is a unit in $\overline{Z}$ which is represented as a fractional function. a is the quotient of the divisors of Φ_1 and Φ_2, hence an integral or fractional rational function of z.

The functions a and E in formula (1) are completely determined by $\overline{A}$, apart from constant factors, which here play the role of the numerical units. We shall call a the "Absolute" of $\overline{A}$. The following theorem then holds:

1. The Absolute of a product is equal to the product of the Absolutes of the factors, and we define:

2. A rational functional is called integral if its Absolute is an integral function of z.

From 1. it follows that the product of two integral functionals in $\overline{Z}$ is again an integral functional. The same follows also for the sum and difference of two integral functionals $\overline{A}_1$, $\overline{A}_2$ in $\overline{Z}$. For if we set:

$$\overline{A}_1 = a_1\frac{E_1}{E}, \quad \overline{A}_2 = \frac{a_2E_2}{E},$$

$$\overline{A}_1 \pm \overline{A}_2 = \frac{a_1E_1 \pm a_2E_2}{E},$$

where a_1, a_2 are integral rational functions of z, and E_1, E_2, E integral units, then the Absolute of $\overline{A}_1 \pm \overline{A}_2$ is the divisor of the integral function $a_1E_1 \pm a_2E_2$, hence likewise an integral function of z. Thus the theorem holds:

3. Sum, difference and product of two integral functionals in $\overline{Z}$ are again integral rational functionals.

By the definition, all units and their reciprocals are to be designated as integral. If the Absolute of an integral rational functional $\overline{A}$ is linear ($= z - c$), then $\overline{A}$ is called a rational prime functional or a prime functional in $\overline{Z}$.

4. Every integral rational functional can be decomposed into prime factors, and indeed, apart from unit factors, in only one way.

If a product of integral functionals in $\overline{Z}$ is divisible by a prime functional, then at least one of the factors is divisible by this prime functional.

§ 178.
Functionals of the field $\overline{\Omega}$.

A functional of the field $\overline{\Omega}$ is an expression of the form:

$$\overline{\omega} = \overline{x}_0 + \overline{x}_1\theta + \cdots + \overline{x}_{n-1}\theta^{n-1}, \tag{1}$$

where θ is the algebraic function that generates the field Ω, and $\overline{x}_0, \overline{x}_1, \ldots, \overline{x}_{n-1}$ are rational functionals. $\overline{\omega}$ is $= 0$ only when all the coefficients $\overline{x}_0, \overline{x}_1, \ldots, \overline{x}_{n-1}$ vanish.

Every functional which, with respect to θ, is of higher than the $(n-1)$th degree can be replaced by the remainder of the division by $f(\theta)$, and so reduced to the $(n-1)$th degree; and the quotient of two expressions (1), in which the denominator is not $= 0$, can be brought into the form (1) as in § 171, 1.

If one takes an arbitrary basis

$$\omega_1, \omega_2, \ldots, \omega_n$$

of the field Ω, then $\overline{\omega}$ can be put into the form

$$\overline{\omega} = \overline{x}_1\omega_1 + \overline{x}_2\omega_2 + \cdots + \overline{x}_n\omega_n \tag{2}$$

as well. If one applies this to the products $\overline{\omega}\omega_i$ and sets

$$\overline{\omega}\omega_i = \overline{x}_{i,1}\omega_1 + \overline{x}_{i,2}\omega_2 + \cdots + \overline{x}_{i,n}\omega_n, \tag{3}$$

in which the $\overline{x}_{i,n}$ are rational functionals, then there results for $\overline{\omega}$ an equation:

$$\begin{vmatrix} \overline{x}_{1,1} - \overline{\omega}, & \overline{x}_{1,2}, & \ldots, \overline{x}_{1,n} \\ \overline{x}_{2,1}, & \overline{x}_{2,2} - \overline{\omega}, & \ldots, \overline{x}_{2,n} \\ \cdot & \cdot & \cdots \\ \overline{x}_{n,1}, & \overline{x}_{n,2}, & \ldots, \overline{x}_{n,n} - \overline{\omega} \end{vmatrix} = 0, \tag{4}$$

and so:

5. Every functional in $\overline{\Omega}$ satisfies an equation

$$\varphi(\overline{\omega}) = 0, \tag{5}$$

in which

$$\varphi(t) = t^n + \overline{A}_1 t_2^{n-1} + \overline{A}_2 t^{n-2}, \cdots + \overline{A}_n \tag{6}$$

is a rational function of degree n of t, whose coefficients $\overline{A}_1, \overline{A}_2, \ldots, \overline{A}_n$ are functionals of the field $\overline{Z}$.

We define the norm and the trace of the functional as in § 172:

$$N(\overline{\omega}) = \begin{vmatrix} \overline{x}_{1,1}, & \overline{x}_{1,2}, & \ldots, & \overline{x}_{1,n} \\ \overline{x}_{2,1}, & x_{2,2}, & \ldots, & \overline{x}_{2,n} \\ \cdot & \cdot & \cdots & \cdot \\ \overline{x}_{n,1}, & \overline{x}_{n,2}, & \ldots, & \overline{x}_{n,n} \end{vmatrix} \tag{7}$$

(8) $$S(\overline{\omega}) = \overline{x}_{1,1} + \overline{x}_{2,2} + \cdots + \overline{x}_{n,n},$$

and the theorems § 172, (6) and (21) hold unchanged for the norms and traces of the functionals as well.

Then, as in § 172, (8),

(9) $$\varphi(t) = N(t - \overline{\omega}),$$

and the discriminant of a system of functionals is defined as in § 173, (1); and if the functional $\overline{\omega}$ is contained in a field $\overline{\Omega}_1$ of degree e that is contained in $\overline{\Omega}$, then $\overline{\omega}$ satisfies an equation of degree e, $\varphi_1(t) = 0$. As in § 172, (19) it is proved that

$$\varphi(t) = [\varphi_1(t)]^f \quad (n = ef)$$

is a power of $\varphi_1(t)$.

§ 179.
Integral Functionals of the Field $\overline{\Omega}$.

6. Definition. A functional $\overline{\omega}$ in $\overline{\Omega}$ is called integral if it satisfies an equation:

(1) $$F(\overline{\omega}) = \overline{\omega}^m + \overline{A}_1\overline{\omega}^{m-1} + \cdots + \overline{A}_{m-1}\overline{\omega} + \overline{A}_m = 0$$

whose coefficients $\overline{A}_1, \ldots, \overline{A}_m$ are integral rational functionals.

From Gauss's theorem (Bd. I, § 2) it follows, since by § 177 one knows the prime factors in $\overline{Z}$, that if $F(t)$ is reducible in $\overline{Z}$, every function dividing $F(t)$ also has integral coefficients, and in particular therefore also the equation of lowest degree $\varphi(\overline{\omega}) = 0$ which $\overline{\omega}$ satisfies.

The necessary and sufficient condition for a functional $\overline{\omega}$ to be integral is therefore that

$$N(t - \overline{\omega})$$

when ordered according to powers of t, has integral rational functionals for coefficients.

As in § 175 one proves the theorems:

7. Sum, difference, product of integral functionals in $\overline{\Omega}$ are again integral functionals.

8. If $\overline{\omega}$ is an integral functional in $\overline{\Omega}$ according to Definition 6. and at the same time rational, then it is an integral functional in $\overline{Z}$ (according to Definition 2.).

9. If $\overline{\omega}$ is an integral functional in $\overline{\Omega}$, then $N(\overline{\omega})$ is an integral functional in $\overline{Z}$, and the absolute of $N(\overline{\omega})$ is called the "absolute norm of $\overline{\omega}$".

The absolute norm of $\overline{\omega}$ is therefore a function in Z.

If we denote the absolute norm by $N_a(\overline{\omega})$, then the theorem holds:

(2) $$N_a(\overline{\omega}\,\overline{\omega}') = N_a(\overline{\omega})\, N_a(\overline{\omega}').$$

§ 180. Divisibility of functionals. Units.

The theorems of Bd. II, § 155 are now to be repeated, with quite slight modifications, to which attention is drawn in what follows. To simplify the expression, integral functionals in $\overline{\Omega}$ shall here be denoted by the small Greek letters, integral functionals in $\overline{Z}$ by the small Latin letters. Then we have:

1. Definition: If β is different from zero, then α is called divisible by β if $\alpha/\beta = \gamma$ is an integral functional in $\overline{\Omega}$.

2. If $\xi, \eta, \ldots$ are arbitrary integral functionals in $\overline{\Omega}$ and $\alpha, \beta, \ldots$ are divisible by δ, then

$$\xi\alpha + \eta\beta + \cdots$$

is also divisible by δ.

3. Definition: An integral functional ε whose reciprocal $1/\varepsilon$ is integral, i. e. a divisor of the number 1, is called a unit in $\overline{\Omega}$. A unit is a divisor of every integral functional. Product and quotient of two units are again units.

4. Two integral functionals α, β which are mutually divisible by one another are called associated. Their quotient $\alpha/\beta = \varepsilon$ is a unit. Two functionals α and $\alpha\varepsilon$ are associated.

5. If α is divisible by β, then every functional associated with α is divisible by every functional associated with β.

6. If two integral functionals are associated with a third, then they are also associated with one another.

7. The norm $N(\alpha)$ is divisible by α.

This follows from equation (6), § 178: $\varphi(\alpha) = 0$, whose last coefficient is $\overline{A}_n = \pm N(\alpha)$. For according to this:

$$\pm N(\alpha) = \alpha(\alpha^{n-1} + \overline{A}_1\alpha^{n-2} + \cdots + \overline{A}_{n-1}).$$

8. An integral functional whose absolute norm is a constant is a unit, and conversely the absolute norm of a unit ε is a constant different from zero.

For if ε is a unit, then

$$N(\varepsilon) \text{ and } N\left(\frac{1}{\varepsilon}\right) = \frac{1}{N(\varepsilon)}$$

are integral functionals in $\overline{Z}$; consequently $N(\varepsilon)$ is a unit in $\overline{Z}$. Conversely $N(\varepsilon)$ is divisible by ε; hence, if $N(\varepsilon)$ is a unit in $\overline{Z}$, ε is a divisor of 1, that is, a unit.

9. There are integral rational functions of z, for example the absolute norm of α, that are divisible by α. If a is an integral rational function of z divisible by α and of the lowest possible degree, then every other integral rational function of z divisible by α is divisible by a.

§ 181. Greatest common divisor.

If $\alpha, \beta, \ldots$ are integral functionals in $\overline{\Omega}$ and $x, y, \ldots$ are variables that do not occur in $\alpha, \beta, \ldots$, then

$$\delta = \alpha x + \beta y + \cdots \tag{1}$$

is, by § 180, 2., likewise an integral functional, and indeed δ is divisible by every common divisor of $\alpha, \beta, \ldots$ If $x_0, y_0, \ldots$ are arbitrary integral functions or functionals in $\overline{Z}$, then, as is now to be proved,

$$\delta_0 = \alpha x_0 + \beta y_0 + \cdots \tag{2}$$

is divisible by δ. For if we denote the absolute norm of δ by D, then

$$N(\delta) = DE(x, y, \ldots) = DE, \tag{3}$$

and $E(x, y, \ldots)$ is a unit in $\overline{Z}$, but at the same time an integral homogeneous function of the variables $x, y, \ldots$

If t denotes a new variable, then

$$N(\delta t - \delta_0) = DE(xt - x_0,\ yt - y_0, \ldots),$$

and hence, if we arrange in descending powers of t and cancel the factor $N(\delta) = DE$ on both sides:

$$N\left(t - \frac{\delta_0}{\delta}\right) = t^n + \overline{C}_1 t^{n-1} + \overline{C}_2 t^{n-2} + \cdots, \tag{4}$$

in which $\overline{C}_1, \overline{C}_2 \ldots$ have no denominator other than E, and consequently are integral functionals in $\overline{Z}$. Accordingly, by the definition of § 179, 6., $\delta_0 : \delta$ is likewise an integral functional, as was to be proved.

Accordingly we obtain the following definitions:

1. The functional $\delta = \alpha x + \beta y + \cdots$, and every functional associated with δ, is the greatest common divisor of $\alpha, \beta, \ldots$

2. If $\alpha x + \beta y$ is a unit, then α and β are called coprime or relatively prime. If there are functionals ξ, η for which $\alpha\xi + \beta\eta$ is a unit, then α, β are relatively prime.

3. If α is relatively prime to β and to γ, then it is also relatively prime to $\beta\gamma$.

For by assumption

$$\varepsilon = \alpha x + \beta y, \quad \varepsilon_1 = \alpha u + \beta v$$

are units (x, y, u, v new variables).

$$\alpha(\alpha u x + \gamma v x + \beta u y) + \beta\gamma v y = \varepsilon\varepsilon_1,$$

and hence, by 3., α and $\beta\gamma$ are relatively prime.

4. If α is relatively prime to β and $\alpha\mu$ is divisible by β, then μ is divisible by β.

For from the assumption it follows that

$$\alpha\mu x + \beta\mu y = \varepsilon\mu,$$

from which, since ε is a unit, the proof results.

§ 182. Prime functionals in $\overline{\Omega}$.

1. Definition. An integral functional π in $\overline{\Omega}$ is called a prime functional if it is not a unit and, apart from by the units, is divisible only by the functionals associated with it.

From this it follows:

2. If a product $\alpha\beta$ is divisible by π, then at least one of the two factors, α or β, is divisible by π.

For if neither α nor β is divisible by π, then both are relatively prime to π, and by § 181, 3. π is also relatively prime to $\alpha\beta$.

3. Every integral functional $\overline{\omega}$ different from zero is divisible by a prime functional.

For if $\overline{\omega}$ is not itself a prime functional, then it is divisible by a functional α different from ω. If

$$\overline{\omega} = \alpha\overline{\omega}',$$

then $\overline{\omega}'$ is not a unit, and

$$N_a(\overline{\omega}) = N_a(\alpha)\, N_a(\overline{\omega}').$$

The absolute norms occurring here are integral functions in z, and the degree of $N_a(\overline{\omega}')$ is smaller than the degree of $N_a(\omega)$. If $\overline{\omega}'$ is still not a prime functional, one can continue in this way and must finally arrive at a prime factor π of $\overline{\omega}$.

In this way one argues further as in Bd. II, § 158, except that here the height of the degree of integral rational functions of the variable z takes the place of the quantity of the integral rational numbers used there. Thus one also obtains the theorems:

4. Every integral functional $\overline{\omega}$ in $\overline{\Omega}$ that is not a unit can be decomposed into a finite number of prime factors, and indeed in only one way, if associated functionals are regarded as not different.

In what follows we must make use of the theorem that an integral rational function of z with numerical coefficients is decomposable into linear factors; in other words, we must presuppose the fundamental theorem of algebra on the existence of roots. From this there follows first of all:

5. The integral rational function of z of lowest degree that is divisible by a prime functional π is a linear function $z - c$.

By § 180, 9. there do exist integral rational functions of z that are divisible by π. If one decomposes such a func-

tion into linear factors, then by 2. one of these linear factors must be divisible by π. Two different linear functions $z - c$ and $z - c'$ cannot be divisible by the same π, since otherwise the constant $c - c'$, which is different from zero, would be divisible by π.

6. Every integral function ω in Ω is congruent to a constant b with respect to the modulus π, i. e. the constant b can be so chosen that $\omega - b$ becomes divisible by π.

For by § 175 the function ω satisfies an equation:

$$\omega^n + a_1\omega^{n-1} + \cdots + a_{n-1}\omega + a_n = 0,$$

in which the $a_1, \ldots, a_n$ are integral rational functions of z. If $a_1^0, \ldots, a_n^0$ are the remainders of these functions on division by $z - c$, and hence constants, it follows that:

$$\omega^n + a_1^0\omega^{n-1} + \cdots + a_{n-1}^0\omega + a_n^0 \equiv 0 \pmod{\pi},$$

and if the function on the left side is decomposed into linear factors:

$$(\omega - b)(\omega - b')(\omega - b'')\cdots \equiv 0 \pmod{\pi}. \tag{1}$$

At least one of these linear factors must therefore be divisible by π, which was to be proved.

From the one-valued decomposability of the functionals into prime factors that has now been established there follow far-reaching consequences, of which the most important shall be adduced here.

If in a functional φ of the field Ω a certain number of the auxiliary variables, $x, y, \ldots$, occur only in the numerator, then

$$\varphi = \varphi(x, y, \ldots)$$

shall be called a holomorphic function of $x, y, \ldots$. Of these the theorem holds:

7. A holomorphic function is an integral functional in $\overline{\Omega}$ only when the coefficients of the ordered function φ are integral functionals.

It is evidently sufficient to establish the theorem for holomorphic functions of one variable

$$\varphi = x^m\varphi_0 + x^{m-1}\varphi_1 + \cdots + x\varphi_{m-1} + \varphi_m \tag{2}$$

since from this it can be proved in general by complete induction.

This proof, however, follows from Gauss's theorem:

If two holomorphic functions

$$\alpha = \alpha_0 x^h + \alpha_1 x^{h-1} + \cdots + \alpha_h,$$
$$\beta = \beta_0 x^k + \beta_1 x^{k-1} + \cdots + \beta_k$$

have a product

$$\gamma = \gamma_0 x^{h+k} + \gamma_1 x^{h+k-1} + \cdots + \gamma_{h+k},$$

whose coefficients are integral functionals, then the γ_0, $\gamma_1, \ldots, \gamma_{h+k}$ can all be divisible by a prime functional π only if either all of $\alpha_0, \alpha_1, \ldots, \alpha_h$, or all of $\beta_0, \beta_1, \ldots, \beta_k$ are divisible by π, which is proved in exactly the same way as in Bd. I, § 2.

If now (2) is an integral functional without $\varphi_0, \varphi_1, \ldots, \varphi_m$ being integral, then one can determine an integral functional μ, and in it a prime factor π, in such a way that

$$\chi = \mu\varphi = \alpha_0 x^m + \alpha_1 x^{m-1} + \cdots + \alpha_m, \tag{3}$$

is divisible by π, while $\alpha_0, \alpha_1, \ldots, \alpha_n$ are not all divisible by π.

If now φ is integral, then it satisfies an equation:

$$E\varphi^m = a_1 E_1 \varphi^{m-1} + a_2 E_2 \varphi^{m-2} + \cdots + a_m E_m \tag{4}$$

and from this, by multiplication by μ^m:

$$E\chi^m = \mu(a_1 E_1 \chi^{m-1} + a_2 E_2 \mu\chi^{m-2} + a_m E_m), \tag{5}$$

in which the $a_1, a_2, \ldots$ are integral functions in Z, and the $E, E_1, \ldots, E_m$ are units in $\bar{Z}$. The units $E, E_1, \ldots, E_m$ may indeed still contain x, but they may be assumed to be holomorphic. If we arrange the right-hand and the left-hand side of (5) according to x, then the coefficients of E are not all divisible by π. Likewise, by the theorem mentioned, the coefficients of χ^m are not all divisible by π, while on the right-hand side all coefficients have the factor μ, and hence also the factor π. That is impossible, and with this our theorem is proved (Bd. II, § 159).

8. A holomorphic integral functional φ is the greatest common divisor of all its coefficients.

If $\alpha, \beta, \ldots$ are the coefficients of φ, then φ is in any case divisible by the greatest common divisor of $\alpha, \beta, \ldots$. If we decompose a linear function $z - c$ into its functional prime factors, then in these prime functionals π we may denote the variables as we please, and since every prime functional must divide some $z - c$, we may assume that the variables in it are different from the variables $x, y, \ldots$ in φ. Accordingly we can

determine a functional $\psi = \varepsilon\varphi$ that is free of the $x, y, \ldots$ and associated with φ. Since φ/ψ is then integral, the coefficients of φ must, by 7., all be divisible by ψ, and consequently all be divisible by φ as well. Which was to be proved.

At the same time this proves:

9. A holomorphic integral functional passes over into an associated one when the variables are denoted differently.

10. Every functional $\bar{\omega}$ in $\bar{\Omega}$ passes over, by multiplication with a unit in $\bar{Z}$, into a holomorphic functional.

For one can first represent $\bar{\omega}$ as the quotient of two integral holomorphic functionals in the form

$$\omega = \frac{\psi}{\varphi}$$

If one puts $N(\varphi) = \varphi\varphi'$, then, as formula (4) shows, in which $a_m E_m = \pm N(\varphi)$, φ' is associated with a holomorphic functional, and it follows:

$$E_m \bar{\omega} = \frac{\psi\varphi'}{a_m}. \tag{6}$$

11. To an integral functional $\bar{\omega}$ one can choose a second one μ in such a way that μ does not become divisible by arbitrarily given prime functionals $\pi_1, \pi_2, \ldots$, and that

$$\bar{\omega}\mu = \alpha \tag{7}$$

becomes a function in Ω.

Let $\bar{\omega}$ be represented, by 10., by a holomorphic functional φ. By 8., its coefficients are all divisible by $\bar{\omega}$, but not all by $\bar{\omega}\pi_1$. There is therefore a function α_1 which is divisible by $\bar{\omega}$, but not by $\bar{\omega}\pi_1$.

Now let one put:

$$\bar{\omega}_1 = \bar{\omega}\pi_2\pi_3\ldots, \quad \bar{\omega}_2 = \bar{\omega}\pi_1\pi_3\ldots, \quad \bar{\omega}_3 = \omega\pi_2\pi_3\ldots$$

and determine:

$$\begin{aligned}
&\alpha_1 \text{ divisible by } \bar{\omega}_1, \text{ not divisible by } \bar{\omega}_1\pi_1,\\
&\alpha_2 \text{ " "} \bar{\omega}_2, \text{ " " "} \bar{\omega}_2\pi_2,\\
&\alpha_3 \text{ " "} \bar{\omega}_3, \text{ " " "} \bar{\omega}_3\pi_3.\\
&\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots
\end{aligned}$$

Then

$$\alpha = \alpha_1 + \alpha_2 + \alpha_3 + \cdots$$

satisfies the requirements of theorem 11.

12. Every integral functional $\bar{\omega}$ is the greatest common divisor of two functions α, β in Ω.

To prove this, we take $\alpha = \bar{\omega}\mu$, divisible by $\bar{\omega}$, then β divisible by $\bar{\omega}$ and $\beta : \bar{\omega}$ relatively prime to μ; then

$$\bar{\omega} = \alpha x + \beta y$$

is the greatest common divisor of α and β.

§ 183. Bases and Basis Forms of the Functionals.

Now let

$$\omega_1, \omega_2, \ldots, \omega_n \tag{1}$$

be a minimal basis of the field Ω (§ 176) and

$$\alpha_1, \alpha_2, \ldots, \alpha_n \tag{2}$$

another basis of Ω. Let the linear substitution by which the α are connected with the ω be:

$$\alpha_s = \sum_i a_{s,i}\omega_i, \tag{3}$$

in which the $a_{s,i}$ are integral functions in Z, whose determinant

$$A = \Sigma \pm a_{1,1}a_{2,2}\ldots a_{n,n} \tag{4}$$

does not vanish identically. If $t_1, t_2, \ldots, t_n$ are variables, then

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 + \cdots + \alpha_n t_n \tag{5}$$

is the greatest common divisor of $\alpha_1, \alpha_2, \ldots, \alpha_n$.

If one substitutes integral functions in Z for $t_1, t_2, \ldots, t_n$, there arise from λ integral functions in Ω, all of which are divisible by λ.

1. One calls λ a basis form and $\alpha_1, \alpha_2, \ldots, \alpha_n$ a basis of the functional λ, if one obtains all numbers in Ω divisible by λ by putting integral functions in Z for t_i.

If the α_i form such a basis, then one can determine integral functions, $g_{r,s}^{(i)}$ in Z, so that

$$\alpha_r \omega_s = \overset{i}{\Sigma} g_{r,s}^{(i)} \alpha_i \tag{6}$$

holds.

From this it follows:

$$\lambda \omega_s = \overset{i}{\Sigma} \alpha_i t_{i,s}, \tag{7}$$

if

$$t_{i,s} = \overset{r}{\Sigma} g_{r,s}^{(i)} t_r \tag{8}$$

is set, and by substituting the expressions (3) for α_i in (7):

$$\lambda \omega_r = \overset{i}{\Sigma} \omega_i \overset{s}{\Sigma} a_{s,i} t_{s,r}. \tag{9}$$

By the definition of the norm (§ 172), $N(\lambda)$ is therefore the determinant formed from the coefficients

$$\overset{s}{\Sigma} a_{s,i} t_{s,r},$$

and by the multiplication theorem for determinants one can decompose it into

$$N(\lambda) = AT \tag{10}$$

where A has the meaning (4), and T is the determinant:

$$T = \Sigma \pm t_{1,1} t_{2,2} \dots t_{n,n}. \tag{11}$$

Here T is a homogeneous function of degree n in the t_i, whose coefficients are integral functions in Z. A is itself an integral function in z. One now proves, as in Bd. II, § 164, that T is a unit. For if this were not the case, then T would have some linear factor $z - c$, and by an elementary theorem on determinants one could determine holomorphic integral functionals y_i in z, not all of them divisible by $z - c$, in such a way that

$$u_s = y_1 t_{s1} + \dots + y_n t_{sn}$$

becomes divisible by $z - c$ (one can even give the y_i constant coefficients, since they can be reduced to their residues with respect to $z - c$).

If one now sets

$$\omega = \omega_1 y_1 + \omega_2 y_2 + \dots + \omega_n y_n,$$

then it follows from (7)

$$\lambda \omega = \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n,$$

and from this it follows, since the α_i are divisible by λ and the u_i by $z - c$, that ω is divisible by $z - c$, which contradicts the definition of the minimal basis. Accordingly it follows from (10) that A is the absolute norm of the functional λ:

$$N_a(\lambda) = \Sigma \pm a_{11} a_{22} \dots a_{nn}. \tag{12}$$

For a prime functional π we can easily find a basis. Not all the elements $\omega_1, \omega_2, \dots, \omega_n$ of a minimal basis can be divisible by π, since by the linear form

$$\omega = x_1 \omega_1 + x_2 \omega_2 + \dots + x_n \omega_n \tag{13}$$

the function "1" must also be representable. Let us assume that ω_1 is not divisible by π. By § 182, 6. there are constants $c_1, c_2, \ldots, c_n$, the first of which may, without prejudice to generality, be assumed $= 1$, and which satisfy the conditions:

$$\omega_1 \equiv 1,\ \omega_2 \equiv c_2,\ \ldots,\ \omega_n = c_n \pmod{\pi}$$

We set:

$$\begin{aligned} \alpha_1 &= (z-c)\,\omega_1, \\ \alpha_2 &= \omega_2 - c_2\,\omega_1, \\ &\cdots\cdots\cdots \\ \alpha_n &= \omega_n - c_n\,\omega_1. \end{aligned} \tag{14}$$

That this is a basis of π one sees at once if one represents the function (13) in this way:

$$\omega = x_2\alpha_2 + \cdots + x_n\alpha_n + (x_1 + c_2x_2 + \cdots + c_nx_n)\,\omega_1\,; \tag{15}$$

since $\alpha_2, \ldots, \alpha_n$ are divisible by π, ω can be divisible by π only if the integral function in Z:

$$x_1 + c_2x_2 + \cdots + c_nx_n$$

is divisible by $(z-c)$, and consequently ω is representable by formula (5). Hence formulae (10) and (14) give:

$$N_a(\pi) = z - c\,, \tag{16}$$

and thus the theorem:

2. The absolute norm of a prime functional is a linear function of z.

This theorem marks an essential difference between the theories of numbers and of functions. In number theory there are prime ideals of various degrees whose norm is a power of a prime. In the theory of functions we have only prime functionals of the first degree. This comes about because by virtue of the fundamental theorem of algebra we can decompose every integral function of one variable, with respect to the modulus π, into linear factors.

The number of prime factors into which $(z-c)$ splits is accordingly n, since $N(z-c) = (z-c)^n$.

If we collect the mutually equal prime factors into powers, then

$$z - c = \varepsilon\,\pi_1^{e_1}\pi_2^{e_2}\pi_3^{e_3}\ldots\,, \tag{17}$$

in which

$$n = e_1 + e_2 + e_3 + \cdots\,, \tag{18}$$

where ε is a unit.

§ 184. Basis Form and Ramification Functional.

By the basis form of the field Ω we shall understand the linear form:

$$\tau = \omega_1t_1 + \omega_2t_2 + \cdots + \omega_nt_n\,, \tag{1}$$

in which $t_1, t_2, \ldots, t_n$ are the functional variables, and whose coefficients $\omega_1, \omega_2, \ldots, \omega_n$ form a minimal basis of Ω. This form is the greatest common divisor of $\omega_1, \omega_2, \ldots, \omega_n$ and consequently a unit. From τ one can derive all integral functions in Ω by putting for the variables $t_1, t_2, \ldots, t_n$ integral functions in Z.

If we put

$$F(t) = N(t-\tau) = t^n + A_1 t^{n-1} + \cdots + A_{n-1}t + A_n\,, \tag{2}$$

then the $A_1, A_2, \ldots, A_n$ are holomorphic integral functionals in $\overline{Z}$, and τ satisfies the equation of degree n:

$$F(\tau) = 0. \tag{3}$$

If π is a prime functional and

$$N_a(\pi) = z - c\,,$$

then by § 182, 6. one can determine the constants $c_1, c_2, \ldots, c_n$ from the congruences

$$\omega_1 \equiv c_1,\ \omega_2 \equiv c_2,\ \ldots,\ \omega_n \equiv c_n \pmod{\pi} \tag{4}$$

and if one puts

$$\tau_0 = c_1 t_1 + c_2 t_2 + \cdots + c_n t_n \tag{5}$$

then τ_0 is a constant functional, and $\tau - \tau_0$ is divisible by π. It is now to be proved:

3. The prime functional π is the greatest common divisor of $\tau - \tau_0$ and $z - c$:

$$\pi = u(\tau - \tau_0) + v(z - c)\,, \tag{6}$$

and if therefore $(z-c)$ is divisible by a higher than the first power of π, then $\tau - \tau_0$ is divisible only by the first power of π.

First of all it follows from the definition § 181 that

$$\tau - \tau_0 = t_1(\omega_1 - c_1) + t_2(\omega_2 - c_2) + \cdots + t_n(\omega_n - c_n) \tag{7}$$

is the greatest common divisor of $(\omega_1 - c_1), (\omega_2 - c_2), \ldots, (\omega_n - c_n)$. If now

$$\omega = x_1\omega_1 + x_2\omega_2 + \cdots x_n\omega_n \equiv 0 \pmod{\pi}\,, \tag{8}$$

then $x_1c_1 + x_2c_2 + \cdots + x_nc_n$ is divisible by π, and, since it is a function in Z, by $(z-c)$. We set:

$$x_1c_1 + x_2c_2 + \cdots + x_nc_n = (z-c)y$$

and obtain from (8)

$$\begin{aligned} \omega &= x_1(\omega_1 - c_1) + x_2(\omega_2 - c_2) + \cdots + x_n(\omega_n - c_n) + (z-c)y \\ &= (\tau - \tau_0) + (z-c)y. \end{aligned} \tag{9}$$

If now $(\tau - \tau_0)$ and $(z-c)$ had the common divisor $\pi\delta$, then by § 182, 11. one could find a function ω in Ω that is divisible by π and coprime to δ. But equation (9) contradicts this, and with that our theorem is proved.

Let

$$\Phi(t) = B_0 t^m + B_1 t^{m-1} + \cdots + B_{m-1}t + B_m \tag{10}$$

be a function of degree m in t, and let $B_0, B_1, \ldots, B_m$ be integral holomorphic functionals in $\overline{Z}$ with the functional variables $t_1, t_2, \ldots, t_n$, the function satisfying the congruence condition

$$\Phi(\tau) \equiv 0 \pmod{z-c} \tag{11}$$

[as, for example, the function $F(t) = N(t-\tau)$ does]. Then

$$\Phi(\tau_0) \equiv 0 \pmod{\pi}\,,$$

and if we divide $\Phi(t)$ by $\tau - \tau_0$, there results

$$\Phi(t) \equiv (t - \tau_0)\Phi_1(t) \pmod{\pi}, \tag{12}$$

in which $\Phi_1(t)$ is a holomorphic integral functional in $\overline{Z}$ (because τ_0 is contained in $\overline{Z}$ as a constant functional). Since now in (12) only rational functions of z occur besides the functional variables, this congruence must also be valid for the modulus $z - c$:

$$\Phi(t) \equiv (t - \tau_0)\Phi_1(t) \pmod{z - c}. \tag{13}$$

If π' is a second prime functional, which may also be identical with π, and if $z - c$ is divisible by $\pi\pi'$, and $\tau \equiv \tau_0' \pmod{\pi'}$, then $(\tau - \tau_0) : \pi$ is relatively prime to π' (by 3.), and because of (11) we have $\Phi_1(\tau_0') \equiv 0 \pmod{\pi'}$. From this one concludes

$$\Phi_1(t) \equiv (t - \tau_0')\Phi_2(t)$$

first for the modulus π', but then, since the functionals lie in $\overline{Z}$, also for the modulus $(z - c)$. We therefore have:

$$\Phi(t) \equiv (t - \tau_0)(t - \tau_0')\Phi_2(t) \pmod{z - c}. \tag{14}$$

From this we infer the following important theorem:

4. Let, according to § 183, (17)

$$z - c = \pi_1\pi_2 \dots \pi_n \tag{15}$$

be decomposed into n (equal or different) prime factors, and let

$$\tau_1, \tau_2, \dots, \tau_n \tag{16}$$

be the constant functionals to which τ, with respect to the moduli

$$\pi_1, \pi_2, \dots, \pi_n \tag{17}$$

becomes congruent.

Let $\Phi(t)$ be an integral holomorphic functional in z of degree m with respect to t, such that the condition

$$\Phi(\tau) \equiv 0 \pmod{z - c} \tag{18}$$

is satisfied. Then:

$$\Phi(t) \equiv (t - \tau_1)(t - \tau_2) \dots (t - \tau_n)\,\Psi(t) \pmod{z - c}, \tag{19}$$

where Ψ is a functional of the same kind of degree $m - n$.

From this it follows:

5. A functional Φ that satisfies the condition (18) cannot be of lower than degree n.

And applied to the function $F(t)$, the formula follows:

$$F(t) = N(t - \tau) \equiv (t - \tau_1)(t - \tau_2) \dots (t - \tau_n) \pmod{z - c}. \tag{20}$$

We can replace the congruence (20) by an equation in the following way:

As in (17) of the preceding section, we collect in the decomposition of $z - c$ into prime factors the equal factors together, and set:

$$\begin{gathered} z - c = \pi_1^{e_1}\pi_2^{e_2} \dots \pi_m^{e_m} \\ e_1 + e_2 + \dots + e_m = n, \end{gathered} \tag{21}$$

then by (20):

$$F(t) = (t-\tau_1)^{e_1}(t-\tau_2)^{e_2}\dots(t-\tau_m)^{e_m} + (z-c)R(t)\,, \tag{22}$$

where $R(t)$ is an integral functional, in the first place in Ω. But formula (22) itself says that $R(t)$ is contained in $\overline{Z}$, and that it is an integral function of the functional variables $t, t_1, t_2, \dots, t_n$.

We can therefore also form the derivative of equation (12) with respect to t, and obtain:

$$\begin{aligned} F'(t) = {} & e_1(t-\tau_1)^{e_1-1}(t-\tau_2)^{e_2}\dots(t-\tau_m)^{e_m} \\ & +e_2(t-\tau_1)^{e_1}(t-\tau_2)^{e_2-1}\dots(t-\tau_m)^{e_m} \\ & \quad\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot\cdot \\ & +e_m(t-\tau_2)^{e_1}(t-\tau_2)^{e_2}\dots(t-\tau_m)^{e_m-1} + (z-c)R'(t) \end{aligned} \tag{23}$$

and herein lies the proof of the following principal theorem, which one obtains by setting $t = \tau_1$.

6. If $e_1 = 1$, then $F'(\tau)$ is not divisible by π_1. Otherwise $F'(\tau)$ is divisible by $\pi_1^{e_1-1}$, but not by $\pi_1^{e_1}$. The same holds of the remaining prime factors of $(z-c)$, and consequently $F'(\tau)$ is coprime to all prime factors $z-c$ in Z which are not divisible by the square of a prime factor in Ω.

There is therefore only a finite number of prime factors $(z-c)$ that are divisible by a higher than the first power of a prime factor in Ω, and $F'(\tau)$, decomposed into prime factors, is the product of all the factors π^{e-1}, when π^e is the highest power of π dividing $N(\pi) = z-c$.

The functional $F'(\tau)$ is called the ramification functional of the field Ω.

For the absolute norm of the ramification functional we obtain:

$$N_a F'(\tau) = \Pi(z-c)^{e_1-1+e_2-1+\cdots}, \tag{24}$$

in which the product Π extends over all linear factors $(z-c)$ that are divisible by a higher than the first power of a prime factor.

We can show further that this absolute norm is nothing other than the field discriminant D.

In order to prove this, we set:

$$\tau^k = u_{k,1}\omega_1 + u_{k,2}\omega_2 + \cdots + u_{k,n}\omega_n, \tag{25}$$

in which the $u_{k,1}, u_{k,2}, \dots, u_{k,n}$ are homogeneous functions of degree k in $t_1, t_2, \dots, t_n$, and the determinant

$$U = \begin{vmatrix} u_{1,0}, & u_{2,0}, & \dots, & u_{n,0} \\ u_{1,1}, & u_{2,1}, & \dots, & u_{n,1} \\ \cdot & \cdot & \cdot & \cdot \\ u_{1,n-1}, & u_{2,n-1}, & \dots, & u_{n,n-1} \end{vmatrix} \tag{26}$$

is a unit.

For if U were not a unit, it would have at least one linear factor $z-c$, and one could determine the integral rational holomorphic functionals $y_0, y_1, \dots, y_{n-1}$ in such a way that

$$u_{r,0}y_0 + u_{r,1}y_1 + \cdots + u_{r,n-1}y_{n-1} \equiv 0 \pmod{z-c}$$

would hold without all the y_s being divisible by $z-c$.

Then from (20) a congruence would follow:

$$y_0 + y_1\tau + y_2\tau^2 + \cdots + y_{n-1}\tau^{n-1} \equiv 0 \pmod{z-c},$$

which by Theorem 5. is not possible.

By § 174, (17) and § 176, (9) we have

$$\begin{aligned} NF'(\tau) &= \pm\Delta(1, \tau, \tau^2, \ldots, \tau^{n-1}) \\ &= \pm U^2\Delta(\omega_1, \omega_2, \ldots, \omega_n) \\ &= \pm U^2\Delta; \end{aligned} \tag{27}$$

thus

$$N_a F'(\tau) = \Delta \tag{28}$$

is the field discriminant, which is accordingly decomposed by (24) into linear factors.

We use the results obtained for the proof of the following theorem as well:

7. If an integral function in Ω

$$\omega = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n \tag{29}$$

is divisible by every prime functional that divides $z-c$, then $S(\omega)$ is divisible by $z-c$.

Here $x_1, x_2, \ldots, x_n$ are integral functions in Z. The function ω arises from τ through the substitution

$$t_1 = x_1,\ t_2 = x_2,\ \ldots,\ t_n = x_n. \tag{30}$$

If therefore ω is divisible by π, and $\omega_1 \equiv c_1, \ldots, \omega_n \equiv c_n \pmod{\pi}$, then

$$x_1c_1 + x_2c_2 + \cdots + x_nc_n \equiv 0 \pmod{\pi}$$

and consequently it is also, as a function in Z, divisible by $z-c$. If therefore $\tau_1, \tau_2, \ldots, \tau_m$ are the constant functionals to which τ is congruent with respect to the prime factors $\pi_1, \pi_2, \ldots, \pi_m$ of $(z-c)$, and if these pass by the substitution (30) into $\tau_1^0, \tau_2^0, \ldots, \tau_m^0$, then all these functions are divisible by $(z-c)$ [because by hypothesis they are divisible by a prime divisor of $(z-c)$]. Now by (22)

$$-S(\tau) \equiv e_1\tau_1 + e_2\tau_2 + \cdots + e_m\tau_m, \tag{31}$$

and hence

$$-S(\omega) \equiv e_1\tau_1^0 + e_2\tau_2^0 + \cdots + e_m\tau_m^0 \equiv 0 \pmod{z-c},$$

as was to be proved.

§ 185. The Fractional Functions in Ω and the Taylor Development.

A fractional function η in the field Ω can be transformed in infinitely many ways into an integral function μ by multiplication with an integral function ν. We then have

$$\eta = \frac{\mu}{\nu}, \tag{1}$$

and μ is called the numerator, ν the denominator of η. These two functions are not completely determined by η; what is determined, however, are the functionals that remain when all common functional factors are removed from numerator and denominator. For if

$$\frac{\mu}{\nu} = \frac{\mu'}{\nu'}; \quad \mu\nu' = \nu\mu', \tag{2}$$

then every prime functional that divides ν but not μ must divide ν'. The functionals $\bar{\alpha}$ and $\bar{\beta}$ thus determined from η we call the numerator functional and denominator functional of η. In order to represent η by functions, one first takes a function ν divisible by $\bar{\beta}$ but otherwise arbitrary, and sets $\nu = \bar{\beta}\bar{\gamma}$; in this $\bar{\gamma}$ may be assumed relatively prime to any functional one pleases. Then $\eta\nu$ is an integral function divisible by $\bar{\gamma}$, and one sets

$$\eta\nu = \mu = \bar{\alpha}\bar{\gamma}.$$

The functional $\bar{\alpha}$ cannot, however, as we shall prove later on, be chosen arbitrarily any longer once $\bar{\beta}$ is given.

The functionals $\bar{\alpha}, \bar{\beta}$ we shall also call, for short, the numerator and the denominator of η.

If π is a prime functional, then there are numbers (constants) a, b, a', b' which satisfy the congruences

$$\mu \equiv a, \quad \nu \equiv b, \quad \mu' \equiv a', \quad \nu' \equiv b' \pmod{\pi}$$

and from (2) it follows that

$$ab' = ba', \quad \frac{a}{b} = \frac{a'}{b'} = c,$$

provided that ν and ν' are not divisible by π. Then $\eta - c$ is a function in Ω in which the numerator, but not the denominator, is divisible by π. If ϱ is an integral or fractional function in which the numerator is divisible by π but not by π^2, and the denominator is likewise not divisible by π, then $(\eta - c) : \varrho$ is a function whose denominator is not divisible by π and whose numerator contains the factor π one time fewer than $(\eta - c)$. We set:

$$\eta - c = \varrho\eta_1.$$

We apply the same consideration to η_1, and so obtain a sequence of equations that may be continued as far as we please:

$$\begin{aligned} \eta &= c + \varrho\eta_1, \\ \eta_1 &= c_1 + \varrho\eta_2, \\ &\cdots \\ \eta_{r-1} &= c_{r-1} + \varrho\eta_r, \end{aligned} \tag{3}$$

from which it follows:

$$\eta = c + c_1\varrho + c_2\varrho^2 + \cdots + c_{r-1}\varrho^{r-1} + \eta_r\varrho^r. \tag{4}$$

The sequence of the constants $c, c_1, c_2, \ldots, c_{r-1}$ is completely determined by η, and $\eta_1, \eta_2, \ldots, \eta_r$ is a sequence of functions in Ω whose denominator is not divisible by π.

One might call this expression the Taylor development of the function η in powers of ϱ. A similar development can also be given for a function η whose denominator is divisible by π, only that negative powers then occur as well: one can namely determine a power ϱ^s in such a way that the denominator of $\varrho^s\eta$ is no longer divisible by π, and if one applies the development (4) to this product, there follows:

$$\eta = c\varrho^{-s} + c_1\varrho^{-s+1} + \cdots + c_{r-1}\varrho^{r-s-1} + \eta_r\varrho^{r-s}. \tag{5}$$

§ 186. **Birational Transformation.**

We now come to a subject which distinguishes the region of algebraic functions above all else from that of algebraic numbers, and the reason for this is that the theory of integral functions and functionals has for functions by no means the character of invariance that it has for numbers; this is birational transformation.

If z_1 is any non-constant function of the field Ω, and $N(t - z_1)$ is the fth power of an irreducible function of degree e (§ 172, 6.), then there subsists between z and z_1 an equation which with respect to z_1 is of degree e, and whose degree with respect to z, once denominators and superfluous factors have been cleared away, shall be denoted by e_1. Let this equation be

$$G(\overset{e}{z_1}, \overset{e_1}{z}) = 0. \tag{1}$$

Among the various equations of this form there is only one irreducible one, and this is of the lowest possible degree both with respect to z and with respect to z_1 (Bd. I, § 20).

If θ is a primitive function of the field Ω, then by § 172, (12)

$$z_1^h\theta^k \qquad \begin{matrix} h = 0, 1, \ldots, e-1 \\ k = 0, 1, \ldots, f-1 \end{matrix} \tag{2}$$

is

a basis of the field Ω, and consequently every function ω in Ω can be represented in the form:

$$\omega = \xi_0 + \xi_1\theta + \cdots + \xi_{f-1}\theta^{f-1}, \tag{3}$$

in which the $\xi_0, \xi_1, \ldots, \xi_{f-1}$ are rational functions of z and z_1, hence numbers of the field determined by (1). If we set

$$n = ef, \quad n_1 = e_1 f, \tag{4}$$

then the functions ξ_i can be represented in each of the two forms:

$$\begin{aligned} \xi &= x_0 + x_1 z_1 + x_2 z^2 + \cdots + x_{e-1} z_1^{e-1}, \\ &= y_0 + y_1 z + y_2 z^2 + \cdots + y_{e_1-1} z^{e_1-1}, \end{aligned} \tag{5}$$

in which the x are rational functions of z, the y rational functions of z_1.

1. Accordingly every function ω can be represented linearly with rational coefficients in z_1 by the n_1 functions:

$$z^r\theta^s \quad \begin{matrix} r = 0, 1, \ldots, e_1 - 1 \\ s = 0, 1, \ldots, f - 1 \end{matrix}, \tag{6}$$

which we denote, in some order, by

$$\eta_1, \eta_2, \ldots, \eta_{n_1} \tag{7}$$

and these functions are linearly independent, i. e. there exists between them no linear relation with rational coefficients in z_1:

$$x_1\eta_1 + x_2\eta_2 + \cdots + x_{n_1}\eta_{n_1} = 0,$$

except when the $x_1, x_2, \ldots, x_{n_1}$ all vanish.

For f is the lowest degree of an equation for θ with rational coefficients in z and z_1 (§ 172).

Let us now consider the functional

$$\tau_1 = t_1\eta_1 + t_2\eta_2 + \cdots + t_{n_1}\eta_{n_1} \tag{8}$$

with the functional variables $t_1, t_2, \ldots, t_{n_1}$; there then results a system of equations:

$$\tau_1^r = x_{1,r}\eta_1 + x_{2,r}\eta_2 + \cdots + x_{n_1,r}\eta_{n_1}, \tag{9}$$

in which r may be any natural number, including 0, and the $x_{i,r}$ are rational holomorphic functionals in z_1. From this there results, by elimination of the η, an equation

$$\Phi(\tau_1) = \tau_1^m + A_1\tau_1^{m-1} + \cdots + A_{m-1}\tau_1 + A_m = 0 \tag{10}$$

of degree at most n_1, with rational functionals in z_1 as coefficients.

It is to be proved that this degree cannot be less than n_1, or, what means the same thing, that $\eta_1, \eta_2, \ldots, \eta_{n_1}$ admit of a rational representation by τ_1. This follows if one differentiates (10) partially with respect to one of the t_ν:

$$\Phi'(\tau_1)\,\eta_\nu + \frac{\partial A_1}{\partial t_\nu}\tau_1^{m-1} + \cdots + \frac{\partial A_{m-1}}{\partial t_\nu}\tau_1 + \frac{\partial A_m}{\partial t_\nu} = 0.$$

Now since $\Phi'(\tau_1)$ does not vanish when m is the lowest possible degree of the equation (10), one obtains from this at once the required representation of η_ν. Hence $m = n_1$ must hold. With this it is proved:

2. The functional τ_1 satisfies an irreducible equation of degree n_1 in z_1.

If therefore we put $r = 0, 1, 2, \ldots, n_1 - 1$ in (9), there results a system of linear equations for $\eta_1, \eta_2, \ldots, \eta_{n_1}$ whose determinant is a non-vanishing functional in z_1, and one can therefore assign to the variable t_i such constant values that the determinant does not vanish even then. If τ_1 passes by this substitution into θ_1, then from the system (9) one can express $\eta_1, \eta_2, \ldots, \eta_{n_1}$ rationally by z_1 and θ_1.

With this, however, it is proved:

3. If z_1 is an arbitrary, non-constant function in Ω, and θ_1 a second function of the same field which satisfies no equation of lower degree than n_1 in z_1, then all functions of the field Ω, and hence also z and θ, are rationally expressible by z_1 and θ_1.

Between any two functions α, β of the field Ω there always subsists an equation with constant coefficients

$$F(\alpha, \beta) = 0, \tag{11}$$

and among all possible equations of this form there is one of the lowest possible degree, both with respect to α and with respect to β.

Twenty-sixth Chapter.
Numerical Values of the Algebraic Functions.

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§ 187. The Point.

The question now arises how a meaning can be assigned to the algebraic functions in the region of numbers. In doing so we presuppose as given the region of the real and imaginary rational and irrational numbers, and within this region the reckoning with the four species. We extend this region by the addition of a number symbol "Infinity": ∞, and we reckon with this symbol too in the well-known manner, so that

$$\frac{1}{\infty} = 0, \quad \frac{1}{0} = \infty, \quad \infty \cdot \infty = \infty \tag{1}$$

holds, while to the symbols

$$\infty \pm \infty, \quad 0 \cdot \infty, \quad \frac{0}{0}, \quad \frac{\infty}{\infty}$$

no meaning attaches (or else, in each individual case as required, an arbitrary numerical value is assigned to them).

The field Ω now consists of all possible algebraic functions $\alpha, \beta, \gamma, \ldots$, which by § 186 can be represented as rational functions of two among them with numerical coefficients. The kind of this representation is at first of no consequence.

We now set up the following definition:

1. If all individuals $\alpha, \beta, \gamma, \ldots$ of the field Ω are replaced by definite numerical values $\alpha_0, \beta_0, \gamma_0, \ldots$ in such a way that

1. $\alpha = \alpha_0$, if α is constant,
2. $(\alpha + \beta)_0 = \alpha_0 + \beta_0$,
3. $(\alpha - \beta)_0 = \alpha_0 - \beta_0$,
4. $(\alpha\beta)_0 = \alpha_0\beta_0$,
5. $\left(\frac{\alpha}{\beta}\right)_0 = \frac{\alpha_0}{\beta_0}$,

so we assign to such a coincidence of values a point $\mathfrak{P}$. We say: $\alpha = \alpha_0$, or α has the value α_0 at the point $\mathfrak{P}$. Two points $\mathfrak{P}$ and $\mathfrak{P}'$ are called different then and only then, when there is at least one function in Ω that has different values at $\mathfrak{P}$ and $\mathfrak{P}'$.

Rules 2. to 5. fail in the exceptional cases, namely 2. and 3., when α_0 and $\beta_0 = \infty$; 4., when $\alpha_0 = 0$, $\beta_0 = \infty$ or $\alpha_0 = \infty$, $\beta_0 = 0$; and 5., when α_0 and β_0 are both $= 0$ or both $= \infty$.

Nevertheless the functions $(\alpha \pm \beta)_0$, $(\alpha\beta)_0$ and $(\alpha : \beta)_0$ must have determinate values in these cases too; these values, however, are not determined directly from prescriptions 2. to 5., but in an indirect way.

The point is accordingly an invariant concept belonging to the field Ω, one that has nothing to do with the accidental circumstance of which of the functions of Ω we regard as the independent variable.

To find all points, one proceeds as follows. If $\mathfrak{P}$ is a point, then there are functions that have a finite value at it, for if a function α is infinite at $\mathfrak{P}$, then $1 : \alpha$ is finite. Such a function is to be taken as the independent variable z and its value at the point $\mathfrak{P}$ denoted by c. All integral functions of z are then likewise finite at $\mathfrak{P}$, for if ω is an integral function that satisfies the equation

$$\omega^n + a_1\omega^{n-1} + \cdots + a_n = 0$$

then

$$1 + \frac{a_1}{\omega} + \frac{a_2}{\omega^2} + \cdots + \frac{a_n}{\omega^n} = 0$$

must be satisfied at $\mathfrak{P}$, and since the $a_1, a_2, \ldots, a_n$, as integral rational functions of z, are finite at $\mathfrak{P}$, ω cannot be $= \infty$.

Every congruence with respect to the modulus $z - c$ must therefore pass over at the point $\mathfrak{P}$ into a correct equation, even when this congruence subsists between holomorphic integral functionals. If, then, τ is the basis form of Ω in Z, the equation must hold, by § 184, (20):

$$(\tau - \tau_1)(\tau - \tau_2)\ldots(\tau - \tau_n) = 0.$$

One of the factors of the left side must therefore vanish in $\mathfrak{P}$. If this is $\tau - \tau_1$, then to this factor there corresponds a prime factor π of $z - c$. Thus in $\mathfrak{P}$ there will be

$$\omega_1 = c_1, \quad \omega_2 = c_2, \ldots, \omega_n = c_n, \tag{2}$$

if $c_1, c_2, \ldots, c_n$ are the numerical values determined by the congruences

$$\omega_1 \equiv c_1,\ \omega_2 \equiv c_2,\ \ldots,\ \omega_n \equiv c_n \pmod{\pi} \tag{3}$$

By this the values of all integral functions of z at the point $\mathfrak{P}$ are determined. Every integral function ω is congruent, with respect to the modulus $z - c$, to an expression

$$a_1\omega_1 + a_2\omega_2 + \cdots + a_n\omega_n$$

with constant coefficient a, and it is then and only then divisible by π, when

$$a_1c_1 + a_2c_2 + \cdots + a_nc_n = 0$$

holds. Accordingly it follows:

2. Among the integral functions of z, only those divisible by π have the value 0 at $\mathfrak{P}$.

Since every fractional function in Ω can be represented in such a way that numerator and denominator are not both divisible by π, the value of every function at $\mathfrak{P}$ is thereby determined.

3. The prime factor π is said to be generated by the point $\mathfrak{P}$. It cannot be generated by a point different from $\mathfrak{P}$.

Conversely, one can also generate a point from every prime factor π in z.

If $z - c$ is the linear function divisible by π, then let the function z be given the value c. Then take a function ϱ that is divisible by π, but not by π^2, and set, in accordance with § 185, (5), for every function η:

$$\eta = a\varrho^m + \eta_1 \varrho^{m+1}, \tag{4}$$

where m is an integral rational number and η_1 a function whose denominator is not divisible by π. To the function ϱ let the value 0 be assigned, and to the function η the value

$$\eta_0 = \infty, \text{ if } m < 0,$$
$$\eta_0 = a, \text{ if } m = 0,$$
$$\eta_0 = 0, \text{ if } m > 0.$$

4. The values of all functions η so determined satisfy the conditions 1. to 5. and therefore constitute a point, which is said to be generated by the prime functional π.

Thus from a single function z and the prime factors of the linear forms $z - c$ one can derive all points at which

z has a finite value. In order to determine the others as well, one must further make use of a second function, say $z_1 = 1 : z$, which has the finite value 0 at the still missing points, which are present in finite number.

The totality of the points forms the absolute Riemann surface[1].

§ 188. Order Numbers.

1. If $\mathfrak{P}$ is a point, then to every function η that receives at this point a finite value different from zero we assign the order 0 in $\mathfrak{P}$.

2. If ω runs through the totality of the functions vanishing in $\mathfrak{P}$, and ϱ is one of these functions for which all

$$\frac{\omega}{\varrho}$$

have finite values, then ϱ has the order 1.

One also says that ϱ becomes infinitely small in $\mathfrak{P}$ in the first order.

If $\frac{\varrho'}{\varrho}$ is neither zero nor infinity, then by this ϱ' likewise has the order 1. Conversely, if ϱ and ϱ' both have the order number 1, then $\varrho : \varrho'$ and $\varrho' : \varrho$ are neither zero nor infinity.

3. If ϱ is of the first order and ω is any function in Ω, then ω has the order number n when $\varrho^{-n}\omega$ becomes neither zero nor infinite in $\mathfrak{P}$.

This determination of the order n is independent of which function of the first order we have taken. If n is positive, ω becomes zero; if n is negative, ω becomes infinite; and if $n = 0$, then ω has in $\mathfrak{P}$ a finite value different from zero.

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[1] According to Riemann's theory of algebraic functions, every point of the closed many-sheeted surface corresponds to a point in our sense. If z is the independent variable, then the Riemann surface is spread out in n sheets over the z-plane. As an absolute Riemann surface one may regard any surface to which the totality of those many-sheeted surfaces can be referred in a one-to-one and continuous manner.

The definition of the order number is therefore attached to the point $\mathfrak{P}$. It is invariant for the field Ω.

That every function receives at every point $\mathfrak{P}$ a definite integer n as its order number now follows from § 185, (5).

Take a function z which remains finite at $\mathfrak{P}$ as independent variable, determine the prime functional π belonging to $\mathfrak{P}$, and choose a function ϱ whose numerator is divisible by π, but not by π^2. This function is of the first order.

Then every function ω can be put into the form:

$$\omega = a\varrho^n + \eta_1\varrho^{n+1}, \tag{1}$$

in which a is a constant different from zero, and n is the order number of ω.

If ϱ has the order 1, then ϱ^n has the order n.

The following theorems hold for the order numbers, and all of them follow easily from (1):

4. The order of a product of two functions is equal to the sum of the orders of the factors. The order of a quotient is equal to the difference of the orders of numerator and denominator.

The order of a sum is equal to the lowest among the orders of the summands. If several summands of the same lowest order number occur, then the order of the sum can be greater, but not smaller.

Of a function which becomes zero at $\mathfrak{P}$ in the nth order one also says that it becomes zero in the first order at n coincident points.

§ 189. Polygons.

1. Definition. Complexes of points, which may also contain the same point several times, are called polygons or many-sided figures (of the field Ω).

The number of points of a polygon is called its order, and a polygon of the mth order is also called briefly an m-gon.

As designation for the polygons the letters of the capital German alphabet $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$ shall serve.

2. By the product $\mathfrak{A}\mathfrak{B}$ of two polygons one understands the polygon that contains the points of $\mathfrak{A}$ and $\mathfrak{B}$ together, and indeed contains a point $\mathfrak{P}$ that occurs r times in $\mathfrak{A}$ and s times in $\mathfrak{B}$, $r+s$ times. The order of a product is equal to the sum of the orders of the factors.

If, then, we denote distinct points by $\mathfrak{P}_1, \mathfrak{P}_2, \ldots$, we can put every polygon $\mathfrak{A}$ in one way into the form:

$$\mathfrak{A} = \mathfrak{P}_1^{r_1}\mathfrak{P}_2^{r_2}\cdots \tag{1}$$

The points are therefore the prime elements in the calculation with polygons. In order to have a representative for the unit as well, the „zero-gon“ $\mathfrak{O}$, which contains no point at all, must be named along with them.

The greatest common divisor of two polygons $\mathfrak{A}, \mathfrak{B}$ contains every point that occurs in $\mathfrak{A}$ and $\mathfrak{B}$, and indeed as often as it occurs in each of the two.

The least common multiple contains every point of $\mathfrak{A}$ and $\mathfrak{B}$, and indeed in the highest of the orders in which it occurs in $\mathfrak{A}$ and $\mathfrak{B}$.

3. If z is a variable of the field Ω and n the degree of the field with respect to z, then z takes every value c at n points.

For if one decomposes $z-c$ into its prime factors [§ 183, (17)]:

$$z - c = \pi_1^{e_1}\pi_2^{e_2}\cdots, \quad e_1 + e_2 + \cdots = n,$$

then each of these prime factors generates a point, e. g. π_1 the point $\mathfrak{P}_1$, at which the function $z-c$ becomes zero in the e_1th order. If one counts this point e_1-fold, one obtains a polygon $\mathfrak{P}_1^{e_1}\mathfrak{P}_2^{e_2}\cdots$ of order n, at whose points the function $z-c$ vanishes.

But the same theorem holds also for the value $c=\infty$, as one concludes from the fact that the degree of the field Ω with respect to $1:z$ is the same as with respect to z.

4. The n points at which z takes a value c are called conjugate with respect to z, and the values $\eta_1, \eta_2, \ldots, \eta_n$ that a function η in Ω takes at n conjugate points are likewise called conjugate with respect to z, and z is said to be of order n. The constants, and only these, have the order 0.

§ 190. Branch Points and Branch Numbers.

According to § 184 there is only a finite number of points at which $z-c$ becomes zero in an order higher than the first. These are called the branch points of Ω in z.

We construct a polygon $\mathfrak{Z}_z$ which contains $(e-1)$ times every point $\mathfrak{P}$ at which $z-c$ or $1:z$ becomes zero in the eth order. This polygon is called the branch polygon of Ω with respect to z, and its order

$$w_z = \Sigma(e-1) \tag{2}$$

is called the branch number with respect to z[40].

In § 184, (20) we obtained the formula:

$$N(t-\tau) \equiv (t-\tau_1)(t-\tau_2)\ldots(t-\tau_n) \pmod{z-c}, \tag{3}$$

in which τ denotes the basis form of the field Ω with respect to z, and $\tau_1, \tau_2, \ldots, \tau_n$ the equal or different constant functionals to which τ becomes congruent with respect to the prime factors of $z-c$, that is to say, the values which τ takes at the points $\mathfrak{P}_1, \mathfrak{P}_2, \ldots, \mathfrak{P}_n$ conjugate with respect to z. At each of these points (3) must pass over into a correct equation, and from this there follow, for example:

$$N(\tau) = \tau_1\tau_2\ldots\tau_n\,, \tag{4}$$

$$S(\tau) = \tau_1 + \tau_2 + \cdots + \tau_n \tag{5}$$

as the values of the rational functionals $N(\tau)$ and $S(\tau)$ for $z=c$. If one substitutes rational functions of z for the functional variables, then corresponding formulae hold for the functions in Ω, provided that none of the exceptional cases $0\,.\,\infty$, $\infty+\infty$ occurs.

If, therefore, $\eta_1, \eta_2, \ldots, \eta_n$ is a basis of Ω, and if $\eta_{i,1}, \eta_{i,2}, \ldots, \eta_{i,n}$ are values of η_i conjugate with respect to z, then by (5) and by the definition of the discriminant in § 173

$$\Delta(\eta_1, \eta_2, \ldots, \eta_n) = (\Sigma \pm \eta_{1,1}\eta_{2,2}\ldots\eta_{n,n})^2. \tag{6}$$

If two functions z and z' in Ω stand in a linear relation:

[40] If one spreads the Riemann surface over the z-plane, then n is the number of sheets lying one above another, and w_z is the number of branch points of this surface.

(7) $$z' = \frac{az+b}{cz+d},$$

in which a, b, c, d are constants whose determinant $ad-bc$ is different from zero, then the branch polygons $\mathfrak{Z}_z$ and $\mathfrak{Z}_{z'}$ are identical, and hence $w_z = w_{z'}$ also; for if at a point $z - z_0$ or $1 : z$ becomes zero in the eth order, then at the same point

$$z' - z'_0 = \frac{(ad-bc)(z-z_0)}{(cz+d)(cz_0+d)}$$

or, when z_0 is infinite, so that $z'_0 = a : c$:

$$z' - z'_0 = \frac{-(ad-bc)}{c(cz+d)},$$

and when $z'_0 = \infty$, so that $cz_0 + d = 0$:

$$\frac{1}{z'} = \frac{cz+d}{az+b}$$

is likewise infinitely small in the cth order.

If we apply this to $z' = 1 : z$, we obtain the branch points at which z has a finite value from the field discriminant D_z according to § 184, (24), (28). There are then still lacking the branch points at which $z = \infty$, and hence $z' = 0$, and these follow in the same way from the vanishing roots of the field discriminant $D_{z'}$. Accordingly we have the theorem:

5. The branching number w_z is equal to the degree of the field discriminant D_z, increased by the number of the vanishing roots of $D_{z'}$. $(z' = 1 : z)$.

§ 191. Polygon quotients and polygon classes.

A function η in Ω has an order number different from zero only at a finite number of points. The sum of the positive order numbers is just as great as the sum of the negative ones, namely equal to the order of the function η (§ 189, 4.). If the order numbers of η are known for every point $\mathfrak{P}$, then the function η is thereby determined up to a constant factor. For if a second function η' has the same order numbers everywhere, then $\eta : \eta'$ has the order number 0 everywhere and is therefore constant (§ 189, 4.).

We form a polygon $\mathfrak{A}$, into which we take up every point at which η has a positive order number as often as this order number indicates, and a second polygon $\mathfrak{B}$, into which we

in corresponding fashion take up the points at which η has a negative order number. Then the polygons $\mathfrak{A}$ and $\mathfrak{B}$ are of equal order, namely of the order of the function η, and by these polygons the function η is determined up to a constant factor.

We can therefore introduce the symbolic notation:

(1) $$\eta = \frac{\mathfrak{A}}{\mathfrak{B}},$$

and call $\mathfrak{A}$ the numerator or the upper polygon, $\mathfrak{B}$ the denominator or the lower polygon of η. According to this definition, $\mathfrak{A}$ and $\mathfrak{B}$ are in the first instance relatively prime to one another. But we wish to extend this notation still further by setting, when $\mathfrak{M}$ is an arbitrary polygon:

(2) $$\frac{\mathfrak{M}\mathfrak{A}}{\mathfrak{M}\mathfrak{B}} = \frac{\mathfrak{A}}{\mathfrak{B}}$$

Then the notation (1) is free of the restriction that $\mathfrak{A}$ and $\mathfrak{B}$ are to be relatively prime. Both polygons are still of equal order, but this order can be greater than the order of η.

With this representation of the functions η the same rules then hold for multiplication and division as in reckoning with numerical fractions in the region of the natural numbers.

In (1) $\mathfrak{A}, \mathfrak{B}$ cannot be arbitrary m-gons, and the investigation of the relation between them is the great question which, in another way, is answered by Abel's theorem.

We set up the following definition:

1. If two n-gons $\mathfrak{A}, \mathfrak{A}'$ can be the upper polygon and the lower polygon of a function η in Ω, then $\mathfrak{A}$ and $\mathfrak{A}'$ are called equivalent.

In symbols: if there is a function η to which the notation

$$\eta = \frac{\mathfrak{A}}{\mathfrak{A}'}$$

belongs, then

$$\mathfrak{A} \sim \mathfrak{A}'. \tag{3}$$

From the formula

$$\eta = \frac{\mathfrak{A}}{\mathfrak{A}'}, \quad \eta' = \frac{\mathfrak{A}}{\mathfrak{A}''}, \quad \frac{\eta'}{\eta} = \frac{\mathfrak{A}'}{\mathfrak{A}''}$$

there results the theorem:

2. If two n-gons are equivalent to a third, then they are also equivalent to one another. Accordingly one unites equivalent polygons $\mathfrak{A}, \mathfrak{A}', \ldots$ into a polygon class A. Every polygon is contained in one class, and in only one class.

There also exist polygons that are equivalent to no other, and that therefore each form a separate class by themselves. These are called isolated polygons.

3. If $\mathfrak{A} \sim \mathfrak{A}'$, $\mathfrak{B} \sim \mathfrak{B}'$, then $\mathfrak{A}\mathfrak{B} \sim \mathfrak{A}'\mathfrak{B}'$.

This follows from

$$\eta = \frac{\mathfrak{A}}{\mathfrak{A}'}, \quad \eta_1 = \frac{\mathfrak{B}}{\mathfrak{B}'}, \quad \eta\eta_1 = \frac{\mathfrak{A}\mathfrak{B}}{\mathfrak{A}'\mathfrak{B}'}. \tag{4}$$

The class C to which the product $\mathfrak{A}\mathfrak{B}$ of the classes A, B belongs therefore contains all products of a polygon of the class A with a polygon of the class B (in certain circumstances still others as well), and is therefore completely determined by the classes A and B. On this rests the multiplication (composition) of the classes, which is expressed in the formula:

$$C = AB = BA. \tag{5}$$

From an equation between three classes A, B, M

$$MA = MB$$

it follows accordingly that $A = B$. For if $\mathfrak{M}\mathfrak{A} \sim \mathfrak{M}\mathfrak{B}$, then by (2) it follows that $\mathfrak{A} \sim \mathfrak{B}$. If $\mathfrak{A} \sim \mathfrak{A}'$ and

$$\mathfrak{A}\mathfrak{B} = \mathfrak{C} \quad \text{and} \quad \mathfrak{A}'\mathfrak{B} = \mathfrak{C}',$$

then $\mathfrak{C} \sim \mathfrak{C}'$ as well. If, then, $\mathfrak{A}$ is contained in $\mathfrak{C}$, every polygon equivalent to $\mathfrak{A}$ is contained in a polygon $\mathfrak{C}'$ equivalent to $\mathfrak{C}$.

4. We call a class C divisible by a class A if a polygon of A is contained in a polygon of C, and we set

$$AB = C. \tag{6}$$

The class B is completely determined by A and C.

§ 192. Families of Polygons.

If $\mathfrak{A}, \mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$ are polygons of a class A, then there are s functions in Ω:

$$\eta_1 = \frac{\mathfrak{A}_1}{\mathfrak{A}}, \quad \eta_2 = \frac{\mathfrak{A}_2}{\mathfrak{A}}, \quad \cdots, \quad \eta_s = \frac{\mathfrak{A}_s}{\mathfrak{A}}. \tag{1}$$

If $\mathfrak{P}$ is any point and ϱ a function which has order 1 at this point, then following § 185 we set

$$\begin{aligned} \eta_1 &= e_1\varrho^m + \sigma_1\varrho^{m+1}, \\ \eta_2 &= e_2\varrho^m + \sigma_2\varrho^{m+1}, \\ &\cdots\cdots\cdots\cdots \\ \eta_s &= e_s\varrho^m + \sigma_s\varrho^{m+1}, \end{aligned} \tag{2}$$

in which $e_1, e_2, \ldots, e_s$ are constants that do not all vanish, and $\sigma_1, \sigma_2, \ldots, \sigma_s$ are functions that are finite at $\mathfrak{P}$; m is the lowest order number at $\mathfrak{P}$ that occurs among the functions η_i.

The totality of the functions

$$\eta = c_1\eta_1 + c_2\eta_2 + \cdots + c_s\eta_s \tag{3}$$

with constants $c_1, c_2, \ldots, c_s$ is called a family. We denote it by

$$(\eta_1, \eta_2, \ldots, \eta_s), \tag{4}$$

and call $\eta_1, \eta_2, \ldots, \eta_s$ a basis of the family. The order of η at $\mathfrak{P}$ is not smaller than m (it is greater when $e_1c_1 + e_2c_2 + \cdots + e_sc_s = 0$). But η can become infinite at no point $\mathfrak{P}$ which is not contained in $\mathfrak{A}$, and also not to a higher order than that to which the point $\mathfrak{P}$ enters into $\mathfrak{A}$.

We can therefore set:

$$\eta = \frac{\mathfrak{A}'}{\mathfrak{A}}, \tag{5}$$

in which $\mathfrak{A}'$ likewise belongs to the class A.

If instead of $\mathfrak{A}$ one takes another polygon $\mathfrak{B}$ of the class A and sets

$$\xi = \frac{\mathfrak{A}}{\mathfrak{B}}, \tag{6}$$

then from 1. it follows that:

$$
\begin{aligned}
\xi\eta_1 = \eta_1' &= \frac{\mathfrak{A}_1}{\mathfrak{B}},\\
\xi\eta_2 = \eta_2' &= \frac{\mathfrak{A}_2}{\mathfrak{B}},\\
&\cdots\cdots\cdots\\
\xi\eta_s = \eta_s' &= \frac{\mathfrak{A}_s}{\mathfrak{B}}
\end{aligned}
\tag{7}
$$

and

$$
\xi\eta = \eta' = c_1\eta_1' + c_2\eta_2' + \cdots + c_s\eta_s' = \frac{\mathfrak{A}'}{\mathfrak{B}}. \tag{8}
$$

Every polygon $\mathfrak{A}'$ generated by the denominator $\mathfrak{A}$ and a system of constants $c_1, c_2, \ldots, c_s$ is therefore also generated by every

denominator equivalent to $\mathfrak{A}$ and the same system of constants, and this polygon $\mathfrak{A}'$ is therefore dependent only on the constants and on the polygons $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$. The totality of these polygons is called a family of polygons with the basis $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$, and the designation

$$
(\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s) \tag{9}
$$

is used for it.

If the polygons $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$ have the greatest common divisor $\mathfrak{M}$, then this divisor is contained in all the polygons of the family, and is called the divisor of the family. But in the family one can determine a polygon $\mathfrak{A}' = \mathfrak{M}\mathfrak{B}$ in such a way that $\mathfrak{B}$ does not contain arbitrarily given points.

For if a point $\mathfrak{P}$ is contained μ times in $\mathfrak{M}$ and ν times in $\mathfrak{A}$, and ϱ is a function that vanishes at $\mathfrak{P}$ to the first order, then in the developments (2):

$$
m = \mu - \nu,
$$

and the development of $\eta = \mathfrak{A}' : \mathfrak{A} = \mathfrak{M}\mathfrak{B} : \mathfrak{A}$ reads

$$
\eta = e\varrho^m + \sigma\varrho^{m+1},
$$

in which

$$
e = e_1c_1 + e_2c_2 + \cdots + e_sc_s.
$$

If, then, one chooses the c_i so that e does not vanish, then $\mathfrak{P}$ is not contained in $\mathfrak{B}$.

A family that has a divisor $\mathfrak{M}$ is called an improper family.

If among the functions $\eta_1, \eta_2, \ldots, \eta_s$ there subsists a relation of the form:

$$
c_1\eta_1 + c_2\eta_2 + \cdots + c_s\eta_s = 0, \tag{10}
$$

in which the constants $c_1, c_2, \ldots, c_s$ are not all $= 0$, then the same relation

$$
c_1\eta_1' + c_2\eta_1' + \cdots + c_s\eta_s' = 0 \tag{11}
$$

subsists between the functions (7). We then call these functions linearly dependent. Correspondingly, the polygons of the family $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$ are also called linearly dependent. One

can then express one of the functions, say η_s, linearly by the others, and the family $(\eta_1, \eta_2, \ldots, \eta_s)$ is identical with $(\eta_1, \eta_2, \ldots, \eta_{s-1})$. The same holds for the families of polygons $(\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s)$ and $(\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_{s-1})$. By repeated application of this reduction one can represent every family by a linearly independent or irreducible basis. Two irreducible bases of one

family have the same number of elements, and this number is called the dimension of the family, and the family is called an s-fold one. If $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s$ are linearly dependent or independent, the same holds of $\mathfrak{M}\mathfrak{A}_1, \mathfrak{M}\mathfrak{A}_2, \ldots, \mathfrak{M}\mathfrak{A}_s$.

In a family of dimension $s > 1$

$$S = (\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s) \tag{12}$$

one can find a polygon that contains an arbitrary point $\mathfrak{P}$ at least once more often than the divisor $\mathfrak{M}$ of this family does.

This purpose is attained by setting up the development (2) for this point and then determining one of the constants $c_1, c_2, \ldots, c_s$ from the equation

$$c_1 e_1 + c_2 e_2 + \cdots + c_s e_s = 0 \tag{13}$$

These polygons then form a family of the $(s-1)$th dimension, whose divisor is divisible by $\mathfrak{M}\mathfrak{P}$.

This theorem can be extended as follows:

5. The polygons of an s-fold family that are divisible by an r-gon $\mathfrak{R}$ form an at least $(s-r)$-fold family.

If this theorem has already been proved for an r-gon, it follows for an $(r+1)$-gon $\mathfrak{R}\mathfrak{P}$ from the preceding one. For by the addition of the point $\mathfrak{P}$ the dimension is either not changed, if $\mathfrak{P}$ divides the divisor of the family reduced by $\mathfrak{R}$, or it is diminished by 1.

The polygon $\mathfrak{R}$ can be so chosen that the dimension of S is reduced exactly to $(s-r)$. For this purpose the points of $\mathfrak{R}$ have to be chosen successively so that each following one is not contained in the divisor of the family reduced by the preceding ones. From this it follows:

6. In an s-fold family there is at least one polygon that is divisible by a given $(s-1)$-gon.

If the class A, to which all polygons of the family S belong, is of the mth order, then the dimension s of the family cannot be greater than $m+1$. For otherwise one could, by 6., find a polygon in S that would be divisible by an $(m+1)$-gon, which is absurd, since S contains only m-gons. Hence the dimension s of the family has a maximum for a given class A. If this maximum is reached, then all polygons of the class A are

are contained in S. For if there is a polygon $\mathfrak{A}_{s+1}$ in A that is not contained in S, then

$$(\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_s, \mathfrak{A}_{s+1})$$

is a family of dimension $s+1$. Thus it is proved:

7. The polygons of a class form a family of a finite dimension s. This number shall be called the dimension of the class.

If in a class C there are polygons that are divisible by a polygon $\mathfrak{A}$ of a class A, then C is divisible by A. If

$$C = AB,$$

then one obtains the dimension of B by selecting from C the family that is divisible by some polygon $\mathfrak{A}$ in A. This dimension therefore depends only on the two classes A and C, and shall be

denoted by (A, C). The symbol for the dimension of a class A is accordingly (O, A), if O is the class of the null polygon. We then have

$$(A, C) = (A, AB) = (O, B), \tag{14}$$

and if a is the order of the class A, then by 5.:

$$(A, C) \geqq (O, C) - a. \tag{15}$$

If we denote by A the proper class which one obtains when one cancels $\mathfrak{M}$ away everywhere, then the improper class with the divisor $\mathfrak{M}$ shall be denoted by $\mathfrak{M}A$.

8. The divisor of an improper class is an isolated polygon.

For if $\mathfrak{M}A$ is an improper class, then a polygon $\mathfrak{A}$ can be found in A that is relatively prime to $\mathfrak{M}$. If $\mathfrak{M}'$ is equivalent to $\mathfrak{M}$, then $\mathfrak{M}'\mathfrak{A}$ is contained in $\mathfrak{M}A$ and consequently divisible by $\mathfrak{M}$. Hence $\mathfrak{M}'$ too is divisible by $\mathfrak{M}$, and therefore identical with $\mathfrak{M}$. There is therefore in the class of $\mathfrak{M}$ only the single polygon $\mathfrak{M}$.

§ 193. Normal bases.

According to the foregoing we can represent a function z of the field Ω of the nth order in the form

$$z = \frac{\mathfrak{U}'}{\mathfrak{U}} \tag{1}$$

in which $\mathfrak{U}$ and $\mathfrak{U}'$ are equivalent n-gons without common divisor. An integral function ω of z is characterised by the fact that it becomes infinite at no point which is not contained in $\mathfrak{U}$. Consequently such a function ω can be represented in the following way:

$$\omega = \frac{\mathfrak{A}}{\mathfrak{U}^r}, \tag{2}$$

in which r is a positive exponent. In (2) the numerator and the denominator may have common divisors. We may, however, assume that $\mathfrak{A}$ is not divisible by $\mathfrak{U}$. Then in (2) the exponent r has the least possible value. This number r shall be called the exponent of ω in z.

1. If we set $z' = 1 : z$, then by (1), (2):

$$z' = \frac{\mathfrak{U}}{\mathfrak{U}'}, \quad \omega' = \omega z'^r = \frac{\mathfrak{A}}{\mathfrak{U}'^r},$$

and if $\mathfrak{A}$ is not divisible by $\mathfrak{U}'$, then r is the exponent of ω' with respect to z'. The assumption that $\mathfrak{A}$ is not divisible by $\mathfrak{U}'$ means that ω is not divisible by z, and from this it follows that ω' too is not divisible by z'.

2. The exponent of an integral rational function of z of degree m

$$x = a_0 + a_1 z + \cdots + a_m z^m$$

is equal to m. For this function becomes infinite of the mth order at every point contained in $\mathfrak{U}$. The exponent of the product $x\omega$ is equal to $r + m$.

3. The exponent of a sum

$$\omega = c_1\omega_1 + c_2\omega_2 + \cdots + c_s\omega_s$$

is at most equal to the greatest of the exponents of $\omega_1, \omega_2, \ldots, \omega_s$.

4. We determine a sequence of integral functions of z

$$\lambda_1, \lambda_2, \lambda_3, \ldots$$

by the following recurrent determination:

1) λ_1 constant (exponent $r_1 = 0$),

2) λ_2 a non-rational integral function of z with the smallest possible exponent r_2,

3) λ_s an integral function of z not contained in the form

$$x_1\lambda_1 + x_2\lambda_2 + \cdots + x_{s-1}\lambda_{s-1} \tag{3}$$

with the smallest possible exponent r_s,

and continue this sequence as long as one can still find functions λ_s that are not contained in the form (3).

By 4., 3) the functions $\lambda_1, \lambda_2, \ldots, \lambda_s$ are linearly independent in the sense that no linear equation with rational coefficients in z subsists between them. Since there are not more than n linearly independent functions, s cannot be greater than n. On the other hand, as long as $s \leqq n$, there are always still integral functions that are not linearly dependent on $\lambda_1, \lambda_2, \ldots, \lambda_{s-1}$, and among them also ones of least exponent. We therefore obtain n functions

$$L = (\lambda_1, \lambda_2, \ldots, \lambda_n), \tag{4}$$

and these form a basis of the field Ω with respect to z. They form a minimal basis; for if this were not the case, then for some $s \leqq n$ a function $x_1\lambda_1 + x_2\lambda_2 + \cdots + x_s\lambda_s$ would have to be divisible by a linear function $z - c$ without x_s being divisible by $z - c$. If one reduces $x_1, x_2, \ldots, x_s$ to their residues with respect to $z - c$, an equation results:

$$c_1\lambda_1 + c_2\lambda_2 + \cdots + c_s\lambda_s = (z - c)\mu,$$

in which the $c_1, c_2, \ldots, c_s$ are constant, c_s different from zero, and μ an integral function. Now μ is not contained in the form (3), and by 2. and 3. its exponent is smaller than the exponent of λ_s, which contradicts the determination 4. 3). Hence:

5. The functions $\lambda_1, \lambda_2, \ldots, \lambda_n$ form a minimal basis of Ω with respect to z. It is called a normal basis.

The exponents $r_1, r_2, r_3, \ldots, r_n$ of these functions satisfy the condition:

$$r_1 = 0, \quad 1 \leqq r_2 \leqq r_3 \ldots \leqq r_n. \tag{5}$$

For if $r_s < r_{s-1}$, then by 3. 3) one would have had to take λ_s in place of λ_{s-1}.

Of the functions λ_i none is divisible by z; for if $\lambda_s = z\mu$, then μ would have a smaller exponent than λ_s and would not be contained in the form (3). Thus μ would have to take the place of λ_s.

6. Accordingly (by 1.)

$$r_1, r_2, \ldots, r_n$$

are the exponents of

$$\lambda'_1 = z'^{r_1}\lambda_1, \quad \lambda'_2 = z'^{r_2}\lambda_2, \ldots, \lambda'_n = z'^{r_n}\lambda_n \tag{6}$$

in regard to $z' = 1 : z$, and these functions are the elements of a normal basis of Ω with respect to z'.

For if we denote by x'_i integral rational functions of z', of degree at most m, then

$$x_i = z^m x'_i \tag{7}$$

is an integral rational function of z. Now if we had

$$\lambda'_s = \sum_{0,\, s-1}^{i} x'_i\lambda'_i,$$

then from (6) and (7) it would follow:

$$z^m \lambda_s = \sum_{0,\, s-1}^{i} x_i z^{r_s - r_i} \lambda_i,$$

and this is impossible, because the λ_i are linearly independent; and there can also be no function λ'_s of lower exponent than r_s, because from it one could derive a function λ_s of lower exponent.

The field discriminants $D_z, D_{z'}$ with respect to the variable z and z' are

$$D_z = \Delta(\lambda_1, \lambda_2, \ldots, \lambda_n),$$
$$D_{z'} = \Delta(\lambda'_1, \lambda'_2, \ldots, \lambda'_n),$$

and from this it follows by (6):

$$D_{z'} = z'^{2(r_1 + r_2 + \cdots + r_n)} D_z. \tag{8}$$

If, then, δ is the degree of D_z, then

$$2(r_1 + r_2 + \cdots + r_n) - \delta$$

is the number of the vanishing roots of $D_{z'}$, and from § 190, 5. it follows:

$$w_z = 2(r_1 + r_2 + \cdots + r_n). \tag{9}$$

7. The branching number w_z is therefore always an even number.

§ 194. Differential quotients.

Where we make no use of continuity, we cannot introduce the differential quotients in the ordinary way. We take two functions α, β from Ω, between which the irreducible equation

$$F(\alpha, \beta) = 0 \tag{1}$$

holds; we denote by $F'(\alpha), F'(\beta)$ the derived functions and define the differential quotient as a function of the field Ω by the formula:

$$\left(\frac{d\alpha}{d\beta}\right) = -\frac{F'(\beta)}{F'(\alpha)}. \tag{2}$$

Let α_0, β_0 be the finite values of the functions α, β at a point $\mathfrak{P}$. Then $F(\alpha_0, \beta_0) = 0$, and we can represent equation (1) in the following way:

$$\begin{aligned} 0 = {} & (\alpha - \alpha_0) F'(\alpha_0) + (\beta - \beta_0) F'(\beta_0) \\ & + \tfrac{1}{2} \Big[(\alpha - \alpha_0)^2 F''(\alpha_0, \alpha_0) + 2(\alpha - \alpha_0)(\beta - \beta_0) F''(\alpha_0, \beta_0) \\ & + (\beta - \beta_0)^2 F''(\beta_0, \beta_0) \Big] + \cdots \end{aligned} \tag{3}$$

The two mutually reciprocal functions

$$\frac{\alpha - \alpha_0}{\beta - \beta_0}, \quad \frac{\beta - \beta_0}{\alpha - \alpha_0}$$

cannot both be infinite at the point $\mathfrak{P}$. Let therefore:

$$\left(\frac{\alpha - \alpha_0}{\beta - \beta_0}\right)_0$$

be the finite value of the first of these quotients. Then it follows from (3), if $F'(\alpha_0)$ is not zero:

$$\left(\frac{\alpha-\alpha_0}{\beta-\beta_0}\right)_0 = -\frac{F'(\beta_0)}{F'(\alpha_0)} = -\left(\frac{F'(\beta)}{F'(\alpha)}\right)_0 = \left(\frac{d\alpha}{d\beta}\right)_0. \tag{4}$$

This equation holds for all points $\mathfrak{P}$, with the possible exception of a finite number[41].

1. A function η in Ω which at all points, with the exception of a finite number, satisfies the condition:

$$\eta_0 = \left(\frac{\alpha-\alpha_0}{\beta-\beta_0}\right)_0 \tag{5}$$

is identical with $\left(\frac{d\alpha}{d\beta}\right)$.

For from (4) and (5) it follows that the function

$$\left(\frac{d\alpha}{d\beta}\right) - \eta$$

has infinitely many zero points and must therefore vanish identically.

Formula (5) can accordingly likewise serve as the definition of the differential quotient.

From this the principal theorems on the differential quotients follow simply:

2. Let α, β, γ be three functions in Ω; then

$$\left(\frac{d\alpha}{d\beta}\right)\left(\frac{d\beta}{d\gamma}\right) = \left(\frac{d\alpha}{d\gamma}\right). \tag{6}$$

For, for infinitely many points,

$$\left(\frac{\alpha-\alpha_0}{\beta-\beta_0}\right)_0\left(\frac{\beta-\beta_0}{\gamma-\gamma_0}\right)_0 = \left(\frac{\alpha-\alpha_0}{\gamma-\gamma_0}\right)_0,$$

and from this Theorem 2 follows as in 1.

Accordingly, to every function α in Ω one can assign a function $d\alpha$ in such a way that, for any two of these functions:

$$\left(\frac{d\alpha}{d\beta}\right) = \frac{d\alpha}{d\beta} \tag{7}$$

becomes an actual quotient of two functions $d\alpha, d\beta$. These functions we call the differentials of α, β. The differentials of the constants, and only these, are equal to zero. The other differentials are all completely determined as soon as one of them has been assumed arbitrarily (for example, constant).

3. If $\alpha, \beta, \gamma, \ldots,$ are arbitrary functions in Ω which satisfy an equation

$$F(\alpha, \beta, \gamma, \ldots) = 0 \tag{8}$$

then

$$F'(\alpha)\,d\alpha + F'(\beta)\,d\beta + F'(\gamma)\,d\gamma + \cdots = 0. \tag{9}$$

To prove this theorem, take a point $\mathfrak{P}$ at which none of the functions $\alpha, \beta, \gamma \ldots$ or $d\alpha, d\beta, d\gamma, \ldots$ becomes infinite or zero and at which the derived functions $F'(\alpha), F'(\beta), F'(\gamma)$ also do not vanish, and arrange, as in (3), the function F according to powers and products of $\alpha-\alpha_0, \beta-\beta_0, \gamma-\gamma_0, \ldots$

[41] Among the points to be excepted are in any case those in which α_0 or β_0 becomes infinite. For at these points the functions $\alpha-\alpha_0$, $\beta-\beta_0$ are not defined at all.

It then turns out that equation (9) is satisfied for infinitely many points, and hence identically. As special cases of (9) there result:

$$
\begin{aligned}
d(\alpha+\beta) &= d\alpha + d\beta,\\
d(\alpha\beta) &= \beta\, d\alpha + \alpha\, d\beta,\\
d\left(\frac{\alpha}{\beta}\right) &= \frac{\beta\, d\alpha - \alpha\, d\beta}{\beta^2}.
\end{aligned}
\tag{10}
$$

The concept of the differential can be carried over without further ado to holomorphic functionals, provided the functional variables are regarded as constants. Theorem 3. and the formulae (10) which follow from it then still hold when $\alpha, \beta, \gamma, \ldots$ are holomorphic functionals and $F(\alpha, \beta, \gamma, \ldots)$ is an integral rational function of $\alpha, \beta, \gamma, \ldots$.

If α is an integral rational function of z, then $d\alpha/dz$ is likewise an integral rational function, namely the derivative of α with respect to z.

§ 195. Representation of the Differential Quotients by Polygon Quotients.

We take any function z of the nth order in Ω and a minimal basis $\omega_1, \omega_2, \ldots, \omega_n$ with respect to z. All integral functions of z of the field Ω are then representable in the form:

$$\omega = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n \tag{1}$$

and arise from the basis functional

$$\tau = t_1\omega_1 + t_2\omega_2 + \cdots + t_n\omega_n \tag{2}$$

when integral rational functions $x_1, x_2, \ldots, x_n$ of z are put for the functional variables $t_1, t_2, \ldots, t_n$.

If we put, as in § 184,

$$N(t-\tau) = F(t,z),$$

then τ satisfies the irreducible rational equation:

$$F(\tau, z) = 0,$$

and one obtains, by § 194, 3.:

$$F'(\tau)\, d\tau + F'(z)\, dz = 0, \tag{3}$$

and here $F'(\tau)$ is the branching functional (§ 184).

By (1) and (2),

$$d\omega = d\tau + \omega_1\, dx_1 + \omega_2\, dx_2 + \cdots + \omega_n\, dx_n, \tag{4}$$

where in $d\tau$ the t_i are to be replaced by the x_i, and from (3) there follows therefore the theorem:

4. If ω is an integral function of z, then

$$F'(\tau)\frac{d\omega}{dz} = -F'(z) + \omega_1\frac{dx_1}{dz} + \cdots + \omega_n\frac{dx_n}{dz} \tag{5}$$

is an integral functional.

On the other hand, $F'(\omega)$ is divisible by $F'(\tau)$, by § 182, 8. If we therefore put:

$$F'(\omega) = \varrho\, F'(\tau), \tag{6}$$

then ϱ is an integral functional, and from (4), when one replaces the t_i by the x_i, there results:

$$
\begin{aligned}
-F'(z) &= F'(\omega)\left(\frac{d\omega}{dz} - \omega_1\frac{dx_1}{dz} - \cdots - \omega_n\frac{dx_n}{dz}\right) \\
&= \varrho\, F'(\tau)\left(\frac{d\omega}{dz} - \omega_1\frac{dx_1}{dz} - \cdots - \omega_n\frac{dx_n}{dz}\right),
\end{aligned} \tag{7}
$$

and from this it follows that $F'(z)$ too (after replacing the t_i by the x_i) is divisible by ϱ.

Accordingly, with regard to the geometric analogy, we call ϱ the functional of the double points in ω, z.

If $\mathfrak{P}$ is a point at which z is finite and $F'(\tau)$ is different from zero, so that it is not a branch point in z, then by 4. an integral function ω of z has a finite differential quotient at $\mathfrak{P}$. A fractional function η of z which is finite at $\mathfrak{P}$ can be represented as a quotient of two integral functions $\omega' : \omega$ whose denominator ω does not vanish at $\mathfrak{P}$.

From this it follows:

$$
\frac{d\eta}{dz} = \frac{1}{\omega^2}\left(\omega'\frac{d\omega}{dz} - \omega\frac{d\omega'}{dz}\right),
$$

and consequently $d\eta : dz$ is likewise finite at the point $\mathfrak{P}$.

In order to recognise the behaviour of an arbitrary differential quotient $d\alpha : d\beta$ at any point $\mathfrak{P}$, take a function z which becomes zero of the first order at $\mathfrak{P}$, and denote by r, s the order numbers of $\alpha - \alpha_0, \beta - \beta_0$ at the point $\mathfrak{P}$. For the case that α becomes infinite at $\mathfrak{P}$, one sets $\alpha_0 = 0$ and obtains a negative order number r. The corresponding statement holds when β becomes infinite.

Then by § 185:

$$
\begin{aligned}
\alpha - \alpha_0 &= z^r\alpha', \\
\beta - \beta_0 &= z^s\beta',
\end{aligned} \tag{8}
$$

where α' and β' are functions in Ω which become neither zero nor infinite at $\mathfrak{P}$.

From this there results, by § 194:

$$
\begin{aligned}
\frac{d\alpha}{dz} &= rz^{r-1}\alpha' + z^r\frac{d\alpha'}{dz}, \\
\frac{d\beta}{dz} &= sz^{s-1}\beta' + z^s\frac{d\beta'}{dz},
\end{aligned}
$$

$$
\frac{\beta - \beta_0}{\alpha - \alpha_0}\frac{d\alpha}{d\beta} = \frac{r + z\dfrac{d\alpha'}{\alpha'\,dz}}{s + z\dfrac{d\beta'}{\beta'\,dz}}, \tag{9}
$$

and if one passes to the point $\mathfrak{P}$:

$$
\left(\frac{\beta - \beta_0}{\alpha - \alpha_0}\frac{d\alpha}{d\beta}\right)_0 = \frac{r}{s}, \tag{10}
$$

which is a finite value different from zero.

From this it follows:

5. The order number of the differential quotient $d\alpha : d\beta$ at any point $\mathfrak{P}$ is equal to the difference of the order numbers of $\alpha - \alpha_0$ and $\beta - \beta_0$.

If $r > s$, or $r < s$, then

$$
\left(\frac{\alpha - \alpha_0}{\beta - \beta_0}\right)_0 = 0 \text{ or } = \infty,
$$

and consequently, by (10), also

$$
\left(\frac{d\alpha}{d\beta}\right)_0 = 0 \text{ or } = \infty.
$$

If $r = s$, then from (10) there follows

$$\left(\frac{d\alpha}{d\beta}\right)_0 = \left(\frac{\alpha - \alpha_0}{\beta - \beta_0}\right)_0. \tag{11}$$

Here every trace of the variable z has dropped out, and this formula holds for every point $\mathfrak{P}$ without exception.

Accordingly the zero points and the points of infinity of the differential quotients can be determined exactly, and these functions can thereby be represented as polygon quotients.

If a is the order number of $\alpha - \alpha_0$ at a point $\mathfrak{P}$ according to § 188, then the branch polygon $\mathfrak{Z}_\alpha$ contains the point $\mathfrak{P}$ $a-1$ times or $-a-1$ times, according as a is positive or negative. If a is negative, then the denominator polygon $\mathfrak{A}$ of α contains the point $\mathfrak{P}$ $(-a)$ times, and consequently in both cases the point $\mathfrak{P}$ is contained $(a-1)$ times in $\mathfrak{Z}_\alpha : \mathfrak{A}^2$. If $\mathfrak{Z}_\beta, \mathfrak{B}, b$ have the corresponding meaning for β, then the quotient:

$$\frac{\mathfrak{Z}_\alpha \mathfrak{B}^2}{\mathfrak{Z}_\beta \mathfrak{A}^2}$$

thus contains the point $\mathfrak{P}$ $(a-b)$ times.

But by 5. the order number of $d\alpha : d\beta$ at $\mathfrak{P}$ is just as great, and from this there results the representation:

$$\frac{d\alpha}{d\beta} = \frac{\mathfrak{Z}_\alpha \mathfrak{B}^2}{\mathfrak{Z}_\beta \mathfrak{A}^2}, \tag{12}$$

in which, to repeat it, $\mathfrak{Z}_\alpha, \mathfrak{Z}_\beta$ are the branch polygons, $\mathfrak{A}, \mathfrak{B}$ the denominators of α, β.

§ 196. Genus of the field Ω.

Since in a polygon quotient numerator and denominator are of the same order, it follows from the last formula [§ 195, (12)]:

$$w_\alpha - 2a = w_\beta - 2b. \tag{1}$$

If, then, we denote by n the order and by w the branching number for an arbitrary variable, which, as we have seen in § 193, is an even number, then the integer:

$$p = \tfrac{1}{2} w - n + 1, \tag{2}$$

is an invariant integer belonging to the field Ω, and it is called the genus of the field. That this number cannot be negative follows from § 193, (5) and (9), according to which

$$p = (r_2 - 1) + (r_3 - 1) + \cdots + (r_{n_1} - 1) \tag{3}$$

holds, so that it consists entirely of summands of which none is negative.

Two functions α, β we call a primitive pair of the field Ω if all functions in Ω can be expressed rationally by α and β. If α is of order m and β of order n, then by § 171 it is necessary and sufficient for a primitive pair that the irreducible equation subsisting between α and β:

$$F(\overset{n}{\alpha}, \overset{m}{\beta}) = 0 \tag{4}$$

be of degree n in α and of degree m in β.

The functions $F'(\alpha)$, $F'(\beta)$ are now to be represented by polygon quotients. We set:

$$\alpha = \frac{\mathfrak{M}}{\mathfrak{A}}, \quad \beta = \frac{\mathfrak{N}}{\mathfrak{B}}, \tag{5}$$

in which $\mathfrak{A}$, $\mathfrak{M}$ are of order m, and $\mathfrak{B}$, $\mathfrak{N}$ of order n. $\mathfrak{A}$ then has no point in common with $\mathfrak{M}$, nor $\mathfrak{B}$ with $\mathfrak{N}$. We set:

$$\begin{aligned} F(\alpha,\beta) &= a_0\alpha^n + a_1\alpha^{n-1} + \cdots + a_n, \\ F'(\alpha) &= na_0\alpha^{n-1} + (n-1)a_1\alpha^{n-2} + \cdots + a_{n-1}, \\ \alpha F'(\alpha) &= -a_1\alpha^{n-1} - 2a_2\alpha^{n-2} - \cdots - na_n. \end{aligned} \tag{6}$$

From the second of these equations one concludes that in the denominator of $F'(\alpha)$ there occur no points that are not contained in $\mathfrak{A}$ or in $\mathfrak{B}$, and from the third it follows that this denominator can at most be $= \mathfrak{A}^{n-2}\mathfrak{B}^m$. We therefore set:

$$F'(\alpha) = \frac{\mathfrak{L}}{\mathfrak{A}^{n-2}\mathfrak{B}^m}. \tag{7}$$

It is now to be proved that $\mathfrak{L}$ is divisible by the branch polygon $\mathfrak{Z}_\beta$.

Let us first assume that $\mathfrak{A}$ and $\mathfrak{B}$ are relatively prime to $\mathfrak{Z}_\beta$; then there is an integral rational function x of β which vanishes at no point of $\mathfrak{Z}_\beta$ and for which $\omega = x\alpha$ becomes an integral function of β. If we set

$$f(\omega) = x^n F(\alpha,\beta),$$

then:

$$f'(\omega) = x^{n-1}F'(\alpha).$$

If τ is the basis form of Ω with respect to β, then by § 182 $f'(\omega)$ is divisible by $f'(\tau)$; and since by assumption β becomes infinite at no branch point, $f'(\tau)$, and consequently $F'(\alpha)$ also, is divisible by the branch polygon $\mathfrak{Z}_\beta$, and our assertion is proved. We set:

$$\mathfrak{L} = \mathfrak{R}\mathfrak{Z}_\beta,$$

$$F'(\alpha) = \frac{\mathfrak{R}\mathfrak{Z}_\beta}{\mathfrak{A}^{n-2}\mathfrak{B}^m}. \tag{8}$$

This formula is thereby proved only under the assumption that neither α nor β becomes infinite at one of the branch points.

But if we make the linear substitution:

$$\begin{aligned} \alpha(\alpha_1+1) &= a\alpha_1, \\ \beta(\beta_1+1) &= b\beta_1, \end{aligned} \tag{9}$$

then by § 190

$$\mathfrak{Z}_\beta = \mathfrak{Z}_{\beta_1},$$

and we can choose the constants a, b so that α_1 and β_1 become infinite at no point of $\mathfrak{Z}_\beta$. We have only to take for a and b any constant values that α and β do not assume at any point of $\mathfrak{Z}_\beta$.

If $F_1(\alpha_1,\beta_1) = 0$ is the rational equation subsisting between α_1, β_1, then:

$$F_1(\alpha_1,\beta_1) = (\alpha_1+1)^n(\beta_1+1)^m F(\alpha,\beta),$$

or also, by (9):

$$\alpha^n\beta^m F_1(\alpha_1,\beta_1) = a^n b^m \alpha_1^n \beta_1^m F(\alpha,\beta);$$

further $d\alpha : d\alpha_1 = \alpha^2 : a\alpha_1^2$, and consequently, having regard to $F = 0$, $F_1 = 0$:

$$\alpha^{n-2}\beta^m F_1'(\alpha_1) = a^{n-1}b^m\alpha_1^{n-2}\beta_1^m F'(\alpha). \tag{10}$$

By (9), α_1, β_1 vanish at the same points and with the same order numbers as α, β, and from this there follow, corresponding to the representations (2) and (4), the polygon representations:

$$\alpha_1 = \frac{\mathfrak{M}}{\mathfrak{A}_1}, \quad \beta_1 = \frac{\mathfrak{N}}{\mathfrak{B}_1}, \tag{11}$$

and from (5), (7) and (10):

$$F_1'(\alpha_1) = \frac{\mathfrak{L}}{\mathfrak{A}_1^{n-2}\mathfrak{B}_1^m}. \tag{12}$$

With this, formula (8) is proved in general.

If one sets, following § 194, (9):

$$F'(\alpha)\,d\alpha + F'(\beta)\,d\beta = 0,$$

then from (8) and § 195, (12) it follows that:

$$F'(\beta) = \frac{\mathfrak{R}\mathfrak{Z}_\alpha}{\mathfrak{A}^n\mathfrak{B}^{m-2}}. \tag{13}$$

Both derivatives $F'(\alpha)$, $F'(\beta)$ therefore vanish at the points of $\mathfrak{R}$, and $\mathfrak{R}$ is called the polygon of the double points in (α, β).

The order of $\mathfrak{A}^n\mathfrak{B}^{m-2}$ is $2n(m-1)$, and from this there results for the order $2r$ of $\mathfrak{R}$ an even number, namely:

$$2r = 2n(m-1) - w_\alpha = 2m(n-1) - w_\beta,$$

and for the genus p there results, by (2):

$$p = (n-1)(m-1) - r. \tag{14}$$

Twenty-seventh Chapter.

Algebraic and Abelian Differentials.

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§ 197. Differentials in Ω.

If z and z_1 are two variables in Ω of the orders n, n_1 with the lower vertices $\mathfrak{U}$, $\mathfrak{U}_1$, with the branching vertices: $\mathfrak{Z}$, $\mathfrak{Z}_1$, and the branching numbers w, w_1, then by § 195, (12)

$$\frac{dz}{dz_1} = \frac{\mathfrak{Z}\mathfrak{U}_1^2}{\mathfrak{Z}_1\mathfrak{U}^2}.$$

If one sets

$$\omega = \frac{\mathfrak{A}\mathfrak{U}^2}{\mathfrak{B}\mathfrak{Z}}, \quad \omega_1 = \frac{\mathfrak{A}\mathfrak{U}_1^2}{\mathfrak{B}\mathfrak{Z}_1}, \tag{1}$$

then it follows from this that

$$\omega\,dz = \omega_1\,dz_1. \tag{2}$$

If $\mathfrak{A}\mathfrak{U}^2$ is equivalent to $\mathfrak{B}\mathfrak{Z}$, then ω is a function in Ω, and ω_1 is likewise a function in Ω. If a, b are the orders of $\mathfrak{A}$ and $\mathfrak{B}$, then [§ 196, (2)]

$$\begin{aligned} b + w &= a + 2n, \\ a &= b + 2p - 2, \end{aligned} \tag{3}$$

where p is the genus of the field Ω.

We now set, in a new symbolic notation,

$$dz = \frac{\mathfrak{Z}}{\mathfrak{U}^2}, \tag{4}$$

$$\omega\, dz = dJ = \frac{\mathfrak{A}}{\mathfrak{B}}, \tag{5}$$

and we call these expressions the differentials belonging to the field Ω.

In the numerator and denominator $\mathfrak{A}$, $\mathfrak{B}$ of a differential, common factors may be added or omitted. If $\mathfrak{A}$ and $\mathfrak{B}$ have no common divisor, then $\mathfrak{A}$ is called the upper vertex, $\mathfrak{B}$ the lower vertex of the differential dJ.

The notation (5) for a differential differs from the similar notation for the functions in Ω in that the degree of the numerator is higher by $(2p-2)$ than that of the denominator.

The differentials $d\alpha$ of the functions in Ω defined in § 194 are special cases of the general differentials (5). For not all of these dJ are differentials of functions in Ω. We distinguish proper differentials $d\alpha$, i.e. those which are differentials of functions in Ω, and improper or Abelian differentials dJ, which are not. For the latter, J itself has no meaning; it acquires one only in the integral calculus, which is not accessible to the purely arithmetical method.

If z is any variable in Ω and dJ a differential, then $dJ : dz$ is a function in α, and $dJ : dz$ is a differential quotient. Here too we distinguish proper and improper differential quotients. The quotient of two differentials $dJ : dJ'$ is likewise always a function in Ω, and can likewise be called a differential quotient.

In order that a polygon quotient $\mathfrak{A} : \mathfrak{B}$ can be the symbol for a differential, it is necessary that the difference $(a-b)$ of the orders of $\mathfrak{A}$ and $\mathfrak{B}$ be equal to $(2p-2)$. But this condition is not sufficient; there is the further requirement that a variable z must exist in Ω for which

$$\mathfrak{U}^2\mathfrak{A} \sim \mathfrak{Z}\mathfrak{B} \tag{6}$$

holds. If this requirement is satisfied, it remains satisfied when z is replaced by any other variable in Ω, and $\mathfrak{A}$ and $\mathfrak{B}$ by equivalent polygons. Thus if $\mathfrak{A}'$, $\mathfrak{A}''$, ... are polygons of the class A determined by $\mathfrak{A}$, then $\mathfrak{A}' : \mathfrak{B}$, $\mathfrak{A}'' : \mathfrak{B}$, ... are likewise differentials.

If

$$A = (\mathfrak{A}_1,\ \mathfrak{A}_2,\ \mathfrak{A}_3,\ \dots)$$

and $\mathfrak{A}_1$, $\mathfrak{A}_2$, $\mathfrak{A}_3$, ... is a basis of A, then the corresponding differential quotients

$$\frac{dJ_1}{dz},\ \frac{dJ_2}{dz},\ \frac{dJ_3}{dz},\ \dots$$

are the basis of a family of functions of finite dimension, and we can call

$$(dJ_1,\ dJ_2,\ dJ_3,\ \dots)$$

the basis of a family of differentials of the same dimension.

Every differential dJ whose lower polygon is $\mathfrak{B}$, or a divisor of $\mathfrak{B}$, can then be represented in the form:

$$dJ = c_1 dJ_1 + c_2 dJ_2 + c_3 dJ_3 + \cdots \tag{7}$$

with constant coefficients.

The simplest differentials are those whose lower polygon is the null polygon, and whose upper polygons $\mathfrak{W}$ are accordingly of order $(2p-2)$. These are called differentials of the first kind and are denoted by dW. They are characterized by the fact that for every variable z the polygons

$\mathfrak{U}^2\mathfrak{W}$ and $\mathfrak{Z}$ must be equivalent, and the $\mathfrak{W}$ accordingly form, assuming their existence, a class of polygons W whose dimension is to be determined.

The polygon $\mathfrak{W}$ is called the fundamental polygon of the differential dW and is named a complete polygon of the first kind. If $\mathfrak{W} = \mathfrak{A}\mathfrak{B}$, then $\mathfrak{A}$ and $\mathfrak{B}$ too are called polygons of the first kind, and indeed complementary polygons of one another.

Every polygon that is not a divisor of a complete polygon of the first kind is said to be of the second kind.

If $\mathfrak{A}$ is a polygon of the first kind, then every polygon $\mathfrak{A}'$ equivalent to $\mathfrak{A}$ is also of the first kind, and we call the class A of $\mathfrak{A}$ a class of the first kind. For if $\mathfrak{A}\mathfrak{B} = \mathfrak{W}$, then $AB = W$, where B is the class of $\mathfrak{B}$, and $\mathfrak{A}'\mathfrak{B} = \mathfrak{W}'$ is equivalent to $\mathfrak{W}$, hence of the first kind.

If, then, q is the number of linearly independent polygons of the first kind that are divisible by a polygon of the class A, then by § 192, (14)

$$q = (A,\ W) = (O,\ B), \tag{8}$$

that is, equal to the dimension of the complementary class.

If A is a class of the second kind, then there is no polygon of the first kind divisible by $\mathfrak{A}$, and

$$(A,\ W) = 0.$$

§ 198. **The Family of Polygons of the First Kind.**

In order to determine the dimension (O, W) of the family of polygons of the first kind and thereby at the same time to establish its existence, we take an arbitrary variable z in Ω with the branch polygon $\mathfrak{Z}$ and the lower polygon $\mathfrak{U}$. If dW is a differential of the first kind, then

$$w = \frac{dW}{dz} = \frac{\mathfrak{W}\mathfrak{U}^2}{\mathfrak{Z}} \tag{1}$$

is a function in Ω, which we call a differential quotient of the first kind.

If $\mathfrak{P}$ is a point at which $(z - z_0)$ becomes infinitely small of order e, then this point occurs $(e-1)$ times in $\mathfrak{Z}$, and consequently $[w(z - z_0)]_0 = 0$. If z becomes infinite in another point $\mathfrak{P}$ in the eth order, then this point occurs e times in $\mathfrak{U}$ and $(e-1)$ times in $\mathfrak{Z}$, hence once in the numerator of wz, and consequently at this point too $(wz)_0 = 0$, and these two requirements, namely:

a) At every point at which z has a finite value z_0, $[w(z - z_0)]_0 = 0$;

b) At a point at which z becomes infinite, $(wz)_0 = 0$, are also sufficient to define a differential quotient of the first kind w.

In order to satisfy the first condition, we take a minimal basis with respect to z:

$$\omega_1,\ \omega_2,\ \ldots,\ \omega_n, \tag{2}$$

put

$$a_{r,\,s} = S(\omega_r\omega_s), \tag{3}$$

so that

$$\Delta = \Sigma \pm a_{1,1}a_{2,2}\ldots a_{n,n} \tag{4}$$

is the field discriminant (with respect to z), and determine a basis

$$\varepsilon_1,\ \varepsilon_2,\ \ldots,\ \varepsilon_n \tag{5}$$

from the equations:

$$\omega_r = \Sigma a_{r,\,i}\varepsilon_i. \tag{6}$$

If (r, s) is a symbol which is $= 0$ when r and s are different, and $= 1$ when $r = s$, then one can determine the rational functions $a'_{r,\,s}$ of z in such a way that

$$(7) \qquad {}^{i}\sum a_{i,\,r} a'_{i,\,s} = (r,\ s)$$

holds, and one obtains from (6):

$$(8) \qquad \varepsilon_r = {}^{i}\sum a'_{i,\,r}\omega_i,$$

and the quantities $a'_{r,\,s}$ have no denominator other than Δ. From this there follows, by reason of (3) and (7):

$$(9) \qquad \begin{aligned} S(\varepsilon_r\omega_s) &= (r,\ s), \\ a'_{r,\,s} &= S(\varepsilon_r\varepsilon_s). \end{aligned}$$

If, conversely, for a system of functions $\eta_1, \eta_2, \ldots, \eta_n$ the condition

$$(10) \qquad S(\eta_r\omega_s) = (r,\ s)$$

is satisfied, then $\eta_r = \varepsilon_r$; for if one sets

$$\eta_r = {}^{i}\sum x_{i,\,r}\varepsilon_i,$$

it follows from (9) and (10) that $x_{r,\,s} = (r,\ s)$. Accordingly the conditions (9) and (6) are completely equivalent.

The quantities ε_i form a basis of Ω, but are not integral functions. The bases (5) and (2) are called complementary to one another.

The quantities

$$(11) \qquad \Delta\varepsilon_1,\ \Delta\varepsilon_2,\ \ldots,\ \Delta\varepsilon_n$$

are integral functions, and if $t_1, t_2, \ldots, t_n$ are functional variables, then

$$(12) \qquad \Delta(\varepsilon_1 t_1 + \varepsilon_2 t_2 + \cdots + \varepsilon_n t_n) = \Delta\varepsilon$$

is the greatest common divisor of the functions (11).

If

$$(13) \qquad r = (z - c_1)(z - c_2)(z - c_3)\ldots$$

is the product of all the linear factors of Δ that are distinct from one another, and

$$(14) \qquad \varkappa = \pi_1\pi_2\pi_3\ldots$$

the product of the distinct prime functionals dividing Δ, then we can put every integral function ϱ of z in Ω that is divisible by r into the form:

$$\varrho = r(x_1\varepsilon_1 + x_2\varepsilon_2 + \cdots + x_n\varepsilon_n),$$

in which $x_1, x_2, \ldots, x_n$ are rational functions of z. Then one obtains by (9)

$$rx_i = S(\varrho\omega_i),$$

and since $\varrho\omega_i$ is divisible by every prime factor of r, by the theorem of § 184, 7. $S(\varrho\omega_i)$ is an integral function in Z divisible by r, and consequently x_i an integral rational function.

From this it follows that

$$\Delta\varrho = r(x_1\Delta\varepsilon_1 + x_2\Delta\varepsilon_2 + \cdots + x_n\Delta\varepsilon_n)$$

is divisible by $r\varepsilon\Delta$, and that accordingly $r\varepsilon$ is an integral functional dividing ϱ.

If π were a prime factor of $r\varepsilon$ not contained in $\varkappa$, then one could assume ϱ to be such that it is not divisible by π,

and ϱ could not be divisible by $r\varepsilon$. In the same way one concludes that each of the prime factors $\pi_1, \pi_2, \ldots$ divides $r\varepsilon$, but none of them more than once; hence, apart from unit factors,

$$r\varepsilon = \varkappa. \tag{15}$$

Thus each of the functions $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ satisfies condition a). Conversely, every function w that satisfies this condition is also contained in the form:

$$w = x_1\varepsilon_1 + x_2\varepsilon_2 + \cdots + x_n\varepsilon_n, \tag{16}$$

in which the $x_1, x_2, \ldots, x_n$ are integral rational functions of z.

For rw is an integral function divisible by $\varkappa$, and consequently, by (9), (16) and § 184, 7.:

$$S(rw\omega_i) = rx_i$$

is an integral function of z divisible by r; consequently

$$x_i = S(w\omega_i). \tag{17}$$

In order, then, to obtain the differential quotients of the first kind, one has to select, among the functions (16), those that satisfy condition b).

To this end we take for (2) the normal basis considered in § 193,

$$\lambda_1, \lambda_2, \ldots, \lambda_n$$

and denote the basis complementary to it by

$$\mu_1, \mu_2, \ldots, \mu_n,$$

which is defined by

$$S(\lambda_r\mu_s) = (r, s) \tag{18}$$

If we then set

$$w = y_1\mu_1 + y_2\mu_2 + \cdots + y_n\mu_n, \tag{19}$$

then, if r_i is the exponent of λ_i, it follows by (17) that

$$\frac{y_i}{z^{r_i-1}} = S\left(wz\frac{\lambda_i}{z^{r_i}}\right).$$

By the definition of r_i, $\lambda_i : z^{r_i}$ is finite for $z = \infty$, and wz vanishes on account of b). Consequently $y_i : z^{r_i-1}$ must vanish for $z = \infty$, that is, y_i can be at most of degree $(r_i - 2)$ and therefore contains at most $r_i - 1$ constant coefficients.

Since $\mu_1 = 1$, $r_1 = 0$ and the degree of y_1 cannot be negative, y_1 must be identically zero, and it follows that

$$S(w) = 0.$$

This is Abel's theorem for the differential quotients of the first kind.

To show that this requirement on the y_i is also sufficient for the fulfilment of b), we regard the functions in Ω as functions of $z' = 1 : z$. For these,

$$\lambda_1' = z'^{r_1}\lambda_1, \quad \lambda_2' = z'^{r_2}\lambda_2, \quad \ldots, \quad \lambda_n' = z'^{r_n}\lambda_n$$

form a minimal basis by § 193, 6., and

$$\mu_1' = z^{r_1}\mu_1, \quad \mu_2' = z^{r_2}\mu_2, \quad \ldots, \quad \mu_n' = z^{r_n}\mu_n$$

is by (10) the basis complementary to it. Consequently $z'\mu' = 0$ for $z' = 0$ [by a), applied to z'], and hence

$$z^{r_i-1}\mu_i = 0,$$

$$zy_i\mu_i = \frac{y_i}{z^{i-2}} z^{r_i-1}\mu_i = 0 \text{ for } z = \infty,$$

if the degree of y_i is not higher than $(r_i - 1)$.

It has thereby been shown that all functions contained in the form (19), in which the y_i are integral rational functions of z of degree at most $r_i - 2$, are differential quotients of the first kind, and that conversely all differential quotients of the first kind can be represented in this form, and the principal theorem follows:

1. The class of differentials of the first kind is of the dimension

$$(O, W) = (r_2 - 1) + (r_3 - 1) + \cdots + (r_n - 1) = p.$$

From this the following further results: if A is any class of polygons of order a and W the principal class of the first kind, then by § 192, (14):

$$(A, W) \underset{>}{=} (O, W) - a = p - a,$$

hence, if $a \underset{<}{=} p - 1$:

$$(A, W) \underset{>}{=} 1.$$

Hence, if a is less than p, there are always polygons of the first kind that are divisible by $\mathfrak{A}$, and from this it follows:

2. Every polygon of the $(p-1)$th and lower order is of the first kind.

By § 192, 5., if m is not greater than p, one can choose an m-gon $\mathfrak{M}$ so that, if M is the class determined by $\mathfrak{M}$,

$$(M, W) = p - m$$

is obtained, and if one here sets $m = p$, it follows that there are classes of order p of the second kind.

§ 199. The Riemann-Roch Theorem.

The theorem known under the name of the Riemann-Roch theorem has the purpose of determining the dimensions of polygon classes.

We consider first a proper polygon class A of order n, take in it two coprime polygons $\mathfrak{A}$, $\mathfrak{A}'$ and set:

$$z = \frac{\mathfrak{A}'}{\mathfrak{A}}.$$

If then $\mathfrak{A}''$ is another polygon of the same class, we set

$$\omega = \frac{\mathfrak{A}''}{\mathfrak{A}}, \quad \frac{\omega}{z} = \frac{\mathfrak{A}''}{\mathfrak{A}'}.$$

Thus ω is an integral function of z, and $\omega : z$ an integral function of $1 : z$. If then ω is not constant, the exponent r of ω is equal to 1. Conversely, every integral function ω of z of exponent 0 or 1 is likewise representable in the form $\mathfrak{A}'' : \mathfrak{A}$.

We thus obtain the whole class A from the totality of the integral functions of z whose exponent is $\leqq 1$. These we obtain easily from the normal basis $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ in z, with the sequence of exponents $r_1, r_2, \ldots, r_n$ (§ 193). If λ_s is the last of these functions whose exponent is equal to 1 or 0, then every integral function whose exponent is $\leqq 1$ is representable in the form

$$\omega = c_0 z + c_1\lambda_1 + c_2\lambda_2 + \cdots + c_s\lambda_s \tag{1}$$

(c_0 must be constant, because by the definition of the normal basis it must be an integral function, and if its exponent were positive, the exponent of ω would be greater than one). Accordingly

$$(O, A) = s + 1, \tag{2}$$

and this number is therefore always $\leqq n + 1$.

The upper limit $n + 1$ is reached only when all $r_2, r_3, \ldots, r_n = 1$ and consequently $p = 0$.

Since at least two polygons are contained in a proper class, so that $(O, A) \geqq 2$ must hold, $n = 1$ can occur only in the case $p = 0$.

A function z which becomes 0 and ∞ at only a single point each, and so has order 1, therefore exists only in the case $p = 0$. Every function in Ω can be rationally represented by such a z[42].

If $r_i > 2$, then by § 198 μ_i and $z\mu_i$ are differential quotients of the first kind. By (1) there are therefore polygons $\mathfrak{W}$, $\mathfrak{W}'$ of the first kind which satisfy the condition:

$$\mu_i = \frac{\mathfrak{A}^2\mathfrak{W}}{\mathfrak{Z}}, \quad \mu_i z = \frac{\mathfrak{A}^2\mathfrak{W}'}{\mathfrak{Z}}, \quad z = \frac{\mathfrak{W}'}{\mathfrak{W}} = \frac{\mathfrak{A}'}{\mathfrak{A}},$$

and since $\mathfrak{A}$, $\mathfrak{A}'$ are coprime, $\mathfrak{A}$ must divide $\mathfrak{W}$, and $\mathfrak{A}'$ must divide $\mathfrak{W}'$. Hence A is a class of the first kind.

If A is a class of the second kind, then all the exponents $r_1, r_2, \ldots, r_n$ are therefore equal to 2 or less than 2, and in particular $r_{s+1} = 2, r_{s+2} = 2, \ldots, r_n = 2$. Consequently

$$p = (r_2 - 1) + (r_3 - 1) + \cdots + (r_n - 1) = n - s,$$

and by (2) we have the theorem:

3. The dimension of a proper class A of the second kind is

$$(O, A) = n - p + 1. \tag{3}$$

If, further, A is a proper class of the first kind and

$$AB = W \tag{4}$$

[42]This function field leads to what the geometers call "unicursal curves".

is the complete class of the first kind, then take a polygon $\mathfrak{B}$ in B and form the differential quotients of the first kind:

$$w = \frac{\mathfrak{A}^3\mathfrak{B}}{\mathfrak{Z}}, \quad zw = \frac{\mathfrak{A}^2\mathfrak{A}'\mathfrak{B}}{\mathfrak{Z}},$$

and this form is preserved when $\mathfrak{B}$ is replaced by an equivalent polygon $\mathfrak{B}'$. The dimension (O, B) is therefore as great as the dimension of the family of differential quotients of the first kind w which have the property that zw too is still of the first kind. For this to be the case, in the expression:

$$w = y_1\mu_1 + y_2\mu_2 + \cdots + y_n\mu_n$$

the degrees of the functions $y_1, y_2, \ldots, y_n$ may rise only to the height $r_1 - 2, r_2 - 2, \ldots, r_n - 2$, and from this it follows:

$$\begin{aligned} (O, B) &= (r_{s+1} - 2) + (r_{s+2} - 2) + \cdots + (r_n - 2), \\ p &= (r_{s+1} - 1) + (r_{s+2} - 1) + \cdots + (r_n - 1), \\ (O, B) &= p - n + s, \end{aligned} \tag{5}$$

and by (2):

$$(O, A) - (O, B) = n - p + 1.$$

Now by § 192, (14) $(O, B) = (A, W)$, and we have the Riemann-Roch theorem for classes of the first kind:

4. The dimension of a proper class A of the first kind is

$$(O, A) = n - p + 1 + (A, W). \tag{6}$$

This formula contains (3) within itself, because for polygons of the second kind $(A, W) = 0$.

Since the dimension of a proper class is at least equal to 2, it follows for a class of the second kind that

$$n \geqq p + 1,$$

and herein is contained the proof of a theorem of Riemann:

5. A function whose order is less than $(p + 1)$ is of the first kind.

It follows further from this that the class W of the polygons of the first kind is a proper one. For suppose W had the divisor $\mathfrak{M}$, and let $W = \mathfrak{M}A$; then $(O, W) = (O, A) = p$. If one takes $\mathfrak{A}$ in A relatively prime to $\mathfrak{M}$, then in W there is only one polygon divisible by $\mathfrak{A}$, namely $\mathfrak{A}\mathfrak{M}$, and consequently $(A, W) = 1$. Accordingly formula (6) gives:

$$n = 2p - 2,$$

hence equal to the order of W. The order of A is therefore just as great as that of W, and consequently $\mathfrak{M} = \mathfrak{O}$.

Theorem 4. holds unchanged for improper classes as well. In order to prove this, let A first be of the first kind and of divisor $\mathfrak{M}$:

$$\begin{aligned} A &= \mathfrak{M}A', \\ AB &= W, \\ A'B' &= W, \\ B' &= \mathfrak{M}B, \quad (\S\,192,\ 8.) \\ (A', W) &= (O, B'). \end{aligned} \tag{7}$$

Let the orders of $\mathfrak{M}, A, B$ be m, a, b, so that $a + b$ is the order of W, i.e.

$$a + b = 2(p - 1). \tag{8}$$

By (7) and § 192 we have

$$(O, B) \geqq (O, B') - m, \tag{9}$$

and by (6), applied to A':

$$(O, A) = (O, A') = a - m - p + 1 + (O, B'),$$

hence by (8):

$$(O, A) - \tfrac{1}{2}a \leqq (O, B) - \tfrac{1}{2}b.$$

But since we can interchange A and B, there results from this

$$(O, A) - \tfrac{1}{2}a = (O, B) - \tfrac{1}{2}b,$$

and if one puts

$$(O, B) = (A, W),$$
$$\tfrac{1}{2}(a - b) = a - p + 1$$

then:

$$(O, A) = a - p + 1 + (A, W)$$

in exact agreement with (6).

In order to derive the corresponding theorem for improper classes of the second kind as well, let

$$A = \mathfrak{M}A' \tag{10}$$

be an improper class of the second kind with the divisor $\mathfrak{M}$.

In a proper class of the second kind, C, whose dimension is greater than the order a of A, there are by § 192, 5. always polygons that are divisible by a polygon of the class A, and C is therefore also divisible by A.

From § 192, (15) it follows:

$$(O, A) \geqq (O, C) - c + a,$$

and from (3):

$$(O, C) = c - p + 1,$$

hence

$$(O, A) \geqq a - p + 1. \tag{11}$$

Now A' is a proper class of the same dimension as A, and hence, if m is the order of $\mathfrak{M}$, by (6):

$$(O, A) = (O, A') = a - m - p + 1 + (A', W), \tag{12}$$

hence by (11):

$$(A', W) \geqq m. \tag{13}$$

From this it follows that A' must be of the first kind.

If $W = A'B'$, then from (13) it follows:

$$(A', W) = (O, B') \geqq m. \tag{14}$$

If now $(O, B') > m$, then in B' one could find a polygon $\mathfrak{MB}$ divisible by $\mathfrak{M}$ (§ 192, 5.), and then

$$\mathfrak{A}'\mathfrak{MB} = \mathfrak{AB} = \mathfrak{W}$$

would be a complete polygon of the first kind; $\mathfrak{A}$ itself would therefore be of the first kind, contrary to the assumption. Hence $(O, B') = (A', W) = m$ and

$$(O, A) = a - p + 1. \tag{15}$$

So here too the Riemann-Roch theorem holds in the form 3.

If A consists of an isolated polygon $\mathfrak{M}$, then $(O, A) = 1$, hence $a = p$, and from this it follows:

6. An isolated polygon of the second kind must be a p-gon, and conversely every p-gon of the second kind is an isolated polygon.

7. If a point $\mathfrak{P}$ divides the divisor of an improper class A of the second kind, and if $A = \mathfrak{P}A'$, then A' must be of the first kind.

For by the theorem (6) and (15) we have:

$$\begin{aligned}(O, A) &= a - p + 1,\\ (O, A') &= a - p + (A', W),\\ (O, A) &= (O, A'),\end{aligned}$$

hence $(A', W) = 1$.

§ 200. Differentials of the second and third kind.

We can now represent invariantly the condition for a differential:

$$dJ = \frac{\mathfrak{A}}{\mathfrak{B}}, \tag{1}$$

which we expressed in § 197, (6) with reference to a variable z in the form:

$$\mathfrak{U}^2\mathfrak{A} \sim \mathfrak{Z}\mathfrak{B},$$

by using the complete polygons of the first kind, according to which

$$\mathfrak{U}^2\mathfrak{W} \sim \mathfrak{Z}$$

holds:

$$\mathfrak{A} \sim \mathfrak{W}\mathfrak{B},$$

or, by introducing the classes:

1. The polygon quotient (1) is a differential, if the classes A, B of $\mathfrak{A}$ and $\mathfrak{B}$ are relatively prime and satisfy the condition:

$$A = WB. \tag{2}$$

If a and b are the order numbers of A and B, then:

$$a - b = 2p - 2, \tag{3}$$

and since A is in any case of the second kind, by § 199, (15):

$$(O, A) = b + p - 1. \tag{4}$$

If B consists of a single point $\mathfrak{P}$, then $(O, A) = p$, and the dimension of A is just as great as that of W; if therefore

$$(\mathfrak{W}_1, \mathfrak{W}_2, \ldots, \mathfrak{W}_p)$$

is a basis of W, then

$$(\mathfrak{P}\mathfrak{W}_1, \mathfrak{P}\mathfrak{W}_2, \ldots, \mathfrak{P}\mathfrak{W}_p)$$

is a basis of A. Hence A is an improper class with the divisor $\mathfrak{P}$, and it follows:

2. A single point $\mathfrak{P}$ cannot be the lower polygon of a differential.

If the class A has a divisor $\mathfrak{M}$ of degree m, then $\mathfrak{M}$ must be contained in the divisor of B; for $\mathfrak{M}$ must divide each of the polygons $\mathfrak{W}\mathfrak{B}$, and consequently, since W is a proper class, each of the $\mathfrak{B}$. If

$$\mathfrak{M}A' = A, \quad \mathfrak{M}B' = B,$$

then

$$(O, A) = (O, A') \tag{5}$$

must hold. But by (3) and (4), applied to A and A', if A' is of the second kind:

$$(O, A) = (O, A') + m,$$

hence $(O, A) > (O, A')$; consequently $m = 0$ must hold.

But A' can be of the first kind only if B is an isolated polygon and $B = \mathfrak{M}$. Then:

$$A' = W, \quad (O, A') = p,$$

so that $(O, A) = p$, $b = 1$ must hold, and we come to the case of theorem 2. Consequently:

3. Every polygon of more than one point can be the lower polygon of a differential.

By a differential of the second kind one understands one whose lower polygon is a power of a single point $\mathfrak{P}$

. A differential whose lower polygon consists of only two simple points is called a differential of the third kind. We prove the theorem:

4. Every differential dJ can be composed linearly and with constant coefficients out of differentials of the first, second and third kind.

This theorem can also be stated, in accordance with (4), by saying that the class A has a basis

$$\mathfrak{A}_1, \mathfrak{A}'_2, \ldots, \mathfrak{A}_{b+p-1} \tag{6}$$

of such a kind that the quotients

$$dJ_r = \frac{\mathfrak{A}_r}{\mathfrak{B}} \tag{7}$$

are differentials of one of the three kinds.

In order to prove it by complete induction, we assume that for some $\mathfrak{B}$ such a basis has been found, and we seek one of the same sort for the lower polygon $\mathfrak{P}\mathfrak{B}$, where $\mathfrak{P}$ is an arbitrary point. Such a basis we may assume in the form:

$$\mathfrak{P}\mathfrak{A}_1, \mathfrak{P}\mathfrak{A}_2, \ldots, \mathfrak{P}\mathfrak{A}_{b+p-1}, \mathfrak{A}_{b+p}, \tag{8}$$

in which $\mathfrak{A}_{b+p}$ must be a polygon equivalent to $\mathfrak{W}\mathfrak{B}\mathfrak{P}$ that does not contain the point $\mathfrak{P}$.

If the point $\mathfrak{P}$ divides $\mathfrak{B}$, we put:

$$\mathfrak{B} = \mathfrak{M}\mathfrak{P}^m,$$

so that $\mathfrak{P}$ no longer divides $\mathfrak{M}$. The polygon $\mathfrak{W}\mathfrak{P}^{m+1}$ then belongs to a proper class, in which therefore a polygon $\mathfrak{N}$ not divisible by $\mathfrak{P}$ exists.

We then put:

$$\mathfrak{A}_{b+p} = \mathfrak{M}\mathfrak{N},$$

and from this we obtain a differential of the second kind:

$$dJ_{b+p} = \frac{\mathfrak{A}_{b+p}}{\mathfrak{B}\mathfrak{P}} = \frac{\mathfrak{N}}{\mathfrak{P}^{m+1}}. \tag{9}$$

If $\mathfrak{P}$ does not divide $\mathfrak{B}$, then one takes a point $\mathfrak{P}_0$ which divides $\mathfrak{B}$. Then $\mathfrak{W}\mathfrak{P}_0\mathfrak{P}$ belongs to a proper class and contains a polygon $\mathfrak{N}$ not divisible by $\mathfrak{P}$. If then

$$\mathfrak{B} = \mathfrak{P}_0\mathfrak{M}, \quad \mathfrak{A}_{b+p} = \mathfrak{M}\mathfrak{N},$$

then

$$dJ_{b+p} = \frac{\mathfrak{A}_{b+p}}{\mathfrak{B}\mathfrak{P}} = \frac{\mathfrak{N}}{\mathfrak{P}_0\mathfrak{P}} \tag{10}$$

is

a differential of the third kind. With this, Theorem 4. is proved, and even with the sharpening that the differentials of the third kind occurring in it contain an arbitrary fixed point $\mathfrak{P}_0$ in the lower polygon.

§ 201. The Residues.

If $\mathfrak{P}$ is a point which divides m times into the lower polygon $\mathfrak{B}$ of a differential dJ ($m \geqq 0$), then one takes a function z in Ω which becomes infinitely large in the first order at $\mathfrak{P}$. Then, by § 197, (1) and § 185, (5), one may put:

$$\frac{dJ}{dz} = a_{m-2}z^{m-2} + a_{m-3}z^{m-3} + \cdots + a_0 + a_{-1}z^{-1} + \eta z^{-2}, \tag{1}$$

in which the a are constants, and η is a function in Ω that is finite at $\mathfrak{P}$.

The coefficient $-a_{-1}$ of $-z^{-1}$ is called the residue of the differential dJ with respect to the point $\mathfrak{P}$.

The residue of dJ at the point $\mathfrak{P}$ can be different from zero only if $m > 0$, that is, if the point $\mathfrak{P}$ occurs in the lower polygon of dJ. Such points shall be called poles of dJ.

1. The residue of a proper differential is equal to zero. For by differentiation of a power of z, z^{-1} can never arise.

If for z one takes a function z_1 which likewise becomes infinite of the first order at $\mathfrak{P}$, then one may put:

$$z = cz_1 + \eta_1, \quad z^{-1} = c^{-1}z_1^{-1} + \eta_2,$$

$$\frac{dz}{dz_1} = c + z_1^{-2}\eta_3, \tag{2}$$

in which the c are constants, and η_1, η_2, η_3 are finite at $\mathfrak{P}$.

If one puts

$$\frac{a_{m-2}z^{m-1}}{m-1} + \frac{a_{m-3}z^{m-2}}{m-2} + \cdots + a_0 = \omega,$$

then from (1) there results:

$$\frac{dJ}{dz_1} = \frac{d\omega}{dz_1} + a_{-1}z^{-1}\frac{dz}{dz_1} + z^{-2}\eta',$$

where η' remains finite at $\mathfrak{P}$, and in it, by 1., only the part

$$a_{-1}z^{-1}\frac{dz}{dz_1}$$

gives a contribution to the term with z_1^{-1}, and by (2) this has the coefficient a_{-1}. From this it follows:

2. The residue of dJ is independent of the choice of the function z.

The following theorem holds further:

3. The sum of all residues of a differential is equal to zero.

We extend the expression $\mathfrak{A}/\mathfrak{B}$ of the differential dJ, if necessary, by the addition of points in the numerator and denominator, so that the mutually distinct points of the lower polygon

$$\mathfrak{P}_1\mathfrak{P}_2\ldots\mathfrak{P}_n \tag{3}$$

form a polygon of a proper class. The poles of dJ are then contained among these $\mathfrak{P}$, and possibly other points as well, for which the residue of dJ is equal to zero.

As denominator of dJ we may then take

$$\mathfrak{B} = \mathfrak{P}_1^{m_1}\mathfrak{P}_2^{m_2}\ldots\mathfrak{P}_n^{m_n} \tag{4}$$

if we add the points taken in excess back again in the numerator.

We take a function z in Ω of the nth order, which becomes infinite at each of the points $\mathfrak{P}_i$ and at each of them only to the first order, and form the development for the point $\mathfrak{P}_i$:

$$\frac{dJ}{dz} = a_{m-2}^{(i)}z^{m-2} + \cdots + a_0^{(i)} + a_{-1}^{(i)}z^{-1} + \eta^{(i)}z^{-2}. \tag{5}$$

If we admit the value 0 for the coefficient $a_r^{(i)}$, we can begin at all the points $\mathfrak{P}_i$ with the same power z^{m-2}. The sum of the residues of dJ is

$$-\sum^{i} a_{-1}^{(i)}.$$

According to § 182, 11. a system of functions $\alpha_1, \alpha_2, \ldots, \alpha_n$ can be determined in such a way that α_r becomes infinitely small in the first order at the point $\mathfrak{P}_r$, while at the remaining points $\mathfrak{P}_i$ it remains finite and different from zero.

If one then puts

$$\alpha_1^m\alpha_2^m\ldots\alpha_n^m = \alpha_i^m\varrho_i,$$

there results

$$\begin{aligned}
&\varrho_1 = 0^m \text{ in } \mathfrak{P}_2, \mathfrak{P}_3, \ldots, \mathfrak{P}_n, \quad \text{finite and not} = 0 \text{ in } \mathfrak{P}_1,\\
&\varrho_2 = 0^m \text{ „ } \mathfrak{P}_1, \mathfrak{P}_3, \ldots, \mathfrak{P}_n, \qquad \text{„ „ „} = 0 \text{ „ } \mathfrak{P}_2,\\
&\quad \ldots\ldots\ldots\ldots\ldots\ldots\\
&\varrho_n = 0^m \text{ „ } \mathfrak{P}_1, \mathfrak{P}_2, \ldots, \mathfrak{P}_{n-1} \qquad \text{„ „ „} = 0 \text{ in } \mathfrak{P}_n.
\end{aligned} \tag{6}$$

If we now understand by $x_1, x_2, \ldots, x_n$ rational functions of z that are not all identically $= 0$, and set:

$$\eta = x_1\varrho_1 + x_2\varrho_2 + \cdots + x_n\varrho_n, \tag{7}$$

then η can be finite for $z = \infty$, i. e. at the points $\mathfrak{P}_1, \mathfrak{P}_2, \ldots, \mathfrak{P}_n$, only if the x_i remain finite for $z = \infty$, that is, are fractions whose numerator is not of higher degree than the denominator.

For if the $x_1, x_2, \ldots, x_n$ are not all finite for $z = \infty$, then there is a positive exponent r such that the products $x_1z^{-r}, x_2z^{-r}, \ldots, x_nz^{-r}$ are finite for $z = \infty$, and at least one of these products, say x_1z^{-r}, is different from zero. Then

$$z^{-r}\eta = x_1z^{-r}\varrho_1 + x_2z^{-r}\varrho_2 + \cdots + x_nz^{-r}\varrho_n$$

is finite and different from zero at the point $\mathfrak{P}_1$, and η can therefore not be finite at $\mathfrak{P}_1$. In the same way one sees that no relation of the form

$$x_1\varrho_1 + x_2\varrho_2 + \cdots + x_n\varrho_n = 0$$

can subsist, i. e. the $\varrho_1, \varrho_2, \ldots, \varrho_n$ form a basis with respect to z.

If therefore we set

$$\varrho_i \frac{dJ}{dz} = x_{i,1}\varrho_1 + x_{i,2}\varrho_2 + \cdots + x_{i,n}\varrho_n, \tag{8}$$

then

$$S\left(\frac{dJ}{dz}\right) = x_{1,1} + x_{2,2} + \cdots + x_{n,n}. \tag{9}$$

According to (5),

$$z^{-m+2}\frac{dJ}{dz}\varrho_i$$

is finite for $z = \infty$, and consequently the same holds of

$$z^{-m+2}x_{i,1}, \quad x^{-m+2}x_{i,2}, \quad \ldots, \quad z^{-m+2}x_{i,n}. \tag{10}$$

The functions $zx_{r,s}$ therefore contain each of the points $\mathfrak{P}$ at most $m-1$ times in the denominator, and

$$z\frac{dJ}{dz}\varrho_1$$

contains each of the points $\mathfrak{P}_2, \mathfrak{P}_3, \ldots$ in the numerator. ($\mathfrak{P}_1$ in general not.)

Now by (8):

$$z\frac{dJ}{dz}\varrho_1 = zx_{1,1}\varrho_1 + zx_{1,2}\varrho_2 + \cdots + zx_{1,n}\varrho_n.$$

This function must therefore vanish in $\mathfrak{P}_2, \mathfrak{P}_3, \ldots$. At the point $\mathfrak{P}_2$, however, $z\,x_{1,1}\varrho_1, z\,x_{1,3}\varrho_3, \ldots, z\,x_{1,n}\varrho_n$ vanish, while ϱ_2 does not vanish. Consequently $z\,x_{1,2}$ must vanish in $\mathfrak{P}_2$, that is, for $z = \infty$, and $z^2x_{1,2}$ must remain finite. We have therefore:

If r is not equal to s, then $z^2x_{r,s}$ is finite for $z = \infty$.

We now define the rational function $u^{(i)}$ by:

$$x_{i,i} = a^{(i)}_{m-2}z^{m-2} + a^{(i)}_{m-3}z^{m-3} + \cdots + a^{(i)}_{-1}z^{-1} + u^{(i)}z^{-2}, \tag{11}$$

in which the constant $a^{(i)}_\nu$ is to be the same as in (5), and we obtain:

$$\frac{dJ}{dz} - x_{i,i} = z^{-2}\left[\eta^{(i)} - u^{(i)}\right], \tag{12}$$

and by (8):

$$\left[\eta^{(i)} - u^{(i)}\right]\varrho_i = z^2x_{i,1}\varrho_1 + \cdots + z^2x_{i,i-1}\varrho_{i-1} + z^2x_{i,i+1}\varrho_{i+1} + \cdots + z^2x_{i,n}\varrho_n.$$

At the point $\mathfrak{P}_i$, $\eta^{(i)}$ is finite and ϱ_i different from zero, while on the right-hand side everything vanishes in $\mathfrak{P}_i$. Consequently $u^{(i)}$ too must be finite in $\mathfrak{P}_i$, and, since it is rational, also for $z = \infty$; and from (9) and (11) it follows that:

$$S\left(\frac{dJ}{dz}\right) = \overset{i}{\Sigma}a^{(i)}_{m-2}z^{m-2} + \cdots + \overset{i}{\Sigma}a^{(i)}_{-1}z^{-1} + \overset{i}{\Sigma}u^{(i)}z^{-2}. \tag{13}$$

By our assumption about z, the lower polygon $\mathfrak{B}$ of dJ contains no point at which z has a finite value.

If $\mathfrak{P}$ is a point at which z has the finite value z_0, then

$$(z - z_0) = \frac{\mathfrak{N}}{\mathfrak{U}}, \quad \frac{dJ}{dz} = \frac{\mathfrak{A}\mathfrak{U}^2}{\mathfrak{Z}\mathfrak{B}}, \quad (z - z_0)\frac{dJ}{dz} = \frac{\mathfrak{U}\mathfrak{M}\mathfrak{A}}{\mathfrak{Z}\mathfrak{B}}, \tag{14}$$

and the point $\mathfrak{P}$ is contained once more often in $\mathfrak{N}$ than in $\mathfrak{Z}$; consequently:

$$\left[(z-z_0)\frac{dJ}{dz}\right]_0 = 0 \quad \text{(in } \mathfrak{P}\text{)}. \tag{15}$$

This is requirement a) in § 198.

If therefore $\lambda_1, \lambda_2, \ldots, \lambda_n$ is the normal basis in z, and $\mu_1, \mu_2, \ldots, \mu_n$ the complementary basis, then:

$$\frac{dJ}{dz} = y_1\mu_1 + y_2\mu_2 + \cdots + y_n\mu_n, \tag{16}$$

in which the y_i are integral rational functions of z, from which it follows that

$$S\left(\frac{dJ}{dz}\right) = y_1 \quad [\S\,198, (18)] \tag{17}$$

is an integral rational function of z.

Since this function can accordingly contain no negative powers of z, the theorem to be proved follows from (13):

$$\overset{i}{\Sigma} a_{-1}^{(i)} = 0,$$

which no longer has anything to do with the particular assumption about the variable z.

As a special application it follows from this that a differential of the second kind has no residue, and that the two residues of the differentials of the third kind are equal and opposite.

A proper differential $d\sigma$, as we have already seen, has no residue, and the residues of the „logarithmic differential“ $d\sigma : \sigma$ are integers, namely the order numbers in z.

We will conclude these considerations with the proof of the following theorem:

4. If we call differentials of the second kind linearly dependent when a linear combination of them and of differentials of the first kind with constant coefficients is equal to a proper differential, then at most $p+1$ differentials of the second kind are always linearly dependent.

To prove this, we take a p-gon of the second kind:

$$\mathfrak{P}_1\mathfrak{P}_2 \ldots \mathfrak{P}_p \tag{18}$$

with p distinct points. If then $\mathfrak{P}$ is a given point, we set:

$$\mathfrak{U} = \mathfrak{P}\mathfrak{P}_1\mathfrak{P}_2 \ldots \mathfrak{P}_r \tag{19}$$

and obtain a class B whose divisor $\mathfrak{P}$ in any case does not divide, since otherwise $\mathfrak{P}_1\mathfrak{P}_2 \ldots \mathfrak{P}_n$ would have to be of the first kind by § 199, 7. We can therefore determine a function in Ω:

$$z = \frac{\mathfrak{A}}{\mathfrak{U}} \tag{20}$$

in whose lower polygon the point $\mathfrak{P}$ is in any case contained. We now form the proper differential

$$dz = \frac{\mathfrak{Z}}{\mathfrak{U}^2}, \tag{21}$$

whose numerator $\mathfrak{Z}$ (the branch polygon with respect to z) the point $\mathfrak{P}$ does not divide, because it divides $\mathfrak{U}$ only simply, and if we express dz, according to § 200, 4., by differentials of the first, second and third

kind, then the third kind cannot occur in this expression, since otherwise the residues could not be $= 0$. This expression therefore gives a linear dependence among the differentials of the second kind, whose lower polygons are $\mathfrak{P}^2, \mathfrak{P}_1^2, \mathfrak{P}_2^2, \ldots, \mathfrak{P}_p^2$.

If one takes, for any positive exponent m:

$$\mathfrak{U} = \mathfrak{P}^{m-1}\mathfrak{P}_1\mathfrak{P}_2 \dots \mathfrak{P}_p,$$

and sets

$$z = \frac{\mathfrak{A}}{\mathfrak{U}}, \quad dz = \frac{\mathfrak{Z}}{\mathfrak{U}^2},$$

then $\mathfrak{Z}$ is divisible by $\mathfrak{P}^{m-2}$, $\mathfrak{U}^2$ by $\mathfrak{P}^{2m-2}$, and consequently the lower polygon of dz by $\mathfrak{P}^m$. One then obtains a linear dependence among the differentials with the lower polygons

$$\mathfrak{P}^m, \mathfrak{P}^{m-1}, \dots, \mathfrak{P}^2, \mathfrak{P}_1^2, \mathfrak{P}_2^2, \dots, \mathfrak{P}_p^2,$$

and with that Theorem 4 is proved in general.

TABLES.

Table I.
Developments of the sixteen ϑ-quotients (p. 88).
ν runs through all integers from $-\infty$ to $+\infty$.

(1) $$\frac{\vartheta_{11}'\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{11}(v)\,\vartheta_{11}(a)} = 2i\sum \frac{e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}-1}.$$

(2) $$\frac{\vartheta_{11}'\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{11}(v)\,\vartheta_{10}(a)} = 2i\sum \frac{(-1)^\nu e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}-1}.$$

(3) $$\frac{\vartheta_{11}'\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{11}(v)\,\vartheta_{01}(a)} = 2i\sum \frac{q^\nu\, e^{\pi iv}\, e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}-1}.$$

(4) $$\frac{\vartheta_{11}'\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{11}(v)\,\vartheta_{00}(a)} = 2i\sum \frac{(-1)^\nu q^\nu\, e^{\pi iv}\, e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}-1}.$$

(5) $$\frac{\vartheta_{11}'\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{11}(a)} = -2i\sum \frac{e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}+1}.$$

(6) $$\frac{\vartheta_{11}'\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{10}(a)} = 2i\sum \frac{(-1)^\nu e^{2\pi i a\nu}}{q^{2\nu}e^{2\pi iv}+1}.$$

(7) $$\frac{\vartheta_{11}'\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{01}(a)} = 2\sum \frac{e^{2\pi i a\nu}\, q^\nu\, e^{\pi iv}}{q^{2\nu}e^{2\pi iv}+1}.$$

(8) $$\frac{\vartheta'_{11}\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{00}(a)} = 2\sum(-1)^{\nu}\frac{e^{2\pi i a\nu}\,q^{\nu}\,e^{\pi iv}}{q^{2\nu}e^{2\pi iv}+1}.$$

(9) $$\frac{\vartheta'_{11}\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{11}(a)} = 2i\sum\frac{e^{(2\nu+1)\pi ia}}{q^{2\nu+1}e^{2\pi iv}-1}.$$

(10) $$\frac{\vartheta'_{11}\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{10}(a)} = -2\sum\frac{(-1)^{\nu}e^{(2\nu+1)\pi ia}}{q^{2\nu+1}e^{2\pi iv}-1}.$$

(11) $$\frac{\vartheta_{11}\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{01}(a)} = 2i\sum\frac{e^{(2\nu+1)\pi ia}\,q^{\frac{2\nu+1}{2}}\,e^{\pi iv}}{q^{2\nu+1}e^{2\pi iv}-1}.$$

(12) $$\frac{\vartheta'_{11}\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{00}(a)} = -2\sum\frac{(-1)^{\nu}e^{(2\nu+1)\pi ia}\,q^{\frac{2\nu+1}{2}}\,e^{\pi iv}}{q^{2\nu+1}e^{2\pi iv}-1}.$$

(13) $$\frac{\vartheta'_{11}\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{11}(a)} = -2i\sum\frac{e^{(2\nu+1)\pi ia}}{q^{2\nu+1}e^{2\pi iv}+1}.$$

(14) $$\frac{\vartheta'_{11}\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{10}(a)} = 2\sum\frac{(-1)^{\nu}e^{(2\nu+1)\pi ia}}{q^{2\nu+1}e^{2\pi iv}+1}.$$

(15) $$\frac{\vartheta'_{11}\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{01}(a)} = 2\sum\frac{e^{(2\nu+1)\pi ia}\,q^{\frac{2\nu+1}{2}}\,e^{\pi iv}}{q^{2\nu+1}e^{2\pi iv}+1}.$$

(16) $$\frac{\vartheta'_{11}\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{00}(a)} = 2i\sum\frac{(-1)^{\nu}e^{(2\nu+1)\pi ia}\,q^{\frac{2\nu+1}{2}}\,e^{\pi iv}}{q^{2\nu+1}e^{2\pi iv}+1}.$$

Table II.

Second form of the development of the sixteen ϑ-quotients (p. 91).

In Tables II, III, IV, V, m runs through the sequence of the positive even numbers, n through the sequence of the positive odd numbers, thus:

$$m = 2, 4, 6, 8, 10, \ldots,$$
$$n = 1, 3, 5, 7, 9, \ldots$$

Likewise m' is to run through the even numbers, n' through the odd numbers.

(1)
$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \operatorname{cotg}\pi v + \operatorname{cotg}\pi a - 2i\sum\left[\frac{q^m e^{m\pi ia}}{e^{-2\pi iv}-q^m} - \frac{q^m e^{-m\pi ia}}{e^{2\pi iv}-q^m}\right].$$

(2)
$$\frac{\vartheta_{11}'\vartheta_{10}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{10}(a)} = \operatorname{cotg}\pi v - \operatorname{tg}\pi a - 2i\sum(-1)^{\frac{m}{2}}\left[\frac{q^m e^{m\pi ia}}{e^{-2\pi iv}-q^m} - \frac{q^m e^{-m\pi ia}}{e^{2\pi iv}-q^m}\right].$$

(3)
$$\frac{\vartheta_{11}'\vartheta_{01}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{01}(a)} = \frac{1}{\sin\pi v} + 2i\sum\left[\frac{q^{\frac{m}{2}} e^{-m\pi ia}e^{\pi iv}}{e^{2\pi iv}-q^m} - \frac{q^{\frac{m}{2}} e^{m\pi ia}e^{-\pi iv}}{e^{-2\pi iv}-q^m}\right].$$

(4)
$$\frac{\vartheta_{11}'\vartheta_{00}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{00}(a)} = \frac{1}{\sin\pi v} + 2i\sum(-1)^{\frac{m}{2}}\left[\frac{q^{\frac{m}{2}} e^{-m\pi ia}e^{\pi iv}}{e^{2\pi iv}-q^m} - \frac{q^{\frac{m}{2}} e^{m\pi ia}e^{-\pi iv}}{e^{-2\pi iv}-q^m}\right].$$

(5)
$$\frac{\vartheta_{11}'\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{11}(a)} = -\operatorname{tg}\pi\,v + \operatorname{cotg}\pi\,a - 2i\sum\left[\frac{q^m\,e^{-m\pi ia}}{e^{2\pi iv}+q^m} - \frac{q^m\,e^{m\pi ia}}{e^{-2\pi iv}+q^m}\right].$$

(6)
$$\frac{\vartheta_{11}'\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{10}(a)} = -\operatorname{tg}\pi\,v + \operatorname{tg}\pi\,a + 2i\sum(-1)^{\frac{m}{2}}\left[\frac{q^m\,e^{-m\pi ia}}{e^{2\pi iv}+q^m} - \frac{q^m\,e^{m\pi ia}}{e^{-2\pi iv}+q^m}\right].$$

(7)
$$\frac{\vartheta_{11}'\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{01}(a)} = \frac{1}{\cos\pi\,v} + 2\sum\left[\frac{q^{\frac{m}{2}}\,e^{m\pi ia}\,e^{\pi iv}}{q^m\,e^{2\pi iv}+1} + \frac{q^{\frac{m}{2}}\,e^{-m\pi ia}\,e^{-\pi iv}}{q^m\,e^{-2\pi iv}+1}\right].$$

(8)
$$\frac{\vartheta_{11}'\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{10}(v)\,\vartheta_{00}(a)} = \frac{1}{\cos\pi\,v} + 2\sum(-1)^{\frac{m}{2}}\left[\frac{q^{\frac{m}{2}}\,e^{m\pi ia}\,e^{\pi iv}}{q^m\,e^{2\pi iv}+1} + \frac{q^{\frac{m}{2}}\,e^{-m\pi ia}\,e^{-\pi iv}}{q^\pi\,e^{-2\pi iv}+1}\right].$$

(9)
$$\frac{\vartheta_{11}'\,\vartheta_{01}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{11}(a)} = \frac{1}{\sin\pi\,a} + 2i\sum\left[\frac{q^n\,e^{-n\pi ia}}{e^{2\pi iv}-q^n} - \frac{q^n\,e^{n\pi ia}}{e^{-2\pi iv}-q^n}\right].$$

(10)
$$\frac{\vartheta_{11}'\,\vartheta_{00}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{10}(a)} = \frac{1}{\cos\pi\,a} + 2\sum(-1)^{\frac{n-1}{2}}\left[\frac{q^n\,e^{-n\pi ia}}{e^{2\pi iv}-q^n} + \frac{'q^n\,e^{n\pi ia}}{e^{-2\pi iv}-q^n}\right].$$

(11)
$$\frac{\vartheta_{11}'\,\vartheta_{11}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{01}(a)} = 2i\sum\left[\frac{q^{\frac{n}{2}}\,e^{-n\pi ia}\,e^{\pi iv}}{e^{2\pi iv}-q^n} - \frac{q^{\frac{n}{2}}\,e^{n\pi ia}\,e^{-\pi iv}}{e^{-2\pi iv}-q^n}\right].$$

(12) $$\frac{\vartheta'_{11}\,\vartheta_{10}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{00}(a)} = 2\sum(-1)^{\frac{n-1}{2}}\left[\frac{q^{\frac{n}{2}}\,e^{-n\pi ia}\,e^{\pi iv}}{e^{2\pi iv}-q} + \frac{q^{\frac{n}{2}}\,e^{n\pi ia}\,e^{-\pi iv}}{e^{-2\pi iv}-q^n}\right].$$

(13) $$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{\pi\vartheta_{00}(v)\vartheta_{11}(a)} = \frac{1}{\sin\pi a} + 2i\sum\left[\frac{q^n e^{n\pi ia}}{e^{-2\pi iv}+q^n} - \frac{q^n e^{-n\pi ia}}{e^{2\pi iv}+q^n}\right].$$

(14) $$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{\pi\vartheta_{00}(v)\vartheta_{10}(a)} = \frac{1}{\cos\pi a} - 2\sum(-1)^{\frac{n-1}{2}}\left[\frac{q^n e^{n\pi ia}}{e^{-2\pi iv}+q^n} + \frac{q^n e^{-n\pi ia}}{e^{2\pi iv}+q^n}\right].$$

(15) $$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{\pi\vartheta_{00}(v)\vartheta_{01}(a)} = 2\sum\left[\frac{q^{\frac{n}{2}}e^{n\pi ia}e^{-\pi iv}}{e^{-2\pi iv}+q^n} + \frac{q^{\frac{n}{2}}e^{-n\pi ia}e^{\pi iv}}{e^{2\pi iv}+q^n}\right].$$

(16) $$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\vartheta_{00}(v)\vartheta_{00}(a)} = -2i\sum(-1)^{\frac{n-1}{2}}\left[\frac{q^{\frac{n}{2}}e^{n\pi ia}e^{-\pi iv}}{e^{-2\pi iv}+q^n} - \frac{q^{\frac{n}{2}}e^{-n\pi ia}e^{\pi iv}}{e^{2\pi iv}+q^n}\right].$$

Table III.
Development of the ϑ-quotients in trigonometric series (p. 92).

(1) $$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \operatorname{cotg}\pi v + \operatorname{cotg}\pi a + 4\sum q^{\frac{mm'}{2}}\sin\pi(ma+m'v).$$

(2) $$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{10}(a)} = \operatorname{cotg}\pi v - \operatorname{tg}\pi a + 4\sum(-1)^{\frac{m}{2}}q^{\frac{mm'}{2}}\sin\pi(ma+m'v).$$

(3) $$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{01}(a)} = \frac{1}{\sin\pi v} + 4\sum q^{\frac{mn}{2}}\sin\pi(ma+nv).$$

(4) $$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{00}(a)} = \frac{1}{\sin\pi v} + 4\sum(-1)^{\frac{m}{2}}q^{\frac{mn}{2}}\sin\pi(ma+nv).$$

(5) $$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{\pi\vartheta_{10}(v)\vartheta_{11}(a)}$$ Interchange of a and v in (2).

(6) $$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\vartheta_{10}(v)\vartheta_{10}(a)} = \operatorname{tg}\pi v + \operatorname{tg}\pi a - 4\sum(-1)^{\frac{m+m'}{2}}q^{\frac{mm'}{2}}\sin\pi(ma+n'v).$$

(7) $$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{\pi\vartheta_{10}(v)\vartheta_{01}(a)} = \frac{1}{\cos\pi v} + 4\sum(-1)^{\frac{n-2}{2}} q^{\frac{mn}{2}} \cos\pi(ma+nv).$$

(8) $$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{\pi\vartheta_{10}(v)\vartheta_{00}(a)} = \frac{1}{\cos\pi v} + 4\sum(-1)^{\frac{m+n-1}{2}} q^{\frac{mn}{2}} \cos\pi(ma+nv).$$

(9) $$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{11}(a)}$$ Interchange of a and v in (3).

(10) $$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{10}(a)}$$ Interchange of a and v in (7).

(11) $$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{01}(a)} = 4\sum q^{\frac{nn'}{2}} \sin\pi(na+n'v).$$

(12) $$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{\pi\,\vartheta_{01}(v)\,\vartheta_{00}(a)} = 4\sum(-1)^{\frac{n-1}{2}} q^{\frac{nn'}{2}} \cos\pi(na+n'v).$$

(13) $$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{11}(a)}$$ Interchange of a and v in (4).

(14) $$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{10}(a)}$$ Interchange of a and v in (8).

(15) $$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{01}(a)}$$ Interchange of a and v in (12).

(16) $$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\,\vartheta_{00}(v)\,\vartheta_{00}(a)} = -4\sum(-1)^{\frac{n+n'}{2}} q^{\frac{nn'}{2}} \sin\pi(na+n'v).$$

Table IV.
Developments of the elliptic functions (p. 163).

(1) $$\frac{2\varkappa K}{\pi}\operatorname{sn} 2Ku = 4\sin\pi v\sum\frac{(1+q^n)q^{\frac{n}{2}}}{1-2q^n\cos 2\pi u+q^{2n}} = 4\sum\frac{q^{\frac{n}{2}}}{1-q^n}\sin n\pi u.$$

$$\frac{2\varkappa K}{\pi}\operatorname{cn}2Ku = 4\cos\pi v\sum(-1)^{\frac{n-1}{2}}\frac{(1-q^n)q^{\frac{n}{2}}}{1-2q^n\cos 2\pi u+q^{2n}} = 4\sum\frac{q^{\frac{n}{2}}}{1+q^n}\cos n\pi u. \tag{2}$$

$$\frac{2K}{\pi}\operatorname{dn}2Ku = 1+4\sum\frac{(-1)^{\frac{n-1}{2}}q^n(\cos 2\pi u-q^n)}{1-2q^n\cos 2\pi u+q^{2n}} = 1+4\sum\frac{q^{\frac{m}{2}}}{1+q^m}\cos m\pi u. \tag{3}$$

$$\frac{2K}{\pi}\frac{1}{\operatorname{sn}2Ku} = \frac{1}{\sin\pi u}+4\sin\pi n\sum\frac{q^{\frac{m}{2}}(1+q^m)}{1-q^m\cos 2\pi u+q^{2m}} = \frac{1}{\sin\pi u}+4\sum\frac{q^n}{1-q^n}\sin\pi nu. \tag{4}$$

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku} = \operatorname{cotg}\pi u+4\sin 2\pi u\sum\frac{q^{\frac{m}{2}}}{1-q^m\cos 2\pi u+q^m} = \operatorname{cotg}\pi u-4\sum\frac{q^m}{1+q^m}\sin\pi mu. \tag{5}$$

$$\frac{2K}{\pi}\frac{\operatorname{dn}2Ku}{\operatorname{sn}2Ku} = \frac{1}{\sin\pi u}+4\sin\pi u\sum\frac{(-1)^{\frac{m}{2}}q^{\frac{m}{2}}(1-q^m)}{1-q^m\cos 2\pi u+q^{2m}} = \frac{1}{\sin\pi u}-4\sum\frac{q^n}{1+q^n}\sin n\pi u. \tag{6}$$

$$\frac{2\varkappa' K}{\pi}\frac{1}{\operatorname{cn}2Ku} = \frac{1}{\cos\pi u}-4\cos\pi u\sum\frac{(-1)^{\frac{m}{2}}q^{\frac{n}{2}}(1+q^m)}{1+2q^m\cos 2\pi u+q^{2m}} = \frac{1}{\cos\pi u}-4\sum(-1)^{\frac{n-1}{2}}\frac{q^n}{1+q^n}\cos n\pi u. \tag{7}$$

$$\frac{2\varkappa' K}{\pi}\frac{\operatorname{sn}Ku}{\operatorname{cn}2Ku} = \operatorname{tg}\pi u+4\sin 2\pi u\sum\frac{(-1)^{\frac{m}{2}}q^m}{1-q^m\cos 2\pi u+q^{2m}} = \operatorname{tg}\pi u+4\sum(-1)^{\frac{m}{2}}\frac{q^m}{1+q^m}\sin m\pi u. \tag{8}$$

$$\frac{2K}{\pi}\frac{\operatorname{dn}2Ku}{\operatorname{cn}2Ku} = \frac{1}{\cos\pi u}+4\cos\pi u\sum\frac{q^{\frac{m}{2}}(1+q^m)}{1+2q^m\cos 2\pi u+q^{2m}} = \frac{1}{\cos\pi u}+4\sum(-1)^{\frac{n-1}{2}}\frac{q^n}{1-q^n}\cos n\pi u. \tag{9}$$

$$\frac{2\varkappa' K}{\pi}\frac{1}{\operatorname{dn}2Ku} = 1-4\sum(-1)^{\frac{n-1}{2}}\frac{q^n(q^n+\cos 2\pi u)}{1+2q^n\cos 2\pi u+q^{2n}} = 1+4\sum(-1)^{\frac{m}{2}}\frac{q^{\frac{m}{2}}}{1+q^m}\cos m\pi u. \tag{10}$$

(11)
$$\frac{2\varkappa\varkappa' K}{\pi}\frac{\operatorname{sn} 2Ku}{\operatorname{dn} 2Ku} = 4\sin\pi u\sum\frac{(-1)^{\frac{n-1}{2}}q^{\frac{n}{2}}(1-q^n)}{1+2q^n\cos 2\pi u+q^{2n}}$$
$$= 4\sum\frac{q^{\frac{n}{2}}}{1-q^n}\sin n\pi u.$$

(12)
$$\frac{2\varkappa K}{\pi}\frac{\operatorname{cn} 2Ku}{\operatorname{dn} 2Ku} = 4\cos\pi u\sum\frac{(-1)^{\frac{n-1}{2}}q^{\frac{n}{2}}(1+q^n)}{1+2q^n\cos 2\pi u+q^{2n}}$$
$$= 4\sum\frac{q^{\frac{n}{2}}}{1+q^n}\cos n\pi u.$$

Table V.
Development of the transcendents of the second kind (p. 164).

(1)
$$\frac{d\log H(2Ku)}{\pi\,du} = \frac{2K}{\pi}Z(2Ku)+\frac{d\log\operatorname{sn} 2Ku}{\pi\,du}$$
$$= \operatorname{cotg}\pi u+4\sin 2\pi u\sum\frac{q^m}{1-2q^m\cos 2\pi u+q^{2m}},$$
$$= \operatorname{cotg}\pi u+4\sum\frac{q^m}{1-q^m}\sin m\pi u.$$

(2)
$$\frac{d\log H(2Ku+K)}{\pi\,du} = \frac{2K}{\pi}Z(2Ku)+\frac{d\log\operatorname{cn} 2Ku}{\pi\,du},$$
$$= -\operatorname{tg}\pi u-4\sin 2\pi u\sum\frac{q^m}{1+2q^m\cos 2\pi u+q^m},$$
$$= -\operatorname{tg}\pi u+4\sum\frac{(-1)^{\frac{m}{2}}q^m}{1-q^m}\sin m\pi u.$$

(3)
$$\frac{d\log\Theta(2Ku)}{\pi\,du} = \frac{2K}{\pi}Z(2Ku),$$
$$= 4\sin 2\pi u\sum\frac{q^n}{1-2q^n\cos 2\pi u+q^{2n}},$$
$$= 4\sum\frac{q^{\frac{m}{2}}\sin\pi mu}{1-q^m}.$$

(4)
$$\frac{d\log\Theta(2Ku+K)}{\pi\,du} = \frac{2K}{\pi}Z(2Ku)+\frac{d\log\operatorname{dn} 2Ku}{\pi\,du},$$
$$= -4\sin 2\pi u\sum\frac{q^n}{1+2q^n\cos 2\pi u+q^{2n}},$$
$$= 4\sum(-1)^{\frac{m}{2}}\frac{q^{\frac{m}{2}}}{1-q^m}\sin m\pi u.$$

Table VI.

List of class invariants.

(To the nineteenth chapter.)

$$f(\sqrt{-1}) = \sqrt[4]{2}, \qquad m = 1$$

$$f_1(\sqrt{-2}) = \sqrt[4]{2}, \qquad m = 2$$

$$f(\sqrt{-3}) = \sqrt[3]{2}, \qquad m = 3$$

$$f_1(\sqrt{-4}) = \sqrt[3]{8}, \qquad m = 4$$

$$f(\sqrt{-5})^4 = 1 + \sqrt{5}, \qquad m = 5$$

$$f_1(\sqrt{-6})^6 = 4 + 2\sqrt{2}, \qquad m = 6$$

$$f(\sqrt{-7}) = \sqrt{2}, \qquad m = 7$$

$$f_1(\sqrt{-8})^8 = 8 + 8\sqrt{2}, \qquad m = 8$$

$$f(\sqrt{-9})^3 = \sqrt[4]{2}\,(1 + \sqrt{3}), \qquad m = 9$$

$$\sqrt{2}\,f_1(\sqrt{-10})^2 = 1 + \sqrt{5}, \qquad m = 10$$

$$f(\sqrt{-11}) = x, \quad x^3 - 2x^2 + 2x - 2 = 0, \qquad m = 11$$

$$f_1(\sqrt{-12})^4 = 2\sqrt[6]{2}\,(1 + \sqrt{3}), \qquad m = 12$$

$$f(\sqrt{-13})^4 = 3 + \sqrt{13}, \qquad m = 13$$

$$f_1(\sqrt{-14})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = 1 + \sqrt{2}, \qquad m = 14$$

$$f(\sqrt{-15})^3 = \sqrt{2}\,(1 + \sqrt{5}), \qquad m = 15$$

$$f_1(\sqrt{-16})^4 = 2\sqrt[4]{8}\,(1 + \sqrt{2}), \qquad m = 16$$

$$f(\sqrt{-17})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = \frac{1 + \sqrt{17}}{2}, \qquad m = 17$$

$$f_1(\sqrt{-18})^3 = \sqrt{2}\,(2 + \sqrt{6}), \qquad m = 18$$

$$f(\sqrt{-19}) = x, \quad x^3 - 2x - 2 = 0, \qquad m = 19$$

$$f_1(\sqrt{-20})^4 = \sqrt{8}\,x, \quad x^2 = \frac{1 + \sqrt{5}}{2}\,(2x + 1), \qquad m = 20$$

$$2f(\sqrt{-21})^{12} = (\sqrt{3} + \sqrt{7})^3(3 + \sqrt{7})^2, \qquad m = 21$$

$$f_1(\sqrt{-22})^2 = \sqrt{2}\,(1 + \sqrt{2}), \qquad m = 22$$

$$f(\sqrt{-23}) = \sqrt{2}\,x, \quad x^3 - x - 1 = 0, \qquad m = 23$$

$$f_1(\sqrt{-24})^{24} = 2^9(1 + \sqrt{2})^2(2 + \sqrt{3})^3(\sqrt{2} + \sqrt{3})^3, \qquad m = 24$$

$$\sqrt[4]{8}\,f(\sqrt{-25}) = 1 + \sqrt{5}, \qquad m = 25$$

$$f_1(\sqrt{-26})^2 = \sqrt{2}\,x, \quad x^3 - x^2 = \frac{3 + \sqrt{13}}{2}\,(x + 1), \qquad m = 26$$

$$f(\sqrt{-27})^3 = 2x, \quad x^3 - 3x^2 - 3x - 1 = 0, \qquad m = 27$$

$$f_1(\sqrt{-28})^4 = 2\sqrt{2}\,(3 + \sqrt{7}), \qquad m = 28$$

$$f(\sqrt{-29})^4 = 2x, \qquad m = 29$$

$$2x^3 - 9x^2 - 8x - 5 = \sqrt{29}\,(x + 1)^2,$$

$$f_1(\sqrt{-30})^6 = 2\sqrt{2}\,(3+\sqrt{10})(2+\sqrt{5}), \qquad m=30$$

$$f(\sqrt{-31}) = \sqrt{2}\,x, \quad x^3 - x^2 - 1 = 0, \qquad m=31$$

$$f_1(\sqrt{-32})^8 = 8x, \qquad m=32$$

$$x^2 - 8(1+\sqrt{2})^2 x - 2(1+\sqrt{2}) = 0,$$

$$\sqrt{2}\,f(\sqrt{-33})^6 = (3+\sqrt{11})(1+\sqrt{3})^3, \qquad m=33$$

$$f_1(\sqrt{-34})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = \frac{3+\sqrt{17}}{2}, \qquad m=34$$

$$f(\sqrt{-35}) = x, \quad x^3 - 2 = (1+\sqrt{5})(x^2 - x), \qquad m=35$$

$$f_1(\sqrt{-36})^3 = \sqrt[8]{8}\,x, \quad x^2 - 4x - 2 = 2\sqrt{3}\,(x+1), \qquad m=36$$

$$f(\sqrt{-37})^4 = 2(6+\sqrt{37}), \qquad m=37$$

$$f_1(\sqrt{-38})^2 = \sqrt{2}\,x, \qquad m=38$$

$$x^3 - 2x^2 - (2x+1)(1+\sqrt{2}) = 0,$$

$$f(\sqrt{-39})^3 = \sqrt{8}\,x, \quad x^2 = \frac{3+\sqrt{13}}{2}\,(x+1), \qquad m=39$$

$$f_1(\sqrt{-40})^8 = 2(1+\sqrt{5})^2(1+\sqrt{2})^2(3+\sqrt{10}), \qquad m=40$$

$$f(\sqrt{-41}) = \sqrt{2}\,x, \qquad m=41$$

$$\left(x + \frac{1}{x}\right)^2 - \frac{5+\sqrt{41}}{2}\left(x + \frac{1}{x}\right) + \frac{7+\sqrt{41}}{2} = 0,$$

$$\sqrt{8}\,f_1(\sqrt{-42})^6 = (2\sqrt{2}+\sqrt{7})(3+\sqrt{7})^3, \qquad m=42$$

$$f(\sqrt{-43}) = x, \quad x^3 - 2x^2 - 2 = 0, \qquad m=43$$

$$f_1(\sqrt{-44})^4 = \sqrt{8}\,x, \qquad m=44$$

$$(x^3 - 6x^2 + 8x - 3) - \sqrt{11}\,(2x^2 - 2x + 1) = 0,$$

$$f(\sqrt{-45})^{12} = 2(2+\sqrt{5})^3(\sqrt{3}+\sqrt{5})^4, \qquad m=45$$

$$f_1(\sqrt{-46})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = 3 + \sqrt{2}, \qquad m=46$$

$$f(\sqrt{-47}) = \sqrt{2}\,x, \qquad m=47$$

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0,$$

$$f_1(\sqrt{-48})^8 = \qquad m=48$$

$$8\,\sqrt[6]{2}\,(1+\sqrt{3})(\sqrt{2}+\sqrt{3})^2(1+\sqrt{2})^2,$$

$$f(\sqrt{-49})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = 2 + \sqrt{7}, \qquad m=49$$

$$f_1(\sqrt{-50}) = \sqrt[4]{2}\,x, \quad x^3 - x^2 = \frac{1+\sqrt{5}}{2}\,(x+1), \qquad m=50$$

$$f(\sqrt{-51})^3 = 2x, \quad x^3 - 4x^2 - x - 1 = \sqrt{17}\,x^2, \qquad m=51$$

$$f_1(\sqrt{-52})^4 = \sqrt{8}\,x, \qquad m=52$$

$$x^2 - 2(4+\sqrt{13})x - \frac{3+\sqrt{13}}{2} = 0,$$

$$f(\sqrt{-55}) = \sqrt{2}\,x, \quad x^2 - x = \frac{1+\sqrt{5}}{2}, \qquad m = 55$$

$$\sqrt{2}\,f(\sqrt{-57})^6 = (1+\sqrt{3})^3(13+3\sqrt{19}), \qquad m = 57$$

$$\sqrt{2}\,f_1(\sqrt{-58})^2 = 5+\sqrt{29}, \qquad m = 58$$

$$f_1(\sqrt{-60})^{12} = \qquad m = 60$$

$$\sqrt{2}^5(1+\sqrt{5})^2(2+\sqrt{3})^3(\sqrt{3}+\sqrt{5})^3,$$

$$f(\sqrt{-63})^3 = \sqrt{8}\,x, \qquad m = 63$$

$$\sqrt{7}\,(x^2 - x + 1) = \sqrt{3}\,(x^2 + 3x - 1),$$

$$f(\sqrt{-67}) = x, \quad x^3 - 2x^2 - 2x - 2 = 0, \qquad m = 67$$

$$\sqrt{8}\,f_1(\sqrt{-70})^2 = (1+\sqrt{5})^2(1+\sqrt{2}), \qquad m = 70$$

$$f(\sqrt{-71}) = \sqrt{2}\,x, \qquad m = 71$$

$$x^7 - 2x^6 - x^5 + x^4 + x^3 + x^2 - x - 1 = 0,$$

$$f_1(\sqrt{-72})^{24} = 2^6(2+\sqrt{6})^4(1+\sqrt{2})^9(2+\sqrt{3})^6, \qquad m = 72$$

$$f(\sqrt{-73})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = \frac{5+\sqrt{73}}{2}, \qquad m = 73$$

$$\sqrt[3]{2}\,f(\sqrt{-75}) = x, \quad x^3 - 2x^2 - 2x - 4 = 2\sqrt{5}\,x, \qquad m = 75$$

$$\sqrt{8}\,f_1(\sqrt{-78})^6 = (3+\sqrt{13})^3(5+\sqrt{26}), \qquad m = 78$$

$$f_1(\sqrt{-82})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = \frac{15+\sqrt{41}}{2} \qquad m = 82$$

$$16f(\sqrt{-85})^4 = (1+\sqrt{5})^4(9+\sqrt{85}), \qquad m = 85$$

$$f_1(\sqrt{-88})^8 = 4(1+\sqrt{2})^2(3+\sqrt{11})^2(7\sqrt{2}+3\sqrt{11}), \qquad m = 88$$

$$f(\sqrt{-91}) = x, \quad x^3 - 2x^2 - x - 2 = \sqrt{13}\,x, \qquad m = 91$$

$$2f(\sqrt{-93})^{12} = (3\sqrt{3}+\sqrt{31})^3(39+7\sqrt{31})^2, \qquad m = 93$$

$$f(\sqrt{-97})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = \frac{9+\sqrt{97}}{2}, \qquad m = 97$$

$$f(\sqrt{-99})^3 = 2x, \qquad m = 99$$

$$x^3 - 13x^2 - 4x - 1 = \sqrt{33}\,(2x^2 + x),$$

$$\sqrt[8]{2}\,f_1(\sqrt{-100}) = x, \qquad m = 100$$

$$x^2 - x - 1 = \sqrt{5}\,(x+1),$$

$$f_1(\sqrt{-102})^6 = \sqrt{2}^3(1+\sqrt{2})^3(3\sqrt{2}+\sqrt{17})^2, \qquad m = 102$$

$$\sqrt{2}^{13}\,f(\sqrt{-105})^6 =$$

$$(1+\sqrt{5})^3(1+\sqrt{3})^3(\sqrt{3}+\sqrt{7})^3(\sqrt{5}+\sqrt{7}), \qquad m = 105$$

$$f_1(\sqrt{-112})^8 = \qquad m = 112$$

$$\sqrt{2}^7(1+\sqrt{2})^4(2\sqrt{2}+\sqrt{7})^2(3+\sqrt{7}),$$

$$8f_1(\sqrt{-120})^{24} = (1+\sqrt{3})^6(1+\sqrt{5})^6(\sqrt{2}+\sqrt{3})^6 \qquad m = 120$$

$$(\sqrt{3}+\sqrt{5})^6(\sqrt{5}+\sqrt{6})^3(3+\sqrt{10})^2,$$

$$\sqrt{2}^7\, f_1(\sqrt{-130})^2 = (1+\sqrt{5})^3(3+\sqrt{13}), \qquad m = 130$$
$$2f(\sqrt{-133})^4 = (3+\sqrt{7})^2(5\sqrt{7}+3\sqrt{19}), \qquad m = 133$$
$$f_1(\sqrt{-142})^2 = \sqrt{2}\,x, \quad x + \frac{1}{x} = 9 + 5\sqrt{2}, \qquad m = 142$$

$$f(\sqrt{-163}) = x, \quad x^3 - 6x^2 + 4x - 2 = 0, \qquad m = 163$$
$$2^9 f(\sqrt{-165})^{12} = (1+\sqrt{5})^6(\sqrt{3}+\sqrt{5})^6 \qquad m = 165$$
$$(3\sqrt{5}+2\sqrt{11})^2(\sqrt{11}+\sqrt{15})^3$$
$$2^6 f_1(\sqrt{-168})^{24} = (1+\sqrt{3})^{12}(1+\sqrt{2})^6, \qquad m = 168$$
$$(3+\sqrt{7})^6(\sqrt{3}+\sqrt{7})^6(\sqrt{6}+\sqrt{7})^3(\sqrt{7}+\sqrt{8})^2,$$
$$f(\sqrt{-175}) = \sqrt{2}\,x, \qquad m = 175$$
$$2x^3 - 4x^2 + x - 3 = \sqrt{5}(2x^2 - x + 1),$$
$$\sqrt{2}^7 f(\sqrt{-177})^6 = (23 + 3\sqrt{59})(1+\sqrt{3})^9, \qquad m = 177$$
$$\sqrt{2}^5 f_1(\sqrt{-190})^2 = (1+\sqrt{5})^3(3+\sqrt{10}), \qquad m = 190$$
$$f(\sqrt{-193})^2 = \sqrt{2}\,x, \qquad x + \frac{1}{x} = 13 + \sqrt{193}, \qquad m = 193$$
$$\sqrt{2}^9 f_1(\sqrt{-210})^6 = \qquad m = 210$$
$$(1+\sqrt{5})^3(\sqrt{2}+\sqrt{3})^3(\sqrt{3}+\sqrt{7})^3(5\sqrt{5}+\sqrt{14}),$$
$$f_1(\sqrt{-232})^8 = \qquad m = 232$$
$$2(1+\sqrt{2})^6(5+\sqrt{29})^2(99+13\sqrt{58}),$$
$$f_1(\sqrt{-240})^{24} = \sqrt{2}^{17}(1+\sqrt{5})^2(2+\sqrt{3})^3(\sqrt{3}+\sqrt{5})^3 \qquad m = 240$$
$$(1+\sqrt{2})^6(3+\sqrt{10})^6(\sqrt{2}+\sqrt{3})^6(\sqrt{5}+\sqrt{6})^6,$$
$$2f(\sqrt{-253})^4 = (5+\sqrt{23})^2(13\sqrt{11}+9\sqrt{23}), \qquad m = 253$$
$$\sqrt{2}^{13} f(\sqrt{-273})^6 = \qquad m = 273$$
$$(3+\sqrt{13})^3(1+\sqrt{3})^3(\sqrt{3}+\sqrt{7})^3(11\sqrt{13}+15\sqrt{7}),$$
$$8f_1(\sqrt{-280})^8 = (1+\sqrt{5})^4(1+\sqrt{2})^2(3+\sqrt{7})^2 \qquad m = 280$$
$$(\sqrt{5}+\sqrt{7})^2(\sqrt{7}+\sqrt{8})^2(5\sqrt{5}+3\sqrt{14}),$$
$$2^6 f_1(\sqrt{-312})^{24} = (1+\sqrt{3})^{18}(3+\sqrt{13})^6(\sqrt{2}+\sqrt{3})^6 \qquad m = 312$$
$$(2\sqrt{3}+\sqrt{13})^6(3\sqrt{3}+\sqrt{26})^3(5+\sqrt{26})^2,$$
$$\sqrt{2}^{15} f_1(\sqrt{-330})^6 = (1+\sqrt{5})^6 \qquad m = 330$$
$$(\sqrt{5}+\sqrt{6})^3(\sqrt{11}+\sqrt{15})^3(\sqrt{10}+\sqrt{11}),$$
$$\sqrt{2}^{19} f(\sqrt{-345})^6 = (1+\sqrt{5})^6 \qquad m = 345$$
$$(1+\sqrt{3})^3(3\sqrt{3}+\sqrt{23})^3(15\sqrt{5}+7\sqrt{23}),$$
$$2^{16} f(\sqrt{-357})^6 = \qquad m = 357$$
$$(\sqrt{3}+\sqrt{7})^{12}(3+\sqrt{7})^6(\sqrt{17}+\sqrt{21})^3(11+\sqrt{119})^2,$$
$$\sqrt{2}^7 f(\sqrt{-385})^2 = \qquad m = 385$$
$$(1+\sqrt{5})^2(3+\sqrt{11})(\sqrt{5}+\sqrt{7})(\sqrt{7}+\sqrt{11}),$$

$$f_1(\sqrt{-408})^{24} = (1+\sqrt{3})^{12}(\sqrt{2}+\sqrt{3})^{12}(1+\sqrt{2})^6 \qquad m = 408$$
$$(7+\sqrt{51})^6(3\sqrt{2}+\sqrt{17})^4(5\sqrt{2}+\sqrt{51})^3,$$
$$\sqrt{2}^9 f_1(\sqrt{-462})^6 = (\sqrt{3}+\sqrt{7})^3 \qquad m = 462$$
$$(\sqrt{6}+\sqrt{7})^3(\sqrt{7}+\sqrt{11})^3(22\sqrt{22}+39\sqrt{7}),$$
$$2^5 f_1(\sqrt{-520})^8 = (1+\sqrt{5})^6(1+\sqrt{2})^4(3+\sqrt{13})^2 \qquad m = 520$$
$$(3+\sqrt{10})^2(5+\sqrt{26})(57+5\sqrt{130}),$$
$$2f_1(\sqrt{-760})^8 = (1+\sqrt{5})^6(3+\sqrt{10})^2(13+3\sqrt{19})^2 \qquad m = 760$$
$$(3\sqrt{2}+\sqrt{19})^2(2\sqrt{5}+\sqrt{19})^2(51\sqrt{10}+37\sqrt{19}),$$
$$2^{12} f_1(\sqrt{-840})^{24} = (\sqrt{3}+\sqrt{5})^{12}(1+\sqrt{2})^6 \qquad m = 840$$
$$(1+\sqrt{3})^6(1+\sqrt{5})^6(3+\sqrt{7})^6(3+\sqrt{10})^6$$
$$(\sqrt{2}+\sqrt{3})^6(\sqrt{5}+\sqrt{7})^6(\sqrt{6}+\sqrt{7})^6$$
$$(\sqrt{14}+\sqrt{15})^3(3\sqrt{3}+2\sqrt{7})^2(5\sqrt{5}+3\sqrt{14})^2,$$
$$2^{24} f_1(\sqrt{-1320})^{24} = (1+\sqrt{3})^{18}(1+\sqrt{5})^{12}(1+\sqrt{2})^6 \qquad m = 1320$$
$$(3+\sqrt{10})^6(3+\sqrt{11})^6(\sqrt{3}+\sqrt{5})^6(\sqrt{5}+\sqrt{6})^6$$
$$(\sqrt{11}+\sqrt{15})^6(3\sqrt{5}+2\sqrt{11})^6(4\sqrt{2}+\sqrt{33})^6$$
$$(3\sqrt{6}+\sqrt{55})^3(\sqrt{10}+\sqrt{11})^2,$$
$$2^{31} f(\sqrt{-1365})^{12} = (\sqrt{3}+\sqrt{7})^{12}(1+\sqrt{5})^6(3+\sqrt{7})^6 \qquad m = 1365$$
$$(3+\sqrt{13})^6(\sqrt{3}+\sqrt{5})^6(2\sqrt{3}+\sqrt{13})^6$$
$$(\sqrt{35}+\sqrt{39})^3(3\sqrt{7}+\sqrt{65})^2,$$
$$2^{21} f_1(\sqrt{-1848})^{24} = (1+\sqrt{3})^{12}(\sqrt{2}+\sqrt{3})^{12} \qquad m = 1848$$
$$(3+\sqrt{7})^{12}(3+\sqrt{11})^6(\sqrt{3}+\sqrt{7})^6(\sqrt{6}+\sqrt{7})^6$$
$$(\sqrt{7}+\sqrt{11})^6(2\sqrt{2}+\sqrt{7})^6(7\sqrt{2}+3\sqrt{11})^6$$
$$(5\sqrt{3}+\sqrt{77})^6(\sqrt{21}+\sqrt{22})^3(22\sqrt{22}+39\sqrt{7})^2.$$

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Corrections to the First Volume.

(Smaller oversights, which the reader easily corrects for himself, are not listed here.)

Page 185, line 4 from the top, read $(m-s)(n-s)+s$ instead of $(m-s)(n-s)$, after some easily evident changes in the last lines of page 184.
„ 186, formula (4) read $n\mu + m\nu$ instead of $n\nu + m\mu$.
„ 195, line 7 from the top, read two instead of three.
„ 198, „ 13 from the top, „ U_1^2 instead of U_1.
„ 215, „ 7 from the bottom, „ $\lambda : r^2$ instead of λr^2.
„ 221, formulae (13) and (14) read r^{-1} instead of r.
„ 246, line 6 from the bottom, O_1 is to be deleted.
„ 320, formula 7, the second $\Phi(z)\theta(z)^2$ to be replaced by $\Phi(x)\theta(x)^2$.
„ 339, line 4 from the bottom, read real instead of imaginary.
„ 445, „ 19 from the top, „ unsolvable instead of solvable.
„ 487, „ 13 from the top, „ $(mod\, p)$ instead of $(mod\, m)$.
„ 541, „ 9 from the top, „ μ instead of m.
„ 552, „ 11 from the top, „ ψ'/π instead of ψ/π.
„ 657, „ 7 from the bottom, after "normal equation" there is to be inserted: with real roots.

Corrections to the Second Volume.

Pages 231, 232. In the application that is here made of Theorem 4, it was overlooked that the invariants themselves belong to the domain of rationality, and hence that the representation of J as a rational function of θ is a pure identity ($J = J$). Accordingly the paragraph from "From this theorem" at the foot of page 231 to "remain undecided" at the top of page 232 is to be deleted.
„ 640, line 14 from the top, read $F'(\tau)$ instead of $F'(\tau')$.
„ 697, „ 13 from the bottom: The mode of reasoning that follows here must be altered as follows:

From (13) there results, by multiplication by g_s and summation:

$$-n\varkappa\Sigma\delta_s g_s + \alpha u < \Sigma g_s\lambda_s < \Sigma\delta_s g_s + \alpha u.$$

If one sets

$$n\varkappa\Sigma\delta_s|g_s| = g,$$

then all the more, since the sum of the absolute values is not smaller than the absolute value of the sum:

$$-g + \alpha u < \Sigma g_s\lambda_s < g + \alpha u. \tag{14}$$

Now set

$$-g + \alpha u = a, \qquad g + \alpha u = a'$$
$$\alpha\delta = a' - a = 2g$$

and one obtains:

$$a < \Sigma g_s\lambda_s < a'. \tag{15}$$

If one replaces u by $u' = u + \delta$, then a passes over into a' (which would not be the case with the formulation given in the text), and if one sets $a'' = a' + \alpha\delta = g + \alpha u'$, and if λ'_s depends on u' in the same way as λ_s does on u, then there results

$$a' < \Sigma g_s\lambda'_s < a''.$$

By continuing this procedure a sequence of integers

$$\eta, \eta', \eta'', \eta''', \ldots$$

is determined in accordance with the following conditions.

From here one can proceed as in the text at the top of page 698. The text on p. 697 becomes correct through the following four corrections:

Page 697, line 13 from the bottom read $nk\Sigma\delta_s|g_s|$ instead of $k\Sigma\delta_s\gamma_s$.

„ 11 from the bottom „ $-g$ instead of $-ng$.

„ 3 from the bottom „ $-g$ „ $-ng$.

„ 1 from the bottom „ $2g$ „ $(n+1)g$.

Corrections to the Third Volume.

Page 28, formula (23) read K^2 instead of K_2.
„ 124, system of formulae (8) read u instead of u_1.
„ 180, line 9 from the bottom read $\alpha^2\beta\delta$ „ $\alpha^2\beta\gamma$.
„ 185, last line and page 186, formula (6), (7) read g_2^2 instead of g_3^2.
„ 333, line 8 from the top read equivalent, if instead of equivalent. If.
„ 340, line 4 from the bottom read a_{22} instead of a_{12}.
„ 394, „ 18 from the top b) the -5 is to be deleted.
„ 525, formula (1) read Ax^2 instead of Ax.

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